

Algebra, Chapter 10

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Contents

10 Homological Algebra	1
10.1 COMPLEMENTS OF LINEAR ALGEBRA	1
10.1.1 Commutative diagrams	1
10.1.2 The snake diagram	3
10.1.3 Flat modules	10
10.1.4 Modules of finite presentation	13
10.1.5 Homomorphisms of a finitely presented module	16
10.1.6 Structure of flat modules	18
10.1.7 Injective modules	21
10.1.8 Injective cogenerator modules	25
10.1.9 Injective envelopes	27
10.1.10 Structure of injective modules	30
10.2 COMPLEXES OF A-MODULES	32
10.2.1 Complexes of A-modules	32
10.2.2 Operations on complexes	37
10.2.3 The connecting homomorphism and the exact homology sequence . . .	39
10.2.4 Homotopies	45
10.2.5 Split complexes	48
10.2.6 Cone and cylinder of a morphism of complexes	50
10.2.7 The cone of an injective morphism; new definition of the connecting homomorphism	54
10.2.8 Euler-Poincaré characteristics	56

10.2.9	Complexes of right modules, complexes of multimodules	60
10.2.10	Example: de Rham complex	61
10.3	RESOLUTIONS	64
10.3.1	Extension of morphisms of complexes	64
10.3.2	Resolutions	66
10.3.3	The canonical free resolution	69
10.3.4	The canonical injective resolution	72
10.3.5	Finitely generated resolutions	74
10.3.6	Minimal projective resolutions	76
10.3.7	Graded resolutions	78
10.3.8	The standard resolution	79
10.3.9	Resolutions and Grothendieck groups	81
10.4	TORSION PRODUCT	84
10.4.1	Tensor product of two complexes	84
10.4.2	Tensor products and homotopy	89
10.4.3	Tensor product by a flat complex bounded on the right	92
10.4.4	Definition and first properties of the torsion product	93
10.4.5	Connecting homomorphisms and exact sequences	98
10.4.6	Flat modules and torsion products	102
10.4.7	Künneth formula	105
10.4.8	Bounded and flat complexes over a Noetherian ring	109
10.4.9	Generalization to complexes of multimodules	111
10.5	EXTENSION MODULES	112
10.5.1	Complexes of homomorphisms	112
10.5.2	Complexes of homomorphisms and homotopies	115
10.5.3	Definition and first properties of extension modules	119
10.5.4	Connecting homomorphisms and exact sequences	123
10.5.5	Projective modules, injective modules, and extension modules	129
10.5.6	Universal coefficient formula	130

10.5.7	Generalization to complexes of multimodules; the canonical isomorphisms	135
10.6	USE OF NON-CANONICAL RESOLUTIONS	137
10.6.1	Computation of the modules $\text{Tor}^A(\mathbf{P}, \mathbf{M})$ and $\text{Ext}_A(\mathbf{M}, \mathbf{N})$	137
10.6.2	Computation of the mappings $\text{Tor}^A(g, f)$ and $\text{Ext}_A(f, h)$	141
10.6.3	Computation of the connecting homomorphisms	142
10.6.4	Finiteness of extension and torsion modules	146
10.6.5	Homomorphisms $\text{Tor}^B(\mathbf{P}, \mathbf{N}) \otimes_A \mathbf{Q} \rightarrow \text{Tor}^B(\mathbf{P}, \mathbf{N} \otimes_A \mathbf{Q})$ and $\text{Ext}_B(\mathbf{M}, \mathbf{N}) \otimes_A \mathbf{Q} \rightarrow \text{Ext}_B(\mathbf{M}, \mathbf{N} \otimes_A \mathbf{Q})$	147
10.6.6	Homomorphisms $\text{Tor}^B(\mathbf{P}, \mathbf{N} \otimes_A \mathbf{Q}) \rightarrow \text{Tor}^A(\mathbf{P} \otimes_B \mathbf{N}, \mathbf{Q})$ and $\text{Ext}_A(\mathbf{Q}, \text{Hom}_B(\mathbf{N}, \mathbf{M})) \rightarrow \text{Ext}_B(\mathbf{N} \otimes_A \mathbf{Q}, \mathbf{M})$	149
10.6.7	Homomorphisms $\mathbf{B} \otimes_A \text{Tor}^A(\mathbf{E}, \mathbf{F}) \rightarrow \text{Tor}^B(\mathbf{E} \otimes_A \mathbf{B}, \mathbf{B} \otimes_A \mathbf{F})$ and $\mathbf{B} \otimes_A \text{Ext}_A(\mathbf{E}, \mathbf{F}) \rightarrow \text{Ext}_B(\mathbf{B} \otimes_A \mathbf{E}, \mathbf{B} \otimes_A \mathbf{F})$	150
10.6.8	Application: homology and cohomology of groups	151
10.7	COMPOSITION PRODUCT	155
10.7.1	The homomorphism $\text{Ext}_A(\mathbf{N}, \mathbf{P}) \otimes \text{Ext}_A(\mathbf{M}, \mathbf{N}) \rightarrow \text{Ext}_A(\mathbf{M}, \mathbf{P})$	155
10.7.2	The seven computations of the composition product	157
10.7.3	The class associated with an exact sequence	159
10.7.4	Properties of the class associated with an exact sequence	161
10.7.5	Relation between exact sequences and elements of $\text{Ext}_A(\mathbf{M}, \mathbf{N})$	166
10.7.6	Composition product and connecting homomorphisms of extension modules	170
10.7.7	The homomorphism $\text{Ext}_A(\mathbf{P}, \mathbf{Q}) \otimes \text{Tor}^A(\mathbf{P}, \mathbf{M}) \rightarrow \text{Tor}^A(\mathbf{Q}, \mathbf{M})$	174
10.7.8	Composition products and connecting homomorphisms of torsion products	177
10.7.9	Computation of composition products by shifting resolutions	180
10.8	HOMOLOGICAL DIMENSION	183
10.8.1	Projective dimension of a module	183
10.8.2	The homomorphism $\text{Tor}_n^A(\mathbf{P}, \mathbf{M}) \rightarrow \text{Hom}_A(\text{Ext}_A^n(\mathbf{M}, \mathbf{A}), \mathbf{P})$	187
10.8.3	Homological dimension of a ring	189

10.8.4	Rings of homological dimension 0	191
10.8.5	Rings of homological dimension 1	192
10.8.6	Homological dimension of polynomial rings	194
10.8.7	Homological dimension of graded modules	197
10.9	KOSZUL COMPLEXES	201
10.9.1	The complexes $K(u), K.(u, C), K'(u, C)$	201
10.9.2	Functoriality	206
10.9.3	Example 1: the complex $S(\mathbf{L}) \otimes_A \Lambda(\mathbf{L})$	208
10.9.4	Example 2: the case of a free module	210
10.9.5	Example 3: the case $L = A$	214
10.9.6	Completely intersecting families	216
10.9.7	A criterion for completely secant sequences	219
10.9.8	Proof of Theorem 1: first part	221
10.9.9	Proof of Theorem 1: second part	223
10.9.10	Extension class associated with a regular sequence	226
Exercises		230
	Exercises for Section 1: Complements of Linear Algebra	230
	Exercises for Section 2: Complexes of A-Modules	241
	Exercises for Section 3: Resolutions	254
	Exercises for Section 4: Torsion Products	267
	Exercises for Section 5: Extension Modules	272
	Exercises for Section 6: Use of Non-Canonical Resolutions	277
	Exercises for Section 7: Composition Product	296
	Exercises for Section 8: Homological Dimension	306
	Exercises for Section 9: Koszul Complexes	316
Index of Notations		324
Terminological Index		329

Chapter 10

Homological Algebra

10.1 COMPLEMENTS OF LINEAR ALGEBRA

In this section, the letter A denotes a ring. Unless expressly stated otherwise, all modules considered are left modules, and all ideals considered are left ideals.

The definitions and results apply to right modules, by considering them as left modules over the opposite ring.

If M is an A -module and if $a \in A$, we denote by a_M the homothety $x \mapsto ax$ of M . We thus have $1_M = Id_M$ (identity map of M); when there is no possible confusion, one sometimes simply writes 1 instead of 1_M .

Finally, we denote by 0 an A -module reduced to its identity element, chosen once and for all (cf. II, p. 8).

10.1.1 Commutative diagrams

Let, for example, B, C, D, E, F be five sets, and let f be a map from E to F , g be a map from B to C , h be a map from D to E , u be a map from B to D , and v be a map from C to E . To summarize a situation of this kind, one often makes use of diagrams; for example, one summarizes the preceding situation by the following diagram (E, II, p. 14):

$$\begin{array}{ccc}
B & \xrightarrow{g} & C \\
u \downarrow & & \\
D & \xrightarrow[h]{\quad} E \xrightarrow[f]{\quad} F.
\end{array} \tag{1}$$

In such a diagram, the group of symbols $E \xrightarrow{f} F$ schematizes the fact that f is a map from E to F . When there can be no ambiguity about f , one suppresses the letter f and simply writes $E \rightarrow F$.

When B, C, D, E, F are groups (resp. A -modules) and f, g, h, u, v are homomorphisms of groups (resp. A -modules), one says, for brevity, that diagram (1) is a diagram of groups (resp. of A -modules).

In principle, a diagram is not a mathematical object, but only a figure, intended to facilitate the reading of a proof. In practice, one often uses diagrams as abbreviating symbols, which avoid naming all the sets and all the maps that one wishes to consider; one thus says “consider diagram (1)” instead of saying: “let B, C, D, E, F be five sets... and v a map from C to E ”; see for example the statement of prop. 1 of no. 2.

Consider, for example, the following diagram:

$$\begin{array}{ccccccc}
B & \xrightarrow{f} & C & \xrightarrow{g} & D & \xrightarrow{h} & E \\
b \downarrow & & c \downarrow & & d \downarrow & & e \downarrow \\
B' & \xrightarrow{f'} & C' & \xrightarrow{g'} & D' & \xrightarrow{h'} & E'.
\end{array} \tag{2}$$

To every path composed of a certain number of segments of the diagram traversed in the direction indicated by the arrows, one associates a map from the set represented by the origin of the first segment to the set represented by the endpoint of the last segment, namely the composite of the maps represented by the various segments traversed. For every vertex of the diagram, for example C , one agrees that there is a path reduced to C and one associates with it the identity map 1_C .

In (2), there are, for example, three paths starting from B and ending at D' ; the corresponding maps are $d \circ g \circ f$, $g' \circ c \circ f$ and $g' \circ f' \circ b$. One says that a diagram is commutative if, for every pair of paths of the diagram having the same origin and the same endpoint, the two corresponding maps

are equal; in particular, if a path has its endpoint coinciding with its origin, the corresponding map must be the identity.

For diagram (2) to be commutative, it is necessary and sufficient that the relations

$$f' \circ b = c \circ f, \quad g' \circ c = d \circ g, \quad h' \circ d = e \circ h; \quad (3)$$

hold; in other words, it is necessary and sufficient that the three square diagrams extracted from (2) be commutative. Indeed, relations (3) imply $d \circ g \circ f = g' \circ c \circ f$ since $d \circ g = g' \circ c$ and $g' \circ c \circ f = g' \circ f' \circ b$ since $c \circ f = f' \circ b$; therefore the three paths starting from B and ending at D' give the same map. One verifies similarly that the four paths starting from B and ending at E' (resp. the three paths starting from C and ending at E') give the same map. Relations (3) mean that the two paths starting from B (resp. C, D) and ending at C' (resp. D', E') give the same map. All the other pairs of vertices of (2) can be joined by at most one path, and diagram (2) is therefore indeed commutative.

In what follows, we shall leave to the reader the task of formulating and verifying analogous results for other types of diagrams.

10.1.2 The snake diagram

PROPOSITION 1. — Consider a commutative diagram of A-modules

$$\begin{array}{ccccc} M & \xrightarrow{u} & N & \xrightarrow{v} & P \\ f \downarrow & & g \downarrow & & h \downarrow \\ M' & \xrightarrow{u'} & N' & \xrightarrow{v'} & P'. \end{array} \quad (4)$$

Suppose that the two rows of (4) are exact. Then:

(i) If h is injective, we have

$$\text{Im}(g) \cap \text{Im}(u') = \text{Im}(u' \circ f) = \text{Im}(g \circ u). \quad (5)$$

(ii) If f is surjective, we have

$$\text{Ker}(g) + \text{Im}(u) = \text{Ker}(v' \circ g) = \text{Ker}(h \circ v). \quad (6)$$

Let us prove (i). It is clear that we have

$$\text{Im}(u' \circ f) = \text{Im}(g \circ u) \subset \text{Im}(g) \cap \text{Im}(u').$$

Conversely, let $y' \in \text{Im}(g) \cap \text{Im}(u')$. There exists $y \in \mathbf{N}$ such that $y' = g(y)$. Since $v' \circ u' = 0$, we have $0 = v'(y') = v'(g(y)) = h(v(y))$, whence $v(y) = 0$ since h is injective. Since (u, v) is an exact sequence, there exists $x \in \mathbf{M}$ such that $y = u(x)$, whence $y' = g(u(x))$.

Let us prove (ii). Since $v \circ u = 0$ and $v' \circ u' = 0$ it is clear that

$$\text{Ker}(g) + \text{Im}(u) \subset \text{Ker}(v' \circ g) = \text{Ker}(h \circ v).$$

Conversely, let $y \in \text{Ker}(v' \circ g)$. Then $g(y) \in \text{Ker}(v')$, and there exists $x' \in M'$ such that $u'(x') = g(y)$ since the sequence (u', v') is exact. Since f is surjective, there exists $x \in M$ such that $f(x) = x'$, whence $g(y) = u'(f(x)) = g(u(x))$; from this we conclude that $y - u(x) \in \text{Ker}(g)$, which completes the proof.

Lemma 1. — Consider a commutative diagram of A-modules

$$\begin{array}{ccc} \mathbf{M} & \xrightarrow{u} & \mathbf{N} \\ f \downarrow & & g \downarrow \\ \mathbf{M}' & \xrightarrow{u'} & \mathbf{N}'. \end{array} \quad (7)$$

Then there exists one and only one homomorphism $u_1 : \text{Ker}(f) \rightarrow \text{Ker}(g)$ and one and only one homomorphism $u_2 : \text{Coker}(f) \rightarrow \text{Coker}(g)$, such that the diagrams

$$\begin{array}{ccc} \text{Ker}(f) & \xrightarrow{u_1} & \text{Ker}(g) \\ i \downarrow & & \\ \mathbf{M} & \xrightarrow{u} & \mathbf{N} \end{array} \quad (8)$$

and

(9)

$$\begin{array}{ccc}
\mathbf{M}' & \xrightarrow{u'} & \mathbf{N}' \\
p \downarrow & & q \downarrow \\
\text{Coker}(f) & \xrightarrow{u_2} & \text{Coker}(g)
\end{array}$$

Let them be commutative, with i and j denoting the canonical injections, and p and q the canonical surjections.

Indeed, if $x \in \text{Ker}(f)$, we have $f(x) = 0$ and $g(u(x)) = u'(f(x)) = 0$, hence $u(x) \in \text{Ker}(g)$, and the existence and uniqueness of u_1 are then immediate. Likewise, we have

$$u'(f(\mathbf{M})) = g(u(\mathbf{M})) \subset g(\mathbf{N}),$$

hence u' gives by passage to quotients a homomorphism

$$u_2 : \text{Coker}(f) \rightarrow \text{Coker}(g),$$

which is the unique homomorphism for which (9) is commutative.

Now start from a commutative diagram (4) of A -modules; by virtue of Lemma 1 there corresponds to it a commutative diagram

$$\begin{array}{ccccc}
\text{Ker}(f) & \xrightarrow{u_1} & \text{Ker}(g) & \xrightarrow{v_1} & \text{Ker}(h) \\
i \downarrow & & j \downarrow & & k \downarrow \\
\mathbf{M} & & u & & \mathbf{N} & v & \mathbf{P} \\
f \downarrow & & g \downarrow & & h \downarrow & & \\
\mathbf{M}' & & u' & & \mathbf{N}' & v' & \mathbf{P}' \\
p \downarrow & & q \downarrow & & r \downarrow & & \\
\text{Coker}(f) & \xrightarrow{u_2} & \text{Coker}(g) & \xrightarrow{v_2} & \text{Coker}(h)
\end{array} \tag{10}$$

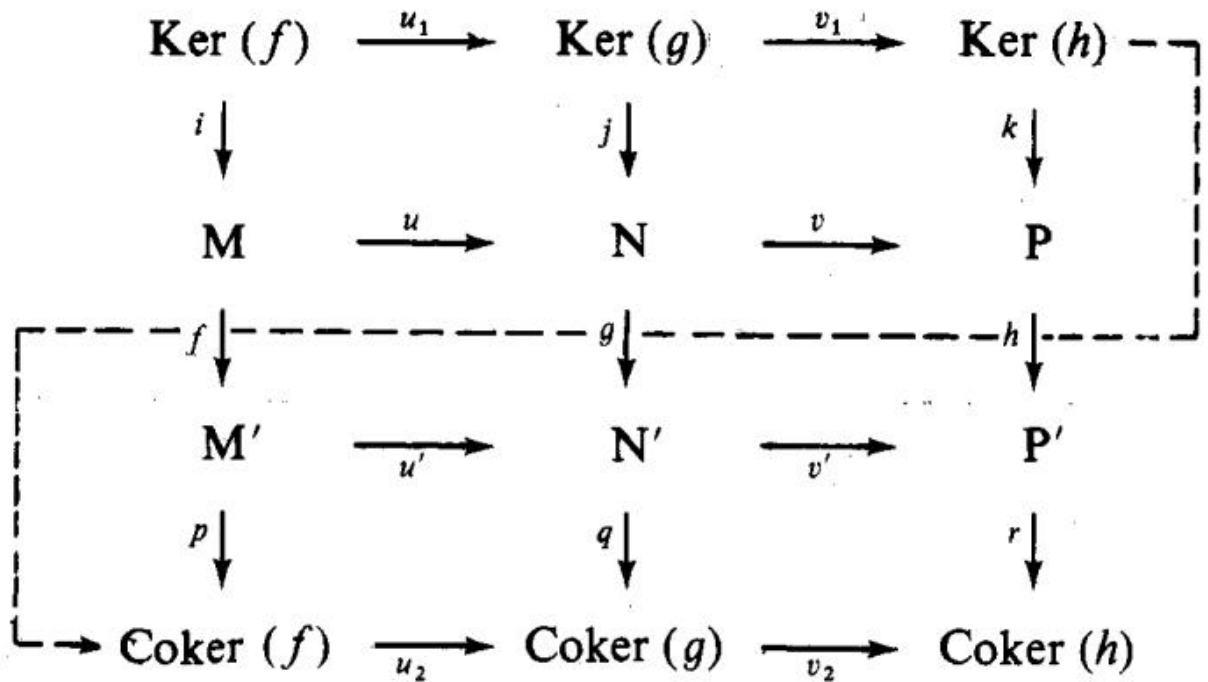
where i, j, k are the canonical injections, p, q, r the canonical surjections, u_1, u_2 (resp. v_1, v_2)

) the homomorphisms deduced from u, u' (resp. v, v') by Lemma 1.

PROPOSITION 2. — Suppose that in the commutative diagram (4), the sequences (u, v) and (u', v') are exact. Then:

- (i) One has $v_1 \circ u_1 = 0$; if u' is injective, the sequence (u_1, v_1) is exact.
- (ii) One has $v_2 \circ u_2 = 0$; if v is surjective, the sequence (u_2, v_2) is exact.
- (iii) Suppose u' injective and v surjective. There then exists one and only one homomorphism $d : \text{Ker}(h) \rightarrow \text{Coker}(f)$ having the following property: if $z \in \text{Ker}(h)$, $y \in \mathbf{N}$ and $x' \in \mathbf{M}'$ satisfy the relations $v(y) = k(z)$ and $u'(x') = g(y)$, we have $d(z) = p(x')$. Moreover, the sequence

(*) $\text{Ker}(f) \xrightarrow{u_1} \text{Ker}(g) \xrightarrow{v_1} \text{Ker}(h) \xrightarrow{d} \text{Coker}(f) \xrightarrow{u_2} \text{Coker}(g) \xrightarrow{v_2} \text{Coker}(h)$ is exact.



Let us prove (i). Since u_1 and v_1 have the same graphs as the restrictions of u and v to $\text{Ker}(f)$ and $\text{Ker}(g)$ respectively, we have $v_1 \circ u_1 = 0$. We have

$$\text{Ker}(v_1) = \text{Ker}(g) \cap \text{Ker}(v) = \text{Ker}(g) \cap \text{Im}(u) = \text{Im}(j) \cap \text{Im}(u).$$

But by prop. 1 (i), we have $\text{Ker}(v_1) = \text{Im}(j \circ u_1) = \text{Im}(u_1)$ if u' is injective. Let us prove (ii). Since u_2 and v_2 come from u and v by passing to quotients, it is clear that $v_2 \circ u_2 = 0$.

Suppose v surjective; since q and p are surjective, we have, by virtue of the hypotheses and of prop. 1 (ii)

$$\begin{aligned}\text{Ker}(v_2) &= q(\text{Ker}(v_2 \circ q)) = q(\text{Ker}(v') + \text{Im}(g)) = q(\text{Ker}(v')) \\ &= q(\text{Im}(u')) = \text{Im}(q \circ u') = \text{Im}(u_2 \circ p) = \text{Im}(u_2).\end{aligned}$$

Finally, let us prove (iii). For $z \in \text{Ker}(h)$ there exists $y \in \mathbf{N}$ such that $v(y) = k(z)$ since v is surjective; moreover, we have $v'(g(y)) = h(k(z)) = 0$ and consequently there exists a unique $x' \in \mathbf{M}'$ such that $u'(x') = g(y)$ since u' is injective. Let us show that the element $p(x') \in \text{Coker}(f)$ is independent of the element $y \in \mathbf{N}$ such that $v(y) = k(z)$. Indeed, if $y_1 \in \mathbf{N}$ is a second element such that $v(y_1) = k(z)$, we have $y_1 = y + u(x)$ where $x \in \mathbf{M}$. Let us show that if $x'_1 \in \mathbf{M}'$ is such that $u'(x'_1) = g(y_1)$, we have $x'_1 = x' + f(x)$; indeed, we have $u'(x' + f(x)) = u'(x') + u'(f(x)) = g(y) + g(u(x)) = g(y + u(x)) = g(y_1)$. Finally, we conclude that $p(x'_1) = p(x') + p(f(x)) = p(x')$. We can therefore set $d(z) = p(x')$ and we have thus defined a map $d : \text{Ker}(h) \rightarrow \text{Coker}(f)$.

If now z_1, z_2 are elements of $\text{Ker}(h)$, if $\lambda_1, \lambda_2 \in \mathbf{A}$ and $z = \lambda_1 z_1 + \lambda_2 z_2$, we take elements y_1 and y_2 of $\mathbf{N}$ such that $v(y_1) = k(z_1)$ and $v(y_2) = k(z_2)$ and we choose for $y \in \mathbf{N}$ the element $\lambda_1 y_1 + \lambda_2 y_2$; it is then immediate that

$$d(z) = \lambda_1 d(z_1) + \lambda_2 d(z_2),$$

so d is a homomorphism.

Suppose that $z = v_1(t)$ for a $t \in \text{Ker}(g)$; we will then take for $y \in \mathbf{N}$ the element $j(t)$. Since $g(j(t)) = 0$, we conclude $d(z) = 0$, hence $d \circ v_1 = 0$. Conversely, suppose that $d(z) = 0$. With the preceding notation, we therefore have $x' = f(x)$, where $x \in \mathbf{M}$. In this case, we have $g(y) = u'(f(x)) = g(u(x))$, or equivalently $g(y - u(x)) = 0$. The element $y - u(x)$ is therefore of the form $j(n)$ for $n \in \text{Ker}(g)$, and we have

$$k(z) = v(y) = v(u(x) + j(n)) = v(j(n)) = k(v_1(n));$$

since k is injective, $z = v_1(n)$, which proves that the sequence (*) is exact at $\text{Ker}(h)$. Finally, we have (still with the same notation)

$$u_2(d(z)) = u_2(p(x')) = q(u'(x')) = q(g(y)) = 0 \quad \text{hence} \quad u_2 \circ d = 0.$$

Conversely, suppose that an element $w = p(x')$ of $\text{Coker}(f)$ is such that

$$u_2(w) = u_2(p(x')) = 0 \quad (\text{with } x' \in \mathbf{M}').$$

We therefore have $q(u'(x')) = 0$, and consequently $u'(x') = g(y)$ for a $y \in \mathbb{N}$; since $v'(u'(x')) = 0$, we have $v'(g(y)) = 0$, hence $h(v(y)) = 0$, in other words $v(y) = k(z)$ for a $z \in \text{Ker}(h)$, and by definition $w = d(z)$, which shows that the sequence (*) is exact at $\text{Coker}(f)$. We saw in (i) that it is exact at $\text{Ker}(g)$ and in (ii) that it is exact at $\text{Coker}(g)$, which completes the proof of (iii).

COROLLARY 1. — Suppose that diagram (4) is commutative and has exact rows. Then:

- (i) If u', f and h are injective, g is injective.
- (ii) If v, f , and h are surjective, then g is surjective.

Assertion (i) follows from assertion (i) of prop. 2: indeed, we have $\text{Ker}(f) = 0$ and $\text{Ker}(h) = 0$, hence $\text{Ker}(g) = 0$.

Assertion (ii) follows from assertion (ii) of prop. 2: indeed, we have $\text{Coker } f = 0$ and $\text{Coker } h = 0$, hence $\text{Coker } g = 0$.

COROLLARY 2. — Suppose that diagram (4) is commutative and has exact rows. Under these conditions:

- (i) If g is injective and if f and v are surjective, then h is injective.
- (ii) If g is surjective and if h and u' are injective, then f is surjective.

To prove (i), consider the diagram

$$\begin{array}{ccccc} u(\mathbf{M}) & \xrightarrow{w} & \mathbf{N} & \xrightarrow{v} & \mathbf{P} \\ f' \downarrow & & g \downarrow & & h \downarrow \\ u'(\mathbf{M}') & \xrightarrow{w'} & \mathbf{N}' & \xrightarrow{v'} & \mathbf{P}' \end{array}$$

where f' is the map having the same graph as the restriction of g to $u(\mathbf{M})$, w and w' the canonical injections; it is clear that this diagram is commutative and has exact rows. Moreover w' is injective, and by hypothesis v is surjective; hence by prop. 2 (iii) we have an exact sequence

$$\text{Ker}(g) \rightarrow \text{Ker}(h) \xrightarrow{d} \text{Coker}(f') ;$$

since g is injective and f' is surjective, we therefore have $\text{Ker}(h) = 0$.

To prove (ii), consider the diagram

$$\begin{array}{ccccc} \mathbf{M} & \xrightarrow{u} & \mathbf{N} & \xrightarrow{w} & v(\mathbf{N}) \\ f \downarrow & & g \downarrow & & h' \downarrow \\ \mathbf{M}' & \xrightarrow{u'} & \mathbf{N}' & \xrightarrow{w'} & v'(\mathbf{N}') \end{array}$$

where this time h' is the map having the same graph as the restriction of h to $v(\mathbf{N})$, and w and w' have respectively the same graphs as v and v' ; this diagram is commutative and its rows are exact. Moreover w is surjective and by hypothesis u' is injective; hence by prop. 2 (iii) we have an exact sequence

$$\text{Ker}(h') \xrightarrow{d} \text{Coker}(f) \rightarrow \text{Coker}(g);$$

since g is surjective and h' is injective, we therefore have $\text{Coker}(f) = 0$.

COROLLARY 3 (Five lemma). — Consider a commutative diagram of A-modules

$$\begin{array}{ccccccccc} \mathbf{M}_1 & \xrightarrow{u_1} & \mathbf{M}_2 & \xrightarrow{u_2} & \mathbf{M}_3 & \xrightarrow{u_3} & \mathbf{M}_4 & \xrightarrow{u_4} & \mathbf{M}_5 \\ f_1 \downarrow & & f_2 \downarrow & & f_3 \downarrow & & f_4 \downarrow & & f_5 \downarrow \\ \mathbf{M}'_1 & \xrightarrow{u'_1} & \mathbf{M}'_2 & \xrightarrow{u'_2} & \mathbf{M}'_3 & \xrightarrow{u'_3} & \mathbf{M}'_4 & \xrightarrow{u'_4} & \mathbf{M}'_5 \end{array}$$

where the rows are exact.

(i) If f_2 and f_4 are injective and f_1 surjective, f_3 is injective.

(ii) If f_2 and f_4 are surjective and f_5 injective, f_3 is surjective.

In particular, if f_1, f_2, f_4 and f_5 are isomorphisms, then so is f_3 . To prove (i), set $\tilde{\mathbf{M}}_2 = \text{Coker}(u_1)$, $\tilde{\mathbf{M}}'_2 = \text{Coker}(u'_1)$ and let us denote by $\tilde{f}_2 : \tilde{\mathbf{M}}_2 \rightarrow \tilde{\mathbf{M}}'_2$ the map deduced from f_2 . It

follows from cor. 2 (i) that $\tilde{f}_2$ is injective. Applying cor. 1 (i) to the diagram

$$\begin{array}{ccccc} \tilde{\mathbf{M}}_2 & \xrightarrow{\tilde{u}_2} & \mathbf{M}_3 & \xrightarrow{u_3} & \mathbf{M}_4 \\ \tilde{f}_2 \downarrow & & f_3 \downarrow & & f_4 \downarrow \\ \tilde{\mathbf{M}}'_2 & \xrightarrow{\tilde{u}'_2} & \mathbf{M}'_3 & \xrightarrow{u'_3} & \mathbf{M}'_4 \end{array}$$

where $\tilde{u}_2$ and $\tilde{u}'_2$ are deduced from u_2 and u'_2 , we see that f_3 is injective.

To prove (ii), set $\tilde{\mathbf{M}}_4 = \text{Ker}(u_4)$, $\tilde{\mathbf{M}}'_4 = \text{Ker}(u'_4)$ and let us denote by $\tilde{f}_4 : \tilde{\mathbf{M}}_4 \rightarrow \tilde{\mathbf{M}}'_4$ the map induced by f_4 . It follows from cor. 2 (ii) that $\tilde{f}_4$ is surjective. Applying cor. 1 (ii) to the diagram

$$\begin{array}{ccccc} \mathbf{M}_2 & \xrightarrow{u_2} & \mathbf{M}_3 & \xrightarrow{\tilde{u}_3} & \tilde{\mathbf{M}}_4 \\ f_2 \downarrow & & f_3 \downarrow & & \tilde{f}_4 \downarrow \\ \mathbf{M}'_2 & \xrightarrow{u'_2} & \mathbf{M}'_3 & \xrightarrow{\tilde{u}'_3} & \tilde{\mathbf{M}}'_4 \end{array}$$

where $\tilde{u}_3$ and $\tilde{u}'_3$ have the same graph as u_3 and u'_3 , we see that f_3 is surjective.

10.1.3 Flat modules

DEFINITION 1. — The A-module E is said to be flat if, for every exact sequence of right A-modules and homomorphisms

$$\mathbf{M}' \xrightarrow{u} \mathbf{M} \xrightarrow{v} \mathbf{M}'', \quad (11)$$

the sequence of Z-linear maps

$$\mathbf{M}' \otimes_{\mathbf{A}} \mathbf{E} \xrightarrow{u \otimes 1} \mathbf{M} \otimes_{\mathbf{A}} \mathbf{E} \xrightarrow{v \otimes 1} \mathbf{M}'' \otimes_{\mathbf{A}} \mathbf{E} \quad (12)$$

is exact.

PROPOSITION 3. — For the A-module E to be flat, it is necessary and sufficient that, for every injective A-homomorphism $u : \mathbf{M}' \rightarrow \mathbf{M}$ of right A-modules, the map $u \otimes 1 : \mathbf{M}' \otimes_{\mathbf{A}} \mathbf{E} \rightarrow \mathbf{M} \otimes_{\mathbf{A}} \mathbf{E}$ be injective.

If E is flat and $u : \mathbf{M}' \rightarrow \mathbf{M}$ injective, the sequence $0 \rightarrow \mathbf{M}' \xrightarrow{u} \mathbf{M}$ is exact, hence also the sequence $0 \rightarrow \mathbf{M}' \otimes_{\mathbf{A}} \mathbf{E} \xrightarrow{u \otimes 1} \mathbf{M} \otimes_{\mathbf{A}} \mathbf{E}$, and $u \otimes 1$ is injective. Conversely, consider the

exact sequence (11); set $M' \cdot 1 = v(M)$, and let $i : M' \cdot 1 \rightarrow M' \cdot$ be the canonical injection

and $p : M \rightarrow M_1''$ the map $m \mapsto v(m)$. The sequence $M' \xrightarrow{u} M \xrightarrow{p} M_1'' \rightarrow 0$

is exact; by II, p. 58, prop. 5, the sequence $M' \otimes_A E \xrightarrow{u \otimes 1} M \otimes_A E \xrightarrow{p \otimes 1} M_1'' \otimes_A E$ is therefore exact. Moreover, we have $v = i \circ p$, hence $(v \otimes 1) = (i \otimes 1) \circ (p \otimes 1)$; if E satisfies the condition of the statement, then $i \otimes 1$ is injective, therefore

$$\text{Ker}(v \otimes 1) = \text{Ker}(p \otimes 1) = \text{Im}(u \otimes 1)$$

and the sequence (12) is exact.

PROPOSITION 4. —(i) Let $(E_i)_{i \in I}$ be a family of A -modules, $E = \bigoplus_{i \in I} E_i$ their direct sum. For the A -module E to be flat, it is necessary and sufficient that each of the E_i be so.

(ii) Let I be a right-filtered preordered set, $(E_\alpha, f_{\beta\alpha})$ an inductive system of A -modules relative to I , $E = \varinjlim E_\alpha$ its inductive limit. If each of the A -modules E_α is flat, then E is flat.

Let $M' \rightarrow M \rightarrow M''$ be an exact sequence of right A -modules.

(i) For the sequence $\bigoplus_{i \in I} (M' \otimes_A E_i) \rightarrow \bigoplus_{i \in I} (M \otimes_A E_i) \rightarrow \bigoplus_{i \in I} (M'' \otimes_A E_i)$ to be exact, it is necessary and sufficient that each of the sequences $M' \otimes_A E_i \rightarrow M \otimes_A E_i \rightarrow M'' \otimes_A E_i$ be exact (II, p. 13, prop. 7), which proves (i) since $\bigoplus (M \otimes_A E_i)$ is canonically identified with $M \otimes_A E$ (II, p. 61, prop. 7).

(ii) By hypothesis, each of the sequences $M' \otimes_A E_i \rightarrow M \otimes_A E_i \rightarrow M'' \otimes_A E_i$ is exact, hence so is the sequence $M' \otimes_A E \rightarrow M \otimes_A E \rightarrow M'' \otimes_A E$, since passage to the inductive limit commutes with tensor product (II, p. 93, prop. 7) and preserves exactness (II, p. 91, prop. 3).

Examples. — 1) It is clear that A_s is a flat A -module; it then follows from prop. 4 (i) that every free A -module, and more generally every projective A -module, is flat (see also II, p. 63, cor. 6).

* Conversely, every flat A -module of finite presentation is projective (no. 5). *

2) By prop. 4 (ii), every A -module which is the inductive limit of a filtered inductive system

of free A -modules is flat. We shall prove a converse in no. 6.

- 3) If A is semisimple, every A -module is projective (VIII, § 5, no. 1, prop. 1), hence flat.
- 4) * If A is an artinian local ring (not necessarily commutative), an A -module is flat if and only if it is free (AC II, § 3, no. 2, cor. 2 of prop. 5). *
- 5) If A is an integral domain, the field of fractions K of A is a flat A -module (II, p. 118, prop. 27).
- 6) * In AC II and III, we shall study two important examples of flat A -modules when A is commutative: the rings of fractions $S^{-1}A$, and when A is noetherian, the separated completions of A for the J -adic topologies. *
- 7) Let $a \in A$ such that the map $a_A : x \mapsto ax$ of A to A is injective (“ a is not a left zero-divisor”). If E is a flat A -module, then the homothety a_E is injective, since it is identified with $a_A \otimes 1 : A_d \otimes_A E \rightarrow A_d \otimes_A E$. In particular, if A is an integral domain, every flat A -module is torsion-free. Conversely, if A is principal, every torsion-free A -module is flat: indeed, if the A -module E is torsion-free, every finitely generated submodule of E is free (VII, § 4, no. 4, cor. 2 to th. 4), and E is a filtering increasing union of flat submodules, hence is flat (prop. 4 (ii)).
- 8) Let B be a ring and $\rho : A \rightarrow B$ a homomorphism. If E is a flat A -module, the B -module $E_{(B)} = B \otimes_A E$ is flat. Indeed, let $u : N' \rightarrow N$ be an injective homomorphism of right B -modules; then $u \otimes_B 1_{E_{(B)}}$ is canonically identified with the homomorphism $u \otimes_A 1_E : N' \otimes_A E \rightarrow N \otimes_A E$, which is injective if E is flat.
- 9) Suppose that $A = K[X, Y]$, where K is a field. Then the maximal ideal m generated by X and Y is a torsion-free A -module, but not flat. Indeed, consider the ring $B = A/(Y)$, which is isomorphic to $K[X]$, hence an integral domain. The B -module $m_{(B)}$ is isomorphic to $m/Ym = (X, Y)/(XY, Y^2)$, in which the class of Y is torsion. Consequently, $m_{(B)}$ is not a flat B -module, hence m is not flat.
- 10) Suppose A is commutative. Let B be the algebra $A[X_1, \dots, X_n]/(P)$, where P is a nonzero

polynomial. For every prime ideal $\mathfrak{p}$ of A , denote by $\kappa(\mathfrak{p})$ the field of fractions of the integral domain $A/\mathfrak{p}$, $E(\mathfrak{p})$ the algebra $\kappa(\mathfrak{p})[X_1, \dots, X_n]$ and $P(\mathfrak{p})$ the image of P in $E(\mathfrak{p})$ under the canonical mapping.

One can show that, for B to be a flat A -module, it suffices that $P(\mathfrak{p}) \neq 0$ for every prime ideal $\mathfrak{p}$ of A . If A is an integral domain, this condition is necessary.

* In geometric language, consider the projection $\pi : \text{Spec}(B) \rightarrow \text{Spec}(A)$. For every $\mathfrak{p} \in \text{Spec}(A)$ the fiber $\pi^{-1}(\mathfrak{p})$ is identified with the subvariety $V_{\mathfrak{p}}$ of the affine space $\mathbf{A}_{\kappa(\mathfrak{p})}^n = \text{Spec}(E(\mathfrak{p}))$ defined by $P(\mathfrak{p})$, and the set F of $\mathfrak{p}$ for which this subvariety is the entire space (i.e. for which $P(\mathfrak{p}) = 0$) is a closed subset of

$\text{Spec}(A)$. The preceding condition means that this closed set is empty, in other words that for every $\mathfrak{p}$ the subvariety $V_{\mathfrak{p}}$ is a hypersurface in $\mathbf{A}_{\kappa(\mathfrak{p})}^n$. *

11) Let S and X be two complex analytic spaces, and $f : X \rightarrow S$ a morphism. We say that f is flat at a point x of X if $\mathcal{O}_{X,x}$, considered as $\mathcal{O}_{S,f(x)}$ -module by means of the homomorphism $f^* : \mathcal{O}_{S,f(x)} \rightarrow \mathcal{O}_{X,x}$, is flat. The set of points of X where f is flat is an open subset of X , and the restriction of f to this open set is an open map. If X and S are connected finite-dimensional analytic manifolds, f is flat (at every point of X) if and only if $f(X)$ is open in S and the fibers $f^{-1}(s)$, for $s \in f(X)$ all have the same dimension. *

10.1.4 Modules of finite presentation

A presentation (or presentation of length 1) of an A -module E is an exact sequence

$$L_1 \rightarrow L_0 \rightarrow E \rightarrow 0 \quad (13)$$

of A -modules where L_0 and L_1 are free.

Every A -module E admits a presentation. Indeed, we know (II, p. 27, prop. 20) that there exists a surjective homomorphism $u : L_0 \rightarrow E$, where L_0 is free; if R is the kernel of u , there likewise exists a surjective homomorphism $v : L_1 \rightarrow R$ where L_1 is free. If one regards v as a homomorphism of L_1 to L_0 , the sequence $L_1 \xrightarrow{v} L_0 \xrightarrow{u} E \rightarrow 0$ is exact by definition, whence our assertion.

If $\rho : A \rightarrow B$ is a ring homomorphism, every presentation (13) of E furnishes a presentation of $E_{(B)} = B \otimes_A E$:

$$\mathbf{B} \otimes_A L_1 \rightarrow \mathbf{B} \otimes_A L_0 \rightarrow \mathbf{B} \otimes_A E \rightarrow 0 \quad (14)$$

by virtue of II, p. 58, prop. 5 and of the fact that $\mathbf{B} \otimes_A L$ is a free B -module when L is free.

One says that a presentation (13) of a module E is finite if the free modules L_0 and L_1 have finite bases. It is clear that if the presentation (13) is finite, so is the presentation (14). One says that E is an A -module of finite presentation if it admits a finite presentation.

PROPOSITION 5. — (i) Every module admitting a finite presentation is finitely generated.

(ii) If A is a left Noetherian ring, every finitely generated A -module admits a finite presentation.

(iii) Every finitely generated projective module admits a finite presentation.

Assertion (i) follows trivially from the definitions. Suppose A is left Noetherian and E is finitely generated. There then exists a surjective homomorphism $u : L_0 \rightarrow E$, where L_0 is a free A -module having a finite basis; the kernel R of u is finitely generated, hence there is a surjective homomorphism $v : L_1 \rightarrow R$ where L_1 is free with finite basis, and the exact sequence $L_1 \xrightarrow{v} L_0 \xrightarrow{u} E \rightarrow 0$ is a finite presentation of E ; whence (ii).

Finally, suppose that E is a finitely generated projective module; it is then a direct summand of a finitely generated free module L_0 (II, p. 40, cor. 1); the kernel R of the surjective homomorphism $L_0 \rightarrow E$ is then isomorphic to a quotient of L_0 , hence is finitely generated, and one finishes as above.

PROPOSITION 6. — Let A be a ring, E an A -module of finite presentation. For every exact sequence

$$0 \rightarrow F \xrightarrow{j} G \xrightarrow{p} E \rightarrow 0$$

where G is finitely generated, the module F is finitely generated.

Let $L_1 \xrightarrow{r} L_0 \xrightarrow{s} E \rightarrow 0$ be a finite presentation; if (e_i) is a basis of L_0 , there exists for

each i an element $g_i \in G$ such that $p(g_i) = s(e_i)$; the homomorphism $u : L_0 \rightarrow G$ such that $u(e_i) = g_i$ for all i is therefore such that $s = p \circ u$. Since $s \circ r = 0$, we have $u(r(L_1)) \subset \text{Ker } p$, and since $\text{Ker } p$ is isomorphic to F , we see that there is a homomorphism $v : L_1 \rightarrow F$ such that the diagram

$$\begin{array}{ccccccc} L_1 & \xrightarrow{r} & L_0 & \xrightarrow{s} & E & \rightarrow & 0 \\ v \downarrow & & u \downarrow & & 1_E \downarrow & & \\ F & \xrightarrow{j} & G & \xrightarrow{p} & E & \rightarrow & 0 \end{array}$$

is commutative. Since j is injective and s surjective, one can apply Proposition 2 of X, p. 4, in other words there is an exact sequence

$$\text{Ker } 1_E \xrightarrow{d} \text{Coker } v \rightarrow \text{Coker } u \rightarrow \text{Coker } 1_E.$$

This shows that $\text{Coker } v$ is isomorphic to $G/u(L_0)$, which is finitely generated by hypothesis. Moreover, we have the exact sequence

$$0 \rightarrow v(L_1) \rightarrow F \rightarrow \text{Coker } v \rightarrow 0$$

and since $v(L_1)$ and $\text{Coker } v$ are finitely generated, so is F (II, p. 17, cor. 5).

PROPOSITION 7. — Let M be an A -module. There exists a right-filtered ordered set I and an inductive system of finitely presented A -modules $(M_\alpha, \varphi_{\beta\alpha})$ relative to I such that M is isomorphic to $\lim M_\alpha$. If M possesses a generating family of n elements, one may assume that the same is true of the M_α .

Consider a presentation

$$A_s^{(K)} \xrightarrow{u} A_s^{(L)} \xrightarrow{v} M \rightarrow 0;$$

let I be the set of pairs $\alpha = (K', L')$, where K' (resp. L') is a finite subset of K (resp. L), such that u induces a map u_α of the submodule $A_s^{K'}$ of $A_s^{(K)}$ in the submodule $A_s^{L'}$ of $A_s^{(L)}$; for $\alpha \in I$, let M_α be the cokernel of u_α and $v_\alpha : A_s^{L'} \rightarrow M_\alpha$ the canonical map, so that we have a commutative diagram with exact rows:

$$A_s^{(K)} \xrightarrow{u} A_s^{(L)} \xrightarrow{i} M \rightarrow 0$$

$$i_\alpha \uparrow, \quad j_\alpha \uparrow, \quad f_\alpha \uparrow$$

$$A_s^{K'} \xrightarrow{u_\alpha} A_s^{L'} \xrightarrow{v_\alpha} M_\alpha \rightarrow 0,$$

where i_α and j_α are the canonical injections, and where f_α is deduced from j_α by passing to quotients. Order the set I by the relation

$$\alpha = (K', L') \leq \beta = (K'', L'') \quad \text{if} \quad K' \subset K'', \quad L' \subset L'';$$

for $\alpha \leq \beta$, let $\varphi_{\beta\alpha} : \mathbf{M}_\alpha \rightarrow \mathbf{M}_\beta$ be the homomorphism deduced by passing to quotients from the inclusion of $A_s^{L'}$ to $A_s^{L''}$. One then immediately verifies that the ordered set I is filtered, and that $(\mathbf{M}_\alpha, \varphi_{\beta\alpha})$ is an inductive system of A -modules and that (φ_α) is an inductive system of A -homomorphisms. Passing to the inductive limit, one obtains a commutative diagram

$$\begin{array}{ccccc} A_s^{(K)} & \xrightarrow{u} & A_s^{(L)} & \xrightarrow{v} & M \\ i \uparrow & & j \uparrow & & \varphi \uparrow \\ \varinjlim A_s^{K'} & \rightarrow & \varinjlim A_s^{L'} & \rightarrow & \varinjlim M_\alpha \rightarrow 0; \end{array} \quad (15)$$

The rows of this diagram are exact (II, p. 91, prop. 3); since i and j are bijective, φ is also bijective (X, p. 7, cor. 3), whence the proposition.

10.1.5 Homomorphisms of a finitely presented module

Let E be an A -module. If I is a filtered preordered set and $(\mathbf{G}_i, u_{ji})$ an inductive system of A -modules relative to I , the canonical maps $G_i \rightarrow \varinjlim G_i$ induce homomorphisms $\text{Hom}_A(E, \mathbf{G}_i) \rightarrow \text{Hom}_A(E, \varinjlim \mathbf{G}_i)$ whence a homomorphism said to be canonical

$$\varinjlim \text{Hom}_A(E, G_i) \rightarrow \text{Hom}_A(E, \varinjlim G_i). \quad (16)$$

Let B be another ring, F a B -module, G an (A, B) -bimodule; we defined in II, p. 75 a canonical homomorphism:

$$\text{Hom}_A(E, G) \otimes_B F \rightarrow \text{Hom}_A(E, G \otimes_B F). \quad (17)$$

PROPOSITION 8. — a) If the A -module E is finitely generated (resp. finitely presented), the canonical homomorphism (16) is injective (resp. bijective).

b) Suppose that the B -module F is flat; if the A -module E is finitely generated (resp. finitely presented), the canonical homomorphism (17) is injective (resp. bijective). Let us prove, for example, b), the proof of a) being analogous. Consider A, B, F, G as fixed, and, for every right A -module E , set

$$T(E) = \text{Hom}_A(E, G) \otimes_B F, \quad T'(E) = \text{Hom}_A(E, G \otimes_B F)$$

and let us denote by v_E the homomorphism (17); for every homomorphism $v : E \rightarrow E'$ of right A -modules, set $T(v) = \text{Hom}(v, 1_G) \otimes 1_F$ and $T'(v) = \text{Hom}(v, 1_G \otimes 1_F)$.

Let $L_1 \xrightarrow{v} L_0 \xrightarrow{w} E \rightarrow 0$ be a presentation of E ; we assume the free module L_0 (resp. the free modules L_0 and L_1) to be finitely generated. The diagram

$$\begin{array}{ccc} & & 0 \rightarrow T(E) \xrightarrow{T(w)} T(E') \\ v_E \downarrow & & \end{array} \quad (18)$$

is commutative, and its second row is exact (II, p. 36, th. 1); moreover, the sequence

$$0 \rightarrow \text{Hom}_A(E, G) \rightarrow \text{Hom}_A(L_0, G) \rightarrow \text{Hom}_A(L_1, G)$$

is exact (loc. cit.), and since F is flat, the first row of (18) is also an exact sequence (X, p. 8, def. 1). That being so, we know that v_{L_0} is bijective (resp. that v_{L_0} and v_{L_1} are bijective) (II, p. 75, prop. 2). If one assumes only v_{L_0} bijective, it follows from (18) that $v_{L_0} \circ T(w) = T'(w) \circ v_E$ is injective, therefore v_E is also. If we suppose that v_{L_0} and v_{L_1} both are bijective, we deduce

from Cor. 2 (ii) of X, p. 6 that v_E is surjective, and since we have just seen that v_E is injective, it is bijective.

COROLLARY. — Every flat module of finite presentation is projective.

Indeed, let E be a flat A -module of finite presentation. Applying (b) to the case $\mathbf{B} = \mathbf{A}$, $\mathbf{G} = {}_s\mathbf{A}_d$, $\mathbf{F} = \mathbf{E}$, we see that the canonical homomorphism

$$\mathrm{Hom}_{\mathbf{A}}(\mathbf{E}, \mathbf{A}) \otimes_{\mathbf{A}} \mathbf{E} \rightarrow \mathrm{Hom}_{\mathbf{A}}(\mathbf{E}, \mathbf{E})$$

is surjective. This implies that E is projective (II, p. 77, Remark 1).

By the preceding corollary and Prop. 5 of X, p. 10, there is an identity between finitely presented flat modules and finitely generated projective modules. On the other hand, there exist finitely generated flat modules that are not finitely presented, hence not projective (cf. X, p. 170, Exercise 17; see however X, p. 169, Exercises 13 and 14).

10.1.6 Structure of flat modules

Lemma 2. — Let I be a right-filtered ordered set, $(E_\alpha, \varphi_{\beta\alpha})$ an inductive system of sets relative to I , E its inductive limit, and $\varphi_\alpha : E_\alpha \rightarrow E$, $\alpha \in I$, the canonical maps. Let $f : I \rightarrow I$ be a map such that $f(\alpha) > \alpha$ for $\alpha \in I$, and suppose given, for each $\alpha \in I$, a set L_α and maps $u_\alpha : E_\alpha \rightarrow L_\alpha$ and $v_\alpha : L_\alpha \rightarrow E_{f(\alpha)}$ such that $v_\alpha \circ u'_\alpha = \varphi_{f(\alpha), \alpha}$. Let J be the ordered set obtained by endowing I with the relation “ $\alpha \leq \beta$ if $\alpha = \beta$ or $f(\alpha) \leq \beta$ ”. If $\alpha, \beta \in J$ with $\alpha \leq \beta$, let $\psi_{\beta\alpha} : L_\alpha \rightarrow L_\beta$ be the map such that $\psi_{\beta\alpha} = \mathrm{Id}$ if $\alpha = \beta$, $\psi_{\beta\alpha} = u_\beta \circ \varphi_{\beta, f(\alpha)} \circ v_\alpha$ if $f(\alpha) \leq \beta$. If $\alpha \in J$, let $\psi_\alpha : L_\alpha \rightarrow E$ be the map $\varphi_{f(\alpha)} \circ v_\alpha$. Then the ordered set J is filtered, $(L_\alpha, \psi_{\beta\alpha})$ is an inductive system relative to J , (ψ_α) is an inductive system of maps, and the map $\psi : \varinjlim L_\alpha \rightarrow E$ deduced from the ψ_α is bijective.

It is clear that J is directed. If $\alpha, \beta \in J$ with $\alpha < \beta$, we have

$$\begin{aligned} \psi_\beta \circ \psi_{\beta\alpha} &= \varphi_{f(\beta)} \circ v_\beta \circ u_\beta \circ \varphi_{\beta, f(\alpha)} \circ v_\alpha \\ &= \varphi_{f(\beta)} \circ \varphi_{f(\beta), \beta} \circ \varphi_{\beta, f(\alpha)} \circ v_\alpha = \varphi_{f(\alpha)} \circ v_\alpha = \psi_\alpha; \end{aligned}$$

likewise, if $\alpha, \beta, \gamma \in J$ with $\alpha < \beta < \gamma$, we have

$$\begin{aligned}
\psi_{\gamma\beta} \circ \psi_{\beta\alpha} &= u_{\gamma} \circ \varphi_{\gamma,f(\beta)} \circ v_{\beta} \circ u_{\beta} \circ \varphi_{\beta,f(\alpha)} \circ v_{\alpha} \\
&= u_{\gamma} \circ \varphi_{\gamma,f(\beta)} \circ \varphi_{f(\beta),\beta} \circ \varphi_{\beta,f(\alpha)} \circ v_{\alpha} = u_{\gamma} \circ \varphi_{\gamma,f(\alpha)} \circ v_{\alpha} = \psi_{\gamma\alpha}.
\end{aligned}$$

Let us prove the last assertion: for each $\alpha \in J$, we have

$$\psi_{\alpha} \circ u_{\alpha} = \varphi_{f(\alpha)} \circ v_{\alpha} \circ u_{\alpha} = \varphi_{f(\alpha)} \circ \varphi_{f(\alpha),\alpha} = \varphi_{\alpha},$$

hence $\varphi_{\alpha}(E_{\alpha}) = \psi_{\alpha}(u_{\alpha}(E_{\alpha})) \subset \psi_{\alpha}(L_{\alpha})$, and ψ is surjective. Let $\alpha \in \mathbf{J}$ and let $x, y \in L_{\alpha}$ with $\psi_{\alpha}(x) = \psi_{\alpha}(y)$, i.e. $\varphi_{f(\alpha)}(v_{\alpha}(x)) = \varphi_{f(\alpha)}(v_{\alpha}(y))$; there exists $\beta \in I$, $\beta \geq f(\alpha)$ such that

$$\varphi_{\beta,f(\alpha)}(v_{\alpha}(x)) = \varphi_{\beta,f(\alpha)}(v_{\alpha}(y)),$$

hence

$$\psi_{\beta,\alpha}(x) = u_{\beta}(\varphi_{\beta,f(\alpha)}(v_{\alpha}(x))) = u_{\beta}(\varphi_{\beta,f(\alpha)}(v_{\alpha}(y))) = \psi_{\beta,\alpha}(y).$$

and ψ is injective.

THEOREM 1 (D. Lazard). — For every A -module E , the following conditions are equivalent:

- (i) E is flat.
- (ii) For every finitely presented A -module P , the canonical homomorphism

$$\mathrm{Hom}_A(P, A) \otimes_A E \rightarrow \mathrm{Hom}_A(P, E)$$

is surjective.

- (iii) For every finitely presented A -module P and every homomorphism $u : P \rightarrow E$ there exists a finitely generated free A -module L and homomorphisms $v : P \rightarrow L$ and $w : L \rightarrow E$ such that $u = w \circ v$.

(iv) There exists a directed ordered set J , an inductive system of finitely generated free modules $(\mathbf{L}_j)_{j \in J}$ and an isomorphism from E to $\lim L_j$.

(v) (iv) $\Rightarrow$ (i): this follows from prop. 4 (ii) of X, p. 8.

(vi) (i) $\Rightarrow$ (ii): this follows from prop. 8b) of X, p. 12.

(vii) (ii) $\Rightarrow$ (iii): let P be a finitely presented A -module and $u : P \rightarrow E$ a homomorphism; by (ii), there exists $v_1, \dots, v_n \in \text{Hom}_A(P, A)$, $w_1, \dots, w_n \in E$ such that $u(x) = \sum v_i(x)w_i$ for all $x \in P$; if $v : P \rightarrow A^n$ is the homomorphism with components (v_i) and $w : A^n \rightarrow E$ the homomorphism $(a_i) \mapsto \sum a_i w_i$, we indeed have $u = w \circ v$.

(viii) (iii) $\Rightarrow$ (iv): suppose (iii) holds, and let $(E_\alpha, \varphi_{\beta, \alpha})$ be an inductive system, relative to a filtered set I , of finitely presented A -modules, with inductive limit E (X, p. 11, prop. 7). Replacing I , if necessary, by the lexicographic product $I \times \mathbb{N}$, with $E_{(\alpha, n)} = E_\alpha$ for every n , we may assume that I has no greatest element. For each $\alpha \in I$, let L_α be a finitely generated free A -module and $u_\alpha : E_\alpha \rightarrow L_\alpha, v'_\alpha : L_\alpha \rightarrow E$ homomorphisms such that $v'_\alpha \circ u_\alpha$ is the canonical map φ_α of E_α into E ; since L_α is finitely generated and free, and I has no greatest element, there exists an index $\beta > \alpha$ and a homomorphism $v''_\alpha : L_\alpha \rightarrow E_\beta$ such that $v'_\alpha = \varphi_\beta \circ v''_\alpha$; since $\varphi_\beta \circ v''_\alpha \circ u_\alpha = \varphi_\beta \circ \varphi_{\beta, \alpha}$ and that E_α is finitely presented, it follows from prop. 8a) of X, p. 12, that there exists $\gamma \geq \beta$ such that $\varphi_{\gamma \beta} \circ v''_\alpha \circ u_\alpha = \varphi_{\gamma \beta} \circ \varphi_{\beta \alpha} = \varphi_{\gamma \alpha}$; set $\gamma = f(\alpha)$ and let v_α be the homomorphism $\varphi_{\gamma \beta} \circ v''_\alpha$ of L_α to $E_{f(\alpha)}$; we have $v_\alpha \circ u_\alpha = \varphi_{f(\alpha), \alpha}$. We may then apply Lemma 2, whence (iv).

COROLLARY. — Suppose A is commutative. For every flat A -module E , the A -modules $T(E), S(E), \Lambda(E), T^n(E), S^n(E), \Lambda^n(E)$ are flat.

Indeed, E is the inductive limit of a filtered system $(\mathbf{L}_j)$ of finitely generated free A -modules, hence $T(E)$ (resp. $S(E)$, etc.) is the inductive limit of the filtered system of free A -modules $T(\mathbf{L}_j)$ (resp. $S(\mathbf{L}_j)$, etc.), hence is flat (cf. III, p. 61, prop. 6, p. 62, th. 1, p. 73, prop. 8, p. 75, th. 1, p. 83, prop. 9, and p. 86, th. 1).

Remark. — Considering in (ii) a finite presentation $A_s^J \xrightarrow{c} A_s^I \rightarrow P \rightarrow 0$ we obtain condition

(ii'), still equivalent to the preceding ones:

(ii'') For every finite matrix $(c_{ij})_{i \in I, j \in J}$ of elements of A , every solution

$$\mathbf{e} = (e_i)_{i \in I} \in E^I$$

of the system of linear and homogeneous equations

$$\sum_{i \in I} c_{ij} e_i = 0, \quad j \in J,$$

can be written $b_1 z_1 + \dots + b_n z_n$, where $b_1, \dots, b_n \in E$ and where, for $r = 1, \dots, n$, $z_r = (z_{r,i})_{i \in I}$ is a solution in A^I of the system of equations

$$\sum_{i \in I} z_{r,i} c_{ij} = 0, \quad j \in J.$$

10.1.7 Injective modules

DEFINITION 2. — The A -module E is said to be injective if, for every exact sequence of A -modules and homomorphisms

$$\mathbf{M}' \xrightarrow{u} \mathbf{M} \xrightarrow{v} \mathbf{M}'', \quad (19)$$

the sequence of Z -linear maps

$$\text{Hom}_A(\mathbf{M}'', E) \xrightarrow{\text{Hom}_A(v, 1)} \text{Hom}_A(\mathbf{M}, E) \xrightarrow{\text{Hom}_A(u, 1)} \text{Hom}_A(\mathbf{M}', E) \quad (20)$$

is exact.

Lemma 3. — For the A -module E to be injective, it is necessary and sufficient that, for every injective A -linear map $u : \mathbf{M}' \rightarrow \mathbf{M}$, the map

$$\text{Hom}_A(u, 1) : \text{Hom}_A(\mathbf{M}, E) \rightarrow \text{Hom}_A(\mathbf{M}', E)$$

be surjective.

If E is injective and if $u : \mathbf{M}' \rightarrow \mathbf{M}$ is injective, then the sequence $0 \rightarrow \mathbf{M}' \xrightarrow{u} \mathbf{M}$ is exact,

hence so is the sequence $\text{Hom}(\mathbf{M}, \mathbf{E}) \xrightarrow{\text{Hom}(u, 1)} \text{Hom}(\mathbf{M}', \mathbf{E}) \rightarrow 0$, and $\text{Hom}(u, 1)$ is surjective. Conversely, consider the exact sequence (19); set $\mathbf{M}_1'' = v(\mathbf{M})$ and let $i : \mathbf{M}_1'' \rightarrow \mathbf{M}''$ be the canonical injection and $p : \mathbf{M} \rightarrow \mathbf{M}_1''$ the map $m \mapsto v(m)$. The sequence $\mathbf{M}' \xrightarrow{u} \mathbf{M} \xrightarrow{p} \mathbf{M}_1'' \rightarrow 0$ is exact; by II, p. 36, th. 1, the sequence

$$\text{Hom}_A(\mathbf{M}_1'', \mathbf{E}) \xrightarrow{\text{Hom}(p, 1)} \text{Hom}_A(\mathbf{M}, \mathbf{E}) \xrightarrow{\text{Hom}(u, 1)} \text{Hom}_A(\mathbf{M}', \mathbf{E})$$

is exact. Moreover, we have $\text{Hom}(v, 1) = \text{Hom}(p, 1) \circ \text{Hom}(i, 1)$. If $\mathbf{E}$ satisfies the condition of the lemma, $\text{Hom}(i, 1)$ is surjective, so the image of $\text{Hom}(v, 1)$ is also that of $\text{Hom}(p, 1)$, and the sequence (20) is exact.

Remark. — Let $\mathbf{E}$ be an injective A -module, $u : \mathbf{M}' \rightarrow \mathbf{M}$ and $f : \mathbf{M}' \rightarrow \mathbf{E}$ homomorphisms of A -modules. If $\text{Ker } u \subset \text{Ker } f$, there exists a homomorphism $g : \mathbf{M} \rightarrow \mathbf{E}$ such that $g \circ u = f$. This follows in fact from what precedes, applied to the injective homomorphism $\mathbf{M}' / \text{Ker } u \rightarrow \mathbf{M}$ deduced from u .

PROPOSITION 9. — Let $(\mathbf{E}_i)_{i \in I}$ be a family of A -modules, $\mathbf{E} = \prod \mathbf{E}_i$ their product. For the A -module $\mathbf{E}$ to be injective, it is necessary and sufficient that each of the $\mathbf{E}_i$ be so.

Let $u : \mathbf{M}' \rightarrow \mathbf{M}$ be an injective homomorphism of A -modules. For the product homomorphism $\prod_{i \in I} \text{Hom}_A(\mathbf{M}, \mathbf{E}_i) \rightarrow \prod_{i \in I} \text{Hom}_A(\mathbf{M}', \mathbf{E}_i)$ to be surjective, it is necessary and sufficient that each of the homomorphisms $\text{Hom}_A(\mathbf{M}, \mathbf{E}_i) \rightarrow \text{Hom}_A(\mathbf{M}', \mathbf{E}_i)$ be so (II, p. 10, prop. 5); this proves the proposition since $\prod_{i \in I} \text{Hom}_A(\mathbf{M}, \mathbf{E}_i)$ is canonically identified with $\text{Hom}_A(\mathbf{M}, \mathbf{E})$.

PROPOSITION 10. — Let $\mathbf{E}$ be an A -module. For $\mathbf{E}$ to be injective, it is necessary and sufficient that, for every ideal $\mathfrak{a}$ of A and every A -homomorphism $f : \mathfrak{a} \rightarrow \mathbf{E}$ there exists $e \in \mathbf{E}$ such that $f(a) = ae$ for all $a \in \mathfrak{a}$.

Suppose $\mathbf{E}$ is injective; let $\mathfrak{a}$ be an ideal of A , $f : \mathfrak{a} \rightarrow \mathbf{E}$ an A -homomorphism, and denote by $i : \mathfrak{a} \rightarrow A$ the canonical injection. Then the map

$$\text{Hom}_A(i, 1) : \text{Hom}_A(\mathfrak{a}, \mathbf{E}) \rightarrow \text{Hom}_A(\mathfrak{a}, \mathbf{E})$$

is surjective (def. 2); if $g \in \text{Hom}_A(A, \mathbf{E})$ is such that $f = g \circ i$, we have

$$f(a) = g(a) = ag(1)$$

for all $a \in \mathfrak{a}$.

Conversely, suppose the condition in the statement holds; let M be an A -module, N a submodule of M , $u : N \rightarrow E$ an A -homomorphism, and let us prove that there exists an A -homomorphism $\bar{u} : M \rightarrow E$ extending u (cf. Lemma 3). Let $\mathcal{P}$ be the set of pairs (P, v) where P is a submodule of M containing N and v a homomorphism from P into E extending u . The set $\mathcal{P}$ ordered by the extension relation is inductive: if (P_j, v_j) is a totally ordered family of elements of $\mathcal{P}$, set $Q = \cup P_j$ and let $w : Q \rightarrow E$ be the unique map inducing v_j over P_j for all j ; then $(Q, w) \in \mathcal{P}$ and (Q, w) majorizes (P_j, v_j) for all j . Then let (P, v) be a maximal element of $\mathcal{P}$ (E, III, p. 20, th. 2); it suffices to prove that $P = M$. Let $x \in M$ and let $\mathfrak{a}$ be the ideal of the $a \in A$ such that $ax \in P$; set $f(a) = v(ax)$ for $a \in \mathfrak{a}$; one thus obtains an A -homomorphism $f : \mathfrak{a} \rightarrow E$. Let then e be an element of E such that $f(a) = ae$ for all $a \in \mathfrak{a}$. Set $P' = P + Ax$ and let $v' : P' \rightarrow E$ be the unique A -homomorphism such that $v'(p + ax) = v(p) + ae$ for $p \in P, a \in A$; then (P', v') belongs to $\mathcal{P}$ and majorizes (P, v) , hence $P' = P$, that is, $x \in P$, which completes the proof.

COROLLARY 1. — If the ring A is left Noetherian, every module that is a direct sum of injective A -modules is injective.

Let $(E_i)_{i \in I}$ be a family of injective A -modules, let E be their direct sum, $\mathfrak{a}$ an ideal of A and $u : \mathfrak{a} \rightarrow E$ an A -homomorphism. Since A is Noetherian, $\mathfrak{a}$ is finitely generated, and consequently the canonical map

$$\varphi : \bigoplus_{i \in I} \text{Hom}_A(\mathfrak{a}, E_i) \rightarrow \text{Hom}_A(\mathfrak{a}, E)$$

is bijective; let (u_i) be the inverse image of u under φ . Since each E_i is injective, and the family (u_i) has finite support, there exists an element $(e_i)_{i \in I}$ of E such that $u_i(a) = ae_i$ for all $a \in \mathfrak{a}$ and for every $i \in I$, hence $u(a) = a((e_i))$ for all $a \in \mathfrak{a}$, and E is injective.

Remark. — If every A -module that is a direct sum of injective A -modules is injective, the ring A is left Noetherian (X, p. 170, Exercise 21).

Suppose A is an integral domain. One says that the A -module E is divisible if the homothety a_E is surjective for every nonzero element a of A .

COROLLARY 2. — Suppose A is an integral domain.

- a) Every injective A -module is divisible.
- b) Every torsion-free (II, p. 115) and divisible A -module is injective.
- c) If A is principal, every divisible A -module is injective.

If $a \in A$ is nonzero, then a_A is injective; on the other hand, for every A -module E , the homothety a_E is canonically identified with

$$\text{Hom}(a_A, 1) : \text{Hom}_A(A, E) \rightarrow \text{Hom}_A(A, E),$$

Therefore, E is divisible if and only if $\text{Hom}(a_E, 1_E)$ is surjective for all $a \in A$ nonzero. The assertion a) therefore follows from Definition 2 (X, p. 15).

Let E be a divisible A -module; suppose A is principal (resp. E is torsion-free) and let us prove that E is injective by applying prop. 10. Let $\mathfrak{a}$ be an ideal of A and $f : \mathfrak{a} \rightarrow E$ an A -homomorphism. Let $x \in \mathfrak{a}$ such that $\mathfrak{a} = Ax$ (resp. such that $x \neq 0$ if $\mathfrak{a} \neq 0$), and let $e \in E$ such that $xe = f(x)$. Let us prove that for every $a \in \mathfrak{a}$, we have

$$f(a) = ae;$$

this is clear if $a \in Ax$, whence the assertion in the case where A is principal; if E is torsion-free and if $xa \in \mathfrak{a}$, we have $xf(a) = f(ax) = axe$, hence $f(a) = ae$ since x is nonzero if $a \neq 0$.

Examples. — 1) If A is an integral domain, the field of fractions K of A is an injective A -module. If A is principal, K/A is an injective A -module.

2) For example, the $\mathbb{Z}$ -modules $\mathbb{Q}$ and $\mathbb{Q}/\mathbb{Z}$ are injective.

3) Let A be a principal ideal ring and let ι be a nonzero element of A . Then $A/\iota A$ is an injective $A/\iota A$ -module (X, p. 170, exercise 20).

10.1.8 Injective cogenerator modules

PROPOSITION 11. — Let B be a ring, F a B -module, and P a (B, A) -bimodule. If F is an injective B -module and P a flat A -module, $\text{Hom}_B(P, F)$ is an injective A -module.

Let $u : M' \rightarrow M$ be an injective homomorphism of A -modules. We have a commutative diagram

$$\begin{array}{ccc} \text{Hom}_A(M, \text{Hom}_B(P, F)) & \xrightarrow{\text{Hom}_A(u, 1)} & \text{Hom}_A(M', \text{Hom}_B(P, F)) \\ \beta \downarrow & & \beta' \downarrow \\ \text{Hom}_B(P \otimes_A M, F) & \xrightarrow{\text{Hom}(1_P \otimes u, 1_F)} & \text{Hom}_B(P \otimes_A M', F) \end{array}$$

where β and β' are the canonical isomorphisms of II, p. 74. Since P is flat over A , the homomorphism $1_P \otimes u : P \otimes_A M' \rightarrow P \otimes_A M$ is injective. Since F is injective, $\text{Hom}(1_P \otimes u, 1_F)$ is surjective, hence also $\text{Hom}_A(u, 1)$, which proves that $\text{Hom}_B(P, F)$ is an injective A -module (X, p. 16, lemma 3).

DEFINITION 3. — The A -module E is said to be a cogenerator if, for every A -module M and every nonzero element x of M , there exists an A -homomorphism $u : M \rightarrow E$ such that $u(x) \neq 0$.

We say that the A -module L is a generator if, for every A -module M and every element x of M , there exists an A -homomorphism $u : L \rightarrow M$ such that $x \in u(L)$. For example, the A -module A_s is a generator.

PROPOSITION 12. — Let E be an injective A -module. For E to be a cogenerator, it is necessary and sufficient that $\text{Hom}_A(S, E) \neq 0$ for every simple A -module S .

The condition is evidently necessary. Conversely, let M be an A -module and x a nonzero element of M ; the submodule Ax of M has a simple quotient S (VIII, § 2, no. 1, prop. 3). If $\text{Hom}_A(S, E) \neq 0$, then $\text{Hom}_A(Ax, E) \neq 0$ and there exists a homomorphism $f : Ax \rightarrow E$ such that $f(x) \neq 0$; since E is injective, f extends to a homomorphism u from M to E and we have $u(x) = f(x) \neq 0$.

Example. — The injective $\mathbb{Z}$ -module $\mathbb{Q}/\mathbb{Z}$ (X, p. 18, example 2) is a cogenerator. Indeed, every simple $\mathbb{Z}$ -module is isomorphic to a module $\mathbb{Z}/p\mathbb{Z}$, $p \neq 0$, and $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Q}/\mathbb{Z})$ is nonzero (it contains, for example, the map induced by passing to quotients from the homomor-

phism $x \mapsto x/p$ from Z to Q).

PROPOSITION 13. — Let B be a ring, F an injective cogenerator B -module, P a (B, A) -bimodule. Suppose P is flat over A and such that $P \otimes_A S \neq 0$ for every simple A -module S (that is, faithfully flat over A in the sense of AC, I). Then the A -module $\text{Hom}_B(P, F)$ is a cogenerator and is injective.

Indeed, $\text{Hom}_B(P, F)$ is injective by Prop. 11. On the other hand, for every simple A -module S , $\text{Hom}_A(S, \text{Hom}_B(P, F))$ is isomorphic to $\text{Hom}_B(P \otimes_A S, F)$ therefore is nonzero since $P \otimes_A S \neq 0$ and the B -module F is a cogenerator; the A -module $\text{Hom}_B(P, F)$ is therefore a cogenerator by Prop. 12.

Corollary 1. — The A -module $E_A = \text{Hom}_Z(A, Q/Z)$ is injective and a cogenerator.

We apply Prop. 13 with $B = Z$, $F = Q/Z$ (example) and $P = A_d$.

For every A -module M , set

$$I^0(M) = E_A^{\text{Hom}(M, E_A)}$$

and let $e_M : M \rightarrow I^0(M)$ be the homomorphism which associates to $m \in M$ the element

$$(\varphi(m))_{\varphi \in \text{Hom}(M, E_A)} \in I^0(M).$$

Then:

COROLLARY 2. — The A -module $I^0(M)$ is injective and the A -homomorphism $e_M : M \rightarrow I^0(M)$ is injective.

Indeed, $I^0(M)$ is injective, since E_A is injective (X, p. 16, prop. 9), and on the other hand e_M is injective since E_A is a cogenerator.

COROLLARY 3. — Every A -module is isomorphic to a submodule of an injective A -module.

COROLLARY 4. — For the A -module E to be injective, it is necessary and sufficient that every injective A -homomorphism $f : E \rightarrow F$ possess an A -linear retraction.

Suppose E is injective and let $f : E \rightarrow F$ be an injective A -homomorphism. Then $\text{Hom}_A(f, 1_E) : \text{Hom}_A(F, E) \rightarrow \text{Hom}_A(E, E)$ is surjective; there therefore exists $r \in \text{Hom}_A(F, E)$

such that $r \circ f = 1_E$ and r is an A -linear retraction of f . Conversely, there exists an injective A -module I and an injective A -homomorphism $f : E \rightarrow I$ (cor. 2); if f possesses an A -linear retraction, E is injective by prop. 9 of X, p. 16.

10.1.9 Injective envelopes

DEFINITION 4. — Let M be an A -module. An injective envelope of M is a pair (I, i) , where I is an injective A -module and $i : M \rightarrow I$ a homomorphism possessing the following property:

- (E) for a submodule P of I to be zero, it is necessary and sufficient that $i^{-1}(P)$ be zero. Note that (E) implies that i is injective. One often identifies M with the submodule $i(M)$ of I , and one then says that I is an injective envelope of M .

Example 1. Suppose A is an integral domain and M is torsion-free. Let K be the field of fractions of A and $i : M \rightarrow K \otimes_A M$ the canonical homomorphism. Then $(K \otimes_A M, i)$ is an injective envelope of M (II, p. 116, prop. 26 and X, p. 17, cor. 2).

THEOREM 2. — Let M be an A -module.

- a) M has injective envelopes.
- b) If (I, i) and (J, j) are two injective envelopes of M , there exists an isomorphism $f : I \rightarrow J$ such that $f \circ i = j$.

Note that the homomorphism f whose existence is asserted in b) is not uniquely determined in general.

- a) One may assume that M is a submodule of an injective A -module E (X, p. 19, cor. 3). Consider the set ordered by inclusion $\mathcal{F}$ of submodules I of E , containing M , and such that the canonical injection $i : M \rightarrow I$ satisfies property (E). Since $\mathcal{F}$ is inductive, it has a maximal element (E, III, p. 20); let I be a maximal element of $\mathcal{F}$. It suffices to prove that I is a direct summand submodule of E . Let N be a submodule of E such that $N \cap I = 0$, and maximal for this property (such an N exists by loc. cit.). The composed homomorphism

$$I \xrightarrow{u} E \xrightarrow{v} E/N,$$

where u and v are the canonical homomorphisms, is injective; since E is injective, there therefore exists a homomorphism $w : E/N \rightarrow E$ such that $w \circ v \circ u = u$, that is, $w \circ v(x) = x$ for $x \in I$. Set $I' = \text{Im}(w) = \text{Im}(w \circ v)$ and let $i' : M \rightarrow I'$ be the canonical injection. Then $I \subset I'$; to complete the proof, it suffices to show that i' satisfies condition (E): this implies that $I = I'$ (maximal character of I), hence that $w \circ v$ is a projector of E over I .

Let P then be a submodule of I' such that $P \cap M = 0$. We have $P = w \circ v(Q)$, where Q is a submodule of E containing N ; moreover

$$Q \cap M = w \circ v(Q \cap M) \subset P \cap M = 0,$$

Thus $Q \cap I = 0$ since $i : M \rightarrow I$ has property (E). By the maximality of N , this implies $Q = N$, that is, $v(Q) = 0$, hence $P = 0$, as was to be shown.

b) Let (I, i) and (J, j) be two injective envelopes of M . Since J is injective, there exists a homomorphism $f : I \rightarrow J$ such that $f \circ i = j$; we have

$$i^{-1}(\text{Ker } f) = \text{Ker } j = 0,$$

hence $\text{Ker } f = 0$ and f is injective. Then $f(I)$ is an injective submodule of J , hence a direct summand; since j satisfies (E), this implies $f(I) = J$ and f is bijective.

Remarks. — 1) Let (I, i) be an injective envelope of M and let $j : M \rightarrow J$ be an injective homomorphism from M into an injective A -module J . By the proof above, there exists an injective homomorphism $f : I \rightarrow J$ such that $f \circ i = j$.

2) Let (I, i) be an injective envelope of M . Identify M with the submodule $i(M)$ of I . For every submodule N of M , there exists an injective submodule of I which is an injective envelope of N (apply Remark 1 to N). Conversely, every injective submodule J of I is an injective envelope of $J \cap M$.

PROPOSITION 14. — Let I be a nonzero injective A -module. The following conditions are equivalent:

(i) I is indecomposable (VII, § 4, no. 8, def. 3);

- (ii) 0 is not the intersection of two nonzero submodules of I ;
- (iii) I is the injective envelope of all of its nonzero submodules;
- (iv) the ring $\text{End}_A(I)$ is local (VIII, § 1, no. 4, def. 4).
- (v) $\Rightarrow$ (iii): let M be a nonzero submodule of I . By Remark 2 above, there exists a submodule I' of I which is an injective envelope of M . Since I' is nonzero and a direct factor of I , we have $I = I'$ if I is indecomposable.
- (vi) $\Rightarrow$ (ii): let E and F be two submodules of I , such that $E \cap F = 0$. If $E \neq 0$, then I is an injective envelope of E by (iii), so " $E \cap F = 0$ " implies " $F = 0$ ".
- (vii) $\Rightarrow$ (i): this is trivial.
- (viii) $\Rightarrow$ (i): this follows from VIII, § 1, no. 6, prop. 13.
- (ix) $\Rightarrow$ (iv): suppose I is indecomposable. Note first that every injective endomorphism f of I is bijective (since $f(I)$ is then a nonzero direct summand of I). Moreover, every endomorphism f of I whose restriction to a nonzero submodule E of I is injective is injective (indeed, since (i) $\Rightarrow$ (iii), I is an injective envelope of E , hence " $E \cap \text{Ker } f = 0$ " implies " $\text{Ker } f = 0$ ". Given that, let f be a non-invertible element of $\text{End}_A(I)$; by VIII, § 1, no. 4, prop. 9, it suffices to prove that $1 - f$ is invertible. Since f is not injective, we have $\text{Ker } f \neq 0$ as the restriction of $1 - f$ to $\text{Ker } f$ is injective, $1 - f$ is injective, hence bijective.

COROLLARY 1. — The relation " I is a class of indecomposable injective A -modules" is collectivizing.

Indeed, by (iii) every indecomposable injective A -module is the injective envelope of a monogenic A -module.

COROLLARY 2. — Let M be an A -module, and let I be an injective envelope of M . For I to be indecomposable, it is necessary and sufficient that 0 not be the intersection of two nonzero submodules of M .

The condition is necessary by Prop. 14 ((i) $\Rightarrow$ (ii)). Conversely, if I is the direct sum of nonzero submodules I_1 and I_2 , we have:

$$I_1 \cap M \neq 0, \quad I_2 \cap M \neq 0 \quad \text{and} \quad (I_1 \cap M) \cap (I_2 \cap M) = 0.$$

Example 2. — If A is commutative and Noetherian, the indecomposable injective A -modules are exactly the injective envelopes of the modules $A/\mathfrak{p}$ where $\mathfrak{p}$ is a prime ideal (X, p. 171, exercise 27).

10.1.10 Structure of injective modules

Lemma 4. — Let M be a nonzero Noetherian A -module, and let I be an injective envelope of M . Then I possesses an indecomposable injective submodule.

We may evidently assume that M is a submodule of I . Let N be a submodule of M such that I is not an injective envelope of N and which is maximal for this property. By remark 2 (X, p. 21), there exists a submodule I_1 of I which is an injective envelope of N ; then I_1 is a direct factor of I , and let J be a complement. We have $J \neq 0$; let us show that J is indecomposable. If J' is a nonzero direct factor submodule of J , we have $J' \cap M \neq 0$ and

$$(J' \cap M) \cap N \subset J' \cap I_1 = 0.$$

The submodule $N' = (J' \cap M) + N$ of M is the direct sum of $J' \cap M$ and N , hence strictly contains N . Moreover, N' is contained in the submodule $J' + I_1$ of J , which is the direct sum of J' and I_1 , hence injective. By the maximal character of N , this implies $J' + I_1 = I$, hence $J' = J$, and J is indecomposable.

Let us denote by $\mathcal{I}$ the set (X, p. 21, cor. 1) of classes of indecomposable injective A -modules.

Recall (X, p. 17, cor. 1) that, if A is left Noetherian, every A -module which is a direct sum of injective A -modules is injective.

THEOREM 3. — Let I be an injective A -module.

- a) If I is the injective envelope of a Noetherian A -module M , I is the direct sum of a finite family of indecomposable (injective) submodules.

- b) If A is left Noetherian, I is the direct sum of a family of indecomposable (injective) submodules.
- c) If I is the direct sum of indecomposable (injective) submodules, there exists a family of cardinals $(a_E)_{E \in \mathcal{J}}$, and only one, such that I is isomorphic to

$$\bigoplus_{E \in \mathcal{J}} E^{(a_E)}.$$

Let us first note that c) follows from prop. 14 (X, p. 21) and VIII, § 1, no. 7, theorem 2. Let us prove a).

Let N be a submodule of M whose injective envelopes are direct sums of a finite family of indecomposable submodules, and maximal for this property (such an N exists since M is Noetherian). By remark 2 (X, p. 21), there exists a submodule I_1 of I which is an injective envelope of N . If $I_1 = I$, the proof is finished; otherwise, let J be a complement of I_1 in I . Then J is an injective envelope of the Noetherian module $J \cap M$ (loc. cit.), hence possesses an indecomposable injective submodule J' (lemma 4). Then $I_1 + J'$ is injective, the direct sum of a finite family of indecomposable submodules, and an injective envelope of the submodule $(I_1 + J') \cap M$ of M which strictly majorizes N , whence a contradiction.

Suppose A is left Noetherian and let us prove b). Let X be the disjoint union of the sets $\text{Hom}_A(E, I)$ for $E \in \mathcal{J}$. To each subset Y of X associate an A -module E_Y and an A -homomorphism $f_Y : E_Y \rightarrow I$ in the following way: Y is the sum of a family $(Y(E))_{E \in \mathcal{J}}$ where $Y(E) \subset \text{Hom}_A(E, I)$, we set

$$E_Y = \bigoplus_{E \in \mathcal{J}} E^{(Y(E))}$$

and the component of f on the direct factor of E_Y corresponding to the element y of $Y(E) \subset \text{Hom}_A(E, I)$ is $y : E \rightarrow I$. Let Y be a subset of X such that f_Y is injective and Y is maximal with respect to this property (such a subset exists by E, III, p. 20); it suffices to prove that f_Y is bijective. Otherwise, let J be a complement of the injective submodule $\text{Im}(f_Y)$ of I ; since J is nonzero, it has a nonzero Noetherian submodule (because A is assumed Noetherian), hence also a nonzero injective submodule J' which is the injective envelope of a Noetherian module.

By a), J' is a direct sum of a nonempty finite family of indecomposable submodules. There therefore exists a nonempty finite subset Y of X such that f_Y maps bijectively $E_{Y'}$ onto J' . Since $\text{Im}(f_Y) \cap J' = 0$, we have $Y \cap Y' = \emptyset$ and $f_{Y \cup Y'}$ is injective; this contradicts the maximality of Y and completes the proof.

10.2 COMPLEXES OF A-MODULES

In this paragraph, A denotes a ring. When we speak of A -modules without further specification, we shall always mean left A -modules.

By graded modules we shall mean graded modules of type Z (II, p. 164). If M is a graded A -module, with grading $(M_n)_{n \in \mathbb{Z}}$, we set $M^n = M_{-n}$ and we say that M_n (resp. M^n) is the homogeneous component of descending degree n (resp. of ascending degree n) of M . If $u : M \rightarrow N$ is a graded homomorphism of degree p of graded A -modules (II, p. 166), we denote by $u_n : M_n \rightarrow M_{n+p}$ (resp. $u^n : M^n \rightarrow M^{n-p}$) the homomorphism deduced from u ; it is called the homogeneous component of descending degree (resp. ascending degree) n of u ; we also say that u is of descending degree p or of ascending degree $-p$.

10.2.1 Complexes of A-modules

DEFINITION 1. — A differential complex of A -modules is a pair (C, d) consisting of a graded A -module C and an endomorphism $d : C \rightarrow C$ graded of descending degree -1 and such that $d \circ d = 0$.

It is also called a complex of A -modules, or an A -complex, or simply a complex. One often writes C instead of (C, d) ; the endomorphism d is called the differential of the complex (C, d) , or, by abuse of language, of C .

If C_n (resp. C^n) is the homogeneous component of descending (resp. ascending) degree n of C , the datum of d is equivalent to that of the sequence of homomorphisms

$$\cdots \rightarrow C_{n+1} \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} C_{n-1} \rightarrow \cdots \quad (1)$$

resp.

$$\dots \rightarrow C^{n-1} \xrightarrow{d^{n-1}} C^n \xrightarrow{d^n} C^{n+1} \rightarrow \dots \quad (1')$$

such that $d_n \circ d_{n+1} = 0$ for all $n \in \mathbf{Z}$ (resp. $d^n \circ d^{n-1} = 0$ for all $n \in \mathbf{Z}$). By abuse of language, the datum of such a sequence of A-modules and homomorphisms will also be called a complex.

As a mnemonic, note that when one "follows the direction of the arrows" in diagrams (1) and (1'), the descending degree decreases and the ascending degree increases.

Every graded A-module will tacitly be regarded as a complex by equipping it with the zero differential; the complexes thus obtained will be called complexes with zero differential. In particular, every A-module M will be endowed with the unique structure of an A-complex such that $M_0 = M^0 = M$. The complex (C, d) is said to be zero if C is reduced to 0. In what follows, we denote by 0 a zero complex, chosen once and for all.

Let us adjoin to the ordered set Z two elements denoted $-\infty$ and $+\infty$; denote by $\overline{\mathbf{Z}}$ the set obtained, and endow it with the order relation extending that of Z and such that $-\infty < n < +\infty$ for all $n \in \mathbf{Z}$; every subset of $\overline{\mathbf{Z}}$ has a lower bound and an upper bound.

Let C be a complex; the right and left bounds¹ of C are the elements $b_d(C)$ and $b_g(C)$ of $\overline{\mathbf{Z}}$ defined by

$$b_d(C) = \inf \{n \in \mathbf{Z}, C_n \neq 0\}, \quad b_g(C) = \sup \{n \in \mathbf{Z}, C_n \neq 0\}.$$

One says that C is zero on the right if $b_d(C) \geq 0$, bounded on the right if $b_d(C) \neq -\infty$, zero on the left if $b_g(C) \leq 0$, bounded on the left if $b_g(C) \neq +\infty$; one says that C is bounded if

$$b_d(C) \neq -\infty, \quad b_g(C) \neq +\infty.$$

The length² of C, denoted by $l(C)$, is the element of $\overline{\mathbf{Z}}$ defined as follows: if C is zero, $l(C) = -\infty$; if C is bounded and nonzero $l(C) = b_g(C) - b_d(C)$; if C is not bounded, $l(C) = +\infty$. * With the conventions of TG, IV, p. 13-17, one always has $l(C) = b_g(C) - b_d(C)$. *

For example, if k consecutive components of C are nonzero, the others being zero, we have

$l(C) = k - 1$ if $k > 0$, $l(C) = -\infty$ if $k = 0$.

We say that the complex (C, d) is free, projective, flat, injective, if each of the modules C_n is so. Note that the complex (C, d) is projective or flat if and only if the module C is (II, p. 39, prop. 3 and X, p. 8, prop. 4), but C can be free without the complex (C, d) being so (since a direct summand of a free module is not always free), just as (C, d) can be injective without C being so (X, p. 170, exercise 21).

Z

Let (C, d) be a complex. We set $\mathbf{Z}(C, d) = \text{Ker}(d)$, $\mathbf{B}(C, d) = \text{Im}(d)$; these are graded submodules of C , called respectively the module of cycles and the module of boundaries of (C, d) ; the homogeneous components of $\mathbf{Z}(C, d)$ and $\mathbf{B}(C, d)$ are denoted $\mathbf{Z}_n(C, d) = \mathbf{Z}^{-n}(C, d)$, $\mathbf{B}_n(C, d) = \mathbf{B}^{-n}(C, d)$; we have $\mathbf{Z}_n(C, d) = \text{Ker}(d_n)$, $\mathbf{B}_n(C, d) = \text{Im}(d_{n+1})$, $\mathbf{Z}^n(C, d) = \text{Ker}(d^n)$, $\mathbf{B}^n(C, d) = \text{Im}(d^{n-1})$.

Since $d \circ d = 0$, we have $\mathbf{B}(C) \subset \mathbf{Z}(C)$; two cycles are said to be homologous if their difference is a boundary; the quotient graded module $H(C, d) = \mathbf{Z}(C, d)/\mathbf{B}(C, d)$ is called the homology module of (C, d) ; its elements are the homology classes; its homogeneous components are denoted $H_n(C, d) = H^{-n}(C, d)$.

Example. — If C has zero differential, we have $\mathbf{Z}(C) = C$, $\mathbf{B}(C) = 0$ and $H(C)$ is canonically identified with C .

We have exact sequences, called canonical:

(I_n)

$$0 \rightarrow \mathbf{Z}_n(C) \rightarrow C_n \xrightarrow{\delta_n} \mathbf{B}_{n-1}(C) \rightarrow 0 \quad ((\text{II}_n))$$

$$0 \rightarrow \mathbf{B}_n(C) \rightarrow \mathbf{Z}_n(C) \rightarrow H_n(C) \rightarrow 0$$

(III_n)

$$0 \rightarrow \mathbf{B}_n(C) \rightarrow C_n \rightarrow C_n/\mathbf{B}_n(C) \rightarrow 0$$

(IV_n)

$$0 \rightarrow H_n(C) \rightarrow C_n/B_n(C) \xrightarrow{\bar{\delta}_n} B_{n-1}(C) \rightarrow 0$$

where δ_n and $\bar{\delta}_n$ are deduced from d_n . By combining (IV_n) and (II_{n-1}), one obtains the exact sequence

$$0 \rightarrow H_n(C) \rightarrow C_n/B_n(C) \rightarrow Z_{n-1}(C) \rightarrow H_{n-1}(C) \rightarrow 0, \quad ((V_n))$$

which is also written, changing n to $-n$

$$0 \rightarrow H^n(C) \rightarrow C^n/B^n(C) \rightarrow Z^{n+1}(C) \rightarrow H^{n+1}(C) \rightarrow 0. \quad ((V^n))$$

DEFINITION 2. — Let (C, d) and (C', d') be two complexes. A morphism¹ from (C, d) to (C', d') is a graded A -homomorphism u of degree 0 from C to C' such that

$$d' \circ u = u \circ d.$$

For every n we therefore have $d'_n \circ u_n = u_{n-1} \circ d_n$ and $d''^n \circ u^n = u^{n+1} \circ d^n$. We have

$$u(Z(C)) \subset Z(C'), \quad u(B(C)) \subset B(C'),$$

and we denote by $Z(u) : Z(C) \rightarrow Z(C')$, $B(u) : B(C) \rightarrow B(C')$, $H(u) : H(C) \rightarrow H(C')$ the homomorphisms of A -modules deduced from them; the homogeneous components of these morphisms are denoted $Z_n(u)$, $Z^n(u)$, ...

If v is another morphism of (C, d) to (C', d') , then $u + v$ is a morphism of (C, d) to (C', d') , and we have

$$Z(u + v) = Z(u) + Z(v), \quad B(u + v) = B(u) + B(v), \quad H(u + v) = H(u) + H(v)$$

Similarly, if A is an algebra over a commutative ring k , and if $\lambda \in k$, then λu is a morphism

of (C, d) to (C', d') and we have

$$\mathbf{Z}(\lambda u) = \lambda \mathbf{Z}(u), \quad \mathbf{B}(\lambda u) = \lambda \mathbf{B}(u), \quad \mathbf{H}(\lambda u) = \lambda \mathbf{H}(u).$$

If $u' : (C', d') \rightarrow (C'', d'')$ is another morphism of complexes, then $u' \circ u$ is a morphism of (C, d) to (C'', d'') and we have

$$\mathbf{Z}(u' \circ u) = \mathbf{Z}(u') \circ \mathbf{Z}(u), \quad \mathbf{B}(u' \circ u) = \mathbf{B}(u') \circ \mathbf{B}(u), \quad \mathbf{H}(u' \circ u) = \mathbf{H}(u') \circ \mathbf{H}(u).$$

It is clear that a bijective morphism is an isomorphism.

DEFINITION 3. — Let (C, d) and (C', d') be two complexes. A homology morphism (or quasi-isomorphism) from (C, d) to (C', d') is a morphism u from (C, d) to (C', d') such that $\mathbf{H}(u)$ is bijective.

Every isomorphism is a homology morphism, and every morphism composed of homology morphisms is a homology morphism.

One says that (C, d) has zero homology if $\mathbf{H}(C) = 0$, that is, if the unique morphism of complexes $0 \rightarrow C$ (resp. $C \rightarrow 0$) is a homology morphism. One says that (C, d) is acyclic in descending degree n (resp. in ascending degree n) if $\mathbf{H}_n(C) = 0$ (resp. $\mathbf{H}^n(C) = 0$).

Let (C, d) be a complex and $p \in \mathbf{Z}$. The p -th translate of (C, d) is the complex $(C(p), d(p))$ obtained as follows: $C(p)$ is the A -module obtained by shifting the grading of C by p (II, p. 163; example 3), so that

$$C(p)_n = C_{n+p}, \quad C(p)^n = C^{n-p};$$

in particular $C(p)_0 = C_p$; one also notes that C is the direct sum of its graded submodules $C_p(-p)$, $p \in \mathbf{Z}$ (resp. $C^p(p)$, $p \in \mathbf{Z}$). One sets $d(p) = (-1)^p d$. We have $\mathbf{Z}(C(p)) = \mathbf{Z}(C)(p)$, $\mathbf{B}(C(p)) = \mathbf{B}(C)(p)$ and $\mathbf{H}(C(p)) = \mathbf{H}(C)(p)$.

For example, d is a morphism of complexes from C to $C(-1)$ and

$$\mathbf{H}(d) : \mathbf{H}(C) \rightarrow \mathbf{H}(C)(-1)$$

is zero.

For every morphism of complexes $u : (C, d) \rightarrow (C', d')$, and every $p \in \mathbf{Z}$, u is also a morphism of $(C(p), d(p))$ to $(C'(p), d'(p))$; it is sometimes denoted $u(p)$ and we have

$$u(p)_n = u_{n+p}, \quad u(p)^n = u^{n-p}.$$

10.2.2 Operations on complexes

On the set $A \times A$, the two operations

$$(a, b) + (a', b') = (a + a', b + b')$$

$$(a, b)(a', b') = (aa', ab' + ba')$$

define a ring structure, denoted $A(\varepsilon)$, with identity element $1 = (1, 0)$; the injection $a \mapsto (a, 0) = a1$ allows A to be identified with a subring of $A(\varepsilon)$; the module $A(\varepsilon)$ is free with basis $\{1, \varepsilon\}$ where $\varepsilon = (0, 1)$; we have $\varepsilon^2 = 0$ and ε is central in $A(\varepsilon)$.

When A is commutative, $A(\varepsilon)$ is a dual number algebra over A (III, p. 15). Endow $A(\varepsilon)$ with the ring grading (II, p. 164) for which $A(\varepsilon)_0 = A.1$, $A(\varepsilon)_{-1} = A.\varepsilon$ and $A(\varepsilon)_n = 0$ for $n \neq 0, -1$. It is clear that giving a structure of A -complex on a set C amounts to giving on C a structure of $A(\varepsilon)$ graded module, the differential d corresponding to the homothety ε_C ; likewise, morphisms of complexes correspond to graded homomorphisms of degree 0 of $A(\varepsilon)$ graded modules. The species of structure of $A(\varepsilon)$ graded modules and of A -complexes are therefore equivalent (E, IV, p. 9-10). We shall use this fact to transport to the theory of complexes the usual notions of the theory of graded modules.

To the notion of graded sub- $A(\varepsilon)$ -module corresponds that of subcomplex: a subcomplex of the complex (C, d) is therefore a graded submodule C' of C such that, for every $n \in \mathbf{Z}$, $d_n(C'_n) \subset C'_{n-1}$; if we denote by d' the graded A -homomorphism from C' to C' induced by d , then (C', d') is a complex structure, called the structure induced by (C, d) . Unless expressly stated otherwise, every subcomplex will be equipped with the induced structure.

We leave it to the reader to similarly spell out the notions of quotient complex, exact sequence of complexes, kernel, cokernel, and image of a morphism of complexes, following the dictionary

below:

graded quotient $A(\epsilon)$ -module

= quotient complex,

kernel, cokernel, image of a graded $A(\epsilon)$ -homomorphism of degree 0 } kernel, cokernel, image of a morphism of complexes,

exact sequence of graded $A(\epsilon)$ -modules and graded homomorphisms of degree 0 } = exact sequence of complexes.

For example, the canonical exact sequences of No. 1 give exact sequences of complexes, called canonical.

(I)

$$0 \rightarrow Z(C) \rightarrow C \xrightarrow{\delta} B(C)(-1) \rightarrow 0, \quad (II)$$

$$0 \rightarrow B(C) \rightarrow Z(C) \rightarrow H(C) \rightarrow 0, \quad (III)$$

$$0 \rightarrow B(C) \rightarrow C \rightarrow C/B(C) \rightarrow 0, \quad (IV)$$

$$0 \rightarrow H(\dot{C}) \rightarrow C/B(C) \xrightarrow{\delta} B(C)(-1) \rightarrow 0, \quad (V)$$

$$0 \rightarrow H(C) \rightarrow C/B(C) \rightarrow Z(C)(-1) \rightarrow H(C)(-1) \rightarrow 0.$$

Similarly, one defines the notions of direct sum of complexes, inductive system of complexes, and inductive limit of an inductive system of complexes.

Let (C_i, d_i) be a family of complexes. One calls product of this family, and one denotes by $\prod_{i \in I} (C_i, d_i)$, the complex (C, d) obtained as follows:

- a) for each $n \in Z$, C_n is the product A -module $\prod_{i \in I} (C_i)_n$ of the homogeneous components $(C_i)_n$ of the given complexes,

b) for each $n \in \mathbb{Z}$, $d_n : C_n \rightarrow C_{n-1}$ is the A -homomorphism with components $(d_i)_n$.

When I is finite, $\prod_{i \in I} (C_i, d_i)$ is equal to $\bigoplus_{i \in I} (C_i, d_i)$. One should note that, in general, the underlying A -module of the product complex $\prod_{i \in I} (C_i, d_i)$ is not the product A -module $\prod_{i \in I} C_i$.

Consider a family (resp. a filtered inductive system) $(C_i)_{i \in I}$. Let C be the direct sum (resp. the inductive limit) of C_i and let $\alpha_i : C_i \rightarrow C$ the canonical homomorphisms. Then the $H(\alpha_i) : H(C_i) \rightarrow H(C)$ define a graded homomorphism of degree 0, called canonical, from $\bigoplus_{i \in I} H(C_i)$ (resp. $\varinjlim_{i \in I} H(C_i)$) in $H(C)$. Likewise, the canonical projections $\prod_{i \in I} C_i \rightarrow C_i$ define a graded homomorphism of degree 0, called canonical, from $H(\prod_{i \in I} C_i)$ to $\prod_{i \in I} H(C_i)$.

PROPOSITION 1. — For every family of complexes $(C_i)_{i \in I}$, the canonical homomorphisms

$$\bigoplus_{i \in I} H(C_i) \rightarrow H\left(\bigoplus_{i \in I} C_i\right), \quad H\left(\prod_{i \in I} C_i\right) \rightarrow \prod_{i \in I} H(C_i) \quad (2)$$

are bijective.

For every filtered inductive system of complexes $(C_i)_{i \in I}$, the canonical homomorphism

$$\varinjlim_{i \in I} H(C_i) \rightarrow H(\varinjlim_{i \in I} C_i) \quad (3)$$

is bijective.

This follows immediately from II, p. 14, cor. 1 to prop. 7, p. 11, cor. to prop. 5, and p. 91, prop. 3.

10.2.3 The connecting homomorphism and the exact homology sequence

In this section, we consider an exact sequence of complexes

$$0 \rightarrow C' \xrightarrow{u} C \xrightarrow{v} C'' \rightarrow 0;$$

denote by the same letter d the differentials of C , C' and C'' .

Let Γ be the set of $x \in C$ such that $dx \in \text{Im}(u)$; for $x \in \Gamma$, we have

$$d(-^1u^1(dx)) = -^1u^1(dd(x)) = 0,$$

hence $\bar{u}^1(dx) \in \mathbf{Z}(\mathbf{C}')$; we also have $dv(x) = v(dx) \in \text{Im}(v \circ u) = 0$, hence $v(x) \in \mathbf{Z}(\mathbf{C}'')$; consider then the linear map $\varphi : \Gamma \rightarrow \mathbf{H}(\mathbf{C}'') \times \mathbf{H}(\mathbf{C}')$ which maps every element $x \in \Gamma$ to the class of $(v(x), \bar{u}^1(dx))$.

Lemma 1. — The image $\varphi(\Gamma)$ of Γ in $\mathbf{H}(\mathbf{C}'') \times \mathbf{H}(\mathbf{C}')$ is the graph of a graded A-homomorphism of degree -1 of $\mathbf{H}(\mathbf{C}'')$ to $\mathbf{H}(\mathbf{C}')$.

- a) If $x \in \Gamma$ and if $v(x) \in \mathbf{B}(\mathbf{C}'')$, then $\bar{u}^1(dx) \in \mathbf{B}(\mathbf{C}')$: there exists indeed $z'' \in \mathbf{C}''$ such that $v(x) = dz''$, then $z \in \mathbf{C}$ such that $z'' = v(z)$, hence $v(x) = v(dz)$, then $t' \in \mathbf{C}'$ such that $x - dz = u(t')$, which gives $dx = u(dt')$, hence $\bar{u}^1(dx) = dt' \in \mathbf{B}(\mathbf{C}')$.
- b) Every element of $\mathbf{Z}(\mathbf{C}'')$ is the image under v of an element x of $\mathbf{C}$ such that $v(dx) = 0$, i.e. $dx \in \text{Im } u$, that is, such that $x \in \Gamma$.
- c) It follows from a) and b) that $\varphi(\Gamma)$ is indeed a functional graph; since φ is bihomogeneous of bidegree $(0, -1)$, this completes the proof.

The graded homomorphism of degree -1 of $\mathbf{H}(\mathbf{C}'')$ in $\mathbf{H}(\mathbf{C}')$ thus defined is called the connecting homomorphism relative to the exact sequence (u, v) ; it is denoted $\partial(u, v)$ or $\partial_{u,v}$ or simply ∂ . Its homogeneous components are denoted

$$\partial_n(u, v) : \mathbf{H}_n(\mathbf{C}'') \rightarrow \mathbf{H}_{n-1}(\mathbf{C}') \quad \text{and} \quad \partial^n(u, v) : \mathbf{H}^n(\mathbf{C}'') \rightarrow \mathbf{H}^{n+1}(\mathbf{C}').$$

By definition, to construct the image of a class $\alpha \in \mathbf{H}_n(\mathbf{C}'')$ by ∂ we choose a cycle $z'' \in \mathbf{Z}_n(\mathbf{C}'')$ in the class α , then an element x of $\mathbf{C}_n$ such that $v(x) = z''$; then dx is of the form $u(t')$, $t' \in \mathbf{C}'_{n-1}$, and $\partial(\alpha)$ is the homology class of t' .

In terms of correspondences, $\partial_n(u, v)$ is therefore obtained from the correspondence $u_{n-1}^{-1} \circ d_n \circ v_n^{-1}$ between $\mathbf{C}''_n$ and $\mathbf{C}'_{n-1}$, by passing to subsets $\mathbf{Z}_n(\mathbf{C}'')$ and $\mathbf{Z}_{n-1}(\mathbf{C}')$, then to their quotients $\mathbf{H}_n(\mathbf{C}'')$ and $\mathbf{H}_{n-1}(\mathbf{C}')$. This shows in particular that, if one replaces $\mathbf{C}, \mathbf{C}', \mathbf{C}'', u, v$ by $\mathbf{C}(p), \mathbf{C}'(p), \mathbf{C}''(p), u(p), v(p)$, we have

$$\partial(u(p), v(p)) = (-1)^p \partial(u, v); \quad (4)$$

likewise, if λ and μ are two invertible elements of the center of A , one has

$$\partial(\lambda u, \mu v) = \lambda^{-1} \mu^{-1} \partial(u, v). \quad (5)$$

One can also relate $\partial(u, v)$ to the snake diagram (X, p. 4). According to loc. cit., prop. 2, the sequences

$$0 \rightarrow Z_n(C') \xrightarrow{Z_n(u)} Z_n(C) \xrightarrow{Z_n(v)} Z_n(C'')$$

and

$$C'_n/B_n(C') \xrightarrow{\bar{u}_n} C_n/B_n(C) \xrightarrow{\bar{v}_n} C''_n/B_n(C') \rightarrow 0,$$

where $\bar{u}_n$ and $\bar{v}_n$ are deduced from u and v , are exact. Using the canonical exact sequences (V_n) , one obtains a commutative diagram with exact rows and columns

$$\begin{array}{ccccccc} 0 & & 0 & & 0 & & \\ \downarrow & & H_n(C') & \xrightarrow{H_n(u)} & H_n(C) & & \\ \downarrow & & C'_n/B_n(C') & \xrightarrow{\bar{u}_n} & C_n/B_n(C) & & \\ \downarrow & & 0 \rightarrow Z_{n-1}(C') & \xrightarrow{Z_{n-1}(u)} & Z_{n-1}(C) & & \\ \downarrow & & H_{n-1}(C') & \xrightarrow{H_{n-1}(u)} & H_{n-1}(C) & & \\ \downarrow & & 0 & & 0 & & 0 \end{array}$$

The homomorphism $H_n(C'') \rightarrow H_{n-1}(C')$ associated with this diagram (loc. cit., prop. 2, (iii)) coincides by construction with $\partial_n(u, v)$. This also implies that the sequence of homomorphisms $(H_n(u), H_n(v), \partial_n(u, v), H_{n-1}(u), H_{n-1}(v))$ is exact; consequently:

THEOREM 1. — The infinite sequence of homomorphisms of A -modules

$$\begin{array}{ccccccc} \dots & \rightarrow & H_{n+1}(C'') & \xrightarrow{\partial_{n+1}(u,v)} & H_n(C') & \xrightarrow{H_n(u)} & H_n(C) \xrightarrow{H_n(v)} H_n(C'') \\ & & \xrightarrow{\partial_n(u,v)} & H_{n-1}(C') & \xrightarrow{H_{n-1}(u)} & H_{n-1}(C) & \xrightarrow{H_{n-1}(v)} H_{n-1}(C'') \xrightarrow{\partial_{n-1}(u,v)} H_{n-2}(C') \rightarrow \dots \end{array}$$

is exact.

This sequence is called the exact homology sequence associated with the exact sequence (u, v) ; it is sometimes written in the form of an exact triangle of A-modules

$$\begin{array}{ccc} & H(C) & \\ & \nearrow \searrow & \\ H(u) & & H(v) \\ & \longleftarrow \partial(u,v) & \\ H(C') & & H(C''). \end{array}$$

COROLLARY 1. — If two of the complexes C, C', C'' have zero homology, then the third also has zero homology. For u (resp. v) to be a homomorphism, it is necessary and sufficient that C'' (resp. C') have zero homology. For $\partial(u, v)$ to be bijective, it is necessary and sufficient that C have zero homology.

COROLLARY 2. — Let u be a morphism of complexes. If $\text{Ker } u$ and $\text{Coker } u$ have zero homology, then u is a homomorphism.

Indeed, let $u : E \rightarrow E'$ a morphism of complexes. If $\text{Ker } u$ (resp. $\text{Coker } u$) has zero homology, then the canonical morphism $E \rightarrow \text{Im } u$ (resp. $\text{Im } u \rightarrow E'$) is a homomorphism by Corollary 1.

PROPOSITION 2. — Consider a commutative diagram of complexes with exact rows

$$\begin{array}{ccccccc} 0 & \rightarrow & C' & \xrightarrow{u} & C & \xrightarrow{v} & C'' \rightarrow 0 \\ f' \downarrow & & & & f \downarrow & & f'' \downarrow \\ 0 & \rightarrow & C'_1 & \xrightarrow{u_1} & C_1 & \xrightarrow{v_1} & C''_1 \rightarrow 0. \end{array}$$

$$\text{Then } \mathbf{H}(f') \circ \partial(u, v) = \partial(u_1, v_1) \circ \mathbf{H}(f'')$$

Let $\alpha'' \in H(C'')$; let z'' be a cycle of class α'' and x an element of C such that $v(x) = z''$; we have

$$\begin{aligned}
(\partial_{u_1, v_1} \circ H(f''))(\alpha'') &= \partial_{u_1, v_1}(\overline{f''(z'')}) = \overline{u_1^{-1}(df(x))} = \overline{f'(u^{-1}(dx))} = \\
&= H(f')(\overline{u^{-1}(dx)}) = (H(f') \circ \partial_{u, v})(\alpha'').
\end{aligned}$$

Example. — Let C be a complex. Consider the canonical exact sequence

$$0 \rightarrow Z(C) \xrightarrow{j} C \xrightarrow{\delta} B(C)(-1) \rightarrow 0, \quad (I)$$

and let $i : B(C) \rightarrow Z(C)$ be the canonical injection. Then the connecting homomorphism $\partial(j, \delta) : H(B(C))(-1) \rightarrow H(Z(C))(-1)$ is identified with $H(i)(-1)$, as one verifies immediately. Since $H(\delta) = 0$, the exact homology sequence associated with (I) decomposes into the short exact sequences

$$0 \rightarrow B_n(C) \xrightarrow{i} Z_n(C) \xrightarrow{H(j)} H_n(C) \rightarrow 0. \quad ((II_n))$$

* Applications:

1) Singular homology

Let A be a ring. For every topological space X , one defines the singular complex $C(X, A)$ of X with coefficients in A as follows:

In $\mathbf{R}^{(N)}$, let (e_n) be the canonical basis; one calls canonical n -simplex the convex hull Δ_n of $\{e_0, \dots, e_n\}$. For $i \in \{0, \dots, n\}$, one defines the affine map $\iota_i : \Delta_{n-1} \rightarrow \Delta_n$ by $\iota_i(e_k) = e_k$ for $k < i$ and $\iota_i(e_k) = e_{k+1}$ for $k \geq i$. One denotes by $C_n(X, A)$ the A -module $A^{\Sigma_n(X)}$, where $\Sigma_n(X)$ is the set of continuous maps from Δ_n to X ; for $n < 0$, we set $C_n = 0$. For $i \in \{0, \dots, n\}$, one defines the linear map $\partial_{n,i} : C_n(X, A) \rightarrow C_{n-1}(X, A)$ by $\partial_{n,i}(e_s) = e_{s \circ \iota_i}$ for $s \in \Sigma_n(X)$, and one sets $d_n = \sum (-1)^t \partial_{n,i}$. One verifies that

$$\dots C_n(X, A) \xrightarrow{d_n} C_{n-1}(X, A) \rightarrow \dots$$

is a complex. Its homology is called the singular homology of X with coefficients in A and is denoted $H(X, A)$ or simply $H(X)$.

If Y is a subspace of X , one denotes by $C(X, Y, A)$ the quotient complex $C(X, A)/C(Y, A)$,

and by $H(X, Y, A)$ its homology. It follows from Theorem I that one has an exact sequence:

$$\cdots \rightarrow H_n(Y, A) \rightarrow H_n(X, A) \rightarrow H_n(X, Y, A) \rightarrow H_{n-1}(Y, A) \rightarrow H_{n-1}(X, A) \rightarrow \cdots$$

2) Finite cellular complexes

Let X be a Hausdorff topological space. A finite cellular decomposition of X is given by an increasing sequence $(X_n)_{n \in \mathbb{Z}}$ of closed subspaces of X satisfying the following conditions: (i) $X_n = \emptyset$ for $n < 0$;

(ii) there exists an N such that $X_N = X$ (hence $X_n = X$ for $n > N$);

(iii) for all n , the space $X_n - X_{n-1}$ has only finitely many connected components, called cells of dimension n ;

(iv) for every n , and for every connected component C of $X_n - X_{n-1}$ there exists a homeomorphism of the open Euclidean ball $\dot{B}_n$ of dimension n (TG, VI, p. 10) onto C which extends to a continuous map from the closed ball into X .

One can show that these conditions imply that $H_n(X_n, X_{n-1}, A)$ is a free A -module Γ_n of rank equal to the number of cells of dimension n , and that $H_i(X_n, X_{n-1}, A) = 0$ for $i \neq n$. We have $C(X_n, X_{n-1}, A) = C(X_n, X_{n-2}, A) / C(X_{n-1}, X_{n-2}, A)$ whence an exact sequence

$$H_n(X_n, X_{n-2}) \rightarrow H_n(X_n, X_{n-1}) \xrightarrow{d_n} H_{n-1}(X_{n-1}, X_{n-2}) \rightarrow H_{n-1}(X_n, X_{n-2}) ,$$

with a connecting homomorphism $d_n : \Gamma_n \rightarrow \Gamma_{n-1}$. We have $d_n \circ d_{n+1} = 0$, which allows one to define a complex $\Gamma : \cdots \rightarrow \Gamma_n \xrightarrow{d_n} \Gamma_{n-1} \rightarrow \cdots$.

The exact sequence

$$H_{n+1}(X_p, X_{p-1}) \rightarrow H_n(X_{p-1}) \rightarrow H_n(X_p) \rightarrow H_n(X_p, X_{p-1}) \rightarrow H_{n-1}(X_{p-1})$$

shows by induction on p that $H_n(X_p) = 0$ for $p < n$, that $H_n(X_n) = \text{Ker}(d_n : \Gamma_n \rightarrow \Gamma_{n-1})$ and that $H_n(X_p) = H_n(\Gamma)$ for $p > n$. In particular $H_n(X)$ is identified with $H_n(\Gamma)$.

Example. — Consider the product of spheres $S_2 \times S_2$ and the complex projective space $P_2(C)$. Let $b \in S_2$; one defines a cell decomposition (Y_n) of $Y = S_2 \times S_2$ by setting

$$Y_0 = Y_1 = \{(b, b)\}, \quad Y_2 = Y_3 = (\{b\} \times S_2) \cup (S_2 \times \{b\}) \quad \text{and} \quad Y_4 = S_2 \times S_2;$$

this decomposition has one cell of dimension 0, two of dimension 2, and one of dimension 4. The differentials of the associated complex are necessarily zero, hence $H_0(Y)$, $H_2(Y)$ and $H_4(Y)$ are free of ranks 1, 2, and 1 respectively, and $H_n(Y) = 0$ for $n \notin \{0, 2, 4\}$.

We obtain a cellular decomposition (Z_n) of $P_2(C)$ by setting

$$Z_0 = Z_1 = \{c\}, \quad Z_2 = Z_3 = P_1(C), \quad Z_4 = P_2(C),$$

the space $P_1(C)$ being embedded in $P_2(C)$ (TG, VIII, p. 20), and c being a point of $P_1(C)$; this decomposition has one cell of dimension 0, one of dimension 2, and one of dimension 4. Here again the differentials of the complex are necessarily zero, and it follows that $H_n(P_2(C))$ is isomorphic to A for $n \in \{0, 2, 4\}$ and 0 otherwise.

Since the homology modules in degree 2 of the two spaces under consideration are free of rank 2 and 1, respectively, these spaces are not homeomorphic. *

10.2.4 Homotopies

DEFINITION 4. — Let (C, d) and (C', d') be two complexes, and let f and g be two morphisms from C to C' . A homotopy connecting f to g is any graded A -homomorphism s of degree 1 from C to C' such that $g - f = d' \circ s + s \circ d$.

We say that f and g are homotopic if there exists a homotopy connecting f to g .

If h is a third morphism of C to C' and if s (resp. t) is a homotopy connecting f to g (resp. g to h), then $s + t$ is a homotopy connecting f to h ; consequently, the relation “ f and g are two homotopic morphisms from C to C' ” is an equivalence relation, whose classes are called the

homotopy classes of morphisms from C to C' .

Given two topological spaces X and Y , and a continuous map $f : X \rightarrow Y$, we define a linear map f_* of the singular complex (cf. no. 3) $C(X, A)$ into $C(Y, A)$ by setting $f_*(e_s) = e_{f \circ s}$ for $s \in \Sigma_n(X)$. This map is a morphism of complexes.

Two continuous maps f and g from X to Y are said to be topologically homotopic if there exists a continuous map h of $[0, 1] \times X$ into Y such that $h(0, x) = f(x)$ and $h(1, x) = g(x)$ for all $x \in X$. We show that, if f and g are topologically homotopic, the morphisms f_* and g_* are homotopic in the sense of Definition 4 above. It is this fact that is the origin of the terminology used in algebra. *

PROPOSITION 3. — If f and g are two homotopic morphisms from C to C' , then $H(f) = H(g)$.

Let s be a homotopy connecting f to g . We have

$$(g - f)(Z(C)) = (d' \circ s + s \circ d)(Z(C)) = (d' \circ s)(Z(C)) \subset B(C'),$$

hence $H(g - f) = 0$ and $H(g) = H(f)$.

COROLLARY. — A morphism homotopic to a homomorphism is a homomorphism.

PROPOSITION 4. — Let C, C', D, D' be four complexes, $f : C \rightarrow C'$, $g : C \rightarrow C'$, $u : D \rightarrow C$, $v : C' \rightarrow D'$ four morphisms. If s is a homotopy relating f to g , then $v \circ s \circ u$ is a homotopy connecting $v \circ f \circ u$ to $v \circ g \circ u$. If f and g are homotopic, then so are $v \circ f \circ u$ and $v \circ g \circ u$.

That is clear.

COROLLARY. — Let C, C', C'' be three complexes, and f and g two morphisms from C to C' , f_1 and g_1 two morphisms from C' to C'' . If s and s_1 are homotopies connecting f to g and f_1 to g_1 respectively, then $s_1 \circ f + g_1 \circ s$ is a homotopy connecting $f_1 \circ f$ to $g_1 \circ g$. If f and f_1 are homotopic to g and g_1 respectively, then $f_1 \circ f$ is homotopic to $g_1 \circ g$.

Indeed, $s_1 \circ f$ connects $f_1 \circ f$ to $g_1 \circ f$ and $g_1 \circ s$ connects $g_1 \circ f$ to $g_1 \circ g$.

DEFINITION 5. — A morphism of complexes $f : C \rightarrow C'$ is called a homotopy equivalence if there exists a morphism $f' : C' \rightarrow C$ such that $f' \circ f$ and $f \circ f'$ are homotopic to 1_C and $1_{C'}$ respectively.

It is clear that f' is then also a homotopy equivalence; one also says that f' is the inverse of f up to homotopy. If f' and f'_1 are both inverses of f up to homotopy, then f' and f'_1 are homotopic (indeed, by the preceding corollary, $f'_1 = f'_1 \circ 1_{C'}$ is homotopic to $f'_1 \circ f \circ f'$, hence to $1_C \circ f' = f'$).

PROPOSITION 5. — A homotopism is a homologism; a morphism composed of homotopisms is a homotopism. A morphism homotopic to a homotopism is a homotopism.

Let $f : C \rightarrow C'$ and $f_1 : C' \rightarrow C''$ be homotopisms of complexes, $f' : C' \rightarrow C$ and $f'_1 : C'' \rightarrow C'$ morphisms inverse up to homotopy. We have

$$H(f') \circ H(f) = H(f' \circ f) = H(1_C) = 1_{H(C)} \quad (\text{prop. 3})$$

and likewise $H(f) \circ H(f') = 1_{H(C')}$, hence $H(f)$ is bijective and f is a homologism. On the other hand, $(f' \circ f'_1) \circ (f_1 \circ f)$ is homotopic to $f' \circ 1_{C'} \circ f$ (prop. 4), hence to 1_C ; similarly, $(f_1 \circ f) \circ (f' \circ f'_1)$ is homotopic to $1_{C''}$ and $f_1 \circ f$ is a homotopism. Finally, if $g : C \rightarrow C'$ is a morphism homotopic to f , $f' \circ g$ is homotopic to $f' \circ f$, hence to 1_C ; similarly, $g \circ f'$ is homotopic to $f \circ f'$, hence to $1_{C'}$ and g is a homotopism.

COROLLARY. — Let C, C', D, D' be four complexes, $f : C \rightarrow C'$ a morphism, $u : D \rightarrow C$ and $v : C' \rightarrow D'$ homotopisms. In order that $v \circ f \circ u$ be a homotopism (resp. a homologism), it is necessary and sufficient that f be one.

If f is a homotopism (resp. a homologism), then $v \circ f \circ u$ is composed of homotopisms (resp. homologisms), hence is one. Conversely, let $\bar{u}$ and $\bar{v}$ be morphisms inverse to u and v up to homotopy; then $\bar{v} \circ (v \circ f \circ u) \circ \bar{u}$ is homotopic to f by prop. 4; whence the conclusion by prop. 5, and the corollary of prop. 3.

We say that the complex C is homotopic to zero if 1_C is homotopic to the zero map, that is, if there exists a graded A -endomorphism s of degree 1 of C such that $1_C = s \circ d + d \circ s$. This also amounts to saying that the unique morphism $0 \rightarrow C$ (resp. $C \rightarrow 0$) is a homotopism. A complex homotopic to zero has zero homology (prop. 5).

Example. — Let $u : M \rightarrow N$ and $v : N \rightarrow P$ be homomorphisms of A -modules such that $v \circ u = 0$; let C be the complex such that $C_2 = M$, $C_1 = N$, $C_0 = P$, $C_i = 0$ for $i \neq 0, 1, 2$, $d_2 = u$, $d_1 = v$, $d_i = 0$ for $i \neq 1, 2$. Then C has zero homology if and only if the

sequence $0 \rightarrow \mathbf{M} \xrightarrow{u} \mathbf{N} \xrightarrow{v} \mathbf{P} \rightarrow 0$ is exact. It is homotopic to zero if and only if this sequence is split. Indeed, saying that C is homotopic to zero means that there exist A -homomorphisms $s : \mathbf{P} \rightarrow \mathbf{N}$ and $t : \mathbf{N} \rightarrow \mathbf{M}$ such that $v \circ s = 1_{\mathbf{P}}, s \circ v + u \circ t = 1_{\mathbf{N}}, t \circ u = 1_{\mathbf{M}}$; this implies that the sequence is split; conversely, if s is an A -linear section of v , we define t by $u \circ t = 1_{\mathbf{N}} - s \circ v$, which is possible since $v \circ (1_{\mathbf{N}} - s \circ v) = v - v \circ s \circ v = 0$.

10.2.5 Split complexes

PROPOSITION 6. — Let (C, d) be a complex. The following conditions are equivalent:

- (i) there exists a homotopism from (C, d) to $(H(C), 0)$
- (ii) there exists a graded A -endomorphism s of C , of degree 1, such that $d = d \circ s \circ d$;
- (iii) $B(C)$ and $Z(C)$ are direct summands of C ;
- (iv) (C, d) is a direct sum of subcomplexes that are either of length 0, or of length 1 with zero homology.
- (v) $\Rightarrow$ (ii): let $\varphi : C \rightarrow H(C)$ be a homotopism; there then exists a morphism of complexes $\psi : H(C) \rightarrow C$ and an endomorphism s of C of degree 1 such that $\psi \circ \varphi = 1_C - s \circ d - d \circ s$. We have $d \circ \psi = \psi \circ 0 = 0$, hence

$$0 = d \circ \psi \circ \varphi = d - d \circ s \circ d - d \circ d \circ s = d - d \circ s \circ d, \quad \text{whence (ii).}$$

(ii) $\Rightarrow$ (iii): let s be as in (ii). Then $d \circ (1_C - s \circ d) = 0$, hence $1_C - s \circ d$ is a projector of C onto $Z(C)$, and $(d \circ s) \circ d = d$, hence $d \circ s$ is a projector of C onto $B(C)$.

(iii) $\Rightarrow$ (iv): for each $n \in \mathbf{Z}$, set $Z_n = Z_n(C)$, $B_n = B_n(C)$ and choose submodules K_n and B'_n of C_n such that $C_n = B'_n \oplus Z_n$, $Z_n = K_n \oplus B_n$. Then

$$E_{(n)} = K_n(-n) \quad \text{and} \quad F_{(n)} = B'_n(-n) \oplus B_{n-1}(1-n)$$

are subcomplexes of (C, d) ; we have $(C, d) = \bigoplus_{n \in \mathbf{Z}} (E_{(n)} \oplus F_{(n)})$; each $E_{(n)}$ is either zero

or of length 0, each $F_{(n)}$ is either zero or of length 1 and with zero homology, whence (iv).

(iv) $\Rightarrow$ (i): it suffices to note that (i) is satisfied when C is of length zero, or of zero homology and length 1.

DEFINITION 6. — A complex C is said to be split if it satisfies the equivalent conditions of prop. 6.

An endomorphism s of C satisfying condition (ii) of prop. 6 is called a splitting of C .

Examples. — 1) A complex with zero differential is split.

2) The complexes homotopic to zero are the split complexes with zero homology, i.e. the complexes C such that $H(C) = 0$ and $Z(C)$ is a direct summand of C .

3) Let $f : \mathbf{M} \rightarrow \mathbf{N}$ be a homomorphism of A -modules and C the complex such that $C_1 = \mathbf{M}, C_0 = \mathbf{N}, C_i = 0$ for $i \neq 0, 1, d_1 = f, d_i = 0$ for $i \neq 1$. Then C is split if and only if $\text{Ker } f$ is a direct summand of $\mathbf{M}$ and $\text{Im } f$ a direct summand of $\mathbf{N}$.

4) The complex C is split as soon as $B(C)$ and $H(C)$ are projective (resp. as soon as $B_n(C)$ and $H_n(C)$ are injective for each n). Indeed, by the exact sequences (I_n) to (IV_n) of no. 1, $Z(C)$ is then a direct summand of C and $B(C)$ a direct summand of $Z(C)$ (resp. $B(C)$ is a direct summand of C and $Z(C)/B(C)$ a direct summand of $C/B(C)$).

5) In particular, if A is principal, a free complex C is split if and only if $H(C)$ is free (that is, $H_n(C)$ free for every $n \in \mathbb{Z}$).

Remark. — a) Suppose that the canonical exact sequence of graded A -modules

$$0 \rightarrow B(C) \rightarrow Z(C) \xrightarrow{\pi} H(C) \rightarrow 0 \quad (\text{II})$$

is split (this holds for example if $H(C)$ is projective, or $B_n(C)$ injective for each n); let $\sigma : H(C) \rightarrow Z(C)$ be a graded A -linear section of π , and let ψ be the homomorphism $x \mapsto \sigma(x)$ from $H(C)$ to C . Then ψ is a homomorphism from $(H(C), 0)$ to C , such that $H(\psi) = 1_{H(C)}$.

b) Suppose that the canonical exact sequence of graded A -modules

$$0 \rightarrow H(C) \xrightarrow{i} C/B(C) \rightarrow B(C)(-1) \rightarrow 0 \quad (IV)$$

is split (this occurs, for example, if $B(C)$ is projective, or $H_n(C)$ is injective for each n); let $\tau : C/B(C) \rightarrow H(C)$ be a graded A -linear retraction of i and φ the homomorphism from C to $H(C)$ which associates to each element of C the image under τ of its class modulo $B(C)$. Then φ is a homomorphism from C to $(H(C), 0)$ such that $H(\varphi) = 1_{H(C)}$.

10.2.6 Cone and cylinder of a morphism of complexes

Let $u : (C', d') \rightarrow (C, d)$ be a morphism of complexes. Let $\text{Cyl}(u)$ and $\text{Con}(u)$ be the graded A -modules $\text{Cyl}(u) = C' \oplus C'(-1) \oplus C$, $\text{Con}(u) = C'(-1) \oplus C$ and let us define graded A -linear maps of degree (-1)

$$\bar{D} : \text{Cyl}(u) \rightarrow \text{Cyl}(u), \quad \bar{D}(x', y', x) = (d'x' + y', -d'y', dx - u(y')),$$

$$D : \text{Con}(u) \rightarrow \text{Con}(u), \quad D(y', x) = (-d'y', dx - u(y')).$$

(Here, and in what follows, we denote by $x, y, \dots$ arbitrary elements of $C, x', y', \dots$ arbitrary elements of C' .)

Lemma 2. — $(\text{Cyl}(u), \bar{D})$ and $(\text{Con}(u), D)$ are complexes of A -modules.

Indeed, we have

$$\begin{aligned} \bar{D} \circ \bar{D}(x', y', x) &= \bar{D}(d'x' + y', -d'y', dx - u(y')) = \\ &= (d'(d'x' + y') - d'y', -d'(-d'y'), d(dx - u(y')) - u(-d'y')) = 0 \end{aligned}$$

since $d' \circ d' = 0$, $d \circ d = 0$ and $d \circ u = u \circ d'$. Likewise $D \circ D = 0$.

DEFINITION 7. — The complexes $\text{Cyl}(u)$ and $\text{Con}(u)$ are called respectively the cylinder and the cone of the morphism u .

Example. — Let $u : M \rightarrow N$ be a homomorphism of A -modules; then the only nonzero homogeneous components of $\text{Cyl}(u)$ and $\text{Con}(u)$ are

$$\text{Cyl}_1(u) = M, \quad \text{Cyl}_0(u) = M \oplus N,$$

$$\text{Con}_1(u) = \mathbf{M}, \quad \text{Con}_0(u) = \mathbf{N},$$

and we have $\overline{D}(m) = (m, -u(m))$, $D(m) = -u(m)$ for $m \in \mathbf{M}$; consequently,

$$H_1(\text{Con}(u)) = \text{Ker}(u), \quad H_0(\text{Con}(u)) = \text{Coker}(u).$$

The complex $\Gamma(s)$ associated with the space $\{s\}$ endowed with its unique cellular decomposition is reduced to the module A and is identified with a subcomplex of $\Gamma(\text{Con}(f))$; denote by $\Gamma(\text{Con}(f), s)$ the quotient complex. The map f defines a morphism of complexes $\Gamma(f) : \Gamma(X) \rightarrow \Gamma(Y)$, and one can show that the complexes $\Gamma(\text{Cyl}(f))$ and $\Gamma(\text{Con}(f), s)$ are respectively identified with $\text{Cyl}(\Gamma(f))$ and $\text{Con}(\Gamma(f))$.

Moreover, and without any hypotheses on X and Y , one associates to f a morphism of complexes $f_* : C(X, A) \rightarrow C(Y, A)$. One can construct injective homotopisms from $\text{Cyl}(f_*)$ to $C(\text{Cyl}(f), A)$ and from $\text{Con}(f_*)$ to $C(\text{Con}(f), \{s\}, A)$.

Let $\tilde{f} : X \rightarrow \text{Cyl}(f)$ be the map which to x associates the image of $(0, x)$, $\alpha : Y \rightarrow \text{Cyl}(f)$ the canonical map and $\beta : \text{Cyl}(f) \rightarrow Y$ the map which associates y to its image in $\text{Cyl}(f)$ for $y \in Y$ and $f(x)$ to the image of (t, x) in $\text{Cyl}(f)$ for $t \in [0, 1]$ and $x \in X$. The map $\tilde{f}$ is a homeomorphism from X onto a closed subset of $\text{Cyl}(f)$, we have $\beta \circ \alpha = \text{Id}_Y$, and $\alpha \circ \beta$ is topologically homotopic to the identity of $\text{Cyl}(f)$. These properties are to be compared with Proposition 7 below.

Consider now the graded A -linear maps of degree 0

$$\tilde{u} : C' \rightarrow \text{Cyl}(u), \quad \tilde{u}(x') = (x', 0, 0),$$

$$\alpha : C \rightarrow \text{Cyl}(u), \quad \alpha(x) = (0, 0, x),$$

$$\beta : \text{Cyl}(u) \rightarrow C, \quad \beta(x', y', x) = u(x') + x,$$

$$\pi : \mathbf{C} \rightarrow \text{Con}(u), \quad \pi(x) = (0, x),$$

$$\tilde{\pi} : \text{Cyl}(u) \rightarrow \text{Con}(u), \quad \tilde{\pi}(x', y', x) = (y', x),$$

$$\delta : \text{Con}(u) \rightarrow \mathbf{C}'(-1), \quad \delta(y', x) = y'.$$

PROPOSITION 7. — a) The maps $\tilde{u}, \alpha, \beta, \pi, \tilde{\pi}, \delta$ are morphisms of complexes: we have $u = \beta \circ \tilde{u}, \pi = \tilde{\pi} \circ \alpha, \beta \circ \alpha = 1_{\mathbf{C}}$.

b) The sequences of morphisms of complexes

(6)

$$0 \rightarrow \mathbf{C}' \xrightarrow{\tilde{u}} \text{Cyl}(u) \xrightarrow{\tilde{\pi}} \text{Con}(u) \rightarrow 0 \quad (7)$$

$$0 \rightarrow \mathbf{C} \xrightarrow{\pi} \text{Con}(u) \xrightarrow{\delta} \mathbf{C}'(-1) \rightarrow 0$$

are exact.

c) The morphisms $\alpha : \mathbf{C} \rightarrow \text{Cyl}(u)$ and $\beta : \text{Cyl}(u) \rightarrow \mathbf{C}$ are mutually inverse homotopisms up to homotopy.

Assertion a) is equivalent to the formulas

$$\tilde{u} \circ d' = \overline{\mathbf{D}} \circ \tilde{u}, \alpha \circ d = \overline{\mathbf{D}} \circ \alpha, \beta \circ \overline{\mathbf{D}} = d \circ \beta, \pi \circ d = \mathbf{D} \circ \pi,$$

$$\tilde{\pi} \circ \overline{\mathbf{D}} = \mathbf{D} \circ \tilde{\pi}, \delta \circ \mathbf{D} = -d' \circ \delta, u = \beta \circ \tilde{u}, \pi = \tilde{\pi} \circ \alpha, \beta \circ \alpha = 1_{\mathbf{C}}$$

which are verified by immediate calculations. Assertion b) is trivial. Let us prove c); on the one hand one has $\beta \circ \alpha = 1_{\mathbf{C}}$; on the other hand if $\sigma : \text{Cyl}(u) \rightarrow \text{Cyl}(u)$ is the graded $\mathbf{A}$ -linear map of degree 1 such that $\sigma(x', y', x) = (0, x', 0)$, one verifies immediately that

$$\overline{D} \circ \sigma + \sigma \circ \overline{D} + \alpha \circ \beta = 1_{\text{Cyl}(u)},$$

whence c).

One can summarize Prop. 7 by the following commutative diagram, where the rows are exact, and where the vertical arrows are homotopisms:

$$\begin{array}{ccccccc} 0 & \rightarrow & C & \xrightarrow{\pi} & \text{Con}(u) & \xrightarrow{\delta} & C'(-1) \rightarrow 0 \\ & & \downarrow \alpha & & & & \downarrow 1 \\ 0 & \rightarrow & C' & \xrightarrow{\tilde{u}} & \text{Cyl}(u) & \xrightarrow{\tilde{\pi}} & \text{Con}(u) \rightarrow 0 \\ & & \downarrow 1 & & & & \downarrow \beta \\ & & C' & \xrightarrow{u} & C & & \end{array}$$

COROLLARY. — For every morphism of complexes $u : C' \rightarrow C$, there exists an injective morphism of complexes $\tilde{u} : C' \rightarrow C_1$ and a homotopism $\beta : C_1 \rightarrow C$ such that $u = \beta \circ \tilde{u}$.

Lemma 3. — a) The connecting homomorphism

$$\partial_{n+1}(\pi, \delta) : H_n(C') \rightarrow H_n(C)$$

relative to the exact sequence (7) is equal to $-H_n(u)$.

b) The connecting homomorphism

$$\partial_n(\tilde{u}, \tilde{\pi}) : H_n(\text{Con}(u)) \rightarrow H_{n-1}(C')$$

relative to the exact sequence (6) is equal to $H_n(\delta)$.

Let $x' \in Z_n(C')$; since $x' = \delta(x', 0)$ and

$$-D(x', 0) = (d'x', u(x')) = (0, u(x')) = \pi(u(x')),$$

$\partial(\pi, \delta)$ maps by definition the class of x' in $H_n(C')$ onto the class of $-u(x')$ in $H_n(C)$, whence a).

Let $(y', x) \in \text{Con}_n(u)$ such that $\mathbf{D}(y', x) = 0$; we then have $(-d'y', dx - u(y')) = 0$. Since

$(y', x) = \tilde{\pi}(0, y', x)$ and

$$\bar{D}(0, y', x) = (y', -d'y', dx - u(y')) = (y', 0, 0) = \tilde{u}(\delta(y', x)),$$

$\partial(\tilde{u}, \tilde{\pi})$ maps by definition the class of (y', x) in $H_n(\text{Con}(u))$ onto the class of $\delta(y', x)$ in $H_{n-1}(C')$, whence $b)$.

PROPOSITION 8. — One has the unbounded exact sequence

$$\dots \rightarrow H_n(C') \xrightarrow{H_n(u)} H_n(C) \xrightarrow{H_n(\pi)} H_n(\text{Con}(u)) \xrightarrow{H_n(\delta)} H_{n-1}(C') \rightarrow \dots \quad (8)$$

Indeed, in view of Lemma 3, a), this follows from Theorem 1 of X, p. 30, applied to the exact sequence (7).

COROLLARY. — For u to be a homologism, it is necessary and sufficient that $\text{Con}(u)$ be of zero homology.

Remark. — Consider the diagram

$$\begin{array}{ccccccc} \dots \rightarrow H_n(C') & \xrightarrow{H_n(\tilde{u})} & H_n(\text{Cyl}(u)) & \xrightarrow{H_n(\tilde{\pi})} & H_n(\text{Con}(u)) & \xrightarrow{\partial_n(\tilde{u}, \tilde{\pi})} & H_{n-1}(C') \rightarrow \dots \\ \downarrow 1 & & \xrightarrow{H_n(\beta)} \downarrow & & \downarrow 1 & & \downarrow 1 \\ \dots \rightarrow H_n(C') & \xrightarrow{H_n(u)} & H_n(C) & \xrightarrow{H_n(\pi)} & H_n(\text{Con}(u)) & \xrightarrow{H_n(\delta)} & H_{n-1}(C') \rightarrow \dots \end{array}$$

where the first line (resp. the second) is the exact homology sequence associated with the exact sequence (6) (resp. (7)). The maps $H_n(\beta)$ are bijective (prop. 7, c)) and the diagram is commutative, since

a) $u = \beta \circ \tilde{u}$ (prop. 7, a)) therefore $H_n(u) = H_n(\beta) \circ H_n(\tilde{u})$ b) $H_n(\beta) = H_n(\alpha)^{-1}$ and $\pi = \tilde{\pi} \circ \alpha$ (prop. 7, a) and c), hence $H_n(\tilde{\pi}) = H_n(\pi) \circ H_n(\beta)$, c) $H_n(\delta) = \partial_n(\tilde{u}, \tilde{\pi})$ (Lemma 3, b)).

10.2.7 The cone of an injective morphism; new definition of the connecting homomorphism

Consider now an exact sequence of complexes

$$0 \rightarrow C' \xrightarrow{u} C \xrightarrow{v} C'' \rightarrow 0. \quad (9)$$

Define a graded A -linear map of degree 0

$$\varphi : \text{Con}(u) \rightarrow C''$$

by $\varphi(y', x) = v(x)$. We then have a commutative diagram of A -modules with exact rows

$$\begin{array}{ccccccc} 0 & \rightarrow & C' & \xrightarrow{\tilde{u}} & \text{Cyl}(u) & \xrightarrow{\tilde{\pi}} & \text{Con}(u) \rightarrow 0 \\ 1 \downarrow & & & & \beta \downarrow & & \varphi \downarrow \\ 0. & \rightarrow & C' & \xrightarrow{u} & C & \xrightarrow{v} & C'' \rightarrow 0. \end{array} \quad (10)$$

PROPOSITION 9. — The maps β and φ are homomorphisms of complexes. For β , this follows from prop. 7, c). We have

$$\begin{aligned} \varphi \circ D(y', x) &= \varphi(-d'y', dx - u(y')) = v(dx - u(y')) \\ &= v(dx) = d''v(x) = d''(\varphi(y', x)), \end{aligned}$$

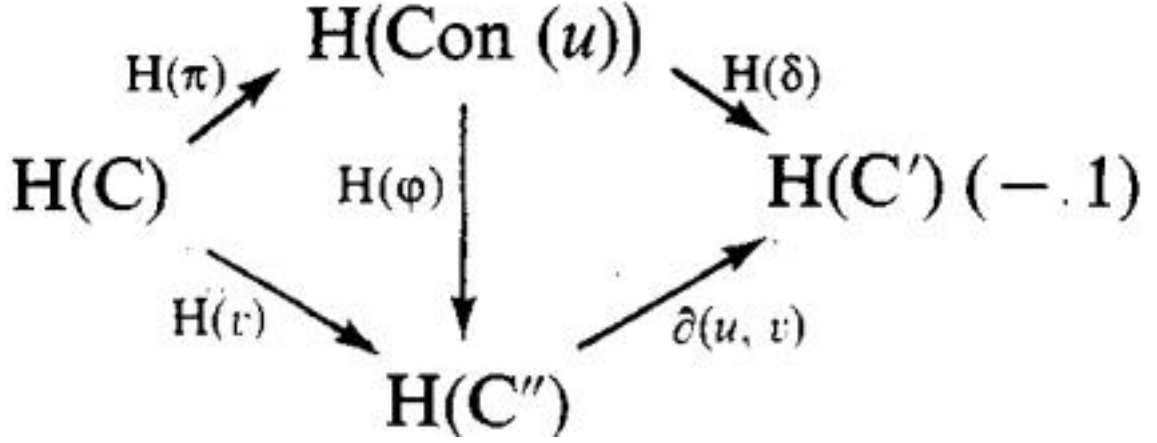
hence φ is indeed a morphism of complexes. On the other hand, φ is surjective and its kernel is identified with the complex $(\mathbf{K}, d_{\mathbf{K}})$ such that $\mathbf{K} = C'(-1) \oplus C'$,

$$d_{\mathbf{K}}(y', x') = (-d'y', d'x' - y');$$

if $d_{\mathbf{K}}(y', x') = 0$, we have $y' = d'x'$, hence $(y', x') = d_{\mathbf{K}}(0, -x')$; it follows that $H(\mathbf{K}) = 0$ and φ is a homomorphism by X, p. 30, cor. 1.

Remark. — The homomorphism β is a homotopism, but φ is not in general a homotopism (cf. X, p. 173, exercise 8).

COROLLARY. — The diagram of graded A -modules



is commutative and $H(\varphi)$ is bijective.

In the commutative diagram (10), we have $H(1_{C'}) \circ \partial(\tilde{u}, \tilde{\pi}) = \partial(u, v) \circ H(\varphi)$ (X, p. 31, prop. 2) and $\partial(\tilde{u}, \tilde{\pi}) = H(\delta)$ (X, p. 38, lemma 3, b)), hence

$$\partial(u, v) \circ H(\varphi) = H(\delta);$$

on the other hand, $H(v) \circ H(\beta) = H(\varphi) \circ H(\tilde{\pi}) = H(\varphi) \circ H(\pi) \circ H(\beta)$ by X, p. 39, remark. Since $H(\beta)$ is bijective, this gives $H(v) = H(\varphi) \circ H(\pi)$.

We therefore have $\partial(u, v) = H(\delta) \circ H(\varphi)^{-1}$, which provides a new definition of the connecting homomorphism $\partial(u, v)$. We also note that if we identify $H(Con(u))$ with $H(C'')$ by $H(\varphi)$, the preceding corollary means that the exact sequence (8) is then identified with the exact homology sequence relative to (9).

10.2.8 Euler-Poincaré characteristics

In this n°, we consider a set C of classes of A -modules that is additive and left exact, that is, satisfying the following two conditions:

- (A) If M and N are two A -modules of type C , then $M \oplus N$ is of type C .
- (B) If $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is an exact sequence of A -modules and if M and M'' are of type $\mathcal{C}$, then M' is of type $\mathcal{C}$.

We say that $\mathcal{C}$ is stable if it satisfies the following conditions, which imply (A) and (G):

- (E) ("C is stable under extensions.") If $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is an exact sequence of

A-modules and if M' and M'' are of type C, then M is of type C.

(F) ("C is stable under kernels and cokernels.") For every homomorphism f of A-modules of type C, the A-modules $\text{Ker } f$ and $\text{Coker } f$ are of type C.

We denote by $K(\mathcal{C})$ the Grothendieck group of $\mathcal{C}$ and $[M]_{\mathcal{C}}$ or $[M]$ the element of $K(\mathcal{C})$ defined by the A-module M (VIII, § 6, no. 2). Let G be a commutative group and φ a homomorphism from $K(\mathcal{C})$ into G .

Examples. — 1) If A is a field, one may take for $\mathcal{C}$ the set of classes of finite-dimensional vector spaces, and for φ the isomorphism of $K(\mathcal{C})$ onto $\mathbb{Z}$ defined by $\varphi([M]) = \dim(M)$.

2) One may take for $\mathcal{C}$ the set of classes of modules of finite length, and for $\varphi : K(\mathcal{C}) \rightarrow \mathbb{Z}$ the homomorphism defined by $\varphi([M]) = \text{long}_A(M)$.

One says that a graded A-module M is of type C if M_n is of type C for every n (for this, it is necessary, if M is bounded, and sufficient, if $\mathcal{C}$ is stable, that the module M be of type C).

DEFINITION 8. — Let M be a bounded graded A-module of type C and let (M_n) be its grading. One calls the φ -characteristic of M , and denotes by $\chi_{\varphi}(M)$ or simply $\chi(M)$, the element $\sum (-1)^n \varphi([M_n])$ of G .

This definition applies in particular when M is the graded module underlying a complex of A-modules.

Examples. — 3) If M is bounded of type C, the same is true of $M(p)$ for every $p \in \mathbb{Z}$, and we have $\chi(M(p)) = (-1)^p \chi(M)$.

4) Let $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ be an exact sequence of graded A-modules and graded homomorphisms of degree 0. If M, M' and M'' are bounded of type $\mathcal{C}$, we have

$$\chi(M) = \chi(M') + \chi(M'').$$

If M and M'' are bounded of type C, the same is true of M' ; if $\mathcal{C}$ is stable and two of the three modules are bounded of type C, the same is true of the third.

5) Let $u : C' \rightarrow C$ be a morphism of bounded complexes of type $\mathcal{C}$. Then $\text{Con}(u)$ is bounded of type $\mathcal{C}$, and we have:

$$\chi(\text{Con}(u)) = \chi(\mathbf{C}) - \chi(\mathbf{C}').$$

- 6) One can take for G the group $K(C)$ itself, and for φ the identity; we then denote by $\chi_{\mathcal{C}}(M)$ the element $\chi_{\varphi}(M) = \sum (-1)^n [M_n]$ of $K(C)$.

Remark. — One calls the Poincaré polynomial of M relative to φ the element $P_M(t) = \sum \varphi([M_n])t^n \in G \otimes \mathbf{Z}[t, t^{-1}]$. We have $P_M(1) = \varphi([M])$ and $P_M(-1) = \chi(M)$.

Lemma 4. — Let C be a bounded complex of type $\mathcal{C}$. If $H(C) = 0$, we have $\chi(C) = 0$. This follows from VIII, § 6, n° 1, cor. of prop. 1.

PROPOSITION 10. — Let C and C' be two bounded complexes of type C . If there exists a homomorphism $u : C' \rightarrow C$, we have $\chi(C) = \chi(C')$.

Indeed, $\text{Con}(u)$ is bounded of type $\mathcal{C}$ and we have $\chi(\text{Con}(u)) = \chi(C) - \chi(C')$; on the other hand, $H(\text{Con}(u)) = 0$ by X, p. 38, cor., hence $\chi(\text{Con}(u)) = 0$ (lemma 4).

PROPOSITION 11. — Let C be a bounded complex of type C .

- a) If $\mathcal{C}$ is stable, $H(C)$ is of type $\mathcal{C}$.
- b) If $H(C)$ is of type C , then so are $B(C)$ and $Z(C)$, and we have $\chi(H(C)) = \chi(C)$.
- c) If $\mathcal{C}$ is stable, for every n the module $\mathbf{Z}_n(C)$ is of type $\mathcal{C}$ as kernel of $d_n : C_n \rightarrow C_{n-1}$, and $H_n(C)$ is of type $\mathcal{C}$ as cokernel of $C_{n+1} \rightarrow \mathbf{Z}_n$. On the other hand, $H_n(C) = 0$ as soon as $C_n = 0$.
- d) Suppose $H(C)$ is of type C . The canonical exact sequences:

$$0 \rightarrow Z_n(C) \rightarrow C_n \rightarrow B_{n-1}(C) \rightarrow 0$$

$$0 \rightarrow B_n(C) \rightarrow Z_n(C) \rightarrow H_n(C) \rightarrow 0$$

show by induction on n starting from the right bound of C that $Z_n(C)$ and $B_n(C)$ are of type C for every n . We then have

$$\chi(C) = \chi(Z(C)) + \chi(B(C)(-1)) = \chi(Z(C)) - \chi(B(C)) = \chi(H(C)).$$

COROLLARY. — If C is stable and C is bounded of type C , the graded module $H(C)$ is bounded of type C and we have $\chi(H(C)) = \chi(C)$.

PROPOSITION 12. — Let $0 \rightarrow C' \rightarrow C \rightarrow C'' \rightarrow 0$ be an exact sequence of complexes. a) If $H(C)$, $H(C')$ and $H(C'')$ are bounded of type C , we have

$$\chi(H(C)) = \chi(H(C')) + \chi(H(C'')).$$

b) If C is stable, and if two of the graded modules $H(C)$, $H(C')$ and $H(C'')$ are bounded of type C , then so is the third.

Part a) follows from lemma 4 applied to the zero-homology complex defined by the homology exact sequence associated with the given exact sequence. Part b) follows, by considering this homology exact sequence, from the following lemma:

Lemma 5. — Let $M \rightarrow N \rightarrow P \rightarrow Q \rightarrow R$ be an exact sequence of A -modules. If C is stable, and if M , N , Q and R are of type C , then the module P is of type C .

Set $N' = \text{Coker}(M \rightarrow N)$ and $Q' = \text{Ker}(Q \rightarrow R)$. The modules N' and Q' are of type $\mathcal{C}$, and we have an exact sequence $0 \rightarrow N' \rightarrow P \rightarrow Q' \rightarrow 0$.

COROLLARY. — Suppose C is stable, and let $u : C' \rightarrow C$ be a morphism of complexes such that $H(C)$ and $H(C')$ are bounded of type C . Then $H(\text{Con}(u))$ is bounded of type C , and we have

$$\chi(H(\text{Con}(u))) = \chi(H(C)) - \chi(H(C')).$$

This follows from prop. 12 applied to the exact sequence of complexes (X, p. 37, prop. 7)

$$0 \rightarrow C \rightarrow \text{Con}(u) \rightarrow C'(-1) \rightarrow 0.$$

Remark. — Let E be a complex, $h : E \rightarrow C$ and $h' : E \rightarrow C'$ homotopisms with C and C' bounded of type $\mathcal{C}$. We then have $\chi(C) = \chi(C')$. Indeed, if h_1 is an inverse of h up to

homotopy, $h' \circ h_1$ is a homotopism, hence a homologism from C to

C' and one can apply prop. 10. Consequently, one can extend definition 8 by setting $\chi(E) = \chi(C)$ as soon as there exists a homotopism from E to a bounded complex C of type C . Propositions 10, 11, 12 and their corollaries generalize to this setting.

Application:

* Let X be a topological space admitting a finite cell decomposition (cf. n° 3). a) Let K and K' be two fields, set $b_i = \dim_K(H_i(X, K))$ and $b'_i = \dim_{K'}(H_i(X, K'))$. One does not necessarily have $b_i = b'_i$, but one has $\sum (-1)^i b_i = \sum (-1)^i b'_i$.

b) Let (X_n) and (X'_n) be two finite cell decompositions of X , and let c_n and c'_n be the number of cells of dimension n in these two decompositions. We have

$$\sum (-1)^i c_i = \sum (-1)^i c'_i.$$

c) With the notations of a) and b), we have $\sum (-1)^i c_i = \sum (-1)^i b_i$.

The properties a) and b) follow from c), and c) follows from prop. 11 applied to the complex Γ described in no. 3, taking for $\mathcal{C}$ the class of finite-dimensional K -vector spaces and for φ the function defined by $\varphi([M]) = \dim_K(M)$ (X, p. 40, example 1). *

10.2.9 Complexes of right modules, complexes of multimodules

A complex of right A -modules is a graded right A -module $(M_n)_{n \in \mathbb{Z}}$ equipped with a graded endomorphism of degree -1 and square zero; it is therefore a complex of modules over the ring A° opposite to A . All the definitions and properties stated in the preceding sections therefore apply to complexes of right modules considered as complexes of modules over the ring A° .

Similarly, if A and B are two rings, a complex of (A, B) -bimodules is a graded (A, B) -bimodule M equipped with an endomorphism d graded of degree (-1) and of square zero; if one endows M with its canonical structure of $A \otimes_{\mathbb{Z}} B^\circ$ -module on the left, d endows it with a structure of $A \otimes_{\mathbb{Z}} B^\circ$ -complex. All definitions and properties stated in the preceding sections therefore apply to complexes of bimodules. Complexes of multimodules are defined analogously.

10.2.10 Example: de Rham complex

In this section, we assume that A is a commutative k -algebra over a commutative ring k . One denotes by $\Omega_{A/k}^1$ the A -module of Kähler differentials of A (III, p. 134), $d^0 : A \rightarrow \Omega_{A/k}^1$ the k -derivation $d_{A/k}$, and $\Omega_{A/k}$ the graded k -algebra $\Lambda_A(\Omega_{A/k}^1)$.

PROPOSITION 13. — There exists a unique k -antiderivation $d : \Omega_{A/k} \rightarrow \Omega_{A/k}$ of degree 1, with square zero, extending the derivation $d^0 : A \rightarrow \Omega_{A/k}^1$.

Let us prove the uniqueness of the antiderivation d . Since $d \circ d = 0$ we have, for $y, x_1, \dots, x_p \in A$:

$$d(y dx_1 \wedge \dots \wedge dx_p) = dy \wedge dx_1 \wedge \dots \wedge dx_p.$$

The A -module $\Omega_{A/k}^p$ being generated by the elements $dx_1 \wedge \dots \wedge dx_p$, this proves the uniqueness of d .

To prove existence, it suffices to construct a k -homomorphism $d^1 : \Omega_{A/k}^1 \rightarrow \Omega_{A/k}^2$ such that $d^1 \circ d^0 = 0$ and

$$d^1(a\omega) = d^0(a) \wedge \omega + ad^1(\omega) \quad \text{for } a \in A, \quad \omega \in \Omega_{A/k}^1. \quad (11)$$

Indeed, it then follows from III, p. 128, prop. 14 (taking into account III, p. 118, remark 2) that there exists an antiderivation $d : \Omega_{A/k} \rightarrow \Omega_{A/k}$ that coincides with d^0 in degree 0 and with d^1 in degree 1. Since d^0 vanishes on A , the antiderivation d is k -linear; since $d^1 \circ d^0 = 0$, we have $d \circ d = 0$ since $\Omega_{A/k}$ is generated as A -algebra by the elements $d^0 a$ for $a \in A$.

To define d^1 , recall (III, p. 133) that $\Omega_{A/k}^1$ is equal to the A -module $\mathfrak{I}/\mathfrak{I}^2$, where $\mathfrak{I}$ is the kernel of the multiplication $m : A \otimes_k A \rightarrow A$. Consider the k -linear map $u : A \otimes_k A \rightarrow \Omega_{A/k}^2$ defined by $u(x \otimes y) = d^0(y) \wedge d^0(x)$. We have

$$u(ax \otimes y - x \otimes ay) = d^0(y) \wedge d^0(ax) - d^0(ay) \wedge d^0(x) = d^0(xy) \wedge d^0(a)$$

for x, y , and a in A , whence

$$u((a \otimes 1 - 1 \otimes a)\xi) = d^0(m(\xi)) \wedge d^0(a), \quad \xi \in A \otimes_k A, \quad a \in A. \quad (12)$$

Since $\mathfrak{I}$ is generated as a left A -module by the elements $(a \otimes 1 - 1 \otimes a)$ for $a \in A$, we deduce that $u(\mathfrak{I}^2) = 0$; therefore u defines by restriction to $\mathfrak{I}$ and passage to the quotient a k -linear map $d^1 : \mathfrak{I}/\mathfrak{I}^2 \rightarrow \Omega_{A/k}^2$.

By making $\xi = b \otimes 1$ in (12), with $b \in A$, we obtain $d^1(bd^0(a)) = d^0(b) \wedge d^0(a)$; it follows that $d^1 \circ d^0 = 0$ and that $d^1(c\omega) = d^0(c) \wedge \omega + cd^1(\omega)$ for $c \in A$ and $\omega = bd^0(a)$. Since $\Omega_{A/k}^1$ is generated as k -module by the elements $bd^0(a)$, for a and b in A , formula (11) is satisfied for all $\omega \in \Omega_{A/k}$, which completes the proof of the proposition.

Sometimes it is said that the elements $\omega \in \Omega_{A/k}^p$ are the exterior differential forms of degree p of A over k and that the antiderivation d is the exterior differential of $\Omega_{A/k}$; the complex $(\Omega_{A/k}, d)$ is called the de Rham complex of A over k , and its homology is the de Rham cohomology of A over k .

Example 1. — Take for A the ring $k[\mathbf{X}_1, \dots, \mathbf{X}_n]$. Then $\Omega_{A/k}^1$ is a free A -module with basis $d\mathbf{X}_1, \dots, d\mathbf{X}_n$ (III, p. 134, example). Consequently, if for every subset $I = \{i_1, \dots, i_p\}$ of $[1, n]$ we set $d\mathbf{X}_I = d\mathbf{X}_{i_1} \wedge \dots \wedge d\mathbf{X}_{i_p}$ (with $i_1 < \dots < i_p$), the A -module $\Omega_{A/k}^p$ admits as a basis the elements $d\mathbf{X}_I$, where I ranges over the set of subsets of $[1, n]$ of cardinality p . We have

$$d(Pd\mathbf{X}_I) = dP \wedge d\mathbf{X}_I = \sum_{i \notin I} (-1)^{n(I,i)} \frac{\partial P}{\partial X_i} d\mathbf{X}_{I \cup \{i\}},$$

where $n(I, i)$ denotes the number of elements of I strictly less than i .

The cycles of $Z^p(\Omega_{A/k})$ are therefore the elements $\omega = \sum_{\text{Card}(I)=p} P_I d\mathbf{X}_I$ such that, for every subset J with $(p+1)$ elements of $[1, n]$, one has:

$$\sum_{i \in J} (-1)^{n(J,i)} \frac{\partial P_{J-\{i\}}}{\partial X_i} = 0.$$

The element ω is a boundary if one can choose, for every subset $J \subset [1, n]$ with $(p-1)$ elements, a polynomial $Q_J \in A$ such that:

$$P_I = \sum_{j \in I} (-1)^{n(I,j)} \frac{\partial Q_{I-\{j\}}}{\partial X_j}.$$

We will see in § 9 that the de Rham complex of A over k is acyclic in degrees > 0 if k is a Q -algebra (X, p. 159, remark 4).

* Example 2. — Suppose that $k = \mathbf{C}$ and $\mathbf{A} = \mathbf{C}[X_1, \dots, X_n]/(P_1, \dots, P_r)$, where the P_i are polynomials in $X_1, \dots, X_n$, such that the set of points of $\mathbf{C}^n$ where all the P_i vanish is an analytic submanifold V of $\mathbf{C}^n$. One can show that the de Rham cohomology of $\mathbf{A}$ over $\mathbf{C}$ is isomorphic to the singular cohomology $H(V, \mathbf{C})$. *

Let M be an $\mathbf{A}$ -module and ∇^0 a k -linear map from M into $\mathbf{M} \otimes_{\mathbf{A}} \Omega_{\mathbf{A}/k}^1$ such that

$$\nabla^0(am) = a\nabla^0(m) + m \otimes da \quad \text{for } a \in \mathbf{A}, \quad m \in M \quad (13)$$

(one sometimes says that ∇^0 is a connection on the $\mathbf{A}$ -module M).

Proposition 14. — (i) There exists a unique k -linear map ∇ of the $\Omega_{\mathbf{A}/k}$ right module $\mathbf{M} \otimes_{\mathbf{A}} \Omega_{\mathbf{A}/k}$ into itself, graded of degree 1, which extends ∇^0 in degree 0 and satisfies the identity:

$$\nabla(x\omega) = (\nabla x)\omega + (-1)^p x(d\omega) \quad \text{for } x \in \mathbf{M} \otimes_{\mathbf{A}} \Omega_{\mathbf{A}/k}^p, \quad \omega \in \Omega_{\mathbf{A}/k}. \quad (14)$$

(ii) The composite mapping $\nabla \circ \nabla$ is $\Omega_{\mathbf{A}/k}$ -linear; in particular the map $\mathbf{R} = \nabla^1 \circ \nabla^0$ of $\mathbf{M}$ to $\mathbf{M} \otimes_{\mathbf{A}} \Omega_{\mathbf{A}/k}^2$ is $\mathbf{A}$ -linear, and we have

$$\nabla \circ \nabla(m \otimes \omega) = R(m) \cdot \omega \quad \text{for } m \in M, \quad \omega \in \Omega_{\mathbf{A}/k}.$$

The homomorphism R is sometimes called the curvature homomorphism of the connection ∇^0 ; if it is zero, the pair $(\mathbf{M} \otimes_{\mathbf{A}} \Omega_{\mathbf{A}/k}, \nabla)$ is a complex, also called the de Rham complex of $(\mathbf{M}, \nabla^0)$ over k .

Let us prove (i). The uniqueness of ∇ is obvious. Let us define a k -homomorphism $\overline{\nabla}$ of $\mathbf{M} \otimes_k \Omega_{\mathbf{A}/k}$ to $\mathbf{M} \otimes_{\mathbf{A}} \Omega_{\mathbf{A}/k}$ by

$$\overline{\nabla}(m \otimes_k \omega) = (\nabla^0 m)\omega + m \otimes d\omega \quad \text{for } m \in \mathbf{M}, \quad \omega \in \Omega_{\mathbf{A}/k}.$$

It follows from (13) that $\overline{\nabla}(am \otimes \omega) = \overline{\nabla}(m \otimes a\omega)$, so that by passing to the quotient we obtain a k -homomorphism ∇ of $\mathbf{M} \otimes_{\mathbf{A}} \Omega_{\mathbf{A}/k}$ into itself, graded of degree 1, extending ∇^0 in degree 0. Let us verify (14): we have, for $m \in \mathbf{M}$, $\alpha \in \Omega_{\mathbf{A}/k}^p$, $\omega \in \Omega_{\mathbf{A}/k}$:

$$\begin{aligned}
\nabla((m \otimes \alpha) \cdot \omega) &= \nabla(m \otimes (\alpha \wedge \omega)) = \nabla^0(m) \cdot (\alpha \wedge \omega) + m \otimes d(\alpha \wedge \omega) \\
&= \nabla^0(m) \alpha \cdot \omega + (m \otimes d\alpha) \omega + (-1)^p (m \otimes \alpha) d\omega \\
&= (\nabla(m \otimes \alpha)) \omega + (-1)^p (m \otimes \alpha) d\omega
\end{aligned}$$

which proves (14) for $x = m \otimes \alpha$; the general case follows from this by linearity.

Let us prove (ii). Let $x \in \mathbf{M} \otimes_{\mathbf{A}} \Omega_{\mathbf{A}/k}^p$, $\omega \in \Omega_{\mathbf{A}/k}$; by repeated application of (14), we obtain:

$$\begin{aligned}
\nabla \circ \nabla(x\omega) &= \nabla(\nabla(x)\omega) + (-1)^p \nabla(xd\omega) \\
&= (\nabla \circ \nabla(x))\omega + (-1)^{p+1} \nabla(x)(d\omega) + (-1)^p \nabla(x)(d\omega) \\
&= (\nabla \circ \nabla(x))\omega,
\end{aligned}$$

which proves the first assertion of (ii); the others follow immediately.

10.3 RESOLUTIONS

We keep the conventions of the preceding paragraph.

10.3.1 Extension of morphisms of complexes

Lemma 1. — Consider a diagram of $\mathbf{A}$ -modules and homomorphisms

$$\begin{array}{ccccc}
\mathbf{M}' & \xrightarrow{\alpha'} & \mathbf{M} & \xrightarrow{\alpha} & \mathbf{M}'' \\
f' \downarrow & \swarrow k/f & \downarrow & \swarrow k'' & \\
\mathbf{N}' & \xrightarrow{\beta'} & \mathbf{N} & \xrightarrow{\beta} & \mathbf{N}''
\end{array}$$

such that $f \circ \alpha' = \beta \circ f'$, $\alpha \circ \alpha' = 0$, $\text{Ker } \beta = \text{Im } \beta'$, and $f = k'' \circ \alpha + \beta \circ k$ and where $\mathbf{M}'$ is projective. There exists an $\mathbf{A}$ -homomorphism $k' : \mathbf{M}' \rightarrow \mathbf{N}'$ such that $f' = k \circ \alpha' + \beta' \circ k'$.

Indeed, set $g = f' - k \circ \alpha'$; we have

$$\beta \circ g = \beta \circ f' - \beta \circ k \circ \alpha' = f \circ \alpha' - \beta \circ k \circ \alpha' = k'' \circ \alpha \circ \alpha' = 0.$$

This implies $\text{Im}(g) \subset \text{Ker}(\beta) = \text{Im}(\beta')$. Since $\mathbf{M}'$ is projective, there therefore exists an

A-homomorphism $k' : \mathbf{M}' \rightarrow \mathbf{N}'$ such that $\beta' \circ k' = g$, whence the lemma.

Lemma 2. — If in the commutative diagram of A-modules and homomorphisms

$$\begin{array}{ccccc} \mathbf{M}' & \xrightarrow{\alpha'} & \mathbf{M} & \xrightarrow{\alpha} & \mathbf{M}'' \\ & & \downarrow u & & \downarrow u'' \\ \mathbf{N}' & \xrightarrow{\beta'} & \mathbf{N} & \xrightarrow{\beta} & \mathbf{N}'' \end{array}$$

we have $\alpha \circ \alpha' = 0$, $\text{Ker } \beta = \text{Im } \beta'$ and if $\mathbf{M}'$ is projective, there exists an A-homomorphism $u' : \mathbf{M}' \rightarrow \mathbf{N}'$ such that $\beta' \circ u' = u \circ \alpha'$.

It suffices to set $k'' = u''$, $k = -u$, $f = 0$, $f' = 0$ and $u' = k'$ in Lemma 1.

Lemma 1 bis. — Consider a diagram of A-modules and homomorphisms

$$\begin{array}{ccccc} \mathbf{M}' & \xrightarrow{\alpha'} & \mathbf{M} & \xrightarrow{\alpha} & \mathbf{M}'' \\ & & \downarrow k' \downarrow f' \downarrow k \downarrow f & & \\ \mathbf{N}' & \xrightarrow{\beta'} & \mathbf{N} & \xrightarrow{\beta} & \mathbf{N}'' \end{array}$$

such that $f \circ \alpha' = \beta \circ f'$, $\text{Ker } \alpha = \text{Im } \alpha'$, $\beta \circ \beta' = 0$, and $f' = k \circ \alpha' + \beta' \circ k'$ and where $\mathbf{N}''$ is injective. There exists an A-homomorphism $k'' : \mathbf{M}'' \rightarrow \mathbf{N}''$ such that $f = k'' \circ \alpha + \beta \circ k$.

Indeed, set $q = f - \beta \circ k$, we have

$$g \circ \alpha' = f \circ \alpha' - \beta \circ k \circ \alpha' = \beta \circ f' - \beta \circ k \circ \alpha' = \beta \circ \beta' \circ k' = 0.$$

This implies $\text{Ker } g \supset \text{Im } \alpha' = \text{Ker } \alpha$. Since $\mathbf{N}''$ is injective, there therefore exists (X, p. 16, remark) an A-homomorphism $k'' : \mathbf{M}'' \rightarrow \mathbf{N}''$ such that $g = k'' \circ \alpha$, whence the lemma.

Lemma 2 bis. — If, in the commutative diagram of A-modules and homomorphisms

$$\begin{array}{ccccc} \mathbf{M}' & \xrightarrow{\alpha'} & \mathbf{M} & \xrightarrow{\alpha} & \mathbf{M}'' \\ u' \downarrow & & u \downarrow & & \\ \mathbf{N}' & \xrightarrow{\beta'} & \mathbf{N} & \xrightarrow{\beta} & \mathbf{N}'' \end{array},$$

we have $\text{Ker } \alpha = \text{Im } \alpha'$, $\beta \circ \beta' = 0$ and if $\mathbf{N}''$ is injective, there exists an A-homomorphism $u'' : \mathbf{M}'' \rightarrow \mathbf{N}''$ such that $u'' \circ \alpha = \beta \circ u$.

It suffices to set $u' = k'$, $u = -k$, $f = 0$, $f' = 0$ and $k'' = u''$ in Lemma 1 bis.

PROPOSITION 1. — Let (P, d_P) and (E, d_E) be two complexes of A -modules and r an integer.

- a) Let $(u_i : P_i \rightarrow E_i)_{i \leq r}$ be a family of homomorphisms such that $d_E \circ u_i = u_{i-1} \circ d_P$ for $i \leq r$. Suppose that P_i is projective for $i > r$ and that $H_i(E) = 0$ for $i \geq r$. Then the family of the u_i extends to a morphism of complexes from P to E ; two such extensions are homotopic.
- b) Let $(u^i : P^i \rightarrow E^i)_{i \leq r}$ be a family of homomorphisms such that $u^i \circ d_P = d_E \circ u^{i-1}$ for $i \leq r$. Suppose that E^i is injective for $i > r$ and that $H^i(P) = 0$ for $i \geq r$. Then the family of the u^i extends to a morphism of complexes from P to E ; two such extensions are homotopic.

Let us prove a). The existence of an extension v of the family $(u_i)_{i \leq r}$ follows immediately from Lemma 2 by induction. Let v' be another extension; set $f = v' - v$, and construct by induction on the integer n a homomorphism $k_n : P_n \rightarrow E_{n+1}$ such that $f_n = d_E \circ k_n + k_{n-1} \circ d_P$. For $i \leq r$, take $k_i = 0$. Let $n \geq r$ and suppose the k_i constructed for $i \leq n$. Consider then the diagram

$$\begin{array}{ccccc} P_{n+1} & \xrightarrow{d_P} & P_n & \xrightarrow{d_P} & P_{n-1} \\ f_{n+1} \downarrow & & k_n \downarrow & f_n \downarrow & k_{n-1} \downarrow \\ E_{n+2} & \xrightarrow{d_E} & E_{n+1} & \xrightarrow{d_E} & E_n \end{array}$$

The hypotheses of Lemma 1 are satisfied; there therefore exists an A -homomorphism $k_{n+1} : P_{n+1} \rightarrow E_{n+2}$ such that $f_{n+1} = d_E \circ k_{n+1} + k_n \circ d_P$, whence a).

The proof of b) is analogous, via Lemmas 1 bis and 2 bis.

10.3.2 Resolutions

In what follows, one always identifies a module with the complex of which it is the degree-zero component and of which all other components are zero.

DEFINITION 1. — Let M be an A -module. A left resolution of M is a pair (P, p) where P is a complex zero to the right and $p : P \rightarrow M$ is a homomorphism. A right resolution of M is a pair

(e, E) where E is a complex zero to the left and $e : M \rightarrow E$ a homomorphism.

One calls the length of the resolution (P, p) (resp. (e, E)) the length of the complex P (resp. E). If (P, p) and (P', p') (resp. (e, E) and (e', E')) are two left (resp. right) resolutions of M , a morphism of complexes $f : P \rightarrow P'$ such that $p' \circ f = p$ (resp. $g : E \rightarrow E'$ such that $g \circ e = e'$) is called a morphism of resolutions.

PROPOSITION 2. — Let P be a complex that is zero on the right and let $p : P \rightarrow M$ be a morphism. For (P, p) to be a left resolution of M , it is necessary and sufficient that the sequence

$$\cdots \rightarrow P_n \xrightarrow{d_p} P_{n-1} \rightarrow \cdots \rightarrow P_1 \xrightarrow{d_p} P_0 \xrightarrow{p_0} M \rightarrow 0 \quad (1)$$

be exact.

Indeed, to say that $p : P \rightarrow M$ is a homomorphism means that $H_i(P) = 0$ for $i > 0$ and that p_0 induces an isomorphism from $\text{Coker}(d_p : P_1 \rightarrow P_0)$ onto M .

Similarly:

PROPOSITION 2bis. — Let E be a complex that is zero on the left and let $e : M \rightarrow E$ be a morphism. For (e, E) to be a right resolution of M , it is necessary and sufficient that the sequence

$$0 \rightarrow M \xrightarrow{e_0} E^0 \xrightarrow{d_E} E^1 \rightarrow \cdots \rightarrow E^n \xrightarrow{d_E} E^{n+1} \rightarrow \cdots \quad (1 \text{ bis})$$

be exact.

By abuse of language, one often says that the sequence (1) (resp. (1 bis)) is a left (resp. right) resolution of M .

DEFINITION 2. — A projective (resp. free, resp. flat) resolution of the A -module M is a left resolution (P, p) of M such that the complex P is projective (resp. free, resp. flat) (X, p. 25). An injective resolution of M is a right resolution (e, E) of M such that the complex E is injective (loc. cit.).

Examples. — 1) Suppose that the ring A is principal; let M be an A -module and $(x_i)_{i \in I}$ a generating family of M . Let L_0 be the free module $A^{(I)}$, (e_i) its canonical basis, and define $p : L_0 \rightarrow M$ by $p(e_i) = x_i$. The morphism p is surjective and its kernel L_1 is a free A -module

by VII, § 3, cor. 2 to th. 1, so the exact sequence

$$0 \rightarrow L_1 \rightarrow L_0 \xrightarrow{P} M \rightarrow 0$$

is a free resolution of M of length 1. If I is finite, L_0 and L_1 are finitely generated.

- 2) Suppose A is commutative; let E be an A -module and u an endomorphism of E . Denote by E_u the $A[X]$ -module obtained by endowing E with the structure defined by

$$(p, x) \mapsto p(u)(x) \quad \text{for } p \in A[X] \quad \text{and} \quad x \in E.$$

By III, p. 106, we have an exact sequence:

$$0 \rightarrow A[X] \otimes_A E \xrightarrow{\psi} A[X] \otimes_A E \xrightarrow{\varphi} E_u \rightarrow 0$$

where $\varphi(p \otimes x) = p.x$ and $\psi(p \otimes x) = Xp \otimes x - p \otimes u(x)$ for $p \in A[X]$ and $x \in E$. This exact sequence is a resolution of length 1 of E_u free (resp. projective, resp. finitely generated) if E is a free (resp. projective, resp. finitely generated) A -module. 3) If A is principal, the exact sequence

$$0 \rightarrow A \rightarrow K \rightarrow K/A \rightarrow 0$$

is an injective resolution of length 1 of the A -module A_s (X, p. 18, example 1).

PROPOSITION 3. — Let $f : \mathbf{M}' \rightarrow \mathbf{M}$ be a homomorphism of A -modules, $p' : \mathbf{P}' \rightarrow \mathbf{M}'$ be a morphism into $\mathbf{M}'$ of a complex that is zero on the right and projective $\mathbf{P}'$, and $p : \mathbf{P} \rightarrow \mathbf{M}$ a left resolution of M . There exists a morphism of complexes $\tilde{f} : \mathbf{P}' \rightarrow \mathbf{P}$ and one only up to homotopy, such that $p \circ \tilde{f} = f \circ p'$.

Consider the complex $\bar{\mathbf{P}}$ defined as follows: $\bar{\mathbf{P}}_n = \mathbf{P}_n$ for $n \neq -1$, $\bar{\mathbf{P}}_{-1} = \mathbf{M}$, $d_{\bar{\mathbf{P}},n} = d_{\mathbf{P},n}$ for $n \neq 0, -1$, $d_{\bar{\mathbf{P}},0} = p_0$, $d_{\bar{\mathbf{P}},-1} = 0$, and the complex $\bar{\mathbf{P}}'$ defined analogously. Applying to the complexes $\bar{\mathbf{P}}$ and $\bar{\mathbf{P}}'$ prop. 1 a) with $r = 0$, $u_i = 0$ for $i < -1$ and $u_{-1} = f$, we obtain prop. 3.

COROLLARY. — Let (P, p) and (P', p') be two projective resolutions of M . There exists one and only one homotopism up to homotopy $\alpha : P' \rightarrow P$ such that $p \circ \alpha = p'$.

Indeed, there exists a morphism $\alpha : P' \rightarrow P$ (resp. $\beta : P \rightarrow P'$) such that $p \circ \alpha = p'$ (resp. $p' \circ \beta = p$). Since $p \circ \alpha \circ \beta = p$ (resp. $p' \circ \beta \circ \alpha = p'$), $\alpha \circ \beta$ is homotopic to 1_P (resp. $\beta \circ \alpha$ is homotopic to $1_{P'}$).

PROPOSITION 3 bis. — Let $g : \mathbf{N} \rightarrow \mathbf{N}'$ be a homomorphism of A -modules, $e' : \mathbf{N}' \rightarrow \mathbf{E}'$ a morphism of $\mathbf{N}'$ into a complex that is zero on the left and injective $\mathbf{E}'$, and $e : \mathbf{N} \rightarrow \mathbf{E}$ a right resolution of $\mathbf{N}$. There exists a morphism of complexes $\tilde{g} : \mathbf{E} \rightarrow \mathbf{E}'$ and one only up to homotopy, such that $\tilde{g} \circ e = e' \circ g$.

This is proved like Prop. 3 with the aid of Prop. 1 b).

COROLLARY. — Let (e, E) and (e', E') be two injective resolutions of $\mathbf{N}$; there exists a homotopism $\alpha : E \rightarrow E'$ and only one up to homotopy such that $\alpha \circ e = e'$.

10.3.3 The canonical free resolution

For every A -module M , denote by $L_0(M)$ the free A -module $A^{(M)}$ with basis M , $(e_m)_{m \in M}$ its canonical basis and $p_M : L_0(M) \rightarrow M$ the homomorphism such that

$$p_M(e_m) = m, \quad m \in M.$$

Set $Z_0(M) = \text{Ker } p_M$ and let $i_M : Z_0(M) \rightarrow L_0(M)$ be the canonical injection. We have an exact sequence

$$0 \rightarrow Z_0(M) \xrightarrow{i_M} L_0(M) \xrightarrow{p_M} M \rightarrow 0. \quad (1)$$

We define a graded module $L(M)$ by setting $L_n(M) = 0$ for $n < 0$ and, by induction on the integer $n > 0$

$$L_n(M) = L_0(Z_{n-1}(M)); \quad Z_n(M) = Z_0(Z_{n-1}(M)). \quad (2)$$

We define A -homomorphisms $d_n^M : L_n(M) \rightarrow L_{n-1}(M)$ by

$$\begin{cases} d_n^{\mathbf{M}} = 0, & n \leq 0, \\ d_1^{\mathbf{M}} = i_{\mathbf{M}} \circ p_{Z_0(\mathbf{M})}, \\ d_n^{\mathbf{M}} = i_{Z_{n-2}(\mathbf{M})} \circ p_{Z_{n-1}(\mathbf{M})}, & n > 1. \end{cases} \quad (3)$$

By construction, we have an exact sequence

$$\rightarrow L_n(\mathbf{M}) \xrightarrow{d_n^{\mathbf{M}}} L_{n-1}(\mathbf{M}) \rightarrow \cdots \rightarrow L_0(\mathbf{M}) \xrightarrow{p_{\mathbf{M}}} \mathbf{M} \rightarrow 0,$$

so that, if one extends $p_{\mathbf{M}}$ as a morphism of complexes

$$p_{\mathbf{M}} : (L(\mathbf{M}), d^{\mathbf{M}}) \rightarrow \mathbf{M},$$

one obtains a free resolution of $\mathbf{M}$, called the canonical free resolution of $\mathbf{M}$.

Let $f : \mathbf{M} \rightarrow \mathbf{N}$ be a homomorphism of A -modules. Denote by

$$L_0(f) : L_0(\mathbf{M}) \rightarrow L_0(\mathbf{N})$$

the unique A -homomorphism such that $L_0(f)(e_m) = e_{f(m)}$ for all $m \in \mathbf{M}$. We have

$$p_{\mathbf{N}} \circ L_0(f) = f \circ p_{\mathbf{M}}. \quad (4)$$

Consequently, $L_0(f)$ induces an A -homomorphism $Z_0(f) : Z_0(\mathbf{M}) \rightarrow Z_0(\mathbf{N})$ and we have

$$i_{\mathbf{N}} \circ Z_0(f) = L_0(f) \circ i_{\mathbf{M}}. \quad (5)$$

Set $L_n(f) = 0$, for $n < 0$ and define by induction on the integer $n > 0$, homomorphisms $L_n(f) : L_n(\mathbf{M}) \rightarrow L_n(\mathbf{N})$ and $Z_n(f) : Z_n(\mathbf{M}) \rightarrow Z_n(\mathbf{N})$ by

$$\begin{cases} L_n(f) = L_0(Z_{n-1}(f)) \\ Z_n(f) = Z_0(Z_{n-1}(f)). \end{cases} \quad (6)$$

PROPOSITION 4. — $L(f) : L(\mathbf{M}) \rightarrow L(\mathbf{N})$ is a morphism of complexes of A -modules; moreover, we have $p_{\mathbf{N}} \circ L(f) = f \circ p_{\mathbf{M}}$.

We need to prove that, for every integer $n > 0$, the formula

$$d_n^{\mathbf{N}} \circ L_n(f) = L_{n-1}(f) \circ d_n^{\mathbf{M}}.$$

We first have

$$\begin{aligned} d_1^{\mathbf{N}} \circ L_1(f) &= i_{\mathbf{N}} \circ p_{Z_0(\mathbf{N})} \circ L_0(Z_0(f)) && \text{(by (3) and (6))} \\ &= i_{\mathbf{N}} \circ Z_0(f) \circ p_{Z_0(\mathbf{M})} && \text{(by (4))} \\ &= L_0(f) \circ i_{\mathbf{M}} \circ p_{Z_0(\mathbf{M})} && \text{(by (5))} \\ &= L_0(f) \circ d_1^{\mathbf{M}} && \text{(by (3)).} \end{aligned}$$

When $n > 1$, we successively have

$$\begin{aligned} d_n^{\mathbf{N}} \circ L_n(f) &= i_{Z_{n-2}(\mathbf{N})} \circ p_{Z_{n-1}(\mathbf{N})} \circ L_0(Z_{n-1}(f)) && \text{(by (3) and (6))} \\ &= i_{Z_{n-2}(\mathbf{N})} \circ Z_{n-1}(f) \circ p_{Z_{n-1}(\mathbf{M})} && \text{(by (4))} \\ &= i_{Z_{n-2}(\mathbf{N})} \circ Z_0(Z_{n-2}(f)) \circ p_{Z_{n-1}(\mathbf{M})} && \text{(by (6))} \\ &= L_0(Z_{n-2}(f)) \circ i_{Z_{n-2}(\mathbf{M})} \circ p_{Z_{n-1}(\mathbf{M})} && \text{(by (5))} \\ &= L_{n-1}(f) \circ d_n^{\mathbf{M}} && \text{(by (3) and (6)).} \end{aligned}$$

We immediately have

$$L(1_{\mathbf{M}}) = 1_{L(\mathbf{M})}. \quad (7)$$

On the other hand, if $g : \mathbf{N} \rightarrow \mathbf{P}$ is a homomorphism of $\mathbf{A}$ -modules, we have

$$L(g \circ f) = L(g) \circ L(f). \quad (8)$$

Indeed, we have for $m \in \mathbf{M}$,

$$L_0(g \circ f)(e_m) = e_{g \circ f(m)} = L_0(g)(e_{f(m)}) = L_0(g) \circ L_0(f)(e_m),$$

hence $L_0(g \circ f) = L(g) \circ L(f)$; therefore $Z_0(g \circ f) = Z_0(g) \circ Z_0(f)$; whence immediately $L_n(g \circ f) = L_n(g) \circ L_n(f)$, for $n \geq 0$, by induction on n , whence (8).

Remark. — If $f, g \in \text{Hom}_{\mathbf{A}}(\mathbf{M}, \mathbf{N})$, one does not have $L(f + g) = L(f) + L(g)$. However,

these two morphisms are homotopic according to X, p. 49, prop. 3.

Let M be a right A -module; denote by A° the opposite ring of A , by M° the underlying A° -module of M , and by $L(M^\circ)$ its canonical free resolution. We denote by $L(M)$ and call the canonical free resolution of M the underlying A -complex $L(M)^\circ$ of $L(M^\circ)$. We thus have

$$L(M^\circ) = L(M)^\circ.$$

10.3.4 The canonical injective resolution

Let F be the A -module $\text{Hom}_{\mathbf{Z}}(A, \mathbf{Q}/\mathbf{Z})$; for every A -module M , we set $I^0(M) = F^{\text{Hom}_A(M, F)}$ and we denote by $e_M : M \rightarrow I^0(M)$ the homomorphism which maps $m \in M$ to the family $(\varphi(m))_{\varphi \in \text{Hom}_A(M, F)}$. By X, p. 19, cor. 2, $I^0(M)$ is an injective A -module and e_M is injective. Set $K^0(M) = \text{Coker } e_M$ and denote by $q_M : I^0(M) \rightarrow K^0(M)$ the canonical projection. We thus have an exact sequence

$$0 \rightarrow M \xrightarrow{e_M} I^0(M) \xrightarrow{q_M} K^0(M) \rightarrow 0.$$

We define a graded A -module $I(M)$ by setting $I^n(M) = 0$ for $n < 0$ and, by induction on the integer $n > 0$,

$$I^n(M) = I^0(K^{n-1}(M)), \quad K^n(M) = K^0(K^{n+1}(M)). \quad (9)$$

We define A -homomorphisms $\delta_M^n : I^n(M) \rightarrow I^{n+1}(M)$ by

$$\begin{cases} \delta_M^n = 0, & n < 0, \\ \delta_M^0 = e_{K^0(M)} \circ q_M, \\ \delta_M^n = e_{K^n(M)} \circ q_{K^{n-1}(M)}, & n > 0. \end{cases} \quad (10)$$

By construction, we have an exact sequence

$$0 \rightarrow M \xrightarrow{e_M} I^0(M) \xrightarrow{\delta_M^0} \dots \rightarrow I^n(M) \xrightarrow{\delta_M^n} I^{n+1}(M) \rightarrow \dots,$$

so that, if one extends e_M as a morphism of complexes

$$e_{\mathbf{M}} : \mathbf{M} \rightarrow (I(\mathbf{M}), \delta_{\mathbf{M}}),$$

we obtain an injective resolution of $\mathbf{M}$, called the canonical injective resolution of $\mathbf{M}$. Let $f : \mathbf{M} \rightarrow \mathbf{N}$ be a homomorphism of A -modules. Denote by $I^0(f)$ the homomorphism from $I^0(\mathbf{M}) = F^{\text{Hom}_A(\mathbf{M}, F)}$ to $I^0(\mathbf{N}) = F^{\text{Hom}_A(\mathbf{N}, F)}$ which maps the family $(x_\varphi)_{\varphi \in \text{Hom}_A(\mathbf{M}, F)}$ to the family $(x_{\psi \circ f})_{\psi \in \text{Hom}_A(\mathbf{N}, F)}$. We have:

$$I^0(f) \circ e_{\mathbf{M}} = e_{\mathbf{N}} \circ f. \quad (11)$$

Consequently, $I^0(f)$ induces a homomorphism $K^0(f) : K^0(\mathbf{M}) \rightarrow K^0(\mathbf{N})$ and we have

$$K^0(f) \circ q_{\mathbf{M}} = q_{\mathbf{N}} \circ K^0(f). \quad (12)$$

Set $I^n(f) = 0$ for $n < 0$ and let us define, by induction on the integer $n > 0$, homomorphisms $I^n(f) : I^n(\mathbf{M}) \rightarrow I^n(\mathbf{N})$ and $K^n(f) : K^n(\mathbf{M}) \rightarrow K^n(\mathbf{N})$ by:

$$\begin{cases} I^n(f) = I^0(K^{n-1}(f)) \\ K^n(f) = K^0(K^{n-1}(f)). \end{cases} \quad (13)$$

PROPOSITION 5. — $I(f) : I(\mathbf{M}) \rightarrow I(\mathbf{N})$ is a morphism of complexes of A -modules; we have $I(f) \circ e_{\mathbf{M}} = e_{\mathbf{N}} \circ f$.

This is proved in a manner analogous to prop. 4.

One has

$$I(1_{\mathbf{M}}) = 1_{I(\mathbf{M})} \quad (14)$$

and for every homomorphism $g : \mathbf{N} \rightarrow \mathbf{P}$ of A -modules

$$I(g \circ f) = I(g) \circ I(f). \quad (15)$$

Remark. — If $f, g \in \text{Hom}_A(\mathbf{M}, \mathbf{N})$, we do not have $I(f + g) = I(f) + I(g)$. However, these two morphisms are homotopic by X, p. 49, prop. 3 bis.

If M is a right A -module, we set $I(M) = I(M^\circ)^\circ$; it is called the canonical injective resolution of M and we have

$$I(M^\circ) = I(M)^\circ.$$

10.3.5 Finitely generated resolutions

It follows in particular from the two preceding sections that every A -module has injective resolutions, free resolutions (hence also projective or flat resolutions). In certain cases, one can be more precise:

Suppose A is left Noetherian and let M be an A -module. Let us construct recursively sequences $(L_n)_{n \geq 0}, (Z_n)_{n \geq 0}, (d_n)_{n \geq 1}$ where, for all $n \geq 0$, L_n is a finitely generated free A -module, Z_n a submodule of L_n and $d_{n+1} : L_{n+1} \rightarrow L_n$ a homomorphism. For this, choose a finite generating family $(m_i)_{i \in I_0}$ of M , set $L_0 = A^{(I_0)}$, define $p : L_0 \rightarrow M$ by $p(e_i) = m_i$ and set $Z_0 = \text{Ker}(p)$. For $n \geq 0$, the modules L_n and Z_n having been constructed, Z_n is of finite type since contained in L_n ; let us choose a finite generating family $(x_{n,i})_{i \in I_{n+1}}$ of Z_n ; set $L_{n+1} = A^{(I_{n+1})}$, define d_{n+1} by $d_{n+1}(e_i) = x_{n,i}$ and set $Z_{n+1} = \text{Ker}(d_{n+1})$. By construction, we have an exact sequence

$$\cdots \rightarrow L_{n+1} \xrightarrow{d_{n+1}} L_n \rightarrow \cdots \rightarrow L_0 \xrightarrow{p} M \rightarrow 0,$$

whence:

PROPOSITION 6. — When A is left Noetherian, every finitely generated A -module M has a free resolution $p : L \rightarrow M$ such that L_n is finitely generated for every integer n . More generally:

PROPOSITION 7. — Let C be an A -complex and let $a \in \mathbb{Z}$ be such that $H_n(C) = 0$ for $n < a$. a) There exists a free A -complex L such that $L_n = 0$ for $n < a$ and a homomorphism $f : L \rightarrow C$.

b) Suppose A is left Noetherian and the A -modules $H_n(C)$, $n \in \mathbb{Z}$, are finitely generated.

There exists a free A -complex L such that $L_n = 0$ for $n < a$ and that L_n be a finitely

generated A-module for every n , and a homomorphism $f : L \rightarrow C$.

Let C' be the subcomplex of C such that $C'_n = C_n$ for $n > a$, $C'_a = Z_a(C)$, $C'_n = 0$ for $n < a$; then the canonical injection from C' into C is a homomorphism. Replacing C by C' , we may therefore assume that $C_n = 0$ for $n < a$. The statement then follows from the iterated application of the following lemma, for $r = a, a + 1, \dots$:

Lemma 3. — Let C be a complex and $r \in \mathbb{Z}$. There exists a complex C' and a homomorphism $f : C' \rightarrow C$ such that $f_n : C'_n \rightarrow C_n$ be an isomorphism for $n < r$ and that C'_r be a free A-module. If A is Noetherian and the A-modules $H_r(C)$ and C_{r-1} are finitely generated, one can impose that C'_r be finitely generated.

- a) First let $h : \mathbf{M} \rightarrow C_r$ be a homomorphism of A-modules; denote by $d = (d_n)$ the differential of C . Let N be the submodule of $\mathbf{M} \times C_{r+1}$ formed by the pairs (m, x) such that $h(m) = d_{r+1}(x)$; we define a complex (C', d') by $C'_n = C_n$ for $n \neq r, r + 1$, $C'_r = \mathbf{M}$, $C'_{r+1} = N$, $d'_n = d_n$ for $n \neq r, r + 1, r + 2$, $d'_r = d_r \circ h$, $d'_{r+1}(m, x) = m$ for $(m, x) \in N$ and $d'_{r+2}(y) = (0, d_{r+2}(y))$ for $y \in C_{r+2}$. Consider also the morphism of complexes $f : C' \rightarrow C$ such that $f_n = 1_{C_n}$ for $n \neq r, r + 1$, $f_r = h$, $f_{r+1}(m, x) = x$.
- b) The complex $\text{Ker } f$ is zero in degree $\neq r, r + 1$ and the differential d'_{r+1} induces an isomorphism from $\text{Ker } f_{r+1}$ onto $\text{Ker } f_r$, hence $H(\text{Ker } f) = 0$.
- c) When the composite map $\mathbf{M} \xrightarrow{h} C_r \rightarrow C_r/B_r(C)$ is surjective, one sees similarly that $H(\text{Coker } f) = 0$, and f is then a homomorphism (X, p. 31, cor. 2).
- d) When A is assumed to be Noetherian and the A-modules $H_r(C)$ and C_{r-1} are finitely generated, then $C_r/B_r(C)$ is finitely generated, by virtue of the exact sequence (X, p. 25)

$$0 \rightarrow H_r(C) \rightarrow C_r/B_r(C) \rightarrow C_{r-1};$$

there then exists a finitely generated free A-module $\mathbf{M}$ and a homomorphism $h : \mathbf{M} \rightarrow C_r$ such that the condition of c) be satisfied; in the general case, there exists a free module $\mathbf{M}$ and a surjective homomorphism $h : \mathbf{M} \rightarrow C_r$. This completes the proof.

10.3.6 Minimal projective resolutions

Let M be an A -module and let

$$\cdots \rightarrow P_n \xrightarrow{d_n} P_{n-1} \rightarrow \cdots \rightarrow P_0 \xrightarrow{d_0} M \rightarrow 0 \quad (P)$$

a resolution of M . We say that (P) is a minimal projective resolution if, for every $n \geq 0$, the homomorphism $\delta_n : P_n \rightarrow \text{Im}(d_n)$ induced by d_n is a projective cover (VIII, § 8, no. 5).

PROPOSITION 8. — Let M be an A -module, and let P and P' be two minimal projective resolutions of M , with $f : P \rightarrow P'$ a morphism of resolutions. Then f is an isomorphism. In particular, any two minimal projective resolutions of M are isomorphic.

Set $\tilde{P}_n = P_n$ for $n \neq -1$ and $\tilde{P}_{-1} = M$; let us define similarly $\tilde{P}'_n$ and set $f_{-1} = 1_M$. Let us prove by induction starting from -1 that $f_n : \tilde{P}_n \rightarrow \tilde{P}'_n$ is an isomorphism for all n . This is evident for $n = -1$; suppose that f_n and f_{n-1} are isomorphisms. It follows from the commutativity of the diagram:

$$\begin{array}{ccc} P_n & \xrightarrow{d_n} & P_{n-1} \\ \downarrow f_n & & \downarrow f_{n-1} \\ P'_n & \xrightarrow{d'_n} & P'_{n-1} \end{array}$$

that f_n induces an isomorphism g_n of $\text{Ker } d_n$ onto $\text{Ker } d'_n$. It then follows from the commutativity of the diagram:

$$\begin{array}{ccc} P_{n+1} & \xrightarrow{\delta_{n+1}} & \text{Ker } d_n \\ \downarrow f_{n+1} & & \downarrow g_n \\ P'_{n+1} & \xrightarrow{\delta'_{n+1}} & \text{Ker } d'_n \end{array}$$

and from VIII, loc. cit., that f_{n+1} is an isomorphism.

COROLLARY. — Let M be an A -module, and P and P' two projective resolutions of M ; suppose that P is minimal. Let $f : P \rightarrow P'$ and $g : P' \rightarrow P$ be two morphisms of resolutions. Then f is injective, g is surjective, and P' is the direct sum of the subcomplexes $\text{Im } f$ and $\text{Ker } g$. Moreover $\text{Ker } g$ has zero homology.

Indeed, $\alpha = g \circ f$ is an automorphism of P (prop. 8). Set $\tilde{f} = f \circ \alpha^{-1}$. We have

$$\text{Im } \tilde{f} = \text{Im } f \quad \text{and} \quad g \circ \tilde{f} = 1_P,$$

which shows that $\mathbf{P}' = \text{Im } \tilde{f} \oplus \text{Ker } g$. Since the sequence

$$0 \rightarrow \text{Ker } g \rightarrow \mathbf{P}' \xrightarrow{g} \mathbf{P} \rightarrow 0$$

is exact and g is a homomorphism, $\text{Ker } g$ has zero homology.

PROPOSITION 9. — Let M be an A -module and (P, p) a projective resolution of M . Let r denote the radical of A . Suppose either that P_n is a finitely generated A -module for all n , or that r is nilpotent. Then for (P, p) to be minimal it is necessary and sufficient that the complex $(A/r) \otimes_A \mathbf{P}$ have zero differential, in other words that

$$d_{n+1}(\mathbf{P}_{n+1}) \subset r\mathbf{P}_n \quad \text{for all } n \geq 0.$$

Suppose that (P, p) is minimal. By VIII, loc. cit., the homomorphism

$$1 \otimes \delta_n : (A/r) \otimes_A \mathbf{P}_n \rightarrow (A/r) \otimes_A \text{Im } d_n$$

is an isomorphism. It then follows from the exact sequence

$$0 \rightarrow \text{Im } d_{n+1} \xrightarrow{j_n} \mathbf{P}_n \xrightarrow{\delta_n} \text{Im } d_n \rightarrow 0$$

that the homomorphism $1 \otimes j_n : (A/r) \otimes_A \text{Im } d_{n+1} \rightarrow (A/r) \otimes_A \mathbf{P}_n$ is zero; since $d_{n+1} = j_n \circ \delta_{n+1}$, we deduce that $1_{A/r} \otimes d_{n+1} = 0$ for $n \geq 0$.

Conversely, suppose that for all $n \geq 1$, $1 \otimes d_n$ is zero, in other words that $\text{Im } d_n = \text{Ker } d_{n-1}$ is contained in $r\mathbf{P}_{n-1}$. Since δ_{n-1} is surjective, it follows from VIII, loc. cit. that δ_{n-1} is a projective cover for $n \geq 1$, hence that $(\mathbf{P}, p)$ is minimal.

PROPOSITION 10. — Suppose that A is a left Noetherian local ring, and let M be a finitely generated A -module. Then M possesses a minimal resolution (P, p) ; for every $n \geq 0$, $\mathbf{P}_n$ is a finitely generated free module.

Indeed, in the construction made in n° 5 (p. 53), one may, by virtue of VIII, loc. cit., take

for (L_0, p) a projective cover of M , and for L_{n+1} a projective cover of $\text{Ker } d_n$. The resolution thus obtained is then minimal.

Remarks. — 1) Let $\mathfrak{m}$ denote the maximal ideal of A and set $k = A/\mathfrak{m}$. Let P be a minimal projective resolution of M , and set $b_n = \dim_k(k \otimes_A P_n)$. Then P_n is a free A -module of rank b_n . It follows from the corollary of prop. 8 that for any other projective resolution P' of M , one has $\dim_k(k \otimes_A P'_n) \geq b_n$, and that equality holds if and only if P' is minimal.

2) By prop. 9, b_n is the dimension over k of $H_n(k \otimes_A P)$, * in other words of $\text{Tor}_n^A(k, M)$.

It is also the dimension over k of $\text{Ext}_A^n(M, k)$ (cf. X, p. 103, remark 3) *.

10.3.7 Graded resolutions

In this section, one supposes that the ring A is equipped with a grading $(A_n)_{n \in \mathbb{Z}}$, such that $A_n = 0$ for $n < 0$. One says that a graded A -module M is bounded below if $M_n = 0$ for n sufficiently small; every finitely generated graded A -module is bounded below.

PROPOSITION 11. — If M is a graded A -module bounded below (resp. if M is a finitely generated graded A -module and if A is left Noetherian), there exists an exact sequence of graded A -modules, unbounded to the left,

$$\cdots \rightarrow L_n \xrightarrow{d_n} L_{n-1} \rightarrow \cdots \rightarrow L_1 \xrightarrow{d_1} L_0 \xrightarrow{d_0} M \rightarrow 0$$

where the L_i are graded free and bounded below (resp. graded free and finitely generated), and where the d_i are graded homomorphisms of degree 0.

If N is a graded A -module bounded below (resp. and finitely generated over Noetherian A) there exists a graded free A -module bounded below (resp. and finitely generated) L and a surjective graded homomorphism $L \rightarrow N$ (II, p. 167, remark 3).

That being so, suppose given an exact sequence of graded A -modules and graded homomorphisms of degree 0

$$L_n \xrightarrow{d_n} L_{n-1} \rightarrow \cdots \rightarrow L_0 \xrightarrow{d_0} M \rightarrow 0,$$

where the $L_i, i = 0, \dots, n$, are graded free and bounded below (resp. graded free and finitely

generated). Then $N = \text{Ker } d_n$ is bounded below (resp. finitely generated); there therefore exists a graded free A -module bounded below (resp. graded free and finitely generated) L_{n+1} and a graded homomorphism $d_{n+1} : L_{n+1} \rightarrow L_n$ of degree 0, such that $\text{Im } d_{n+1} = N$; the sequence

$$L_{n+1} \xrightarrow{d_{n+1}} L_n \xrightarrow{d_n} L_{n-1} \rightarrow \cdots \rightarrow L_0 \xrightarrow{d_0} M \rightarrow 0$$

is then exact. The proposition then follows from the preceding by induction on n .

10.3.8 The standard resolution

In this section, one supposes that the ring A is an (associative and unital) algebra over a commutative ring k . For $n \geq 0$ we denote by $\mathbf{B}_n$ the tensor product over k of $(n+2)$ modules equal to A . We regard it as an (A, A) -bimodule by endowing it with the left (resp. right) A -module structure induced by the left (resp. right) A -module structure of the first (resp. last) factor of the tensor product.

For $n \geq 1$, we define bimodule homomorphisms d_n^i (for $0 \leq i \leq n$) and d_n from B_n to B_{n-1} , by the formulas:

$$d_n^i(x_0 \otimes \cdots \otimes x_{n+1}) = x_0 \otimes \cdots \otimes x_i x_{i+1} \otimes \cdots \otimes x_{n+1}, \quad 0 \leq i \leq n,$$

$$d_n = \sum_{i=0}^n (-1)^i d_n^i.$$

It is clear that

$$d_{n-1}^i \circ d_n^j = d_{n-1}^{j-1} \circ d_n^i \quad \text{for } i < j$$

and consequently

$$d_{n-1} \circ d_n = \sum_{0 \leq i < j \leq n} (-1)^{i+j} d_{n-1}^i \circ d_n^j + \sum_{0 \leq j \leq i \leq n-1} (-1)^{i+j} d_{n-1}^i \circ d_n^j = 0.$$

Therefore, if we set $\mathbf{B}_n = 0$ for $n < 0$ and $d_n = 0$ for $n \leq 0$, the sequence $(\mathbf{B}_n, d_n)$ defines a complex of (A, A) -bimodules (X, p. 43), which will be denoted $\mathbf{B}(A)$. For every left A -module

$\mathbf{M}$ we denote by $\mathbf{B}(\mathbf{A}, \mathbf{M})$ the complex formed by the $\mathbf{B}_n \otimes_{\mathbf{A}} \mathbf{M}$ and the $d_n \otimes 1_{\mathbf{M}}$, in other words the tensor product complex $\mathbf{B}(\mathbf{A}) \otimes_{\mathbf{A}} \mathbf{M}_*$; it is a complex of left $\mathbf{A}$ -modules.

We define an $\mathbf{A}$ -linear map $\varepsilon_{\mathbf{M}}$ from $\mathbf{B}_0(\mathbf{A}, \mathbf{M}) = (\mathbf{A} \otimes_k \mathbf{A}) \otimes_{\mathbf{A}} \mathbf{M}$ to $\mathbf{M}$ by the formula $\varepsilon_{\mathbf{M}}(a \otimes b \otimes m) = abm$ for $a, b \in \mathbf{A}, m \in \mathbf{M}$. We have $\varepsilon_{\mathbf{M}} \circ d_1 = 0$, so that the graded homomorphism $\bar{\varepsilon}_{\mathbf{M}} : \mathbf{B}(\mathbf{A}, \mathbf{M}) \rightarrow \mathbf{M}$, which coincides with $\varepsilon_{\mathbf{M}}$ in degree 0, is a morphism of complexes of $\mathbf{A}$ -modules.

PROPOSITION 12. — The map $\bar{\varepsilon}_{\mathbf{M}} : \mathbf{B}(\mathbf{A}, \mathbf{M}) \rightarrow \mathbf{M}$ is a homotopism of complexes of k -modules. In particular, the complex $\mathbf{B}(\mathbf{A}, \mathbf{M})$ is split over k , and $(\mathbf{B}(\mathbf{A}, \mathbf{M}), \bar{\varepsilon}_{\mathbf{M}})$ is a left resolution of the $\mathbf{A}$ -module $\mathbf{M}$.

For $n \geq 0$, let us define a k -linear map $s_n : \mathbf{B}_n \rightarrow \mathbf{B}_{n+1}$ by the formula:

$$s_n(x_0 \otimes \cdots \otimes x_{n+1}) = 1 \otimes x_0 \otimes \cdots \otimes x_{n+1} \quad \text{for } x_0, \dots, x_{n+1} \in A.$$

It is a homomorphism of right $\mathbf{A}$ -modules, which satisfies the identities:

$$d_{n+1}^i \circ s_n = s_{n-1} \circ d_n^{i-1} \quad \text{for } n \geq 1, \quad 1 \leq i \leq n+1,$$

$$d_{n+1}^0 \circ s_n = 1_{\mathbf{B}_n} \quad \text{for } n \geq 1,$$

and consequently we have

$$d_{n+1} \circ s_n + s_{n-1} \circ d_n = 1_{\mathbf{B}_n} \quad \text{for } n \geq 1. \quad (16)$$

Moreover, we have

$$d_1 \circ s_0(x_0 \otimes x_1) = x_0 \otimes x_1 - 1 \otimes x_0 x_1 \quad \text{for } x_0, x_1 \in A. \quad (17)$$

Let us denote by $\eta : \mathbf{A} \rightarrow \mathbf{A} \otimes_k \mathbf{A}$ the map defined by $\eta(a) = 1 \otimes a$, and $\bar{\eta} : \mathbf{A} \rightarrow \mathbf{B}(\mathbf{A})$ the morphism of complexes which coincides with η in degree 0. It is clear that $\bar{\varepsilon}_{\mathbf{A}} \circ \bar{\eta} = 1_{\mathbf{A}}$; formulas (16) and (17) show that $d \circ s + s \circ d = 1_{\mathbf{B}(\mathbf{A})} - \bar{\eta} \circ \bar{\varepsilon}_{\mathbf{A}}$. Setting $\bar{\eta}_{\mathbf{M}} = \bar{\eta} \otimes 1_{\mathbf{M}}$, $d_{\mathbf{M}} = d \otimes 1_{\mathbf{M}}$ and $s_{\mathbf{M}} = s \otimes 1_{\mathbf{M}}$, we deduce that $\bar{\varepsilon}_{\mathbf{M}} \circ \bar{\eta}_{\mathbf{M}} = 1_{\mathbf{M}}$ and $d_{\mathbf{M}} \circ s_{\mathbf{M}} + s_{\mathbf{M}} \circ d_{\mathbf{M}} = 1_{\mathbf{B}(\mathbf{A}, \mathbf{M})} - \bar{\eta}_{\mathbf{M}} \circ \bar{\varepsilon}_{\mathbf{M}}$.

In other words, (X, p. 33, def. 5), $\bar{\varepsilon}_M$ is a homotopism of complexes of k -modules. The other assertions of the proposition follow immediately.

DEFINITION 3. — The left resolution $(B(A, M), \bar{\varepsilon}_M)$ of M is called the standard resolution of the A -module M .

If A and M are projective (resp. free, resp. flat) k -modules, the standard resolution $B(A, M)$ is a projective (resp. free, resp. flat) resolution of M .

10.3.9 Resolutions and Grothendieck groups

If $\mathcal{C}$ is a set of classes of A -modules, we shall say that a left resolution $(\mathbf{P}, p)$ is bounded of type $\mathcal{C}$ if the complex $\mathbf{P}$ is bounded of type $\mathcal{C}$ (X, p. 41).

THEOREM 1. — Let $\mathcal{C}_0$ and $\mathcal{C}$ be two additive and left exact classes of A -modules such that $\mathcal{C}_0 \subset \mathcal{C}$ and that every A -module of type $\mathcal{C}$ has a bounded left resolution of type $\mathcal{C}_0$. Then the homomorphism $\alpha : K(\mathcal{C}_0) \rightarrow K(\mathcal{C})$ deduced from the inclusion of $\mathcal{C}_0$ into $\mathcal{C}$ is bijective; if M is an A -module of type $\mathcal{C}$ and P a bounded left resolution of M of type $\mathcal{C}_0$, we have $\alpha^{-1}([M]_{\mathcal{C}}) = \chi_{\mathcal{C}_0}(P)$ (X, p. 41, example 6).

Lemma 4. — Let $f : \mathbf{M}' \rightarrow \mathbf{M}$ be a homomorphism of A -modules of type $\mathcal{C}$, and $p : \mathbf{P} \rightarrow \mathbf{M}$ a left resolution of $\mathbf{P}$ bounded of type $\mathcal{C}_0$. There exists a left resolution $p' : \mathbf{P}' \rightarrow \mathbf{M}'$ bounded of type $\mathcal{C}_0$ and a morphism of complexes $u : \mathbf{P}' \rightarrow \mathbf{P}$ such that $p \circ u = f \circ p'$.

Let us reason by induction on the length n of $\mathbf{P}$, the assertion being trivial when this length is < 0 . Consider the mapping $g : \mathbf{M}' \times \mathbf{P}_0 \rightarrow \mathbf{M}$ such that

$$g(x, y) = f(x) - p_0(y) \quad \text{for } x \in \mathbf{M}', y \in \mathbf{P}_0,$$

and its kernel K ; the A -module K is of type $\mathcal{C}$ since g is surjective and $\mathbf{M}' \times \mathbf{P}_0$ and $\mathbf{M}$ are of type $\mathcal{C}$. Let $h : P'_0 \rightarrow K$ be a surjective homomorphism where P'_0 is of type $\mathcal{C}_0$; denote by $p'_0 : P'_0 \rightarrow \mathbf{M}'$ (resp. $u_0 : P'_0 \rightarrow \mathbf{P}_0$) the composite homomorphism of h and the projection $K \rightarrow \mathbf{M}$ (resp. $K \rightarrow \mathbf{P}_0$); the homomorphism p'_0 is surjective and we have a commutative diagram

$$\begin{array}{ccc}
& \mathbf{P}'_0 \xrightarrow{u_0} \mathbf{P}_0 & \\
p'_0 \downarrow & & \downarrow p_0 \\
& \mathbf{M}' \xrightarrow{f} \mathbf{M}. &
\end{array}$$

It then suffices to apply the induction hypothesis to the homomorphism

$$\text{Ker } p'_0 \rightarrow \text{Ker } p_0$$

deduced from u_0 .

Lemma 5. — Consider a commutative diagram

$$\begin{array}{ccc}
& \mathbf{P}' \xrightarrow{u} \mathbf{P} & \\
p' \downarrow & & \downarrow p \\
0 \rightarrow \mathbf{M}' \xrightarrow{f} \mathbf{M} \rightarrow \mathbf{M}'' \rightarrow 0 & &
\end{array}$$

where $(\mathbf{P}, p)$ (resp. $(\mathbf{P}', p')$) is a left resolution of $\mathbf{M}$ (resp. $\mathbf{M}'$), and where the bottom horizontal row is an exact sequence. There exists a homomorphism $p'' : \text{Con}(u) \rightarrow \mathbf{M}''$.

Indeed, the exact sequence (X, p. 37, prop. 7)

$$0 \rightarrow \mathbf{P} \xrightarrow{\pi} \text{Con}(u) \xrightarrow{\delta} \mathbf{P}'(-1) \rightarrow 0$$

gives an exact homology sequence

$$\begin{aligned}
& \rightarrow H_n(\mathbf{P}) \rightarrow H_n(\text{Con}(u)) \rightarrow H_{n-1}(\mathbf{P}') \rightarrow \dots \\
& \dots \rightarrow H_1(\text{Con}(u)) \rightarrow H_0(\mathbf{P}') \xrightarrow{\partial} H_0(\mathbf{P}) \rightarrow H_0(\text{Con}(u)) \rightarrow 0.
\end{aligned}$$

According to X, p. 38, lemma 3 a), we have $\partial = -H_0(u)$. Since $H_n(\mathbf{P}) = 0 = H_n(\mathbf{P}')$ for $n > 0$ and $H_0(u) : H_0(\mathbf{P}') \rightarrow H_0(\mathbf{P})$ is identified with $f : \mathbf{M}' \rightarrow \mathbf{M}$, we conclude that $H_n(\text{Con}(u)) = 0$ for $n > 0$ and that $H_0(\text{Con}(u))$ is isomorphic to $\mathbf{M}''$, whence the lemma.

Now let us prove the theorem.

- a) Let $\mathbf{M}$ be an A -module of type C. For every left resolution $(\mathbf{P}, p)$ of $\mathbf{M}$, bounded of type C_0 , the element $\chi_{\mathcal{C}_0}(\mathbf{P})$ of $\mathbf{K}(\mathcal{C}_0)$ depends only on $\mathbf{M}$. Indeed, let $(\mathbf{P}_1, p_1)$ and $(\mathbf{P}_2, p_2)$

be two resolutions of this type. Consider the resolution

$$(\mathbf{P}_1 \times \mathbf{P}_2, p_1 \times p_2)$$

of the A-module $\mathbf{M} \times \mathbf{M}$ and the homomorphism $\Delta : x \mapsto (x, x)$ of $\mathbf{M}$ to $\mathbf{M} \times \mathbf{M}$. By Lemma 4, there exists a resolution $(\mathbf{Q}, q)$ of $\mathbf{M}$ bounded of type $\mathcal{C}_0$ and a commutative diagram

$$\begin{array}{ccc} \mathbf{Q} & \xrightarrow{u} & \mathbf{P}_1 \times \mathbf{P}_2 \\ q \downarrow & & \downarrow p_1 \times p_2 \\ \mathbf{M} & \xrightarrow{\Delta} & \mathbf{M} \times \mathbf{M}; \end{array}$$

we deduce a commutative diagram

$$\begin{array}{ccc} \mathbf{Q} & \xrightarrow{u \circ pr_i} & \mathbf{P}_i \\ q \downarrow & & \downarrow p_i \\ \mathbf{M} & \xrightarrow{l_M} & \mathbf{M}, \end{array} \quad i = 1, 2.$$

By lemma 5, $\text{Con}(u \circ pr_i)$ has zero homology, hence $u \circ pr_i$ is a homomorphism and $\chi_{\mathcal{C}_0}(\mathbf{Q}) = \chi_{\mathcal{C}_0}(\mathbf{P}_i)$ (X, p. 41, prop. 10); it follows that $\chi_{\mathcal{C}_0}(\mathbf{P}_1) = \chi_{\mathcal{C}_0}(\mathbf{P}_2)$ as announced.

- b) For every A-module $\mathbf{M}$ of type C, let $\varphi(\mathbf{M}) \in K(\mathcal{C}_0)$ be the common value of $\chi_{\mathcal{C}_0}(P)$ for all bounded left resolutions P of $\mathbf{M}$ of type $\mathcal{C}_0$. Let us show that the function $\varphi : \mathcal{C} \rightarrow K(\mathcal{C}_0)$ is additive. Let therefore

$$0 \rightarrow \mathbf{M}' \xrightarrow{f} \mathbf{M} \rightarrow \mathbf{M}'' \rightarrow 0$$

be an exact sequence of A-modules of type C. By Lemma 4, there exists a commutative diagram

$$\begin{array}{ccc} \mathbf{P}' & \xrightarrow{u} & \mathbf{P} \\ p' \downarrow & & \downarrow p \\ 0 \rightarrow \mathbf{M}' & \xrightarrow{f} & \mathbf{M} \rightarrow \mathbf{M}'' \rightarrow 0 \end{array}$$

where (P, p) and (P', p') are bounded left resolutions of type $\mathcal{C}_0$. Then we have

$$\varphi(\mathbf{M}) = \chi_{\mathcal{C}_0}(\mathbf{P}), \quad \varphi(\mathbf{M}') = \chi_{\mathcal{C}_0}(\mathbf{P}')$$

and by Lemma 5

$$\varphi(\mathbf{M}'') = \chi_{\mathcal{C}_0}(\text{Con}(u)) = \chi_{\mathcal{C}_0}(\mathbf{P}) - \chi_{\mathcal{C}_0}(\mathbf{P}') = \varphi(\mathbf{M}) - \varphi(\mathbf{M}');$$

which is what we wanted to prove.

- c) Let then $\beta : K(\mathcal{C}) \rightarrow K(\mathcal{C}_0)$ be the homomorphism such that, with the preceding notation, we have $\beta([M]_{\mathcal{C}}) = \chi_{\mathcal{C}_0}(\mathbf{P})$. Since p is a homomorphism, we have $\chi_{\mathcal{C}}(\mathbf{P}) = [M]_{\mathcal{C}}$, hence $\alpha \circ \beta([M]_{\mathcal{C}}) = \alpha(\chi_{\mathcal{C}_0}(\mathbf{P})) = \chi_{\mathcal{C}}(\mathbf{P}) = [M]_{\mathcal{C}}$ and $\alpha \circ \beta = 1_{K(\mathcal{C})}$. If M is of type $\mathcal{C}_0$, then $(M, 1_M)$ is a resolution of M , hence $\varphi(M) = [M]_{\mathcal{C}_0}$ and $\beta \circ \alpha = 1_{K(\mathcal{C}_0)}$, which completes the proof.

We shall apply this theorem to modules of "finite projective dimension" in § 8 (X, p. 137).

10.4 TORSION PRODUCT

In §§ 4 to 8, we denote by k a commutative ring, and by A an associative unital k -algebra. The role of k is auxiliary in nature; we have principally in view the following three particular cases:

- a) one considers an arbitrary ring A , sets $k = \mathbb{Z}$ and endows A with its natural structure of $\mathbb{Z}$ -algebra,
- b) one considers an arbitrary ring A , takes for k the center of A ,
- c) one considers a commutative ring A and takes $k = A$.

10.4.1 Tensor product of two complexes

Let (C, d) be a complex of right A -modules and (C', d') a complex of left A -modules.

Endow the k -module $C \otimes_A C'$ with the grading such that

$$(C \otimes_A C')_n = \sum_{p+q=n} (C_p \otimes C'_q)$$

and let D denote the unique k -linear endomorphism of degree (-1) of $C \otimes_A C'$ such that

$$D(x \otimes x') = dx \otimes x' + (-1)^p x \otimes d'x', \quad x \in C_p, y \in C'_q, p, q \in \mathbf{Z}. \quad (1)$$

One has $D \circ D = 0$ since, with the notations of (1),

$$D^2(x \otimes x') = ddx \otimes x' + (-1)^{p-1} dx \otimes d'x' + (-1)^p dx \otimes d'x' - x \otimes d'd'x'.$$

The complex of k -modules $(C \otimes_A C', D)$ is called the tensor product complex of the complexes (C, d) and (C', d') .

Remarks. — 1) When C' is reduced to $C'_0 = \mathbf{M}$, then $(C \otimes_A C')_n = C_n \otimes_A \mathbf{M}$ and $D = d \otimes 1_M$; for example $C \otimes_A A_s$ is naturally identified with C . Similarly, when C is reduced to $C_0 = P$, then $(C \otimes_A C')_n = P \otimes_A C'_n$ and $D = 1_P \otimes d$.

2) For every integer r , we have $(C \otimes_A C')(r) = C(r) \otimes_A C'$, but $(C \otimes_A C')(r)$ and $C \otimes_A C'(r)$ do not in general have the same differential.

Let p, q be two integers, $x \in Z_p(C)$, $x' \in Z_q(C')$; then the element $x \otimes x'$ of $C_p \otimes C'_q$ belongs to $Z_{p+q}(C \otimes C')$ by (1); moreover, if $y \in C_{p+1}$, $y' \in C'_{q+1}$, we have

$$(x + dy) \otimes (x' + d'y') = x \otimes x' + D(y \otimes x' + (-1)^p (x + dy) \otimes y')$$

by passing to quotients, we deduce a k -linear map, called canonical

$$\gamma_{p,q}(C, C') : H_p(C) \otimes_A H_q(C') \rightarrow H_{p+q}(C \otimes_A C');$$

if we equip $H(C) \otimes_A H(C')$ with the grading such that

$$(H(C) \otimes H(C'))_n = \sum_{p+q=n} H_p(C) \otimes_A H_q(C'),$$

the $\gamma_{p,q}$ define a graded k -linear map of degree 0

$$\gamma(C, C') : H(C) \otimes_A H(C') \rightarrow H(C \otimes_A C').$$

Proposition 6 of II, p. 59, is reformulated as follows:

PROPOSITION 1. — If the complexes C and C' are zero on the right, then $C \otimes_A C'$ is zero on the right, and the canonical k -linear map

$$\gamma_{0,0}(C, C') : H_0(C) \otimes_A H_0(C') \rightarrow H_0(C \otimes_A C')$$

is bijective.

Let $u : (C, d) \rightarrow (C_1, d_1)$ be a morphism of complexes of right A -modules and $u' : (C', d') \rightarrow (C'_1, d'_1)$ a morphism of complexes of left A -modules; then $u \otimes u' : C \otimes_A C' \rightarrow C_1 \otimes_A C'_1$ is a morphism of complexes of k -modules; indeed, it is graded of degree 0, and if one denotes by D and D_1 the differentials of $C \otimes C'$ and $C_1 \otimes C'_1$ we have, for $p, q \in \mathbb{Z}, x \in C_p, x' \in C_q$,

$$\begin{aligned} (u \otimes u')(D(x \otimes x')) &= u(dx) \otimes u'(x') + (-1)^p u(x) \otimes u'(d'x') = \\ &= d_1 u(x) \otimes u'(x') + (-1)^p u(x) \otimes d'_1 u'(x') = D_1(u(x) \otimes u'(x')). \end{aligned}$$

Moreover, the following diagram is commutative:

$$\begin{array}{ccc} H(C) \otimes_A H(C') & \xrightarrow{\gamma(C, C')} & H(C \otimes_A C') \\ & \downarrow H(u) \otimes H(u') & \\ H(C_1) \otimes_A H(C'_1) & \xrightarrow{\gamma(C_1, C'_1)} & H(C_1 \otimes_A C'_1). \end{array}$$

Let A° be the opposite k -algebra of A , and let C° (resp. C'°) be the complex C (resp. C') viewed as a complex of left (resp. right) A° -modules.

$$\sigma(C, C') : C \otimes_A C' \rightarrow C'^\circ \otimes_{A^\circ} C^\circ$$

the unique graded k -linear map of degree 0 such that, for $x \in C_p, x' \in C'_q, p, q \in \mathbb{Z}$, one has

$$\sigma(\mathbf{C}, \mathbf{C}')(x \otimes x') = (-1)^{pq} x' \otimes x.$$

PROPOSITION 2. — The map $\sigma(\mathbf{C}, \mathbf{C}') : \mathbf{C} \otimes_{\mathbf{A}} \mathbf{C}' \rightarrow \mathbf{C}'^{\circ} \otimes_{\mathbf{A}^{\circ}} \mathbf{C}^{\circ}$ is an isomorphism of complexes of k -modules, whose inverse isomorphism is $\sigma(\mathbf{C}'^{\circ}, \mathbf{C}^{\circ})$.

Since the maps $\sigma(\mathbf{C}, \mathbf{C}')$ and $\sigma(\mathbf{C}^{\circ}, \mathbf{C}^{\circ})$ are inverses of each other, it suffices to prove that $\sigma(\mathbf{C}, \mathbf{C}')$ is a morphism of complexes. Now, for

$$x \in C_p = C_p^{\circ}, \quad x' \in C'_q = C_q'^{\circ}, \quad p, q \in \mathbf{Z},$$

we have, writing D^c for the differential of $\mathbf{C}' \otimes_{\mathbf{A}^0} \mathbf{C}$,

$$\begin{aligned} \sigma(\mathbf{C}, \mathbf{C}') \circ D(x \otimes x') &= \sigma(\mathbf{C}, \mathbf{C}')(dx \otimes x' + (-1)^p x \otimes d'x') = \\ &= (-1)^{(p+1)q} x' \otimes dx + (-1)^{p+p(q+1)} d'x' \otimes x = (-1)^{pq} d'x' \otimes x + (-1)^{pq+q} x' \otimes dx \\ &= (-1)^{pq} D^{\circ}(x' \otimes x) = D^{\circ} \circ \sigma(\mathbf{C}, \mathbf{C}')(x \otimes x'); \end{aligned}$$

this gives $\sigma(\mathbf{C}, \mathbf{C}') \circ D = D^{\circ} \circ \sigma(\mathbf{C}, \mathbf{C}')$, whence the desired assertion.

The isomorphism $\sigma(\mathbf{C}, \mathbf{C}') : \mathbf{C} \otimes_{\mathbf{A}} \mathbf{C}' \rightarrow \mathbf{C}^{\circ} \otimes_{\mathbf{A}^{\circ}} \mathbf{C}'^{\circ}$ is called the commutation isomorphism of the tensor product of the complexes $\mathbf{C}$ and $\mathbf{C}'$.

If $u : \mathbf{C} \rightarrow \mathbf{C}_1$ and $v : \mathbf{C}' \rightarrow \mathbf{C}'_1$ are two morphisms of complexes as above, we have a commutative diagram:

$$\begin{array}{ccc} \mathbf{C} \otimes_{\mathbf{A}} \mathbf{C}' & \xrightarrow{\sigma(\mathbf{C}, \mathbf{C}')} & \mathbf{C}'^{\circ} \otimes_{\mathbf{A}^{\circ}} \mathbf{C}^{\circ} \\ & \downarrow u \otimes u' & \\ \mathbf{C}_1 \otimes_{\mathbf{A}} \mathbf{C}'_1 & \xrightarrow{\sigma(\mathbf{C}_1, \mathbf{C}'_1)} & \mathbf{C}'_1{}^{\circ} \otimes_{\mathbf{A}^{\circ}} \mathbf{C}_1^{\circ}. \end{array}$$

Suppose for the remainder of this number that the ring $\mathbf{A}$ is commutative (cf. no. 9 for the general case).

Let $\mathbf{C}, \mathbf{C}', \mathbf{C}''$ be three complexes of $\mathbf{A}$ -modules; the canonical homomorphism of $\mathbf{A}$ -modules (III, p. 64)

$$\varphi : (C \otimes_A C') \otimes_A C'' \rightarrow C \otimes_A (C' \otimes_A C'')$$

is an isomorphism of complexes, as one verifies immediately using the definitions.

More generally, let $(\mathbf{C}^{(i)}, d^{(i)})_{i \in I}$ be a family of complexes of A-modules, where the set I is finite and totally ordered; for simplicity of notation, we will identify I with the interval $[1, r]$ of $\mathbb{N}$. Endow the A-module $C = \bigotimes_{i=1}^r C^{(i)}$ with the grading such that

$$C_n = \sum_{p_1+p_2+\dots+p_r=n} (C^{(1)})_{p_1} \otimes (C^{(2)})_{p_2} \otimes \dots \otimes (C^{(r)})_{p_r},$$

and let us define a graded A-endomorphism of degree (-1) of C by

$$D(x_1 \otimes \dots \otimes x_r) = \sum_{j=1}^r (-1)^{p_1+\dots+p_{j-1}} x_1 \otimes \dots \otimes x_{j-1} \otimes d_j x_j \otimes x_{j+1} \otimes \dots \otimes x_r$$

where $x_i \in (C^{(i)})_{p_i}$ for $i = 1, \dots, n$. Then (C, D) is a complex of A-modules called the tensor product complex of the family $(\mathbf{C}_i, d_i)$. For every strictly increasing sequence $r_0, \dots, r_k$ of $[0, r]$ such that $r_0 = 0, r_k = r$, the canonical associativity isomorphism

$$\bigotimes_{j=0}^{k-1} \left(\bigotimes_{i=r_j+1}^{r_{j+1}} \mathbf{C}^{(i)} \right) \rightarrow \bigotimes_{i=1}^r \mathbf{C}^{(i)}$$

is an isomorphism of complexes.

We define as above a graded homomorphism of degree 0

$$\gamma((\mathbf{C}^{(i)})) : \bigotimes_{i \in I} H(\mathbf{C}^{(i)}) \rightarrow H\left(\bigotimes_{i \in I} \mathbf{C}^{(i)}\right).$$

Remarks. — 3) One can define the tensor product of a finite family of complexes without endowing the index set with a total order (X, p. 185, exercise 3).

- 4) Suppose that each $\mathbf{C}^{(i)}$ is endowed with a graded algebra structure compatible with its grading and such that the $d^{(i)}$ are antiderivations (III, p. 117). Let us then equip $\bigotimes_{i \in I} \mathbf{C}^{(i)}$ with the structure of left graded tensor product algebra of the given structures (III, p. 49).

Then D is an antiderivation. Indeed, using the associativity of the tensor product, one may assume that $I = \{1, 2\}$; let then $p_1, q_1, p_2, q_2 \in \mathbf{Z}$, $x_1 \in (C^{(1)})_{p_1}$, $y_1 \in (C^{(1)})_{q_1}$, $x_2 \in (C^{(2)})_{p_2}$, $y_2 \in (C^{(2)})_{q_2}$; we have

$$\begin{aligned}
& (D(x_1 \otimes x_2))(y_1 \otimes y_2) + (-1)^{p_1+p_2}(x_1 \otimes x_2)(D(y_1 \otimes y_2)) = \\
& = (dx_1 \otimes x_2 + (-1)^{p_1}x_1 \otimes dx_2)(y_1 \otimes y_2) + \\
& \quad + (-1)^{p_1+p_2}(x_1 \otimes x_2)(dy_1 \otimes y_2 + (-1)^{q_1}y_1 \otimes dy_2) = \\
& = (-1)^{p_2q_1}(dx_1)y_1 \otimes x_2y_2 + (-1)^{p_1+(p_2-1)q_1}x_1y_1 \otimes (dx_2)y_2 + \\
& \quad + (-1)^{p_1+p_2+p_2(q_1-1)}x_1dy_1 \otimes x_2y_2 + (-1)^{p_1+p_2+q_1+p_2q_1}x_1y_1 \otimes x_2dy_2 \\
& = (-1)^{p_2q_1}[(dx_1)y_1 + (-1)^{p_1}x_1dy_1] \otimes x_2y_2 + \\
& \quad + (-1)^{p_1+q_1+p_2q_1}x_1y_1 \otimes ((dx_2)y_2 + (-1)^{p_2}x_2dy_2) \\
& = (-1)^{p_2q_1}[d(x_1y_1) \otimes x_2y_2 + (-1)^{p_1+q_1}x_1y_1 \otimes d(x_2y_2)] \\
& = (-1)^{p_2q_1}D(x_1y_1 \otimes x_2y_2) = D((x_1 \otimes x_2)(y_1 \otimes y_2)).
\end{aligned}$$

10.4.2 Tensor products and homotopy

PROPOSITION 3. — Let C, C_1 be two complexes of right A -modules, C', C'_1 two complexes of left A -modules, and $u : C \rightarrow C_1, v : C \rightarrow C_1, u' : C' \rightarrow C'_1, v' : C' \rightarrow C'_1$ be morphisms of complexes.

- a) If u and u' are homotopic to v and v' respectively, then the two morphisms $u \otimes u'$ and $v \otimes v'$ from $C \otimes_A C'$ to $C_1 \otimes_A C'_1$ are homotopic.
- b) If u and u' are homotopisms, $u \otimes u'$ is a homotopism.
- c) If C or C' is homotopic to zero, $C \otimes_A C'$ is homotopic to zero.

Let us denote by the same letter d the differentials of the complexes C, C_1, C', C'_1 and by D the differentials of the complexes $C \otimes_A C'$ and $C_1 \otimes_A C'_1$.

If u (resp. u') is homotopic to v (resp. v'), there exists a graded homomorphism of degree 1 $s : C \rightarrow C_1$ (resp. $s' : C' \rightarrow C'_1$) such that

$$u - v = ds + sd \quad (\text{resp. } u' - v' = ds' + s'd). \quad (2)$$

Let $S : C \otimes_A C' \rightarrow C_1 \otimes_A C'_1$ be the unique graded homomorphism of degree 1 such that, for $x \in C_p, y \in C'_q, p, q \in \mathbb{Z}$, one has

$$S(x \otimes y) = s(x) \otimes u'(y) + (-1)^p v(x) \otimes s'(y). \quad (3)$$

We then have, with the preceding notation:

$$\begin{aligned} (DS + SD)(x \otimes y) &= D(sx \otimes u'y) + (-1)^p D(vx \otimes s'y) + S(dx \otimes y) + \\ &\quad + (-1)^p S(x \otimes dy) = \\ &= dsx \otimes u'y + (-1)^{p+1} sx \otimes du'y + (-1)^p dvx \otimes s'y + vx \otimes ds'y + \\ &\quad + sdx \otimes u'y + (-1)^{p-1} vdx \otimes s'y + (-1)^p sx \otimes u'dy + vx \otimes s'dy \\ &= (ds + sd)(x) \otimes u'y + vx \otimes (ds' + s'd)(y) \\ &= (ux - vx) \otimes u'y + vx \otimes (u'y - v'y) = ux \otimes u'y - vx \otimes v'y. \end{aligned}$$

This gives $DS + SD = u \otimes u' - v \otimes v'$, whence a).

Let us prove b). If u and u' are homotopisms, there exist homomorphisms of complexes $\alpha : C_1 \rightarrow C$ and $\alpha' : C'_1 \rightarrow C'$ such that $u \circ \alpha, \alpha \circ u, u' \circ \alpha', \alpha' \circ u'$ are homotopic respectively to $Id_{C_1}, Id_C, Id_{C'_1}$ and $Id_{C'}$. Then $(u \otimes u') \circ (\alpha \otimes \alpha')$, which is equal to $(u \circ \alpha) \otimes (u' \circ \alpha')$, is homotopic by a) to $Id_{C_1} \otimes Id_{C'_1} = Id_{C_1 \otimes C'_1}$, whereas $(\alpha \otimes \alpha') \circ (u \otimes u')$ is homotopic to $Id_{C \otimes C'}$, whence b). Finally, c) follows from b) applied to the case where C_1 or C'_1 is zero.

COROLLARY 1. — Let C' be a split complex of left A -modules such that $H(C')$ is flat. For every complex C of right A -modules, the canonical map

$$\gamma(C, C') : H(C) \otimes_A H(C') \rightarrow H(C \otimes_A C')$$

is bijective.

According to X, p. 35, def. 6, there exists a homotopism $u' : C' \rightarrow H(C')$. By prop. 3, $1_C \otimes u' : C \otimes_A C' \rightarrow C \otimes_A H(C')$ is a homotopism; since

$$H(1_C \otimes u') \circ \gamma(C, C') = \gamma(C, H(C')) \circ (1_C \otimes H(u')),$$

and $\mathbf{H}(1_{\mathbf{C}} \otimes u')$ and $\mathbf{H}(u')$ are bijective, it suffices to prove that $\gamma(\mathbf{C}, \mathbf{H}(\mathbf{C}'))$ is bijective, and we are reduced to the case where $\mathbf{C}'$ is flat and has zero differential. In this case the canonical exact sequences

(I)

$$0 \rightarrow \mathbf{Z}(\mathbf{C}) \xrightarrow{i} \mathbf{C} \xrightarrow{\delta} \mathbf{B}(\mathbf{C}) \rightarrow 0 \quad (\text{II})$$

$$0 \rightarrow \mathbf{B}(\mathbf{C}) \xrightarrow{j} \mathbf{Z}(\mathbf{C}) \xrightarrow{\pi} \mathbf{H}(\mathbf{C}) \rightarrow 0$$

give exact sequences:

$$\begin{aligned} 0 \rightarrow \mathbf{Z}(\mathbf{C}) \otimes_A \mathbf{C}' &\xrightarrow{i \otimes 1} \mathbf{C} \otimes_A \mathbf{C}' \xrightarrow{\delta \otimes 1} \mathbf{B}(\mathbf{C}) \otimes_A \mathbf{C}' \rightarrow 0 \\ 0 \rightarrow \mathbf{B}(\mathbf{C}) \otimes_A \mathbf{C}' &\xrightarrow{j \otimes 1} \mathbf{Z}(\mathbf{C}) \otimes_A \mathbf{C}' \xrightarrow{\pi \otimes 1} \mathbf{H}(\mathbf{C}) \otimes_A \mathbf{C}' \rightarrow 0. \end{aligned}$$

Since $d = i \circ j \circ \delta$, we have $\mathbf{D} = d \otimes 1_{\mathbf{C}'} = (i \otimes 1) \circ (j \otimes 1) \circ (\delta \otimes 1)$, which shows that the canonical maps $\mathbf{Z}(\mathbf{C}) \otimes_A \mathbf{C}' \rightarrow \mathbf{Z}(\mathbf{C} \otimes_A \mathbf{C}')$ and $\mathbf{B}(\mathbf{C}) \otimes_A \mathbf{C}' \rightarrow \mathbf{B}(\mathbf{C} \otimes_A \mathbf{C}')$ are bijective, hence also $\gamma(\mathbf{C}, \mathbf{C}')$ by passing to quotients.

COROLLARY 2. — Let $\mathbf{N}$ be a flat left A -module. For every complex $\mathbf{C}$ of right A -modules, the canonical homomorphisms

$$\gamma_n(\mathbf{C}, \mathbf{N}) : \mathbf{H}_n(\mathbf{C}) \otimes_A \mathbf{N} \rightarrow \mathbf{H}_n(\mathbf{C} \otimes_A \mathbf{N})$$

are bijective.

COROLLARY 3. — Let $\mathbf{C}'$ be a complex of left A -modules such that $\mathbf{B}(\mathbf{C}')$ and $\mathbf{H}(\mathbf{C}')$ are projective. For every complex $\mathbf{C}$ of right A -modules, the map $\gamma(\mathbf{C}, \mathbf{C}')$ is bijective.

Indeed, $\mathbf{C}'$ is split (X, p. 35, example 4) and $\mathbf{H}(\mathbf{C}')$ is projective; one can therefore apply Corollary 1.

Remarks. — 1) Using the commutation isomorphisms, one deduces from Corollaries 1, 2, and 3 the analogous statements obtained by interchanging the roles of the two arguments of the tensor products.

2) We shall see below (X, p. 79, cor. 4) that the conclusion of cor. 1 is also true when one assumes that C' and $H(C')$ are flat and C' is bounded on the right.

10.4.3 Tensor product by a flat complex bounded on the right

Lemma 1. — Let C be a complex of right A -modules and E a complex of left A -modules. Suppose that $H(C) = 0$ and that E is flat and bounded on the right. Then $H(C \otimes_A E) = 0$.

For $k \in \mathbf{Z}$, let $T^{(k)}$ be the subcomplex of $C \otimes_A E$ such that

$$T_n^{(k)} = \sum_{\substack{p+q=n \\ q \leq k}} C_p \otimes_A E_q;$$

then $T^{(k-1)} \subset T^{(k)}$ and we have an exact sequence of complexes

$$0 \longrightarrow T^{(k-1)} \xrightarrow{i_k} T^{(k)} \xrightarrow{\pi} C \otimes_A E_k(-k) \longrightarrow 0$$

where i_k is the canonical injection and where π projects the preceding direct sum onto its factor $C_{n-k} \otimes_A E_k = (C \otimes_A E_k(-k))_n$. By cor. 2 above, we have $H(C \otimes_A E_k(-k)) = 0$, hence i_k is a homomorphism. We have $T^{(k)} = 0$ for k small enough, since E is bounded on the right, hence $H(T^{(k)}) = 0$ for all k by induction on k . Finally, the canonical morphism $\lim_A T^{(k)} \rightarrow C \otimes_A E$ is obviously an isomorphism, hence $H(C \otimes_A E) = 0$ (X, p. 28, prop. 1).

Lemma 2. — If $u : C \rightarrow C'$ is a morphism of complexes of right A -modules and E is a complex of left A -modules, then the complexes $\text{Con}(u) \otimes_A E$ and $\text{Con}(u \otimes 1_E)$ are isomorphic.

By definition, $\text{Con}(u) \otimes_A E$ is the graded module $(C'(-1) \oplus C) \otimes_A E$ endowed with the differential D such that, for $x \in C_p$, $y' \in C'(-1)_p = C'_{p-1}$, $z \in E_q$, one has

$$D((y', x) \otimes z) = (-dy', dx - u(y')) \otimes z + (-1)^p (y', x) \otimes dz, \quad (4)$$

while $\text{Con}(u \otimes 1_E)$ is the graded module $(C' \otimes_A E)(-1) \oplus (C \otimes_A E)$ equipped with the differential D_1 such that, for $x \in C_p$, $y' \in C'_{p-1}$, $z \in E_q$, one has

$$D_1(y' \otimes z, x \otimes z) = (-dy' \otimes z - (-1)^{p-1}y' \otimes dz, dx \otimes z + (-1)^p x \otimes dz - u(y') \otimes z)$$

$$= (-dy' \otimes z, (dx - u(y')) \otimes z) + (-1)^p (y' \otimes dz, x \otimes dz)$$

whence the assertion.

PROPOSITION 4. — Let $u : C \rightarrow C'$ be a homomorphism of complexes of right A -modules, and E a complex of left A -modules, flat and bounded on the right. Then

$$u \otimes 1_E : C \otimes_A E \rightarrow C' \otimes_A E$$

is a homomorphism of complexes of k -modules.

Indeed, according to X, p. 38, cor., u (resp. $u \otimes 1_E$) is a homomorphism if and only if $H(\text{Con}(u)) = 0$ (resp. $H(\text{Con}(u \otimes 1_E)) = 0$). We then conclude by Lemmas 1 and 2.

Remark. — Using the commutation isomorphisms, one deduces from the preceding statements the analogous statements obtained by interchanging the roles of the two arguments of the tensor products.

10.4.4 Definition and first properties of the torsion product

For every A -module E , we denote by $p_E : L(E) \rightarrow E$ the canonical free resolution of E (X, p. 50).

DEFINITION 1. — Let M be a right A -module and N a left A -module. The torsion product of M and N is the graded k -module

$$\text{Tor}^A(M, N) = H(L(M) \otimes_A L(N)). \quad (4)$$

The homogeneous components of $\text{Tor}^A(M, N)$ are denoted

$$\text{Tor}_n^A(M, N) = H_n(L(M) \otimes_A L(N)). \quad (5)$$

Since $L(M)$ and $L(N)$ are zero on the right, we have

$$\mathrm{Tor}_n^A(\mathbf{M}, \mathbf{N}) = 0 \quad \text{for } n < 0. \quad (6)$$

Remark 1. — We shall see below (X, p. 107, prop. 6) finiteness properties of the $\mathrm{Tor}^A(\mathbf{M}, \mathbf{N})$ modules. For example, if A is commutative Noetherian and M and N are finitely generated A -modules, then each A -module $\mathrm{Tor}_n^A(\mathbf{M}, \mathbf{N})$ is finitely generated.

Let $f : \mathbf{M} \rightarrow \mathbf{M}'$ be a homomorphism of right A -modules and $g : \mathbf{N} \rightarrow \mathbf{N}'$ be a homomorphism of left A -modules; we set $\mathrm{Tor}^A(f, g) = H(L(f) \otimes_A L(g))$; it is a homomorphism of graded k -modules

$$\mathrm{Tor}^A(f, g) : \mathrm{Tor}^A(\mathbf{M}, \mathbf{N}) \rightarrow \mathrm{Tor}^A(\mathbf{M}', \mathbf{N}')$$

whose homogeneous components are denoted

$$\mathrm{Tor}_n^A(f, g) : \mathrm{Tor}_n^A(\mathbf{M}, \mathbf{N}) \rightarrow \mathrm{Tor}_n^A(\mathbf{M}', \mathbf{N}').$$

By prop. 1 of X, p. 62, the canonical homomorphism

$$\gamma_{0,0} : H_0(L(M)) \otimes_A H_0(L(N)) \rightarrow H_0(L(M) \otimes_A L(N))$$

is bijective; using the isomorphisms $\mathbf{M} \rightarrow \mathbf{H}_0(\mathbf{L}(\mathbf{M}))$ and $\mathbf{N} \rightarrow \mathbf{H}_0(\mathbf{L}(\mathbf{N}))$ we derive an isomorphism, called canonical

$$\gamma_{\mathbf{M},\mathbf{N}} : \mathbf{M} \otimes_A \mathbf{N} \rightarrow \mathrm{Tor}_0^A(\mathbf{M}, \mathbf{N}). \quad (7)$$

We shall always identify $\mathrm{Tor}_0^A(\mathbf{M}, \mathbf{N})$ with $\mathbf{M} \otimes_A \mathbf{N}$ by this isomorphism. Then the k -linear map $\mathrm{Tor}_0^A(f, g)$ is identified with $f \otimes g$.

Remark 2. — The morphism of complexes $p_M \otimes p_N : \mathbf{L}(\mathbf{M}) \otimes_A \mathbf{L}(\mathbf{N}) \rightarrow \mathbf{M} \otimes_A \mathbf{N}$ induces on the homology of degree 0 the isomorphism

$$\gamma_{\mathbf{M},\mathbf{N}}^{-1} : \mathrm{Tor}_0^A(\mathbf{M}, \mathbf{N}) \rightarrow \mathbf{M} \otimes_A \mathbf{N}$$

inverse of $\gamma_{M,N}$.

One has $L(1_M) = 1_{L(M)}$, $L(1_N) = i_{L(N)}$ hence by passing to homology:

$$\mathrm{Tor}^A(1_M, 1_N) = 1_{\mathrm{Tor}^A(M,N)}. \quad (8)$$

If $f' : M' \rightarrow M''$ (resp. $g' : N' \rightarrow N''$) is a homomorphism of right (resp. left) A -modules, we have $L(g' \circ g) = L(g') \circ L(g)$ and $L(f' \circ f) = L(f') \circ L(f)$, hence

$$\mathrm{Tor}^A(f' \circ f, g' \circ g) = \mathrm{Tor}^A(f', g') \circ \mathrm{Tor}^A(f, g). \quad (9)$$

Consider the morphisms of k -complexes

$$L(M) \otimes_A N \xleftarrow{1 \otimes p_N} L(M) \otimes_A L(N) \xrightarrow{p_M \otimes 1} M \otimes_A L(N)$$

and the k -homomorphisms they induce in homology:

$$H(L(M) \otimes_A N) \xleftarrow{\psi_M(N)} \mathrm{Tor}^A(M, N) \xrightarrow{\bar{\psi}_N(M)} H(M \otimes_A L(N));$$

by prop. 4 of X, p. 67, $1 \otimes p_N$ and $p_M \otimes 1$ are homomorphisms. Hence:

Proposition 5. — The k -homomorphisms

$$\begin{aligned} \psi_M(N) : \mathrm{Tor}^A(M, N) &\rightarrow H(L(M) \otimes_A N) \\ \bar{\psi}_N(M) : \mathrm{Tor}^A(M, N) &\rightarrow H(M \otimes_A L(N)) \end{aligned}$$

are bijective.

COROLLARY. — If M or N is flat, $\mathrm{Tor}_i^A(M, N) = 0$ for $i \geq 0$.

Suppose N (resp. M) is flat; then $p_M \otimes 1 : L(M) \otimes_A N \rightarrow M \otimes_A N$ (resp. $1 \otimes p_N : M \otimes_A L(N) \rightarrow M \otimes_A N$) is a homomorphism (X, p. 67, prop. 4), hence $H_i(L(M) \otimes_A N)$ (resp. $H_i(M \otimes_A L(N))$) is zero for $i > 0$.

Remark 3. — If $g : N \rightarrow N'$ is a homomorphism of left A -modules, then

$$(1_{L(M)} \otimes g) \circ (1_{L(M)} \otimes 1_N) = (1_{L(M)} \otimes 1_N) \circ (1_{L(M)} \otimes L(g)),$$

therefore the diagram

$$\begin{array}{ccc} \mathrm{Tor}^{\mathbf{A}}(\mathbf{M}, \mathbf{N}) & \xrightarrow{\psi_{\mathbf{M}}(\mathbf{N})} & \mathrm{H}(\mathrm{L}(\mathbf{M}) \otimes_{\mathbf{A}} \mathbf{N}) \\ \mathrm{Tor}^{\mathbf{A}}(1, g) \downarrow & & \downarrow \mathrm{H}(1 \otimes g) \\ \mathrm{Tor}^{\mathbf{A}}(\mathbf{M}, \mathbf{N}') & \xrightarrow{\psi_{\mathbf{M}}(\mathbf{N}')} & \mathrm{H}(\mathrm{L}(\mathbf{M}) \otimes_{\mathbf{A}} \mathbf{N}') \end{array}$$

is commutative.

Likewise, if $f : \mathbf{M} \rightarrow \mathbf{M}'$ is a homomorphism of right $\mathbf{A}$ -modules, we have a commutative diagram:

$$\begin{array}{ccc} \mathrm{Tor}^{\mathbf{A}}(\mathbf{M}, \mathbf{N}) & \xrightarrow{\bar{\psi}_{\mathbf{N}}(\mathbf{M})} & \mathrm{H}(\mathbf{M} \otimes_{\mathbf{A}} \mathrm{L}(\mathbf{N})) \\ \mathrm{Tor}^{\mathbf{A}}(f, 1) \downarrow & & \downarrow \mathrm{H}(f \otimes 1) \\ \mathrm{Tor}^{\mathbf{A}}(\mathbf{M}', \mathbf{N}) & \xrightarrow{\bar{\psi}_{\mathbf{N}}(\mathbf{M}')} & \mathrm{H}(\mathbf{M}' \otimes_{\mathbf{A}} \mathrm{L}(\mathbf{N})). \end{array}$$

PROPOSITION 6. — The map $(f, g) \mapsto \mathrm{Tor}^{\mathbf{A}}(f, g) :$

$$\mathrm{Hom}_{\mathbf{A}}(\mathbf{M}, \mathbf{M}') \times \mathrm{Hom}_{\mathbf{A}}(\mathbf{N}, \mathbf{N}') \rightarrow \mathrm{Hom}_k(\mathrm{Tor}^{\mathbf{A}}(\mathbf{M}, \mathbf{N}), \mathrm{Tor}^{\mathbf{A}}(\mathbf{M}', \mathbf{N}'))$$

is k -bilinear.

Let $f \in \mathrm{Hom}_{\mathbf{A}}(\mathbf{M}, \mathbf{M}')$, $g_1, g_2 \in \mathrm{Hom}_{\mathbf{A}}(\mathbf{N}, \mathbf{N}')$, $\lambda_1, \lambda_2 \in k$. Then the morphisms

$$\lambda_1(\mathrm{L}(f) \otimes g_1) + \lambda_2(\mathrm{L}(f) \otimes g_2) \quad \text{and} \quad \mathrm{L}(f) \otimes (\lambda_1 g_1 + \lambda_2 g_2)$$

of $\mathrm{L}(\mathbf{M}) \otimes_{\mathbf{A}} \mathbf{N}$ in $\mathrm{L}(\mathbf{M}) \otimes_{\mathbf{A}} \mathbf{N}'$ coincide; by prop. 5 and remark 3, we therefore have

$$\mathrm{Tor}^{\mathbf{A}}(f, \lambda_1 g_1 + \lambda_2 g_2) = \lambda_1 \mathrm{Tor}^{\mathbf{A}}(f, g_1) + \lambda_2 \mathrm{Tor}^{\mathbf{A}}(f, g_2). \quad (10)$$

We reason similarly for the mapping $f \mapsto \mathrm{Tor}^{\mathbf{A}}(f, g)$.

COROLLARY. — Let $\lambda \in k$. If λ annihilates $\mathbf{M}$ or $\mathbf{N}$, it annihilates $\mathrm{Tor}^{\mathbf{A}}(\mathbf{M}, \mathbf{N})$.

Indeed, $\lambda.1_{\mathrm{Tor}(\mathbf{M}, \mathbf{N})} = \mathrm{Tor}(\lambda.1_{\mathbf{M}}, 1_{\mathbf{N}}) = \mathrm{Tor}(1_{\mathbf{M}}, \lambda.1_{\mathbf{N}})$.

PROPOSITION 7. — Let I and J be two sets, $(\mathbf{M}_{\alpha})_{\alpha \in I}$ a family of right $\mathbf{A}$ -modules, $(\mathbf{N}_{\beta})_{\beta \in J}$ a family of left $\mathbf{A}$ -modules. The homomorphism

$$\bigoplus_{\alpha \in I, \beta \in J} \text{Tor}^A(M_\alpha, N_\beta) \rightarrow \text{Tor}^A\left(\bigoplus_{\alpha \in I} M_\alpha, \bigoplus_{\beta \in J} N_\beta\right)$$

deduced from the canonical homomorphisms $M_\alpha \rightarrow \bigoplus M_\alpha$ and $N_\beta \rightarrow \bigoplus N_\beta$ is bijective. It suffices to prove that for every right module M (resp. every left module N), the canonical homomorphism

$$\bigoplus_{\beta \in J} \text{Tor}^A(M, N_\beta) \rightarrow \text{Tor}^A(M, \bigoplus_{\beta \in J} N_\beta)$$

(resp. $\bigoplus_{\alpha \in I} \text{Tor}^A(\mathbf{M}_\alpha, \mathbf{N}) \rightarrow \text{Tor}^A(\bigoplus_{\alpha \in I} \mathbf{M}_\alpha, \mathbf{N})$) is bijective. This follows from what precedes, from Proposition 1 of X, p. 28, and from the canonical isomorphisms:

$$\begin{aligned} \bigoplus_{\beta} (L(M) \otimes_A N_\beta) &\rightarrow L(M) \otimes_A (\bigoplus_{\beta} N_\beta), \\ \bigoplus_{\alpha} (M_\alpha \otimes_A L(N)) &\rightarrow (\bigoplus_{\alpha} M_\alpha) \otimes_A L(N). \end{aligned}$$

An analogous argument gives:

PROPOSITION 8. — Let I (resp. J) be a preordered set filtered to the right, $((M_\alpha), (u_{\alpha'\alpha}))$ (resp. $((N_\beta), (v_{\beta'\beta}))$) an inductive system of right (resp. left) A -modules relative to I (resp. J). The homomorphism of graded k -modules

$$\lim_{(\alpha, \beta) \in I \times J} \text{Tor}^A(M_\alpha, N_\beta) \rightarrow \text{Tor}^A\left(\lim_{\alpha \in I} M_\alpha, \lim_{\beta \in J} N_\beta\right),$$

deduced from canonical A -homomorphisms $\mathbf{M}_\alpha \rightarrow \varinjlim \mathbf{M}_\alpha$ and $\mathbf{N}_\beta \rightarrow \varinjlim \mathbf{N}_\beta$ is bijective.

In particular, taking $J = I$ and noting that the $(\alpha, \alpha), \alpha \in I$ form a cofinal subset of $I \times I$, we obtain:

COROLLARY. — Let I be a preordered set filtered to the right, $(\mathbf{M}_i, u_{ji})$ (resp. $(\mathbf{N}_i, v_{ji})$) an inductive system of right (resp. left) A -modules relative to I . The homomorphism of graded k -modules

$$\varinjlim \text{Tor}^A(M_i, N_i) \rightarrow \text{Tor}^A(\varinjlim M_i, \varinjlim N_i),$$

deduced from canonical A -homomorphisms $\mathbf{M}_i \rightarrow \varinjlim \mathbf{M}_i$ and $\mathbf{N}_j \rightarrow \varinjlim \mathbf{N}_j$ is bijective.

Let M be a right A -module, N a left A -module, A° the opposite ring of A , M° the left A° -module underlying M , N° the right A° -module underlying N . We have $L(M^\circ) = L(M)^\circ$ and $L(N^\circ) = L(N)^\circ$, whence a commutation isomorphism (X, p. 63, prop. 2)

$$\sigma(L(M), L(N)) : L(M) \otimes_A L(N) \rightarrow L(N^\circ) \otimes_{A^\circ} L(M^\circ).$$

Passing to homology, $\sigma(L(\mathbf{M}), L(\mathbf{N}))$ induces a graded isomorphism of degree 0 $\sigma_{\mathbf{M}, \mathbf{N}} : \text{Tor}^A(\mathbf{M}, \mathbf{N}) \rightarrow \text{Tor}^{A^\circ}(\mathbf{N}^\circ, \mathbf{M}^\circ)$ called the commutation isomorphism of torsion products.

Note that $\sigma_{\mathbf{N}^\circ, \mathbf{M}^\circ} \circ \sigma_{\mathbf{M}, \mathbf{N}} = \text{Id}_{\text{Tor}(\mathbf{M}, \mathbf{N})}$ and that $\sigma_{\mathbf{M}, \mathbf{N}}$ induces on the degree 0 terms the commutation homomorphism of the tensor product. On the other hand, if $f : \mathbf{M} \rightarrow \mathbf{M}'$ and $g : \mathbf{N} \rightarrow \mathbf{N}'$ are homomorphisms of A -modules, we have

$$\text{Tor}^{A^\circ}(g, f) \circ \sigma_{\mathbf{M}, \mathbf{N}} = \sigma_{\mathbf{M}', \mathbf{N}'} \circ \text{Tor}^A(f, g).$$

10.4.5 Connecting homomorphisms and exact sequences

Let M be a right A -module. Recall that for every left A -module N , we defined in the preceding number (X, p. 69, prop. 5) an isomorphism

Let

$$\psi_{\mathbf{M}}(\mathbf{N}) : \text{Tor}^A(\mathbf{M}, \mathbf{N}) \rightarrow H(L(\mathbf{M}) \otimes_A \mathbf{N}). \quad (\epsilon)$$

$$0 \rightarrow \mathbf{N}' \xrightarrow{u} \mathbf{N} \xrightarrow{v} \mathbf{N}'' \rightarrow 0$$

an exact sequence of left A -modules; the sequence of k -complexes

$$({}^M\mathcal{E}) \quad 0 \rightarrow L(M) \otimes_A \mathbf{N}' \xrightarrow{1 \otimes u} L(M) \otimes_A \mathbf{N} \xrightarrow{1 \otimes v} L(M) \otimes_A \mathbf{N}'' \rightarrow 0$$

is then exact (X, p. 66, lemma 1); let

$$\hat{c}({}^M\mathcal{E}) : H(L(M) \otimes_A \mathbf{N}'') \rightarrow H(L(M) \otimes_A \mathbf{N}')$$

the corresponding connecting homomorphism (X, p. 29).

DEFINITION 2. — A connecting homomorphism of torsion products, relative to the module M and to the exact sequence $\mathcal{E}$, the composite homomorphism

$$\partial(M, \mathcal{E}) = \psi_M(N')^{-1} \circ \partial({}^M\mathcal{E}) \circ \psi_M(N'') : \text{Tor}^A(M, N'') \rightarrow \text{Tor}^A(M, N').$$

This is a graded k -homomorphism of degree (-1) , whose homogeneous components are denoted $\partial_n(M, \mathcal{E}) : \text{Tor}_n^A(M, N'') \rightarrow \text{Tor}_{n-1}^A(M, N')$.

THEOREM 1. — The left-unbounded sequence of homomorphisms of k -modules

$$\begin{array}{ccccccc} \cdots & \rightarrow & \text{Tor}_n^A(M, N') & \xrightarrow{\text{Tor}_n^A(1, u)} & \text{Tor}_n^A(M, N) & \xrightarrow{\text{Tor}_n^A(1, v)} & \text{Tor}_n^A(M, N'') \\ & & \xrightarrow{\hat{c}_n(M, \mathcal{E})} & \text{Tor}_{n-1}^A(M, N') & \xrightarrow{\text{Tor}_{n-1}^A(1, u)} & \cdots & \xrightarrow{\text{Tor}_1^A(1, v)} \text{Tor}_1^A(M, N'') \\ & & \xrightarrow{\hat{c}_1(M, \mathcal{E})} & M \otimes_A N' & \xrightarrow{1 \otimes u} & M \otimes_A N & \xrightarrow{1 \otimes v} M \otimes_A N'' \rightarrow 0 \end{array}$$

is exact.

Consider indeed the diagram

$$\begin{array}{ccccccccc} \text{Tor}(M, N') & \xrightarrow{\text{Tor}(1, u)} & \text{Tor}(M, N) & \xrightarrow{\text{Tor}(1, v)} & \text{Tor}(M, N'') & \xrightarrow{\partial(M, \mathcal{E})} & \text{Tor}(M, N') & \xrightarrow{\text{Tor}(1, u)} & \text{Tor}(M, N) \\ & & \psi_{M(N')} \downarrow & \psi_{M(N)} \downarrow & \psi_{M(N'')} \downarrow & \psi_{M(N')} \downarrow & \psi_{M(N')} \downarrow & & \\ H(L(M) \otimes N') & \xrightarrow{H(1 \otimes u)} & H(L(M) \otimes N) & \xrightarrow{H(1 \otimes v)} & H(L(M \otimes N'')) & \xrightarrow{\partial^{(M)} \mathcal{E}} & H(L(M) \otimes N') & \rightarrow & H(L(M) \otimes N). \end{array}$$

It is commutative by (X, p. 69, remark 3) and def. 2. On the other hand, the bottom row is exact (X, p. 30, th. 1), and the various ψ_M are bijective (X, p. 69, prop. 5).

COROLLARY 1. — If $\text{Tor}_1^A(M, N'') = 0$, the sequence

$$0 \rightarrow M \otimes_A N' \xrightarrow{1 \otimes u} M \otimes_A N \xrightarrow{1 \otimes v} M \otimes_A N'' \rightarrow 0$$

is exact.

COROLLARY 2. — Let $0 \rightarrow C' \xrightarrow{u} C \xrightarrow{v} C'' \rightarrow 0$ be an exact sequence of complexes of left A -modules and E a complex of right A -modules. If C'' or E is flat, the sequence

$$0 \rightarrow E \otimes_A C' \xrightarrow{1 \otimes u} E \otimes_A C \xrightarrow{1 \otimes v} E \otimes_A C'' \rightarrow 0$$

is exact.

Indeed, $\text{Tor}_1^A(E, C'') = 0$ according to X, p. 69, cor. to prop. 5.

Example. — Let $\mathfrak{a}$ be an ideal of A . The exact sequence

$$0 \rightarrow \mathfrak{a} \rightarrow A_{\mathfrak{s}} \rightarrow A/\mathfrak{a} \rightarrow 0$$

of left A -modules gives rise to an exact sequence of torsion products, in which the terms $\text{Tor}_i^A(\mathbf{M}, A)$ are zero for $i > 0$. From this, we deduce isomorphisms

$$\text{Tor}_{i+1}^A(\mathbf{M}, A/\mathfrak{a}) \rightarrow \text{Tor}_i^A(\mathbf{M}, \mathfrak{a}), \quad i > 0$$

and an exact sequence

$$0 \rightarrow \text{Tor}_1^A(\mathbf{M}, A/\mathfrak{a}) \rightarrow \mathbf{M} \otimes_A \mathfrak{a} \rightarrow \mathbf{M} \otimes_A A \rightarrow \mathbf{M} \otimes A/\mathfrak{a} \rightarrow 0 :$$

it follows that $\text{Tor}_1^A(\mathbf{M}, A/\mathfrak{a})$ is identified with the kernel of the canonical homomorphism $\mathbf{M} \otimes_A \mathfrak{a} \rightarrow \mathbf{M}$.

For example, taking for $\mathbf{M}$ a module of the form A_d/b where b is a right ideal of A , one obtains an isomorphism of $\text{Tor}_1^A(A/b, A/\mathfrak{a})$ onto $(\mathfrak{a} \cap b)/ba$.

PROPOSITION 9. — Let $f : \mathbf{M} \rightarrow \mathbf{M}_1$ be a homomorphism of right A -modules and

(ε)

$$\begin{array}{ccc} & & 0 \rightarrow N' \xrightarrow{u} N \xrightarrow{v} N'' \rightarrow 0 \\ g' \downarrow & & \\ & & 0 \rightarrow N'_1 \xrightarrow{u_1} N_1 \xrightarrow{v_1} N''_1 \rightarrow 0 \\ & & ((\mathcal{E}_1)) \end{array}$$

a commutative diagram with exact rows of homomorphisms of left A -modules. The diagram of k -modules

$$\begin{array}{ccc}
\mathrm{Tor}^{\mathbf{A}}(\mathbf{M}, \mathbf{N}'') & \xrightarrow{\hat{c}(\mathbf{M}, \mathcal{E})} & \mathrm{Tor}^{\mathbf{A}}(\mathbf{M}, \mathbf{N}') \\
& \mathrm{Tor}^{\mathbf{A}}(f, g'') \downarrow & \\
\mathrm{Tor}^{\mathbf{A}}(\mathbf{M}_1, \mathbf{N}_1'') & \xrightarrow{\hat{c}(\mathbf{M}_1, \mathcal{E}_1)} & \mathrm{Tor}^{\mathbf{A}}(\mathbf{M}_1, \mathbf{N}_1')
\end{array}$$

is commutative.

This follows from X, p. 31, prop. 2, applied to the commutative diagram

$$\begin{array}{ccccccc}
0 \rightarrow & \mathrm{L}(\mathbf{M}) \otimes_{\mathbf{A}} \mathbf{N}' & \xrightarrow{1 \otimes u} & \mathrm{L}(\mathbf{M}) \otimes_{\mathbf{A}} \mathbf{N} & \xrightarrow{1 \otimes v} & \mathrm{L}(\mathbf{M}) \otimes_{\mathbf{A}} \mathbf{N}'' \rightarrow 0 \\
& \mathrm{L}(f) \otimes g' \downarrow & & \mathrm{L}(f) \otimes g \downarrow & & \mathrm{L}(f) \otimes g'' \downarrow & \\
0 \rightarrow & \mathrm{L}(\mathbf{M}_1) \otimes_{\mathbf{A}} \mathbf{N}_1' & \xrightarrow{1 \otimes u_1} & \mathrm{L}(\mathbf{M}_1) \otimes_{\mathbf{A}} \mathbf{N}_1 & \xrightarrow{1 \otimes v_1} & \mathrm{L}(\mathbf{M}_1) \otimes_{\mathbf{A}} \mathbf{N}_1'' \rightarrow 0.
\end{array}$$

Analogously, if $\mathbf{N}$ is a left $\mathbf{A}$ -module and

$$0 \rightarrow \mathbf{M}' \xrightarrow{r} \mathbf{M} \xrightarrow{s} \mathbf{M}'' \rightarrow 0 \quad (\mathrm{F})$$

an exact sequence of right $\mathbf{A}$ -modules, one defines connecting homomorphisms

$$\partial(\mathcal{F}, \mathbf{N}) : \mathrm{Tor}^{\mathbf{A}}(\mathbf{M}'', \mathbf{N}) \rightarrow \mathrm{Tor}^{\mathbf{A}}(\mathbf{M}', \mathbf{N})$$

$$\partial_n(\mathcal{F}, \mathbf{N}) : \mathrm{Tor}_n^{\mathbf{A}}(\mathbf{M}'', \mathbf{N}) \rightarrow \mathrm{Tor}_{n-1}^{\mathbf{A}}(\mathbf{M}', \mathbf{N})$$

by $\partial(\mathcal{F}, \mathbf{N}) = \overline{\psi}_{\mathbf{N}}(\mathbf{M}')^{-1} \circ \partial(\mathcal{F}^{\mathbf{N}}) \circ \overline{\psi}_{\mathbf{N}}(\mathbf{M}'')$, where $\partial(\mathcal{F}^{\mathbf{N}})$ is the connecting homomorphism of the exact sequence

$$(\mathcal{F}^{\mathbf{N}})$$

$$0 \rightarrow \mathbf{M}' \otimes_{\mathbf{A}} \mathrm{L}(\mathbf{N}) \rightarrow \mathbf{M} \otimes_{\mathbf{A}} \mathrm{L}(\mathbf{N}) \rightarrow \mathbf{M}'' \otimes_{\mathbf{A}} \mathrm{L}(\mathbf{N}) \rightarrow 0$$

deduced from $\mathcal{F}$, and we have:

THEOREM 1 bis. — The left-unbounded sequence of homomorphisms of $\mathbf{k}$ -modules

$$\begin{aligned} &\rightarrow \operatorname{Tor}_n^A(\mathbf{M}', \mathbf{N}) \xrightarrow{\operatorname{Tor}_n^A(r, 1)} \operatorname{Tor}_n^A(\mathbf{M}, \mathbf{N}) \xrightarrow{\operatorname{Tor}_n^A(s, 1)} \operatorname{Tor}_n^A(\mathbf{M}'', \mathbf{N}) \xrightarrow{\partial_n(\mathcal{F}, \mathbf{N})} \operatorname{Tor}_{n-1}^A(\mathbf{M}', \mathbf{N}) \\ &\cdots \rightarrow \operatorname{Tor}_1^A(\mathbf{M}'', \mathbf{N}) \xrightarrow{\partial_1(\mathcal{F}, \mathbf{N})} \mathbf{M}' \otimes_A \mathbf{N} \xrightarrow{r \otimes 1} \mathbf{M} \otimes_A \mathbf{N} \xrightarrow{s \otimes 1} \mathbf{M}'' \otimes_A \mathbf{N} \rightarrow 0 \end{aligned}$$

is exact.

We leave it to the reader to state and prove the properties analogous to the corollaries of Thm.

1 and to Prop. 9. Moreover:

PROPOSITION 10. — Let (F°) denote the exact sequence of left A -modules

$$0 \rightarrow \mathbf{M}'^\circ \xrightarrow{r} \mathbf{M}^\circ \xrightarrow{s} \mathbf{M}''^\circ \rightarrow 0.$$

The diagram

$$\begin{array}{ccc} \operatorname{Tor}^A(\mathbf{M}'', \mathbf{N}) & \xrightarrow{\partial(\mathcal{F}, \mathbf{N})} & \operatorname{Tor}^A(\mathbf{M}', \mathbf{N}) \\ \sigma_{\mathbf{M}', \mathbf{N}} \downarrow & & \downarrow \sigma_{\mathbf{M}', \mathbf{N}} \\ \operatorname{Tor}^{A^\circ}(\mathbf{N}^\circ, \mathbf{M}''^\circ) & \xrightarrow{\partial(\mathbf{N}^\circ, \mathcal{F}^\circ)} & \operatorname{Tor}^{A^\circ}(\mathbf{N}^\circ, \mathbf{M}'^\circ) \end{array}$$

is commutative.

Indeed, this follows from X, p. 31, prop. 2, applied to the commutative diagram

$$\begin{array}{ccccccc} 0 \rightarrow \mathbf{M}' \otimes_A \mathbf{L}(\mathbf{N}) & \xrightarrow{r \otimes 1} & \mathbf{M} \otimes_A \mathbf{L}(\mathbf{N}) & \xrightarrow{s \otimes 1} & \mathbf{M}'' \otimes_A \mathbf{L}(\mathbf{N}) & \rightarrow & 0 \\ \sigma(\mathbf{M}', \mathbf{L}(\mathbf{N})) \downarrow & & \sigma(\mathbf{M}, \mathbf{L}(\mathbf{N})) \downarrow & & \sigma(\mathbf{M}'', \mathbf{L}(\mathbf{N})) \downarrow & & \\ 0 \rightarrow \mathbf{L}(\mathbf{N}^\circ) \otimes_{A^\circ} \mathbf{M}'^\circ & \xrightarrow{1 \otimes r} & \mathbf{L}(\mathbf{N}^\circ) \otimes_{A^\circ} \mathbf{M}^\circ & \xrightarrow{1 \otimes s} & \mathbf{L}(\mathbf{N}^\circ) \otimes_{A^\circ} \mathbf{M}''^\circ & \rightarrow & 0. \end{array}$$

We shall see other commutation relations later (cf. X, p. 131, cor. 1).

10.4.6 Flat modules and torsion products

THEOREM 2. — Let E be a right A -module. The following conditions are equivalent:

- (i) E is flat;
- (ii) for every left A -module F , and every integer $n > 0$, one has

$$\mathrm{Tor}_n^A(\mathbf{E}, \mathbf{F}) = 0;$$

(iii) for every monogenic left A -module of finite presentation F , one has

$$\mathrm{Tor}_1^A(\mathbf{E}, \mathbf{F}) = 0;$$

(iv) for every finitely generated left ideal $\mathfrak{a}$ of A , the canonical map $E \otimes_A \mathfrak{a} \rightarrow E$ is injective;

(v) for every exact sequence of right A -modules, of the form

$$0 \rightarrow \mathbf{G} \xrightarrow{v} \mathbf{H} \xrightarrow{w} \mathbf{E} \rightarrow 0,$$

and every left A -module F , the sequence

$$0 \rightarrow \mathbf{G} \otimes_A F \xrightarrow{v \otimes 1} \mathbf{H} \otimes_A F \xrightarrow{w \otimes 1} \mathbf{E} \otimes_A F \rightarrow 0$$

is exact.

(i) $\Rightarrow$ (ii): this is the cor. to prop. 5 of X, p. 69.

(ii) $\Rightarrow$ (iii): this is trivial.

(iii) $\Leftrightarrow$ (iv): every monogenic left A -module of finite presentation is isomorphic to a quotient $A/\mathfrak{a}$, where $\mathfrak{a}$ is a finitely generated left ideal, so that (iii) is equivalent to (iv) by X, p. 72, example.

(iv) $\Rightarrow$ (i): by X, p. 8, prop. 3, X, p. 72, th. 1, E is flat as soon as $\mathrm{Tor}_1^A(\mathbf{E}, \mathbf{F}) = 0$ for every left A -module F . If (iii) is satisfied, this is so as soon as F is monogenic and of finite presentation. By X, p. 11, prop. 7, every A -module (resp. every monogenic A -module) is a filtered inductive limit of modules of finite presentation (resp. of monogenic modules of finite presentation); one therefore sees, by X, p. 70, prop. 8, that it suffices to prove that if $\mathrm{Tor}_1^A(\mathbf{E}, \mathbf{F}) = 0$ when F is monogenic, then this is so as soon as F is finitely generated. Let us therefore reason by induction on the cardinality of a generating family $(f_1, \dots, f_n)$

of F ; the exact sequence

$$0 \rightarrow Af_1 \rightarrow F \rightarrow F/Af_1 \rightarrow 0$$

gives rise to an exact sequence

$$\mathrm{Tor}_1^A(E, Af_1) \rightarrow \mathrm{Tor}_1^A(E, F) \rightarrow \mathrm{Tor}_1^A(E, F/Af_1),$$

so that $\mathrm{Tor}_1^A(E, F) = 0$ since $\mathrm{Tor}_1^A(E, Af_1) = 0$ and $\mathrm{Tor}_1^A(E, F/Af_1) = 0$ by the induction hypothesis.

(i) $\Rightarrow$ (v): this is cor. 2 to th. 1 (X, p. 72).

(ii) $\Rightarrow$ (iii): the exact sequence (X, p. 50)

$$0 \rightarrow Z_0(E) \xrightarrow{i_E} L_0(E) \xrightarrow{p_E} E \rightarrow 0$$

gives rise, for every left A -module F , to an exact sequence

$$0 \rightarrow \mathrm{Tor}_1^A(E, F) \rightarrow Z_0(E) \otimes_A F \xrightarrow{i_E \otimes 1} L_0(E) \otimes_A F \xrightarrow{p_E \otimes 1} E \otimes_A F \rightarrow 0.$$

If (v) is satisfied, one has $\mathrm{Tor}_1^A(E, F) = 0$ whence (iii).

COROLLARY 1. — Let $0 \rightarrow E' \rightarrow E \rightarrow E'' \rightarrow 0$ be an exact sequence of right A -modules. Suppose that E'' is flat. Then for E to be flat, it is necessary and sufficient that E' be flat.

Let F be a left A -module. Since $\mathrm{Tor}_i^A(E'', F) = 0$ for $i = 1, 2$ (th. 2, (i) $\Rightarrow$ (ii)), one has an exact sequence

$$0 \rightarrow \mathrm{Tor}_1^A(E', F) \rightarrow \mathrm{Tor}_1^A(E, F) \rightarrow 0$$

whence the assertion (th. 2, (i) $\Leftrightarrow$ (iii)).

COROLLARY 2. — Let $0 \rightarrow E_n \rightarrow E_{n-1} \rightarrow \cdots \rightarrow E_1 \rightarrow 0$ be an exact sequence of right A -modules. If E_i is flat for $i = 1, \dots, n-1$, then E_n is flat.

10.4.7 Künneth formula

In this number, we consider a complex (C, d) of right A -modules and a complex (C', d') of left A -modules. Consider the canonical exact sequences

$$0 \rightarrow Z(C) \xrightarrow{j} C \xrightarrow{\delta} B(C)(-1) \rightarrow 0, \quad (I)$$

$$0 \rightarrow B(C) \xrightarrow{i} Z(C) \xrightarrow{p} H(C) \rightarrow 0; \quad (II)$$

we deduce from δ a k -homomorphism

$$H(\delta \otimes 1) : H(C \otimes_A C') \rightarrow H(B(C) \otimes_A C')(-1);$$

we deduce from (II) a connecting homomorphism

$$\partial(II, H(C')) : \text{Tor}_1^A(H(C), H(C')) \rightarrow B(C) \otimes_A H(C');$$

if we equip $\text{Tor}_1^A(H(C), H(C'))$ with the grading whose homogeneous component of degree n is $\bigoplus_{p+q=n} \text{Tor}_1^A(H_p(C), H_q(C'))$, this connecting homomorphism is graded of degree 0. Moreover, there is a canonical homomorphism (X, p. 62)

$$\gamma(B(C), C') : B(C) \otimes_A H(C') \rightarrow H(B(C) \otimes_A C').$$

With these notations:

THEOREM 3. — Suppose the A -modules $B(C)$ and $Z(C)$ are flat. There exists a unique homomorphism of graded k -modules, of degree -1,

$$\alpha : H(C \otimes_A C') \rightarrow \text{Tor}_1^A(H(C), H(C'))$$

thereby making the diagram commutative

$$\begin{array}{ccc}
H(C \otimes_A C') & \xrightarrow{\alpha} & \text{Tor}_1^A(H(C), H(C'))(-1) \\
H(\delta \otimes 1) \downarrow & & \downarrow \hat{c}(\Pi, H(C')) \\
H(B(C) \otimes_A C')(-1) & \xleftarrow{\gamma(B(C), C')} & (B(C) \otimes_A H(C'))(-1).
\end{array}$$

The sequence of graded k -modules

$$0 \rightarrow H(C) \otimes_A H(C') \xrightarrow{\gamma(C, C')} H(C' \otimes_A C') \xrightarrow{\alpha} \text{Tor}_1^A(H(C), H(C'))(-1) \rightarrow 0$$

is exact.

Thus for each n , we have an exact sequence

$$\begin{aligned}
0 \rightarrow \bigoplus_{p+q=n} H_p(C) \otimes_A H_q(C') & \xrightarrow{\gamma_n(C, C')} H_n(C \otimes_A C') \\
& \xrightarrow{\alpha_n} \bigoplus_{p+q=n-1} \text{Tor}_1^A(H_p(C), H_q(C')) \rightarrow 0.
\end{aligned} \tag{11}$$

For simplicity set $\mathbf{B} = \mathbf{B}(C)$, $\mathbf{Z} = \mathbf{Z}(C)$, $\mathbf{H} = \mathbf{H}(C)$ and $\mathbf{H}' = \mathbf{H}(C')$:

Since B is flat, we deduce from (I) an exact sequence (X, p. 72, cor. 2)

$$0 \rightarrow Z \otimes_A C' \xrightarrow{j \otimes 1} C \otimes_A C' \xrightarrow{\delta \otimes 1} (B \otimes_A C')(-1) \rightarrow 0. \tag{12}$$

Lemma 3. — The connecting homomorphism $H(B \otimes_A C') \rightarrow H(Z \otimes_A C')$ associated with the exact sequence (12) is equal to $H(i \otimes 1)$.

Indeed, let $a \in Z(B \otimes_A C')$; since B is flat, a belongs to the image of $B \otimes_A Z(C')$, hence can be written as $\sum_{\lambda} da_{\lambda} \otimes b_{\lambda}$, with $a_{\lambda} \in C$, $b_{\lambda} \in C'$, $db_{\lambda} = 0$. The image of

the class of a under the sought homomorphism is by definition the class of $\mathbf{D}(\sum a_{\lambda} \otimes b_{\lambda}) = \sum da_{\lambda} \otimes b_{\lambda} = (i \otimes 1)(a)$, whence the lemma.

The exact homology sequence associated with (12) is therefore

$$\begin{aligned}
H(B \otimes_A C') & \xrightarrow{H(i \otimes 1)} H(Z \otimes_A C') \xrightarrow{H(j \otimes 1)} H(C \otimes_A C') \\
& \xrightarrow{H(\delta \otimes 1)} H(B \otimes_A C')(-1) \xrightarrow{H(i \otimes 1)} H(Z \otimes_A C')(-1).
\end{aligned}$$

Moreover, since Z is flat, we deduce from (II) an exact sequence of graded k -modules

$$0 \rightarrow \text{Tor}_1^A(H, H') \xrightarrow{\hat{c}(\Pi, H')} B \otimes_A H' \xrightarrow{i \otimes 1} Z \otimes_A H' \xrightarrow{p \otimes 1} H \otimes_A H' \rightarrow 0;$$

finally, we have the canonical homomorphisms of no. 1

$$\gamma_B = \gamma(B, C') : B \otimes_A H' \rightarrow H(B \otimes_A C')$$

$$\gamma_Z = \gamma(Z, C') : Z \otimes_A H' \rightarrow H(Z \otimes_A C')$$

$$\gamma_C = \gamma(C, C') : H \otimes_A H' \rightarrow H(C \otimes_A C'),$$

whence a diagram of graded k -modules, with exact rows

$$\begin{array}{ccccccc} B \otimes H' & \xrightarrow[\gamma_B]{i \otimes 1} & Z \otimes H' & \xrightarrow[\gamma_Z]{p \otimes 1} & H \otimes H' & \rightarrow & 0 \\ H(B \otimes C') & \xrightarrow{H(i \otimes 1)} & H(Z \otimes C') & \xrightarrow{H(j \otimes 1)} & H(C \otimes C') & \xrightarrow{H(\delta \otimes 1)} & H(B \otimes C')(-1) \xrightarrow{H(i \otimes 1)} H(Z \otimes C')(-1) \\ 0 \rightarrow \text{Tor}_1^A(H, H')(-1) & \xrightarrow{\delta(\Pi, H')} & (B \otimes H')(-1) & \xrightarrow[i \otimes 1]{\gamma_B} & (Z \otimes H')(-1), & & \end{array}$$

which is commutative by definition of homomorphisms γ . But, since the complexes B and Z are split and flat, γ_B and γ_Z are bijective (X, p. 65, cor. 1). From this one deduces, on the one hand, that γ_C is injective with image equal to $\text{Ker } H(\delta \otimes 1)$, and on the other hand that $\gamma_B \circ \partial(\Pi, H')$ is injective, with image equal to $\text{Im } H(\delta \otimes 1)$. The theorem follows immediately from this.

COROLLARY 1. — If $B(C)$ and $Z(C)$ are flat, then for every left A -module N and every integer n there is an exact sequence

$$0 \rightarrow H_n(C) \otimes_A N \xrightarrow{\gamma_n} H_n(C \otimes_A N) \xrightarrow{\alpha_n} \text{Tor}_1^A(H_{n-1}(C), N) \rightarrow 0. \quad (13)$$

COROLLARY 2. — Suppose $B(C)$ and $B(C')$ are projective and $Z(C)$ is flat. Then the sequences of k -modules (11) and (13) are exact and split.

This follows from the theorem and the following lemma:

Lemma 4. — If $B(C)$ and $B(C')$ are projective, then the canonical homomorphism

$$\gamma(C, C') : H(C) \otimes_A H(C') \rightarrow H(C \otimes_A C')$$

has a k -linear retraction.

Indeed, according to X, p. 65, remark b), there exist homomorphisms $\varphi : C \rightarrow H(C)$ and $\varphi' : C' \rightarrow H(C')$ such that $H(\varphi) = 1_{H(C)}$ and $H(\varphi') = 1_{H(C')}$. In the commutative diagram

$$\begin{array}{ccc} H(C) \otimes_A H(C') & \xrightarrow{\gamma(C, C')} & H(C \otimes_A C') \\ & \downarrow H(\varphi) \otimes H(\varphi') & \\ H(C) \otimes_A H(C') & \xrightarrow{\gamma(H(C), H(C'))} & H(C) \otimes_A H(C') \end{array}$$

$H(\varphi) \otimes H(\varphi')$ and $\gamma(H(C), H(C'))$ are the identity, whence the assertion.

COROLLARY 3 (“universal coefficient formula”). — Suppose the ring A is principal. If the complexes C and C' are free, then the sequences of A -modules (11) are exact and split; if the complex C is free, then the sequences of A -modules (13) are exact and split for every A -module N .

Indeed, $B(C)$, $Z(C)$, and $B(C')$ are submodules of the free modules C , C , C' , hence are free (VII, § 3, cor. 2 to th. 1), and one applies cor. 2.

COROLLARY 4 (“Künneth formula”). — Suppose C is bounded above, and C and $H(C)$ are flat; then the canonical homomorphism

$$\gamma(C, C') : H(C) \otimes_A H(C') \rightarrow H(C \otimes_A C')$$

is bijective.

By the theorem, it suffices to prove that $B(C)$ and $Z(C)$ are flat. Now we have exact sequences

$$0 \rightarrow B_n(C) \rightarrow Z_n(C) \rightarrow H_n(C) \rightarrow 0$$

$$0 \rightarrow Z_n(\mathbf{C}) \rightarrow \mathbf{C}_n \rightarrow \mathbf{B}_{n-1}(\mathbf{C}) \rightarrow 0,$$

whence, by X, p. 75, cor. 1, the implications $(\mathbf{B}_{n-1}(\mathbf{C}) \text{ is flat}) \Rightarrow (\mathbf{Z}_n(\mathbf{C}) \text{ is flat}) \Rightarrow (\mathbf{B}_n(\mathbf{C}) \text{ is flat})$; we conclude by noting that $\mathbf{B}_n(\mathbf{C}) = 0$ for n small enough.

COROLLARY 5. — Let $u : \mathbf{C} \rightarrow \mathbf{C}'$ a homomorphism of complexes of right A -modules, flat and bounded on the right. For every complex E of left A -modules, the morphism $u \otimes 1_E : \mathbf{C} \otimes_A E \rightarrow \mathbf{C}' \otimes_A E$ is a homomorphism.

Indeed, $\text{Con}(u)$ is a flat complex, bounded on the right and with zero homology; we therefore have $H(\text{Con}(u) \otimes_A E) = 0$ by cor. 4, hence $H(\text{Con}(u \otimes 1_E)) = 0$ (X, p. 67, lemma 2), and $u \otimes 1_E$ is a homomorphism.

10.4.8 Bounded and flat complexes over a Noetherian ring

PROPOSITION 11. — Suppose A is left Noetherian, and let C be a bounded complex of flat left A -modules such that $H(C)$ is a finitely generated A -module. Let a and b be two integers such that $a \leq b$ and that $H_n(C) = 0$ for $n < a$, $C_n = 0$ for $n > b$. There exists a complex P of left A -modules such that P_n is projective and finitely generated for each n , and that $P_n = 0$ for $n \notin [a, b]$, and a homomorphism $u : P \rightarrow C$. Moreover, for every complex E of right A -modules, the homomorphism

$$H(1_E \otimes u) : H(E \otimes_A P) \rightarrow H(E \otimes_A C) \quad \text{is bijective.}$$

By X, p. 53, prop. 7, there exists a complex (L, d) such that L_n is free and finitely generated for each n , and zero when $n < a$, and a homomorphism $f : L \rightarrow C$. Let P be the quotient complex L/L' , where $L'_n = 0$ for $n < b$, $L'_n = L_n$ for $n > b$, $L'_b = B_b(L)$. Since $C_n = 0$ for $n > b$, $f(L') = 0$, so f factors through a morphism of complexes $u : P \rightarrow C$.

$$\begin{array}{ccccccc}
\cdots & \longrightarrow & L_{b+1} & \xrightarrow{d_{b+1}} & L_b & \xrightarrow{d_b} & L_{b-1} \longrightarrow \cdots \\
& & \downarrow & & \downarrow & & \parallel \\
& & 0 & \longrightarrow & P_b & \longrightarrow & P_{b-1} \longrightarrow \cdots \\
& & \downarrow & & \downarrow^{u_b} & & \downarrow^{u_{b-1}} \\
& & 0 & \longrightarrow & C_b & \longrightarrow & C_{b-1} \longrightarrow \cdots
\end{array}$$

Since f is a homomorphism, we have $H(\text{Con}(f)) = 0$ whence an exact sequence

$$\cdots \rightarrow L_{b+1} \xrightarrow{d_{b+1}} L_b \rightarrow L_{b-1} \oplus C_b \rightarrow L_{b-2} \oplus C_{b-1} \rightarrow \cdots$$

Thus we have an exact sequence

$$0 \rightarrow P_b \rightarrow L_{b-1} \oplus C_b \rightarrow L_{b-2} \oplus C_{b-1} \rightarrow \cdots$$

This shows, on the one hand, that the cone of u has zero homology, so that u is a homomorphism, and on the other hand that the module P_b is flat (X, p. 76, cor. 2); since P_b is of finite type as a quotient of L_{b+1} it is projective (X, p. 13, cor.). The pair (P, u) therefore satisfies the required condition. The last assertion follows from X, p. 79, cor. 5.

* Example. — Let A be a Noetherian commutative ring, X a proper and flat A -scheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -module, flat over A . There exists a bounded complex P formed of finitely generated projective A -modules such that for every A -module M , $H(X, \mathcal{F} \otimes_A M)$ is naturally identified with $H(P \otimes_A M)$. Indeed, let $\mathcal{U}$ be a covering of X by a finite number of affine open sets, $\mathcal{E}(\mathcal{U}, \mathcal{F})$ the associated Čech complex: One shows that $H^i(\mathcal{E}(\mathcal{U}, \mathcal{F}))$ is isomorphic to the A -module $H^i(X, \mathcal{F})$, and that the latter is finitely generated; moreover, for every A -module M , the complex $\mathcal{E}(\mathcal{U}, \mathcal{F}) \otimes_A M$ is isomorphic to $\mathcal{E}(\mathcal{U}, \mathcal{F} \otimes_A M)$. By applying prop. 11 to the complex $\mathcal{E}(\mathcal{U}, \mathcal{F})$ (which is bounded), one obtains a complex P which answers the question.

For every point y of $\text{Spec}(A)$ $\kappa(y)$ the residue field of A at y , $X_y = X \otimes_A \kappa(y)$ the fibre of X above y , $\mathcal{F}_y = \mathcal{F} \otimes_A \kappa(y)$, and set $h^p(y) = \dim_{\kappa(y)} H^p(X_y, \mathcal{F}_y)$ for $p \geq 0$. One easily deduces from the existence of the complex P the following results:

- (i) the function h^p is upper semi-continuous on $\text{Spec}(A)$;

(ii) the function $\sum_{p \geq 0} (-1)^p h^p$ is locally constant on $\text{Spec}(\mathbf{A})$.*

10.4.9 Generalization to complexes of multimodules

Let B and B' be two rings, C a complex of (B, A) -bimodules, C' a complex of (A, B') -bimodules (X, p. 43); then $(C \otimes_A C', D)$ (X, p. 61) is a complex of (B, B') -bimodules and the canonical homomorphism

$$\gamma : H(C) \otimes_A H(C') \rightarrow H(C \otimes_A C')$$

is compatible with the structures of (B, B') -bimodules of the two members.

If B'' is a third ring, and C'' a complex of (B', B'') -bimodules, the canonical homomorphism (II, p. 64, prop. 8)

$$(C \otimes_A C') \otimes_{B'} C'' \rightarrow C \otimes_A (C' \otimes_{B''} C'')$$

is an isomorphism of complexes of (B, B'') -bimodules.

More generally, we leave it to the reader to develop the theory of tensor products of finite, totally ordered families of complexes of multimodules, following the model of no. 1 (X, p. 63) and of II, pp. 65 to 72 (associativity and commutativity isomorphisms, ...).

Let B and B' be two rings, M a (B, A) -bimodule, N an (A, B') -bimodule: then $L(M) \otimes_A L(N)$ is a complex of (B, B') -bimodules, so that $\text{Tor}^A(M, N)$ is endowed with a natural structure of graded (B, B') -bimodule; on the degree 0 term, this structure coincides with that of $M \otimes_A N$.

If $\lambda \in \mathbf{B}$, $\lambda' \in \mathbf{B}'$, and if we denote by $\lambda_M, \lambda'_N, \lambda_T, \lambda'_T$, the homotheties $x \mapsto \lambda x, y \mapsto y \lambda', z \mapsto \lambda z, z \mapsto z \lambda'$ of $\mathbf{M}, \mathbf{N}, \text{Tor}^A(\mathbf{M}, \mathbf{N}), \text{Tor}^A(\mathbf{M}, \mathbf{N})$ respectively, then

$$\lambda_T = \text{Tor}^A(\lambda_M, 1_N), \quad \lambda'_T = \text{Tor}^A(1_M, \lambda'_N),$$

which provides another description of the bimodule structure of $\text{Tor}^A(M, N)$. We leave it to the reader to generalize nos. 5 and 7 to the case of complexes of multimodules.

10.5 EXTENSION MODULES

We keep the general notation of § 4. We further agree that, unless expressly stated otherwise, all modules considered are left modules, and all complexes considered are complexes of left modules.

10.5.1 Complexes of homomorphisms

Let (C, d) and (C', d') be two A -complexes. Consider the graded k -module $\text{Homgr}_A(C, C')$ (II, pp. 174–175): for $n \in \mathbf{Z}$ $\text{Homgr}_A(C, C')_n$ is the k -module of graded A -linear maps of degree n from C to C' ; in other words, $\text{Homgr}_A(C, C')$ is canonically identified with the A -module

$$\bigoplus_{n \in \mathbf{Z}} \prod_{p \in \mathbf{Z}} \text{Hom}_A(C_p, C'_{p+n}) = \bigoplus_{n \in \mathbf{Z}} \prod_{p+q=n} \text{Hom}_A(C_p, C'_q).$$

Let us define k -linear maps.

$$D_n : \text{Homgr}_A(C, C')_n \rightarrow \text{Homgr}_A(C, C')_{n-1}, \quad n \in \mathbf{Z},$$

by

$$D_n(f) = d' \circ f - (-1)^n f \circ d; \tag{1}$$

we have

$$\begin{aligned} D_{n-1} \circ D_n(f) &= D_{n-1}(d' \circ f - (-1)^n f \circ d) = d' \circ d' \circ f - (-1)^n d' \circ f \circ d \\ &\quad - (-1)^{n-1} d' \circ f \circ d - f \circ d \circ d = 0. \end{aligned}$$

Then $(\text{Homgr}_A(C, C'), D)$ is a complex of k -modules called the complex of homomorphisms from C to C' .

For example, $\text{Homgr}_A(A, C')$ is canonically identified with C' . Let us also note that, for every $n \in \mathbf{Z}$, we have $\text{Homgr}_A(C, C')(n) = \text{Homgr}_A(C, C'(n))$.

The elements of $Z_n(\text{Homgr}_A(C, C'))$ are the graded homomorphisms f of (descending) degree n of C to C' such that $d' \circ f = (-1)^n f \circ d$, that is, the morphisms of complexes of C to $C'(n)$, or again of $C(p)$ to $C'(p+n)$ for p any fixed. These are said to be the morphisms of complexes of (descending) degree n of C to C' ; if $f, g \in Z_n(\text{Homgr}_A(C, C'))$ and $s \in \text{Homgr}_A(C, C')_{n+1}$, then the condition $g - f = Ds$ means that s is a homotopy connecting the morphisms f and g of C to $C'(n)$, so that $H_n(\text{Homgr}_A(C, C'))$ is the k -module of homotopy classes of morphisms of degree (descending) n of C to C' .

Let $\alpha \in H_n(\text{Homgr}_A(C, C'))$ and $p \in \mathbf{Z}$. Let us represent α by $f \in Z_n(\text{Homgr}_A(C, C'))$; then f is a morphism of complexes of C to $C'(n)$, hence $\mathbf{H}_p(f)$ is a homomorphism of $\mathbf{H}_p(C)$ to $\mathbf{H}_p(C'(n)) = \mathbf{H}_{p+n}(C')$; since $\mathbf{H}_p(f)$ depends only on the homotopy class α of f (**X, p. 33, prop. 3**), we deduce a canonical homomorphism of k -modules

$$H_n(\text{Homgr}_A(C, C')) \rightarrow \text{Hom}_A(\mathbf{H}_p(C), \mathbf{H}_{p+n}(C')),$$

whence a graded k -linear map of degree 0, called canonical

$$\lambda(C, C') : H(\text{Homgr}_A(C, C')) \rightarrow \text{Homgr}_A(H(C), H(C')).$$

The homogeneous components of $\lambda(C, C')$ will often be denoted:

$$\lambda^n(C, C') : H^n(\text{Homgr}_A(C, C')) \rightarrow \prod_{p+q=n} \text{Hom}_A(\mathbf{H}_p(C), \mathbf{H}^q(C')).$$

PROPOSITION 1. — If C is zero on the right and C' zero on the left, then $\text{Homgr}_A(C, C')$ is zero on the left, and the canonical k -linear map

$$\lambda^0(C, C') : H^0(\text{Homgr}_A(C, C')) \rightarrow \text{Hom}_A(\mathbf{H}_0(C), \mathbf{H}^0(C'))$$

is bijective.

We have exact sequences

$$0 \rightarrow H^0(C') \xrightarrow{i} C'^0 \xrightarrow{d'^0} C'^1$$

$$C_1 \xrightarrow{d_1} C_0 \xrightarrow{p} H_0(C) \rightarrow 0.$$

On the other hand $\text{Homgr}_A^0(C, C')$ is identified with $\text{Hom}_A(C_0, C'^0)$, $Z^0(\text{Homgr}_A(C, C'))$ then being identified with the set of $f : C_0 \rightarrow C'^0$ such that $d'^0 \circ f = 0$, $f \circ d_1 = 0$; $B^0(\text{Homgr}_A(C, C'))$ is zero; finally the map λ^0 associates to the class of f modulo $\{0\}$ the homomorphism $\varphi : H_0(C) \rightarrow H^0(C')$ such that $f = i \circ \varphi \circ p$, whence the proposition.

Let $u : \tilde{C} \rightarrow C$ and $u' : C' \rightarrow \tilde{C}'$ be morphisms of complexes; then the canonical homomorphism $\text{Homgr}_A(u, u') : \text{Homgr}_A(C, C') \rightarrow \text{Homgr}_A(\tilde{C}, \tilde{C}')$, defined by $f \mapsto u' \circ f \circ u$, is a morphism of complexes, as follows immediately from formula (1). Moreover, the following diagram is commutative

$$\begin{array}{ccc} H(\text{Homgr}_A(C, C')) & \xrightarrow{\lambda(C, C')} & \text{Homgr}_A(H(C), H(C')) \\ & \downarrow H(\text{Homgr}_A(u, u')) & \\ H(\text{Homgr}_A(\tilde{C}, \tilde{C}')) & \xrightarrow{\lambda(\tilde{C}, \tilde{C}')} & \text{Homgr}_A(H(\tilde{C}), H(\tilde{C}')). \end{array}$$

PROPOSITION 2. — a) Let $C' \xrightarrow{u} C \xrightarrow{v} C''$ be an exact sequence of A-complexes, P a projective complex, E an injective complex (X, p. 25). Then the sequences

$$\text{Homgr}_A(P, C') \xrightarrow{\text{Homgr}(1, u)} \text{Homgr}_A(P, C) \xrightarrow{\text{Homgr}(1, v)} \text{Homgr}_A(P, C'')$$

and

$$\text{Homgr}_A(C'', E) \xrightarrow{\text{Homgr}(v, 1)} \text{Homgr}_A(C, E) \xrightarrow{\text{Homgr}(u, 1)} \text{Homgr}_A(C', E)$$

are exact sequences of complexes of k-modules.

b) Let $0 \rightarrow C' \xrightarrow{u} C \xrightarrow{v} C'' \rightarrow 0$ be a sequence of A-complexes which is split as an exact sequence of graded A-modules (this is the case, for example, if C' is injective, or if C'' is projective). Then for every complex E , the sequences

$$0 \rightarrow \text{Homgr}_A(E, C') \xrightarrow{\text{Homgr}(1, u)} \text{Homgr}_A(E, C) \xrightarrow{\text{Homgr}(1, v)} \text{Homgr}_A(E, C'') \rightarrow 0$$

$$0 \rightarrow \text{Homgr}_{\mathbf{A}}(C'', E) \xrightarrow{\text{Homgr}(v, 1)} \text{Homgr}_{\mathbf{A}}(C, E) \xrightarrow{\text{Homgr}(u, 1)} \text{Homgr}_{\mathbf{A}}(C', E) \rightarrow 0$$

are exact sequences of complexes of k -modules.

In the case a), we note that the sequences

$$\text{Hom}_{\mathbf{A}}(P_p, C'_q) \rightarrow \text{Hom}_{\mathbf{A}}(P_p, C_q) \rightarrow \text{Hom}_{\mathbf{A}}(P_p, C''_q)$$

and

$$\text{Hom}_{\mathbf{A}}(C''_q, E_p) \rightarrow \text{Hom}_{\mathbf{A}}(C_q, E_p) \rightarrow \text{Hom}_{\mathbf{A}}(C'_q, E_p)$$

are exact for all $p, q \in \mathbf{Z}$, and we apply II, p. 10, prop. 5 and II, p. 13, prop. 7. The proof of b) is analogous.

10.5.2 Complexes of homomorphisms and homotopies

PROPOSITION 3. — Let $C, \tilde{C}, C', \tilde{C}'$ be four $\mathbf{A}$ -complexes, $u : \tilde{C} \rightarrow C, v : \tilde{C} \rightarrow C, u' : C' \rightarrow \tilde{C}'$ and $v' : C' \rightarrow \tilde{C}'$ four morphisms of complexes.

- a) If u and u' are homotopic to v and v' respectively, then the two morphisms $\text{Homgr}_{\mathbf{A}}(u, u')$ and $\text{Homgr}_{\mathbf{A}}(v, v')$ of $\text{Homgr}_{\mathbf{A}}(C, C')$ to $\text{Homgr}_{\mathbf{A}}(\tilde{C}, \tilde{C}')$ are homotopic.
- b) If u and u' are homotopisms, $\text{Homgr}_{\mathbf{A}}(u, u')$ is a homotopism.
- c) If C or C' is homotopic to zero, $\text{Homgr}_{\mathbf{A}}(C, C')$ is homotopic to zero.

Let us denote by the same letter d the differentials of the complexes C, C_1, C', C'_1 , and by D the differentials of $\text{Homgr}_{\mathbf{A}}(C, C')$ and $\text{Homgr}_{\mathbf{A}}(C_1, C'_1)$. If u (resp. u') is homotopic to v (resp. v'), there exists a graded homomorphism of degree 1,

$$w : C_1 \rightarrow C \quad (\text{resp. } w' : C' \rightarrow C'_1)$$

such that

$$u - v = dw + wd \quad (\text{resp. } u' - v' = dw' + w') \quad (2)$$

Let $W : \text{Homgr}_A(C, C') \rightarrow \text{Homgr}_A(C_{-1}, C_{-1})$ be the graded homomorphism of degree 1 such that, for $f \in \text{Homgr}_A(C, C')_n$, $n \in \mathbb{Z}$, we have

$$W(f) = w'fu + (-1)^n v'fw. \quad (3)$$

We then have

$$\begin{aligned} (DW + WD)(f) &= D[w'fu + (-1)^n v'fw] + W[df - (-1)^n fd] \\ &= dw'fu - (-1)^{n+1} w'fud + (-1)^n dv'fw + v'fwd \\ &\quad + w'dfu + (-1)^{n+1} v'dfw - (-1)^n w'fdu + v'fdw \\ &= (dw' + w'd)fu + v'f(wd + dw) \\ &= (u' - v')fu + v'f(u - v) = u'fu - v'fv; \end{aligned}$$

which is written $DW + WD = \text{Homgr}_A(u, u') - \text{Homgr}_A(v, v')$, whence a).

Let us prove b). If u and u' are homotopisms, let $\alpha : C \rightarrow \tilde{C}$ and $\alpha' : \tilde{C}' \rightarrow C'$ be morphisms of complexes such that $u \circ \alpha$, $\alpha \circ u$, $u' \circ \alpha'$, $\alpha' \circ u'$ are homotopic respectively to Id_C , $Id_{\tilde{C}}$, $Id_{\tilde{C}'}$, $Id_{C'}$. Then $\text{Homgr}(u, u') \circ \text{Homgr}(\alpha, \alpha')$, which is equal to $\text{Homgr}(\alpha \circ u, u' \circ \alpha')$, is homotopic by a) to

$$\text{Homgr}_A(Id_C, Id_{C'}) = Id_{\text{Homgr}(C, C')} :$$

likewise $\text{Homgr}(\alpha, \alpha') \circ \text{Homgr}(u, u')$ is homotopic to $Id_{\text{Homgr}(\tilde{C}, \tilde{C})}$, whence b).

Finally c) follows from b) (applied to the case where $\tilde{C}$ or $\tilde{C}'$ is zero).

COROLLARY 1. — If C is split and $H(C)$ is projective (resp. if C' is split and $H_n(C')$ injective for each n), then the canonical homomorphism

$$\lambda(C, C') : H(\text{Homgr}_A(C, C')) \rightarrow \text{Homgr}_A(H(C), H(C'))$$

is bijective.

Suppose for example that C' is split and $H(C')$ is injective for each n , the case where C

is split and $H(C)$ is projective being proved analogously. By X, p. 35, def. 6, there exists a homotopism $u' : C' \rightarrow H(C')$; by prop. 3, $\text{Homgr}_A(1, u')$ is a homotopism of $\text{Homgr}_A(C, C')$ on $\text{Homgr}_A(C, H(C'))$; since

$$\text{Homgr}_A(1, H(u')) \circ \lambda(C, C') = \lambda(C, H(C')) \circ H(\text{Homgr}_A(1, u'))$$

and that $\text{Homgr}_A(1, H(u'))$ and $H(\text{Homgr}_A(1, u'))$ are bijective, it suffices for us to prove that $\lambda(C, H(C'))$ is bijective, which brings us back to the case where C' is injective and has zero differential.

Then the canonical exact sequences (X, p. 28)

$$0 \rightarrow B(C) \xrightarrow{i} C \xrightarrow{p} C/B(C) \rightarrow 0 \quad (\text{III})$$

$$0 \rightarrow H(C) \xrightarrow{j} C/B(C) \xrightarrow{\delta} B(C) \rightarrow 0 \quad (\text{IV})$$

give exact sequences (X, p. 83, prop. 2, a))

$$0 \rightarrow \text{Homgr}_A(C/B(C), C') \xrightarrow{P} \text{Homgr}_A(C, C') \xrightarrow{I} \text{Homgr}_A(B(C), C') \rightarrow 0$$

$$0 \rightarrow \text{Homgr}_A(B(C), C') \xrightarrow{\Delta} \text{Homgr}_A(C/B(C), C') \xrightarrow{J} \text{Homgr}_A(H(C), C') \rightarrow 0.$$

Since $d_C = i \circ \delta \circ p$ the differential D of $\text{Homgr}(C, C')$ is given by $D_n = (-1)^{n+1} P_n \circ \Delta_n \circ I_n$; we then have

$$Z(\text{Homgr}_A(C, C')) = \text{Ker}(P \circ \Delta \circ I) = \text{Ker } I = \text{Im } P,$$

$$B(\text{Homgr}_A(C, C')) = \text{Im}(P \circ \Delta \circ I) = P(\text{Im } \Delta) = P(\text{Ker } J);$$

whence an isomorphism $\varphi : H(\text{Homgr}_A(C, C')) \rightarrow \text{Homgr}_A(H(C), C')$ such that, if $a \in$

$\text{Homgr}_A(C/B(C), C')$, then the image under φ of the class of $P(a)$ is $J(a)$; one immediately verifies that φ identifies with the canonical homomorphism λ .

COROLLARY 2. — Suppose $B(C)$ and $H(C)$ are projective (resp. $B_n(C')$ and $H_n(C')$ injective for each n). Then $\lambda(C, C')$ is bijective.

Indeed C (resp. C') is then split by X , p. 35, example 4, and we apply Cor. 1.

COROLLARY 3. — Let M be a projective (resp. injective) A -module. For every complex C of A -modules and every integer n , the canonical homomorphism

$$H^n(\text{Homgr}_A(M, C)) \rightarrow \text{Hom}_A(M, H^n(C))$$

$$(\text{ resp. } H^n(\text{Homgr}_A(C, M)) \rightarrow \text{Hom}_A(H_n(C), M)) \text{ is bijective.}$$

Lemma 1. — a) If C or C' is bounded on the right, if C is projective and if $H(C') = 0$, then $H(\text{Homgr}_A(C, C')) = 0$.

b) If C or C' is bounded on the left, if C' is injective and if $H(C) = 0$, then $H(\text{Homgr}_A(C, C')) = 0$.

Let $f \in Z_n(\text{Homgr}_A(C, C'))$; f is therefore a morphism of complexes of C to $C'(n)$; in the case a (resp. b), f_m is zero for m small enough (resp. large enough). By X , p. 47, prop. 1, f is then homotopic to zero, hence belongs to $B_n(\text{Homgr}_A(C, C'))$ whence the conclusion.

PROPOSITION 4. — Let $u : \mathbb{C}' \rightarrow \mathbb{C}$ be a homologism of complexes, $\mathbb{P}$ a projective complex, $\mathbb{E}$ an injective complex.

a) If $\mathbb{P}$ is bounded above, or if C and C' are bounded above, then

$$\text{Homgr}_A(1, u) : \text{Homgr}_A(\mathbb{P}, C') \rightarrow \text{Homgr}_A(\mathbb{P}, C)$$

is a homologism.

b) If $\mathbb{E}$ is bounded below, or if C and C' are bounded below, then

$$\text{Homgr}_A(u, 1) : \text{Homgr}_A(C, E) \rightarrow \text{Homgr}_A(C', E)$$

is a homologism.

Suppose first that u is injective, and set $C'' = \text{Coker } u$. Since u is a homologism, C'' has zero homology. On the other hand, we have exact sequences (prop. 2)

$$0 \rightarrow \text{Homgr}_A(P, C') \xrightarrow{\text{Homgr}(1, u)} \text{Homgr}_A(P, C) \rightarrow \text{Homgr}_A(P, C'') \rightarrow 0$$

$$0 \rightarrow \text{Homgr}_A(C'', E) \rightarrow \text{Homgr}_A(C, E) \xrightarrow{\text{Homgr}(u, 1)} \text{Homgr}_A(C', E) \rightarrow 0$$

by lemma 1, $\text{Homgr}_A(P, C'')$ has zero homology in the case a), $\text{Homgr}_A(C'', E)$ has zero homology in the case b), whence the conclusion.

In the general case, there exists (X, p. 38, cor. to prop. 7) a complex $\tilde{C}'$, which is bounded on the right (resp. on the left) when C and C' are, an injective morphism $\tilde{u} : C' \rightarrow \tilde{C}'$ and a homotopism $\beta : \tilde{C}' \rightarrow C$ such that $u = \beta \circ \tilde{u}$. Then $\tilde{u}$ is a homologism (X, p. 34, cor. to prop. 5); by the preceding, $\text{Homgr}_A(1_P, \tilde{u})$ (resp. $\text{Homgr}_A(\tilde{u}, 1_E)$) is a homologism in the case a) (resp. b). On the other hand $\text{Homgr}_A(1_P, \beta)$ (resp. $\text{Homgr}_A(\beta, 1_E)$) is a homotopism (prop. 3): then $\text{Homgr}_A(1_P, u)$ (resp. $\text{Homgr}_A(u, 1_E)$) is composed of two homologisms, hence is a homologism.

10.5.3 Definition and first properties of extension modules

For every A -module E , we denote by $p_E : L(E) \rightarrow E$ (resp. $e_E : E \rightarrow I(E)$) the canonical free (resp. injective) resolution, cf. X, p. 50 (resp. p. 52).

DEFINITION 1. — Let M and N be two A -modules. One calls the extension module of N by M the graded k -module

$$\text{Ext}_A(\mathbf{M}, \mathbf{N}) = H(\text{Homgr}_A(L(\mathbf{M}), I(\mathbf{N}))). \quad (4)$$

The homogeneous components of $\text{Ext}_A(M, N)$ are denoted

$$\mathrm{Ext}_{\mathbf{A}}^n(\mathbf{M}, \mathbf{N}) = \mathrm{H}^n(\mathrm{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{M}), \mathbf{I}(\mathbf{N}))). \quad (5)$$

Since $\mathbf{L}(\mathbf{M})$ (resp. $\mathbf{I}(\mathbf{N})$) is zero on the right (resp. left), we have

$$\mathrm{Ext}_{\mathbf{A}}^n(\mathbf{M}, \mathbf{N}) = 0 \quad \text{for } n < 0. \quad (6)$$

Remark 1. — We shall see below (X, p. 107, prop. 6) finiteness properties of the modules $\mathrm{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N})$. For example, if $\mathbf{A}$ is commutative Noetherian, and if $\mathbf{M}$ and $\mathbf{N}$ are finitely generated $\mathbf{A}$ -modules, each $\mathbf{A}$ -module $\mathrm{Ext}_{\mathbf{A}}^n(\mathbf{M}, \mathbf{N})$ is finitely generated.

Let $f : \mathbf{M}' \rightarrow \mathbf{M}$ and $g : \mathbf{N} \rightarrow \mathbf{N}'$ homomorphisms of $\mathbf{A}$ -modules, we set

$$\mathrm{Ext}_{\mathbf{A}}(f, g) = \mathrm{H}(\mathrm{Homgr}_{\mathbf{A}}(\mathbf{L}(f), \mathbf{I}(g)));$$

it is a degree 0 homomorphism of graded k -modules

$$\mathrm{Ext}_{\mathbf{A}}(f, g) : \mathrm{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}) \rightarrow \mathrm{Ext}_{\mathbf{A}}(\mathbf{M}', \mathbf{N}'),$$

whose homogeneous components are denoted

$$\mathrm{Ext}_{\mathbf{A}}^n(f, g) : \mathrm{Ext}_{\mathbf{A}}^n(\mathbf{M}, \mathbf{N}) \rightarrow \mathrm{Ext}_{\mathbf{A}}^n(\mathbf{M}', \mathbf{N}').$$

By prop. 1 of X, p. 82, the canonical homomorphism

$$\lambda^0(\mathbf{L}(\mathbf{M}), \mathbf{I}(\mathbf{N})) : \mathrm{H}^0(\mathrm{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{M}), \mathbf{I}(\mathbf{N}))) \rightarrow \mathrm{Hom}_{\mathbf{A}}(\mathrm{H}_0(\mathbf{L}(\mathbf{M})), \mathrm{H}^0(\mathbf{I}(\mathbf{N})))$$

is bijective; using the isomorphisms $\mathbf{M} \rightarrow \mathrm{H}_0(\mathbf{L}(\mathbf{M}))$ and $\mathrm{H}^0(\mathbf{I}(\mathbf{N})) \rightarrow \mathbf{N}$, we deduce an isomorphism said to be canonical

$$\lambda_{\mathbf{M}, \mathbf{N}} : \mathrm{Ext}_{\mathbf{A}}^0(\mathbf{M}, \mathbf{N}) \rightarrow \mathrm{Hom}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}). \quad (7)$$

One will always identify $\mathrm{Ext}_{\mathbf{A}}^0(\mathbf{M}, \mathbf{N})$ with $\mathrm{Hom}_{\mathbf{A}}(\mathbf{M}, \mathbf{N})$ by this isomorphism. Then the

map k -linear $\text{Ext}_{\mathbf{A}}^0(f, g)$ is identified with $\text{Hom}_{\mathbf{A}}(f, g)$.

Remark 2. — The morphism of complexes

$$\text{Homgr}_{\mathbf{A}}(p_{\mathbf{M}}, e_{\mathbf{N}}) : \text{Hom}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}) \rightarrow \text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{M}), \mathbf{I}(\mathbf{N}))$$

induces on the homology of degree 0 the isomorphism

$$\lambda_{\mathbf{M}, \mathbf{N}}^{-1} : \text{Hom}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}) \rightarrow \text{Ext}_{\mathbf{A}}^0(\mathbf{M}, \mathbf{N})$$

inverse of $\lambda_{\mathbf{M}, \mathbf{N}}$.

One has $L(1_{\mathbf{M}}) = 1_{L(\mathbf{M})}$, $I(1_{\mathbf{N}}) = 1_{I(\mathbf{N})}$, hence by passing to homology

$$\text{Ext}_{\mathbf{A}}(1_{\mathbf{M}}, 1_{\mathbf{N}}) = 1_{\text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N})}. \quad (8)$$

If $f' : M'' \rightarrow M'$ and $g' : N' \rightarrow N''$ are homomorphisms of $\mathbf{A}$ -modules, we have $L(f \circ f') = L(f) \circ L(f')$ and $I(g' \circ g) = I(g') \circ I(g)$, hence

$$\text{Ext}_{\mathbf{A}}(f \circ f', g' \circ g) = \text{Ext}_{\mathbf{A}}(f', g') \circ \text{Ext}_{\mathbf{A}}(f, g). \quad (9)$$

Consider the morphisms of k -complexes

$$\text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{M}), \mathbf{N}) \xrightarrow{\text{Homgr}_{\mathbf{A}}(1, e_{\mathbf{N}})} \text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{M}), \mathbf{I}(\mathbf{N})) \xleftarrow{\text{Homgr}_{\mathbf{A}}(p_{\mathbf{M}}, 1)} \text{Homgr}_{\mathbf{A}}(\mathbf{M}, \mathbf{I}(\mathbf{N})),$$

and the homomorphisms they induce in homology

$$\text{H}(\text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{M}), \mathbf{N})) \xrightarrow{\varphi_{\mathbf{M}}(\mathbf{N})} \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}) \xleftarrow{\overline{\varphi}_{\mathbf{N}}(\mathbf{M})} \text{H}(\text{Homgr}_{\mathbf{A}}(\mathbf{M}, \mathbf{I}(\mathbf{N}))).$$

By prop. 4 of X, p. 86, $\text{Homgr}_{\mathbf{A}}(1, e_{\mathbf{N}})$ and $\text{Homgr}_{\mathbf{A}}(p_{\mathbf{M}}, 1)$ are homomorphisms, hence:

PROPOSITION 5. — The k -homomorphisms

$$\varphi_{\mathbf{M}}(\mathbf{N}) : \text{H}(\text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{M}), \mathbf{N})) \rightarrow \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N})$$

$$\text{and } \overline{\varphi}_{\mathbf{N}}(\mathbf{M}) : \text{H}(\text{Homgr}_{\mathbf{A}}(\mathbf{M}, \mathbf{I}(\mathbf{N}))) \rightarrow \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}) \text{ are bijective.}$$

COROLLARY. — If M is projective (resp. if N is injective), then $\text{Ext}_A^i(M, N) = 0$ for $i > 0$.

Indeed,

$$\text{Homgr}_A(1, e_N) : \text{Hom}_A(M, N) \rightarrow \text{Homgr}_A(M, I(N))$$

(resp. $\text{Homgr}_A(p_M, 1) : \text{Hom}_A(M, N) \rightarrow \text{Homgr}_A(L(M), N)$) is then a homomorphism (X, p. 86, prop. 4), whence the conclusion.

Remarks. — 3) If $g : N \rightarrow N'$ is a homomorphism of A -modules, then

$$\text{Homgr}_A(1_{L(M)}, g) \circ \text{Homgr}_A(1_{L(M)}, e_N) = \text{Homgr}_A(1_{L(M)}, e_{N'}) \circ \text{Homgr}_A(1_{L(M)}, I(g))$$

therefore the diagram

$$\begin{array}{ccc} & & H(\text{Homgr}_A(L(M), N)) \xrightarrow{\varphi_M(N)} \text{Ext}_A(M, N) \\ H(\text{Homgr}_A(1_{L(M)}, g)) \Big\downarrow & & \\ & & H(\text{Homgr}_A(L(M), N')) \xrightarrow{\varphi_M(N')} \text{Ext}_A(M, N') \end{array}$$

is commutative.

4) Likewise, if $f : M' \rightarrow M$ is a homomorphism of A -modules, the diagram

$$\begin{array}{ccc} H(\text{Homgr}_A(M, I(N))) & \xrightarrow{\bar{\varphi}_N(M)} & \text{Ext}_A(M, N) \\ H(\text{Homgr}_A(f, 1_{I(N)})) \Big\downarrow & & \Big\downarrow \text{Ext}_A(f, 1_N) \\ H(\text{Homgr}_A(M', I(N))) & \xrightarrow{\bar{\varphi}_N(M')} & \text{Ext}_A(M', N) \end{array}$$

is commutative.

PROPOSITION 6. — The map $(f, g) \mapsto \text{Ext}_A(f, g) :$

$$\text{Hom}_A(M', M) \times \text{Hom}_A(N, N') \rightarrow \text{Homgr}_k(\text{Ext}_A(M, N), \text{Ext}_A(M', N'))$$

is k -bilinear.

Let $f \in \text{Hom}_{\mathbf{A}}(\mathbf{M}', \mathbf{M})$, $g_1, g_2 \in \text{Hom}_{\mathbf{A}}(\mathbf{N}, \mathbf{N}')$, $\lambda_1, \lambda_2 \in k$; then the morphisms $\text{Homgr}_{\mathbf{A}}(\mathbf{L}(f), \lambda_1 g_1 + \lambda_2 g_2)$ and $\lambda_1 \text{Homgr}_{\mathbf{A}}(\mathbf{L}(f), g_1) + \lambda_2 \text{Homgr}_{\mathbf{A}}(\mathbf{L}(f), g_2)$ of $\text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{M}), \mathbf{N})$ to $\text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{M}), \mathbf{N}')$ coincide; therefore, by prop. 5 and remark 3,

$$\text{Ext}_{\mathbf{A}}(f, \lambda_1 g_1 + \lambda_2 g_2) = \lambda_1 \text{Ext}_{\mathbf{A}}(f, g_1) + \lambda_2 \text{Ext}_{\mathbf{A}}(f, g_2). \quad (10)$$

We reason similarly for the mapping $f \mapsto \text{Ext}_{\mathbf{A}}(f, g)$.

COROLLARY. — Let $\lambda \in k$. If λ annihilates $\mathbf{M}$ or $\mathbf{N}$, it annihilates $\text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N})$.

Indeed, $\lambda 1_{\text{Ext}(\mathbf{M}, \mathbf{N})} = \text{Ext}(\lambda 1_{\mathbf{M}}, 1_{\mathbf{N}}) = \text{Ext}(1_{\mathbf{M}}, \lambda 1_{\mathbf{N}})$.

PROPOSITION 7. — Let I and J be two sets, $(\mathbf{M}_{\alpha})_{\alpha \in I}$ and $(\mathbf{N}_{\beta})_{\beta \in J}$ two families of $\mathbf{A}$ -modules; the homomorphism $\text{Ext}_{\mathbf{A}}(\bigoplus_{\alpha \in I} \mathbf{M}_{\alpha}, \prod_{\beta \in J} \mathbf{N}_{\beta}) \rightarrow \prod_{\beta \in J, \alpha \in I} \text{Ext}_{\mathbf{A}}(\mathbf{M}_{\alpha}, \mathbf{N}_{\beta})$ deduced from the canonical homomorphisms $\mathbf{M}_{\alpha} \rightarrow \bigoplus \mathbf{M}_{\alpha}$ and $\prod \mathbf{N}_{\beta} \rightarrow \mathbf{N}_{\beta}$ is bijective.

It suffices to prove that for every $\mathbf{A}$ -module $\mathbf{M}$ (resp. $\mathbf{N}$), the homomorphisms

$$\text{Ext}(\mathbf{M}, \prod \mathbf{N}_{\beta}) \rightarrow \prod \text{Ext}(\mathbf{M}, \mathbf{N}_{\beta}) \quad (\text{resp. } \text{Ext}(\bigoplus \mathbf{M}_{\alpha}, \mathbf{N}) \rightarrow \prod \text{Ext}(\mathbf{M}_{\alpha}, \mathbf{N}))$$

are bijective. Now this follows from what precedes, from prop. 1 of X, p. 28, and from the canonical isomorphisms $\text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{M}), \prod \mathbf{N}_{\beta}) \rightarrow \prod \text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{M}), \mathbf{N}_{\beta})$ and $\text{Homgr}_{\mathbf{A}}(\bigoplus \mathbf{M}_{\alpha}, \mathbf{I}(\mathbf{N})) \rightarrow \prod \text{Homgr}_{\mathbf{A}}(\mathbf{M}_{\alpha}, \mathbf{I}(\mathbf{N}))$.

Remark 5. — Let $\mathbf{P}$ and $\mathbf{Q}$ be two right $\mathbf{A}$ -modules. One defines $\text{Ext}_{\mathbf{A}}(\mathbf{P}, \mathbf{Q})$ by

$$\begin{aligned} \text{Ext}_{\mathbf{A}}(\mathbf{P}, \mathbf{Q}) &= \text{H}(\text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{P}), \mathbf{I}(\mathbf{Q}))) = \text{H}(\text{Homgr}_{\mathbf{A}^{\circ}}(\mathbf{L}(\mathbf{P}^{\circ}), \mathbf{I}(\mathbf{Q}^{\circ}))) = \\ &= \text{Ext}_{\mathbf{A}^{\circ}}(\mathbf{P}^{\circ}, \mathbf{Q}^{\circ}). \end{aligned}$$

All definitions and propositions in this paragraph therefore apply to right $\mathbf{A}$ -modules by viewing them as left modules over the ring $\mathbf{A}^{\circ}$.

10.5.4 Connecting homomorphisms and exact sequences

Let $\mathbf{M}$ be an $\mathbf{A}$ -module. Recall that for every $\mathbf{A}$ -module $\mathbf{N}$, we defined in the preceding number an isomorphism of k -modules

$$\varphi_{\mathbf{M}}(\mathbf{N}) : H(\text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{M}), \mathbf{N})) \rightarrow \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}).$$

Let

$$0 \rightarrow \mathbf{N}' \xrightarrow{u} \mathbf{N} \xrightarrow{v} \mathbf{N}'' \rightarrow 0 \quad ((\mathcal{E}))$$

an exact sequence of $\mathbf{A}$ -modules; the sequence of k -complexes

$$\begin{aligned} (\mathbf{M}^{\mathcal{E}}) \quad 0 \rightarrow \text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{M}), \mathbf{N}') &\xrightarrow{\text{Homgr}(1, u)} \text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{M}), \mathbf{N}) \\ &\xrightarrow{\text{Homgr}(1, v)} \text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{M}), \mathbf{N}'') \rightarrow 0 \end{aligned}$$

is then exact (X, p. 83, prop. 2, a)), i.e.

$$\partial_{(\mathbf{M}^{\mathcal{E}})} : H(\text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{M}), \mathbf{N}'')) \rightarrow H(\text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{M}), \mathbf{N}'))$$

the corresponding connecting homomorphism (X, p. 29).

DEFINITION 2. — A connecting homomorphism of extension modules relative to the module $\mathbf{M}$ and the exact sequence $(\mathcal{E})$ is called the composite homomorphism

$$\delta(\mathbf{M}, \mathcal{E}) = \varphi_{\mathbf{M}}(\mathbf{N}') \circ \partial_{(\mathbf{M}^{\mathcal{E}})} \circ \varphi_{\mathbf{M}}(\mathbf{N}'')^{-1} : \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}'') \rightarrow \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}').$$

This is a k -graded homomorphism of ascending degree 1, whose homogeneous components are denoted $\delta^n(\mathbf{M}, \mathcal{E}) : \text{Ext}_{\mathbf{A}}^n(\mathbf{M}, \mathbf{N}'') \rightarrow \text{Ext}_{\mathbf{A}}^{n+1}(\mathbf{M}, \mathbf{N}')$.

THEOREM 1. — The right-unbounded sequence of homomorphisms of k -modules

$$\begin{aligned} 0 \longrightarrow \text{Hom}_{\mathbf{A}}(M, N') &\xrightarrow{\text{Hom}(1, u)} \text{Hom}_{\mathbf{A}}(M, N) \xrightarrow{\text{Hom}(1, v)} \text{Hom}_{\mathbf{A}}(M, N'') \\ \xrightarrow{\delta(M, \mathcal{E})} \text{Ext}_{\mathbf{A}}^1(M, N') &\rightarrow \dots \xrightarrow{\delta^{n-1}(M, \mathcal{E})} \text{Ext}_{\mathbf{A}}^n(M, N') \xrightarrow{\text{Ext}^n(1, u)} \text{Ext}_{\mathbf{A}}^n(M, N) \\ \xrightarrow{\text{Ext}^n(1, v)} \text{Ext}_{\mathbf{A}}^n(M, N'') &\xrightarrow{\delta^n(M, \mathcal{E})} \text{Ext}_{\mathbf{A}}^{n+1}(M, N') \rightarrow \dots \end{aligned}$$

is exact.

Consider indeed the diagram on page X. 91.

It is commutative by Remark 3 of X, p. 88 and Def. 2; moreover, the bottom row is exact

(X, p. 30, Th. 1), and the vertical arrows are bijective (X, p. 88, prop. 5).

COROLLARY. — If $\text{Ext}_A^1(M, N') = 0$, the sequence

$$0 \rightarrow \text{Hom}_A(M, N') \xrightarrow{\text{Hom}(1, u)} \text{Hom}_A(M, N) \xrightarrow{\text{Hom}(1, v)} \text{Hom}_A(M, N'') \rightarrow 0$$

is exact.

PROPOSITION 8. — Let $f : \mathbf{M}_1 \rightarrow \mathbf{M}$ be a homomorphism of A-modules and

$$0 \rightarrow N' \xrightarrow{u} N \xrightarrow{v} N'' \rightarrow 0 \quad ((\mathcal{E}))$$

$$\begin{array}{ccc} g' \downarrow & g \downarrow & g'' \downarrow \end{array}$$

$$0 \rightarrow N'_1 \xrightarrow{u_1} N_1 \xrightarrow{v_1} N''_1 \rightarrow 0 \quad ((\mathcal{E}_1))$$

$$\begin{array}{ccccccc}
\text{Ext}(\mathbf{M}, \mathbf{N}') & \xrightarrow{\text{Ext}(\mathbf{L}, \mathbf{v})} & \text{Ext}(\mathbf{M}, \mathbf{N}) & \xrightarrow{\text{Ext}(\mathbf{L}, \mathbf{v})} & \text{Ext}(\mathbf{M}, \mathbf{N}'') & \xrightarrow{\delta(\mathbf{M}, \mathcal{E}')} & \text{Ext}(\mathbf{M}, \mathbf{N}') \\
\uparrow \phi_{\mathbf{M}}(\mathbf{N}') & & \uparrow \phi_{\mathbf{M}}(\mathbf{N}) & & \uparrow \phi_{\mathbf{M}}(\mathbf{N}'') & & \uparrow \phi_{\mathbf{M}}(\mathbf{N}') \\
\mathbf{H}(\text{Homgr}(\mathbf{L}(\mathbf{M}), \mathbf{N}')) & \xrightarrow{\mathbf{H}(\text{Homgr}(\mathbf{L}, \mathbf{v}))} & \mathbf{H}(\text{Homgr}(\mathbf{L}(\mathbf{M}), \mathbf{N})) & \xrightarrow{\mathbf{H}(\text{Homgr}(\mathbf{L}, \mathbf{v}))} & \mathbf{H}(\text{Homgr}(\mathbf{L}(\mathbf{M}), \mathbf{N}'')) & \xrightarrow{\delta(\mathbf{M}, \mathcal{E}')} & \mathbf{H}(\text{Homgr}(\mathbf{L}(\mathbf{M}), \mathbf{N}')) \\
& & & & & & \uparrow \mathbf{H}(\text{Homgr}(\mathbf{L}, \mathbf{v})) \\
& & & & & & \mathbf{H}(\text{Homgr}(\mathbf{L}(\mathbf{M}), \mathbf{N}'))
\end{array}$$

a commutative diagram of $\mathbf{A}$ -modules with exact rows. The diagram of $\mathbf{k}$ -modules

$$\begin{array}{ccc}
\text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}'') & \xrightarrow{\delta(\mathbf{M}, \mathcal{E})} & \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}') \\
\text{Ext}(f, g'') \Big|_{\downarrow} & & \\
\text{Ext}_{\mathbf{A}}(\mathbf{M}_1, \mathbf{N}_1'') & \xrightarrow{\delta(\mathbf{M}, \mathcal{E}_1)} & \text{Ext}_{\mathbf{A}}(\mathbf{M}_1, \mathbf{N}_1')
\end{array}$$

is commutative.

This follows from X, p. 31, prop. 2 applied to the commutative diagram.

$$\begin{array}{ccccccc}
0 & \xrightarrow{\text{Homgr}_A(L(M), N')} & \xrightarrow{\text{Homgr}_A(L(M), N)} & \xrightarrow{\text{Homgr}_A(L(M), N'')} & 0 \\
\text{Homgr}(L(f), g') \Big|_{\downarrow} & \text{Homgr}(1, u') & \Big|_{\downarrow} & \text{Homgr}(L(f), g) & \Big|_{\downarrow} & \text{Homgr}(L(f), g'') \\
0 & \xrightarrow{\text{Homgr}_A(L(M_1), N'_1)} & \xrightarrow{\text{Homgr}_A(L(M_1), N_1)} & \xrightarrow{\text{Homgr}_A(L(M_1), N''_1)} & 0
\end{array}$$

Let N be an A-module, and

$$0 \rightarrow \mathbf{M}' \xrightarrow{r} \mathbf{M} \xrightarrow{s} \mathbf{M}'' \rightarrow 0 \quad ((\mathcal{F}))$$

an exact sequence of A-modules; the sequence of complexes

$$\begin{array}{c}
(\mathcal{F}_N) \quad 0 \rightarrow \text{Homgr}_A(\mathbf{M}'', \mathbf{I}(\mathbf{N})) \xrightarrow{\text{Homgr}(s, 1)} \text{Homgr}_A(\mathbf{M}, \mathbf{I}(\mathbf{N})) \\
\quad \quad \quad \xrightarrow{\text{Homgr}(r, 1)} \text{Homgr}_A(\mathbf{M}', \mathbf{I}(\mathbf{N})) \rightarrow 0
\end{array}$$

is exact (X, p. 83, prop. 2, a)); let

$$\partial(\mathcal{F}_N) : H(\text{Homgr}_A(\mathbf{M}', \mathbf{I}(\mathbf{N}))) \rightarrow H(\text{Homgr}_A(\mathbf{M}'', \mathbf{I}(\mathbf{N})))$$

the corresponding connecting homomorphism.

DEFINITION 3. — One calls connecting homomorphism of extension modules relative to the exact sequence $(\mathcal{F})$ and to the module N, the composite homomorphism

$$\delta(\mathcal{F}, N) : \overline{\varphi}_N(\mathbf{M}'') \circ \partial(\mathcal{F}_N) \circ \overline{\varphi}_N(\mathbf{M}')^{-1} : \text{Ext}_A(\mathbf{M}', N) \rightarrow \text{Ext}_A(\mathbf{M}'', N).$$

This is a k -graded homomorphism of ascending degree 1, whose homogeneous components are denoted $\delta^n(\mathcal{F}, N) : \text{Ext}_A^n(\mathbf{M}', N) \rightarrow \text{Ext}_A^{n+1}(\mathbf{M}'', N)$.

We then prove, as above, the following statements:

THEOREM 2. — The right-unbounded sequence of homomorphisms of k -modules

$$\begin{array}{ccccccc}
0 & \longrightarrow & \text{Hom}_A(M'', N) & \xrightarrow{\text{Hom}(s, 1)} & \text{Hom}_A(M, N) & \xrightarrow{\text{Hom}(r, 1)} & \text{Hom}_A(M', N) \\
& & \xrightarrow{\delta^\circ(\mathcal{F}, N)} & \text{Ext}_A^1(M'', N) & \rightarrow \dots & \xrightarrow{\delta^{n-1}(\mathcal{F}, N)} & \text{Ext}_A^n(M'', N) \xrightarrow{\text{Ext}^n(s, 1)} \text{Ext}_A^n(M, N) \\
& & \xrightarrow{\text{Ext}^n(r, 1)} & \text{Ext}_A^n(M', N) & \xrightarrow{\delta^n(\mathcal{F}, N)} & \text{Ext}_A^{n+1}(M'', N) & \rightarrow \dots
\end{array}$$

is exact.

COROLLARY. — If $\text{Ext}_A^1(M'', N) = 0$, the sequence

$$0 \longrightarrow \text{Hom}_A(M'', N) \xrightarrow{\text{Hom}(s, 1)} \text{Hom}_A(M, N) \xrightarrow{\text{Hom}(r, 1)} \text{Hom}_A(M', N) \longrightarrow 0$$

is exact.

PROPOSITION 9. — Let $g : N \rightarrow N_1$ be a homomorphism of A-modules and

$$(\mathcal{F}_1)$$

$$0 \longrightarrow \mathbf{M}'_1 \xrightarrow{r_1} \mathbf{M}_1 \xrightarrow{s_1} \mathbf{M}''_1 \longrightarrow 0$$

$$f' \downarrow \quad f \downarrow \quad f'' \downarrow$$

$$0 \longrightarrow \mathbf{M}' \xrightarrow{r} \mathbf{M} \xrightarrow{s} \mathbf{M}'' \longrightarrow 0 \quad ((\mathcal{F}))$$

a commutative diagram of A-modules with exact rows. The diagram of k-modules

$$\begin{array}{ccc}
\text{Ext}_A(\mathbf{M}', N) & \xrightarrow{\delta(\mathcal{F}, N)} & \text{Ext}_A(\mathbf{M}'', N) \\
& \text{Ext}_A(f', g) \downarrow & \\
\text{Ext}_A(\mathbf{M}'_1, N_1) & \xrightarrow{\delta(\mathcal{F}_1, N_1)} & \text{Ext}_A(\mathbf{M}''_1, N_1)
\end{array}$$

is commutative.

10.5.5 Projective modules, injective modules, and extension modules

PROPOSITION 10. — Let M be an A -module. The following conditions are equivalent:

- (i) M is projective.
- (ii) $\text{Ext}_A^i(M, N) = 0$ for every A -module N and for every integer $i > 0$.
- (iii) $\text{Ext}_A^1(M, N) = 0$ for every A -module N .
- (iv) There exists an exact sequence

$$0 \rightarrow K \xrightarrow{u} P \xrightarrow{r} M \rightarrow 0,$$

where P is projective, and where $\text{Ext}_A^1(M, K) = 0$.

- (i) $\Rightarrow$ (ii): this is the corollary of prop. 5 of X, p. 88.
- (ii) $\Rightarrow$ (iii): this is trivial.
- (iii) $\Rightarrow$ (iv): this is clear since M is a quotient of a free module P .
- (iv) $\Rightarrow$ (i): since $\text{Ext}_A^1(M, K) = 0$ the canonical map

$$\text{Hom}_A(M, P) \rightarrow \text{Hom}_A(M, M)$$

is surjective (X, p. 90, corollary); there therefore exists an A -linear section of v and M is isomorphic to a direct factor of P , hence is projective.

PROPOSITION 11. — Let N be an A -module. The following conditions are equivalent:

- (i) N is injective.
- (ii) $\text{Ext}_A^i(M, N) = 0$ for every A -module M and every integer $i > 0$.
- (iii) $\text{Ext}_A^1(M, N) = 0$ for every A -module M .
- (iv) There exists an exact sequence

$$0 \rightarrow N \xrightarrow{u} I \xrightarrow{v} C \rightarrow 0,$$

where I is injective and where $\text{Ext}_{\mathbf{A}}^1(C, N) = 0$;

(v) $\text{Ext}_{\mathbf{A}}^1(\mathbf{M}, N) = 0$ for every monogenic A -module $\mathbf{M}$.

(w) $\Rightarrow$ (ii): this is the corollary of prop. 5 of X, p. 88.

(ii) $\Rightarrow$ (iii) $\Rightarrow$ (v): this is trivial.

(iii) $\Rightarrow$ (iv): this is clear since N is a submodule of an injective module (X, p. 19, cor. 3).

(iv) $\Rightarrow$ (i): since $\text{Ext}_{\mathbf{A}}^1(C, N) = 0$, the canonical homomorphism

$$\text{Hom}_{\mathbf{A}}(I, N) \rightarrow \text{Hom}_{\mathbf{A}}(N, N)$$

is surjective (X, p. 93, corollary); there therefore exists an A -linear retraction of u and N is isomorphic to a direct factor of I , hence is injective (X, p. 16, prop. 9).

(v) $\Rightarrow$ (i): if $\mathfrak{a}$ is an ideal of A , one has $\text{Ext}_{\mathbf{A}}^1(A/\mathfrak{a}, N) = 0$; the canonical map $\text{Hom}_{\mathbf{A}}(A, N) \rightarrow \text{Hom}_{\mathbf{A}}(\mathfrak{a}, N)$ is therefore surjective and N is injective (X, p. 16, prop. 10).

10.5.6 Universal coefficient formula

In this section, we consider two complexes of A -modules (C, d) and (C', d') . Consider the canonical exact sequences:

$$0 \rightarrow Z(C) \xrightarrow{j} C \xrightarrow{\delta} B(C)(-1) \rightarrow 0, \quad (\text{I})$$

(II_p)

$$0 \rightarrow B_p(C) \xrightarrow{i} Z_p(C) \xrightarrow{p} H_p(C) \rightarrow 0;$$

we deduce from δ a k -homomorphism

$$H(\text{Homgr}(\delta, 1)) : H(\text{Homgr}_{\mathbf{A}}(B(C), C'))(1) \rightarrow H(\text{Homgr}_{\mathbf{A}}(C, C'));$$

one deduces from (Π_p) connecting homomorphisms:

$$\delta(\Pi_p, H^q(C')) : \text{Hom}_{\mathbf{A}}(B_p(C), H^q(C')) \rightarrow \text{Ext}_{\mathbf{A}}^1(H_p(C), H^q(C')).$$

whence, by passing to the product, homomorphisms of k -modules

$$\varphi^n : \text{Homgr}_{\mathbf{A}}^n(B(C), H(C')) \rightarrow \prod_{p+q=n} \text{Ext}_{\mathbf{A}}^1(H_p(C), H^q(C'))$$

One also has canonical homomorphisms (X, p. 82)

$$\lambda^n(B(C), C') : H^n(\text{Homgr}_{\mathbf{A}}(B(C), C')) \rightarrow \text{Homgr}_{\mathbf{A}}^n(B(C), H(C')).$$

With these notations:

THEOREM 3. — Suppose the $\mathbf{A}$ -modules $B(C)$ and $Z(C)$ are projective. There exists, for each n , a unique homomorphism of k -modules

$$\beta^n : \prod_{p+q=n-1} \text{Ext}_{\mathbf{A}}^1(H_p(C), H^q(C')) \rightarrow H^n(\text{Homgr}_{\mathbf{A}}(C, C'))$$

thereby making the diagram commutative

$$\begin{aligned} \text{Homgr}_{\mathbf{A}}^{n-1}(B(C), H(C')) &\xleftarrow{\lambda^{n-1}(B(C), C')} H^{n-1}(\text{Homgr}_{\mathbf{A}}(B(C), C')) \\ \prod_{p+q=n-1} \text{Ext}_{\mathbf{A}}^1(H_p(C), H^q(C')) &\xrightarrow{\beta^n} H^n(\text{Homgr}_{\mathbf{A}}(C, C')). \end{aligned}$$

The sequences of graded k -modules

$$\begin{aligned} 0 \rightarrow \prod_{p+q=n-1} \text{Ext}_{\mathbf{A}}^1(H_p(C), H^q(C')) &\xrightarrow{\beta^n} H^n(\text{Homgr}_{\mathbf{A}}(C, C')) \\ &\xrightarrow{\lambda^n(C, C')} \prod_{p+q=n} \text{Hom}_{\mathbf{A}}(H_p(C), H^q(C')) \rightarrow 0 \end{aligned} \quad (11)$$

are exact.

Remark. — One can prove an analogous statement, assuming $\mathbf{B}_n(C')$ and $C'/\mathbf{B}_n(C')$ injective for each n .

For simplicity set $\mathbf{B} = \mathbf{B}(\mathbf{C})$, $\mathbf{Z} = \mathbf{Z}(\mathbf{C})$, $\mathbf{H} = \mathbf{H}(\mathbf{C})$ and $\mathbf{H}' = \mathbf{H}(\mathbf{C}')$. Since $\mathbf{B}$ is projective, from (I) one derives an exact sequence

$$(12) \quad 0 \rightarrow \text{Homgr}_{\mathbf{A}}(\mathbf{B}, \mathbf{C}') \xrightarrow{\text{Homgr}(\delta, 1)} \text{Homgr}_{\mathbf{A}}(\mathbf{C}, \mathbf{C}') \xrightarrow{\text{Homgr}(j, 1)} \text{Homgr}_{\mathbf{A}}(\mathbf{Z}, \mathbf{C}') \rightarrow 0.$$

Lemma 2. — The connecting homomorphism

$$H^n(\text{Homgr}_{\mathbf{A}}(\mathbf{Z}, \mathbf{C}')) \rightarrow H^n(\text{Homgr}_{\mathbf{A}}(\mathbf{B}, \mathbf{C}'))$$

associated with (12) is equal to $(-1)^{n+1} H(\text{Homgr}(i, 1))$.

Indeed, let $a \in Z^n(\text{Homgr}_{\mathbf{A}}(\mathbf{Z}, \mathbf{C}'))$; this is a morphism of complexes of ascending degree n of $\mathbf{Z}$ to $\mathbf{C}'$, whose values therefore lie in $\mathbf{Z}(\mathbf{C}')$. Since the exact sequence (I) is split ($\mathbf{B}$ being projective), a extends to an element b of $\text{Homgr}_{\mathbf{A}}^n(\mathbf{C}, \mathbf{Z}(\mathbf{C}'))$. By definition, the image of the class of a under the connecting homomorphism in question is the class in $H^n(\text{Homgr}_{\mathbf{A}}(\mathbf{B}, \mathbf{C}'))$ of the homomorphism u of $\mathbf{B}(\mathbf{C})$ to $\mathbf{C}'$ such that for $x \in \mathbf{C}$, one has

$$u(dx) = Db(x) = d'b(x) - (-1)^n b(dx) = (-1)^{n+1} b(dx) = (-1)^{n+1} a(dx),$$

whence the assertion.

The exact homology sequence associated with (12) therefore gives the exact sequence

$$\rightarrow H^n(\text{Homgr}_{\mathbf{A}}(\mathbf{Z}, \mathbf{C}')) \xrightarrow{H(\text{Homgr}(i, 1))} H^n(\text{Homgr}_{\mathbf{A}}(\mathbf{B}, \mathbf{C}'))$$

$$\xrightarrow{H(\text{Homgr}(\delta, 1))} H^{n+1}(\text{Homgr}_{\mathbf{A}}(\mathbf{C}, \mathbf{C}')) \xrightarrow{H(\text{Homgr}(j, 1))} H^{n+1}(\text{Homgr}_{\mathbf{A}}(\mathbf{Z}, \mathbf{C}')) \rightarrow \dots$$

Moreover, since $\mathbf{Z}$ is projective, one derives from (II_p) exact sequences

$$0 \rightarrow \text{Hom}_{\mathbf{A}}(\mathbf{H}_p, \mathbf{H}'^q) \rightarrow \text{Hom}_{\mathbf{A}}(\mathbf{Z}_p, \mathbf{H}'^q) \rightarrow \text{Hom}_{\mathbf{A}}(\mathbf{B}_p, \mathbf{H}'^q) \rightarrow \text{Ext}_{\mathbf{A}}^1(\mathbf{H}_p, \mathbf{H}'^q) \rightarrow 0,$$

whence, by passing to products, exact sequences

$$\begin{aligned} 0 \rightarrow \text{Homgr}_{\mathbf{A}}^n(\mathbf{H}, \mathbf{H}') &\rightarrow \text{Homgr}_{\mathbf{A}}^n(\mathbf{Z}, \mathbf{H}') \rightarrow \text{Homgr}_{\mathbf{A}}^n(\mathbf{B}, \mathbf{H}') \\ &\rightarrow \prod_{p+q=n} \text{Ext}_{\mathbf{A}}^1(\mathbf{H}_p, \mathbf{H}'^q) \rightarrow 0. \end{aligned}$$

Finally, one has the canonical homomorphisms of no. 1:

$$\begin{aligned} \lambda_{\mathbf{B}} &= \lambda(\mathbf{B}, \mathbf{C}') : \mathbf{H}(\text{Homgr}_{\mathbf{A}}(\mathbf{B}, \mathbf{C}')) \rightarrow \text{Homgr}_{\mathbf{A}}(\mathbf{B}, \mathbf{H}'), \\ \lambda_{\mathbf{Z}} &= \lambda(\mathbf{Z}, \mathbf{C}') : \mathbf{H}(\text{Homgr}_{\mathbf{A}}(\mathbf{Z}, \mathbf{C}')) \rightarrow \text{Homgr}_{\mathbf{A}}(\mathbf{Z}, \mathbf{H}'), \\ \lambda_{\mathbf{C}} &= \lambda(\mathbf{C}, \mathbf{C}') : \mathbf{H}(\text{Homgr}_{\mathbf{A}}(\mathbf{C}, \mathbf{C}')) \rightarrow \text{Homgr}_{\mathbf{A}}(\mathbf{H}, \mathbf{H}'), \end{aligned}$$

whence the diagram with exact rows on page X. 97.

This diagram is commutative by construction of the homomorphisms λ . Moreover, since the complexes $\mathbf{B}$ and $\mathbf{Z}$ are split and projective, $\lambda_{\mathbf{B}}$ and $\lambda_{\mathbf{Z}}$ are bijective (X, p. 84, cor. 1). From this one deduces, on the one hand, that $\lambda_{\mathbf{C}}^n$ is surjective, with kernel equal to $\text{Im } \mathbf{H}^n(\text{Homgr}(\delta, 1))$, and on the other hand, that $\varphi^{n-1} \circ \lambda_{\mathbf{B}}^{n-1}$ is surjective, with kernel equal to $\text{Ker } \mathbf{H}^n(\text{Homgr}(\delta, 1))$. The theorem follows immediately from this.

COROLLARY 1. — Suppose $\mathbf{B}(\mathbf{C})$ and $\mathbf{Z}(\mathbf{C})$ are projective and $\mathbf{B}^n(\mathbf{C}')$ is injective for each n . Then the exact sequences (11) are split.

This follows from the theorem and the following lemma:

Lemma 3. — If $\mathbf{B}(\mathbf{C})$ is projective and $\mathbf{B}_n(\mathbf{C}')$ injective for each n , the canonical homomorphism $\lambda(\mathbf{C}, \mathbf{C}') : \mathbf{H}(\text{Homgr}_{\mathbf{A}}(\mathbf{C}, \mathbf{C}')) \rightarrow \text{Homgr}_{\mathbf{A}}(\mathbf{H}(\mathbf{C}), \mathbf{H}(\mathbf{C}'))$ has a section k -linear.

Indeed, according to X, p. 35, remarks a) and b), there exist homomorphisms

$$\varphi : \mathbf{C} \rightarrow \mathbf{H}(\mathbf{C}) \quad \text{and} \quad \varphi' : \mathbf{H}(\mathbf{C}') \rightarrow \mathbf{C}'$$

such that $\mathbf{H}(\varphi) = 1_{\mathbf{H}(\mathbf{C})}$ and $\bar{\mathbf{H}}(\varphi') = 1_{\mathbf{H}(\mathbf{C}')}.$

In the commutative diagram

$$\begin{array}{ccc} \mathbf{H}(\text{Homgr}_{\mathbf{A}}(\mathbf{H}(\mathbf{C}), \mathbf{H}(\mathbf{C}')) & \xrightarrow{\mathbf{H}(\text{Homgr}(\varphi, \varphi'))} & \mathbf{H}(\text{Homgr}_{\mathbf{A}}(\mathbf{C}, \mathbf{C}')) \\ \lambda(\mathbf{H}(\mathbf{C}), \mathbf{H}(\mathbf{C}')) \Big\downarrow & & \\ \text{Homgr}_{\mathbf{A}}(\mathbf{H}(\mathbf{C}), \mathbf{H}(\mathbf{C}')) & \xrightarrow{\text{Homgr}(\mathbf{H}(\varphi), \mathbf{H}(\varphi'))} & \text{Homgr}_{\mathbf{A}}(\mathbf{H}(\mathbf{C}), \mathbf{H}(\mathbf{C}')), \end{array}$$

$\lambda(H(C), H(C'))$ is bijective and $\text{Homgr}(H(\varphi), H(\varphi'))$ is the identity, whence the assertion.
 $\text{Homgr}^{n-1}(Z, H') \rightarrow \text{Homgr}^{n-1}(B, H') \rightarrow \Phi^{n-1} \rightarrow \prod_{p+q=n-1} \text{Ext}^1(H_p, H'_q) \rightarrow 0$
 $H^{n-1}(\text{Homgr}(A, C')) \rightarrow H^{n-1}(\text{Homgr}(A, B, C')) \rightarrow H^n(\text{Homgr}(A, C)) \rightarrow H^n(\text{Homgr}(A, C)) \rightarrow H^n(\text{Homgr}(A, B, C))$
 $0 \rightarrow \text{Homgr}^n(H, H') \rightarrow \text{Homgr}^n(Z, H') \rightarrow \text{Homgr}^n(B, H')$

COROLLARY 2. — If $B(C)$ and $Z(C)$ are projective, then, for every A -module N and every integer n , there is a split exact sequence:

$$(13) \quad 0 \rightarrow \text{Ext}_A^1(H_{n-1}(C), N) \xrightarrow{\beta^n} H^n(\text{Homgr}_A(C, N)) \xrightarrow{\lambda^n} \text{Hom}_A(H_n(C), N) \rightarrow 0.$$

COROLLARY 3 (“universal coefficient formula”). — Suppose A is principal and C is free. Then for every A -module N and every integer n , there is a split exact sequence (13).

Indeed, $B(C)$ and $Z(C)$ are free as submodules of the free module C (VII, § 3, cor. 2 to th. 1).

COROLLARY 4. — If C is bounded on the right and if C and $H(C)$ are projective, then

$$\lambda(C, C') : H(\text{Homgr}_A(C, C')) \rightarrow \text{Homgr}_A(H(C), H(C'))$$

is bijective.

By the theorem, it suffices to prove that $B(C)$ and $Z(C)$ are projective. Now we have exact sequences

$$0 \rightarrow B_n(C) \rightarrow Z_n(C) \rightarrow H_n(C) \rightarrow 0$$

$$0 \rightarrow Z_n(C) \rightarrow C_n \rightarrow B_{n-1}(C) \rightarrow 0,$$

hence $(B_{n-1}(C) \text{ is projective}) \Rightarrow (Z_n(C) \text{ is projective}) \Rightarrow (B_n(C) \text{ is projective})$. We conclude by noting that $B_n(C) = 0$ for n small enough.

10.5.7 Generalization to complexes of multimodules; the canonical isomorphisms

Let B, B' be two rings, C a complex of (A, B) -bimodules, C' a complex of (A, B') -bimodules; then $(\text{Homgr}_A(C, C'), D)$ is a complex of (B, B') -bimodules, and the canonical homomorphism $\lambda : H(\text{Homgr}_A(C, C')) \rightarrow \text{Homgr}_A(H(C), H(C'))$ is a homomorphism of (B, B') -bimodules.

If M is an (A, B) -bimodule and N an (A, B') -bimodule, then $\text{Homgr}_A(L(M), I(N))$ is a complex of (B, B') -bimodules, so that $\text{Ext}_A(M, N)$ is endowed with a natural structure of graded (B, B') -bimodule; on the degree 0 term, this structure coincides with that of $\text{Hom}_A(M, N)$ (II, p. 35).

If $\lambda \in \mathbf{B}, \lambda' \in \mathbf{B}'$ and if we denote $\lambda_M, \lambda'_N, \lambda_E, \lambda'_E$ the homotheties $x \mapsto x\lambda, y \mapsto y\lambda', z \mapsto \lambda z, z \mapsto z\lambda'$ of $M, N, \text{Ext}_A(M, N), \text{Ext}_A(M, N)$ respectively, we then have

$$\lambda_E = \text{Ext}_A(\lambda_M, 1), \quad \lambda'_E = \text{Ext}(1, \lambda'_N),$$

which provides another description of the bimodule structure of $\text{Ext}_A(M, N)$. We leave it to the reader to generalize nos. 4 and 6 to the case of complexes of multimodules.

Let C, C' , and C'' be complexes of A -modules. Composition of maps defines a graded homomorphism of degree zero:

$$\text{Homgr}_A(C', C'') \otimes_k \text{Homgr}_A(C, C') \rightarrow \text{Homgr}_A(C, C''). \quad (14)$$

Let B, B', E be rings, C, C', C'' of complexes of (B', A) -bimodules, (A, E) -bimodules, (B, E) -bimodules respectively. By restriction of the canonical isomorphism of II, p. 73, one obtains a bijective homomorphism of (B, B') -bimodules:

$$\text{Homgr}_E(C \otimes_A C', C'') \rightarrow \text{Homgr}_A(C, \text{Homgr}_E(C', C'')). \quad (15)$$

Finally, let B be a ring, C a complex of right B -modules, C' a complex of right A -modules, C'' a complex of (B, A) -bimodules. We deduce from the canonical homomorphisms (II, p. 75)

$$C_p \otimes_B \text{Hom}_A (C'_q, C''_r) \rightarrow \text{Hom}_A (C'_q, C_p \otimes_B C''_r)$$

a graded homomorphism of degree zero:

$$C \otimes_B \text{Homgr}_A (C', C'') \rightarrow \text{Homgr}_A (C', C \otimes_B C''). \quad (16)$$

This homomorphism is bijective when C is a finitely generated projective module (II, p. 75, prop. 2).

PROPOSITION 12. — The homomorphisms (14), (15), (16) are morphisms of complexes.

Let us prove it, for example, for the homomorphism (14). Denote

$$\kappa : \text{Homgr}_A (C', C'') \otimes_k \text{Homgr}_A (C, C') \rightarrow \text{Homgr}_A (C, C'')$$

this homomorphism. Let $f \in \text{Homgr}_A (C', C')_p$ and $g \in \text{Homgr}_A (C, C')_q$; we then have by definition $\kappa(f \otimes g) = f \circ g$ Moreover:

$$\begin{aligned} D(f \otimes g) &= Df \otimes g + (-1)^p f \otimes Dg = (d'' \circ f) \otimes g - (-1)^p (f \circ d') \otimes g + \\ &\quad + (-1)^p f \otimes (d' \circ g) - (-1)^{p+q} f \otimes (g \circ d), \end{aligned}$$

whence

$$\begin{aligned} \kappa(D(f \otimes g)) &= d'' \circ f \circ g - (-1)^p f \circ d' \circ g + \\ &\quad + (-1)^p f \circ d' \circ g - (-1)^{p+q} f \circ g \circ d \\ &= d'' \circ f \circ g - (-1)^{p+q} f \circ g \circ d = D(f \circ g) = D(\kappa(f \otimes g)). \end{aligned}$$

One proves similarly that the homomorphisms (15) and (16) are morphisms of complexes.

From the morphism (14) we deduce homomorphisms of k -modules (X, p. 62)

$$H^p (\text{Homgr}_A (C', C'')) \otimes_k H^q (\text{Homgr}_A (C, C')) \rightarrow H^{p+q} (\text{Homgr}_A (C, C'')). \quad (17)$$

Taking $C = A$, we see that the homomorphism:

$$\text{Homgr}_A(C', C'') \otimes_k C' \rightarrow C'' \quad (18)$$

which maps $f \otimes x$ to $f(x)$ is a morphism of complexes of left A -modules; associated to it is a canonical homomorphism (X, p. 80) of graded A -modules $\gamma : H(\text{Homgr}_A(C', C'')) \otimes_k H(C') \rightarrow H(C'')$, which corresponds to the canonical homomorphism of k -modules

$$\lambda : H(\text{Homgr}_A(C', C'')) \rightarrow \text{Homgr}_A(H(C'), H(C'')) .$$

10.6 USE OF NON-CANONICAL RESOLUTIONS

We resume the conventions of § 4.

10.6.1 Computation of the modules $\text{Tor}^A(P, M)$ and $\text{Ext}_A(M, N)$

Let M, N be left A -modules, P a right A -module. Let moreover $a : R \rightarrow M$ be a left resolution of M , $b : S \rightarrow P$ a left resolution of P , and $c : N \rightarrow E$ a right resolution of N .

By X, p. 49, prop. 3 and 3 bis, there exist morphisms of complexes $\alpha : L(M) \rightarrow R$, $\beta : L(P) \rightarrow S$, $\gamma : E \rightarrow I(N)$ such that $a \circ \alpha = p_M$, $b \circ \beta = p_P$, $\gamma \circ c = e_N$; and the homotopy classes of α , β , γ depend only on the given resolutions. By X, p. 64, prop. 3 and X, p. 84, prop. 3, the homotopy classes of the morphisms

$$\beta \otimes \alpha : L(P) \otimes_A L(M) \rightarrow S \otimes_A R,$$

$$\text{Homgr}_A(\alpha, \gamma) : \text{Homgr}_A(R, E) \rightarrow \text{Homgr}_A(L(M), I(N))$$

depend only on the given resolutions, whence, by passing to homology, graded k -homomorphisms of degree 0

$$\psi(S, R) : \text{Tor}^A(P, M) \rightarrow H(S \otimes_A R),$$

$$\varphi(R, E) : H(\text{Homgr}_A(R, E)) \rightarrow \text{Ext}_A(M, N),$$

independent of the choices of α, β, γ .

For example, taking for a, b, c the identity maps of M, P, N respectively, one obtains the homomorphisms $\psi(P, M) : \text{Tor}^A(P, M) \rightarrow P \otimes_A M$ and $\varphi(M, N) : \text{Hom}_A(M, N) \rightarrow \text{Ext}_A(M, N)$ introduced in X, p. 68, remark 2) and X, p. 87, remark 2).

THEOREM 1. — a) If one of the resolutions R or S is flat, then $\psi(S, R)$ is an isomorphism of k -graded modules.

b) If R is projective or if E is injective, $\varphi(R, E)$ is an isomorphism of k -graded modules.

c) Suppose for example R is flat, and choose α and β as above. The homomorphism $\beta \otimes \alpha$ is the composite of the morphisms

$$L(P) \otimes_A L(M) \xrightarrow{l_{L(P)} \otimes \alpha} L(P) \otimes_A R \xrightarrow{\beta \otimes 1_R} S \otimes_A R.$$

Since $L(P)$ (resp. R) is flat and α (resp. β) is a homology isomorphism, $1_{L(P)} \otimes \alpha$ (resp. $\beta \otimes 1_R$) is one, by prop. 4 of X, p. 67. Hence $\beta \otimes \alpha$ is a homology isomorphism and $\psi(S, R) = H(\beta \otimes \alpha)$ is bijective.

b) One argues similarly, using prop. 4 of X, p. 86.

COROLLARY. — If R is a flat resolution of M , the homomorphism

$$\psi(P, R) : \text{Tor}^A(P, M) \rightarrow H(P \otimes_A R)$$

is bijective. If R is a projective resolution of M , the homomorphism

$$\varphi(R, N) : H(\text{Homgr}_A(R, N)) \rightarrow \text{Ext}_A(M, N)$$

is bijective. If E is an injective resolution of N , the homomorphism

$$\varphi(M, E) : H(\text{Homgr}_A(M, E)) \rightarrow \text{Ext}_A(M, N)$$

is bijective.

Remark. — The diagram of k -modules

$$\begin{array}{ccc} \mathrm{Tor}^A(P, M) & \xrightarrow{\psi(S, R)} & H(S \otimes_A R) \\ & \sigma_{P, M} \downarrow & \\ \mathrm{Tor}^{A^\circ}(M^\circ, P^\circ) & \xrightarrow{\psi(R^\circ, S^\circ)} & H(R^\circ \otimes_{A^\circ} S^\circ), \end{array}$$

where $\sigma_{P, M}$ and $\sigma(S, R)$ are the commutation isomorphisms (X, p. 71 and 63), is commutative: it is indeed deduced, by passing to homology, from the commutative diagram of complexes

$$\begin{array}{ccc} L(P) \otimes_A L(M) & \xrightarrow{\beta \otimes \alpha} & S \otimes_A R \\ \sigma(L(P), L(M)) \downarrow & & \downarrow \sigma(S, R) \\ L(M^\circ) \otimes_{A^\circ} L(P^\circ) & \xrightarrow{\alpha \otimes \beta} & R^\circ \otimes_{A^\circ} S^\circ \end{array}$$

(“the morphisms ψ are compatible with the commutation isomorphisms”).

Similarly, let $a_1 : \mathbb{R}_1 \rightarrow \mathbb{M}$, $b_1 : \mathbb{S}_1 \rightarrow \mathbb{P}$, $c_1 : \mathbb{N} \rightarrow \mathbb{E}_1$ be morphisms of complexes, where $\mathbb{R}_1$ and $\mathbb{S}_1$ are projective and zero on the right, and $\mathbb{E}_1$ injective and zero on the left. By X, p. 49, prop. 3 and 3 bis, there exist morphisms of complexes

$$\alpha_1 : \mathbb{R}_1 \rightarrow L(M), \quad \beta_1 : \mathbb{S}_1 \rightarrow L(P), \quad \gamma_1 : \mathbb{I}(\mathbb{N}) \rightarrow E_1$$

such that $p_M \circ \alpha_1 = a_1, p_P \circ \beta_1 = b_1, \gamma_1 \circ e_N = c_1$, whence morphisms of complexes

$$\beta_1 \otimes \alpha_1 : S_1 \otimes_A R_1 \rightarrow L(P) \otimes_A L(M),$$

$$\mathrm{Homgr}_A(\alpha_1, \gamma_1) : \mathrm{Homgr}_A(L(M), \mathbb{I}(\mathbb{N})) \rightarrow \mathrm{Homgr}_A(R_1, E_1),$$

and, by passing to homology, graded k -linear maps of degree 0:

$$\begin{aligned} \psi'(S_1, R_1) &: H(S_1 \otimes_A R_1) \rightarrow \mathrm{Tor}^A(P, M) \\ \varphi'(R_1, E_1) &: \mathrm{Ext}_A(M, N) \rightarrow H(\mathrm{Homgr}_A(R_1, E_1)) \end{aligned}$$

which one verifies as above are independent of the choice of α_1 , β_1 , γ_1 .

PROPOSITION 1. — If a_1, b_1, c_1 are homology isomorphisms, $\psi'(S_1, R_1)$ and $\varphi'(R_1, E_1)$ are the inverse bijections of the bijections $\psi(S_1, R_1)$ and $\varphi(R_1, E_1)$ respectively. Indeed, $f = (\beta \otimes \alpha) \circ (\beta_1 \otimes \alpha_1)$ is a morphism of the complex $S_1 \otimes_A R_1$ into itself, and one has $(b_1 \circ \alpha_1) \circ f = f$. By X, p. 49, prop. 3 and 3 bis, f is a homotopy equivalence, hence $H(f) = 1$ and

$$\psi(S_1, R_1) \circ \psi'(S_1, R_1) = H(\beta \otimes \alpha) \circ H(\beta_1 \otimes \alpha_1) = H(f) = 1;$$

likewise $\psi'(S_1, R_1) \circ \psi(S_1, R_1) = 1$. One argues analogously for the maps φ and φ' .

Examples. — 1) Let a be an element of A such that the map $\varphi : x \mapsto xa$ from A into itself be injective (a is not a right zero divisor). Using the resolution

$$0 \rightarrow A_s \xrightarrow{\varphi} A_s \rightarrow A/Aa \rightarrow 0$$

we see that for every right A -module M , we have

$$\mathrm{Tor}_i^A(M, A/Aa) = 0 \quad \text{for } i > 1$$

and that the k -module $\mathrm{Tor}_1^A(M, A/Aa)$ is isomorphic to $\mathrm{Ker}(a_M)$.

Similarly, for every left A -module M , we have

$$\mathrm{Ext}_A^i(A/Aa, M) = 0 \quad \text{for } i > 1$$

and the k -module $\mathrm{Ext}_A^1(A/Aa, M)$ is isomorphic to M/aM .

2) Suppose A is an integral domain; let K be the field of fractions of A and M an A -module.

Using the flat resolution

$$0 \rightarrow A \rightarrow K \rightarrow K/A \rightarrow 0$$

(X, p. 9, example 5), we see that $\mathrm{Tor}_i^A(K/A, M) = 0$ for $i > 1$; moreover, taking into account II, p. 116, prop. 26, (ii), the A -module $\mathrm{Tor}_1^A(K/A, M)$ is isomorphic to the torsion submodule of M .

3) Suppose that A is a Noetherian local ring, denote its maximal ideal by $\mathfrak{m}$, and set $\kappa = A/\mathfrak{m}$. Let M be a finitely generated A -module and P a minimal projective resolution of M (X, p. 54). For every $n \geq 0$, the κ vector spaces $\text{Tor}_n^A(\kappa, M)$ and $\text{Ext}_A^n(M, \kappa)$ are finite-dimensional, of dimension equal to the rank of the free A -module P_n ; indeed, the complexes $\kappa \otimes_A P$ and $\text{Homgr}_A(P, \kappa)$ have zero differential.

10.6.2 Computation of the mappings $\text{Tor}^A(g, f)$ and $\text{Ext}_A(f, h)$

Let $f : M \rightarrow M'$, $h : N' \rightarrow N$ be homomorphisms of left A -modules, $g : P \rightarrow P'$ a homomorphism of right A -modules, $a : R \rightarrow M$, $a' : R' \rightarrow M'$, $b : S \rightarrow P$, $b' : S' \rightarrow P'$ left resolutions of M, M', P, P' , respectively, $c : N \rightarrow E$, $c' : N' \rightarrow E'$ right resolutions of N and N' , $\tilde{f} : R \rightarrow R'$, $\tilde{g} : S \rightarrow S'$, $\tilde{h} : E' \rightarrow E$ morphisms of complexes such that

$$a' \circ \tilde{f} = f \circ a, b' \circ \tilde{g} = g \circ b, \tilde{h} \circ c' = c \circ h.$$

PROPOSITION 2. — The following two diagrams are commutative:

$$\begin{array}{ccc} & & \text{Tor}^A(P, M) \xrightarrow{\psi(S, R)} H(S \otimes_A R) \\ \text{Tor}^A(g, f) \downarrow & & \\ & & \text{Tor}^A(P', M') \xrightarrow{\psi(S', R')} H(S' \otimes_A R') \\ & & \\ & \text{H}(\text{Homgr}_A(R', E')) \xrightarrow{\varphi(R', E')} \text{Ext}_A(M', N') \\ & \text{H}(\text{Homgr}_A(\tilde{f}, \tilde{h})) \downarrow \\ & \text{H}(\text{Homgr}_A(R, E)) \xrightarrow{\varphi(R, E)} \text{Ext}_A(M, N). \end{array}$$

Let $\alpha : L(M) \rightarrow R$, $\alpha' : L(M') \rightarrow R'$, $\gamma : L(P) \rightarrow S$, $\gamma' : L(P') \rightarrow S'$ morphisms of complexes such that

$$\alpha \circ \alpha = p_M, \alpha' \circ \alpha' = p_{M'}, b \circ \gamma = p_P, b' \circ \gamma' = p_{P'}.$$

By definition, $H(\tilde{g} \otimes \tilde{f}) \circ \psi(S, R)$ is equal to

$$H(\tilde{g} \otimes \tilde{f}) \circ H(\gamma \otimes \alpha) = H((\tilde{g} \circ \gamma) \otimes (\tilde{f} \circ \alpha)),$$

while $\psi(S', R') \circ \text{Tor}^A(g, f)$ is equal to

$$H(\gamma' \otimes \alpha') \circ H(L(g) \otimes L(f)) = H((\gamma' \circ L(g)) \otimes (\alpha' \circ L(f))).$$

On the other hand, $\alpha' \circ L(f)$ and $\tilde{f} \circ \alpha$ are two morphisms from $L(M)$ into R' such that $\alpha' \circ (\alpha' \circ L(f)) = p_M \circ L(f) = f \circ p_M = f \circ \alpha \circ \alpha = \alpha' \circ (\tilde{f} \circ \alpha)$. By X, p. 49, prop. 3, $\alpha' \circ L(f)$ and $\tilde{f} \circ \alpha$ are homotopic; likewise $\gamma' \circ L(g)$ and $\tilde{g} \circ \gamma$ are homotopic, and consequently so are $(\gamma' \circ L(g)) \otimes (\alpha' \circ L(f))$ and $(\tilde{g} \circ \gamma) \otimes (\tilde{f} \circ \alpha)$ by prop. 3 of X, p. 64. We therefore have

$$\begin{aligned} H(\tilde{g} \otimes \tilde{f}) \circ \psi(S, R) &= H((\tilde{g} \circ \gamma) \otimes (\tilde{f} \circ \alpha)) = H((\gamma' \circ L(g)) \otimes (\alpha' \circ L(f))) \\ &= \psi(S', R') \circ \text{Tor}^A(g, f). \end{aligned}$$

We argue analogously for the second diagram.

Remark. — Consider similarly commutative diagrams of morphisms of complexes.

$$\begin{array}{ccccc} R_1 & \xrightarrow{a_1} & M & S_1 & \xrightarrow{b_1} & P & N' & \xrightarrow{c'_1} & E' \\ & & \tilde{f} \downarrow & & f \downarrow & & g \downarrow & & \\ R'_1 & \xrightarrow{a'_1} & M' & S'_1 & \xrightarrow{b'_1} & P' & N & \xrightarrow{c_1} & E_1 \end{array}$$

where the complexes R_1, R'_1, S_1, S'_1 are projective and zero on the right, and where the complexes E_1, E'_1 are injective and zero on the left. Then

$$\begin{aligned} \text{Tor}^A(g, f) \circ \psi'(S_1, R_1) &= \psi'(S'_1, R'_1) \circ H(\tilde{g} \otimes \tilde{f}) \\ \varphi'(R_1, E_1) \circ \text{Ext}_A(f, h) &= H(\text{Homgr}_A(\tilde{f}, \tilde{h})) \circ \varphi'(R'_1, E'_1) \end{aligned}$$

as is shown by an argument analogous to that of Prop. 2.

10.6.3 Computation of the connecting homomorphisms

Consider a commutative diagram

(1)

$$\begin{array}{ccc}
 & 0 \rightarrow R' \xrightarrow{\tilde{u}} R \xrightarrow{\tilde{v}} R'' \rightarrow 0 \\
 a' \downarrow & & \\
 & 0 \rightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \rightarrow 0 \\
 & & (2)
 \end{array}$$

where the first row (1) is an exact sequence of complexes of left A -modules, the second row (2) is an exact sequence of left A -modules, and where the vertical arrows are left resolutions.

PROPOSITION 3. — a) Let P be an A -module, $b : S \rightarrow P$ a left resolution of P ; suppose that the sequence of complexes of k -modules

$$0 \rightarrow S \otimes_A R' \xrightarrow{l_s \otimes \tilde{u}} S \otimes_A R \xrightarrow{l_s \otimes \tilde{v}} S \otimes_A R'' \rightarrow 0 \quad (3)$$

be exact. Then the following diagram is commutative:

$$\begin{array}{ccc}
 \text{Tor}^A(P, M'') & \xrightarrow{\partial(P, (2))} & \text{Tor}^A(P, M') \\
 \psi(S, R'') \downarrow & & \downarrow \psi(S, R') \\
 H(S \otimes_A R'') & \xrightarrow{\partial((3))} & H(S \otimes_A R').
 \end{array}$$

b) Let N be a left A -module, $c : N \rightarrow E$ a right resolution of N ; suppose that the sequence of complexes of k -modules

(4)

$0 \rightarrow \text{Homgr}_A(R'', E) \xrightarrow{\text{Homgr}_A(\tilde{t}, 1)} \text{Homgr}_A(R, E) \xrightarrow{\text{Homgr}_A(\tilde{\mu}, 1)} \text{Homgr}_A(R', E) \rightarrow 0$ be exact. Then the following diagram is commutative:

$$\begin{array}{ccc}
 H(\text{Homgr}_A(R', E)) & \xrightarrow{\delta((4))} & H(\text{Homgr}_A(R'', E)) \\
 \varphi(R', E) \downarrow & & \\
 \text{Ext}_A(M', N) & \xrightarrow{\partial((2), N)} & \text{Ext}_A(M'', N).
 \end{array}$$

Let us prove for example a). Let $\beta : L(P) \rightarrow S$ be a morphism of complexes such that $b \circ \beta = p_P$; consider the diagram of k -complexes

$$\begin{array}{c}
0 \rightarrow S \otimes_A R' \xrightarrow{1 \otimes \tilde{u}} S \otimes_A R \xrightarrow{1 \otimes \tilde{v}} S \otimes_A R'' \rightarrow 0 \\
\left. \begin{array}{ccc} \beta \otimes 1_R & & \beta \otimes 1_R \end{array} \right| \quad \left. \begin{array}{ccc} \beta \otimes 1_R & & \beta \otimes 1_R \end{array} \right| \\
0 \rightarrow L(P) \otimes_A R' \xrightarrow{1 \otimes \tilde{u}} L(P) \otimes_A R \xrightarrow{1 \otimes \tilde{v}} L(P) \otimes_A R'' \rightarrow 0 \\
\left. \begin{array}{ccc} 1 \otimes a' & & 1 \otimes a \end{array} \right| \quad \left. \begin{array}{ccc} 1 \otimes a & & 1 \otimes a'' \end{array} \right| \\
0 \rightarrow L(P) \otimes_A M' \xrightarrow{1 \otimes u} L(P) \otimes_A M \xrightarrow{1 \otimes v} L(P) \otimes_A M'' \rightarrow 0.
\end{array}$$

It is commutative with exact rows (by the hypothesis for the first row, and by the fact that $L(P)$ is flat for the other two). We therefore have a commutative diagram (X, p. 31, prop. 2, and X, p. 72, def. 2).

$$\begin{array}{ccc}
H(S \otimes_A R'') & \xrightarrow{\partial((3))} & H(S \otimes_A R') \\
H(\beta \otimes 1) \uparrow & & \uparrow H(\beta \otimes 1) \\
H(L(P) \otimes_A R'') & \rightarrow & H(L(P) \otimes_A R') \\
H(1 \otimes a'') \downarrow & & \downarrow H(1 \otimes a') \\
H(L(P) \otimes_A M'') & \rightarrow & H(L(P) \otimes_A M') \\
\psi_P(M'') \uparrow & & \uparrow \psi_P(M') \\
\text{Tor}(P, M'') & \xrightarrow{\partial(P, (2))} & \text{Tor}(P, M').
\end{array}$$

By X, p. 67, prop. 4, $H(1 \otimes a'')$ and $H(1 \otimes a')$ are bijective; on the other hand, by definition of the homomorphisms ψ , we have $H(\beta \otimes 1) \circ \psi(L(P), R'') = \psi(S, R'')$ and $H(1 \otimes a'') \circ \psi(L(P), R'') = \psi(L(P), M'') = \psi_P(M'')$, hence

$$\psi(S, R'') = H(\beta \otimes 1) \circ H(1 \otimes a'')^{-1} \circ \psi_P(M'');$$

likewise, $\psi(S, R') = H(\beta \otimes 1) \circ H(1 \otimes a')^{-1} \circ \psi_P(M')$, and the desired assertion $\partial((3)) \circ \psi(S, R'') = \psi(S, R') \circ \partial(P, (2))$ follows from the commutativity of the preceding diagram.

Remarks. — 1) Using the commutation isomorphisms, one deduces from a) the analogous statement obtained by interchanging the roles of the two arguments of the tensor product.

2) With the notations of a), suppose either S is flat, or R, R', R'' flat; then on the one hand the sequence (3) is exact (X, p. 72, cor. 2) and one can apply Prop. 3; on the other hand

$\psi(S, R')$ is bijective (th. 1), hence

$$\partial(P, (2)) = \psi(S, R')^{-1} \circ \partial((3)) \circ \psi(S, R'').$$

3) With the notations of b), suppose either E is injective, or R, R', R'' projective; then on the one hand the sequence (4) is exact (X, p. 83, prop. 2) and one can apply prop. 3; on the other hand, $\varphi(R', E)$ is bijective (th. 1); therefore

$$\delta((2), N) = \varphi(R'', E) \circ \hat{\partial}((4)) \circ \varphi(R', E)^{-1}.$$

Consider now a commutative diagram

(5)

$$\begin{array}{ccccccc} 0 & \rightarrow & N' & \xrightarrow{r} & N & \xrightarrow{s} & N'' \rightarrow 0 \\ c' \downarrow & & & & c \downarrow & & c'' \downarrow \\ 0 & \rightarrow & E' & \xrightarrow{\tilde{r}} & E & \xrightarrow{\tilde{s}} & E'' \rightarrow 0 \end{array} \quad (6)$$

whose first row (5) is an exact sequence of left A -modules, the second row (6) an exact sequence of complexes of left A -modules, and where the vertical arrows are right resolutions.

Reasoning as in Prop. 3, one proves the following proposition:

PROPOSITION 4. — Let M be a left A -module, $a : R \rightarrow M$ a left resolution of M such that the sequence

$$(7) \quad 0 \rightarrow \text{Homgr}_A(R, E') \xrightarrow{\text{Homgr}(\tilde{\mathcal{F}}, 1)} \text{Homgr}_A(R, E) \xrightarrow{\text{Homgr}(\tilde{\mathcal{F}}, 1)} \text{Homgr}_A(R, E'') \rightarrow 0$$

is exact. Then the following diagram is commutative.

$$\begin{array}{ccc} H(\text{Homgr}_A(R, E'')) & \xrightarrow{\partial((7))} & H(\text{Homgr}_A(R, E')) \\ \varphi(R, E'') \downarrow & & \\ \text{Ext}_A(M, N'') & \xrightarrow{\delta(M, (5))} & \text{Ext}_A(M, N'). \end{array}$$

Remarks. — 4) If R is projective or if E, E', E'' are injective, the sequence (7) is exact (X, p. 83, prop. 2); moreover, $\varphi(R, E'')$ is bijective (th. 1); hence

$$\delta(M, (5)) = \varphi(R, E') \circ \partial((7)) \circ \varphi(R, E'')^{-1}.$$

- 5) We leave it to the reader to state and prove the propositions analogous to Props. 3 and 4 and relating to homomorphisms. ψ_1 and φ_1 .

10.6.4 Finiteness of extension and torsion modules

Let M be a left A -module, I a filtered preordered set, (N_i, u_{ji}) an inductive system of left A -modules relative to I , $N = \varinjlim N_i$ its inductive limit, $u_i : N_i \rightarrow N, i \in I$, the canonical map. Then $(\text{Ext}_A(M, N_i), \text{Ext}_A(1_M, u_{ji}))$ is an inductive system of k -modules and $(\text{Ext}_A(1_M, u_i))$ an inductive system of maps, whose inductive limit is a homomorphism of graded k -modules, called canonical

$$\varinjlim_{i \in I} \text{Ext}_A(M, N_i) \rightarrow \text{Ext}_A\left(M, \varinjlim_{i \in I} N_i\right). \quad (8)$$

PROPOSITION 5. — If A is left Noetherian and M is a finitely generated A -module, the canonical homomorphism (8) is bijective.

Indeed, let $(X, p, 53, \text{prop. } 6) p : L \rightarrow M$ be a free resolution of M such that L_n be finitely generated for each n . The canonical morphism from k -complexes

$$u : \varinjlim \text{Homgr}_A(L, N_i) \rightarrow \text{Homgr}_A(L, \varinjlim N_i)$$

is bijective, hence so is the homomorphism

$$\varinjlim H(\text{Homgr}_A(L, N_i)) \rightarrow H(\text{Homgr}_A(L, \varinjlim N_i))$$

deduces homomorphisms $H(\text{Homgr}(1, u_i))$ (X, p. 28, prop. 1). We then conclude by prop. 2 (X, p. 103) and th. 1 (X, p. 100).

PROPOSITION 6. — Let B be a ring and N an (A, B) -bimodule, which is a Noetherian B -module (resp. of finite length).

- a) Suppose A is right Noetherian and let M be a finitely generated right A -module. Then the

B-modules (X, p. 81) $\text{Tor}_n^A(M, N)$ are Noetherian (resp. of finite length).

b) Suppose A is left Noetherian and let M be a finitely generated left A -module. Then the B-modules (X, p. 98) $\text{Ext}_A^n(M, N)$ are Noetherian (resp. of finite length).

Choose a free resolution $p : \mathbf{L} \rightarrow \mathbf{M}$ such that each of the A -modules $\mathbf{L}_n$ be finitely generated (X, p. 53, prop. 6), and let C be the complex of B-modules $\mathbf{L} \otimes_A \mathbf{N}$ in the case a), $\text{Homgr}_A(\mathbf{L}, \mathbf{N})$ in the case b). Each of the B-modules C_n is isomorphic to a product of a finite number of copies of N , hence is Noetherian (resp. of finite length); the same is therefore true of the modules $H_n(C)$. Now, according to X, p. 100, th. 1, these are isomorphic to the $\text{Tor}_n^A(\mathbf{M}, \mathbf{N})$ in the case a), to the $\text{Ext}_A^{-n}(\mathbf{M}, \mathbf{N})$ in the case b).

COROLLARY. — Let $\rho : A \rightarrow B$ be a homomorphism of Noetherian commutative rings, M a finitely generated A -module, N a B -module. If N is a finitely generated (resp. finite length) B -module, then so are the B-modules $\text{Tor}_n^A(M, N)$ and $\text{Ext}_A^n(M, N)$.

10.6.5 Homomorphisms $\text{Tor}^B(\mathbf{P}, \mathbf{N}) \otimes_A \mathbf{Q} \rightarrow \text{Tor}^B(\mathbf{P}, \mathbf{N} \otimes_A \mathbf{Q})$ and $\text{Ext}_B(\mathbf{M}, \mathbf{N}) \otimes_A \mathbf{Q} \rightarrow \text{Ext}_B(\mathbf{M}, \mathbf{N} \otimes_A \mathbf{Q})$

Let B be a ring, N a (B, A) -bimodule, M a left B -module, P a right B -module, and Q a left A -module.

According to X, p. 62, we have a homomorphism

$$\gamma_1 : H(L(P) \otimes_B N) \otimes_A Q \rightarrow H(L(P) \otimes_B N \otimes_A Q);$$

moreover (X, p. 69, prop. 5), we have isomorphisms

$$\begin{aligned} \psi_1(N) : \text{Tor}^B(P, N) \otimes_A Q &\rightarrow H(L(P) \otimes_B N) \otimes_A Q, \\ \psi_1(N \otimes_A Q) : \text{Tor}^B(P, N \otimes_A Q) &\rightarrow H(L(P) \otimes_B N \otimes_A Q). \end{aligned}$$

The canonical degree-0 graded homomorphism

$$\text{Tor}^B(P, N) \otimes_A Q \rightarrow \text{Tor}^B(P, N \otimes_A Q) \tag{9}$$

is defined as the composite $\psi_P(\mathbf{N} \otimes_A \mathbf{Q})^{-1} \circ \gamma_1 \circ (\psi_P(\mathbf{N}) \otimes 1_Q)$.

Similarly, we deduce from the canonical morphism of complexes

$$\alpha : \text{Homgr}_B(L(M), N) \otimes_A Q \rightarrow \text{Homgr}_B(L(M), N \otimes_A Q)$$

a homomorphism $H(\alpha)$; we have the canonical homomorphism (X, p. 62)

$$\gamma_2 : H(\text{Homgr}_B(L(M), N)) \otimes_A Q \rightarrow H(\text{Homgr}_B(L(M), N \otimes_A Q)),$$

and isomorphisms (X, p. 88, prop. 5)

$$\varphi_M(N) : H(\text{Homgr}_B(L(M), N)) \otimes_A Q \rightarrow \text{Ext}_B(M, N) \otimes_A Q,$$

$$\varphi_M(N \otimes_A Q) : H(\text{Homgr}_B(L(M), N \otimes_A Q)) \rightarrow \text{Ext}_B(M, N \otimes_A Q).$$

The canonical degree-0 graded homomorphism

$$\text{Ext}_B(\mathbf{M}, \mathbf{N}) \otimes_A \mathbf{Q} \rightarrow \text{Ext}_B(\mathbf{M}, \mathbf{N} \otimes_A \mathbf{Q}) \quad (10)$$

is defined as the composite

$$\varphi_M(N \otimes_A Q) \circ H(\alpha) \circ \gamma_2 \circ (\varphi_M(N) \otimes 1_Q)^{-1}.$$

PROPOSITION 7. — a) If the A-module Q is flat, the homomorphism (9) is bijective. b) If the A-module Q is finitely generated projective, the homomorphism (10) is bijective.

c) If the A-module Q is flat, the ring B is Noetherian, and the B-module M is finitely generated, then the homomorphism (10) is bijective.

d) If Q is flat, γ_1 is bijective (X, p. 66, cor. 2).

e) If Q is finitely generated projective, it is flat, hence γ_2 is bijective, and moreover α is bijective (II, p. 75, prop. 2, a)).

f) Under the hypotheses of c), γ_2 is bijective since Q is flat. Moreover (X, p. 53, prop. 6), there exists a resolution L of M such that each L_n be finitely generated free; let $u :$

$L(M) \rightarrow L$ be a homotopy equivalence (X, p. 49, cor. to prop. 3); in the commutative diagram,

$$\begin{array}{ccc} \text{Homgr}_B(L(M), N) \otimes_A Q & \xrightarrow{\alpha} & \text{Homgr}_B(L(M), N \otimes_A Q) \\ \uparrow & & \uparrow \\ \text{Homgr}_B(L, N) \otimes_A Q & \xrightarrow{\bar{\alpha}} & \text{Homgr}_B(L, N \otimes_A Q). \end{array}$$

the vertical arrows deduced from u are homotopy equivalences (X, p. 64, prop. 3 and p. 83, prop. 3) and $\bar{\alpha}$ is bijective (II, p. 75, prop. 2 (ii)); hence $H(\alpha)$ is bijective, and the homomorphism (10) is bijective.

10.6.6 Homomorphisms $\text{Tor}^B(P, N \otimes_A Q) \rightarrow \text{Tor}^A(P \otimes_B N, Q)$ and $\text{Ext}_A(Q, \text{Hom}_B(N, M)) \rightarrow \text{Ext}_B(N \otimes_A Q, M)$

Keep the previous notation, and suppose N is flat over A . Then the morphism $N \otimes_A L(Q) \xrightarrow{1 \otimes p_Q} N \otimes_A Q$ is a homomorphism (X, p. 67, prop. 4), whence homomorphisms

$$\psi(P, N \otimes_A L(Q)) : \text{Tor}^B(P, N \otimes_A Q) \rightarrow H(P \otimes_B N \otimes_A L(Q))$$

$$\varphi(N \otimes_A L(Q), M) : H(\text{Homgr}_B(N \otimes_A L(Q), M)) \rightarrow \text{Ext}_B(N \otimes_A Q, M).$$

Using then the isomorphisms

$$\begin{aligned} \bar{\psi}_Q(P \otimes_B N) &: \text{Tor}^A(P \otimes_B N, Q) \rightarrow H(P \otimes_B N \otimes_A L(Q)), \\ \beta &: \text{Homgr}_A(L(Q), \text{Hom}_B(N, M)) \rightarrow \text{Homgr}_B(N \otimes_A L(Q), M), \\ \varphi_Q(\text{Hom}_B(N, M)) &: H(\text{Homgr}_A(L(Q), \text{Hom}_B(N, M))) \rightarrow \text{Ext}_A(Q, \text{Hom}_B(N, M)), \end{aligned}$$

From this, we deduce canonical graded homomorphisms of degree 0:

(11)

$$\mathrm{Tor}^{\mathbf{B}}(\mathbf{P}, \mathbf{N} \otimes_{\mathbf{A}} \mathbf{Q}) \rightarrow \mathrm{Tor}^{\mathbf{A}}(\mathbf{P} \otimes_{\mathbf{B}} \mathbf{N}, \mathbf{Q}) \quad (12)$$

$$\mathrm{Ext}_{\mathbf{A}}(\mathbf{Q}, \mathrm{Hom}_{\mathbf{B}}(\mathbf{N}, \mathbf{M})) \rightarrow \mathrm{Ext}_{\mathbf{B}}(\mathbf{N} \otimes_{\mathbf{A}} \mathbf{Q}, \mathbf{M}).$$

PROPOSITION 8. — a) If $\mathbf{N}$ is flat over $\mathbf{A}$ and over $\mathbf{B}$, the homomorphism (11) is bijective.
b) If $\mathbf{N}$ is flat over $\mathbf{A}$ and projective over $\mathbf{B}$, the homomorphism (12) is bijective.

Indeed, $\mathbf{N} \otimes_{\mathbf{A}} \mathbf{L}(\mathbf{Q})$ is isomorphic to a direct sum of copies of $\mathbf{N}$, hence is a flat (resp. projective) $\mathbf{B}$ -module when the $\mathbf{B}$ -module $\mathbf{N}$ is flat (resp. projective); one then applies th. 1 (X, p. 100).

10.6.7 Homomorphisms $\mathbf{B} \otimes_{\mathbf{A}} \mathrm{Tor}^{\mathbf{A}}(\mathbf{E}, \mathbf{F}) \rightarrow \mathrm{Tor}^{\mathbf{B}}(\mathbf{E} \otimes_{\mathbf{A}} \mathbf{B}, \mathbf{B} \otimes_{\mathbf{A}} \mathbf{F})$ and $\mathbf{B} \otimes_{\mathbf{A}} \mathrm{Ext}_{\mathbf{A}}(\mathbf{E}, \mathbf{F}) \rightarrow \mathrm{Ext}_{\mathbf{B}}(\mathbf{B} \otimes_{\mathbf{A}} \mathbf{E}, \mathbf{B} \otimes_{\mathbf{A}} \mathbf{F})$

In this section, we assume that $\mathbf{A}$ is commutative; we are given a ring homomorphism $\rho : \mathbf{A} \rightarrow \mathbf{B}$ such that $\rho(\mathbf{A})$ be contained in the center of $\mathbf{B}$, and let $\mathbf{E}$ and $\mathbf{F}$ be two $\mathbf{A}$ -modules. We have a canonical isomorphism of complexes of $\mathbf{A}$ -modules

$$u : \mathbf{B} \otimes_{\mathbf{A}} (\mathbf{L}(\mathbf{E}) \otimes_{\mathbf{A}} \mathbf{L}(\mathbf{F})) \rightarrow (\mathbf{L}(\mathbf{E}) \otimes_{\mathbf{A}} \mathbf{B}) \otimes_{\mathbf{B}} (\mathbf{B} \otimes_{\mathbf{A}} \mathbf{L}(\mathbf{F}));$$

on the other hand, since $\mathbf{L}(\mathbf{E}) \otimes_{\mathbf{A}} \mathbf{B}$ and $\mathbf{B} \otimes_{\mathbf{A}} \mathbf{L}(\mathbf{F})$ are free $\mathbf{B}$ -complexes, there is a canonical homomorphism of graded $\mathbf{A}$ -modules (X, p. 102)

$$\begin{aligned} \psi'(\mathbf{L}(\mathbf{E}) \otimes_{\mathbf{A}} \mathbf{B}, \mathbf{B} \otimes_{\mathbf{A}} \mathbf{L}(\mathbf{F})) : H((\mathbf{L}(\mathbf{E}) \otimes_{\mathbf{A}} \mathbf{B}) \otimes_{\mathbf{B}} (\mathbf{B} \otimes_{\mathbf{A}} \mathbf{L}(\mathbf{F}))) \\ \rightarrow \mathrm{Tor}^{\mathbf{B}}(\mathbf{E} \otimes_{\mathbf{A}} \mathbf{B}, \mathbf{B} \otimes_{\mathbf{A}} \mathbf{F}) \end{aligned}$$

Finally, we have a homomorphism (X, p. 62)

$$\gamma : \mathbf{B} \otimes_{\mathbf{A}} \mathrm{Tor}^{\mathbf{A}}(\mathbf{E}, \mathbf{F}) \rightarrow H(\mathbf{B} \otimes_{\mathbf{A}} \mathbf{L}(\mathbf{E}) \otimes_{\mathbf{A}} \mathbf{L}(\mathbf{F})).$$

The canonical homomorphism of graded $\mathbf{B}$ -modules

$$B \otimes_A \text{Tor}^A(E, F) \rightarrow \text{Tor}^B(E \otimes_A B, B \otimes_A F) \quad (13)$$

is defined as the composite $\psi'(L(E) \otimes_A B, B \otimes_A L(F)) \circ H(u) \circ \gamma$.

PROPOSITION 9. — If B is flat over A , then the homomorphism (13) is bijective.

Indeed, $\psi'(L(E) \otimes_A B, B \otimes_A L(F))$ is bijective (X, p. 102, prop. 1) and γ is bijective (X, p. 66, cor. 2).

Suppose B is flat over A . Substituting in the homomorphism (12) E for Q , B for N , and $B \otimes_A F$ for M , we obtain a homomorphism

$$\text{Ext}_A(E, B \otimes_A F) \rightarrow \text{Ext}_B(B \otimes_A E, B \otimes_A F)$$

which is bijective by Prop. 8. Substituting E for M , F for N , A for B , B for Q in the homomorphism (10), and interchanging the factors of the tensor products, we obtain a homomorphism of B -modules

$$B \otimes_A \text{Ext}_A(E, F) \rightarrow \text{Ext}_A(E, B \otimes_A F),$$

whence by composition a homomorphism, called canonical

$$B \otimes_A \text{Ext}_A(E, F) \rightarrow \text{Ext}_B(B \otimes_A E, B \otimes_A F). \quad (14)$$

PROPOSITION 10. — The homomorphism (14) is bijective in the following cases:

- a) B is a finitely generated projective A -module;
- b) B is a flat A -module, A is Noetherian, and E is a finitely generated A -module. This follows from prop. 7 (X, p. 108).

10.6.8 Application: homology and cohomology of groups

Let G be a group, $\mathbf{Z}^{(G)}$ its algebra over $\mathbf{Z}$ (III, p. 19). Recall (cf. III, p. 20, example) that if M is a commutative group, it is equivalent to giving an action of G on M (that is, a homomorphism $\tau : G \rightarrow \text{Aut}(M)$), or a structure of $\mathbf{Z}^{(G)}$ left module on the additive group M . In particular,

one will consider the group Z as a $\mathbf{Z}^{(G)}$ -module on the left, endowing it with the trivial action.

DEFINITION 1. — Let M be a $\mathbf{Z}^{(G)}$ left (resp. right) module, n be an integer ≥ 0 . The group $\text{Ext}_{\mathbf{Z}^{(G)}}^n(\mathbf{Z}, M)$ (resp. $\text{Tor}_n^{\mathbf{Z}^{(G)}}(M, \mathbf{Z})$) is denoted $H^n(G, M)$ (resp. $H_n(G, M)$) and called n -th cohomology group (resp. homology group) of G with coefficients in M .

The standard resolution (X, p. 58) $B(\mathbf{Z}^{(G)}, Z)$ is a free resolution of the $\mathbf{Z}^{(G)}$ -module Z ; it follows that the groups $H^n(G, M)$ (resp. $H_n(G, M)$) are identified with the homology groups of the complex:

$$\text{Hom}_{\mathbf{Z}^{(G)}}(B(\mathbf{Z}^{(G)}, Z), M) \quad (\text{resp. } M \otimes_{\mathbf{Z}^{(G)}} B(\mathbf{Z}^{(G)}, Z)).$$

Using the canonical isomorphism of $(\mathbf{Z}^{(G)})^{\otimes n}$ onto $\mathbf{Z}^{(G^n)}$ (III, p. 36) and the properties of extension of scalars (II, p. 82), we conclude that $H^n(G, M)$ is canonically isomorphic to the homology group of ascending degree n of the complex $C(G, M)$ defined in the following way: $C^n(G, M) = 0$ for $n < 0$; for $n \geq 0$, $C^n(G, M)$ is the $\mathbf{Z}$ -module of maps from G^n to M ; for $n \geq 0$, the differential $d^n : C^n(G, M) \rightarrow C^{n+1}(G, M)$ is given by

$$\begin{aligned} (d^n f)(g_0, \dots, g_n) &= g_0 \cdot f(g_1, \dots, g_n) + \sum_{i=0}^{n-1} (-1)^{i+1} f(g_0, \dots, g_i g_{i+1}, \dots, g_n) \\ &\quad + (-1)^{n+1} f(g_0, \dots, g_{n-1}) \end{aligned}$$

whatever f in $C^n(G, M)$ and $g_0, \dots, g_n$ in G .

Similarly, $H_n(G, M)$ is identified with the homology group of degree n of the complex $C'(G, M)$, where $C'_n(G, M) = M \otimes_{\mathbf{Z}} \mathbf{Z}^{(G^n)}$ for $n \geq 0$, $C'_n(G, M) = 0$ for $n < 0$, the differential $d_n : C'_n(G, M) \rightarrow C'_{n-1}(G, M)$ being defined by:

$$\begin{aligned} d_n(m \otimes e_{g_1, \dots, g_n}) &= m \cdot g_1 \otimes e_{g_2, \dots, g_n} + \sum_{i=1}^{n-1} (-1)^i m \otimes e_{g_1, \dots, g_i g_{i+1}, \dots, g_n} \\ &\quad + (-1)^n m \otimes e_{g_1, \dots, g_{n-1}} \end{aligned}$$

whatever $n \geq 1$, m in M and $g_1, \dots, g_n$ in G .

Examples. — 1) It follows directly from the definition that $H^0(G, M)$ is isomorphic to the submodule of elements of M invariant under the action of G , and $H^0(G, M)$ to the quotient module of M by the submodule generated by the elements $m \cdot g - m$ for $m \in M, g \in G$.

2) It follows from the preceding that $H^1(G, M)$ is isomorphic to the $\mathbb{Z}$ -module $Z^1(G, M)/B^1(G, M)$, where $Z^1(G, M)$ is the $\mathbb{Z}$ -module of functions f from G to M satisfying:

$$f(g_1 g_2) = g_1 \cdot f(g_2) + f(g_1) \quad \text{for all } g_1, g_2 \text{ in } G,$$

and $B^1(G, M)$ is the sub $\mathbb{Z}$ -module of $Z^1(G, M)$ consisting of f for which there exists an element m of M such that:

$$f(g) = g \cdot m - m \quad \text{for every } g \in G.$$

One sometimes says that $Z^1(G, M)$ is the $\mathbb{Z}$ -module of crossed homomorphisms from G to M , and $B^1(G, M)$ the submodule of principal crossed homomorphisms.

Let us denote by $\tau : G \rightarrow \text{Aut}(M)$ the homomorphism deduced from the action of G ; consider the external semidirect product $M \rtimes_\tau G$ and the extension $\xi_\tau : M \xrightarrow{i} M \rtimes_\tau G \xrightarrow{p} G$ (I, p. 64). Let $e : G \rightarrow M \rtimes_\tau G$ be a map such that $p \circ e = 1_G$; we have $e = (f, 1_G)$, where $f \in C^1(G, M)$. For e to be a homomorphism (that is, a section of the extension ξ_τ), it is necessary and sufficient that $f \in Z^1(G, M)$. For two sections of ξ_τ to be conjugate by an element of $i(M)$, it is necessary and sufficient that the corresponding crossed homomorphisms have the same class in $H^1(G, M)$.

When G acts trivially on M , we have $B^1(G, M) = 0$ and $H^1(G, M)$ is isomorphic to the $\mathbb{Z}$ -module of group homomorphisms from G to M .

3) Similarly $H^2(G, M)$ is isomorphic to the $\mathbb{Z}$ -module $Z^2(G, M)/B^2(G, M)$, where $Z^2(G, M)$ is the $\mathbb{Z}$ -module of maps f from $G \times G$ to M , satisfying:

$$g_1 \cdot f(g_2, g_3) - f(g_1 g_2, g_3) + f(g_1, g_2 g_3) - f(g_1, g_2) = 0$$

whatever g_1, g_2, g_3 in $\mathbf{G}$, and $\mathbf{B}^2(\mathbf{G}, \mathbf{M})$ is the sub- $\mathbf{Z}$ -module of $\mathbf{Z}^2(\mathbf{G}, \mathbf{M})$ consisting of f for which there exists a mapping h of $\mathbf{G}$ to $\mathbf{M}$ such that:

$$f(g_1, g_2) = g_1 \cdot h(g_2) - h(g_1 g_2) + h(g_1)$$

whatever g_1, g_2 in $\mathbf{G}$.

We thus recover the definition of the group $H^2(G, M)$ given in VIII, App. It follows in particular that there exists a canonical isomorphism of $H^2(G, M)$ on the group of extension classes of $\mathbf{G}$ by $\mathbf{M}$ (loc. cit.).

- 4) Let $\mathbf{M}$ be a $\mathbf{Z}$ -module, which we consider as a $\mathbf{Z}^{(G)}$ -module on the right by letting $\mathbf{G}$ act trivially. The group $H_1(\mathbf{G}, \mathbf{M})$ is isomorphic to the quotient of $\mathbf{M} \otimes_{\mathbf{Z}} \mathbf{Z}^{(G)}$ by the sub- $\mathbf{Z}$ -module generated by the elements $m \otimes (e_{g_1 g_2} - e_{g_1} - e_{g_2})$ for m in $\mathbf{M}$, g_1, g_2 in $\mathbf{G}$; it follows easily that $H_1(\mathbf{G}, \mathbf{M})$ is isomorphic to $\mathbf{M} \otimes_{\mathbf{Z}} (\mathbf{G}/(\mathbf{G}, \mathbf{G}))$.

Let us denote by σ the anti-automorphism of $\mathbf{Z}^{(G)}$ defined by $\sigma(e_g) = e_{g^{-1}}$ for $g \in \mathbf{G}$. Every $\mathbf{Z}^{(G)}$ -module on the left can be considered as a $\mathbf{Z}^{(G)}$ -module on the right by means of σ , and conversely. This allows us, for example, to define the groups $H_q(\mathbf{G}, \mathbf{M})$ for a $\mathbf{Z}^{(G)}$ -module on the left $\mathbf{M}$, by setting $H_q(\mathbf{G}, \mathbf{M}) = H_q(\mathbf{G}, \sigma_*(\mathbf{M})) = \text{Tor}_q^{\mathbf{Z}^{(G)}}(\mathbf{Z}, \mathbf{M})$.

Lemma 1. — Let $\mathbf{M}$ be a $\mathbf{Z}$ -module; denote by $\mathbf{M}^G$ the group $\text{Hom}_{\mathbf{Z}}(\mathbf{Z}^{(G)}, \mathbf{M})$ equipped with its natural $\mathbf{Z}^{(G)}$ -module structure on the left. Then:

$$H^i(\mathbf{G}, \mathbf{M}^G) = 0 \quad \text{for } i \geq 1.$$

It follows in fact from prop. 8, b) (X, p. 109), applied with $\mathbf{A} = \mathbf{N} = \mathbf{Z}^{(G)}$ and $\mathbf{B} = \mathbf{Q} = \mathbf{Z}$, that we have a canonical isomorphism:

$$\text{Ext}_{\mathbf{Z}^{(G)}}(\mathbf{Z}, \mathbf{M}^G) \rightarrow \text{Ext}_{\mathbf{Z}}(\mathbf{Z}, \mathbf{M})$$

whence the lemma.

PROPOSITION 11. — Let $\mathbf{L}$ be a finite Galois extension of a commutative field $\mathbf{K}$, with Galois group $\mathbf{G}$.

- a) One has $H^i(\mathbf{G}, \mathbf{L}) = 0$ for $i \geq 1$
- b) One has $H^1(\mathbf{G}, \mathbf{L}^*) = 0$
- c) The group $H^2(\mathbf{G}, \mathbf{L}^*)$ is canonically isomorphic to the group $\text{Br}(\mathbf{K}, \mathbf{L})$ (VIII, § 13).

The normal basis theorem (V, §10, no. 9, th. 6) shows that $\mathbf{L}$ is isomorphic as $\mathbf{Z}^{(G)}$ -module to $\mathbf{K}^G = \text{Hom}_{\mathbf{Z}}(\mathbf{Z}^{(G)}, \mathbf{K})$; the assertion *a*) then follows from Lemma 1. In view of Example 2, the assertion *b*) follows from V, §10, no. 5, cor. 1 to prop. 9; finally, the assertion *c*) was proved in VIII, §13.

10.7 COMPOSITION PRODUCT

We resume the general conventions of § 4.

10.7.1 The homomorphism $\text{Ext}_{\mathbf{A}}(\mathbf{N}, \mathbf{P}) \otimes \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}) \rightarrow \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{P})$

Let $\mathbf{M}$ and $\mathbf{N}$ be two left $\mathbf{A}$ -modules. Consider the homomorphisms $p_{\mathbf{M}} : \mathbf{L}(\mathbf{M}) \rightarrow \mathbf{M}$, $e_{\mathbf{M}} : \mathbf{M} \rightarrow \mathbf{I}(\mathbf{M})$ and $e_{\mathbf{M}} \circ p_{\mathbf{M}} : \mathbf{L}(\mathbf{M}) \rightarrow \mathbf{I}(\mathbf{M})$; by prop. 4 of X, p. 86, we deduce from this a bijective homomorphism

$$a_{\mathbf{M}, \mathbf{N}} = H(\text{Homgr}_{\mathbf{A}}(e_{\mathbf{M}} \circ p_{\mathbf{M}}, 1)) : H(\text{Homgr}_{\mathbf{A}}(\mathbf{I}(\mathbf{M}), \mathbf{I}(\mathbf{N}))) \rightarrow \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}).$$

Recall moreover (X, p. 82) that $H^n(\text{Homgr}_{\mathbf{A}}(\mathbf{I}(\mathbf{M}), \mathbf{I}(\mathbf{N})))$ is the set of homotopy classes of morphisms of ascending degree n from the complex $\mathbf{I}(\mathbf{M})$ into the complex $\mathbf{I}(\mathbf{N})$. For example, if $f \in \text{Hom}_{\mathbf{A}}(\mathbf{M}, \mathbf{N})$ the homotopy class of $\mathbf{I}(f)$ is sent by $a_{\mathbf{M}, \mathbf{N}}$ to f .

If $\mathbf{P}$ is a third left $\mathbf{A}$ -module, we have, according to X, p. 99, a canonical k -homomorphism

$$H(\text{Homgr}_{\mathbf{A}}(\mathbf{I}(\mathbf{N}), \mathbf{I}(\mathbf{P}))) \otimes_k H(\text{Homgr}_{\mathbf{A}}(\mathbf{I}(\mathbf{M}), \mathbf{I}(\mathbf{N}))) \rightarrow H(\text{Homgr}_{\mathbf{A}}(\mathbf{I}(\mathbf{M}), \mathbf{I}(\mathbf{P})))$$

from which we deduce by transport of structure via the isomorphisms $a_{\mathbf{N}, \mathbf{P}}, a_{\mathbf{M}, \mathbf{N}}, a_{\mathbf{M}, \mathbf{P}}$ a k -homomorphism (composition homomorphism):

$$c_{\mathbf{M},\mathbf{N},\mathbf{P}} : \text{Ext}_{\mathbf{A}}(\mathbf{N}, \mathbf{P}) \otimes_k \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}) \rightarrow \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{P}).$$

This corresponds to a k -bilinear map.

$$\text{Ext}_{\mathbf{A}}(\mathbf{N}, \mathbf{P}) \times \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}) \rightarrow \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{P}) \quad (1)$$

which decomposes into homogeneous components

$$\text{Ext}_{\mathbf{A}}^i(\mathbf{N}, \mathbf{P}) \times \text{Ext}_{\mathbf{A}}^j(\mathbf{M}, \mathbf{N}) \rightarrow \text{Ext}_{\mathbf{A}}^{i+j}(\mathbf{M}, \mathbf{P}). \quad (2)$$

If $u \in \text{Ext}_{\mathbf{A}}(\mathbf{N}, \mathbf{P})$, $v \in \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N})$, the image of (u, v) under (1) is called the composition product of u and v and is denoted $u \circ v$. If g (resp. f) is a morphism of complexes of ascending degree j (resp. i) from $\mathbf{I}(\mathbf{M})$ to $\mathbf{I}(\mathbf{N})$ (resp. from $\mathbf{I}(\mathbf{N})$ to $\mathbf{I}(\mathbf{P})$), of homotopy class $\bar{g}$ (resp. $\bar{f}$), then the composition product $a_{\mathbf{N},\mathbf{P}}(\bar{f}) \circ a_{\mathbf{M},\mathbf{N}}(\bar{g})$ is the image under $a_{\mathbf{M},\mathbf{P}}$ of the homotopy class of the morphism $f \circ g$ from $\mathbf{I}(\mathbf{M})$ into $\mathbf{I}(\mathbf{P})$.

Example 1. — If $u \in \text{Hom}_{\mathbf{A}}(\mathbf{N}, \mathbf{P})$, $v \in \text{Hom}_{\mathbf{A}}(\mathbf{M}, \mathbf{N})$, $u \circ v$ is the composite of u and of v .

Example 2. — If $u \in \text{Hom}_{\mathbf{A}}(\mathbf{N}, \mathbf{P})$, $v \in \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N})$, then

$$u \circ v = \text{Ext}_{\mathbf{A}}(1_{\mathbf{M}}, u)(v) \in \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{P});$$

likewise, if $u \in \text{Ext}_{\mathbf{A}}(\mathbf{N}, \mathbf{P})$, $v \in \text{Hom}_{\mathbf{A}}(\mathbf{M}, \mathbf{N})$, then

$$u \circ v = \text{Ext}_{\mathbf{A}}(v, 1_{\mathbf{P}})(u) \in \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{P}).$$

This follows from the definitions and remarks in X, p. 88.

If $\mathbf{Q}, \mathbf{M}, \mathbf{N}, \mathbf{P}$ are four left $\mathbf{A}$ -modules, and if

$$u \in \text{Ext}_{\mathbf{A}}(\mathbf{N}, \mathbf{P}), \quad v \in \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}), \quad w \in \text{Ext}_{\mathbf{A}}(\mathbf{Q}, \mathbf{M}),$$

then $(u \circ v) \circ w = u \circ (v \circ w)$: the composition product is associative; we will therefore write the composites of several elements without parentheses. In particular, according to Example 2:

Example 3. — Let $\mathbf{M}, \mathbf{N}, \mathbf{M}', \mathbf{N}'$ be four left $\mathbf{A}$ -modules. If $u \in \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N})$, $f \in$

$\text{Hom}_A(\mathbf{M}', \mathbf{M})$, $g \in \text{Hom}_A(\mathbf{N}, \mathbf{N}')$, then

$$\text{Ext}_A(f, g)(u) = g \circ u \circ f \in \text{Ext}_A(\mathbf{M}', \mathbf{N}'). \quad (3)$$

This gives a new proof of the k -bilinearity of the mapping $(f, g) \rightarrow \text{Ext}_A(f, g)$ (X, p. 88, prop. 6).

10.7.2 The seven computations of the composition product

Let M, M' , and M'' be three left A -modules, $a : R \rightarrow M$, $a' : R' \rightarrow M'$ and $a'' : R'' \rightarrow M''$ projective resolutions, $c : M \rightarrow E$, $c' : M' \rightarrow E'$ and $c'' : M'' \rightarrow E''$ injective resolutions. It follows from X, p. 100, th. 1 and p. 103, prop. 2, that the diagram:

$$\begin{array}{ccccc} \text{H}(\text{Homgr}_A(M, E')) & \rightarrow & \text{H}(\text{Homgr}_A(R, E')) & \rightarrow & \text{H}(\text{Homgr}_A(R, M')) \\ & \uparrow & \xrightarrow{\varphi(M, E')} & \downarrow \varphi(R, E') & \uparrow \\ \text{H}(\text{Homgr}_A(E, E')) & \xrightarrow{\varphi(E, E')} & \text{Ext}_A(M, M') & \xleftarrow{\varphi(R, R')} & \text{H}(\text{Homgr}_A(R, R')) \end{array}$$

where the unlabeled arrows are canonically deduced from c , a , c' , a' , is commutative, and that all the arrows are isomorphisms, which gives five descriptions of $\text{Ext}_A(\mathbf{M}, \mathbf{M}')$. One obtains likewise five descriptions of $\text{Ext}_A(\mathbf{M}', \mathbf{M}'')$, and as many of $\text{Ext}_A(\mathbf{M}, \mathbf{M}'')$.

Consider now the seven composition homomorphisms

$$\text{H}(\text{Homgr}_A(C', C'')) \otimes_k \text{H}(\text{Homgr}_A(C, C')) \rightarrow \text{H}(\text{Homgr}_A(C, C'')) ,$$

where one takes successively for (C, C', C'') the seven triples (R, R', R'') , (R, R', M'') , (R, R', E'') , (R, M', E'') , (R, E', E'') , (M, E', E'') , (E, E', E'') .

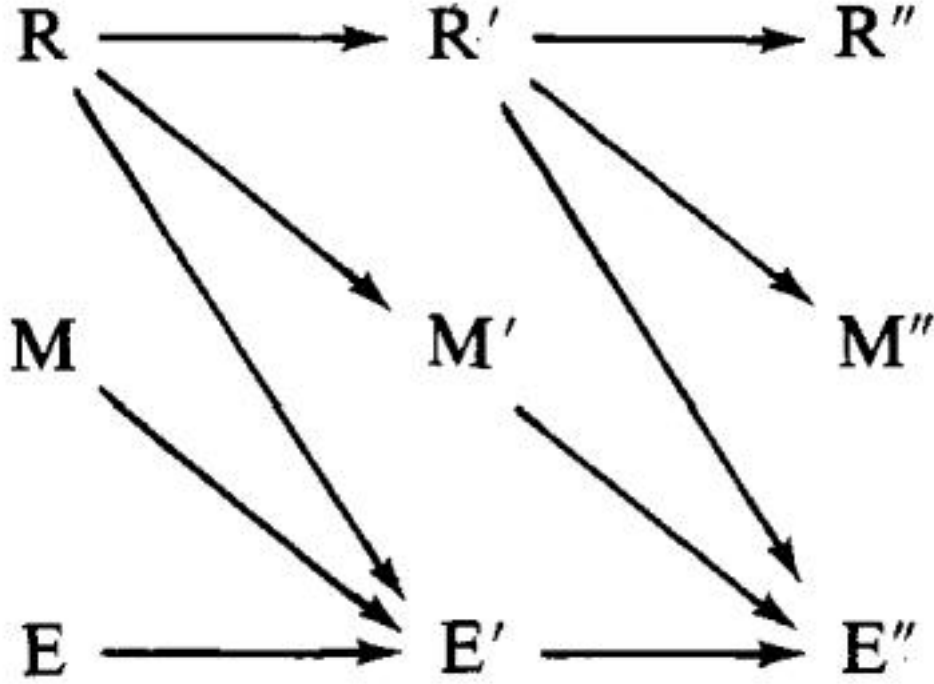


FIG. 1.

Identifying $H(\text{Homgr}_A(C, C'))$ with $\text{Ext}(M, M')$ by the above isomorphism, and likewise for $H(\text{Homgr}_A(C', C''))$ and $H(\text{Homgr}_A(C, C''))$, one obtains seven homomorphisms

$$\text{Ext}_A(M', M'') \otimes_k \text{Ext}_A(M, M') \rightarrow \text{Ext}_A(M, M'') .$$

These seven homomorphisms coincide, and are independent of the choice of resolutions. In particular, they coincide with the homomorphism defined in no. 2, via the triple $(I(M), I(M'), I(M''))$. This follows indeed from the interpretation of the modules $H(\text{Homgr}(C, C'))$ as the module of homotopy classes of morphisms of complexes from C to C' , and from the fact that if in a diagram of complexes

$$\begin{array}{ccccc} C & \xrightarrow{f} & C' & \xrightarrow{g} & C'' \\ \alpha \downarrow & & \alpha' \downarrow & & \alpha'' \downarrow \\ C_1 & \xrightarrow{f_1} & C'_1 & \xrightarrow{g_1} & C''_1 \end{array}$$

$\alpha'' \circ g$ is homotopic to $g_1 \circ \alpha'$ and $\alpha' \circ f$ homotopic to $f_1 \circ \alpha$, then $\alpha'' \circ g \circ f$ is homotopic to $g_1 \circ f_1 \circ \alpha$ (X, p. 33, prop. 4 and cor.).

In what follows, we shall use, depending on the case, one or another of the seven preceding

constructions of composition homomorphisms.

10.7.3 The class associated with an exact sequence

PROPOSITION 1. — Let (C, d) and (C', d') be two complexes of left A -modules and n, p, q three integers such that $p \geq q$. For $p \geq i \geq q-1$, let $f_i : C_i \rightarrow C'_{i+n+1}$ be a homomorphism of A -modules such that $f_p \circ d = 0, f_i \circ d = d' \circ f_{i+1}$ for $p > i \geq q-1$, and $d' \circ f_{q-1} = 0$ (see fig. 2).

$$\begin{array}{ccccccccccccccc}
 C_{p+1} & \xrightarrow{d} & C_p & \xrightarrow{d} & C_{p-1} & \xrightarrow{d} & \dots & \xrightarrow{d} & C_q & \xrightarrow{d} & C_{q-1} & \xrightarrow{d} & C_{q-2} \\
 \downarrow 0 & & \downarrow f_p & \searrow \alpha & \downarrow f_{p-1} & & & & \downarrow f_q & \searrow \beta & \downarrow f_{q-1} & & \downarrow 0 \\
 C'_{p+n+2} & \xrightarrow{d'} & C'_{p+n+1} & \xrightarrow{d'} & C'_{p+n} & \xrightarrow{d'} & \dots & \xrightarrow{d'} & C'_{q+n+1} & \xrightarrow{d'} & C'_{q+n} & \xrightarrow{d'} & C'_{q+n-1}
 \end{array}$$

FIG. 2.

Set $\alpha = f_{p-1} \circ d = d' \circ f_p$, $\beta = f_{q-1} \circ d = d' \circ f_q$, and let a (resp. b) be the graded A -homomorphism of degree n from C to C' whose only nonzero bihomogeneous component is α (resp. β). Then one has $a \in Z_n(\text{Homgr}_A(C, C'))$, $b \in Z_n(\text{Homgr}_A(C, C'))$ and $a - (-1)^{(n+1)(p-q)}b \in B_n(\text{Homgr}_A(C, C'))$.

One has $d' \circ \alpha = d' \circ d' \circ f_p = 0$, $\alpha \circ d = f_{p-1} \circ d \circ d = 0$, hence

$$a \in Z_n(\text{Homgr}_A(C, C'));$$

likewise $b \in Z_n(\text{Homgr}_A(C, C'))$. Set $\varepsilon = (-1)^{n+1}$. One has, in the complex $\text{Homgr}_A(C, C')$ the relations

$$Df_{p-1} = d' \circ f_{p-1} - \varepsilon f_{p-1} \circ d = f_{p-2} \circ d - \varepsilon \alpha$$

$$Df_i = d' \circ f_i - \varepsilon f_i \circ d = f_{i-1} \circ d - \varepsilon f_i \circ d \quad (p-1 > i > q)$$

$$Df_q = d' \circ f_q - \varepsilon f_q \circ d = \beta - \varepsilon f_q \circ d,$$

hence

$$\sum_{i=1}^{p-q} \varepsilon^i Df_{p-i} = \varepsilon^{p-q} \beta - \alpha,$$

which proves the lemma.

Consider two left A -modules M and N and an exact sequence of A -modules

$$0 \rightarrow N \rightarrow R_n \rightarrow R_{n-1} \rightarrow \dots \rightarrow R_1 \rightarrow M \rightarrow 0. \quad (4)$$

By props. 3 and 3 bis of X, p. 49, there exists a commutative diagram:

(5)

$$\begin{array}{ccccccccccccccc} 0 & \longrightarrow & N & \xrightarrow{e_N} & I^0(N) & \longrightarrow & \dots & \longrightarrow & I^{n-1}(N) & \xrightarrow{\delta^{n-1}} & I^n(N) & \xrightarrow{\delta^n} & I^{n+1}(N) \\ \uparrow & & \uparrow I_N & & \uparrow v^0 & & & & \uparrow v^{n-1} & & \uparrow v^n & & \uparrow \\ 0 & \longrightarrow & N & \longrightarrow & R_n & \longrightarrow & \dots & \longrightarrow & R_1 & \longrightarrow & M & \longrightarrow & 0 \\ \uparrow & & \uparrow u_n & & \uparrow u_{n-1} & & & & \uparrow u_0 & & \uparrow I_M & & \uparrow \\ L_{n+1}(M) & \xrightarrow{d_{n+1}} & L_n(M) & \xrightarrow{d_n} & L_{n-1}(M) & \longrightarrow & \dots & \xrightarrow{d_1} & L_0(M) & \xrightarrow{p_M} & M & \longrightarrow & 0 \end{array}$$

Consider the two elements b and a of $\text{Homgr}_A(L(M), I(N))$ whose only nonzero bihomogeneous components are

$$b^n = e_N \circ u_n : L_n(M) \rightarrow I^0(N) \quad \text{and} \quad a^n = v^n \circ p_M : L_0(M) \rightarrow I^n(N)$$

respectively.

PROPOSITION 2. — One has $a, b \in Z^n(\text{Homgr}_A(L(M), I(N)))$. Moreover, the classes $\bar{a}$ and $\bar{b}$ of a and b in $H^n(\text{Homgr}_A(L(M), I(N))) = \text{Ext}_A^n(M, N)$ depend only on the exact sequence (4) and are equal.

By prop. 1, applied to the two extreme rows of (5) and to the composed vertical arrows, with $p = n, q = 0$, we have $a, b \in Z^n(\text{Homgr}_A(I(M), L(N)))$ and

$$a - b = a - (-1)^{(n+1)n} b \in B^n(\text{Homgr}_A(L(M), I(N))).$$

Since a (resp. b) is independent of the choice of u (resp. v), the element $\bar{a} = \bar{b}$ of $\text{Ext}_A^n(M, N)$ is independent of the choice of morphisms u and v , whence the proposition.

DEFINITION 1. — The element θ of $\text{Ext}_{\mathbf{A}}^n(\mathbf{M}, \mathbf{N})$ defined by $\theta = (-1)^{n(n+1)/2} \bar{a} = (-1)^{n(n+1)/2} \bar{b}$ is called the class associated with the exact sequence (4) .

Remarks. — 1) Let (P, p) be a projective resolution of M . By X, p. 49, prop. 3, there exists a commutative diagram

$$\begin{array}{ccccccc} 0 & \rightarrow & N & \rightarrow & R_n & \rightarrow & \cdots \rightarrow R_1 \rightarrow M \rightarrow 0 \\ & & \tilde{u}_n \uparrow & & \tilde{u}_{n-1} \uparrow & & \tilde{u}_0 \uparrow & \uparrow_M \\ & & P_n & \rightarrow & P_{n-1} & \rightarrow & \cdots \rightarrow P_0 \xrightarrow{p} M \rightarrow 0. \end{array}$$

With the notations of § 6, θ is the image under $\varphi(P, N)$ of the homotopy class of the morphism $P \rightarrow N$ defined by $(-1)^{n(n+1)/2} \tilde{u}_n$. Similarly, if (e, E) is an injective resolution of N , there exists a commutative diagram

$$\begin{array}{ccccccc} & & & & 0 & \rightarrow & N \rightarrow E^0 \rightarrow \cdots \rightarrow E^{n-1} \rightarrow E^n \\ 1_N \uparrow & & \tilde{v}^0 \uparrow & & & & & & 0 \rightarrow N \rightarrow R_n \rightarrow \cdots \rightarrow R_1 \rightarrow M \end{array}$$

and θ is the image under $\varphi(\mathbf{M}, \mathbf{E})$ of the homotopy class of the morphism $\mathbf{M} \rightarrow \mathbf{E}$ defined by $(-1)^{n(n+1)/2} v^n$. This follows from the construction of θ and definitions of $\varphi(\mathbf{P}, \mathbf{N})$ and $\varphi(\mathbf{M}, \mathbf{E})$.

2) When $n = 0$, the exact sequence (4) is written $0 \rightarrow \mathbf{N} \xrightarrow{f} \mathbf{M} \rightarrow 0$, and the associated class is $f^{-1} \in \text{Hom}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}) = \text{Ext}_{\mathbf{A}}^0(\mathbf{M}, \mathbf{N})$.

10.7.4 Properties of the class associated with an exact sequence

PROPOSITION 3. — Let

(6)

$$0 \rightarrow P \rightarrow S_m \rightarrow S_{m-1} \rightarrow \cdots \rightarrow S_1 \xrightarrow{\lambda_n} N \rightarrow 0 \quad (7)$$

$$0 \rightarrow N \xrightarrow{\mu} R_n \rightarrow R_{n-1} \rightarrow \cdots \rightarrow R_1 \rightarrow M \rightarrow 0$$

two exact sequences of left A -modules of respective classes $\theta \in \text{Ext}_A^m(\mathbf{N}, \mathbf{P})$ and $\theta' \in \text{Ext}_A^n(\mathbf{M}, \mathbf{N})$. The class in $\text{Ext}_A^{m+n}(\mathbf{M}, \mathbf{P})$ associated with the exact sequence

$$0 \rightarrow \mathbf{P} \rightarrow \mathbf{S}_m \rightarrow \dots \rightarrow \mathbf{S}_1 \xrightarrow{\mu \circ \lambda} \mathbf{R}_n \rightarrow \dots \rightarrow \mathbf{R}_1 \rightarrow \mathbf{M} \rightarrow 0 \quad (8)$$

is the composition product $\theta \circ \theta'$.

Let us choose commutative diagrams

$$\begin{array}{ccccccc} 0 & \rightarrow & \mathbf{P} & \rightarrow & \mathbf{I}^0(\mathbf{P}) & \rightarrow & \dots \rightarrow \mathbf{I}^{m-1}(\mathbf{P}) \xrightarrow{\delta_P^{m-1}} \mathbf{I}^m(\mathbf{P}) \\ & & & & & & \\ 0 & \rightarrow & \mathbf{P} & \rightarrow & \mathbf{S}_m & \rightarrow & \dots \rightarrow \mathbf{S}_1 \xrightarrow{\lambda} \mathbf{N} \rightarrow 0, \\ & & & & & & \\ 0 & \rightarrow & \mathbf{N} & \xrightarrow{e_N} & \mathbf{I}^0(\mathbf{N}) & \rightarrow & \dots \rightarrow \mathbf{I}^{n-1}(\mathbf{N}) \rightarrow \mathbf{I}^n(\mathbf{N}) \\ & & \uparrow 1_N & & v^0 \uparrow & & v^{n-1} \uparrow & v^n \uparrow \\ 0 & \rightarrow & \mathbf{N} & \xrightarrow{\mu} & \mathbf{R}_n & \rightarrow & \dots \rightarrow \mathbf{R}_1 \rightarrow \mathbf{M} \rightarrow 0. \end{array}$$

Since $\mathbf{I}^m(\mathbf{P})$ is injective, there exists a homomorphism $h^0 : \mathbf{I}^0(\mathbf{N}) \rightarrow \mathbf{I}^m(\mathbf{P})$ such that $w^m = h^0 \circ e_N$; by X, p. 49, prop. 3 bis, h^0 extends to a morphism of complexes $h : \mathbf{I}(\mathbf{N}) \rightarrow \mathbf{I}(\mathbf{P})(-m)$. Then $w^m = h^0 \circ e_N = h^0 \circ v^0 \circ \mu$, hence

$$\delta_1^{m-1} \circ w^{m-1} = w^m \circ \lambda = h^0 \circ v^0 \circ (\mu \circ \lambda),$$

and the following diagram is commutative:

$$\begin{array}{ccccccccccc} 0 & \rightarrow & \mathbf{P} & \rightarrow & \mathbf{I}^0(\mathbf{P}) & \rightarrow & \dots \rightarrow \mathbf{I}^{m-1}(\mathbf{P}) & \rightarrow & \mathbf{I}^m(\mathbf{P}) & \rightarrow & \mathbf{I}^{m+1}(\mathbf{P}) & \rightarrow & \dots \rightarrow \mathbf{I}^{m+n}(\mathbf{P}) \\ 0 & \rightarrow & \mathbf{P} & \rightarrow & \mathbf{S}_m & \rightarrow & \dots \rightarrow \mathbf{S}_1 & \xrightarrow{\mu \circ \lambda} & \mathbf{R}_n & \rightarrow & \mathbf{R}_{n-1} & \rightarrow & \dots \rightarrow \mathbf{M} \rightarrow 0 \end{array}$$

where $t^0 = h^0 \circ v^0$, $t^1 = (-1)^m h^1 \circ v^1, \dots, t^i = (-1)^{mi} h^i \circ v^i, \dots, t^n = (-1)^{mn} h^n \circ v^n$. The class θ associated with (6) is that of $(-1)^{m(m+1)/2} w^m \in \text{Homgr}_A^m(\mathbf{N}, \mathbf{I}(\mathbf{P}))$, hence corresponds under the isomorphism $a_{\mathbf{N}, \mathbf{P}}$ to the class of $(-1)^{m(m+1)/2} h \in \text{Homgr}_A^m(\mathbf{I}(\mathbf{N}), \mathbf{I}(\mathbf{P}))$; the class θ' associated with (7) is that of $(-1)^{n(n+1)/2} v^n \in \text{Homgr}_A^m(\mathbf{M}, \mathbf{I}(\mathbf{N}))$, the class associated with (8) is that of $(-1)^{(m+n)(m+n+1)/2} t^n \in \text{Homgr}_{m+n}^m(\mathbf{M}, \mathbf{I}(\mathbf{P}))$ whence the conclusion, by the definition of the composition product (X, p. 114) and the formula

$$m(m+1)/2 + n(n+1)/2 = (m+n)(m+n+1)/2 - mn.$$

4. — Consider a commutative diagram of A-modules with exact

rows

$$\begin{array}{ccccccccccc} 0 & \longrightarrow & N & \longrightarrow & R_n & \longrightarrow & R_{n-1} & \longrightarrow & \dots & \longrightarrow & R_1 & \longrightarrow & M & \longrightarrow & 0 \\ & & \downarrow g & & \downarrow & & \downarrow & & & & \downarrow & & \downarrow f & & \\ 0 & \longrightarrow & N' & \longrightarrow & R'_n & \longrightarrow & R'_{n-1} & \longrightarrow & \dots & \longrightarrow & R'_1 & \longrightarrow & M' & \longrightarrow & 0. \end{array}$$

Let θ (resp. θ') be the class of the first (resp. second) row in

$$\text{Ext}_A^n(M, N) \quad (\text{resp. } \text{Ext}_A^n(M', N')).$$

In $\text{Ext}_A^n(M, N')$, one has $\theta' \circ f = g \circ \theta$.

Indeed, consider a commutative diagram

$$\begin{array}{ccccccccccc} L_n(M) & \xrightarrow{d_n} & L_{n-1}(M) & \longrightarrow & \dots & \longrightarrow & L_0(M) & \xrightarrow{p_M} & M & \longrightarrow & 0 \\ & & \downarrow u_n & & \downarrow & & \downarrow & & \downarrow 1 & & \\ 0 & \longrightarrow & N & \longrightarrow & R_n & \longrightarrow & \dots & \longrightarrow & R_1 & \longrightarrow & M & \longrightarrow & 0 \\ & & \downarrow g & & \downarrow & & \downarrow & & \downarrow f & & & & \\ 0 & \longrightarrow & N' & \longrightarrow & R'_n & \longrightarrow & \dots & \longrightarrow & R'_1 & \longrightarrow & M' & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow v^n & & & & \\ 0 & \longrightarrow & N' & \xrightarrow{e_{N'}} & I^0(N') & \xrightarrow{\delta^0} & \dots & \longrightarrow & I^{n-1}(N') & \xrightarrow{\delta^{n-1}} & I^n(N') \end{array}$$

By definition $\theta' \circ f$ is the class of $(-1)^{n(n+1)/2} v^n \circ f \circ p_M \in \text{Homgr}^n(L(M), I(N'))$, whereas $g \circ \theta$ is the class of $(-1)^{n(n+1)/2} e_{N'} \circ g \circ u_n$. By Lemma 1, applied to the two extreme rows of the diagram, these two classes are equal.

COROLLARY 1. — Consider a commutative diagram with exact rows

$$\begin{array}{ccccccccccc} 0 & \longrightarrow & N & \longrightarrow & R_n & \longrightarrow & \dots & \longrightarrow & R_1 & \longrightarrow & M & \longrightarrow & 0 \\ & & \downarrow 1 & & & & \downarrow & & & & \downarrow 1 & & \\ 0 & \longrightarrow & N & \longrightarrow & R'_n & \longrightarrow & \dots & \longrightarrow & R'_1 & \longrightarrow & M & \longrightarrow & 0; \end{array}$$

the two rows of the diagram have the same associated class in $\text{Ext}_{\mathbf{A}}^n(\mathbf{M}, \mathbf{N})$.

COROLLARY 2. — Let

$$0 \rightarrow \mathbf{N} \xrightarrow{f_{n-1}} \mathbf{R}_n \xrightarrow{f_n} \mathbf{R}_{n-1} \rightarrow \dots \xrightarrow{f_2} \mathbf{R}_1 \xrightarrow{f_1} \mathbf{M} \rightarrow 0$$

be an exact sequence, $\theta \in \text{Ext}_{\mathbf{A}}^n(\mathbf{M}, \mathbf{N})$ the associated class, $a_1, \dots, a_{n+1}$ invertible elements of k . The class associated with the exact sequence

$$0 \rightarrow \mathbf{N} \xrightarrow{a_{n+1}f_{n+1}} \mathbf{R}_n \xrightarrow{a_nf_n} \mathbf{R}_{n-1} \rightarrow \dots \xrightarrow{a_2f_2} \mathbf{R}_1 \xrightarrow{a_1f_1} \mathbf{M} \rightarrow 0$$

is $(a_1^{-1}a_2^{-1} \dots a_{n+1}^{-1})\theta$.

Indeed, we have a commutative diagram

$$\begin{array}{ccccccc} 0 \rightarrow \mathbf{N} & \xrightarrow{a_{n+1}f_{n+1}} & \mathbf{R}_n & \rightarrow \dots \rightarrow & \mathbf{R}_2 & \xrightarrow{a_2f_2} & \mathbf{R}_1 \xrightarrow{a_1f_1} \mathbf{M} \rightarrow 0 \\ & \downarrow a_1 \dots a_{n+1} & \downarrow a_1 \dots a_n & & \downarrow a_1 a_2 & \downarrow a_1 & \downarrow 1 \\ 0 \rightarrow \mathbf{N} & \xrightarrow{f_{n+1}} & \mathbf{R}_n & \rightarrow \dots \rightarrow & \mathbf{R}_2 & \xrightarrow{f_2} & \mathbf{R}_1 \xrightarrow{f_1} \mathbf{M} \rightarrow 0. \end{array}$$

and we apply the proposition.

COROLLARY 3. — Let $0 \rightarrow \mathbf{N} \xrightarrow{f_{n+1}} \mathbf{R}_n \xrightarrow{f_n} \dots \rightarrow \mathbf{R}_1 \xrightarrow{f_1} \mathbf{M} \rightarrow 0$ be an exact sequence, θ its class in $\text{Ext}_{\mathbf{A}}^n(\mathbf{M}, \mathbf{N})$, $u : \mathbf{M}' \rightarrow \mathbf{M}$ and $v : \mathbf{N} \rightarrow \mathbf{N}'$ be two homomorphisms of $\mathbf{A}$ -modules.

a) The element $v \circ \theta$ of $\text{Ext}_{\mathbf{A}}^n(\mathbf{M}, \mathbf{N}')$ is equal to the class of the exact sequence

$$0 \rightarrow \mathbf{N}' \xrightarrow{f'_{n+1}} \mathbf{R}'_n \xrightarrow{f'_n} \mathbf{R}_{n-1} \xrightarrow{f_{n-1}} \dots \rightarrow \mathbf{R}_1 \rightarrow \mathbf{M} \rightarrow 0,$$

where $\mathbf{R}'_n$ is the $\mathbf{A}$ -module quotient of $\mathbf{R}_n \oplus \mathbf{N}'$ by the submodule formed by the pairs $(f_{n+1}(x), -v(x))$ for $x \in \mathbf{N}$, and where f'_{n+1} (resp. f'_n) is deduced from the canonical injection (resp. from $(f_n, 0)$) by passing to quotients.

b) The element $\theta \circ u$ of $\text{Ext}_{\mathbf{A}}^n(\mathbf{M}', \mathbf{N})$ is the class of the exact sequence

$$0 \rightarrow \mathbf{N} \rightarrow \mathbf{R}_n \rightarrow \dots \rightarrow \mathbf{R}_2 \xrightarrow{f''_2} \mathbf{R}'_1 \xrightarrow{f''_1} \mathbf{M}' \rightarrow 0,$$

where R'_1 is the fiber product $R_1 \times_M M'$, that is to say (I, p. 44) the submodule of $R_1 \times M'$ formed by the pairs (x, y) such that $f_1(x) = u(y)$, and where f'_2 (resp. f'_1) is deduced from $(f_2, 0)$ (resp. from the second projection).

Let us prove for example *a*). Let z be an element of $\mathbf{R}'_n$ such that $f'_n(z) = 0$; if z is the class of a pair (x, y) , with $x \in \mathbf{R}_n, y \in \mathbf{N}'$, we have $f_n(x) = 0$ so that there exists an element $t \in \mathbf{N}$ such that $x = f_{n+1}(t)$. We then have $z = f'_{n+1}(y + v(t))$ which proves that $\text{Ker } f'_n = \text{Im } f'_{n+1}$. The injectivity of f'_{n+1} follows from that of the f_{n+1} .

Let $j : \mathbf{R}_n \rightarrow \mathbf{R}'_n$ be the homomorphism deduced from the canonical injection; we have a commutative diagram of exact sequences:

$$\begin{array}{ccccccc} 0 \rightarrow \mathbf{N} & \xrightarrow{f_{n+1}} & \mathbf{R}_n & \xrightarrow{f_n} & \mathbf{R}_{n-1} & \rightarrow \cdots \rightarrow \mathbf{M} \rightarrow 0 \\ & & \downarrow v & \downarrow j & \downarrow 1 & & \\ 0 \rightarrow \mathbf{N}' & \xrightarrow{f'_{n+1}} & \mathbf{R}'_n & \xrightarrow{f'_n} & \mathbf{R}'_{n-1} & \rightarrow \cdots \rightarrow \mathbf{M} \rightarrow 0; \end{array}$$

assertion *a*) then follows from the proposition.

The proof of *b* is analogous.

Remark. — Let $\theta \in \text{Ext}^n_{\mathbf{A}}(\mathbf{M}, \mathbf{N})$, resp. $\theta' \in \text{Ext}^n_{\mathbf{A}}(\mathbf{M}', \mathbf{N}')$, be the class of an exact sequence

$$0 \rightarrow \mathbf{N} \xrightarrow{f_{n+1}} \mathbf{R}_n \rightarrow \cdots \rightarrow \mathbf{R}_1 \xrightarrow{f_1} \mathbf{M} \rightarrow 0,$$

$$\text{resp. } 0 \rightarrow \mathbf{N}' \xrightarrow{f'_{n+1}} \mathbf{R}'_n \rightarrow \cdots \rightarrow \mathbf{R}'_1 \xrightarrow{f'_1} \mathbf{M}' \rightarrow 0.$$

Let $i_{\mathbf{N}}, i_{\mathbf{N}'}$, be the canonical injections of $\mathbf{N}$ and $\mathbf{N}'$ into $\mathbf{N} \oplus \mathbf{N}'$, $q_{\mathbf{M}}, q_{\mathbf{M}'}$, the projections of $\mathbf{M} \oplus \mathbf{M}'$ onto $\mathbf{M}$ and $\mathbf{M}'$. Consider the homomorphism

$$m = \text{Ext}(q_{\mathbf{M}}, i_{\mathbf{N}}) \oplus \text{Ext}(q_{\mathbf{M}'}, i_{\mathbf{N}'})$$

of $\text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}) \oplus \text{Ext}_{\mathbf{A}}(\mathbf{M}', \mathbf{N}')$ to $\text{Ext}_{\mathbf{A}}(\mathbf{M} \oplus \mathbf{M}', \mathbf{N} \oplus \mathbf{N}')$. The element

$$m(\theta, \theta') = i_{\mathbf{N}} \circ \theta \circ q_{\mathbf{M}} + i_{\mathbf{N}'} \circ \theta' \circ q_{\mathbf{M}'}$$

is the class of the exact sequence

$$0 \rightarrow N \oplus N' \xrightarrow{f_{n+1} \oplus f'_{n-1}} R_n \oplus R'_n \rightarrow \dots \rightarrow R_1 \oplus R'_1 \xrightarrow{f_1 \oplus f'_1} M \oplus M' \rightarrow 0.$$

Indeed, if one denotes this class by θ'' , it follows from prop. 4 that one has

$$\theta'' \circ i_M = i_N \circ \theta = m(\theta, \theta') \circ i_M \quad \text{and} \quad \theta'' \circ i_{M'} = i_{N'} \circ \theta = m(\theta, \theta') \circ i_{M'};$$

according to X, p. 89, prop. 7, this implies $\theta'' = m(\theta, \theta')$.

10.7.5 Relation between exact sequences and elements of $\text{Ext}_A(M, N)$

THEOREM 1. — Let n be an integer ≥ 1 , and M and N two A -modules.

a) Every element of $\text{Ext}_A^n(M, N)$ is the class of an exact sequence (X, p. 117, def. 1).

b) Let $0 \rightarrow N \xrightarrow{f_{n+1}} R_n \xrightarrow{f_n} \dots \rightarrow R_1 \xrightarrow{f_1} M \rightarrow 0$

and

$$0 \rightarrow N \xrightarrow{f'_{n+1}} R'_n \xrightarrow{f'_n} \dots \rightarrow R'_1 \xrightarrow{f'_1} M \rightarrow 0$$

of exact sequences, θ and θ' the associated classes. The following conditions are equivalent:

(i) $\theta = \theta'$;

(ii) there exists a commutative diagram with exact rows:

$$\begin{array}{ccccccc} 0 & \rightarrow & N & \xrightarrow{f_{n+1}} & R_n & \rightarrow & \dots \rightarrow R_1 \xrightarrow{f_1} M \rightarrow 0 \\ & & & & & & \\ 0 & \rightarrow & N & \rightarrow & R''_n & \rightarrow & \dots \rightarrow R'_1 \rightarrow M \rightarrow 0 \\ & & & & & & \\ 0 & \rightarrow & N & \xrightarrow{f'_{n+1}} & R'_n & \rightarrow & \dots \rightarrow R'_1 \xrightarrow{f'_1} M \rightarrow 0; \end{array}$$

(iii) there exists a commutative diagram with exact rows:

$$\begin{aligned}
0 &\rightarrow N \xrightarrow{f_{n+1}} R_n \rightarrow \cdots \rightarrow R_1 \xrightarrow{f_1} M \rightarrow 0 \\
0 &\rightarrow N \rightarrow R''_n \rightarrow \cdots \rightarrow R''_1 \rightarrow M \rightarrow 0 \\
0 &\rightarrow N \xrightarrow{f'_{n+1}} R'_n \rightarrow \cdots \rightarrow R'_1 \xrightarrow{f'_1} M \rightarrow 0.
\end{aligned}$$

Let us prove a). Let $\alpha \in \text{Ext}_{\mathbf{A}}^n(\mathbf{M}, \mathbf{N})$, and let $\mathbf{P}$ be a projective resolution of $\mathbf{M}$. Let $a : \mathbf{P}(n) \rightarrow \mathbf{N}$ be a morphism of complexes representing α ; its unique nonzero component is an $\mathbf{A}$ -homomorphism $u : P_n \rightarrow \mathbf{N}$ which satisfies $u \circ d_{n+1} = 0$, hence factors through $u = \bar{u} \circ \delta_n$, where $\delta_n : P_n \rightarrow Z_{n-1}$ is the map induced by d_n (we set $Z_{n-1} = \text{Im } d_n$) and $\bar{u}$ is an $\mathbf{A}$ -homomorphism of Z_{n-1} to $\mathbf{N}$. By Remark 1, p. 117, the class $\theta \in \text{Ext}_{\mathbf{A}}^n(\mathbf{M}, Z_{n-1})$ of the exact sequence

$$0 \rightarrow Z_{n-1} \rightarrow P_{n-1} \rightarrow \cdots \rightarrow P_0 \rightarrow M \rightarrow 0$$

is equal to the homotopy class of the morphism $(-1)^{n(n+1)/2} \delta_n$. We therefore have

$$\alpha = (-1)^{n(n+1)/2} \bar{u} \circ \theta,$$

which, according to Cor. 3, p. 120, allows one to represent α as the class of an exact sequence.

It follows from cor. 1 of X, p. 120 that (ii) $\Rightarrow$ (i) and that (iii) $\Rightarrow$ (i). Suppose (i) is satisfied, and let $\mathbf{P}$ be a projective resolution of $\mathbf{M}$. There exists a commutative diagram

$$\begin{array}{ccccccc}
0 & \rightarrow & N & \xrightarrow{f_{n+1}} & R_n & \rightarrow & R_{n-1} \rightarrow \cdots \rightarrow R_1 \rightarrow M \rightarrow 0 \\
& & u_n \uparrow & & \uparrow u_{n-1} & & \uparrow u_{n-2} & & \uparrow & \uparrow_{I_M} \\
P_n & \xrightarrow{d_n} & P_{n-1} & \rightarrow & P_{n-2} & \rightarrow & \cdots \rightarrow P_0 \rightarrow M \rightarrow 0 \\
& & u'_n \downarrow & & \downarrow u'_{n-1} & & \downarrow u'_{n-2} & & \downarrow & \downarrow_{I_M} \\
0 & \rightarrow & N & \xrightarrow{f'_{n+1}} & R'_n & \rightarrow & R'_{n-1} \rightarrow \cdots \rightarrow R'_1 \rightarrow M \rightarrow 0.
\end{array}$$

The morphisms of $\mathbf{P}(n)$ to $\mathbf{N}$ defined by u_n and u'_n are homotopic, since they both belong to the class $(-1)^{n(n+1)/2} \theta$, hence $u'_n - u_n$ is of the form $w \circ d_n$, where $w : P_{n-1} \rightarrow \mathbf{N}$ is an $\mathbf{A}$ -homomorphism. By replacing u'_{n-1} by $u'_{n-1} - f'_{n+1} \circ w$ and u'_n by u_n one reduces to the case where $u_n = u'_n$. This allows one to construct a new commutative diagram with exact rows:

$$\begin{array}{ccccccc}
0 \rightarrow N \rightarrow R_n \rightarrow R_{n-1} \rightarrow \cdots \rightarrow R_1 \rightarrow M \rightarrow 0 \\
\qquad \qquad \qquad 1_N \uparrow \qquad \qquad \qquad \uparrow v \quad \uparrow u_{n-2} \quad \uparrow \quad \uparrow 1_M \\
0 \rightarrow N \rightarrow N' \rightarrow P_{n-2} \rightarrow \cdots \rightarrow P_0 \rightarrow M \rightarrow 0 \\
\qquad \qquad \qquad 1_N \downarrow \qquad \qquad \qquad \downarrow v' \quad \downarrow u'_{n-2} \quad \downarrow \quad \downarrow 1_M \\
0 \rightarrow N \rightarrow R'_n \rightarrow R'_{n-1} \rightarrow \cdots \rightarrow R'_1 \rightarrow M \rightarrow 0
\end{array}$$

where N' is the quotient of $P_{n-1} \oplus N$ by the submodule formed by the pairs $(d_n(x), -u_n(x))$ for $x \in P_n$, and where v (resp. v') is defined by passing to the quotient from the map $u_{n-1} \oplus f_{n+1}$ (resp. $u'_{n-1} \oplus f'_{n+1}$). Condition (ii) is therefore satisfied.

Suppose again that condition (i) is satisfied, and let E be an injective resolution of N . There exists a commutative diagram

$$\begin{array}{ccccccc}
0 \rightarrow N \rightarrow R_n \rightarrow \cdots \rightarrow R_1 \rightarrow M \rightarrow 0 \\
\qquad \qquad \qquad 1_N \downarrow \qquad \qquad \qquad \downarrow v_0 \qquad \qquad \qquad v_{n-1} \downarrow \quad \downarrow v_n \\
0 \rightarrow N' \rightarrow E^0 \xrightarrow{\delta^0} \cdots \rightarrow E^{n-1} \xrightarrow{\delta^{n-1}} E^n \\
\qquad \qquad \qquad 1_N \uparrow \qquad \qquad \qquad \uparrow v'_0 \qquad \qquad \qquad v'_{n-1} \uparrow \quad \uparrow v'_n \\
0 \rightarrow N \rightarrow R'_n \rightarrow \cdots \rightarrow R'_1 \rightarrow M \rightarrow 0
\end{array}$$

and one shows as above that one may assume $v'_n = v_n$. We then have a commutative diagram with exact rows

$$\begin{array}{ccccccc}
0 \rightarrow N \rightarrow R_n \rightarrow \cdots \rightarrow R_2 \rightarrow R_1 \rightarrow M \rightarrow 0 \\
\qquad \qquad \qquad \downarrow 1_N \qquad \qquad \qquad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow 1_M \\
0 \rightarrow N \rightarrow E^0 \rightarrow \cdots \rightarrow E^{n-2} \rightarrow M' \rightarrow M \rightarrow 0 \\
\qquad \qquad \qquad \uparrow 1_N \qquad \qquad \qquad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow 1_M \\
0 \rightarrow N \rightarrow R'_n \rightarrow \cdots \rightarrow R'_2 \rightarrow R'_1 \rightarrow M \rightarrow 0
\end{array}$$

with $M' = M \times_{R_1} E_{n-1}$ (cf. X, p. 120, cor. 3, b)). Condition (iii) is therefore satisfied, which completes the proof of the theorem.

Remark 1. — If the ring A is Noetherian, and if the A -modules M and N are finitely generated, it follows from the proof of a) that every element of $\text{Ext}_A^n(M, N)$ is the class associated with an exact sequence $0 \rightarrow N \rightarrow R_n \rightarrow \cdots \rightarrow R_1 \rightarrow M \rightarrow 0$ where the R_i are finitely generated.

COROLLARY. — Let $0 \rightarrow \mathbf{N} \xrightarrow{f} \mathbf{R} \xrightarrow{g} \mathbf{M} \rightarrow 0$ and $0 \rightarrow \mathbf{N} \xrightarrow{f'} \mathbf{R}' \xrightarrow{g'} \mathbf{M} \rightarrow 0$ be two exact sequences, θ and θ' the associated classes in $\text{Ext}^1(\mathbf{M}, \mathbf{N})$. For $\theta = \theta'$, it is necessary and sufficient that there exist an $\mathbf{A}$ -homomorphism $h : \mathbf{R} \rightarrow \mathbf{R}'$ making the diagram

$$\begin{array}{ccccc} \mathbf{R} & & f & & g \\ & \searrow & & \searrow & \\ \mathbf{N} & & h & & \mathbf{M} \\ & \swarrow & & \swarrow & \\ f' & & \mathbf{R}' & & g' \end{array}$$

commutative. Such a homomorphism is necessarily an isomorphism.

The condition is sufficient by cor. 1 of prop. 4. If $\theta = \theta'$, we have a commutative diagram with exact rows:

$$\begin{array}{ccccccc} 0 & \rightarrow & \mathbf{N} & \rightarrow & \mathbf{R} & \rightarrow & \mathbf{M} \rightarrow 0 \\ & & \uparrow 1_{\mathbf{N}} & & \downarrow h' & & \downarrow 1_{\mathbf{M}} \\ 0 & \rightarrow & \mathbf{N} & \rightarrow & \mathbf{R}'' & \rightarrow & \mathbf{M} \rightarrow 0 \\ & & \uparrow 1_{\mathbf{N}} & & \downarrow h'' & & \downarrow 1_{\mathbf{M}} \\ 0 & \rightarrow & \mathbf{N} & \rightarrow & \mathbf{R}' & \rightarrow & \mathbf{M} \rightarrow 0. \end{array}$$

The morphisms h' and h'' are isomorphisms by X, p. 7, cor. 3, and $h = h'' \circ h'^{-1}$ answers the question. The last assertion follows from loc. cit.

Remark 2. — Theorem 1 gives a description of $\text{Ext}_{\mathbf{A}}^n(\mathbf{M}, \mathbf{N})$ as a set of equivalence classes of exact sequences; it is easy to describe the group law obtained on this set by transport of structure. Indeed, let θ (resp. θ') be the class of an exact sequence $0 \rightarrow \mathbf{N} \xrightarrow{f_{n+1}} \mathbf{R}_n \xrightarrow{f_n} \dots \rightarrow \mathbf{R}_1 \rightarrow \mathbf{M} \rightarrow 0$ (resp. $0 \rightarrow \mathbf{N} \xrightarrow{f'_{n+1}} \mathbf{R}'_n \xrightarrow{f'_n} \dots \rightarrow \mathbf{R}'_1 \rightarrow \mathbf{M} \rightarrow 0$). Let $\Delta : \mathbf{M} \rightarrow \mathbf{M} \oplus \mathbf{M}$ and $\nabla : \mathbf{N} \oplus \mathbf{N} \rightarrow \mathbf{N}$ be the $\mathbf{A}$ -linear maps defined by $\Delta(x) = (x, x)$ for $x \in \mathbf{M}$ and $\nabla(y, z) = y + z$ for $y, z \in \mathbf{N}$. Consider the map

$$m : \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}) \oplus \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}) \rightarrow \text{Ext}_{\mathbf{A}}(\mathbf{M} \oplus \mathbf{M}, \mathbf{N} \oplus \mathbf{N})$$

defined in the remark, p. 121. With the notations of loc. cit., we have $\nabla \circ i_{\mathbf{N}} = 1_{\mathbf{N}}$ and $q_{\mathbf{M}} \circ \Delta = 1_{\mathbf{M}}$, and consequently $\theta + \theta' = \nabla \circ m(\theta, \theta') \circ \Delta$. In view of loc. cit. and cor. 3, p. 120, this yields an exact sequence of class $\theta + \theta'$: if for example $n \geq 2$, one may take the

sequence

$$0 \rightarrow N \rightarrow R''_n \rightarrow R_{n-1} \oplus R'_{n-1} \xrightarrow{f_{n-1} \oplus f'_{n-1}} \cdots \rightarrow R_2 \oplus R'_2 \rightarrow R'_1 \rightarrow M \rightarrow 0$$

where R''_n is the quotient of $R_n \oplus R'_n$ by the submodule formed by the pairs

$$(f_{n+1}(x), -f'_{n+1}(x)) \quad \text{for } x \in N,$$

and where $R'_1 = R_1 \times_M R'_1$

10.7.6 Composition product and connecting homomorphisms of extension modules

PROPOSITION 5. — Let

$$0 \rightarrow M' \xrightarrow{f} M \xrightarrow{g} M'' \rightarrow 0 \quad (8)$$

be an exact sequence of left A -modules, $\theta \in \text{Ext}_A^1(M'', M')$ the associated class, N a left A -module, n an integer.

- a) The connecting homomorphism $\delta^n(N, \mathcal{E}) : \text{Ext}_A^n(N, M'') \rightarrow \text{Ext}_A^{n+1}(N, M')$ is the composition product $\alpha \mapsto \theta \circ \alpha$ by θ .
- b) The connecting homomorphism $\delta^n(\mathcal{E}, N) : \text{Ext}_A^n(M', N) \rightarrow \text{Ext}_A^{n+1}(M'', N)$ is the composition product $\alpha \mapsto (-1)^{n+1} \alpha \circ \theta$ by $(-1)^{n+1} \theta$.
- c) Consider a commutative diagram

$$\begin{array}{ccccccc} 0 & \rightarrow & M' & \xrightarrow{f} & M & \xrightarrow{g} & M'' \rightarrow 0 \\ & & & & & & \\ 0 & \rightarrow & M' & \xrightarrow{e_{M'}} & I^0(M') & \xrightarrow{\delta^0} & I^1(M'). \end{array}$$

By definition, θ is the class of $-v^1 \in \text{Homgr}_A^1(M'', I(M'))$. Let moreover $\alpha \in \text{Ext}_A^n(N, M'')$, represented by an element a of $\text{Homgr}_A^n(L(N), M'')$. By construction, $\delta^n(\alpha)$ is obtained as follows: one lifts $a^n \in \text{Hom}_A(L_n(N), M'')$ to

$$b \in \text{Hom}_{\mathbf{A}}(L_n(\mathbf{N}), \mathbf{M}),$$

and $\delta^n(\alpha)$ is the class of $e_{\mathbf{M}'} \circ c$ where $c \in \text{Hom}_{\mathbf{A}}(L_{n+1}(\mathbf{N}), \mathbf{M}')$ is such that

$$f \circ c = Db = (-1)^{n+1} b \circ d_{n+1}.$$

We thus have a commutative diagram

$$\begin{array}{ccccccc} L_{n+1}(N) & \xrightarrow{d_{n+1}} & L_n(N) & & & & \\ & (-1)^{n+1} c \downarrow & & & & & \\ 0 \rightarrow M' & \xrightarrow{f} & M & \xrightarrow{g} & M'' & \rightarrow & 0 \\ & & 1_{M'} \downarrow & & & & \\ 0 \rightarrow M' & \xrightarrow{e_{M'}} & I^0(M') & \xrightarrow{\delta^0} & I^1(M') & & \end{array}$$

But, in $\text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{N}), \mathbf{I}(\mathbf{M}'))$, we have

$$D(v^0 \circ b) = \delta^0 \circ v^0 \circ b - (-1)^n v^0 \circ b \circ d_{n+1} = v^1 \circ a^n + e_{M'} \circ c.$$

The classes of $e_{\mathbf{M}'} \circ c$ and $(-v^1) \circ a$ in $\text{Ext}_{\mathbf{A}}^{n+1}(\mathbf{N}, \mathbf{M}')$ are equal, whence a).

b) Consider a commutative diagram

$$\begin{array}{ccccccc} L_1(M'') & \xrightarrow{d_1} & L_0(M'') & \xrightarrow{p_{M''}} & M'' & \rightarrow & 0 \\ & & \downarrow u_1 & & & & \\ 0 \rightarrow M' & \xrightarrow{f} & M & \xrightarrow{g} & M'' & \rightarrow & 0. \end{array}$$

By definition, θ is the class of $-u_1 \in \text{Homgr}_{\mathbf{A}}^1(L(\mathbf{M}''), \mathbf{M}')$. Let moreover $\alpha \in \text{Ext}^n(\mathbf{M}', \mathbf{N})$ represented by an element a of $\text{Homgr}_{\mathbf{A}}^n(\mathbf{M}', \mathbf{I}(\mathbf{N}))$. By construction, $\delta^n(\alpha)$ is obtained as follows: one extends $a^n \in \text{Hom}_{\mathbf{A}}(\mathbf{M}', I^n(\mathbf{N}))$ to

$$b \in \text{Hom}_{\mathbf{A}}(\mathbf{M}, I^n(\mathbf{N}))$$

and $\delta^n(\alpha)$ is the class of $c \circ p_{M''}$, where $c \in \text{Hom}_{\mathbf{A}}(\mathbf{M}'', I^{n+1}(\mathbf{N}))$ is such that

$$g \circ c = Db = \delta^{n+1} \circ b.$$

We thus have a commutative diagram

$$\begin{array}{ccccccc}
 & & L_1(M'') & \xrightarrow{d_1} & L_0(M'') & \xrightarrow{p_{M''}} & M'' \rightarrow 0 \\
 u_1 \downarrow & & & & & & \\
 & & 0 \rightarrow M' & \xrightarrow{f} & M & \xrightarrow{g} & M'' \rightarrow 0 \\
 & a^n \downarrow & & & & & \downarrow c \\
 & & I^n(N) & \xrightarrow{\delta^n} & I^{n+1}(N) & &
 \end{array}$$

But, in $\text{Homgr}_A(\mathbf{L}(\mathbf{M}''), \mathbf{I}(\mathbf{N}))$, we have

$$D(b \circ u_0) = \delta^n \circ b \circ u_0 - (-1)^n b \circ u_0 \circ c = c \circ p - (-1)^n a^n \circ u_1.$$

The classes of $c \circ p$ and of $(-1)^{n+1} a^n \circ (-u_1)$ in $\text{Ext}_A^{n+1}(\mathbf{M}'', \mathbf{N})$ are therefore equal, whence b).

COROLLARY 1. — a) The connecting homomorphism $\text{Hom}_A(\mathbf{M}'', \mathbf{M}'') \rightarrow \text{Ext}_A^1(\mathbf{M}'', \mathbf{M}')$ sends $1_{M''}$ to θ .

b) The connecting homomorphism $\text{Hom}_A(\mathbf{M}', \mathbf{M}') \rightarrow \text{Ext}_A^1(\mathbf{M}'', \mathbf{M}')$ sends $1_{M'}$ to $-\theta$.

COROLLARY 2. — Consider two exact sequences of left A-modules

$$0 \rightarrow \mathbf{M}' \rightarrow \mathbf{M} \rightarrow \mathbf{M}'' \rightarrow 0$$

$$0 \rightarrow \mathbf{N}' \rightarrow \mathbf{N} \rightarrow \mathbf{N}'' \rightarrow 0.$$

Then the composed homomorphisms of connecting homomorphisms

$$\text{Ext}_A^n(\mathbf{M}', \mathbf{N}'') \rightarrow \text{Ext}_A^{n+1}(\mathbf{M}'', \mathbf{N}'') \rightarrow \text{Ext}_A^{n+2}(\mathbf{M}'', \mathbf{N}')$$

and

$$\text{Ext}_{\mathbf{A}}^n(M', N'') \rightarrow \text{Ext}_{\mathbf{A}}^{n+1}(M', N') \rightarrow \text{Ext}_{\mathbf{A}}^{n+2}(M'', N')$$

are opposite.

Indeed, if θ_1, θ_2 are the classes associated with the given exact sequences, and if $\alpha \in \text{Ext}_{\mathbf{A}}^n(M', M'')$ the images of α are respectively

$$\theta_2 \circ ((-1)^{n+1} \alpha \circ \theta_1) \quad \text{and} \quad (\theta_2 \circ \alpha) \circ ((-1)^{n+2} \theta_1).$$

Consider an exact sequence of left $\mathbf{A}$ -modules

$$0 \rightarrow N \rightarrow R_n \xrightarrow{f_n} R_{n-1} \xrightarrow{f_{n-1}} \dots \rightarrow R_1 \xrightarrow{f_1} M \rightarrow 0 \quad (\text{S})$$

and set $K_0 = M, K_i = \text{Ker} f_i, i = 1, \dots, n-1, K_n = N$. We therefore have exact sequences

$$0 \rightarrow \mathbf{K}_i \rightarrow \mathbf{R}_i \rightarrow \mathbf{K}_{i-1} \rightarrow 0, \quad 1 \leq i \leq n, \quad (9)$$

to which are associated, for every left $\mathbf{A}$ -module P , connecting homomorphisms

$$\begin{aligned} \text{Ext}_{\mathbf{A}}^m(P, \mathbf{K}_{i-1}) &\rightarrow \text{Ext}_{\mathbf{A}}^{m+1}(P, \mathbf{K}_i), \\ \text{Ext}_{\mathbf{A}}^m(\mathbf{K}_i, P) &\rightarrow \text{Ext}_{\mathbf{A}}^{m+1}(\mathbf{K}_{i-1}, P), \end{aligned}$$

whence by composition of the iterated connecting homomorphisms, associated with (S)

$$\begin{aligned} \delta^m(P, \mathcal{S}) : \text{Ext}_{\mathbf{A}}^m(P, M) &\rightarrow \text{Ext}_{\mathbf{A}}^{m+n}(P, N) \\ \delta^m(\mathcal{S}, P) : \text{Ext}_{\mathbf{A}}^m(N, P) &\rightarrow \text{Ext}_{\mathbf{A}}^{m+n}(M, P). \end{aligned}$$

COROLLARY 3. — If $\theta \in \text{Ext}_{\mathbf{A}}^n(\mathbf{M}, \mathbf{N})$ is the class of the exact sequence $(\mathcal{S})$, we have

$$\delta^m(P, \mathcal{S})(\alpha) = \theta \circ \alpha, \quad \delta^m(\mathcal{S}, P)(\beta) = (-1)^{mn+n(n+1)/2} \beta \circ \theta.$$

If $\theta_i \in \text{Ext}_{\mathbf{A}}^1(\mathbf{K}_{i-1}, \mathbf{K}_i)$ is the class associated with the exact sequence (9), we have by prop.

$$\delta^m(\mathbf{P}, \mathcal{S})(\alpha) = \theta_n \circ \dots \circ \theta_2 \circ \theta_1 \circ \alpha$$

$$\delta^m(\mathcal{S}, \mathbf{P})(\beta) = (-1)^{(m+1)+\dots+(m+n)} \beta \circ \theta_n \circ \dots \circ \theta_1.$$

Moreover, by prop. 3 (X, p. 118), we have $\theta = \theta_n \circ \dots \circ \theta_1$. The corollary follows immediately from this, and from the relation (E, III, p. 44)

$$(m+1) + \dots + (m+n) = mn + n(n+1)/2.$$

COROLLARY 4. — If each module $\mathbf{R}_i$, $i = 1, \dots, n$, is injective (resp. projective), the map $\alpha \mapsto \theta \circ \alpha$ (resp. $\alpha \mapsto \alpha \circ \theta$) from $\text{Ext}_{\mathbf{A}}^m(\mathbf{P}, \mathbf{M})$ to $\text{Ext}_{\mathbf{A}}^{m+n}(\mathbf{P}, \mathbf{N})$ (resp. from $\text{Ext}_{\mathbf{A}}^m(\mathbf{N}, \mathbf{P})$ to $\text{Ext}_{\mathbf{A}}^{m+n}(\mathbf{M}, \mathbf{P})$) is bijective for every $\mathbf{A}$ -module $\mathbf{P}$ and every integer $m > 0$. This follows indeed from cor. 3 and the exact sequences

$$\text{Ext}_{\mathbf{A}}^{m+i-1}(\mathbf{P}, \mathbf{R}_i) \rightarrow \text{Ext}_{\mathbf{A}}^{m+i-1}(\mathbf{P}, \mathbf{K}_{i-1}) \rightarrow \text{Ext}_{\mathbf{A}}^{m+i}(\mathbf{P}, \mathbf{K}_i) \rightarrow \text{Ext}_{\mathbf{A}}^{m+i}(\mathbf{P}, \mathbf{R}_i)$$

$$\left(\text{resp. } \text{Ext}_{\mathbf{A}}^{m+i-1}(\mathbf{R}_i, \mathbf{P}) \rightarrow \text{Ext}_{\mathbf{A}}^{m+i-1}(\mathbf{K}_i, \mathbf{P}) \rightarrow \text{Ext}_{\mathbf{A}}^{m+i}(\mathbf{K}_{i-1}, \mathbf{P}) \rightarrow \text{Ext}_{\mathbf{A}}^{m+i}(\mathbf{R}_i, \mathbf{P}) \right),$$

whose extreme terms are zero by hypothesis.

Remark. — The definitions and propositions of nos. 3 to 6 apply to right $\mathbf{A}$ -modules, considered as left modules over the opposite ring $\mathbf{A}^\circ$ of $\mathbf{A}$.

10.7.7 The homomorphism $\text{Ext}_{\mathbf{A}}(\mathbf{P}, \mathbf{Q}) \otimes \text{Tor}^{\mathbf{A}}(\mathbf{P}, \mathbf{M}) \rightarrow \text{Tor}^{\mathbf{A}}(\mathbf{Q}, \mathbf{M})$

Let $\mathbf{M}$ be a left $\mathbf{A}$ -module, and let $\mathbf{P}$ and $\mathbf{Q}$ be two right $\mathbf{A}$ -modules. Consider the homomorphism $\text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{P}), \mathbf{L}(\mathbf{Q})) \otimes_k (\mathbf{L}(\mathbf{P}) \otimes_{\mathbf{A}} \mathbf{L}(\mathbf{M})) \rightarrow \mathbf{L}(\mathbf{Q}) \otimes_{\mathbf{A}} \mathbf{L}(\mathbf{M})$ which to $f \otimes (x \otimes y)$ associates $f(x) \otimes y$. By X, p. 99, this is a morphism of complexes. From this one deduces a graded k -linear map of degree 0

$$\mathbf{H}(\text{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{P}), \mathbf{L}(\mathbf{Q}))) \otimes_k \text{Tor}^{\mathbf{A}}(\mathbf{P}, \mathbf{M}) \rightarrow \text{Tor}^{\mathbf{A}}(\mathbf{Q}, \mathbf{M})$$

hence, via the isomorphism $\varphi(\mathbf{L}(\mathbf{P}), \mathbf{L}(\mathbf{Q}))$ of § 6 (X, p. 100, th. 1), a graded k -linear map of degree 0

$$\mathrm{Ext}_{\mathbf{A}}(\mathbf{P}, \mathbf{Q}) \otimes_k \mathrm{Tor}^{\mathbf{A}}(\mathbf{P}, \mathbf{M}) \rightarrow \mathrm{Tor}^{\mathbf{A}}(\mathbf{Q}, \mathbf{M}), \quad (10)$$

corresponding to k -bilinear maps

$$c_{\mathbf{P}, \mathbf{Q}; \mathbf{M}} : \mathrm{Ext}_{\mathbf{A}}^n(\mathbf{P}, \mathbf{Q}) \times \mathrm{Tor}_m^{\mathbf{A}}(\mathbf{P}, \mathbf{M}) \rightarrow \mathrm{Tor}_{m-n}^{\mathbf{A}}(\mathbf{Q}, \mathbf{M}); \quad (11)$$

the image of the pair (α, γ) by $c_{\mathbf{P}, \mathbf{Q}; \mathbf{M}}$ is called the composition product of α and γ and is denoted $\alpha \circ \gamma$.

By construction, $\alpha \circ \gamma$ is obtained as follows: one represents α by a morphism of complexes $f : \mathbf{L}(\mathbf{P}) \rightarrow \mathbf{L}(\mathbf{Q})(-n)$, γ by an element $z \in Z_m(\mathbf{L}(\mathbf{P}) \otimes_{\mathbf{A}} \mathbf{L}(\mathbf{M}))$, and $\alpha \otimes \gamma$ is the class of the element

$$(f \otimes 1)(z) \in Z_m(\mathbf{L}(\mathbf{Q})(-n) \otimes_{\mathbf{A}} \mathbf{L}(\mathbf{M})) = Z_{m-n}(\mathbf{L}(\mathbf{Q}) \otimes_{\mathbf{A}} \mathbf{L}(\mathbf{M})).$$

For example, if $\alpha \in \mathrm{Hom}_{\mathbf{A}}(\mathbf{P}, \mathbf{Q})$, then $\alpha \circ \gamma = \mathrm{Tor}(\alpha, 1)(\gamma)$.

Remarks. — 1) If one uses the isomorphisms ψ of X, p. 69, one can also define the composition product by the commutative diagram

$$\begin{array}{ccc} \mathrm{Ext}_{\mathbf{A}}^n(\mathbf{P}, \mathbf{Q}) \times \mathrm{Tor}_m^{\mathbf{A}}(\mathbf{P}, \mathbf{M}) & \xrightarrow{c_{\mathbf{P}, \mathbf{Q}; \mathbf{M}}} & \mathrm{Tor}_{m-n}^{\mathbf{A}}(\mathbf{Q}, \mathbf{M}) \\ & \downarrow \bar{a}_{\mathbf{P}, \mathbf{Q}} \times \psi_{\mathbf{P}}(\mathbf{M}) & \\ \mathrm{H}^n(\mathrm{Homgr}_{\mathbf{A}}(\mathbf{L}(\mathbf{P}), \mathbf{L}(\mathbf{Q}))) \times \mathrm{H}_m(\mathbf{L}(\mathbf{P}) \otimes_{\mathbf{A}} \mathbf{M}) & \longrightarrow & \mathrm{H}_{m-n}(\mathbf{L}(\mathbf{Q}) \otimes_{\mathbf{A}} \mathbf{M}); \end{array}$$

in other words, we represent α by a morphism f from $\mathbf{L}(\mathbf{P})$ to $\mathbf{L}(\mathbf{Q})(-n)$, γ by a cycle $x \in \mathbf{L}_m(\mathbf{P}) \otimes_{\mathbf{A}} \mathbf{M}$, and $\alpha \circ \gamma$ is the cycle class of

$$(f_m \otimes 1_{\mathbf{M}})(x) \in \mathbf{L}_{m-n}(\mathbf{Q}) \otimes_{\mathbf{A}} \mathbf{M}.$$

2) One can also use the resolutions $\mathbf{I}(\mathbf{P})$ and $\mathbf{I}(\mathbf{Q})$.

Similarly, if $\mathbf{N}$ is a second left $\mathbf{A}$ -module, one defines a composition product $(\mu, \gamma) \mapsto \mu \circ \gamma$

, denoted

$$c_{P;M,N} : \text{Ext}_A^r(M, N) \times \text{Tor}_m^A(P, M) \rightarrow \text{Tor}_{m-r}^A(P, N)$$

by the commutative diagram

$$\begin{array}{ccc} \text{Ext}_A^r(\mathbf{M}, \mathbf{N}) \times \text{Tor}_m^A(\mathbf{P}, \mathbf{M}) & \xrightarrow{c_{P;M,N}} & \text{Tor}_{m-r}^A(\mathbf{P}, \mathbf{N}) \\ & \downarrow 1 \times \sigma_{P,M,r} & \\ \text{Ext}_{A^\circ}^r(\mathbf{M}^\circ, \mathbf{N}^\circ) \times \text{Tor}_m^{A^\circ}(\mathbf{M}^\circ, \mathbf{P}^\circ) & \xrightarrow{c_{M^\circ, N^\circ; P^\circ}} & \text{Tor}_{m-r}^{A^\circ}(\mathbf{N}^\circ, \mathbf{P}^\circ) \end{array} \quad (12)$$

where σ denotes the commutation isomorphisms (X, p. 71).

If $\mu \in \text{Ext}_A^r(\mathbf{M}, \mathbf{N})$ is the class of the morphism $g : L(\mathbf{M}) \rightarrow L(\mathbf{N})(-r)$, and if $\gamma \in \text{Tor}_m^A(\mathbf{P}, \mathbf{M})$ is the cycle class of $z = \sum_{i,j} z_{ij}$, where $z_{ij} \in L_i(\mathbf{P}) \otimes_A L_j(\mathbf{M})$, $\mu \circ \gamma$ is therefore the class of the cycle $\sum_{i,j} (-1)^{ir} (1 \otimes g)(z_{ij})$.

One can also represent γ by a cycle $y \in P \otimes L_m(\mathbf{M})$, and $\mu \circ \gamma$ is the cycle class of $(1 \otimes g)(y) \in P \otimes L_{m-r}(\mathbf{M})$.

PROPOSITION 6. — Let K, M, N be left A -modules, and P, Q, R right A -modules,

$$\alpha \in \text{Ext}_A^n(P, Q), \beta \in \text{Ext}_A^p(Q, R), \lambda \in \text{Ext}_A^r(K, M), \mu \in \text{Ext}_A^s(M, N), \gamma \in \text{Tor}_m^A(P, K).$$

Then

(13)

$$(\beta \circ \alpha) \circ \gamma = \beta \circ (\alpha \circ \gamma) \text{ in } \text{Tor}_{m-p-n}^A(R, K), \quad (14)$$

$$(\mu \circ \lambda) \circ \gamma = \mu \circ (\lambda \circ \gamma) \text{ in } \text{Tor}_{n-r-s}^A(P, N), \quad (15)$$

$$\alpha \circ (\lambda \circ \gamma) = (-1)^{nr} \lambda \circ (\alpha \circ \gamma) \text{ in } \text{Tor}_{m-p-r}^A(Q, M).$$

Formulas (13) and (14) follow immediately from the definitions. Let us prove (15). Let $z = \sum z_{ij}, z_{ij} \in \mathbf{L}_i(\mathbf{P}) \otimes \mathbf{L}_j(\mathbf{K})$ a cycle representing $\gamma, f : \mathbf{L}(\mathbf{P}) \rightarrow \mathbf{L}(\mathbf{Q})(-n)$ and $g : \mathbf{L}(\mathbf{K}) \rightarrow \mathbf{L}(\mathbf{M})(-r)$ morphisms representing α and λ . Then $\lambda \circ (\alpha \circ \gamma)$ is the class of $\sum (-1)^{(i-n)r} (f \otimes g)(z_{ij})$ and $\alpha \circ (\lambda \circ \gamma)$ is the class of

$$\sum (-1)^{ir} (f \otimes g)(z_{ij}), \quad \text{whence (15).}$$

10.7.8 Composition products and connecting homomorphisms of torsion products

PROPOSITION 7. — a) Let

$$0 \rightarrow \mathbf{P}' \xrightarrow{f} \mathbf{P} \xrightarrow{g} \mathbf{P}'' \rightarrow 0 \quad (8)$$

be an exact sequence of right A-modules, $\theta \in \text{Ext}_A^1(\mathbf{P}'', \mathbf{P}')$ the associated class, M a left A-module. The connecting homomorphism

$$\delta_n(\mathcal{E}, \mathbf{M}) : \text{Tor}_n^A(\mathbf{P}'', \mathbf{M}) \rightarrow \text{Tor}_{n-1}^A(\mathbf{P}', \mathbf{M}) \quad \text{is the map} \quad \gamma \mapsto \theta \circ \gamma.$$

b) Let

$$0 \rightarrow \mathbf{M}' \rightarrow \mathbf{M} \rightarrow \mathbf{M}'' \rightarrow 0 \quad ((\mathcal{E}_1))$$

be an exact sequence of left A-modules, $\theta_1 \in \text{Ext}_A^1(\mathbf{M}'', \mathbf{M}')$ the associated class, P a right A-module. The connecting homomorphism

$$\delta_n(\mathbf{P}, \mathcal{E}_1) : \text{Tor}_n^A(\mathbf{P}, \mathbf{M}'') \rightarrow \text{Tor}_{n-1}^A(\mathbf{P}, \mathbf{M}') \quad \text{is the map} \quad \gamma \mapsto \theta_1 \circ \gamma.$$

Let $\gamma \in \text{Tor}_n^A(\mathbf{P}'', \mathbf{M})$ the class of a cycle $z'' \in \dot{Z}_n(\mathbf{L}(\mathbf{P}'') \otimes_A \mathbf{L}(\mathbf{M}))$, and let

$$\begin{array}{ccccccc}
0 & \longrightarrow & \mathbf{P}' & \xrightarrow{f} & \mathbf{P} & \xrightarrow{g} & \mathbf{P}'' \longrightarrow 0 \\
& & \uparrow_{u_1} & & \uparrow_{u_0} & & \uparrow_1 \\
& & \mathbf{L}_1(\mathbf{P}'') & \xrightarrow{d_1} & \mathbf{L}_0(\mathbf{P}'') & \xrightarrow{p_0''} & \mathbf{P}'' \longrightarrow 0
\end{array}$$

a commutative diagram. We will denote $p' : \mathbf{L}(\mathbf{P}') \rightarrow \mathbf{P}'$ and $p'' : \mathbf{L}(\mathbf{P}'') \rightarrow \mathbf{P}''$ the canonical morphisms of complexes. By definition, $\delta(\gamma) \in \text{Tor}_{n-1}^{\mathbf{A}}(\mathbf{P}', \mathbf{M})$ is obtained as follows: one chooses $x \in \mathbf{P} \otimes \mathbf{L}_n(\mathbf{M})$ such that $(g \otimes 1)(x) = (p'' \otimes 1)(z'')$ and $\delta(\gamma)$ is the class of cycles $z' \in Z_{n-1}(\mathbf{L}(\mathbf{P}') \otimes \mathbf{L}(\mathbf{M}))$ such that

$$(f \otimes 1)(p' \otimes 1)(z') = (1 \otimes d_n)(x).$$

For $0 \leq i \leq n$ z_i'' the component of z'' to $\mathbf{L}_i(\mathbf{P}'') \otimes \mathbf{L}_{n-i}(\mathbf{M})$; we have

$$0 = D z'' = \sum_i (d_i \otimes 1 + (-1)^i \otimes d_{n-i}) (z_i''),$$

donc $(d_i \otimes 1)(z_i'') = (-1)^i \otimes d_{n-i+1}(z_{i-1}'')$ and in particular

$$(d_1 \otimes 1)(z_1'') = -1 \otimes d_n(z_0'').$$

Let us then choose $x = (u_0 \otimes 1)(z_0'')$: we indeed have

$$(g \otimes 1)(x) = (p_0'' \otimes 1) z_0'' = (p'' \otimes 1)(z'').$$

Since

$$\begin{aligned}
(1 \otimes d_n)(x) &= (u_0 \otimes 1)(1 \otimes d_n)(z_0'') = -(u_0 \otimes 1)(d_1 \otimes 1)(z_1'') \\
&= -(f \otimes 1)(u_1 \otimes 1)(z_1''),
\end{aligned}$$

it follows that $\delta(\gamma)$ is the class of cycles $z' \in Z_{n-1}(\mathbf{L}(\mathbf{P}') \otimes_{\mathbf{A}} \mathbf{L}(\mathbf{M}))$ such that $(p' \otimes 1)(z') = -(u_1 \otimes 1)(z_1'')$. But, by definition, the class θ corresponds under the isomorphism $\text{Ext}_A^1(\mathbf{P}', \mathbf{P}') \rightarrow H^1(\text{Homgr}_A(\mathbf{L}(\mathbf{P}''), \mathbf{P}'))$ to the class of the morphism $f : \mathbf{L}(\mathbf{P}')(1) \rightarrow \mathbf{P}'$ defined by $-u_1$, and the product $\theta \circ \gamma$ is the class of cycles

$$\bar{z}' \in Z_{n-1}(L(P') \otimes_A L(M)) \quad \text{tels que} \quad (p \otimes 1)(\bar{z}') = f(z'') = -(u_1 \otimes 1)(z_1''),$$

which completes the proof of *a*). The assertion *b*) follows from *a*) by the commutation isomorphisms.

COROLLARY 1. — Let $0 \rightarrow P' \rightarrow P \rightarrow P'' \rightarrow 0$ be an exact sequence of right A -modules, $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ an exact sequence of left A -modules. Then the homomorphisms composed of connecting homomorphisms

$$\text{Tor}_n^A(P'', M'') \rightarrow \text{Tor}_{n-1}^A(P'', M') \rightarrow \text{Tor}_{n-2}^A(P', M')$$

and

$$\text{Tor}_n^A(P'', M'') \rightarrow \text{Tor}_{n-1}^A(P', M'') \rightarrow \text{Tor}_{n-2}^A(P', M')$$

are opposite.

Indeed, if θ and θ_1 are the classes associated with the given exact sequences, and if $\gamma \in \text{Tor}_n^A(P'', M'')$ the images of γ are respectively $\theta \circ (\theta_1 \circ \gamma)$ and $\theta_1 \circ (\theta \circ \gamma)$, hence are opposite by prop. 6.

Let us take up the notations of X, p. 127 and consider the sequence (S) of left A -modules and the connecting homomorphisms associated with the exact sequences (9).

$$\text{Tor}_m^A(P, \mathbf{K}_{i-1}) \rightarrow \text{Tor}_{m-1}^A(P, \mathbf{K}_i);$$

we deduce by composing the iterated connecting homomorphisms

$$\partial_m(P, \mathcal{S}) : \text{Tor}_m^A(P, M) \rightarrow \text{Tor}_{m-n}^A(P, N).$$

Then by prop. 7 and prop. 3 of X, p. 118:

Corollary 2. — If $\theta \in \text{Ext}_A^n(\mathbf{M}, \mathbf{N})$ is the class associated with the exact sequence $(\mathcal{S})$, we have $\partial_m(P, \mathcal{S})(\alpha) = \theta \circ \alpha$ for all $\alpha \in \text{Tor}_m^A(P, \mathbf{M})$.

COROLLARY 3. — If all the modules $\mathbb{R}_i$, $i = 1, \dots, n$, are flat, the map $\alpha \mapsto \theta \circ \alpha$ of $\text{Tor}_{m+n}^{\mathbf{A}}(\mathbf{P}, \mathbf{M})$ to $\text{Tor}_m^{\mathbf{A}}(\mathbf{P}, \mathbf{N})$ is bijective for every right $\mathbf{A}$ -module $\mathbf{P}$ and every integer $m > 0$.

This follows from Cor. 2 and the exact sequences.

$$\text{Tor}_{m+n-i+1}^{\mathbf{A}}(\mathbf{P}, \mathbf{R}_i) \rightarrow \text{Tor}_{m+n-i+1}^{\mathbf{A}}(\mathbf{P}, \mathbf{K}_{i-1}) \xrightarrow{\hat{c}} \text{Tor}_{m+n-i}^{\mathbf{A}}(\mathbf{P}, \mathbf{K}_i) \rightarrow \text{Tor}_{m+n-i}^{\mathbf{A}}(\mathbf{P}, \mathbf{R}_i)$$

where the extreme terms are zero by hypothesis.

Likewise, if

$$0 \rightarrow \mathbf{Q} \rightarrow \mathbf{S}_n \rightarrow \mathbf{S}_{n-1} \rightarrow \dots \rightarrow \mathbf{S}_1 \rightarrow \mathbf{P} \rightarrow 0 \quad ((\mathcal{S}_1))$$

is an exact sequence of right $\mathbf{A}$ -modules, and $\mathbf{M}$ is a left $\mathbf{A}$ -module, one defines iterated connecting homomorphisms

$$\partial^m(\mathcal{S}_1, \mathbf{M}) : \text{Tor}_m^{\mathbf{A}}(\mathbf{P}, \mathbf{M}) \rightarrow \text{Tor}_{m-n}^{\mathbf{A}}(\mathbf{Q}, \mathbf{M})$$

and we have:

COROLLARY 4. — If $\theta_1 \in \text{Ext}_{\mathbf{A}}^n(\mathbf{P}, \mathbf{Q})$ is the class associated with the exact sequence $(\mathcal{S}_1)$, we have $\partial^m(\mathcal{S}_1, \mathbf{M})(\alpha) = \theta_1 \circ \alpha$ for all $\alpha \in \text{Tor}_m^{\mathbf{A}}(\mathbf{P}, \mathbf{M})$.

10.7.9 Computation of composition products by shifting resolutions

Let

$$0 \rightarrow \mathbf{M} \xrightarrow{f} \mathbf{K}_n \rightarrow \mathbf{K}_{n-1} \rightarrow \dots \rightarrow \mathbf{K}_1 \xrightarrow{g} \mathbf{M}' \rightarrow 0 \quad (16)$$

be an exact sequence of left $\mathbf{A}$ -modules and $\theta \in \text{Ext}_{\mathbf{A}}^n(\mathbf{M}', \mathbf{M})$ the associated class.

Let $a : (\mathbf{R}, d) \rightarrow \mathbf{M}$ be a left resolution of $\mathbf{M}$; we thus have an exact sequence

$$\rightarrow \mathbf{R}_k \xrightarrow{d_k} \mathbf{R}_{k-1} \rightarrow \dots \xrightarrow{d_1} \mathbf{R}_0 \xrightarrow{a_0} \mathbf{M} \rightarrow 0.$$

and by translation of n (X, p. 26) an exact sequence

$$\rightarrow R_k \xrightarrow{(-1)^n d_k} R_{k-1} \rightarrow \dots \xrightarrow{(-1)^n d_1} R_0 \xrightarrow{(-1)^n a_0} M \rightarrow 0. \quad (17)$$

From (16) and (17) we deduce an exact sequence

$$\rightarrow R_k \xrightarrow{(-1)^n d_k} R_{k-1} \rightarrow \dots \xrightarrow{(-1)^n d_1} R_0 \xrightarrow{(-1)^n f \circ a_0} K_n \rightarrow K_{n-1} \rightarrow \dots \rightarrow K_1 \rightarrow M' \rightarrow 0$$

whence a resolution $\mathbf{R}'$ of $\mathbf{M}'$; denote by $\varphi : \mathbf{R}' \rightarrow \mathbf{R}(-n)$ the morphism such that $\varphi_k = 1_{\mathbf{R}_{k-n}}$ for $k \geq n$.

If N is a left A -module and P is a right A -module, we therefore have homomorphisms

$$H(1_P \otimes \varphi) : H(P \otimes_A R') \rightarrow H(P \otimes_A R)(-n)$$

$$H(\text{Homgr}_A(\varphi, 1_N)) : H(\text{Homgr}_A(R, N))(n) \rightarrow H(\text{Homgr}_A(R', N)).$$

Let k be an integer.

PROPOSITION 8. — a) The following diagram, where $h_\theta(\alpha) = \theta \circ \alpha$, is commutative

$$\begin{array}{ccc} \text{Tor}_{k+n}^A(P, M') & \xrightarrow{h_\theta} & \text{Tor}_k^A(P, M) \\ \psi_{k+n}(P, R') \Big|_{\downarrow} & & \Big|_{\downarrow} \psi_k(P, R) \\ H_{k+n}(P \otimes_A R') & \xrightarrow{H_{k+n}(1 \otimes \varphi)} & H_k(P \otimes_A R) \end{array}$$

b) The following diagram, where $\delta_\theta(\beta) = \beta \circ \theta$, is commutative

$$\begin{array}{ccc} H^k(\text{Homgr}_A(R, N)) & \xrightarrow{H^{k+n}(\text{Homgr}_A(\varphi, 1))} & H^{k+n}(\text{Homgr}_A(R', N)) \\ & \varphi^k(R, N) \Big|_{\downarrow} & \\ \text{Ext}_A^k(M, N) & \xrightarrow{\delta_\theta} & \text{Ext}_A^{k+n}(M', N). \end{array}$$

Let $\alpha : L(M) \rightarrow R$ be a morphism of complexes such that $a \circ \alpha = p_M$ and let

$$\begin{array}{ccc}
& & L_n(M') \rightarrow L_{n-1}(M') \rightarrow \cdots \rightarrow L_0(M) \\
u_n \downarrow & & \downarrow u_{n-1} \\
& & 0 \rightarrow M \xrightarrow{f} K_n \rightarrow \cdots \rightarrow K_1 \rightarrow
\end{array}$$

a commutative diagram; let us choose a homomorphism $v_n : L_n(M') \rightarrow L_0(M)$ such that $p_M \circ v_n = (-1)^n u_n$; by X, p. 47, prop. 1, a), v_n extends to a morphism of complexes $v : L(M') \rightarrow L(M)(-n)$, and θ is the image under the canonical isomorphism $H^n(\text{Homgr}_A(L(M'), L(M))) \rightarrow \text{Ext}_A^n(M', M)$ of the class of v (X, p. 117, remark 1). One defines a morphism of complexes $\beta : L(M') \rightarrow R'$ by $\beta_p = u_p$ for $p \leq n-1$, $\beta_p = \alpha_{p-n} \circ v_p$ for $p \geq n$, and we have

$$\varphi \circ \beta = \alpha(-n) \circ v.$$

On the other hand, by definition of the maps φ and ψ , we have

$$\begin{aligned}
\psi_k(P, R) &= H_k(p_P \otimes \alpha), & \varphi^k(R, N) &= H^k(\text{Homgr}_A(\alpha, e_N)), \\
\psi_{k+n}(P, R') &= H_{k+n}(p_P \otimes \beta), & \varphi^{k+n}(R', \tilde{N}) &= H^{k+n}(\text{Homgr}_A(\beta, e_N)).
\end{aligned}$$

Finally, by definition of the composition product, one has

$$h_\theta = H(1_{L(\mathbb{P})} \otimes v), \quad \delta_\theta = H(\text{Homgr}_A(v, 1_{I(\mathbb{N})}))$$

Consequently, one has the equalities

$$\begin{aligned}
\psi_k(P, R) \circ h_\theta &= H_k(p_P \otimes \alpha) \circ H_k(1_{L(\mathbb{P})} \otimes v) = H_k(p_P \otimes (\alpha \circ v)) = H_{k+n}(p_P \otimes (\varphi \circ \beta)) \\
&= H_{k+n}(1 \otimes \varphi) \circ H_{k+n}(p_P \circ \beta) = H_{k+n}(1 \otimes \varphi) \circ \psi_{k+n}(P, R'),
\end{aligned}$$

whence a); the proof of b) is analogous.

Remark. — By the commutation isomorphisms, we deduce from a) an analogous statement in the case of an exact sequence (16) of right A-modules.

Now let $b : M' \rightarrow E'$ be a right resolution of M' ; we thus have an exact sequence

$$0 \rightarrow \mathbf{M}' \xrightarrow{b^0} E'^0 \xrightarrow{\delta^0} E'^1 \rightarrow \dots \rightarrow E'^k \xrightarrow{\delta^k} E'^{k+1}$$

whence an exact sequence

$$0 \rightarrow M \xrightarrow{f} K_n \rightarrow K_{n-1} \rightarrow \dots \rightarrow K_1 \xrightarrow{(-1)^n b^0 \circ g} E'^0 \xrightarrow{(-1)^n \delta^0} E'^1 \rightarrow \dots$$

corresponding to a right resolution E of M ; denote $\sigma : E'(n) \rightarrow E$ the morphism such that $\sigma^k = 1_{E'_{k-n}}$ for $k \geq n$. We therefore have homomorphisms

$$H(\text{Homgr}_A(1_N, \sigma)) : H(\text{Homgr}_A(N, E'))(n) \rightarrow H(\text{Homgr}_A(N, E)).$$

PROPOSITION 9. — The following diagram, where $\gamma_\theta(\alpha) = \theta \circ \alpha$, is commutative:

$$\begin{array}{ccc} H^k(\text{Homgr}_A(N, E')) & \xrightarrow{H^{k+n}(\text{Homgr}_A(1_N, \sigma))} & H^{k+n}(\text{Homgr}_A(N, E)) \\ & \varphi^k(N, E') \downarrow & \\ \text{Ext}_A^k(N, M') & \xrightarrow{\gamma_\theta} & \text{Ext}_A^{k+n}(N, M). \end{array}$$

This is proved in a manner analogous to prop. 8.

10.8 HOMOLOGICAL DIMENSION

In this paragraph, we resume the conventions of § 5.

10.8.1 Projective dimension of a module

DEFINITION 1. — Let M be an A -module. One calls projective dimension of M , and denotes by $\text{dp}_A(M)$, the lower bound in $\mathbb{Z}$ of the lengths of projective resolutions of M (X, p. 48).

We therefore have $\text{dp}_A(0) = -\infty$, $\text{dp}_A(\mathbf{M}) \geq 0$ if $\mathbf{M} \neq 0$. For $\mathbf{M}$ to be projective, it is necessary and sufficient that $\text{dp}_A(\mathbf{M}) \leq 0$.

Lemma 1. — If $\text{dp}_A(M) < n < +\infty$, we have $\text{Ext}_A^n(M, N) = 0$ for every A -module N and $\text{Tor}_n^A(P, M) = 0$ for every right A -module P .

This follows immediately from the fact that M has a projective resolution of length $< n$ and

of X, p. 100, th. 1.

PROPOSITION 1. — Let M be an A -module and n an integer ≥ 0 . The following conditions are equivalent:

- (i) $\text{dp}_A(\mathbf{M}) \leq n$ (i.e. (def. 1), M has a projective resolution of length $\leq n$)
- (ii) $\text{Ext}_A^r(\mathbf{M}, \mathbf{N}) = 0$ for every A -module N and every integer $r > n$;
- (iii) $\text{Ext}_A^{n+1}(\mathbf{M}, \mathbf{N}) = 0$ for every A -module N ;
- (iv) for every exact sequence

$$0 \rightarrow K \rightarrow P_{n-1} \rightarrow \cdots \rightarrow P_0 \rightarrow M \rightarrow 0$$

where the P_i are projective, K is projective.

(i) $\Rightarrow$ (ii): this follows from Lemma 1.

(ii) $\Rightarrow$ (iii): this is trivial.

(iii) $\Rightarrow$ (iv): in the situation of (iv), one has for every A -module N an isomorphism of $\text{Ext}_A^1(\mathbf{K}, \mathbf{N})$ over $\text{Ext}_A^{n+1}(\mathbf{M}, \mathbf{N})$ (X, p.128, cor. 4) ; if (iii) is satisfied, one has

$$\text{Ext}_A^1(\mathbf{K}, \mathbf{N}) = 0$$

for all N and K is projective (X, p. 93, prop. 10).

(iv) $\Rightarrow$ (i): consider the exact sequence (X, p. 50)

$$0 \rightarrow Z_{n-1}(M) \rightarrow L_{n-1}(M) \rightarrow \cdots \rightarrow L_0(M) \rightarrow M \rightarrow 0.$$

If (iv) is satisfied, $Z_{n-1}(M)$ is projective and M has a projective resolution of length $\leq n$.

COROLLARY 1. — Let $(\mathbf{M}_i)_{i \in E}$ be a family of A -modules. We have

$$\text{dp}_A \left(\bigoplus_{i \in E} \mathbf{M}_i \right) = \sup_{i \in E} \text{dp}_A(\mathbf{M}_i).$$

This follows from the equivalence of conditions (i) and (iii) of Prop. 1, and from Prop. 7 of X, p. 89.

In the following statement, we agree that $\pm\infty + 1 = \pm\infty - 1 = \pm\infty$.

COROLLARY 2. — Let

$$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$$

be an exact sequence of A-modules.

a) We have

$$\text{dp}_A(M) \leq \sup(\text{dp}_A(M'), \text{dp}_A(M'')) .$$

Equality holds as soon as $\text{dp}_A(M'') \neq \text{dp}_A(M') + 1$

b) We have

$$\text{dp}_A(M'') \leq \sup(\text{dp}_A(M), \text{dp}_A(M') + 1) .$$

Equality holds as soon as $\text{dp}_A(M) \neq \text{dp}_A(M')$.

$$\text{dp}_A(M') \leq \sup(\text{dp}_A(M), \text{dp}_A(M'') - 1)$$

Equality holds as soon as $\text{dp}_A(M) \neq \text{dp}_A(M'')$.

Let us prove, for example, part a), the proofs of parts b) and c) being analogous.

$$\geq 0.$$

$$\begin{aligned} \text{Ext}_A^{n+1}(M'', N) &\rightarrow \text{Ext}_A^{n+1}(M, N) \rightarrow \text{Ext}_A^{n+1}(M', N) \rightarrow \text{Ext}_A^{n+2}(M'', N) \\ &\rightarrow \text{Ext}_A^{n+2}(M, N). \end{aligned}$$

If $\text{dp}_A(M'), \text{dp}_A(M'') \leq n$, then $\text{Ext}_A^{n+1}(M', N) = 0$ and $\text{Ext}_A^{n+1}(M'', N) = 0$ (prop. 1) therefore $\text{Ext}_A^{n+1}(M, N) = 0$ and $\text{dp}_A(M) \leq n$ (prop. 1), so that

$$\mathrm{dp}_A(\mathbf{M}) \leq \sup(\mathrm{dp}_A(\mathbf{M}'), \mathrm{dp}_A(\mathbf{M}'')) .$$

If $\mathrm{dp}_A(\mathbf{M}) < \sup(\mathrm{dp}_A(\mathbf{M}'), \mathrm{dp}_A(\mathbf{M}''))$ then necessarily $\mathrm{dp}_A(\mathbf{M}) < +\infty$. For every $n > \mathrm{dp}_A(\mathbf{M})$ and for every A -module N , we have

$$\mathrm{Ext}_A^{n+1}(\mathbf{M}', N) \neq 0 \Leftrightarrow \mathrm{Ext}_A^{n+2}(\mathbf{M}'', N) \neq 0,$$

by the preceding exact sequence; by prop. 1, this immediately implies $\mathrm{dp}_A(\mathbf{M}'') = \mathrm{dp}_A(\mathbf{M}') + 1$, since one of the quantities $\mathrm{dp}_A(\mathbf{M}')$, $\mathrm{dp}_A(\mathbf{M}'')$ is $> \mathrm{dp}_A(\mathbf{M})$.

Example. — Let a be an element of A which is neither invertible nor a right zero divisor. Then $\mathrm{dp}_A(A/Aa) = 1$.

Indeed, by the exact sequence $0 \rightarrow A_s \xrightarrow{\varphi} A_s \rightarrow A/Aa \rightarrow 0$, where $\varphi(x) = xa$, we have $\mathrm{dp}_A(A/Aa) \leq 1$. If $\mathrm{dp}_A(A/Aa) < 1$, then A/Aa is projective, and there exists an A -linear map $\psi : A_s \rightarrow A_s$ such that $\psi \circ \varphi = 1d$; this implies

$$1 = \psi(\varphi(1)) = \psi(a) = a.\psi(1)$$

and a is invertible.

PROPOSITION 2. — Suppose A is left noetherian. Let M be a finitely generated A -module and n an integer ≥ 0 . The following conditions are equivalent:

(i) $\mathrm{dp}_A(\mathbf{M}) \leq n$.

(i bis) M has a projective resolution P of length $\leq n$ such that P_i is a finitely generated A -module for each i .

(ii) $\mathrm{Ext}_A^r(\mathbf{M}, \mathbf{N}) = 0$ for every finitely generated A -module N and every integer $r > n$.

(iii) $\mathrm{Ext}_A^{n+1}(\mathbf{M}, \mathbf{N}) = 0$ for every finitely generated A -module N .

(iv) $\mathrm{Tor}_r^A(\mathbf{P}, \mathbf{M}) = 0$ for every right A -module P and every $r > n$.

(v) $\mathrm{Tor}_{n+1}^A(A/\mathfrak{a}, \mathbf{M}) = 0$ for every finitely generated right ideal $\mathfrak{a}$ of A .

(i bis) $\Rightarrow$ (i): this is trivial.

(i) $\Rightarrow$ (ii): this follows from Lemma 1.

(ii) $\Rightarrow$ (iii): this is trivial.

(iii) $\Rightarrow$ (i): by (iii) and X, p. 107, prop. 5, we have $\text{Ext}_A^{n+1}(\mathbf{M}, \mathbf{N}) = 0$ for every A-module N, whence (i) by prop. 1.

(iv) $\Rightarrow$ (iv): this follows from lemma 1.

(v) $\Rightarrow$ (v): this is trivial.

(vi) $\Rightarrow$ (i bis): let $(\mathbf{L}, d)$ be a free resolution of $\mathbf{M}$ such that $\mathbf{L}_r$ be finitely generated for every r (X, p. 53, prop. 6). Set $\mathbf{K} = \mathbf{Z}_{n-1}(\mathbf{L})$; then $\mathbf{K}$ is finitely generated as a submodule of $\mathbf{L}_{n-1}$ and we have an exact sequence

$$0 \rightarrow \mathbf{K} \rightarrow \mathbf{L}_{n-1} \rightarrow \mathbf{L}_{n-2} \rightarrow \cdots \rightarrow \mathbf{L}_1 \rightarrow \mathbf{L}_0 \rightarrow \mathbf{M} \rightarrow 0. \quad (1)$$

By (v) and X, p. 131, cor. 3, we have $\text{Tor}_1^A(\mathbf{A}/\mathfrak{a}, \mathbf{K}) = 0$ for every finitely generated right ideal $\mathfrak{a}$ of A. According to th. 2 of X, p. 74, the A-module K is flat; since it is finitely generated, hence finitely presented (X, p. 10, prop. 5), it is projective (X, p. 13, cor.), so (1) is a projective resolution of M.

COROLLARY. — Suppose A is left noetherian and let $\mathcal{C}_0$ (resp. $\mathcal{C}$) be the set of classes of finitely generated projective A-modules (resp. of finitely generated A-modules of finite projective dimension). Then the homomorphism of Grothendieck groups $K(\mathcal{C}_0) \rightarrow K(\mathcal{C})$ is bijective.

This follows from X, p. 58, th. 1 (note that $\mathcal{C}_0$ and $\mathcal{C}$ are left exact by cor. 2).

10.8.2 The homomorphism $\text{Tor}_n^A(\mathbf{P}, \mathbf{M}) \rightarrow \text{Hom}_A(\text{Ext}_A^n(\mathbf{M}, \mathbf{A}), \mathbf{P})$

Let M be a left A-module, P a right A-module, and n an integer ≥ 0 . The k-bilinear map (X, p. 129)

$$c_{\mathbf{P}; \mathbf{M}, \mathbf{A}_s} : \text{Ext}_A^n(\mathbf{M}, \mathbf{A}) \times \text{Tor}_n^A(\mathbf{P}, \mathbf{M}) \rightarrow \mathbf{P} \otimes_A \mathbf{A}$$

corresponds to a k-linear map

$$\mathrm{Tor}_n^A(P, M) \rightarrow \mathrm{Hom}_k(\mathrm{Ext}_A^n(M, A), P); \quad (2)$$

Moreover, if one equips $\mathrm{Ext}_A^n(M, A)$ of the structure of right A -module arising from the bimodule structure of A , the image of (2) consists of A -linear maps, as is verified immediately; one deduces from this a k -homomorphism, said to be canonical

$$\mathrm{Tor}_n^A(P, M) \rightarrow \mathrm{Hom}_A(\mathrm{Ext}_A^n(M, A), P). \quad (3)$$

PROPOSITION 3. — a) If $\mathrm{dp}_A(\mathbf{M}) \leq n$, the canonical homomorphism (3) is injective. b) If $\mathrm{dp}_A(\mathbf{M}) \leq n$, if A is left noetherian and if M is finitely generated, the canonical homomorphism (3) is bijective.

Let us reason by induction on n . If $n = 0$, M is projective, the homomorphism (3) reduces to the canonical homomorphism $\mathbf{P} \otimes_A \mathbf{M} \rightarrow \mathrm{Hom}_A(\mathbf{M}^*, \mathbf{P})$ of II, p. 77, and the proposition follows from loc. cit., corollary. If $n > 0$, let

$$0 \rightarrow \mathbf{N} \rightarrow \mathbf{L} \rightarrow \mathbf{M} \rightarrow 0 \quad (4)$$

be an exact sequence of A -modules where L is free (and finitely generated in case b)); then $\mathrm{dp}_A(\mathbf{N}) \leq n - 1$ (X, p. 135, cor. 2, c)) and N is finitely generated in case b).

Let $\theta \in \mathrm{Ext}_A^1(\mathbf{M}, \mathbf{N})$ be the class associated with the exact sequence (4) (X, p. 117, def. 1). Let us denote by

$$u_n : \mathrm{Ext}_A^{n-1}(\mathbf{N}, A) \rightarrow \mathrm{Ext}_A^n(\mathbf{M}, A) \quad v_n : \mathrm{Tor}_n^A(\mathbf{P}, \mathbf{M}) \rightarrow \mathrm{Tor}_{n-1}^A(\mathbf{P}, \mathbf{N})$$

$$(\alpha \circ \theta) \circ \beta = \alpha \circ (\theta \circ \beta)$$

the maps defined by $u_n(\alpha) = \alpha \circ \theta$ and $v_n(\beta) = \theta \circ \beta$. We have for all $\alpha \in \mathrm{Ext}_A^{n-1}(\mathbf{N}, A)$, $\beta \in \mathrm{Tor}_n^A(\mathbf{P}, \mathbf{M})$ (X, p. 129, prop. 6), so that the diagram

$$\begin{array}{ccc}
\mathrm{Tor}_n^A(P, M) & \longrightarrow & \mathrm{Hom}_A(\mathrm{Ext}_A^n(M, A), P) \\
\downarrow v_n & & \mathrm{Hom}(u_n, 1) \downarrow \\
\mathrm{Tor}_{n-1}^A(P, N) & \longrightarrow & \mathrm{Hom}_A(\mathrm{Ext}_A^{n-1}(N, A), P),
\end{array} \tag{5}$$

where the horizontal arrows are the canonical homomorphisms, is commutative. If $n = 1$, we thus have a commutative diagram:

$$\begin{array}{ccc}
0 & & 0 \\
\downarrow & & \downarrow \\
\mathrm{Tor}_1^A(P, M) & \longrightarrow & \mathrm{Hom}_A(\mathrm{Ext}_A^1(M, A), P) \\
\downarrow v_1 & & \mathrm{Hom}(u_1, 1) \downarrow \\
P \otimes_A N & \longrightarrow & \mathrm{Hom}_A(N^*, P) \\
\downarrow & & \downarrow \\
P \otimes_A L & \longrightarrow & \mathrm{Hom}_A(L^*, P)
\end{array}$$

where the columns are exact; we deduce the result in this case. If $n \geq 2$, the maps u_n and v_n are bijective (X, p. 128, cor. 4 and p. 131, cor. 3). By the induction hypothesis, the canonical homomorphism

$$\mathrm{Tor}_{n-1}^A(P, N) \rightarrow \mathrm{Hom}_A(\mathrm{Ext}_A^{n-1}(N, A), P)$$

is injective (resp. bijective); diagram (5) shows that the same holds for the canonical homomorphism $\mathrm{Tor}_n^A(P, M) \rightarrow \mathrm{Hom}_A(\mathrm{Ext}_A^n(M, A), P)$, which completes the proof.

10.8.3 Homological dimension of a ring

DEFINITION 2. — One calls the homological dimension of A and denotes by $\mathrm{dh}(A)$ the least upper bound in $\overline{\mathbb{Z}}$ of the set of integers n for which there exist two A -modules M and N such that $\mathrm{Ext}_A^n(M, N) \neq 0$.

One has $\text{dh}(0) = -\infty$, $\text{dh}(A) \geq 0$ if $A \neq 0$. One will see below that $\text{dh}(A) = 1$ if A is principal and is not a field, and that, if K is a commutative field

$$\text{dh}(K[X_1, \dots, X_n]) = n.$$

PROPOSITION 4. — Let n be an integer ≥ 0 . The following conditions are equivalent: (i) $\text{dh}(A) \leq n$,

(ii) for every A -module M , one has $\text{dp}_A(\mathbf{M}) \leq n$,

(ii') for every finitely generated A -module M , one has $\text{dp}_A(\mathbf{M}) \leq n$,

(iii) for every exact sequence

$$0 \rightarrow K \rightarrow P_{n-1} \rightarrow P_{n-2} \rightarrow \dots \rightarrow P_0$$

where the P_i are projective, K is projective,

(iv) for every exact sequence

$$I^0 \rightarrow I^1 \rightarrow \dots \rightarrow I^{n-1} \rightarrow N \rightarrow 0$$

where the I^i are injective, N is injective,

(v) every A -module has an injective resolution of length $\leq n$.

The equivalence of conditions (i), (ii) and (iii) follows from prop. 1. We obviously have (ii) $\Rightarrow$ (ii'). It therefore suffices for us to prove (ii') $\Rightarrow$ (iv) $\Rightarrow$ (v) $\Rightarrow$ (i).

(ii') $\Rightarrow$ (iv): with the notations of (iv), let K be the kernel of $I^0 \rightarrow I^1$. By X, p. 128, cor. 4, we have for every A -module M an isomorphism $\text{Ext}_A^1(\mathbf{M}, \mathbf{N}) \rightarrow \text{Ext}_A^{n+1}(\mathbf{M}, \mathbf{K})$; it then follows from (ii') that $\text{Ext}_A^1(\mathbf{M}, \mathbf{N})$ is zero for every finitely generated A -module M . By X, p. 93, prop. 11, this implies that N is injective, whence (iv).

(iv) $\Rightarrow$ (v): let M be an A -module. Applying (iv) to the exact sequence

$$0 \rightarrow M \rightarrow I^0(M) \rightarrow I^1(M) \rightarrow \cdots \rightarrow I^{n-1}(M) \rightarrow K^{n-1}(M) \rightarrow 0$$

of X, p. 52, we conclude that $K^{n-1}(M)$ is injective, whence (v).

(v) $\Rightarrow$ (i): this follows from X, p. 100, th. 1.

Remarks. — 1) If $\text{dh}(A) \leq n < +\infty$, we have $\text{Tor}_{n+1}^A(P, M) = 0$ for every A-module M and every right A-module P, since $\text{dp}_A(M) \leq n$ (cf. lemma 1).

2) For $\text{dh}(A)$ to be finite, it is necessary and sufficient that $\text{dp}_A(M)$ be finite for every nonzero A-module M. This indeed follows from what precedes and from X, p. 135, cor. 1.

COROLLARY. — Suppose A is left noetherian and let n be an integer > 0 . The following conditions are equivalent:

- (i) $\text{dh}(A) \leq n$,
- (ii) for every pair of finitely generated A-modules M and N, we have $\text{Ext}_A^{n+1}(M, N) = 0$,
- (iii) for every finitely generated left A-module M and every finitely generated right A-module P, we have $\text{Tor}_{n+1}^A(P, M) = 0$.

This follows from props. 2 and 4.

Remark. — By the equivalence of (i) and (iii), we have $\text{dh}(A) = \text{dh}(A^\circ)$ if A is right and left noetherian. This equality is not satisfied in general (X, p. 204, exercise 20).

PROPOSITION 5. — Suppose A is left noetherian and of finite homological dimension and let $\mathcal{C}_0$ (resp. $\mathcal{C}$) be the set of isomorphism classes of finitely generated projective A-modules (resp. of finitely generated A-modules). Then the canonical homomorphism of Grothendieck groups $K(\mathcal{C}_0) \rightarrow K(\mathcal{C})$ is bijective.

This follows from X, p. 137, cor.

10.8.4 Rings of homological dimension 0

PROPOSITION 6. — The following conditions are equivalent:

- (i) every A -module is projective,
- (ii) every A -module is injective,
- (iii) every ideal of A is an injective module,
- (iv) $\text{dh}(\mathbf{A}) \leq 0$,
- (v) A is semisimple,
- (vi) A is noetherian and every A -module is flat,
- (vii) every complex of A -modules is split,
- (viii) every exact sequence of A -modules is split.

By prop. 4, we have $(i) \Leftrightarrow (ii) \Leftrightarrow (iv)$; by cor. 1 to prop. 4, we have $(vi) \Rightarrow (iv)$. By X, p. 35, example 4, we have $(i) \Rightarrow (vii)$; since $(ii) \Rightarrow (iii)$ and $(vii) \Rightarrow (viii)$ are trivial and since $(viii) \Rightarrow (i)$ follows from II, p. 39, prop. 4, it remains to prove $(iii) \Rightarrow (v)$ and $(v) \Rightarrow (vi)$; the last assertion follows from VIII, § 5, no. 1, props. 1 and 2; finally, if every ideal of A is injective, it is a direct summand in A , whence $(iii) \Rightarrow (v)$.

10.8.5 Rings of homological dimension 1

PROPOSITION 7. — The following conditions are equivalent:

- (i) $\text{dh}(\mathbf{A}) \leq 1$
- (ii) every submodule of a projective module is projective,
- (ii') every ideal of A is projective,
- (iii) every quotient of an injective A -module is injective,
- (iv) for every projective complex C , there exists a homologism $\varphi : C \rightarrow H(C)$ such that $H(\varphi) = 1_{H(C)}$,
- (v) for every injective complex C , there exists a homologism $\psi : H(C) \rightarrow C$ such that $H(\psi) = 1_{H(C)}$.

(vi) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii): this follows from prop. 4 of X, p. 138.

(vii) $\Rightarrow$ (iv): let C be a projective complex. If (ii) holds, the submodule $B(C)$ of C is projective, whence (iv) by X, p. 35, remark b).

(viii) $\Rightarrow$ (v): let C be an injective complex. If (iii) holds, the quotient $B_n(C)$ of C_{n+1} is injective for every n , whence (v) by X, p. 35, remark a).

(ix) $\Rightarrow$ (ii): let P be a projective A -module, M a submodule of P , $i : \mathbf{M} \rightarrow \mathbf{P}$ the canonical injection. Let $p : \mathbf{L} \rightarrow \mathbf{M}$ be a surjective homomorphism of a free module L onto M . Consider the projective complex C such that $C_1 = \mathbf{L}$, $C_0 = \mathbf{P}$, $C_i = 0$ for $i \neq 0, 1$, $d_1 = i \circ p$. If (iv) is satisfied, let $\varphi : C \rightarrow H(C)$ be a homologism such that $H(\varphi) = 1_{H(C)}$. Since $H_1(C) = \text{Ker } p$, φ_1 is a projector of L onto $\text{Ker } p$, the exact sequence

$$0 \rightarrow \text{Ker } p \rightarrow L \xrightarrow{p} M \rightarrow 0$$

is split and M is isomorphic to a direct summand of L , hence is projective.

(v) $\Rightarrow$ (iii): let I be an injective module, M a quotient of I , $\pi : I \rightarrow M$ the canonical projection. Let $i : M \rightarrow J$ be an injective homomorphism of M into an injective module J . Consider the injective complex C such that $C^0 = I$, $C^1 = J$, $C^i = 0$ for $i \neq 0, 1$, $d^0 = i \circ \pi$. If (v) is satisfied, let $\psi : H(C) \rightarrow C$ be a homologism such that $H(\psi) = 1_{H(C)}$. Since

$$H^1(C) = \text{Coker } i,$$

ψ^1 is a section of the canonical projection $J \rightarrow \text{Coker } i$, and M is a direct summand in J , hence injective.

(ii) $\Rightarrow$ (ii'): this is trivial.

(ii') $\Rightarrow$ (ii): this follows from VII, § 3, cor. 1 to th. 1.

Example. — If A is principal, then $\text{dh}(A) \leq 1$.

Remark. — If A is (commutative) integral, the preceding conditions are also equivalent to the following:

Integral rings satisfying these conditions are called Dedekind rings (cf. X, p. 204, exercise 12 and AC, VII, § 2, no. 2, th. 1).

COROLLARY. — Let A be a ring of homological dimension ≤ 1 , C a complex of projective A -modules, $\tilde{C}$ a complex of projective right A -modules, C' a complex of injective A -modules, P a right A -module, M a left A -module, n an integer. We then have split exact sequences

$$\begin{aligned} 0 \rightarrow \bigoplus_{p+q=n} H_p(\tilde{C}) \otimes_A H_q(C) &\xrightarrow{\gamma} H_n(\tilde{C} \otimes_A C) \xrightarrow{\alpha} \bigoplus_{p+q=n-1} \text{Tor}_1^A(H_p(\tilde{C}), H_q(C)) \rightarrow 0, \\ 0 \rightarrow \prod_p \text{Ext}_A^1(H_p(C), H^{n-p-1}(C')) &\xrightarrow{\beta} H^n(\text{Homgr}_A(C, C')) \\ &\xrightarrow{\lambda_p} \prod_p \text{Homgr}_A(H_p(C), H^{n-p}(C')) \rightarrow 0, \\ 0 \rightarrow P \otimes_A H_n(C) &\xrightarrow{\gamma_p} H_n(P \otimes_A C) \xrightarrow{\alpha_p} \text{Tor}_1^A(P, H_{n-1}(C)) \rightarrow 0, \\ 0 \rightarrow \text{Ext}_A^1(H_{n-1}(C), M) &\xrightarrow{\beta_p} H^n(\text{Homgr}_A(C, M)) \xrightarrow{\lambda_p} \text{Hom}_A(H_n(C), M) \rightarrow 0. \end{aligned}$$

Since $Z(C)$, $B(C)$, $Z(\tilde{C})$ and $B(\tilde{C})$ are projective and $B(C')$ injective, this follows from X, p. 78, cor. 2, p. 96, cor. 1 and p. 98, cor. 2.

10.8.6 Homological dimension of polynomial rings

Lemma 2. — Let $\rho : A \rightarrow A'$ be a ring homomorphism, M an A -module, M' a A' -module. If the A -module A'_d is flat, we have $\text{dp}_{A'}(A' \otimes_A M) \leq \text{dp}_A(M)$. If the A -module A'_s is projective, we have $\text{dp}_A(M') \leq \text{dp}_{A'}(M')$.

The first assertion is clear if $\text{dp}_A(M) = \pm\infty$; if $\text{dp}_A(M) = n \in \mathbb{N}$, there exists an exact sequence of A -modules

$$0 \rightarrow P_n \rightarrow P_{n-1} \rightarrow \cdots \rightarrow P_0 \rightarrow M \rightarrow 0$$

where the P_i are projective; the sequence of A' -modules

$$0 \rightarrow A' \otimes_A P_n \rightarrow A' \otimes_A P_{n-1} \rightarrow \cdots \rightarrow A' \otimes_A P_0 \rightarrow A' \otimes_A M \rightarrow 0$$

is exact, since A'_d is flat, and the A' -modules $A' \otimes_A P_i$ are projective (II, p. 89, cor.); hence $\text{dp}_{A'}(A' \otimes_A M) \leq n = \text{dp}_A(M)$. The second assertion is clear if $\text{dp}_{A'}(M') = \pm\infty$; if

$\text{dp}_{\mathbf{A}'}(\mathbf{M}') = m \in \mathbf{N}$, there exists an exact sequence of $\mathbf{A}'$ -modules

$$0 \rightarrow P'_m \rightarrow P'_{m-1} \rightarrow \dots \rightarrow P'_0 \rightarrow M' \rightarrow 0,$$

where the P'_i are projective; the sequence of underlying A -modules is exact. On the other hand, each P'_i is a direct summand of an A' -submodule of a module $A_s'^{(I)}$, hence is a projective A -module; we therefore have $\text{dp}_A(\mathbf{M}') \leq m = \text{dp}_{\mathbf{A}'}(\mathbf{M}')$.

Lemma 3. — Suppose A is commutative and let M be an $A[X]$ -module.

- a) We have $\text{dp}_A(\mathbf{M}) \leq \text{dp}_{A[X]}(\mathbf{M}) \leq \text{dp}_A(\mathbf{M}) + 1$.
- b) If the homothety X_M is injective, we have $\text{dp}_A(M/XM) \leq \text{dp}_{A[X]}(M)$.
- c) If $\mathbf{XM} = 0$, we have $\text{dp}_A(\mathbf{M}) + 1 = \text{dp}_{A[X]}(\mathbf{M})$.
- d) We have an exact sequence of $A[X]$ -modules (III, p. 106 and VII, § 5, no. 1)

$$0 \rightarrow A[X] \otimes_A M \rightarrow A[X] \otimes_A M \rightarrow M \rightarrow 0;$$

Assertion (a) then follows from X, p. 135, cor. 2 and from Lemma 2.

- b) If $\text{dp}_{A[X]}(\mathbf{M}) = \pm\infty$ the assertion is trivial. If $\mathbf{M}$ is projective and nonzero, then the A -module $\mathbf{M}/\mathbf{XM}$ is identified with $A \otimes_{A[X]} \mathbf{M}$, hence is projective, and we have $\text{dp}_A(\mathbf{M}/\mathbf{XM}) \leq 0 = \text{dp}_{A[X]}(\mathbf{M})$. Let us reason by induction on $\text{dp}_{A[X]}(\mathbf{M}) = n$, assumed > 0 . Consider an exact sequence of $A[X]$ -modules $0 \rightarrow \mathbf{N} \rightarrow \mathbf{L} \rightarrow \mathbf{M} \rightarrow 0$, where $\mathbf{L}$ is a free $A[X]$ -module; applying to the diagram

$$\begin{array}{ccccccc} 0 & \rightarrow & \mathbf{N} & \rightarrow & \mathbf{L} & \rightarrow & \mathbf{M} \rightarrow 0 \\ & & X_N \downarrow & & X_L \downarrow & & X_M \downarrow \\ 0 & \rightarrow & \mathbf{N} & \rightarrow & \mathbf{L} & \rightarrow & \mathbf{M} \rightarrow 0 \end{array}$$

prop. 2 of X, p. 4, we see that X_N is injective and that we have the exact sequence

$$0 \rightarrow \mathbf{N}/X\mathbf{N} \rightarrow \mathbf{L}/X\mathbf{L} \rightarrow \mathbf{M}/X\mathbf{M} \rightarrow 0.$$

Since L is free over $A[X]$, L/XL is free over A and we have

$$\mathrm{dp}_{A[X]}(\mathbf{N}) = n - 1, \quad \mathrm{dp}_A(\mathbf{M}/\mathbf{XM}) \leq 1 + \mathrm{dp}_A(\mathbf{N}/\mathbf{XN});$$

as $\mathrm{dp}_A(\mathbf{N}/\mathbf{XN}) \leq n - 1$ by the induction hypothesis, we deduce $\mathrm{dp}_A(\mathbf{M}/\mathbf{XM}) \leq n$, which is what had to be proved.

c) The assertion is trivial if $\mathrm{dp}_{A[X]}(\mathbf{M}) = \pm\infty$, and also if $\mathrm{dp}_{A[X]}(\mathbf{M}) = 0$ (which is impossible since $\mathbf{XM} = 0$). We may therefore assume $\mathrm{dp}_{A[X]}(\mathbf{M}) = n > 0$. Considering as above an exact sequence $0 \rightarrow \mathbf{N} \rightarrow \mathbf{L} \rightarrow \mathbf{M} \rightarrow 0$, where $\mathbf{L}$ is a free $A[X]$ -module, we obtain an exact sequence of A -modules

$$0 \rightarrow \mathbf{M} \rightarrow \mathbf{N}/\mathbf{XN} \rightarrow \mathbf{L}/\mathbf{XL} \rightarrow \mathbf{M} \rightarrow 0.$$

By b), we have $\mathrm{dp}_A(\mathbf{N}/\mathbf{XN}) \leq \mathrm{dp}_{A[X]}(\mathbf{N}) = \mathrm{dp}_{A[X]}(\mathbf{M}) - 1 = n - 1$; since $\mathrm{dp}_A(\mathbf{L}/\mathbf{XL}) = 0$, we deduce from the preceding exact sequence, applying X, p. 135, cor. 2 twice, that $\mathrm{dp}_A(\mathbf{M}) \leq n - 1$. But, by a), we have

$$\mathrm{dp}_A(\mathbf{M}) \geq \mathrm{dp}_{A[X]}(\mathbf{M}) - 1 = n - 1,$$

whence c).

THEOREM 1. — Suppose A is commutative. Then

$$\mathrm{dh}(A[X]) = \mathrm{dh}(A) + 1.$$

For every $A[X]$ -module M , we have (lemma 3)

$$\mathrm{dp}_{A[X]}(\mathbf{M}) \leq \mathrm{dp}_A(\mathbf{M}) + 1 \leq \mathrm{dh}(A) + 1$$

hence $\mathrm{dh}(A[X]) \leq \mathrm{dh}(A) + 1$; conversely, if $\mathbf{M}$ is an A -module, let $\overline{\mathbf{M}}$ be the $A[X]$ -module obtained by endowing $\mathbf{M}$ with the structure for which $\mathbf{XM} = 0$, then (lemma 3)

$$\mathrm{dp}_A(\mathbf{M}) = \mathrm{dp}_{A[X]}(\overline{\mathbf{M}}) - 1 \leq \mathrm{dh}(A[X]) - 1,$$

hence $\mathrm{dh}(A) \leq \mathrm{dh}(A[X]) - 1$.

COROLLARY 1. — Suppose A is commutative. We have

$$\mathrm{dh}(A[X_1, \dots, X_n]) = \mathrm{dh}(A) + n.$$

This follows from the theorem by induction on n .

COROLLARY 2. — Let K be a commutative field (resp. a principal ideal ring $*$ or a Dedekind ring $*$ which is not a field). Then $\mathrm{dh}(K[X_1, \dots, X_n])$ is equal to n (resp. $n + 1$). This follows from the fact that $\mathrm{dh}(K) = 0$ (resp. $\mathrm{dh}(K) = 1$).

10.8.7 Homological dimension of graded modules

In this section, we assume that A is a graded ring with degrees ≥ 0 . We denote by $(\mathbf{A}_n)_{n \in \mathbb{Z}}$ its grading; we thus have $A_n = 0$ for $n < 0$, A_0 is a subring of A , $J_0 = \bigoplus_{n>0} A_n$ is a two-sided ideal of A and the quotient graded ring A/J_0 is identified with A_0 .

Lemma 4. — Let M be a graded A -module bounded below (X, p. 56). If $\mathbf{A}_0 \otimes_A \mathbf{M} = 0$, then $\mathbf{M} = 0$.

Since $A_0 \otimes_A M$ is isomorphic to M/J_0M , this is none other than II, p. 171, prop. 6.

Lemma 5. — Let M be a graded A -module bounded below, and let

$$s : A \otimes_{A_0} M/J_0M \rightarrow M$$

be a graded A -homomorphism such that $1 \otimes_A s : \mathbf{A}_0 \otimes_A (A \otimes_{A_0} M/J_0M) \rightarrow \mathbf{A}_0 \otimes_A \mathbf{M}$ is the canonical isomorphism. Then s is surjective. If $\mathrm{Tor}_1^A(\mathbf{A}_0, \mathbf{M}) = 0$, s is bijective.

We have an exact sequence

$$0 \rightarrow \mathrm{Ker} s \rightarrow A \otimes_{A_0} M/J_0M \xrightarrow{s} M \rightarrow \mathrm{Coker} s \rightarrow 0$$

and the graded A -modules $\mathrm{Ker} s$ and $\mathrm{Coker} s$ are bounded below. From the exact sequence

$\mathbf{A}_0 \otimes_{\mathbf{A}} (\mathbf{A} \otimes_{\mathbf{A}_0} \mathbf{M} / \mathbf{J}_0 \mathbf{M}) \xrightarrow{1 \otimes s} \mathbf{A}_0 \otimes_{\mathbf{A}} \mathbf{M} \rightarrow \mathbf{A}_0 \otimes_{\mathbf{A}} \text{Coker } s \rightarrow 0$, we deduce that $\mathbf{A}_0 \otimes_{\mathbf{A}} \text{Coker } s = 0$, hence s is surjective (lemma 4). We then have an exact sequence

$$\text{Tor}_1^{\mathbf{A}}(\mathbf{A}_0, \mathbf{M}) \rightarrow \mathbf{A}_0 \otimes_{\mathbf{A}} \text{Ker } s \rightarrow \mathbf{A}_0 \otimes_{\mathbf{A}} (\mathbf{A} \otimes_{\mathbf{A}_0} \mathbf{M} / \mathbf{J}_0 \mathbf{M}) \xrightarrow{1 \otimes s} \mathbf{A}_0 \otimes_{\mathbf{A}} \mathbf{M}.$$

If $\text{Tor}_1^{\mathbf{A}}(\mathbf{A}_0, \mathbf{M}) = 0$, then $\mathbf{A}_0 \otimes_{\mathbf{A}} \text{Ker } s = 0$ and s is injective (lemma 4).

PROPOSITION 8. — Let $\mathbf{M}$ be a graded $\mathbf{A}$ -module bounded below.

- a) The following conditions are equivalent:
- (i) $\mathbf{M}$ is isomorphic to a graded $\mathbf{A}$ -module of the form $\mathbf{A} \otimes_{\mathbf{A}_0} \mathbf{N}$, where $\mathbf{N}$ is a graded projective (resp. graded free) $\mathbf{A}_0$ -module;
 - (ii) $\mathbf{M}$ is a projective (resp. free) graded $\mathbf{A}$ -module;
 - (iii) $\mathbf{M} / \mathbf{J}_0 \mathbf{M}$ is a projective (resp. graded free) $\mathbf{A}_0$ -module and $\text{Tor}_1^{\mathbf{A}}(\mathbf{A}_0, \mathbf{M}) = 0$.
- b) Suppose moreover that $\mathbf{M}$ has a generating system consisting of homogeneous elements of bounded degrees. Then the following conditions are equivalent:
- (i) The graded $\mathbf{A}$ -module $\mathbf{M}$ possesses a finite composition series whose quotients are isomorphic to graded $\mathbf{A}$ -modules of the form $\mathbf{A} \otimes_{\mathbf{A}_0} \mathbf{N}$, where $\mathbf{N}$ is a flat graded $\mathbf{A}_0$ -module;
 - (ii) $\mathbf{M}$ is a flat $\mathbf{A}$ -module;
 - (iii) $\mathbf{M} / \mathbf{J}_0 \mathbf{M}$ is a flat $\mathbf{A}_0$ -module and $\text{Tor}_1^{\mathbf{A}}(\mathbf{A}_0, \mathbf{M}) = 0$.

In each of the two cases, we evidently have (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii). We must therefore prove (iii) $\Rightarrow$ (i).

- a) The graded $\mathbf{A}$ -module $\mathbf{A} \otimes_{\mathbf{A}_0} \mathbf{M} / \mathbf{J}_0 \mathbf{M}$ is a projective $\mathbf{A}$ -module since $\mathbf{M} / \mathbf{J}_0 \mathbf{M}$ is a projective $\mathbf{A}_0$ -module. The canonical homomorphism of $\mathbf{A}$ -modules

$$p : \mathbf{M} \rightarrow \mathbf{M} / \mathbf{J}_0 \mathbf{M}$$

is surjective; there therefore exists a graded $\mathbf{A}$ -homomorphism of degree zero

$$s : \mathbf{A} \otimes_{\mathbf{A}_0} \mathbf{M}/J_0\mathbf{M} \rightarrow \mathbf{M}$$

such that $p \circ s(a \otimes x) = ax$ for $a \in A$ and $x \in M/J_0M$.

By Lemma 5, s is an isomorphism of A -modules from $\mathbf{A} \otimes_{\mathbf{A}_0} \mathbf{M}/J_0\mathbf{M}$ over $\mathbf{M}$, whence (i).

- b) By the hypothesis on M , there exist integers a, b such that $a \leq b$ such that M is generated by $\oplus M_i$. Let us reason by induction on the positive integer $b - a$.

If $b - a = 0$ then M is generated by M_a and the canonical $\mathbf{A}_0$ -homomorphism $M_a \rightarrow M/J_0M$ is bijective; one then deduces from the A -homomorphism $\mathbf{A} \otimes_{\mathbf{A}_0} M_a \rightarrow \mathbf{M}$ defined by the structure of M as an A -module a graded A -homomorphism

$$s : \mathbf{A} \otimes_{\mathbf{A}_0} \mathbf{M}/J_0\mathbf{M} \rightarrow \mathbf{M}$$

satisfying the condition of Lemma 5. Then, by Lemma 5, s is bijective, whence (i).

In the general case, let $\mathbf{M}^{(a)}$ be the (graded) sub- A -module of M generated by M_a and M' the quotient $\mathbf{M}/\mathbf{M}^{(a)}$. We have an exact sequence

$$0 \rightarrow \mathbf{M}^{(a)} \xrightarrow{f} \mathbf{M} \xrightarrow{g} \mathbf{M}' \rightarrow 0,$$

whence, since $\text{Tor}_1^A(\mathbf{A}_0, \mathbf{M}) = 0$ by hypothesis, exact sequences

(6)

$$\begin{aligned} \text{Tor}_2^A(\mathbf{A}_0, \mathbf{M}') &\rightarrow \text{Tor}_1^A(\mathbf{A}_0, \mathbf{M}^{(a)}) \rightarrow 0 \\ 0 &\rightarrow \text{Tor}_1^A(\mathbf{A}_0, \mathbf{M}') \rightarrow M^{(a)}/J_0M^{(a)} \xrightarrow{1 \otimes f} M/J_0M \xrightarrow{1 \otimes g} M'/J_0M' \rightarrow 0. \end{aligned} \quad (7)$$

But the canonical homomorphism $M_a \rightarrow M^{(a)}/J_0M^{(a)}$ is bijective. It follows that the homomorphism $1 \otimes f : M^{(a)}/J_0M^{(a)} \rightarrow M/J_0M$ is injective and that its image is a sub- A_0 -module direct summand of M/J_0M . It then follows from the exact sequence (7) that $\text{Tor}_1^A(\mathbf{A}_0, \mathbf{M}') = 0$ and that the A_0 -module M'/J_0M' is flat since it is isomorphic to a direct summand of M/J_0M . By the induction hypothesis (which applies to M' since the latter is generated by the M'^i for $a < i \leq b$), M' satisfies condition (i), hence is flat. One then deduces from the exact sequence

(6) that $\text{Tor}_1^A(A_0, M^{(a)}) = 0$, but $M^{(a)}/J_0 M^{(a)}$ is identified with M_a which is a flat A_0 -module (as a direct summand submodule of $M/J_0 M$); by what has already been proved, the graded A -module $M^{(a)}$ is isomorphic to $A \otimes_{A_0} M_a$ hence also satisfies (i), which completes the proof.

COROLLARY 1. — Let M be a finitely generated graded A -module. If the A_0 -module $M/J_0 M$ is projective (resp. graded free, resp. flat) and if $\text{Tor}_1^A(A_0, M) = 0$ then the A -module M is projective (resp. graded free, resp. flat).

Corollary 2. — Suppose that every projective A_0 -module is free (resp. that A be noetherian and that every finitely generated projective A_0 -module be free). Let M be a graded A -module bounded below (resp. a finitely generated graded A -module) and let n be an integer ≥ 0 such that $\text{dp}_A(M) \leq n$. There exists an exact sequence of graded A -modules and graded homomorphisms of degree 0

$$0 \rightarrow L_n \rightarrow L_{n-1} \rightarrow \cdots \rightarrow L_0 \rightarrow M \rightarrow 0$$

where the L_i are graded free and bounded below (resp. graded free and finitely generated).

Indeed, there exists (X, p. 56, prop. 11) an exact sequence of graded A -modules and homomorphisms of degree 0

$$0 \rightarrow L_n \rightarrow L_{n-1} \rightarrow \cdots \rightarrow L_0 \rightarrow M \rightarrow 0$$

where the L_i are bounded below (resp. finitely generated) for $0 \leq i \leq n$ and graded free for $0 \leq i \leq n-1$.

Since $\text{dp}_A(M) \leq n$ the A -module L_n is projective; therefore $A_0 \otimes_A L_n$ is a projective A_0 -module, hence graded free; since L_n is bounded below and $A_0 \otimes_A L_n$ graded free, L_n is graded free (prop. 7).

COROLLARY 3 (Hilbert's syzygy theorem). — Suppose that A_0 is a commutative field and that A is generated as A_0 -algebra by n homogeneous elements of degrees > 0 algebraically independent. For every graded A -module M bounded below (resp. of finite type), there exists an exact sequence of graded A -modules and graded homomorphisms of degree 0

$$0 \rightarrow L_n \rightarrow L_{n-1} \rightarrow \cdots \rightarrow L_0 \rightarrow M \rightarrow 0,$$

where the L_i are graded free and bounded below (resp. and finitely generated).

Indeed, $\text{dh}(\mathbf{A}) = n$ by theorem 1 of X, p. 143, and one applies cor. 2.

Remark. — Cor. 2 also applies to the following cases:

- a) $\mathbf{A}_0$ is principal and $\mathbf{A} = \mathbf{A}_0[\mathbf{X}_1, \dots, \mathbf{X}_{n-1}]$;
- b) $\mathbf{A}_0$ is a regular local noetherian ring of dimension r and $\mathbf{A} = \mathbf{A}_0[\mathbf{X}_1, \dots, \mathbf{X}_n, r]$. *

COROLLARY 4. — Assume $\mathbf{A}_0$ semisimple. Let $\mathbf{M}$ be a graded $\mathbf{A}$ -module bounded below and n an integer ≥ 0 . For $\text{dp}_{\mathbf{A}}(\mathbf{M}) \leq n$, it is necessary and sufficient that $\text{Tor}_{n+1}^{\mathbf{A}}(\mathbf{A}_0, \mathbf{M}) = 0$.

If $\text{dp}_{\mathbf{A}}(\mathbf{M}) \leq n$, then $\text{Tor}_{n+1}^{\mathbf{A}}(\mathbf{A}_0, \mathbf{M}) = 0$ (X, p. 135, lemma 1). Conversely, let $0 \rightarrow \mathbf{K} \rightarrow \mathbf{L}_{n-1} \rightarrow \cdots \rightarrow \mathbf{L}_1 \rightarrow \mathbf{L}_0 \rightarrow \mathbf{M} \rightarrow 0$ be an exact sequence of graded $\mathbf{A}$ -modules bounded below such that $\mathbf{L}_0, \dots, \mathbf{L}_{n-1}$ are graded free (X, p. 56, prop. 11); by cor. 3 of X, p. 131, the equality $\text{Tor}_{n+1}^{\mathbf{A}}(\mathbf{A}_0, \mathbf{M}) = 0$ implies $\text{Tor}_1^{\mathbf{A}}(\mathbf{A}_0, \mathbf{K}) = 0$; since $\mathbf{K}/\mathbf{J}_0\mathbf{K}$ is a projective $\mathbf{A}_0$ -module since $\mathbf{A}_0$ is semisimple (X, p. 140, prop. 6), $\mathbf{K}$ is projective by prop. 8 (X, p. 144), and $\text{dp}_{\mathbf{A}}(\mathbf{M}) \leq n$.

COROLLARY 5. — Suppose the ring $\mathbf{A}_0$ semisimple. If $\text{Tor}_{n+1}^{\mathbf{A}}(\mathbf{A}_0, \mathbf{A}_0) = 0$, we have $\text{dp}_{\mathbf{A}}(\mathbf{M}) \leq n$ for every graded $\mathbf{A}$ -module bounded below.

Let $\mathbf{A}^\circ$ denote the opposite graded ring of $\mathbf{A}$: we have $(\mathbf{A}^\circ)_0 = (\mathbf{A}_0)^\circ$, so $(\mathbf{A}^\circ)_0$ is semisimple (VIII, § 5, n° 1, remark 3). Since $\text{Tor}_{n+1}^{\mathbf{A}^\circ}(\mathbf{A}_0^\circ, \mathbf{A}_0^\circ) = 0$, we have $\text{dp}_{\mathbf{A}}(\mathbf{A}_0) \leq n$ by Cor. 4: this implies $\text{Tor}_{n+1}^{\mathbf{A}^\circ}(\mathbf{M}_0, \mathbf{A}_0^\circ) = 0$ for every $\mathbf{A}$ -module $\mathbf{M}$, hence $\text{Tor}_{n+1}^{\mathbf{A}}(\mathbf{A}_0, \mathbf{M}) = 0$ and we apply Cor. 4.

10.9 KOSZUL COMPLEXES

In this paragraph, all rings considered are commutative.

10.9.1 The complexes $\mathbf{K}(u)$, $\mathbf{K}_*(u, C)$, $\mathbf{K}'(u, C)$

Let $\mathbf{A}$ be a ring, $\mathbf{L}$ an $\mathbf{A}$ -module, $u : \mathbf{L} \rightarrow \mathbf{A}$ a linear form, and $\Lambda(\mathbf{L})$ the exterior algebra of the $\mathbf{A}$ -module $\mathbf{L}$. For $x \in \Lambda(\mathbf{L})$ let $d_u(x)$ denote the interior product $x \sqsubseteq u$ (III, p. 161, example).

According to loc. cit., p. 162, formula (60), we have

$$d_u(e_1 \wedge \cdots \wedge e_n) = \sum_{i=1}^n (-1)^{i+1} u(e_i) e_1 \wedge \cdots \wedge e_{i-1} \wedge e_{i+1} \wedge \cdots \wedge e_n \quad (1)$$

for $e_1, \dots, e_n$ in L . According to III, p. 164 and 165, the map $d_u : \Lambda(L) \rightarrow \Lambda(L)$ is an antiderivation of degree (-1) and of square zero. It is the unique antiderivation of the A -algebra $\Lambda(L)$ which extends $u : \Lambda^1(L) \rightarrow \Lambda^0(L)$.

DEFINITION 1. — The complex $(\Lambda(L), d_u)$ is denoted $\mathbf{K}^A(u)$ or $\mathbf{K}(u)$.

Be careful that $\mathbf{K}_n(u) = \Lambda^n(L) = \mathbf{K}^{-n}(u)$. It is clear that $\mathbf{K}(u)$ is zero on the right and that $H_0(\mathbf{K}(u)) = \text{Coker}(u) = A/\mathfrak{q}$ where $\mathfrak{q}$ is the ideal $u(L)$ of A .

For every complex C of A -modules, we set

$$\mathbf{K}^A(u, C) = C \otimes_A \mathbf{K}^A(u), \quad \mathbf{K}_*(u, C) = \text{Homgr}_A(\mathbf{K}^A(u), C).$$

$$H^A(u, C) = H(C \otimes_A \mathbf{K}^A(u)) \quad H_A^\bullet(u, C) = H(\text{Homgr}_A(\mathbf{K}^A(u), C)),$$

$$H_r^A(u, C) = H_r(C \otimes_A \mathbf{K}^A(u)), \quad H_A^r(u, C) = H^r(\text{Homgr}_A(\mathbf{K}^A(u), C)).$$

We therefore have canonical homomorphisms of A -modules (X, p. 62 and p. 82)

$$\gamma_0 : H_0(C) \otimes_A A/\mathfrak{q} \rightarrow H_0^A(u, C),$$

$$\lambda^0 : H_A^0(u, C) \rightarrow \text{Hom}_A(A/\mathfrak{q}, H^0(C)).$$

Lemma 1. — If the complex C is zero on the right (resp. on the left), then $\mathbf{K}^A(u, C)$ (resp. $\mathbf{K}_A^*(u, C)$) is zero on the right (resp. on the left), and γ_0 (resp. λ^0) is bijective.

This follows from X, p. 62, prop. 1 and p. 82, prop. 1.

PROPOSITION 1. — Let $x \in L$; denote by $\mathbf{R}_x : y \mapsto x \wedge y$ the left multiplication by x in the algebra $\Lambda(L)$. Then $d_u \circ \mathbf{R}_x + \mathbf{R}_x \circ d_u = u(x) \cdot 1_{\Lambda(L)} = u(x)_{\Lambda(L)}$.

Indeed, $(d_u \circ \mathbf{R}_x + \mathbf{R}_x \circ d_u)(y) = d_u(x \wedge y) + x \wedge d_u(y)$; since d_u is a derivation, $d_u(x \wedge y) +$

$$x \wedge d_u(y) = d_u(x) \wedge y = u(x).y .$$

COROLLARY 1. — If u is surjective, $\mathbf{K}(u)$ is homotopic to zero (X, p. 34) as is $\mathbf{K}^A(u, C)$ and $\mathbf{K}'(u, C)$ for every complex C .

Indeed, there exists $x \in \mathbf{L}$ such that $u(x) = 1$. Then $\mathbf{K}(u)$ is homotopic to zero by prop. 1, hence also $\mathbf{K}^A(u, C)$ (X, p. 64, prop. 3) and $\mathbf{K}_A^*(u, C)$ (X, p. 83, prop. 3).

COROLLARY 2. — Let C be a complex, $\text{Ann}(C)$ its annihilator. Then $q + \text{Ann}(C)$ annihilates $H^A(u, C)$ and $H_A^*(u, C)$.

For every $\lambda \in q$, the homothety $\lambda_{\mathbf{K}(u)}$ is homotopic to zero by the proposition, hence also $1_C \otimes \lambda_{\mathbf{K}(u)}$ and $\text{Homgr}(\lambda_{\mathbf{K}(u)}, 1_C)$ by X, p. 64, prop. 3 and X, p. 83, prop. 3; it follows that λ annihilates $H(u, C)$ and $H'(u, C)$. If $\lambda \in \text{Ann}(C)$, then $1_{\mathbf{K}(u)} \otimes \lambda_C$ and $\text{Homgr}(1_{\mathbf{K}(u)}, \lambda_C)$ are zero.

Suppose L is projective (resp. $\mathbf{K}(u)$ acyclic in degrees > 0). Then the complex $\Lambda(L)$ is projective by III, p. 87, cor. 2 (resp. is a resolution of A/q); by X, p. 102 (resp. p. 100), we therefore have, for every A -module M , homomorphisms

(2)

$$H_r^A(u, M) \rightarrow \text{Tor}_r^A(A/q, M), \quad \text{Ext}_A^r(A/q, M) \rightarrow H_A^r(u, M) \quad (3)$$

$$\text{Tor}_r^A(A/q, M) \rightarrow H_r^A(u, M), \quad H^r(u, M) \rightarrow \text{Ext}_A^r(A/q, M).$$

If L is projective and $\mathbf{K}(u)$ acyclic in degrees > 0 , the homomorphisms (2) and (3) above are bijective and inverses of each other (X, p. 102, prop. 1).

PROPOSITION 2. — Let $(\mathbf{L}_i)_{i \in I}$ be a family of A -modules, where the set I is finite and totally ordered. Let u be a linear form on $\bigoplus_{i \in I} \mathbf{L}_i$, u_i its restriction to $\mathbf{L}_i$.

The canonical isomorphism of A -algebras (III, p. 84)

$$g : \bigotimes_{i \in I} \Lambda(\mathbf{L}_i) \rightarrow \Lambda\left(\bigoplus_{i \in I} \mathbf{L}_i\right)$$

is an isomorphism of the complex $\bigotimes \mathbf{K}(u_i)$ (X, p. 63) on the complex $\mathbf{K}(u)$.

Indeed, according to X, p. 64, remark 4, the differential D of the complex $\otimes_{i \in I} \mathbf{K}(u_i)$ is an antiderivation; the antiderivations d_u and $g \circ D \circ g^{-1}$ of $\Lambda(\oplus L_i)$ coincide on $\oplus L_i$ with the map $x \mapsto u(x) \cdot 1$ of $\oplus L_i$ to $\Lambda(\oplus L_i)$, hence they are equal (III, p. 128, cor.).

Let C and C' be two complexes of A -modules. We have (X, p. 63 and p. 99) canonical isomorphisms of complexes

$$C \otimes_A (C' \otimes_A \mathbf{K}(u)) \rightarrow (C \otimes_A C') \otimes_A \mathbf{K}(u)$$

$$\text{Homgr}_A(C', \text{Homgr}_A(\mathbf{K}(u), C)) \rightarrow \text{Homgr}_A(C' \otimes_A \mathbf{K}(u), C),$$

that is, isomorphisms

(4)

$$C \otimes_A \mathbf{K}^A(u, C') \rightarrow \mathbf{K}^A(u, C \otimes_A C') \quad (5)$$

$$\text{Homgr}_A(C', \mathbb{K}_A^*(u, C)) \rightarrow \text{Homgr}_A(\mathbb{K}_A^*(u, C'), C).$$

In (4) and (5), take $C' = \mathbf{K}(u')$, where $u' : L' \rightarrow A$ is a linear form on an A -module L' and note that $\mathbf{K}^A(u, \mathbf{K}(u'))$ which is equal by definition to $\mathbf{K}(u') \otimes_A \mathbf{K}(u)$ identifies, according to prop. 2, with $\mathbf{K}(u' \oplus u)$ where $u' \oplus u : L' \oplus L \rightarrow A$ is the linear form $(x', x) \mapsto u'(x') + u(x)$. We then obtain isomorphisms of complexes

(6)

$$\mathbf{K}_\bullet^A(u' \oplus u, C) \rightarrow \mathbf{K}_\bullet^A(u, \mathbf{K}_\bullet^A(u', C)) \quad (7)$$

$$\mathbf{K}_A^\bullet(u', \mathbf{K}_A^\bullet(u, C)) \rightarrow \mathbf{K}_A^\bullet(u' \oplus u, C).$$

By passing to homology, we deduce isomorphisms of A -modules

$$H_r^A(u' \oplus u, C) \rightarrow H_r^A(u, K_\bullet^A(u', C)), \quad r \in \mathbf{Z},$$

$$H_A^r(u', K_\bullet^A(u, C)) \rightarrow H_A^r(u' \oplus u, C), \quad r \in \mathbf{Z}.$$

Finally, note that the homomorphism deduced from the product in the algebra $\Lambda(L)$

$$m : K^A(u) \otimes_A K^A(u) \rightarrow K^A(u)$$

is a morphism of complexes (since d_u is an antiderivation). Assuming L is free of rank n and composing with the morphism of complexes $K^A(u) \rightarrow \Lambda^n L(-n)$ which is the identity in degree n from this one deduces a morphism of complexes

$$\chi : K^A(u) \otimes_A K^A(u) \rightarrow \Lambda^n L(-n);$$

to this morphism there corresponds canonically, according to X, p. 99, prop. 12, a morphism of complexes

$$\varphi : K^A(u) \rightarrow \text{Homgr}_A(K^A(u), \Lambda^n L(-n))$$

which is bijective (III, p. 87, formula (20)). For every complex C , one deduces from this a composed isomorphism

$$\begin{aligned} K_\bullet^A(u, C) &= C \otimes_A K^A(u) \xrightarrow{1 \otimes \varphi} C \otimes \text{Homgr}_A(K^A(u), \Lambda^n L(-n)) \rightarrow \\ &\rightarrow \text{Homgr}_A(K^A(u), C \otimes_A \Lambda^n L(-n)) = K_\bullet^A(u, C \otimes_A \Lambda^n L(-n)). \end{aligned}$$

By passing to homology, one therefore has canonical isomorphisms

$$H_r^A(u, C) \rightarrow H_A^{n-r}(u, C \otimes_A \Lambda^n L), \quad r \in \mathbf{Z}. \quad (8)$$

Remarks. — 1) * The preceding remains valid when L is projective of rank n . *

2) Since L is free of rank n , $\Lambda^n L$ is isomorphic to A , we have non-canonical isomorphisms

$$H_r^A(u, C) \rightarrow H_A^{n-r}(u, C) .$$

10.9.2 Functoriality

Let $f : C \rightarrow C'$ be a morphism of complexes. We denote by

$$K_{\bullet}^A(u, f) : K_{\bullet}^A(u, C) \rightarrow K_{\bullet}^A(u, C'),$$

$$K_A^{\bullet}(u, f) : K_A^{\bullet}(u, C) \rightarrow K_A^{\bullet}(u, C'),$$

the morphisms of complexes $f \otimes 1_{K(u)}$ and $\text{Homgr}_A(1_{K(u)}, f)$.

We denote by $H_{\bullet}^A(u, f) : H_{\bullet}^A(u, C) \rightarrow H_{\bullet}^A(u, C')$, $H_A^{\bullet}(u, f) : H_A^{\bullet}(u, C) \rightarrow H_A^{\bullet}(u, C')$, $H_r^A(u, f) : H_r^A(u, C) \rightarrow H_r^A(u, C')$, $H_A^r(u, f) : H_A^r(u, C) \rightarrow H_A^r(u, C')$ the induced morphisms in homology. The map $f \mapsto K_{\bullet}^A(u, f)$ is linear; if $g : C' \rightarrow C''$ is another morphism of complexes, we have $K_{\bullet}^A(u, g \circ f) = K_{\bullet}^A(u, g) \circ K_{\bullet}^A(u, f)$; similarly for $K_A^{\bullet}, H_{\bullet}^A, H_A^{\bullet}, H_r^A, H_A^r$.

Let $0 \rightarrow C' \xrightarrow{f} C \xrightarrow{g} C'' \rightarrow 0$ be an exact sequence of complexes.

a) Suppose L is flat; then $\Lambda(L)$ is flat (X, p. 15, cor.). The sequence

$$0 \rightarrow K_{\bullet}^A(u, C') \xrightarrow{K_{\bullet}^A(u, f)} K_{\bullet}^A(u, C) \xrightarrow{K_{\bullet}^A(u, g)} K_{\bullet}^A(u, C'') \rightarrow 0$$

is then exact, and gives rise (X, p. 30) to an exact homology sequence

$$\dots \rightarrow H_n^A(u, C') \xrightarrow{H_n^A(u, f)} H_n^A(u, C) \xrightarrow{H_n^A(u, g)} H_n^A(u, C'') \xrightarrow{\hat{\sigma}_n} H_{n-1}^A(u, C') \rightarrow \dots$$

b) Suppose L is projective; then $\Lambda(L)$ is projective. The sequence

$$0 \rightarrow K_A^{\bullet}(u, C') \xrightarrow{K_A^{\bullet}(u, f)} K_A^{\bullet}(u, C) \xrightarrow{K_A^{\bullet}(u, g)} K_A^{\bullet}(u, C'') \rightarrow 0$$

is then exact, and gives rise to an exact homology sequence

$$\dots \rightarrow H_A^n(u, C') \xrightarrow{H_A^n(u, f)} H_A^n(u, C) \xrightarrow{H_A^n(u, g)} H_A^n(u, C'') \xrightarrow{\partial^n} H_A^{n+1}(u, C') \rightarrow \dots$$

Let $\rho : A \rightarrow A'$ be a ring homomorphism, L' be the A' -module $A' \otimes_A L$, $u' : L' \rightarrow A'$ the linear form $1 \otimes u$. The canonical bijective homomorphism (III, p. 83, prop. 8)

$$\psi : \Lambda_{A'}(A' \otimes_A L) \rightarrow A' \otimes_A \Lambda_A(L)$$

is an isomorphism of complexes of A' -modules. We deduce:

1) for every complex of A' -modules C' , an isomorphism of complexes of A -modules

$$\mathbf{K}_{\bullet}^{A'}(u', C') \rightarrow \mathbf{K}_{\bullet}^A(u, C'),$$

(9)

composed of the diagram

$$C' \otimes_{A'} (\Lambda_{A'}(A' \otimes_A L)) \xrightarrow{1_{C'} \otimes \psi} C' \otimes_{A'} A' \otimes_A \Lambda_A(L) \xrightarrow{\varphi} C' \otimes_A \Lambda_A(L)$$

where φ is the canonical bijection (III, p. 85, prop. 14);

2) for every complex of A -modules C , an isomorphism of complexes of A' -modules

$$\mathbf{K}_{\bullet}^A(u, A' \otimes_A C) \rightarrow A' \otimes_A \mathbf{K}_{\bullet}^A(u, C),$$

whence homomorphisms of A' -modules

$$A' \otimes_A H_n^A(u, C) \rightarrow H_n^{A'}(u', A' \otimes_A C),$$

which are bijective when A' is flat over A (X, p. 66, cor. 2).

Let L' be an A -module, $u' : L' \rightarrow A$ a linear form, $f : L \rightarrow L'$ an A -homomorphism such that $u' \circ f = u$. It follows from III, p. 161, formula (55), that the homomorphism $\Lambda(f) : \Lambda(L) \rightarrow \Lambda(L')$ satisfies $d_u \circ \Lambda(f) = \Lambda(f) \circ d_{u'}$, hence defines a morphism of complexes $\Lambda(u) : \mathbf{K}^A(u) \rightarrow \mathbf{K}^A(u')$. If C is an A -complex, we deduce morphisms of complexes

$$1_C \otimes \Lambda(u) : \mathbf{K}^A(u, C) \rightarrow \mathbf{K}^A(u', C) \quad \text{and} \quad \text{Homgr}(\Lambda(u), 1_C) : \mathbf{K}_A^*(u', C) \rightarrow \mathbf{K}_A^*(u, C)$$

If f is bijective, all these morphisms are isomorphisms.

10.9.3 Example 1: the complex $\mathbf{S}(L) \otimes_A \Lambda(L)$

Let A be a ring, L an A -module, $\mathbf{S}(L)$ its symmetric algebra, $\mathbf{S}(L) \otimes_A L$ be the $\mathbf{S}(L)$ -module obtained by extension of scalars, $u : \mathbf{S}(L) \otimes_A L \rightarrow \mathbf{S}(L)$ the linear form such that $u(s \otimes x) = sx$ for $s \in \mathbf{S}(L)$, $x \in L$. By the canonical isomorphism of $\mathbf{S}(L)$ -modules (III, p. 83, prop. 8)

$$\Lambda_{\mathbf{S}(L)}(\mathbf{S}(L) \otimes_A L) \rightarrow \mathbf{S}(L) \otimes_A \Lambda(L),$$

the differential of the complex $\mathbf{K}^{\mathbf{S}(L)}(u)$ is transported to the map

$$d : \mathbf{S}(L) \otimes_A \Lambda(L) \rightarrow \mathbf{S}(L) \otimes_A \Lambda(L)$$

such that, for $x_1, \dots, x_p, y_1, \dots, y_q$ in L , we have

$$\begin{aligned} d((x_1 \dots x_p) \otimes (y_1 \wedge \dots \wedge y_q)) \\ = \sum_{i=1}^q (-1)^{i+1} y_i x_1 \dots x_p \otimes (y_1 \wedge \dots \wedge y_{i-1} \wedge y_{i+1} \wedge \dots \wedge y_q). \end{aligned}$$

Note that d maps $\mathbf{S}^p(L) \otimes \Lambda^q(L)$ to $\mathbf{S}^{p+1}(L) \otimes \Lambda^{q-1}(L)$, so that the complex of A -modules $\mathbf{S}(L) \otimes \Lambda(L)$ decomposes as the direct sum of the complexes described by the following diagrams:

$$(\mathcal{E}_n) : 0 \rightarrow \mathbf{S}^0 L \otimes_A \Lambda^n L \rightarrow \mathbf{S}^1 L \otimes_A \Lambda^{n-1} L \rightarrow \dots \rightarrow \mathbf{S}^n L \otimes_A \Lambda^0 L \rightarrow 0, n \in \mathbf{N}.$$

If the A -module L is the direct sum of a finite family $(\mathbf{L}_i)_{i \in I}$ where I is totally ordered, the canonical bijection

$$\bigotimes_{i \in I}^{\varepsilon} (\mathbf{S}(L_i) \otimes_A \Lambda(L_i)) \rightarrow \mathbf{S}(L) \otimes_A \Lambda(L)$$

is an isomorphism of complexes of A -modules (this follows from prop. 2 of X, p. 148 or from formula (9) above).

PROPOSITION 3. — If the A -module L is flat, then the sequences $(\mathcal{E}_n)$ above are exact for $n > 0$.

a) Let us first note that, if p_L is the composite homomorphism

$$\mathbf{S}(L) \otimes \Lambda(L) \xrightarrow{\alpha} \mathbf{S}^0(L) \otimes \Lambda^0(L) \xrightarrow{\beta} A,$$

where α is the tensor product of the canonical projections and β the canonical isomorphism; it remains to prove that $\mathbf{H}(p_L)$ is bijective.

b) If $L = 0$ or if $L = A$, the proposition is evident.

c) Suppose L is free of finite rank; write it as a direct sum $L_1 \oplus \dots \oplus L_n$ of free A -modules of rank 1. By the remark preceding the proposition, the complex $\mathbf{S}(L) \otimes \Lambda(L)$ is isomorphic to the tensor product of the n free complexes $\mathbf{S}(L_i) \otimes \Lambda(L_i)$ whose homology is free by b). By X, p. 79, cor. 4, the canonical homomorphism

$$\gamma : \bigoplus_{i=1}^n \mathbf{H}(\mathbf{S}(L_i) \otimes \Lambda(L_i)) \rightarrow \mathbf{H}(\mathbf{S}(L) \otimes \Lambda(L))$$

is bijective. By b) $\mathbf{H}(p_{L_i})$ is bijective for all i . Since $\bigotimes_{i=1}^n \mathbf{H}(p_{L_i}) = \mathbf{H}(p_L) \circ \gamma$,

$\mathbf{H}(p_L)$ is bijective.

d) In the general case, L is the inductive limit of a filtered inductive system $(L_i)_{i \in I}$ of free modules of finite rank (X, p. 14, th. 1). Since the canonical bijective homomorphism

$$\varinjlim \mathbf{S}(L_i) \otimes \Lambda(L_i) \rightarrow \mathbf{S}(L) \otimes \Lambda(L)$$

is an isomorphism of complexes, the proposition follows from X, p. 28, prop. 1.

Remarks. — 1) We shall see below (X, p. 158, example) another proof of part c) above.

2) If A is a $\mathbb{Q}$ -algebra, the conclusion of Prop. 3 remains true without any hypothesis on L (cf. X, p. 206, Exercise 1).

Let G be a group and $\rho : G \rightarrow \mathbf{GL}(L)$ a linear representation of G in a flat A -module L . Then the $(\mathcal{E}_n)$ are exact sequences of linear representations. Suppose L is finitely generated projective, and denote by $R_A(G)$ the ring of representations of G in finitely generated projective A -modules. It follows from Prop. 3 that we have in $R_A(G)$ the relations

$$\sum_{i=0}^n (-1)^i [S^i(L)] [\Lambda^{n-i}(L)] = 0, \quad n > 0. \quad (10)$$

If we consider the formal series

$$s(T) = \sum_{i=0}^{\infty} [S^i(L)] T^i \in R_A(G)[[T]],$$

$$\lambda(T) = \sum_{i=0}^{\infty} [\Lambda^i(L)] T^i \in R_A(G)[[T]],$$

relations (10) are written

(11)

$$s(T) \lambda(-T) = 1_*.$$

10.9.4 Example 2: the case of a free module

Let k be a ring, $\mathbf{M}$ a k -module, $\mathbf{I}$ a set and p be an integer ≥ 0 . A map $m : \mathbf{I}^p \rightarrow \mathbf{M}$ is said to be alternating if it satisfies the following two conditions:

a) for every permutation $\sigma \in \mathfrak{S}_p$ and every sequence $(\alpha_1, \dots, \alpha_p) \in \mathbf{I}^p$, we have

$$m(\alpha_{\sigma(1)}, \dots, \alpha_{\sigma(p)}) = \varepsilon_{\sigma} m(\alpha_1, \dots, \alpha_p),$$

b) for every sequence $(\alpha_1, \dots, \alpha_p) \in \mathbb{I}^p$ such that two of the indices $\alpha_1, \dots, \alpha_p$ are equal, one has $m(\alpha_1, \dots, \alpha_p) = 0$.

(In the case where I is a k -module and m is multilinear, one recovers the notion introduced in III, p. 80.)

Suppose I is finite and denote by $C_I^p(M)$ the k -module of alternating maps from I^p to M .

Let L_0 be a k -module, $(e_i)_{i \in I}$ a family of elements of L_0 ; we define two k -linear maps

$$g : \text{Hom}_k(\Lambda^p L_0, M) \rightarrow C_I^p(M)$$

$$h : C_I^p(M) \rightarrow M \otimes_k \Lambda^p L_0$$

as follows: if $f \in \text{Hom}_k(\Lambda^p L_0, M)$, we set

$$g(f)(\alpha_1, \dots, \alpha_p) = f(e_{\alpha_1} \wedge \dots \wedge e_{\alpha_p});$$

let $m \in C_I^p(M)$, we define $h(m) \in M \otimes_k \Lambda^p L_0$. For every element $(\alpha_1, \dots, \alpha_p)$

of I^p the element $m(\alpha_1, \dots, \alpha_p) \otimes (e_{\alpha_1} \wedge \dots \wedge e_{\alpha_p})$ of $\Lambda^p L_0 \otimes_k M$ is zero if $\text{Card} \{\alpha_1, \dots, \alpha_p\} < p$ and is independent of the order of the indices $\alpha_1, \dots, \alpha_p$ if

$$\text{Card} \{\alpha_1, \dots, \alpha_p\} = p.$$

It depends only on the subset $J = (\alpha_1, \dots, \alpha_p)$ of I ; denote it by $h_J(m)$; we have $h_J(m) = 0$ if $\text{Card}(J) < p$; we then set:

$$h(m) = \sum_J h_J(m),$$

where J ranges over the subsets of I having p elements.

Lemma 2. — If the k -module L_0 is free with basis $(e_i)_{i \in I}$, the k -linear maps g and h are bijective.

This follows from III, p. 79, th. 1.

Now let M be a k -module, and $x = (x_i)_{i \in I}$ a family of k -endomorphisms of M , pairwise commuting. Consider the polynomial ring $A = k[(X_i)_{i \in I}]$ and endow M with the A -module structure such that $P(X_i)m = P(x_i)m$ for $P \in A$ and $m \in M$. Let, on the other hand, L be

the free A -module A^I , $(e_i)_{i \in I}$ its canonical basis and $u : L \rightarrow A$ the linear form which maps e_i over X_i for all $i \in I$. Consider the complexes of k -modules $\mathbf{K}^A(u, \mathbf{M})$ and $\mathbf{K}_A^\bullet(u, \mathbf{M})$; we have canonical isomorphisms

$$\begin{aligned}\mathbf{K}_A^P(u, \mathbf{M}) &= \text{Hom}_A(\Lambda_A^P(A^I), \mathbf{M}) \rightarrow \text{Hom}_k(\Lambda_k^P(k^I), \mathbf{M}), \\ \mathbf{M} \otimes_k \Lambda_k^P(k^I) &\rightarrow \mathbf{M} \otimes_A \Lambda_A^P(A^I) = \mathbf{K}_P^A(u, \mathbf{M});\end{aligned}$$

whence, by composition with the isomorphisms g and h , isomorphisms of k -modules

$$\begin{aligned}\theta^P &: \mathbf{K}_A^P(u, \mathbf{M}) \rightarrow C_1^P(\mathbf{M}), \\ \theta_P &: C_1^P(\mathbf{M}) \rightarrow \mathbf{K}_P^A(u, \mathbf{M}).\end{aligned}$$

We denote by

$$\begin{aligned}\hat{c}^P &: C_1^P(\mathbf{M}) \rightarrow C_1^{P+1}(\mathbf{M}), \\ \hat{c}_P &: C_1^P(\mathbf{M}) \rightarrow C_1^{P-1}(\mathbf{M}),\end{aligned}$$

the k -homomorphisms obtained by transporting the differentials of $\mathbf{K}_A^\bullet(u, \mathbf{M})$ and $\mathbf{K}_A^A(u, \mathbf{M})$ via the isomorphisms θ . We have, for example:

$$(\partial^P m)(\alpha_1, \dots, \alpha_{p+1}) = \sum_{j=1}^{p+1} (-1)^{j+1} x_{\alpha_j} m(\alpha_1, \dots, \alpha_{j-1}, \alpha_{j+1}, \dots, \alpha_{p+1}). \quad (12)$$

The complex formed by the $C_I^P(\mathbf{M})$ and the ∂^P (resp. the ∂_P) is denoted $\mathbf{K}'(\mathbf{x}, \mathbf{M})$ (resp. $\mathbf{K}(\mathbf{x}, \mathbf{M})$) and is called the ascending (resp. descending) Koszul complex

associated with the module $\mathbf{M}$ and the sequence of endomorphisms $(x_1, \dots, x_n)$. We therefore have isomorphisms of complexes of k -modules

$$\begin{aligned}\theta^\bullet &: \mathbf{K}_A^\bullet(u, \mathbf{M}) \rightarrow \mathbf{K}^\bullet(x, \mathbf{M}), \\ \theta_\bullet &: \mathbf{K}_\bullet(x, \mathbf{M}) \rightarrow \mathbf{K}_A^A(u, \mathbf{M}).\end{aligned}$$

Remark. — Conversely, let B be a k -algebra, L a free B -module with basis $(e_i)_{i \in I}$, and $\mathbf{M}$ a B -module. The data of a linear form $u : L \rightarrow B$ is equivalent to that of a family $x = (x_i)_{i \in I}$ of elements of B , via the relation $x_i = u(e_i)$. The complex of k -modules underlying $\mathbf{K}_B^\bullet(u, \mathbf{M})$ (resp. $\mathbf{K}^B(u, \mathbf{M})$) is then identified, via the isomorphism $\theta^\bullet$ (resp. $\theta_\bullet$) with the Koszul complex

$\mathbf{K}'(x, M)$ (resp. $\mathbf{K}^\bullet(x, M)$). For example $\mathbf{K}^B(u)$ is identified with $\mathbf{K}'(x, B)$.

We introduce, as in no. 1 (X, p. 147), the notations $H_*(x, M)$, $H^*(x, M)$, etc., and all the results of nos. 1 and 2 apply mutatis mutandis, the module A^I being free. For example, we have isomorphisms

$$\begin{aligned} H_0(x, M) &\rightarrow M/(x)M \\ H^0(x, M) &\rightarrow \text{Hom}_A(A/(x), M), \end{aligned}$$

where (x) denotes the ideal $\sum \mathbf{A}x_i$ of A . Also for example, if $\mathbf{K}(\mathbf{x}, \mathbf{A})$ is acyclic in degrees > 0 , we have isomorphisms

$$\begin{aligned} H_r(x, M) &\rightarrow \text{Tor}_r^A(k, M), \\ \text{Ext}_A^r(k, M) &\rightarrow H^r(x, M). \end{aligned}$$

Finally, suppose I is (finite and) totally ordered, for example $I = \{1, \dots, n\}$; let us identify $\Lambda^n(\mathbf{A}^I)$ to A by means of the basis element $e_1 \wedge \dots \wedge e_n$ and let us translate the isomorphism $\mathbf{K}_p^A(u, M) \rightarrow \mathbf{K}_A^{n-p}(u, M)$ of X, p. 149. It becomes, by transport by (θ_p) and (θ^{n-p}) , the isomorphism

$$C_I^p(\mathbf{M}) \rightarrow C_I^{n-p}(\mathbf{M})$$

which associates with $m \in C_I^p(M)$ the element $\check{m}$ of $C_I^{n-p}(M)$ such that

$$m(\alpha_1, \dots, \alpha_p) = \check{m}(\beta_1, \dots, \beta_{n-p}) \quad (13)$$

if $(\alpha_1, \dots, \alpha_p, \beta_1, \dots, \beta_{n-p})$ is an even permutation of $\{1, \dots, n\}$. Let us also note that when $I = \{1, \dots, n\}$, one can identify $C_I^p(M)$ with the set of families $m(\alpha_1, \dots, \alpha_p)$ of elements of M where $\alpha_1 < \alpha_2 < \dots < \alpha_p$; formula (12) remains valid, as does relation (13).

Example. — Take $\mathbf{M} = k[\mathbf{T}_1, \dots, \mathbf{T}_n]$; the Koszul complex $\mathbf{K}(\partial/\partial \mathbf{T}, \mathbf{M})$ associated with the sequence of endomorphisms $(\partial/\partial \mathbf{T}_1, \dots, \partial/\partial \mathbf{T}_n)$ is identified with the de Rham complex of $k[x_1, \dots, x_n]$ over k (X, p. 44): to $m \in \mathbf{C}_{\{1, \dots, n\}}^p(\mathbf{M})$, one associates the differential form

$$\omega(m) = \sum_{\alpha_1 < \dots < \alpha_p} m(\alpha_1, \dots, \alpha_p) dx_{\alpha_1} \wedge \dots \wedge dx_{\alpha_p},$$

cf. formula (12) and example 1, p. 44.

10.9.5 Example 3: the case $L = A$

If $L = A$, set $u(1) = x \in A$. The complex $\mathbf{K}(u)$ then has length 1, and we have $\mathbf{K}_0(u) = \mathbf{K}_1(u) = A$ and $d_1(a) = xa$, so for every A -module M , $\mathbf{K}_0(u, M)$, $\mathbf{K}_1(u, M)$, $\mathbf{K}^0(u, M)$ and $\mathbf{K}^1(u, M)$ are identified with M , the differentials

$$d_1 : \mathbf{K}_1(u, M) \rightarrow \mathbf{K}_0(u, M) \quad \text{and} \quad d^0 : \mathbf{K}^0(u, M) \rightarrow \mathbf{K}^1(u, M)$$

being $m \mapsto xm$. We therefore have isomorphisms

$$\begin{aligned} H_0(x, M) &\rightarrow A/xA \otimes_A M \leftarrow H^1(x, M), \\ H_1(x, M) &\rightarrow \text{Hom}_A(A/xA, M) \leftarrow H^0(x, M). \end{aligned}$$

Lemma 3. — Let K be a complex such that $\mathbf{K}_i = 0$ for $i \neq 0, 1$ and let C be a complex and p an integer.

a) If K is flat, we have for all $p \in \mathbb{N}$ an exact sequence

$$0 \rightarrow H_0(\mathbf{K} \otimes_A H_p(C)) \rightarrow H_p(\mathbf{K} \otimes_A C) \rightarrow H_1(\mathbf{K} \otimes_A H_{p-1}(C)) \rightarrow 0.$$

b) If K is projective, we have for every $p \in \mathbb{N}$ an exact sequence

$$\begin{aligned} 0 \rightarrow H^1(\text{Homgr}_A(\mathbf{K}, H^{p-1}(C))) &\rightarrow H^p(\text{Homgr}_A(\mathbf{K}, C)) \\ &\rightarrow H^0(\text{Homgr}_A(\mathbf{K}, H^p(C))) \rightarrow 0. \end{aligned}$$

Let us prove *a*, the proof of *b* being analogous. For every i , denote by $\mathbf{K}_{(i)}$ the complex $\mathbf{K}_i(-i)$. We have an exact sequence of complexes, split as a sequence of A -modules

$$0 \rightarrow \mathbf{K}_{(0)} \xrightarrow{\alpha} \mathbf{K} \xrightarrow{\beta} \mathbf{K}_{(1)} \rightarrow 0;$$

the sequence

$$0 \rightarrow K_{(0)} \otimes_A C \xrightarrow{\alpha \otimes 1} K \otimes_A C \xrightarrow{\beta \otimes 1} K_{(1)} \otimes_A C \rightarrow 0 \quad (14)$$

is exact, and since K is flat, the homomorphisms

$$\gamma_{0,p}(K_{(0)}, C) : K_{(0)} \otimes_A H_p(C) \rightarrow H_p(K_{(0)} \otimes_A C)$$

$$\gamma_{1,p-1}(K_{(1)}, C) : K_{(1)} \otimes_A H_{p-1}(C) \rightarrow H_{p-1}(K_{(1)} \otimes_A C)$$

are bijective (X, p. 66, cor. 2). Let us compute the connecting homomorphism $\partial(\alpha \otimes 1, \beta \otimes 1)$; by definition, it maps the class of the cycle $\sum a_i \otimes b_i$ to that of $\sum da_i \otimes b_i$, which means that

$$\partial(\alpha \otimes 1, \beta \otimes 1) \circ \gamma(K_{(1)}, C) = \gamma(K_{(0)}, C) \circ (d_K \otimes 1).$$

The exact homology sequence associated with (14) therefore takes the form

$$\begin{aligned} K_1 \otimes H_p(C) &\xrightarrow{d_K \otimes 1} K_0 \otimes H_p(C) \rightarrow H_p(K \otimes C) \\ &\rightarrow K_1 \otimes H_{p-1}(C) \xrightarrow{d_K \otimes 1} K_0 \otimes H_{p-1}(C), \end{aligned}$$

whence a).

Applying lemma 3, a) to the complex $K(u)$, and using the commutation isomorphisms, we obtain:

PROPOSITION 4. — For every complex C , we have exact sequences.

$$0 \rightarrow A/xA \otimes_A H_p(C) \rightarrow H_p(x, C) \rightarrow \text{Hom}_A(A/xA, H_{p-1}(C)) \rightarrow 0.$$

Corollary 1. — For $H_p(x, C) = 0$ it is necessary and sufficient that the homothety with ratio x to $H_p(C)$ be surjective and the homothety with ratio x to $H_{p-1}(C)$ be injective.

COROLLARY 2. — Let $x = (x_1, \dots, x_n)$ be a sequence of elements of A , M an A -module, x' the sequence $(x_1, \dots, x_{n-1})$. We have exact sequences

$$0 \rightarrow A/x_n A \otimes_A H_p(x', M) \rightarrow H_p(x, M) \rightarrow \text{Hom}_A(A/x_n A, H_{p-1}(x', M)) \rightarrow 0.$$

Corollary 3. — For $H_i(\mathbf{x}, \mathbf{M})$ to be zero for all $i > 0$, it is necessary and sufficient that the homothety with ratio x_n to $H_i(\mathbf{x}', \mathbf{M})$ be bijective for $i > 0$, and that the homothety with ratio x_n to $M/(\mathbf{x}') M$ be injective.

10.9.6 Completely intersecting families

Let A be a ring, M an A -module, $x = (x_i)_{i \in I}$ a family of elements of A .

DEFINITION 2. — The family x is said to be completely secant for M if one has $H_i(\mathbf{x}, \mathbf{M}) = 0$ for $i > 0$.

If I is finite, it amounts to the same thing (X, p. 150) to say that one has $\mathbf{H}^i(\mathbf{x}, \mathbf{M}) = 0$ for $i < \text{Card}(I)$.

The following proposition allows us to give examples of completely secant families.

PROPOSITION 5. — Let $x = (x_1, \dots, x_n)$ be a sequence of elements of A . If for $i = 1, \dots, n$ the homothety of ratio x_i in the A -module $M/(x_1 M + \dots + x_{i-1} M)$ is injective, the sequence x is completely secant for M .

A sequence satisfying the conditions of the proposition is said to be regular for M , or M -regular. Note that this property generally depends on the order of the x_i . For example, the sequence $(1, 0)$ is always M -regular, whereas the sequence $(0, 1)$ is M -regular only if M is zero. On the other hand, whether a sequence is completely secant does not depend on the order of the terms.

Let us prove the proposition by induction on n , the case $n = 0$ being immediate. Set $x' = (x_1, \dots, x_{n-1})$; if the sequence x is M -regular, the sequence x' is M -regular and multiplication by x_n to $M/(\mathbf{x}')M$ is injective; by the induction hypothesis, we have $H_i(\mathbf{x}', M) = 0$ for $i > 0$; it then follows from cor. 3 of X, p. 157, that $H_i(\mathbf{x}, M) = 0$ for $i > 0$.

Example. — Let k be a ring; take $\mathbf{A} = k[\mathbf{X}_1, \dots, \mathbf{X}_n]$ and $x = (\mathbf{X}_1, \dots, \mathbf{X}_n)$. The sequence x is $\mathbf{A}$ -regular and the proposition gives back acyclicity in degrees > 0 of the complex $\mathbf{S}_k(k^n) \otimes_k$

$\Lambda_k(k^n)$ (cf. X, p. 152, prop. 3).

Similarly in the ring of formal power series $\hat{\mathbf{A}} = k[[\mathbf{X}_1, \dots, \mathbf{X}_n]]$, the sequence $(\mathbf{X}_1, \dots, \mathbf{X}_n)$ is $\hat{\mathbf{A}}$ -regular, hence completely secant for $\hat{\mathbf{A}}$.

PROPOSITION 6. — a) If $\sum_{i \in I} x_i \mathbf{A} = \mathbf{A}$, the family $(x_i)_{i \in I}$ is completely secant for $\mathbf{M}$.

b) Let $x = (x_1, \dots, x_n)$ be a sequence of elements of $\mathbf{A}$. Let $(a_{ij}) \in \mathbf{GL}_n(\mathbf{A})$; set

$$y_i = \sum_j a_{ij} x_j.$$

If the sequence x is completely secant for $\mathbf{M}$, so is the sequence $(y_1, \dots, y_n)$. c) Let $k_1, \dots, k_n$ be integers ≥ 1 ; for the sequence $(x_1^{k_1}, \dots, x_n^{k_n})$ to be completely secant for $\mathbf{M}$, it is necessary and sufficient that the sequence x be completely secant for $\mathbf{M}$. Assertion a) follows from cor. 1, p. 148.

Let us prove b). Let $f : \mathbf{A}^n \rightarrow \mathbf{A}^n$ be the isomorphism defined by the matrix ${}^t(a_{ij})$; it follows from X, p. 151, that $1_{\mathbf{M}} \otimes \Lambda(f)$ is an isomorphism of the complex $\mathbf{K}(\mathbf{y}, \mathbf{M})$ on the complex $\mathbf{K}(x, \mathbf{M})$, whence b).

To prove c), it suffices evidently to show that if k is an integer ≥ 1 , the sequence $(x_1, \dots, x_{n-1}, x_n^k)$ is completely secant for $\mathbf{M}$ if and only if the same holds for the sequence x . Let $x' = (x_1, \dots, x_{n-1})$; by cor. 3, p. 157, the first condition (resp. the second) means that the homothety of ratio x_n^k (resp. x_n) is bijective in $H_i(x', \mathbf{M})$ for $i \geq 1$ and injective in $\mathbf{M}/(x') \mathbf{M}$. These two conditions are clearly equivalent, whence c).

Remark. — 1) The assertion analogous to c) for regular sequences is true (X, p. 207, exercise 5).

PROPOSITION 7. — a) Let $\mathbf{N}$ be a flat $\mathbf{A}$ -module. If the family x is completely secant for $\mathbf{M}$, it is so for $\mathbf{M} \otimes_{\mathbf{A}} \mathbf{N}$.

b) Let $0 \rightarrow \mathbf{M}' \rightarrow \mathbf{M} \rightarrow \mathbf{M}'' \rightarrow 0$ be an exact sequence of $\mathbf{A}$ -modules. If the family x is completely secant for $\mathbf{M}'$ and for $\mathbf{M}''$, it is so for $\mathbf{M}$.

The complex $\mathbf{K}(\mathbf{x}, \mathbf{M} \otimes_{\mathbf{A}} \mathbf{N})$ is isomorphic by definition to $\mathbf{K}(\mathbf{x}, \mathbf{M}) \otimes_{\mathbf{A}} \mathbf{N}$; since $\mathbf{N}$ is flat, we deduce an isomorphism $H_i(\mathbf{x}, \mathbf{M}) \otimes_{\mathbf{A}} \mathbf{N} \rightarrow H_i(\mathbf{x}, \mathbf{M} \otimes_{\mathbf{A}} \mathbf{N})$ (X, p. 66, cor. 2), whence a).

The complex $\mathbf{K}(\mathbf{x})$ being flat, we have an exact sequence of complexes

$$0 \rightarrow \mathbf{K}.(x, M') \rightarrow \mathbf{K}.(x, M) \rightarrow \mathbf{K}.(x, M'') \rightarrow 0;$$

assertion b) follows from the associated exact homology sequence.

Remarks. — 2) The assertions analogous to a) and b) for regular sequences are immediate.

- 3) If the family x is completely secant for A , the complex $K.(x, A)$ defines a free resolution of the A -module A/x , with $x = \sum_{i \in I} x_i A$; we therefore have for every integer $i \geq 0$ and for every A -module M isomorphisms

$$\text{Ext}_{\mathbf{A}}^{n-i}(A/x, \mathbf{M}) \rightarrow H^{n-i}(x, \mathbf{M}) \rightarrow H_i(x, \mathbf{M}) \rightarrow \text{Tor}_i^{\mathbf{A}}(A/x, \mathbf{M}). \quad (15)$$

- 4) We say that the sequence $x = (x_1, \dots, x_n)$ is M -coregular if (denoting by $(x_i)_M$ the homothety with ratio x_i in M) the homothety of ratio x_i in the module

$$\text{Ker}(x_1)_M \cap \dots \cap \text{Ker}(x_{i-1})_M$$

is surjective for $i = 1, \dots, n$. We then have $H_i(x, \mathbf{M}) = 0$ for $i < n$; this is proved in the same manner as prop. 5.

$$A = k[D_1, \dots, D_n],$$

$\mathbf{Q}$ -algebra,

$M = k[T_1, \dots, T_n]$, endowed with the structure of an A -module such that $D_i P = \partial P / \partial T_i$ for all $P \in M$ ($1 \leq i \leq n$). One verifies immediately that the sequence $(D_1, \dots, D_n)$ is M -coregular; taking into account the example on p. 155, one deduces that the de Rham complex of $k[T_1, \dots, T_n]$ over k is acyclic in degrees > 0 .

10.9.7 A criterion for completely secant sequences

Let A be a ring, M an A -module, and x an ideal of A . The x -adic topology on M is the topology compatible with the group structure of M for which the set of submodules $x^r M$ ($r \geq 0$) is a fundamental system of neighborhoods of zero (TG, III, p. 5, example). This topology is separated if and only if

$$\bigcap_{r \geq 0} x^r M = 0.$$

Suppose now that the ideal x is generated by a sequence $x = (x_1, \dots, x_n)$

of elements of A . Consider the graded A -module $\bigoplus_{r \geq 0} x^r M$ and the graded A -homomorphism of degree 0

$$a_M^x : A[X_1, \dots, X_n] \otimes_A M \rightarrow \bigoplus_{r \geq 0} x^r M$$

such that $a_M^x(\mathbf{P} \otimes m) = \mathbf{P}(x_1, \dots, x_n) m$ if $\mathbf{P}$ is a homogeneous polynomial in $X_1, \dots, X_n$ and $m \in M$. Let $\mathfrak{d}$ be the ideal of $A[X_1, \dots, X_n]$ generated by the elements $(x_i X_j - x_j X_i)$ for $1 \leq i < j \leq n$. We have $\mathbf{P}(x_1, \dots, x_n) = 0$ if $\mathbf{P} \in \mathfrak{d}$, so that a_M^x gives, by passing to the quotient, a graded A -homomorphism of degree 0

$$\alpha_M^x : (A[X_1, \dots, X_n] / \mathfrak{d}) \otimes_A M \rightarrow \bigoplus_{r \geq 0} x^r M.$$

By taking the tensor product with A/x , we deduce from α_M^x a graded A -homomorphism of degree 0

$$\beta_M^x : (A/x)[X_1, \dots, X_n] \otimes_A M \rightarrow \bigoplus_{r \geq 0} (x^r M / x^{r+1} M).$$

Homomorphisms a_M^x , α_M^x and β_M^x are surjective.

THEOREM 1. — Consider the following conditions:

- (i) The sequence x is M -regular (X, p. 158).
- (ii) The sequence x is completely secant for M (X, p. 157, def. 2).

(iii) One has $\mathbf{H}_1(x, \mathbf{M}) = 0$.

(iv) L' homomorphism $\alpha_{\mathbf{M}}^x : (\mathbf{A}[\mathbf{X}_1, \dots, \mathbf{X}_n]/\mathfrak{d}) \otimes_{\mathbf{A}} \mathbf{M} \rightarrow \bigoplus x^r \mathbf{M}$ is bijective.

(v) The homomorphism $\beta_{\mathbf{M}}^x : (\mathbf{A}/x)[\mathbf{X}_1, \dots, \mathbf{X}_n] \otimes_{\mathbf{A}} \mathbf{M} \rightarrow \bigoplus_{r \geq 0} (x^r \mathbf{M}/x^{r+1} \mathbf{M})$ is bijective.

We then have the implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Leftrightarrow$ (iv) $\Rightarrow$ (v). If for $1 \leq i \leq n$ the $\mathbf{A}$ -module $\mathbf{M}/(x_1 \mathbf{M} + \dots + x_{i-1} \mathbf{M})$ is separated for the x -adic topology, conditions (i) through (v) are equivalent.

The theorem will be proved in nos. 8 and 9.

COROLLARY 1. — If $\mathbf{A}$ is Noetherian, if the $\mathbf{A}$ -module $\mathbf{M}$ is finitely generated, and if the x_i belong to the radical of $\mathbf{A}$, conditions (i) through (v) of the theorem are equivalent. Indeed, on each of the modules $\mathbf{M}/(x_1 \mathbf{M} + \dots + x_{i-1} \mathbf{M})$ the x -adic topology is separated (AC III, § 3, no. 3, prop. 6).

COROLLARY 2. — Suppose that $\mathbf{A}$ is a positively graded ring, $\mathbf{M}$ a graded $\mathbf{A}$ -module bounded below, and the x_i homogeneous elements of degree > 0 of $\mathbf{A}$. Then conditions (i) through (v) of the theorem are equivalent.

The x -adic topology is indeed separated for any graded $\mathbf{A}$ -module $\mathbf{N}$ that is bounded below, since if $N_n = 0$ for $n < n_0$ we have $x^a \mathbf{N} \subset \sum_{j \geq n_0+a} N_j$ for all $a \geq 0$.

COROLLARY 3. — Suppose that the modules $\mathbf{M}/(x_1 \mathbf{M} + \dots + x_{i-1} \mathbf{M})$ are separated for the x -adic topology ($1 \leq i \leq n$); let p be an integer between 1 and n . For the sequence $\mathbf{x}$ to be completely secant for $\mathbf{M}$, it is necessary and sufficient that the sequence $(x_1, \dots, x_p)$ be completely secant for $\mathbf{M}$ and that the sequence $(x_{p+1}, \dots, x_n)$ be completely secant for $\mathbf{M}/(x_1 \mathbf{M} + \dots + x_p \mathbf{M})$.

Indeed, the corollary is obvious if in the statement one replaces "completely secant sequences" by "regular sequences"; and the two notions coincide here by the theorem.

Remark. — Let $x = (x_1, \dots, x_n)$ be a sequence of elements of $\mathbf{A}$ such that $H_1(x, \mathbf{A}) = 0$; then the kernel of the surjective homomorphism $u : A^n \rightarrow x$ such that $u(\sum a_i e_i) = \sum a_i x_i$ is generated by the elements $X_j e_i - X_i e_j$; consequently, the $\mathbf{A}$ -algebra $A[X_1, \dots, X_n]/\mathfrak{d}$ is isomorphic to the symmetric algebra $S_A(x)$ (III, p. 69, prop. 4). It therefore follows from Theorem 1 that the algebra homomorphism $S_A(x) \rightarrow \bigoplus_n x^n$ deduced from the canonical injection of x to $\bigoplus_n x^n$ is

an isomorphism. The same holds for the homomorphism $S_A(x/x^2) \rightarrow \bigoplus_n x^n/x^{n+1}$.

10.9.8 Proof of Theorem 1: first part

The implication (i) $\Rightarrow$ (ii) has already been proved (X, p. 157, prop. 5). The implication (ii) $\Rightarrow$ (iii) is evident; likewise for (iv) $\Rightarrow$ (v), since β_M^x is identified with $\alpha_M^x \otimes 1_{A/x}$.

To show that (iv) implies (iii), consider the homomorphism $(\alpha_M^x)_1$ induced on the degree-1 components. Let E denote the graded A -module $A[X_1, \dots, X_n]$; the A -module E_1 is free with basis $X_1, \dots, X_n$ and $\mathfrak{d}_1$ is the sub- A -module of E_1 generated by the elements $(x_i X_j - x_j X_i)$ for $1 \leq i < j \leq n$. Consequently, $((E/\mathfrak{d}) \otimes_A M)_1$ is identified with $K_1(x, M)/B_1(K(x, M))$, the homomorphism $(\alpha_M^x)_1$ being identified with the map from $K_1(x, M)/B_1(K(x, M))$ over $B_0(K(x, M))$ induced by d_1 . Thus the vanishing of $H_1(x, M)$ is equivalent to the injectivity of $(\alpha_M^x)_1$, whence the implication (iv) $\Rightarrow$ (iii).

To prove that (iii) implies (iv), we shall use the following lemma:

Lemma 4. — Let A_0 be the ring $\mathbb{Z}[T_1, \dots, T_n]$, and let $u : A_0[X_1, \dots, X_n] \rightarrow A_0[U]$ be the homomorphism of A_0 -algebras such that $u(X_i) = T_i U$. The kernel of u is the ideal $\mathfrak{d}_0$ of $A_0[X_1, \dots, X_n]$ generated by the elements $(T_i X_j - T_j X_i)$ for $1 \leq i < j \leq n$. If $\mathfrak{t}$ is the ideal of A_0 generated by $(T_1, \dots, T_n)$, u induces an isomorphism

$$\bar{u} : A_0[X_1, \dots, X_n]/\mathfrak{d}_0 \rightarrow \bigoplus_{r \geq 0} t^r.$$

It obviously suffices to prove the first assertion. For every sequence of natural integers $\alpha = (\alpha_1, \dots, \alpha_n)$ and every integer $k \in [0, n]$, denote by $P_{\alpha, k}$ the monomial

$$T_1^{\alpha_1} \dots T_k^{\alpha_k} X_{k+1}^{\alpha_{k+1}} \dots X_n^{\alpha_n};$$

let N be the sub- $\mathbb{Z}$ -module of $A_0[X_1, \dots, X_n]$ generated by the $P_{\alpha, k}$ for $\alpha \in N^n$ and $0 \leq k \leq n$. We shall show that the restriction of u to N is injective and that $A_0[X_1, \dots, X_n] = \mathfrak{d}_0 + N$; since one has $\mathfrak{d}_0 \subset \text{Ker } u$, this will imply the lemma.

Observe that N is generated by the set S consisting of $P_{0,0} = 1$ and of those $P_{\alpha, k}$ for which $\alpha_k \neq 0$. To prove the injectivity of the restriction of u to N , it suffices to show that two distinct

elements of S have as images under u distinct monomials in $A_0[U]$. Now one has $u(P_{\alpha,k}) = T^\alpha U^{\sum_{i \geq k} \alpha_i}$, so that the equality $u(P_{\alpha,k}) = u(P_{\alpha',k'})$ implies $\alpha = \alpha'$ and $\sum_{i \geq k} \alpha_i = \sum_{i \geq k'} \alpha_i$. Suppose that $P_{\alpha,k}$ and $P_{\alpha',k'}$ belong to S . If $\alpha = 0$, one then has $k = k' = 0$; if $\alpha \neq 0$, we obtain $k = k'$ since α_k and $\alpha_{k'}$ are nonzero, hence the result.

Let us show that every monomial $T^\alpha X^\beta \in A_0[X_1, \dots, X_n]$ is congruent modulo $\mathfrak{d}_0$ to a $P_{\eta,k}$. For every sequence $\lambda \in \mathbb{N}^n$, we denote by $i(\lambda)$ (resp. $j(\lambda)$) the smallest (resp. the largest) integer $k \in [1, n]$ such that $\lambda_k \neq 0$. In the set of monomials $T^\gamma X^\delta$ that are congruent to $T^\alpha X^\beta$ modulo $\mathfrak{d}_0$ let us choose one for which the rational integer $j(\gamma) - i(\delta)$ is minimal; let us show that we then have $j(\gamma) - i(\delta) < 0$. Suppose that we have $j(\gamma) \geq i(\delta)$; denote by $j = j(\gamma)$, $i = i(\delta)$, and let $\varepsilon = \inf(\gamma_j, \delta_i)$. Since $(T_j^\varepsilon X_i^\varepsilon - T_i^\varepsilon X_j^\varepsilon)$ is divisible by $(T_j X_i - T_i X_j)$, hence belongs to $\mathfrak{d}_0$, we see that $T^\gamma X^\delta$ is congruent modulo $\mathfrak{d}_0$ to $T^{\gamma'} X^{\delta'}$, where $\gamma'_i = \gamma_i + \varepsilon$, $\gamma'_j = \gamma_j - \varepsilon$, $\gamma'_k = \gamma_k$ for $k \neq i, j$, and $\delta'_i = \delta_i - \varepsilon$, $\delta'_j = \delta_j + \varepsilon$, $\delta'_k = \delta_k$ for $k \neq i, j$. Since γ'_j or δ'_i is zero, we have $j(\gamma') - i(\delta') < j(\gamma) - i(\delta)$ which contradicts the minimal character of $j(\gamma) - i(\delta)$.

Consequently we have $j(\gamma) < i(\delta)$, whence $T^\gamma X^\delta \in \mathbb{N}$, which completes the proof of the lemma.

Let us prove that (iii) implies (iv). Consider the ring $A_0 = \mathbb{Z}[T_1, \dots, T_n]$ and the ideal $\mathfrak{t}$ of A_0 generated by $T_1, \dots, T_n$. Endow M with the structure of A_0 -module for which $T_i m = x_i m$ for $m \in M$, $1 \leq i \leq n$. By X, p. 155, $H_i(x, M)$ is canonically identified with $H_i(T, M)$.

Since the sequence T is regular for A_0 it follows from implication (i) $\Rightarrow$ (ii) that the complex $K. (T, A_0)$ defines a free resolution of the A_0 -module $A_0/\mathfrak{t}$. Note that this one identifies with $\mathbb{Z}$, equipped with the structure of A_0 -module such that $T_i \mathbb{Z} = 0$ for $1 \leq i \leq n$. Thus condition (iii) is equivalent to $\text{Tor}_1^{A_0}(M, \mathbb{Z}) = 0$.

Let us show that (iii) implies $\text{Tor}_1^{A_0}(M, A_0/t^r) = 0$ for all $r \geq 1$. This follows from the preceding for $r = 1$. In the general case, consider the exact sequence

$$0 \rightarrow t^r/t^{r+1} \rightarrow A_0/t^{r+1} \rightarrow A_0/t^r \rightarrow 0. \quad (16)$$

The A_0 -module t^r/t^{r+1} is isomorphic to a finite product of copies of $\mathbb{Z}$; we therefore have $\text{Tor}_1^{A_0}(M, t^r/t^{r+1}) = 0$. From this, by induction on r , we deduce that

$$\mathrm{Tor}_1^{A_0}(M, A_0/t^r) = 0 \quad \text{for all } r.$$

The exact sequence (16) therefore yields, by tensor product with M , an exact sequence

$$0 \rightarrow (t^r/t^{r+1}) \otimes_{A_0} M \rightarrow M/x^{r+1}M \rightarrow M/x^rM \rightarrow 0 \quad (17)$$

whence an isomorphism of $(t^r/t^{r+1}) \otimes_{A_0} M$ over $x^rM/x^{r+1}M$. Considering the exact sequence $0 \rightarrow t^{r+1} \rightarrow t^r \rightarrow t^r/t^{r+1} \rightarrow 0$, one then sees by induction on r that the map $m_r : t^r \otimes_{A_0} M \rightarrow x^rM$, induced by the action of A_0 to M is an isomorphism.

To prove (iv), it remains to observe that the diagram

$$\begin{array}{ccc} (A_0[X_1, \dots, X_n]/\mathfrak{d}_0) \otimes_{A_0} M & \xrightarrow{\bar{u} \otimes 1_M} & \bigoplus_{r \geq 0} (t^r \otimes_{A_0} M) \\ \downarrow e & & \\ {}'(A[X_1, \dots, X_n]/\mathfrak{d}) \otimes_A M & \xrightarrow{\alpha_M^*} & \bigoplus_{r \geq 0} x^rM \end{array}$$

where e is the canonical isomorphism of extension of scalars (II, p. 85, prop. 6), is commutative, and to apply Lemma 4.

10.9.9 Proof of Theorem 1: second part

Consider again the exact sequence

$$0 \rightarrow t^r/t^{r+1} \xrightarrow{i_r} A_0/t^{r+1} \xrightarrow{p_r} A_0/t^r \rightarrow 0 \quad (16)$$

and the associated exact sequence of torsion modules

$$\begin{aligned} \mathrm{Tor}_1^{A_0}(A_0/t^{r+1}, M) &\rightarrow \mathrm{Tor}_1^{A_0}(A_0/t^r, M) \\ &\rightarrow (t^r/t^{r+1}) \otimes_{A_0} M \xrightarrow{i, \otimes 1_M} (A_0/t^{r+1}) \otimes_{A_0} M \xrightarrow{p, \otimes 1_M} (A_0/t^r) \otimes_{A_0} M \rightarrow 0. \end{aligned} \quad (18)$$

The kernel of $p_r \otimes 1_M$ is identified with $x^rM/x^{r+1}M$; moreover, t^r/t^{r+1} is annihilated by the T_i and is identified with the homogeneous component of degree r of A_0 , so that the

homomorphism $(t^r/t^{r+1}) \otimes_{A_0} M \rightarrow x^r M/x^{r+1} M$ deduced from $i_r \otimes 1_M$ is identified with the homogeneous component of degree r of the homomorphism β_M^x . It therefore follows from the exact sequence (18) that condition (v) is equivalent to

(v'): the homomorphism $\text{Tor}_1^{A_0}(p_r, 1_M) : \text{Tor}_1^{A_0}(A_0/t^{r+1}, M) \rightarrow \text{Tor}_1^{A_0}(A_0/t^r, M)$ is surjective for all $r \geq 1$.

It remains for us to prove the implication (v) $\Rightarrow$ (i) when the modules

$$M/(x_1 M + \dots + x_{i-1} M)$$

are separated for the x -adic topology ($1 \leq i \leq n$). Let $\overline{M}$ be the A -module $M/x_1 M$. By definition, the sequence x is regular for M if and only if $(x_1)_M$ is injective and the sequence $x' = (x_2, \dots, x_n)$ is regular for $\overline{M}$. Reasoning by induction on n , it therefore suffices to prove that, if M is separated for the x -adic topology and if β_M^x is bijective, then $(x_1)_M$ is injective and $\beta_M^{x'}$ is bijective. Now the bijectivity of β_M^x implies in particular that the homothety with ratio x_1 to $\bigoplus_r x^r M/x^{r+1} M$ is injective, hence that $\text{Ker } (x_1)_M \subset \bigcap_i x^i M$ and consequently that $(x_1)_M$ is injective if the x -adic topology on M is separated.

We are thus reduced to proving that if $(x_1)_M$ is injective and if M satisfies the condition (v'), then $\overline{M}$ satisfies the condition (v') relative to the sequence x' .

By hypothesis, we have an exact sequence

$$0 \rightarrow M \xrightarrow{(x_1)_M} M \rightarrow \overline{M} \rightarrow 0;$$

let us set $\overline{A}_0 = A_0/T_1 A_0$, $\overline{t} = t\overline{A}_0$. Let $q : L \rightarrow M$ be a free resolution of the A_0 -module M ; since the homothety of ratio T_1 is injective in A_0 , it is injective in L , and we have an exact sequence of complexes

$$0 \rightarrow L \xrightarrow{(x_1)_L} L \rightarrow \overline{L} \rightarrow 0$$

with $\overline{L} = L/x_1 L$, and a commutative diagram

$$\begin{array}{ccccccc}
0 & \longrightarrow & \mathbf{L} & \xrightarrow{(x_1)_L} & \mathbf{L} & \longrightarrow & \overline{\mathbf{L}} \longrightarrow 0 \\
& & \downarrow q & & \downarrow q & & \downarrow \bar{q} \\
0 & \longrightarrow & \mathbf{M} & \xrightarrow{(x_1)_M} & \mathbf{M} & \longrightarrow & \overline{\mathbf{M}} \longrightarrow 0.
\end{array}$$

Since q is a homomorphism, $\bar{q} : \overline{\mathbf{L}} \rightarrow \overline{\mathbf{M}}$ is a free resolution of the $\overline{\mathbf{A}}_0$ -module $\overline{\mathbf{M}}$ (cor. 1, p. 30).
For every $\overline{\mathbf{A}}_0$ -module $\mathbf{P}$, there is a canonical isomorphism

$$\mathbf{P} \otimes_{\mathbf{A}_0} \mathbf{L} \rightarrow \mathbf{P} \otimes_{\overline{\mathbf{A}}_0} \overline{\mathbf{L}},$$

whence, by passing to homology, an isomorphism

$$\varphi_P : \text{Tor}_1^{\mathbf{A}_0}(\mathbf{P}, \mathbf{M}) \rightarrow \text{Tor}_1^{\overline{\mathbf{A}}_0}(\mathbf{P}, \overline{\mathbf{M}}).$$

If $u : \mathbf{P} \rightarrow \mathbf{P}'$ is a $\overline{\mathbf{A}}$ -homomorphism, then

$$\varphi_{P'} \circ \text{Tor}_1^{\mathbf{A}_0}(u, 1_M) = \text{Tor}_1^{\overline{\mathbf{A}}_0}(u, 1_M) \circ \varphi_P.$$

That being so, suppose condition (v') is satisfied for $\mathbf{M}$, and let us prove that it holds for $\overline{\mathbf{M}}$.
Let r be an integer ≥ 1 ; we have a commutative diagram with exact rows

$$\begin{array}{ccccccc}
0 & \longrightarrow & \mathbf{A}_0/t^r & \xrightarrow{T_1} & \mathbf{A}_0/t^{r+1} & \longrightarrow & \overline{\mathbf{A}}_0/\bar{t}^{r+1} \longrightarrow 0 \\
& & \downarrow p_{r-1} & & \downarrow p_r & & \downarrow \bar{p}_r \\
0 & \longrightarrow & \mathbf{A}_0/t^{r-1} & \xrightarrow{T_1} & \mathbf{A}_0/t^r & \longrightarrow & \overline{\mathbf{A}}_0/\bar{t}^r \longrightarrow 0
\end{array}$$

from which we deduce a commutative diagram with exact rows

$$\begin{array}{ccccccc}
\text{Tor}_1^{\mathbf{A}_0}(\mathbf{A}_0/t^{r+1}, \mathbf{M}) & \longrightarrow & \text{Tor}_1^{\mathbf{A}_0}(\overline{\mathbf{A}}_0/\bar{t}^{r+1}, \mathbf{M}) & \longrightarrow & \mathbf{M}/x^r \mathbf{M} & \xrightarrow{x_1} & \mathbf{M}/x^{r+1} \mathbf{M} \\
\text{Tor}_1(p_r, 1) & & \downarrow & & \text{Tor}_1(\bar{p}_r, 1) & & \downarrow \\
\text{Tor}_1^{\mathbf{A}_0}(\mathbf{A}_0/t^r, \mathbf{M}) & \longrightarrow & \text{Tor}_1^{\mathbf{A}_0}(\overline{\mathbf{A}}_0/\bar{t}^r, \mathbf{M}) & \longrightarrow & \mathbf{M}/x^{r-1} \mathbf{M} & \xrightarrow{x_1} & \mathbf{M}/x^r \mathbf{M}.
\end{array}$$

Now multiplication by x_1 defines an injection of $\mathbf{M}/x^r \mathbf{M}$ to $\mathbf{M}/x^{r+1} \mathbf{M}$: this follows immediately, by induction on r , from the exact sequence

$$0 \rightarrow x^{r-1}M/x^rM \rightarrow M/x^rM \rightarrow M/x^{r-1}M \rightarrow 0$$

and from the injectivity of the homothety with ratio x_1 to $\bigoplus_{r \geq 0} (x^rM/x^{r+1}M)$. Condition (v) therefore implies that the homomorphism $\text{Tor}_1^{\bar{A}_0}(\bar{p}_r, 1_M)$ is surjective for all $r \geq 1$. By composition with the isomorphisms $(\varphi_{\bar{A}_0/\bar{t}}^{-r+1})^{-1}$ and $\varphi_{\bar{A}_0/\bar{t}}^{-r}$, we deduce that the homomorphism

$$\text{Tor}_1^{\bar{A}_0}(\bar{p}_r, 1_M) : \text{Tor}_1^{\bar{A}_0}(\bar{A}_0/\bar{t}^{r+1}, M) \rightarrow \text{Tor}_1^{\bar{A}_0}(\bar{A}_0/\bar{t}^r, M)$$

is surjective for $r \geq 1$, whence condition (v') for $\bar{M}$. This completes the proof of the theorem.

10.9.10 Extension class associated with a regular sequence

Let A be a ring, M an A -module, $x = (x_1, \dots, x_n)$ a sequence of elements of A . Denote by M_i the A -module $M/(x_1M + \dots + x_{i-1}M)$ for $i = 0, \dots, n+1$, so that M_0 and M_1 identify with M and that $M_{n+1} = M/(x)M$. Let

$$\bar{x}_i : M_{i-1} \rightarrow M_i, \quad i = 1, \dots, n,$$

denote the A -homomorphism composed of the homothety of M_{i-1} of ratio x_i and of the canonical projection of M_{i-1} over M_i . Finally, let us denote by $p : M_n \rightarrow M/(x)M$ the canonical projection. The diagram

$$0 \rightarrow M \xrightarrow{\bar{x}_1} M_1 \xrightarrow{\bar{x}_2} M_2 \rightarrow \dots \xrightarrow{\bar{x}_n} M_n \xrightarrow{p} M/(x)M \rightarrow 0 \quad (19)$$

is an exact sequence if and only if the sequence x is M -regular. Suppose henceforth that the sequence x is regular for M . The element $\theta_x \in \text{Ext}_A^n(M/(x)M, M)$ associated with the exact sequence (19) is also said to be associated with the M -regular sequence x .

Let i be an integer, $1 \leq i \leq n$. Note that the sequence (19) can be decomposed into the two exact sequences

(20)

$$0 \rightarrow M \xrightarrow{\bar{x}_1} M_1 \xrightarrow{\bar{x}_2} M_2 \rightarrow \dots \xrightarrow{\bar{x}_i} M_i \rightarrow M/(x_1 M + \dots + x_i M) \rightarrow 0 \quad (21)$$

$$0 \rightarrow M/(x_1 M + \dots + x_i M) \xrightarrow{\bar{x}_{i+1}} M_{i+1} \rightarrow \dots$$

$$\rightarrow M_n \xrightarrow{P} M/(x)M \rightarrow 0$$

which are none other than the exact sequences associated with the sequence $(x_1, \dots, x_i)$ which is regular for M and to the sequence $(x_{i+1}, \dots, x_n)$ which is regular for $M/(x_1 M + \dots + x_i M)$. Denoting

$$\theta_{(x_1, \dots, x_i)} \in \text{Ext}_{\mathbf{A}}^i(\mathbf{M}/(x_1 \mathbf{M} + \dots + x_i \mathbf{M}), \mathbf{M})$$

$$\theta_{(x_{i+1}, \dots, x_n)} \in \text{Ext}_{\mathbf{A}}^{n-i}(\mathbf{M}/(x) \mathbf{M}, \mathbf{M}/(x_1 \mathbf{M} + \dots + x_i \mathbf{M}))$$

the extension classes associated with (20) and (21), we have, according to X, p. 118, prop. 3

$$\theta_{(x_1, \dots, x_n)} = \theta_{(x_1, \dots, x_i)} \circ \theta_{(x_{i+1}, \dots, x_n)}. \quad (22)$$

Moreover, by prop. 5 (X, p. 157), the Koszul complex $K.(x, M)$ is acyclic in degrees $\neq n$, whence an exact sequence

$$0 \rightarrow M \xrightarrow{\partial^0} K^1(x, M) \xrightarrow{\partial^1} K^2(x, M) \xrightarrow{\partial^2} \dots \rightarrow K^n(x, M) \xrightarrow{q} M/(x)M \rightarrow 0 \quad (23)$$

where we have identified $K^0(x, \mathbf{M})$ to $\mathbf{M}$ and where q maps each element $m \in K^n(x, \mathbf{M})$ on the class in $\mathbf{M}/(x)\mathbf{M}$ of $m(1, 2, \dots, n) \in \mathbf{M}$.

PROPOSITION 8. — Suppose the sequence x is regular for M . The element of $\text{Ext}_{\mathbf{A}}^n(M/(x)M, M)$ associated with the exact sequence (23) is $(-1)^{n(n+1)/2} \cdot \theta_x$.

For $i = 0, 1, \dots, n$ let us define an A -linear map

$$a^i : \mathbf{K}^i(x, \mathbf{M}) \rightarrow \mathbf{M}_i = \mathbf{M}/(x_1 \mathbf{M} + \cdots + x_{i-1} \mathbf{M})$$

as follows: if $m \in \mathbf{K}^i(x, \mathbf{M})$, $a^i(m)$ is the class in $\mathbf{M}_i$ of the element $m(1, 2, \dots, i)$ of $\mathbf{M}$. It is clear that a^0 is the identity map of $\mathbf{M}$ and that $p \circ a^n = q$. Moreover $a^{i+1} \circ \partial^i(m)$ is the image in $\mathbf{M}_{i+1}$ of the element

$$\sum_{k=1}^{i+1} (-1)^{k+1} x_k m(1, 2, \dots, k-1, k+1, \dots, i+1).$$

Since x_k annihilates $\mathbf{M}_{i+1}$ for $k = 1, \dots, i$, $a^{i+1} \circ \partial^i(m)$ is also the image of

$$(-1)^i x_{i+1} m(1, 2, \dots, i),$$

hence

$$a^{i+1} \circ \partial^i = (-1)^i \bar{x}_{i+1} \circ a^i.$$

By X, p. 120, cor. 1 and 2, the element of $\text{Ext}_A^n(\mathbf{M}/(\mathbf{x})\mathbf{M}, \mathbf{M})$ associated with (23) is equal to $\prod_{i=1}^n (-1)^i \cdot \theta_x$, whence the assertion.

COROLLARY. — Suppose moreover that the modules $\mathbf{M}/(x_1 \mathbf{M} + \cdots + x_{i-1} \mathbf{M})$ are separated for the $(\mathbf{x})$ -adic topology, and let $(a_{ij}) \in \mathbf{GL}_n(\mathbf{A})$. Set

$$y_i = \sum_j a_{ij} x_j \text{ and } \mathbf{y} = (y_1, \dots, y_n).$$

Then the sequence $\mathbf{y}$ is regular for $\mathbf{M}$, and one has $\theta_y = \det(a_{ij})^{-1} \theta_x$.

Indeed, the sequence $\mathbf{y}$ is regular for $\mathbf{M}$ by prop. 6 (X, p. 158) and theorem 1 (X, p. 160); the last assertion follows from prop. 8, and from prop. 4 of X, p. 119.

PROPOSITION 9. — Suppose the sequence $\mathbf{x}$ is regular for $\mathbf{M}$. If $\mathbf{N}$ is an $\mathbf{A}$ -module such that $(\mathbf{x})\mathbf{N} = 0$, one has $\text{Ext}_A^i(\mathbf{N}, \mathbf{M}) = 0$ for $i < n$, and the map $\alpha \rightarrow \theta_x \circ \alpha$ from $\text{Hom}_A(\mathbf{N}, \mathbf{M}/(\mathbf{x})\mathbf{M})$ to $\text{Ext}_A^n(\mathbf{N}, \mathbf{M})$ (which is also the iterated connecting homomorphism associated with (19), cf. X, p. 127, cor. 3) is bijective.

It remains to prove that the homomorphism $\psi^i : \alpha \mapsto \theta_x \circ \alpha$ of $\text{Ext}_A^{i-n}(\mathbf{N}, \mathbf{M}/(\mathbf{x})\mathbf{M})$ to

$\text{Ext}_{\mathbf{A}}^i(\mathbf{N}, \mathbf{M})$ is bijective for $i \leq n$. Let us reason by induction on n , the assertion being trivial for $n = 0$. Set $\mathbf{M}_1 = \mathbf{M}/x_1\mathbf{M}$, $\mathbf{x}' = (x_2, \dots, x_n)$, so that $\mathbf{x}'$ is $\mathbf{M}_1$ -regular. By the induction hypothesis, the homomorphism

$$\overline{\psi}^{i-1} : \alpha \mapsto \theta_{x'} \circ \alpha$$

of $\text{Ext}_{\mathbf{A}}^{i-n}(\mathbf{N}, \mathbf{M}/(\mathbf{x})\mathbf{M})$ to $\text{Ext}_{\mathbf{A}}^{i-1}(\mathbf{N}, \mathbf{M}_1)$ is bijective for $i < n$. Moreover, consider the exact sequence

$$0 \rightarrow \mathbf{M} \xrightarrow{(x_1)_{\mathbf{M}}} \mathbf{M} \rightarrow \mathbf{M}_1 \rightarrow 0;$$

the connecting homomorphism $\text{Ext}_{\mathbf{A}}^i(\mathbf{N}, \mathbf{M}_1) \rightarrow \text{Ext}_{\mathbf{A}}^{i+1}(\mathbf{N}, \mathbf{M})$ associated is $\varphi^i : \beta \mapsto \theta_{x_1} \circ \beta$ (X, p. 125, prop. 5);

since we have $\text{Ext}^i(1_{\mathbf{N}}, (x_1)_{\mathbf{M}}) = \text{Ext}^i((x_1)_{\mathbf{N}}, 1_{\mathbf{M}}) = 0$, we deduce exact sequences

$$0 \rightarrow \text{Ext}_{\mathbf{A}}^i(\mathbf{N}, \mathbf{M}) \rightarrow \text{Ext}_{\mathbf{A}}^i(\mathbf{N}, \mathbf{M}_1) \xrightarrow{\varphi^i} \text{Ext}_{\mathbf{A}}^{i+1}(\mathbf{N}, \mathbf{M}) \rightarrow 0.$$

Since $\text{Ext}_{\mathbf{A}}^i(\mathbf{N}, \mathbf{M}_1) = 0$ for $i < n - 1$, we deduce that $\text{Ext}_{\mathbf{A}}^{i+1}(\mathbf{N}, \mathbf{M}) = 0$ for $i < n - 1$, that is, $i + 1 < n$; it follows that φ^i is bijective for $i < n$. Since $\psi^i(\alpha) = \theta_x \circ \alpha = \theta_{x_1} \circ \theta_{x'} \circ \alpha = \varphi^{i-1} \circ \overline{\psi}^i(\alpha)$ for $\alpha \in \text{Ext}_{\mathbf{A}}^{i-n}(\mathbf{N}, \mathbf{M}/(\mathbf{x})\mathbf{M})$, ψ^i is indeed bijective for $i \leq n$.

Exercises

Exercises for Section 1: Complements of Linear Algebra

1) Let

$$\mathbf{M} \rightarrow \mathbf{N} \xrightarrow{t} \mathbf{P} \rightarrow \mathbf{Q}$$

$$\begin{array}{ccc} \downarrow u & \downarrow v & \downarrow w \\ \downarrow & \downarrow & \downarrow s \end{array}$$

$$\mathbf{M}' \rightarrow \mathbf{N}' \xrightarrow{t'} \mathbf{P}' \rightarrow \mathbf{Q}'$$

a commutative diagram of A -modules where the rows are exact; suppose that u is surjective and s injective. Show that:

$$\text{Ker}(w) = t(\text{Ker}(v)) \quad \text{Im}(v) = t'^{-1}(\text{Im}(w)).$$

2) Let $u : \mathbf{M} \rightarrow \mathbf{N}, v : \mathbf{N} \rightarrow \mathbf{P}$ two homomorphisms of A -modules. Show that there exists an exact sequence:

$$0 \rightarrow \text{Ker}(u) \rightarrow \text{Ker}(v \circ u) \rightarrow \text{Ker}(v) \rightarrow \text{Coker}(u) \rightarrow \text{Coker}(v \circ u) \rightarrow \text{Coker } v \rightarrow 0.$$

3) In the diagram of A -modules:

$$\begin{array}{ccc}
P & \leftarrow & M \\
\downarrow & R^\leftarrow & \downarrow \\
N & \leftarrow & Q
\end{array}$$

Does the commutativity of the two triangles extracted from the diagram necessarily imply that of the diagram?

4) Let k a field, $B = k[T]$, let A be the subring $k[T^2, T^3] \subset k[T]$. Show that the A -module B is torsion-free, but is not a flat A -module.

5) Let k a field, $C = k[X, Y]$; consider the subrings $A = k[X^4, Y^4]$, $B = k[X^4, X^3Y, XY^3, Y^4]$ and $\tilde{B} = k[X^4, X^3Y, X^2Y^2, XY^3, Y^4]$ of C . The inclusions $A \subset B \subset \tilde{B}$ make it possible to define on B and $\tilde{B}$ A -module structures. Show that $\tilde{B}$ is a flat A -module, while the A -module B is not flat.

6) Show that an A -module E is flat if and only if it satisfies the following condition (cf. X, p. 15, remark):

(ii'') For every finite family $(c_i)_{i \in I}$ of elements of A , every solution $e = (e_i)_{i \in I}$ of the equation $\sum_{i \in I} c_i e_i = 0$ can be written $b_1 z_1 + \dots + b_n z_n$, where $b_1, \dots, b_n \in E$ and where, for $r = 1, \dots, n$, $z_r = (z_{r,i})_{i \in I}$ is a solution of the equation $\sum_{i=1}^n z_{r,i} c_i = 0$.

7) Let E be a monoid and S a subsemigroup of E that is cancellative and satisfies the hypotheses of Exercise 17 of III, p. 121, so that there exists a "monoid of fractions" $\bar{E}$ and an injective homomorphism $\varepsilon : E \rightarrow \bar{E}$. Let A be a commutative ring. From ε one deduces a homomorphism of monoid algebras $u : A^{(E)} \rightarrow A^{(\bar{E})}$ which allows one to regard $A^{\bar{E}}$ as a $A^{(E)}$ -module on the left. Show that this module is flat over $A^{(E)}$ (write $A^{\bar{E}}$ as a filtered inductive limit of free $A^{(E)}$ -modules of rank 1).

* 8) Let A be the ring of continuous real-valued functions on the interval $[0, 1]$.

a) Show that the ideal of functions vanishing at 0 is a flat A -module (show that it is a filtered inductive limit of the ideals Af , where f is a function which vanishes only at 0).

b) Show that the ideal I of functions vanishing in a neighborhood of 0 is a projective A -

module (show that there exists a sequence $(f_n)_{n \geq 1}$, $f_n \in \mathbf{I}$, such that $\text{Supp}(f_n) \subset \left[\frac{1}{n+2}, \frac{1}{n} \right]$ and $\sum_n f_n(x) = 1$ for $x \neq 0$; deduce

that every element h of $\mathbf{I}$ can be written $h = \sum_n {}^{t_h} (f_n h) e_n$ with $e_n = \sum_{r \leq n+1}^n f_r$, and use II, p. 46, prop. 12. *

9) Let $\mathbf{P}$ be an $\mathbf{A}$ -module. Show that the following conditions are equivalent:

α) For a sequence $\mathbf{M}' \xrightarrow{u} \mathbf{M} \xrightarrow{v} \mathbf{M}''$ of right $\mathbf{A}$ -modules to be exact, it is necessary and sufficient that the sequence

$$\mathbf{M}' \otimes_{\mathbf{A}} \mathbf{P} \xrightarrow{u \otimes 1} \mathbf{M} \otimes_{\mathbf{A}} \mathbf{P} \xrightarrow{v \otimes 1} \mathbf{M}'' \otimes_{\mathbf{A}} \mathbf{P}$$

be exact.

β) $\mathbf{P}$ is flat, and for every nonzero right $\mathbf{A}$ -module $\mathbf{M}$, one has $\mathbf{M} \otimes_{\mathbf{A}} \mathbf{P} \neq (0)$.

γ) $\mathbf{P}$ is flat, and for all right $\mathbf{A}$ -modules $\mathbf{M}, \mathbf{N}$, the canonical homomorphism

$$\text{Hom}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}) \rightarrow \text{Hom}_{\mathbf{Z}}(\mathbf{M} \otimes_{\mathbf{A}} \mathbf{P}, \mathbf{N} \otimes_{\mathbf{A}} \mathbf{P})$$

is injective.

δ) $\mathbf{P}$ is flat, and for every maximal right ideal $\mathfrak{m}$ of $\mathbf{A}$, one has $\mathfrak{m}\mathbf{P} \neq \mathbf{P}$.

One then says that $\mathbf{P}$ is a faithfully flat $\mathbf{A}$ -module.

10) a) Show that a faithfully flat module is faithful (II, p. 28).

b) The $\mathbf{Z}$ -module $\mathbf{Q}$ is faithful and flat, but not faithfully flat.

c) Let (p_n) the strictly increasing sequence of prime numbers, and let $\mathbf{A}$ be the product ring $\prod_n \mathbf{Z}/p_n \mathbf{Z}$.

Show that the direct sum $\mathbf{E}$ of the $\mathbf{Z}/p_n \mathbf{Z}$ is a faithful projective $\mathbf{A}$ -module which is not faithfully flat (observe that $\mathbf{E}$ is an ideal of $\mathbf{A}$ such that $\mathbf{E}^2 = \mathbf{E}$).

11) Let $0 \rightarrow \mathbf{P}' \rightarrow \mathbf{P} \rightarrow \mathbf{P}'' \rightarrow 0$ an exact sequence of $\mathbf{A}$ -modules. Suppose that $\mathbf{P}'$ and $\mathbf{P}''$ are flat, and that one of them is faithfully flat. Show that $\mathbf{P}$ is faithfully flat.

12) Let $\rho : A \rightarrow B$ a ring homomorphism that makes B an A faithfully flat right module; let M be an A module.

a) Show that for M to be finitely generated (resp. of finite presentation), it is necessary and sufficient that the same be true of the B -module $B \otimes_A M$.

b) We assume that $\rho(A)$ is contained in the center of B . Show that M is flat (resp. faithfully flat, resp. finitely generated projective) if and only if the B -module $B \otimes_A M$ is.

13) Show that every finitely generated flat module P over an integral domain A is projective (consider a presentation $0 \rightarrow R \xrightarrow{L} L \xrightarrow{P} 0$, where L is finitely generated free; choose a finitely generated submodule R' of R such that R/R' be torsion, and apply Theorem 1, p. 14).

¶ 14) a) Let $0 \rightarrow \mathbf{R} \rightarrow \mathbf{L} \rightarrow \mathbf{E} \rightarrow 0$ an exact sequence of A -modules, where L is a free A -module; let (e_α) a basis of L . Show that the following conditions are equivalent:

β) For every $x \in \mathbb{R}$, if a_x is the right ideal generated by the components of x on the basis (e_α) , we have $x \in a_x \mathbb{R}$.

γ) For every $x \in \mathbb{R}$, there exists a homomorphism $u_x : \mathbb{L} \rightarrow \mathbb{R}$ such that $u_x(x) = x$.

δ) For every finite sequence $(x_i)_{1 \leq i \leq n}$ of elements of R , there exists a homomorphism $u : L \rightarrow R$ such that $u(x_i) = x_i$ for $1 \leq i \leq n$ (Use Thm. 1, p. 14.)

b) Let r be the radical of A , and let $0 \rightarrow R \rightarrow L \rightarrow E \rightarrow 0$ an exact sequence of A -modules, such that L is free. Suppose that E is flat and that R is contained in rL . Show that then $R = 0$ (with the notations of a), observe that a_x is a finitely generated ideal and that we have $a_x = a_x r$).

c) Let E be a flat finitely generated A -module; suppose there exists a two-sided ideal b of A contained in the radical of A , such that E/bE is a free (A/b) -module. Show that E is then a free A -module (observe that there exists a finitely generated free A -module L such that L/bL is isomorphic to E/bE , and apply b)). In particular, every finitely generated flat module over a local ring is free.

15) Let A be a ring, a an element of A . Show that the following properties are equivalent: α)

$$a \in aAa .$$

β) Aa is a direct summand in the module A_s .

γ) A_s/Aa is a flat A -module.

δ) For every right ideal b of A , we have $b \cap Aa = ba$.

16) Let A be a ring. Show that the following properties are equivalent:

α) Every element $a \in A$ satisfies the equivalent properties of Exercise 15.

β) Every finitely generated left ideal is a direct summand of A_s .

γ) Every left A -module is flat.

δ) Every right A -module is flat.

We then say that A is an absolutely flat ring (cf. VIII, § 8, Exercise 15).

17) Let A be an absolutely flat ring (Exercise 16).

a) Let P be a projective A -module. Show that every finitely generated submodule of P is a direct summand of P (reduce to the case where P is finitely generated free).

b) Show that every projective A -module is a direct sum of cyclic submodules, isomorphic to cyclic ideals of A . (Use Kaplansky's theorem (II, p. 183, exercise 2) to reduce to the case where P admits a countable family of generators, then use a).)

c) Show that every ideal of A generated by a countable family of generators is projective.

d) Give an example of an absolutely flat ring A and a finitely generated A -module that is not projective.

18) Let M be an A -module. Show that the following conditions are equivalent:

α) M is finitely generated (resp. finitely presented).

β) For every inductive system (N_i, u_{ji}) of A -modules relative to a filtered preordered set I , the canonical homomorphism $\varinjlim \text{Hom}_A(M, N_i) \rightarrow \text{Hom}_A(M, \varinjlim N_i)$ is injective (resp. bijective).

γ) For every family $(P_j)_{j \in J}$ of right A -modules, the canonical homomorphism

$$\left(\prod_{j \in J} P_j \right) \otimes_A M \rightarrow \prod_{j \in J} (P_j \otimes_A M)$$

(II, p. 61) is surjective (resp. bijective).

δ) For every set I, the canonical homomorphism $A_d^1 \otimes_A M \rightarrow M^1$ is surjective (resp. bijective).

19) Let A and B be two rings, $\varphi : A \rightarrow B$ a homomorphism, M a B-module. We equip $B \otimes_A M$ and $\text{Hom}_A(B, M)$ with left B-module structures induced from the B-module structure of B. One says that M is projective relative to A, or φ -projective, if the canonical surjection $B \otimes_A M \rightarrow M$ admits a B-linear section; one says that M is injective relative to A, or φ -injective, if the canonical injection

$$M \rightarrow \text{Hom}_A(B, M)$$

admits a B-linear retraction.

- a) If M is projective (resp. injective) as an A-module and φ -projective (resp. φ -injective), then M is a projective (resp. injective) B-module.
- b) For every A-module N, the B-module $B \otimes_A N$ (resp. $\text{Hom}_A(B, N)$) is φ -projective (resp. φ -injective).
- c) We assume that $\varphi(A)$ is contained in the center of B. Let N be an A-module, P a B-module φ -projective. Show that the B-module $P \otimes_A N$ (resp. the right B-module $\text{Hom}_A(P, N)$) is φ -projective (resp. φ -injective).

20) Let A be a principal ideal ring, a a nonzero element of A, $B = A/Aa$. Show that the B-module B_s is injective.

21) Show that the following conditions are equivalent:

- α) A is Noetherian.
- β) Every A-module that is an inductive limit of injective A-modules is injective.
- γ) Every A-module that is a direct sum of injective A-modules is injective.

(To prove that γ) implies α), let $(a_n)_{n \geq 1}$ an increasing sequence of left ideals of A, $a = \bigcup_n a_n$,

I_n an injective envelope of a/a_n . The natural homomorphism from a to $\bigoplus I_n$ is the restriction of a homomorphism $f : A \rightarrow \bigoplus I_n$; consider $f(1)$.

22) Let A be a principal ideal ring, K its field of fractions, π an extremal element of A . Show that the π -primary component of the A -module K/A is an injective envelope of the A -module $A/\pi^n A$ for every $n \geq 1$. ¶ 23) Let I be an injective A -module, E the ring $\text{End}_A(I)$, r the set of elements u of E such that I is an injective envelope of $\text{Ker}(u)$.

- a) Show that r is a two-sided ideal and that the ring E/r is absolutely flat (exercise 16). (Let u be an element of E , N a submodule of I such that $N \cap \text{Ker } u = 0$ and maximal for this property; show that there exists $v \in E$ such that $vu(x) = x$ for all $x \in N$, and deduce that $uvu - u \in r$.)
- b) Show that r is the radical of E (the inclusion $r(E) \subset r$ follows from a); show that for every $u \in r$, $1 - u$ is bijective).
- c) Assume that I is a direct sum of a finite family of indecomposable injectives. Show that the ring E/r is semisimple.

Let $i : A \rightarrow I$ an injective envelope of the A -module A_s . We set $E = \text{End}_A(I)$ and $B = \text{End}_E(I)$, so that B is the bicommutant of the A -module I (VIII, § 3, def. 1). We consider A as a subring of B . One denotes $j : B \rightarrow I$ and $p : E \rightarrow I$ the maps defined by $j(b) = b(i(1))$ and $p(u) = u(i(1))$ for $b \in B$, $u \in E$.

- a) Show that p is a surjection and j an injection.
- b) Show that the following conditions are equivalent:
 - α) The map p is bijective.
 - β) The map j is bijective.
 - γ) The A -module B is injective.

If these conditions are satisfied, the mapping $j^{-1} \circ p$ is an isomorphism from the ring E onto the opposite ring of B .

- c) Let $\mathfrak{F}$ the set of left ideals a of A such that $a \cap b \neq (0)$ for every nonzero left ideal b . One denotes $s(A)$ the set of elements a of A for which there exists $a \in \mathfrak{F}$ such that $aa = 0$. Show that $s(A)$ is a two-sided ideal of A .

d) Show that the following conditions are equivalent:

α) $s(\mathbf{A}) = 0$.

β) The ring E is semiprimitive.

γ) The ring B is absolutely flat.

If these conditions are satisfied, then the equivalent properties of b) are satisfied.

(Let $s(I)$ the set of elements x of I for which there exists $a \in \mathfrak{F}$ such that $ax = 0$. Using Exercise 23, prove that $p^{-1}(s(I))$ is equal to the radical r of E , whence the implication $\beta) \Rightarrow \alpha$). Then show that α implies $r = \text{Ker}(p) = 0$, which entails β and γ (in view of Exercise 23) as well as condition α of b). Finally show that γ implies α .)

e) If A is left Noetherian, the ideal $s(A)$ is nilpotent (first prove that every element a of $s(A)$ is nilpotent, by considering the left annihilators of the a^n ; then use Exercise 8, b) of VIII, § 9). ¶ 25) Assume that the ring A is left Noetherian and contains no nonzero nilpotent two-sided ideals. a) Prove that the ring B defined in Exercise 24 is semisimple (“Goldie’s theorem”; use Exercise 23, c) and Exercise 24, d) and e)). If moreover (0) is a prime two-sided ideal in A (VIII, § 8, Exercise 19), show that B is simple.

f) One identifies A with a subring of B . Prove the following properties:

(i) Every element of A that is left cancellable is invertible in B (hence cancellable in A).

(ii) Every element b of B can be written $s^{-1}a$, with $s \in A$, $a \in A$ and s cancellable.

One sometimes says that B is the total ring of left fractions of A .

(To prove (ii), reduce to the case where B is simple. Show then that the ideal (0) of A is prime. Let a be an ideal of F , a an element of a for which the right ideal aB is maximal among the ideals $a'B$ ($a' \in a$), and x an element of B such that $axa = a$. Prove that $B(1 - xa) \subset aB$, whence

$$(B(1 - ax) \cap a) \cdot (B(1 - xa) \cap a) = (0);$$

deduce that a is cancellable. Conclude by taking for a the set of all $a \in A$ such that $ab \in A$

.) c) Assume that every nonzero element of A is cancellable. Show that B is a field, which is a

left field of fractions for A (cf. I, p. 155, Exercise 15).

¶ 26) a) Show that the following conditions are equivalent:

α) The ring A is right Noetherian, and the A -module A_s is injective.

α^0) A is left Noetherian and the A -module A_d is injective.

β) The ring A is self-injective (VIII, § 2, Exercise 11).

(Prove that α) implies that every right ideal r satisfies $r = r^0$. Then show that $r(A)$ is nilpotent, and deduce that A is right Artinian using Exercise 23. Next prove that the dual of a simple left A -module is simple. Deduce that A is left Noetherian and finally self-injective. b) Show that the ring A is self-injective if and only if every injective A -module is projective and every projective A -module is injective.

¶ 27) Assume the ring A is left Noetherian.

a) Let I be an indecomposable injective A -module. Show that the set of annihilator ideals of a nonzero submodule of I admits a greatest element $\mathfrak{A}(I)$, which is a prime two-sided ideal (VIII, § 8, Exercise 19).

b) Show that for every prime two-sided ideal $\mathfrak{p}$ of A , there exists an indecomposable injective A -module I such that $\mathfrak{A}(I) = \mathfrak{p}$ (consider an indecomposable direct factor of the injective envelope of $A/\mathfrak{p}$).

c) Assume A is commutative. Show that every indecomposable injective A -module I is the injective envelope of $A/U(I)$. Deduce that the map $I \mapsto U(I)$ defines a bijection from the set S of isomorphism classes of indecomposable injective A -modules onto the set of prime ideals of A . The inverse bijection associates to the ideal $\mathfrak{p}$ the injective envelope of $A/\mathfrak{p}$.

d) Show that for the map $I \mapsto \mathfrak{A}(I)$ from $\mathfrak{A}$ onto the set of prime two-sided ideals of A to be bijective, it suffices that A satisfy the following condition (H):

(H) For every left ideal $\mathfrak{a}$ of A , there exist elements $x_1, \dots, x_k$ of $A/\mathfrak{a}$ such that

$$\text{Ann}(A/\mathfrak{a}) = \bigcap_i \text{Ann}(x_i).$$

e) Show that condition (H) is satisfied in the following cases:

- α) All left ideals of A are two-sided.
- β) A is Artinian.
- γ) The center Z of A is a Noetherian ring and A is a finitely generated Z -module.
- f) Let k be a field, $\mathbb{Q}^+$ the additive monoid of rational numbers ≥ 0 , A the algebra $k^{(\mathbb{Q}^+)}$, a a nonzero element of A , I the injective envelope of A/Aa . Show that the A -module I is indecomposable, but contains no submodule isomorphic to A/p for p prime (note that the set of ideals of A is totally ordered and that A contains only two prime ideals).
- 28) Let A be a Noetherian ring, M an A -module, I an injective envelope of M , a direct sum of a family $(I_\alpha)_{\alpha \in J}$ of indecomposable submodules. We denote by $\text{Ass}(M)$ the set of prime two-sided ideals $\mathfrak{p}(I_\alpha)$, for $\alpha \in J$ (exercise 27); if $p \in \text{Ass}(M)$, we say that p is associated with M .
- a) We have $\text{Ass}(M) \neq \emptyset$ if $M \neq 0$; if M is finitely generated, the set $\text{Ass}(M)$ is finite.
- b) Show that a prime two-sided ideal p is associated with M if and only if M contains a submodule N such that $\text{Ann}(N') = p$ for every nonzero submodule N' of N . If A is commutative, p is associated with M if and only if M contains a submodule isomorphic to A/p .
- c) Let M' be a submodule of M . Show that we have

$$\text{Ass}(M') \subset \text{Ass}(M) \subset \text{Ass}(M') \cup \text{Ass}(M/M').$$

- d) We set $J_p = \sum_{\mathfrak{p}(I_\alpha)=p} I_\alpha$ and $Q(p) = \left(\sum_{q \neq p} J_q \right) \cap M$. Show that we have $\bigcap_{p \in \text{Ass}(M)} Q(p) = 0$ and $\bigcap_{p \in E} Q(p) \neq 0$ for every subset E of $\text{Ass}(M)$ distinct from $\text{Ass}(M)$. For every $p \in \text{Ass}(M)$, we have

$$\text{Ass}(M/Q(p)) = \{p\}.$$

- e) Let a be an element of the center of A . Show that for the homothety of ratio a to be injective in M , it is necessary and sufficient that a belong to none of the prime ideals associated with M .

Assume that the ring A is commutative and Noetherian. Let $\mathfrak{a}$ be an ideal of A .

- a) Let H be an A -module such that every element of H is annihilated by a power of $\mathfrak{a}$. Show that for H to be injective, it suffices that for every integer $n \geq 1$ every finitely generated A -module M annihilated by $\mathfrak{a}^n$ and every submodule M' of M , the natural homomorphism $\text{Hom}_A(M, H) \rightarrow \text{Hom}_A(M', H)$ be surjective.

(Use the fact that for every ideal $\mathfrak{b}$ of A , there exists an integer $k \geq 0$ such that $\mathfrak{b} \cap \mathfrak{a}^k \subset \mathfrak{b}\mathfrak{a}$ (AC, III, § 3, no. 1, cor. 2).)

- b) Let M be an injective A -module. For $n \geq 1$, we denote by H_n the submodule of M consisting of the elements annihilated by $\mathfrak{a}^n$; set $H = \bigcup H_n$. Show that H is an injective A -module.
- c) Show that every element of the injective envelope of $A/\mathfrak{a}$ is annihilated by a power of $\mathfrak{a}$.
- * 30) Assume the ring A is local, commutative, and Noetherian; denote its maximal ideal by $\mathfrak{m}$, and $k = A/\mathfrak{m}$. Let I be an A -module such that every element of I is annihilated by a power of $\mathfrak{m}$. For every A -module M , denote by $T(M)$ the A -module $\text{Hom}_A(M, I)$ and $c_M : M \rightarrow T(T(M))$ the homomorphism defined by

$$(c_M(m))(u) = u(m) \quad \text{for } m \in M, \quad u \in T(M).$$

Show that the following properties are equivalent:

- α) For every finite-length A -module M , the A -module $T(M)$ is of finite length and the homomorphism $c_M : M \rightarrow T(T(M))$ is bijective.
- β) For every A -module M of finite length, one has $\text{long } T(M) = \text{long } M$.
- γ) The A -module I is injective, and the A -modules k and $T(k)$ are isomorphic.
- δ) The A -module I is isomorphic to the injective envelope of k .

When these conditions are satisfied, one sometimes says that I is a dualizing module for the ring A . (Show that property γ) is equivalent to each of the other three: to see that α) or β) implies γ), use Exercise 29.)

- 31) With the hypotheses of the previous exercise, let B be a local, commutative, Noetherian

ring, and let $\rho : A \rightarrow B$ a ring homomorphism that makes B a finitely generated A -module. If I is a dualizing module for A , show that the B -module $\text{Hom}_A(B, I)$ is dualizing for B . In particular, for every ideal a of A , the submodule of I consisting of elements annihilated by a is a dualizing module for A/a .

- 32) a) We keep the hypotheses of Exercise 30; we further assume that A contains a field k_0 such that k is a finite-dimensional k_0 -vector space. Show that the A -module

$$I = \varinjlim_n \text{Hom}_{k_0}(A/m^n, k_0)$$

is dualizing for A .

- b) Let k a field of characteristic zero, B the ring of formal power series $k[[T_1, \dots, T_n]]$. We endow the group $\mathbf{I} = k[X_1, \dots, X_n]$ with a B -module structure by setting $T_i \cdot \mathbf{P} = \partial \mathbf{P} / \partial X_i$ for all $\mathbf{P} \in k[X_1, \dots, X_n]$. Show that $\mathbf{I}$ is an injective envelope of k .

Exercises for Section 2: Complexes of A-Modules

- 1) Assume the ring A is principal; let C be a complex of free A -modules.
 - a) Show that C is the direct sum of a family of complexes $({}^p \mathbf{E})_{p \in \mathbb{Z}}$ satisfying: ${}^p \mathbf{E}_n = 0$ for $n \neq p, p-1$; $Z_n({}^p \mathbf{E}) = 0$.
 - b) We further assume that C_n is finitely generated for all $n \in \mathbb{Z}$. Show that C is a direct sum of complexes having only either one nonzero component, or two consecutive nonzero components, these components being isomorphic to A .
- 2) Let C and C' be two complexes of A -modules; suppose that the complexes C and $B(C)$ (resp. C' and $Z(C')$) are projective (resp. injective). Show that every morphism of complexes from $H(C)$ to $H(C')$ is induced by a morphism from C to C' .
- 3) Let C be a complex of A -modules; one regards C as an $A(\varepsilon)$ -module (X, p. 27). We denote by

$$i : \mathbf{A} \rightarrow \mathbf{A}(\varepsilon)$$

the canonical injection.

Show that the following conditions are equivalent:

- (i) C is i -projective (X, p. 170, exercise 19).
 - (ii) It is i -injective (loc. cit.).
 - (iii) The complex C is homotopic to zero.
 - (iv) There exists a graded A -module M and an isomorphism of graded $A(\varepsilon)$ -modules $A(\varepsilon) \otimes_A M \rightarrow C$.
 - (v) There exists a graded A -module N and an isomorphism of graded $A(\varepsilon)$ -modules $C \rightarrow \text{Hom}_A(A(\varepsilon), N)$.
- 4) a) Let C be a right-bounded A -complex, such that C and $H(C)$ are projective. Show that C is split.
- b) On the ring $B = A(\varepsilon)$ (X, p. 27), we consider the complex C defined by $C_n = B$ for all n , and $d(b) = \varepsilon b$ for $b \in B$. The complex C is free with zero homology, but not split.
- 5) Let C be a complex, and let s be a splitting of C , $\delta : C \rightarrow C$ a graded homomorphism of degree -1 , such that $(d + \delta) \circ (d + \delta) = 0$. One denotes C' the complex $(C, d + \delta)$. Assume that the A -linear map θ from C to C defined by $\theta = 1_C + \delta s$ is bijective.
- a) In order for $s\theta^{-1}$ to be a splitting of C' , it is necessary and sufficient that $\delta(Z(C)) \subset \theta(B(C))$.
- b) Assume the conditions of a) are satisfied. Show that there exist direct sum decompositions: $C = B(C) \oplus \text{Ker}(ds) = B(C') \oplus \text{Ker}(ds)$. If p (resp. p') is the projector with image $\text{Ker}(ds)$ and kernel $B(C)$ (resp. $B(C')$), we have $p' = p\theta^{-1}$.
- c) We further assume that the A -linear map λ from C to C defined by $\lambda = 1_C + s\delta$ is bijective. Show that we have $Z(C') = \lambda^{-1}(Z(C))$ and $C = Z(C) \oplus \text{Im}(sd) = Z(C') \oplus \text{Im}(sd)$. If q (resp. q') is the projector with image $Z(C)$ (resp. $Z(C')$) and kernel $\text{Im}(sd)$, we have $q' = \lambda^{-1}q$.

- 6) We keep the notation from Exercise 5; we further assume that A is commutative, that the complexes C and $C'/B(C')$ are flat and that there exists an ideal m of A satisfying $\delta(C) \subset mC$ and $\bigcap_n (B(C) + m^n C) = B(C)$ (This last condition is always satisfied if A is a local Noetherian ring with maximal ideal m and if C_n is a finitely generated A -module for every n , cf. AC, III, § 3, n° 3, prop. 6). Then show that $s\theta^{-1}$ is a splitting of C' .

(The flatness hypotheses imply $\text{Im}(d + \delta) \cap m^n C = (d + \delta)(m^n C)$ for all n ; if $x \in Z(C)$, construct by induction elements $y_n \in C$, $z_n \in Z(C) \cap m^n C$ such that $\delta(x) = \theta d(y_n) + \delta(z_n)$. Conclude using exercise 5, a).)

- 7) Let C and C' be two complexes, s_1 (resp. s') a splitting of C (resp. C'), $u : C' \rightarrow C$ a morphism of complexes. We define an A -endomorphism σ of $\text{Con}(u)$, graded of degree -1 , by setting

$$\sigma(y', x) = (-s'(y'), s(x) - sus'(y')) .$$

Show that in order for σ to be a splitting of $\text{Con}(u)$, it is necessary and sufficient that we have $H(u) = 0$.

- 8) Let $0 \rightarrow M' \xrightarrow{\mu} M \xrightarrow{\nu} M'' \rightarrow 0$ an exact sequence of A -modules, which can be viewed as a sequence

exact sequence of complexes (zero in degrees $\neq 0$). Show that the morphism of complexes $\varphi : \text{Con}(u) \rightarrow M''(X, p. 39)$ is a homotopy equivalence if and only if the exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is split. 9) Let $u : C' \rightarrow C$ be a morphism of A -complexes. Denote by $v : C \rightarrow \text{Coker}(u)$ the quotient map. Consider the graded A -homomorphisms

$$\alpha : (\text{Ker } u)(-1) \rightarrow \text{Con}(u) \quad \text{and} \quad \varphi : \text{Con}(u) \rightarrow \text{Coker}(u)$$

defined by $\alpha(x') = (x', 0)$, $\varphi(x', x) = v(x)$ for $x' \in C'$, $x \in C$. Show that α, φ are morphisms of complexes and that they give rise to an exact sequence

$$\cdots \rightarrow H_{n-1}(\text{Ker } u) \xrightarrow{H(\alpha)} H_n(\text{Con}(u)) \xrightarrow{H(\varphi)} H_n(\text{Coker } u) \xrightarrow{\delta} H_{n-2}(\text{Ker } u) \rightarrow \cdots$$

where δ is the composite of the connecting homomorphisms relative to the two exact sequences

$$0 \rightarrow \text{Ker } u \rightarrow C' \rightarrow \text{Im } u \rightarrow 0 \quad \text{and} \quad 0 \rightarrow \text{Im } u \rightarrow C \rightarrow \text{Coker } u \rightarrow 0.$$

10) Let $u : C \rightarrow C'$ a morphism of A-complexes, P a projective A-module, n be an integer ≥ 0 ,

$$h : P \rightarrow H_n(\text{Con}(u))$$

an A-homomorphism. Show that there exists a complex $\tilde{C}$, an exact sequence of complexes

$$0 \rightarrow C \xrightarrow{i} \tilde{C} \rightarrow P(-n) \rightarrow 0$$

and a morphism $\tilde{u} : \tilde{C} \rightarrow C'$ such that $\tilde{u} \circ i = u$ and that the morphism $C(i) : \text{Con}(u) \rightarrow \text{Con}(\tilde{u})$ induced by i satisfies the following properties:

- α) The map $H_p(C(i)) : H_p(\text{Con}(u)) \rightarrow H_p(\text{Con}(\tilde{u}))$ is bijective for $p \neq n, n+1$;
- β) There exists an exact sequence

$$0 \rightarrow H_{n+1}(\text{Con}(u)) \xrightarrow{H_{n+1}(C(i))} H_{n+1}(\text{Con}(\tilde{u})) \rightarrow P \xrightarrow{h} H_n(\text{Con}(u)) \xrightarrow{H_n(C(i))} H_n(\text{Con}(\tilde{u})) \rightarrow 0.$$

11) We call a bicomplex of A-modules a triple (C, d', d'') , where C is a graded A-module of type $\mathbf{Z} \times \mathbf{Z}$, $d' : C \rightarrow C$ a graded A-endomorphism of degree $(-1, 0)$ and $d'' : C \rightarrow C$ a graded A-endomorphism of degree $(0, -1)$, satisfying $d' \circ d' = d'' \circ d'' = d' \circ d'' + d'' \circ d' = 0$

We will also use the ascending graduation defined by $C^{p,q} = C_{-p,-q}$ for $p, q \in \mathbf{Z}$.

We denote by $Z'(C)$, $Z''(C)$, $B'(C)$, $B''(C)$ the sub-bicomplexes $\text{Ker}(d')$, $\text{Ker}(d'')$,

$\text{Im}(d')$, $\text{Im}(d'')$, and $H'(C)$ (resp. $H''(C)$) the bicomplex $Z'(C)/B'(C)$ (resp. $Z''(C)/B''(C)$).

- a) We set $d = d' + d''$. Show that when C is equipped with the total graduation associated with the given bigraduation (defined by $C_n = \bigoplus_{p+q=n} C_{p,q}$, cf. II, p. 164), the pair (C, d) is a complex. This is called

the complex associated with the bicomplex (C, d', d'') ; its homology is denoted $H(C)$.

- b) Let (C, d', d'') and (K, δ', δ'') be two bicomplexes. A morphism from (C, d', d'') to (K, δ', δ'') is a graded A -homomorphism u of degree $(0, 0)$ from C to K satisfying: $\delta' \circ u = u \circ d'$ and $\delta'' \circ u = u \circ d''$. Show that u induces a morphism of the associated complexes.

- c) Let u, v two morphisms of bicomplexes from C to K . One calls a homotopy connecting u to v a pair (s', s'') of A -homomorphisms from C to K , graded of degrees $(1, 0)$ and $(0, 1)$ respectively, such that

$$g - f = d' \circ s' + s' \circ d' + d'' \circ s'' + s'' \circ d''; \quad s' \circ d'' + d'' \circ s' = 0; \quad s'' \circ d' + d' \circ s'' = 0.$$

Show that $s' + s''$ is a homotopy connecting the associated complex morphisms induced by u and v . 12) Let (C, d', d'') a bicomplex, such that one has $H''_{p,-p}(C) = 0$ for all $p \in \mathbf{Z}$, $H'_{-q,q+1}(C) = 0$ for $q \geq 0$, $H'_{r,-r-1}(C) = 0$ for $r \geq 0$, and that there exist positive integers a and b such that $C_{a,-a} = C_{-b,b} = 0$. Prove that the A -module $H'_{0,0}(C)$ is zero.

- 13) Let I be a finite set; we denote by $(e_i)_{i \in I}$ the canonical basis of the $\mathbf{Z}$ -module $\mathbf{Z}^{(I)}$. We call an I -complex (resp. I -precomplex) of A -modules a graded A -module C of type $\mathbf{Z}^I$, equipped with a family of A -endomorphisms $(d_i)_{i \in I}$, such that d_i be graded of degree $(-e_i)$ satisfying:

$$d_i \circ d_i = 0 \quad \text{for every } i \in I;$$

$$d_i \circ d_j + d_j \circ d_i = 0 \text{ (resp. } d_i \circ d_j = d_j \circ d_i) \quad \text{for } i, j \in I.$$

If $\text{Card}(I) = 1$, an I-complex is a complex; if $\text{Card}(I) = 2$, an I-complex is a bicomplex (exercise 11). If $(C, (d_i))$ and $(C', (d'_i))$ are two I-complexes (resp. I-precomplexes), a morphism of $(C, (d_i))$ to $(C', (d'_i))$ is a graded homomorphism of degree zero $u : C \rightarrow C'$ such that $d'_i \circ u = u \circ d_i$ for all $i \in I$.

- a) Let $(C, (d_i))$ an I-complex. If one equips C with the total grading (of type Z) deduced from its grading, show that $(C, \sum_{i \in I} d_i)$ is a complex of A-modules, called the complex associated with the I-complex $(C, (d_i))$.

Show that a morphism of I-complexes induces a morphism of the associated complexes.

- b) Let C be an I-precomplex, and let R be a total order relation on I , denoted $\leq$. For $k = (k_i)_{i \in I}, x \in C_k$ and $i \in I$, we set

$$\delta_i(x) = (-1)^{\sum_{j < i} k_j} d_i(x). \text{ Show that } (C, (\delta_i)) \text{ is an I-complex, denoted } C_R.$$

- c) Let R_α, R_β two total order relations on I , denoted $\leq_\alpha$ and $\leq_\beta$. Let $u_{\beta\alpha}$ the endomorphism of C defined by $u(x) = (-1)^{\varepsilon(k)} x$ for $x \in C_k$, with $\varepsilon(k) = \sum_{\substack{i <_\alpha j \\ j <_\beta i}} k_i k_j$. Show that $u_{\beta\alpha}$ is a morphism of I-complexes of C_{R_α} to C_{R_β} .

- d) Let S be the set of total order relations on I ; we endow S with the trivial preorder (that is, the preorder such that $R_\alpha \leq R_\beta$ whatever $R_\alpha, R_\beta \in S$). Show that the system $(C_{R_\alpha}, u_{\alpha\beta})$ is a projective system of I-complexes, relative to S ; its limit C is an I-complex, called the I-complex associated with the I-precomplex C . Show that every morphism of I-precomplexes induces a morphism of the associated I-complexes.

A spectral sequence of A-modules is defined as a sequence of A-complexes $(E_r, d_r)_{r \geq 2}$ and of mappings $\alpha_r : H(E_r) \rightarrow E_{r+1}$ such that:

- (i) The grading of the complex E_r is the total grading associated with a bigrading

$$E_r = \bigoplus_{p, q \in \mathbb{Z}} E_r^{p, q} (r \geq 2);$$

we have $d_r(E_r^{pq}) \subset E_r^{p+r, q-r+1}$.

(ii) The maps α_r are isomorphisms of bigraded A-modules.

We will also use the descending bigrading of E_r : we denote $E_{pq}^r = E_r^{-p, -q}$.

a) Here we denote by $Z_{r+1}(E_r)$ (resp. $B_{r+1}(E_r)$) the bigraded A-module $\text{Ker } d_r$ (resp. $\text{Im } d_r$). Let

$$p_r : E_r \rightarrow E_r / \mathbf{B}_{r+1}(E_r)$$

the quotient map and $i_r : \mathbf{E}_{r+1} \rightarrow \mathbf{E}_r / \mathbf{B}_{r+1}(\mathbf{E}_r)$ the composite map of α_r^{-1} and the natural injection. For every $r \geq 2$, we define by induction on k the bigraded submodules $\mathbf{Z}_{r+k}(\mathbf{E}_r)$ and $\mathbf{B}_{r+k}(\mathbf{E}_r)$ of $\mathbf{E}_r$ by setting:

$$\mathbf{Z}_{r+k}(\mathbf{E}_r) = p_r^{-1}(i_r(\mathbf{Z}_{r+k}(\mathbf{E}_{r+1}))) ; \quad \mathbf{B}_{r+k}(\mathbf{E}_r) = p_r^{-1}(i_r(\mathbf{B}_{r+k}(\mathbf{E}_{r+1}))) .$$

Prove that we have the inclusions:

$$0 \subset \mathbf{B}_{r+1}(\mathbf{E}_r) \subset \mathbf{B}_{r+2}(\mathbf{E}_r) \subset \cdots \subset \mathbf{Z}_{r+2}(\mathbf{E}_r) \subset \mathbf{Z}_{r+1}(\mathbf{E}_r) \subset \mathbf{E}_r$$

and that $\mathbf{E}_k$ is isomorphic to $\mathbf{Z}_k(\mathbf{E}_r) / \mathbf{B}_k(\mathbf{E}_r)$ for $k \geq r+1$

We set $Z_\infty(E_2) = \bigcap_{r \geq 3} Z_r(E_2)$, $B_\infty(E_2) = \bigcup_{r \geq 3} B_r(E_2)$ and $E_\infty = Z_\infty(E_2) / B_\infty(E_2)$. We say that the spectral sequence is regular if the decreasing sequence $(Z_r(E_2^{pq}))_{r \geq 2}$ is stationary for every pair (p, q) .

b) Let $E = (E_r, d_r, \alpha_r)_{r \geq 2}$ and $E' = (E'_r, d'_r, \alpha'_r)_{r \geq 2}$ two spectral sequences of A-modules; a morphism from E to E' is a sequence of morphisms of A-complexes $u_r : E_r \rightarrow E'_r$, compatible with the bigradings, such that $\alpha'_r \circ H(u_r) = u_{r+1} \circ \alpha_r$ for all $r \geq 2$. Show that for every $r \geq 2$ and for every $k \geq r+1$, u_r induces a homomorphism of $Z_k(E_r)$ (resp. $B_k(E_r)$) in $Z_k(E'_r)$ (resp. $B_k(E'_r)$). In particular, u_2 induces homomorphisms $Z_\infty(u_2) : Z_\infty(E_2) \rightarrow Z_\infty(E'_2)$, $B_\infty(u_2) : B_\infty(E_2) \rightarrow B_\infty(E'_2)$ and $u_\infty : E_\infty \rightarrow E'_\infty$; all these homomorphisms are compatible with the bigradings.

c) We assume that u_2 is an isomorphism. Show that then u_r is an isomorphism for all $r \geq 2$ as well as $\mathbf{B}_\infty(u_2)$. If moreover the sequences E and E' are regular, $\mathbf{Z}_\infty(u_2)$ and u_∞ are isomorphisms.

d) Let $G = \bigoplus_{n \in \mathbb{Z}} G^n$ a graded A -module, equipped with a decreasing sequence $(F^p G)_{p \in \mathbb{Z}}$ of graded sub- A -modules

such that $\bigcap_p F^p G = 0$ and $\bigcup_p F^p G = G$. We say that the spectral sequence E approaches $(G, (F^p G)_{p \in \mathbb{Z}})$

(or simply G) if there exist isomorphisms $\beta^{p,q} : E_\infty^{p,q} \rightarrow (F^p G)^{p+q} / (F^{p+1} G)^{p+q}$ for all

$$(p, q) \in \mathbb{Z} \times \mathbb{Z}.$$

One says that the spectral sequence E converges to G if it approaches G and if moreover:

- (i) The spectral sequence E is regular;
- (ii) For every integer n , there exists an integer p such that $(F^p G)^n = 0$.

Suppose that E (resp. E') converges to G (resp. G'). Let $u : E \rightarrow E'$ a morphism of spectral sequences, $v : G \rightarrow G'$ a graded homomorphism of degree 0, such that $v(F^p G) \subset F^p G'$ for all p and that the induced homomorphisms $\bar{v}^{p,n} : (F^p G)^n / (F^{p+1} G)^n \rightarrow (F^p G')^n / (F^{p+1} G')^n$ satisfy $\bar{v}^{p,p+q} \circ \beta^{p,q} = \beta'^{p,q} \circ u_\infty^{p,q}$ for every pair (p, q) . Show that if $u_2^{p,q}$ is an isomorphism for every pair (p, q) , then v is an isomorphism.

15) Let E be a spectral sequence of A -modules, which approaches a graded A -module G (exercise 14).

a) Assume $E_2^{p,q} = 0$ for $p < 0$. Show that the maps α_r induce for every q injective A -homomorphisms:

$$E_\infty^{0,q} \rightarrow \dots \rightarrow E_3^{0,q} \rightarrow E_2^{0,q}$$

whence, by composition with $(\beta^{0,q})^{-1}$ and with the passage to the quotient map, an A -homomorphism $\gamma^q : G^q \rightarrow E_2^{0,q}$.

b) Assume $E_2^{pq} = 0$ for $q < 0$. Show that the maps α_r^{-1} induce surjective homomorphisms:

$$E_2^{p,0} \rightarrow E_3^{p,0} \rightarrow \dots \rightarrow E_\infty^{p,0}$$

whence, by composition with $\beta^{p,0}$, an A-homomorphism $\delta^p : E_2^{p,0} \rightarrow G^p$ c) We assume that the hypotheses of a) and b) are satisfied, that is, $E_2^{pq} = 0$ if $p < 0$ or $q < 0$. Show that the sequence

$$0 \rightarrow E_2^{1,0} \xrightarrow{\delta^1} G^1 \xrightarrow{\gamma^1} E_2^{0,1} \xrightarrow{d_2} E_2^{2,0} \xrightarrow{\delta^2} G^2$$

is exact.

d) Assume $E_2^{pq} = 0$ if $p > 0$ or $q > 0$; define an exact sequence analogous to that of c).

e) One assumes that there exist integers r, d , with $d \geq r \geq 2$, such that $E_r^{pq} = 0$ if $p \neq 0, d$. Define an exact sequence:

$$\dots \rightarrow E_r^{d,n-d} \rightarrow G^n \rightarrow E_r^{0,n} \rightarrow E_r^{d,n+1-d} \rightarrow G^{n+1} \rightarrow \dots$$

If likewise $E_r^{pq} = 0$ for $q \neq 0, d$, show that there exists an exact sequence:

$$\dots \rightarrow E_r^{n,0} \rightarrow G^n \rightarrow E_r^{n-d,d} \rightarrow E_r^{n+1,0} \rightarrow \dots$$

f) Assume $E_2^{pq} = 0$ if $q \neq 0$ (resp. $p \neq 0$). Show that the homomorphism

$$\delta^n : E_2^{n,0} \rightarrow G^n \text{ (resp. } \gamma^n : G^n \rightarrow E_2^{0,n})$$

is bijective for every n .

¶ 16) Let C be an A-complex, $(F^p C)_{p \in \mathbb{Z}}$ a decreasing sequence of subcomplexes of C .

a) For $p, q, r \in \mathbb{Z}$, let M_r^{pq} the set of elements x of $(F^p C)^{p+q}$ such that $dx \in F^{p+r} C$; for $r \geq 0$, we set $E_r^{pq} = M_r^{pq} / (d(M_{r-1}^{p-r+1, q+r-2}) + M_r^{p+1, q-1})$ and we denote by $d_r : E_r^{pq} \rightarrow E_r^{p+r, q-r+1}$ the homomorphism deduced from d by passing to quotients. Show that there

exists for $r \geq 0$ an isomorphism $\alpha_r : H(E_r) \rightarrow E_{r+1}$, so that the sequence $(E_r, d_r)_{r \geq 2}$, equipped with the maps α_r , is a spectral sequence. We have

$$E_0^{pq} = (F^p C)^{p+q} / (F^{p+1} C)^{p+q}.$$

- b) We assume that $\bigcup_p F^p C = C$ and that for every n , there exists an integer p such that $(F^p C)^n = 0$. Show that the spectral sequence defined in a) converges (exercise 14) to the graded A -module $H(C)$, equipped with the sequence of graded submodules $F^p H(C) = \text{Im } H(i_p)$ where $i_p : F^p C \rightarrow C$ is the canonical injection.
- c) If $F^0 \bar{C} = C$ the A -module E_r^{pq} is zero for $p < 0$; if $(F^p C)^n = 0$ for $p > n$, we have $E_r^{pq} = 0$ for $q < 0$. d) Let C' be a second complex, $(F^p C')_{p \in \mathbb{Z}}$ a decreasing sequence of subcomplexes of C' , E' the associated spectral sequence. Let $u : C \rightarrow C'$ be a morphism of complexes such that $u(F^p C) \subset F^p C'$ for all p . Show that u induces a morphism of spectral sequences $\tilde{u} : E \rightarrow E'$.
- d) With the notations of d), let $v : C \rightarrow C'$ be a second morphism such that $v(F^p C) \subset F^p C'$ for all p , and let h a homotopy relating u to v . Let k be an integer ≥ 0 such that $h(F^p C) \subset F^{p-k} C'$ for all p . Show that $\tilde{u}_r = \tilde{v}_r$ for $r > k$.

17) Let C be a bicomplex (X, p. 174, exercise 11).

- a) One defines two decreasing sequences $({}'F^p C)_{p \in \mathbb{Z}}$ and $({}''F^q C)_{q \in \mathbb{Z}}$ of subcomplexes of the complex associated with C by setting

$$({}'F^p C)^n = \bigoplus_{i \geq p} C^{i, n-i} \quad ({}''F^q C)^n = \bigoplus_{j \geq q} C^{n-j, j}.$$

Let $'E$ and $''E$ be the corresponding spectral sequences (exercise 16, a)). Show that for every pair (p, q) the A -module $'E_2^{pq}$ (resp. $''E_2^{pq}$) is isomorphic to $H'^p(H'^q(C))$ (resp. $H''^p(H''^q(C))$).

- b) Suppose there exists an integer n such that $C^{pq} = 0$ for $p \geq n$ or for $q \leq n$. Show that the spectral sequence $'E$ converges to $H(C)$. If likewise there exists an integer m such that

$C^{pq} = 0$ for $p \leq m$ or for $q \geq m$, the spectral sequence mE converges to $H(C)$.

c) We assume that $C^{pq} = 0$ for $p < 0$ or $q < 0$. Show that the homomorphism $\delta^p : E_2^{p0} \rightarrow H^p(C)$ (exercise 15) comes, by passing to homology, from the canonical injection of complexes " $Z^{\wedge\{0\}}(C) \rightarrow C$ ". Likewise the homomorphism " $E_{-2}^{\wedge\{p0\}} \rightarrow H^{\wedge\{p\}}(C)$ " comes from the injection " $Z^{\wedge\{0\}}(C) \rightarrow C$ ".

d) Let $\tilde{C}$ a second bicomplex, mE and nE the two associated spectral sequences. Show that every morphism of bicomplexes $u : C \rightarrow \tilde{C}$ induces morphisms of spectral sequences ${}^m u : {}^mE \rightarrow {}^mE$ and ${}^n u : {}^nE \rightarrow {}^nE$. Show that if two morphisms u, v of C to $\tilde{C}$ are homotopic, the associated morphisms ${}^m u$ and ${}^m v$ (resp. ${}^n u$ and ${}^n v$) are equal (use exercise 16, e)).

18) Let E be a spectral sequence (exercise 14), such that the A -module E_2 has finite length. Let $\mathcal{C}$ be the set of isomorphism classes of A -modules of finite length; show that one has $\chi_{\mathcal{C}}(E_2) = \chi_{\mathcal{C}}(E_r) = \chi_{\mathcal{C}}(E_{\infty})$ for all $r \geq 2$ (X, p. 41). If moreover E converges to a graded A -module G , the latter has finite length and one has $\chi_{\mathcal{C}}(G) = \chi_{\mathcal{C}}(E_2)$.

¶ 19) One calls a simplicial scheme a pair (K, F) , where K is a set and F a set of finite subsets of K , called faces of K , such that every subset of a face is a face and every point of K belongs to at least one face. A simplicial map from a simplicial scheme (K, F) to a simplicial scheme (L, G) is a map from K to L which carries every face of K to a face of L . a) Let n be an integer ≥ 0 . One calls an n -simplex of the simplicial scheme K every sequence $(\alpha_0, \dots, \alpha_n)$ of elements of K such that the α_i belong to a single face of K . One denotes by $S_n(K, A)$ the free A -module generated by the n -simplexes for $n \geq 0$, and one sets $S_n(K, A) = 0$ for $n < 0$. One defines an A -homomorphism $d_n : S_n(K, A) \rightarrow S_{n-1}(K, A)$ by setting $d_n = 0$ for $n \leq 0$ and $d_n(\alpha_0, \dots, \alpha_n) = \sum_{i=0}^n (-1)^i (\alpha_0, \dots, \theta_i, \dots, \alpha_n)$

for all n -simplex $(\alpha_0, \dots, \alpha_n)$ (the sign over a letter means that it is to be omitted). Show that $d_{n-1} \circ d_n = 0$ for all n , so that $(S(K, A), d)$ is a complex of A -modules. One defines in the same way an A -complex $C(K, A)$ by setting $C^n(K, A) = \text{Hom}_Z(S_n(K, Z), A)$, $\delta^n = \text{Hom}(d_n, 1_A)$ for all n . For $n \geq 0$, one sometimes denotes $H_n(K, A)$ (resp. $H^n(K, A)$) the A -module $H_n(S(K, A))$ (resp. $H^n(C(K, A))$). b) Let K, L be two simplicial schemes, $f : K \rightarrow L$ a simplicial map. Show that f induces morphisms of complexes $S(f) : S(K, A) \rightarrow S(L, A)$ and $C(f) : C(K, A) \rightarrow C(L, A)$

$C(K, A)$, whence A -homomorphisms $f_* : H(K, A) \rightarrow H(L, A)$ and $f^* : H'(L, A) \rightarrow H'(K, A)$.

- c) Let $g : \mathbf{K} \rightarrow \mathbf{L}$ a second simplicial map. We say that f and g are simplicially homotopic if for every face σ of $\mathbf{K}$, the set $f(\sigma) \cup g(\sigma)$ is a face of $\mathbf{L}$. Show that the morphisms $\mathbf{S}(f)$ and $\mathbf{S}(g)$ (resp. $\mathbf{C}(f)$ and $\mathbf{C}(g)$) are then homotopic (define a homotopy h by setting

$$h(\alpha_0, \dots, \alpha_n) = \sum_{i=0}^n (-1)^i (f(\alpha_0), \dots, f(\alpha_i), g(\alpha_i), \dots, g(\alpha_n))$$

for every n -simplex $(\alpha_0, \dots, \alpha_n)$.

- d) A simplicial scheme K is said to be conical if there exists a point s of K such that for every face σ of K , $\sigma \cup \{s\}$ be a face. Show that there then exist homotopisms from $\mathbf{S}(K, A)$ and from $\mathbf{C}(K, A)$ onto A . e) Let $D(K, A)$ denote the subcomplex of $\mathbf{S}(K, A)$ generated by the simplices $(\alpha_0, \dots, \alpha_n)$ for which two of the α_i are equal, and by the elements of the form $(\alpha_0, \dots, \alpha_n) - \varepsilon(\sigma) \cdot (\alpha\sigma(0), \dots, \alpha\sigma(n))$ for every n -simplex $(\alpha_0, \dots, \alpha_n)$ and every permutation σ of $\{0, \dots, n\}$. Show that $D(K, A)$ is a subcomplex of $\mathbf{S}(K, A)$, and that the quotient map $p : \mathbf{S}(K, A) \rightarrow \mathbf{S}(K, A)/D(K, A)$ is a homotopy equivalence (a homotopy inverse is obtained by choosing a total order on K and associating to the class modulo $D(K, A)$ of a simplex $(\alpha_0, \dots, \alpha_n)$ the simplex $\varepsilon(\sigma) \cdot (\alpha\sigma(0), \dots, \alpha\sigma(n))$, where σ is the permutation such that $\alpha\sigma(0) < \dots < \alpha\sigma(n)$, if all the α_i are distinct, and 0 otherwise).

20) Let R be a complete discrete valuation ring (AC, VI, § 3, no. 6) with residue field k of characteristic $p > 0$. Assume that the field of fractions of R has characteristic 0. Let v denote the normalized valuation of R , $\mathfrak{m}$ the maximal ideal of R , and R^ the multiplicative group of units of R . For every integer $n \geq 0$ Let $R^*(n)$ be the subgroup of R^* consisting of the elements x such that $v(x - 1) \geq n$. a) Let r be an integer, $r > v(p)/(p - 1)$. Show that the series

$$\exp(x) = \sum_{i=0}^{+\infty} x^i / i!$$

converges for $x \in \mathfrak{m}^r$ and defines an isomorphism from the additive group $\mathfrak{m}^r$ onto the group $\mathbf{R}^*(r)$. b) Suppose that k is a finite field with q elements. Let n be an integer ≥ 1 , μ_n the subgroup of elements x of $\mathbf{R}^*$ such that $x^n = 1$. Show that the groups μ_n and $\mathbf{R}^*/(\mathbf{R}^*)^n$ are

finite and that one has

$$\text{Card} (R^*/R^{*n}) / \text{Card} (\mu_n) = qv(n).$$

(Let $\mathcal{C}$ the set of classes of finite $\mathbf{Z}$ -modules, $\varphi : K(\mathcal{C}) \rightarrow \mathbf{Q}^*$ be the homomorphism defined by

$$\varphi([G]) = \text{Card} (G) \quad \text{for} \quad G \in \mathcal{C},$$

$C^{(r)}$ the complex of $\mathbf{Z}$ -modules such that $C_p^{(r)} = 0$ for $p \neq 0, 1$, $C_0^{(r)} = C_1^{(r)} = R^*(r)$, $d_1(x) = x^n$ for $x \in R^*(r)$. Prove that $\varphi(H(C^{(r)}))$ is independent of r , then use a).

- 21) Assume that the ring A is a commutative algebra over a commutative ring k ; the A -module $D_k(A)$ of k -derivations of A is then identified with the dual of the A -module $\Omega_{A/k}^1$, which defines on $\Omega_{A/k}$ a structure of $\Lambda_A(D_k(A))$ left module (cf. III, p. 165). Prove the formula

$$\begin{aligned} (X_0 \wedge \cdots \wedge X_p)]d\omega &= \sum_{i=0}^p (-1)^i X_i ((X_0 \wedge \cdots \wedge \hat{X}_i \wedge \cdots \wedge X_p)]\omega) \\ &+ \sum_{0 \leq i < j \leq p} (-1)^{i+j} ([X_i, X_j] \wedge X_0 \wedge \cdots \wedge \hat{X}_i \wedge \cdots \wedge \hat{X}_j \wedge \cdots \wedge X_p)]d\omega. \end{aligned}$$

where $\omega \in \Omega_{A/k}^p$, $X_0, \dots, X_p \in D_k(A)$ and where the sign $\wedge$ on a letter means that it is to be omitted. (Reduce to the case $p = 1$.)

- 22) We keep the conventions of the preceding exercise; if M is an A -module, ∇ a connection on M and if $X \in D_k(A)$, we denote by ∇_X the k -endomorphism $(1_M \otimes X) \circ \nabla$ of M .

- a) Let $\mathbf{M}_1, \mathbf{M}_2$ two A -modules, equipped with connections ${}^1\nabla, {}^2\nabla$; show that one defines a connection ∇ over $\mathbf{M}_1 \oplus \mathbf{M}_2$ (resp. $\mathbf{M}_1 \otimes_A \mathbf{M}_2$, resp. $\text{Hom}_A(\mathbf{M}_1, \mathbf{M}_2)$) by setting $\nabla(m_1 + m_2) = {}^1\nabla(m_1) + {}^2\nabla(m_2)$ for $m_1 \in \mathbf{M}_1, m_2 \in \mathbf{M}_2$ (resp. $\nabla_X(m_1 \otimes m_2) = {}^1\nabla_X(m_1) \otimes m_2 + m_1 \otimes {}^2\nabla_X(m_2)$ for $X \in D_k(A), m_1 \in \mathbf{M}_1$ and $m_2 \in \mathbf{M}_2$, resp. $\nabla_X(u) = {}^2\nabla_X \circ u - u \circ {}^1\nabla_X$ for $X \in D_k(A), u \in \text{Hom}_A(\mathbf{M}_1, \mathbf{M}_2)$).

- b) Let M be an A -module, $\nabla : M \rightarrow M \otimes_A \Omega_{A/k}^1$ a connection on M . If $X, Y \in D_k(A)$, we denote by $R(X, Y)$ the A -endomorphism of M such that $R(X, Y)(m) = \langle R(m), X \wedge Y \rangle$ for all $m \in M$. Prove the formula: $R(X, Y) = [\nabla X, \nabla Y] - \nabla[X, Y]$ (use Exercise 21).

- c) Assume the A -module M is projective, so that the curvature homomorphism R is identified with an element of $\text{End}_A(M) \otimes_A \Omega_{A/k}^2$. If one equips $\text{End}_A(M)$ with the connection ∇ defined in a), show that one has $\nabla^2 R = 0$.

Exercises for Section 3: Resolutions

- 1) Let a, b two elements of A , such that the left annihilator of a (resp. b) is the ideal Ab (resp. Aa). Show that the sequence

$$\cdots \rightarrow A_s \xrightarrow{\delta_a} A_s \xrightarrow{\delta_b} A_s \xrightarrow{\delta_a} A_s \rightarrow \cdots \rightarrow A_s \xrightarrow{\delta_a} A_s \rightarrow A/Aa \rightarrow 0,$$

where δ_a (resp. δ_b) denotes right multiplication by a (resp. b), defines a free resolution of the A -module A/Aa . Show that this construction applies in the following two cases:

- α) Let A_0 a ring; we take $A = A_0(\varepsilon)$ (X, p. 27), $a = b = \varepsilon$.
 β) Let G be a finite cyclic group, σ a generator of G , k a commutative ring. We take

$$A = k(G), \quad a = 1 - e_\sigma, \quad b = \sum_{g \in G} e_g.$$

- 2) We assume the ring A is commutative. Let B be an A -algebra; for $n \geq 0$, we denote by $\mathcal{A}^n(B)$ the tensor product over A of $(n+1)$ modules equal to B . We set $\mathcal{A}^n(B) = 0$ for $n < 0$ and $\mathcal{A}(B) = \bigoplus \mathcal{A}^n(B)$. We define a graded A -endomorphism of ascending degree $+1$ of $\mathcal{A}(B)$ by setting:

$$d(b_0 \otimes \cdots \otimes b_n) = \sum_{i=0}^{n+1} (-1)^i b_0 \otimes \cdots \otimes b_{i-1} \otimes 1_B \otimes b_i \otimes \cdots \otimes b_n \quad \text{for } b_0, \dots, b_n \in B.$$

- a) Show that $d \circ d = 0$, so that $(\mathcal{A}(B), d)$ is a complex of A -modules. If M is an A -module, we denote by $\mathcal{A}(B, M)$ the complex such that $\mathcal{A}^n(B, M) = \mathcal{A}^n(B) \otimes_A M$, with differential $d \otimes 1_M$. b) Suppose that the algebra B is a faithfully flat A -module (X, p. 169, exercise 9). Show that the complex $\mathcal{A}(B, M)$ defines a right resolution of the

A-module M . (It suffices to verify that the complex $\mathbf{B} \otimes_{\mathbf{A}} \mathcal{A}(\mathbf{B}, \mathbf{M})$ defines a resolution of $\mathbf{B} \otimes_{\mathbf{A}} \mathbf{M}$; construct an A -linear homotopism between these two complexes.)

- 3) Let B be a ring, a a two-sided ideal of B , b a left ideal of B containing a . Denote by A the ring B/a , by $p : \mathbf{B} \rightarrow \mathbf{A}$ the quotient homomorphism and by b' the ideal $p(b) \subset \mathbf{A}$. a) Consider the sequence of canonical injections:

$$\cdots \rightarrow a^n b \rightarrow a^n \rightarrow a^{n-1} b \rightarrow a^{n-1} \rightarrow \cdots \rightarrow a \rightarrow b \rightarrow B.$$

Show that the sequence obtained by tensor product with B/a

$$\cdots \rightarrow a^n b / a^{n+1} b \rightarrow a^n / a^{n+1} \rightarrow \cdots \rightarrow a / a^2 \rightarrow b / a b \rightarrow \mathbf{A}$$

defines a left resolution of the A -module A/b' .

- b) Suppose that the ideals a and b are free left A -modules. Show that the preceding resolution is a free resolution of the A -module A/b' .
- c) Let G be a group, k a commutative ring, $A = k^{(G)}$ the algebra of G over k , $\varepsilon : A \rightarrow k$ the homomorphism of k -algebras such that $\varepsilon(e_g) = 1$ for all $g \in G$. One endows k with the structure of A -module (on the left) associated with the homomorphism ε . Let $(g_i)_{i \in I}$ be a generating family of G , F the free group constructed on I , $B = k^{(F)}$, $\pi : F \rightarrow G$ the homomorphism such that $\pi(i) = g_i$ for $i \in I$, $p : B \rightarrow A$ the homomorphism of k -algebras deduced from π . Show that the ideals $a = \text{Ker}(p)$ and $b = \text{Ker}(\varepsilon \circ p)$ are free left B -modules; deduce from this a free resolution of the $k^{(G)}$ -module k .

(According to I, p. 147, exercise 20, the group $\text{Ker}(\pi)$ admits a basic family $(r_\alpha)_{\alpha \in J}$; show that the elements $(e_i r_\alpha - 1)$, for $\alpha \in J$ (resp. $(e_i - 1)$, for $i \in I$) form a basis of the left B -module a (resp. b .)

- d) With the notations of c), we set $\mathbf{R} = \text{Ker}(\pi)$; let $\varphi : \mathbf{R} \rightarrow \mathbf{R}/(\mathbf{R}, \mathbf{R})$ be the quotient map. We endow the k -module $k \otimes_{\mathbf{Z}} \mathbf{R}/(\mathbf{R}, \mathbf{R})$ of the A -module structure such that

$$e_{\pi(f)}(1 \otimes \varphi(r)) = 1 \otimes \varphi(frf^{-1})$$

for all $r \in \mathbb{R}$, $f \in \mathbb{F}$. Show that there exists an exact sequence of A -modules:

$$0 \rightarrow k \otimes_{\mathbb{Z}} R/(R, R) \rightarrow A^{(I)} \rightarrow A \rightarrow k \rightarrow 0.$$

4) Assume the ring A is commutative.

Let G be a group, and K a simplicial scheme (X, p. 177, exercise 19). An action of G on K is a homomorphism from G into the group of bijective simplicial maps from K to itself. a) An action of G on K defines an action of G on the A -module $S(K, A)$ (loc. cit.); show that this action makes $S(K, A)$ into a complex of $A^{(G)}$ -modules.

b) Assume that G acts freely on the underlying set of K . Show that the $A^{(G)}$ -modules $S_n(K, A)$ are free.

c) Assume that the simplicial scheme K is conical. Show that the $A^{(G)}$ -complex $S(K, A)$ defines a left resolution of A , equipped with the structure of $A^{(G)}$ -module for which $e_g a = a$ for all $g \in G$, $a \in A$.

d) Let $|G|$ the simplicial scheme $(G, \mathcal{F})$, where $\mathcal{F}$ is the set of finite subsets of G . The group G acts by left translations on the simplicial scheme $|G|$; show that the $A^{(G)}$ -complex $S(|G|, A)$ is a free resolution of the $A^{(G)}$ -module A .

e) Let $B(A^{(G)}, A)$ the standard resolution of the $A^{(G)}$ -module A ; for $n \geq 0$, the $A^{(G)}$ -module $B_n(A^{(G)}, A)$ is identified with $(A^{(G)})^{\otimes(n+1)}$. Show that the A -linear homomorphism $u : S(|G|, A) \rightarrow B(A^{(G)}, A)$, defined by

$$u(g_0, \dots, g_n) = g_0 \otimes g_0^{-1} g_1 \otimes \dots \otimes g_{n-1}^{-1} g_n \quad \text{for } g_0, \dots, g_n \text{ in } G,$$

is an isomorphism of $A^{(G)}$ -complexes.

5) a) Let $0 \rightarrow K \rightarrow P \rightarrow M \rightarrow 0$, $0 \rightarrow K' \rightarrow P' \rightarrow M \rightarrow 0$ two exact sequences of A -modules, with P and P' projective. Show that there exists a commutative diagram:

$$\begin{array}{ccccccc}
0 & \rightarrow & K \oplus P' & \rightarrow & P \oplus P' & \rightarrow & M \rightarrow 0 \\
& & \downarrow v & & \downarrow u & \downarrow l_M & \\
0 & \rightarrow & P \oplus K' & \rightarrow & P \oplus P' & \rightarrow & M \rightarrow 0
\end{array}$$

where the horizontal lines are exact, and where u and v are isomorphisms (cf. II, p. 183, exercise 4). b) Let n be an integer ≥ 0 , (P, p) and (P', p') be two left resolutions of the A -module M , such that $P_j = P'_j = 0$ for $j > n$ and that the A -modules P_i and P'_i be projective for $0 \leq i \leq n-1$. Show that the A -modules $\bigoplus_{i \geq 0} (P_{2i} \oplus P'_{2i+1})$ and $\bigoplus_{i \geq 0} (P'_{2i} \oplus P_{2i+1})$ are isomorphic.

- c) Under the assumptions of b), show that there exist two projective complexes R, R' of zero homology, such that $R_i = R'_i = 0$ for $i \notin [0, n]$, and an isomorphism of resolutions $u : P \oplus R \rightarrow P' \oplus R'$.

¶(6). Let M be a left A -module. One calls a presentation of length n , or an n -presentation of M , an exact sequence

$$L_n \rightarrow L_{n-1} \rightarrow \cdots \rightarrow L_1 \rightarrow L_0 \rightarrow M \rightarrow 0$$

where L_i is a free left A -module ($0 \leq i \leq n$). The presentation is said to be finite if all the L_i are finitely generated free modules.

If M is a finitely generated left A -module, one denotes by $\lambda(M)$ the least upper bound (finite or equal to $+\infty$) of the integers $n \geq 0$ such that M admits a finite n -presentation. If M is not finitely generated, one sets $\lambda(M) = -1$.

- a) Let $0 \rightarrow P \rightarrow N \rightarrow M \rightarrow 0$ an exact sequence of left A -modules. One then has $\lambda(N) \geq \inf(\lambda(P), \lambda(M))$.
- b) Let $L_n \xrightarrow{d_n} L_{n-1} \rightarrow \cdots \rightarrow L_0 \rightarrow M \rightarrow 0$ a finite n -presentation of M ; if $\lambda(M) > n$ show that $\text{Ker}(d_n)$ is a finitely generated A -module (use exercise 5, b)).
- c) Show that, under the hypotheses of a), one has

$$\lambda(\mathbf{M}) \geq \inf(\lambda(\mathbf{N}), \lambda(\mathbf{P}) + 1).$$

(If $n \leq \inf(\lambda(\mathbf{N}), \lambda(\mathbf{P}) + 1)$, show by induction on n that $\lambda(\mathbf{M}) \geq n$ reasoning as in a and using b .)

d) Show that, under the hypotheses of a), one has

$$\lambda(\mathbf{P}) \geq \inf(\lambda(\mathbf{N}), \lambda(\mathbf{M}) - 1).$$

(Same method as in c .) Deduce that, if $\lambda(\mathbf{N}) = +\infty$, then $\lambda(\mathbf{M}) = \lambda(\mathbf{P}) + 1$.

e) Deduce from a), c) and d) that, if $\mathbf{N} = \mathbf{M} \oplus \mathbf{P}$, we have

$$\lambda(\mathbf{N}) = \inf(\lambda(\mathbf{M}), \lambda(\mathbf{P})).$$

In particular, for $\mathbf{N}$ to admit a finite presentation it is necessary and sufficient that the same be true for $\mathbf{M}$ and $\mathbf{P}$. f) Let N_1, N_2 be two submodules of an A -module $\mathbf{M}$. Suppose that N_1 and N_2 admit a finite presentation. For $N_1 + N_2$ to admit a finite presentation, it is necessary and sufficient that $N_1 \cap N_2$ be finitely generated.

7) a) With the notations of exercise 6, show that, if $\mathbf{M}$ is a projective module, one has

$$\lambda(\mathbf{M}) = -1 \quad \text{or} \quad \lambda(\mathbf{M}) = +\infty.$$

If A is a left Noetherian ring, then, for every A -module $\mathbf{M}$, one has $\lambda(\mathbf{M}) = -1$ or $\lambda(\mathbf{M}) = +\infty$. b) If a is a left ideal of a ring A , which is not finitely generated, A_s/a is a monogenic A -module which does not admit a finite presentation, in other words $\lambda(A_s/a) = 0$ (cf. X, p. 11, prop. 6).

c) Give an example of a monogenic left ideal a of a ring A such that A_s/a (which is of finite presentation) admits a dual which is not a finitely generated right A -module.

d) Let K be a commutative field, E the vector space $\mathbf{K}^{(\mathbf{N})}$, (e_n) the canonical basis of E , T the

tensor algebra of E , of which a basis is therefore formed by the finite products $e_{i_1} e_{i_2} \dots e_{i_k}$ ($k \geq 0, i_j \in \mathbb{N}$ for all j). For a given integer n let b be the two-sided ideal of T generated by the products $e_1 e_0, e_2 e_1, \dots, e_n e_{n-1}$ and $e_{n+k} e_n$ for all $k \geq 1$; let A be the quotient ring T/b , and for every integer m , let a_m the canonical image of e_m in A . Show that, if $\mathbf{M} = \mathbf{A}_s / \mathbf{A}a_0$, we have $\lambda(\mathbf{M}) = n$ (observe that, for $m \leq n-1$ the left annihilator of a_m is $\mathbf{A}a_{m+1}$ and use exercise 6, b)).

- 8) Let C be a commutative ring, E, F two C -modules. Show that one has $\lambda(E \otimes_C F) \geq \inf(\lambda(E), \lambda(F))$.
- 9) Let $\rho : A \rightarrow B$ a ring homomorphism, M an A -module.
 - a) If B is a flat right A -module and if M admits a finite n -presentation, the B -module $B \otimes_A M$ admits a finite n -presentation.
 - b) If B is a faithfully flat right A -module (X, p. 169, exercise 9), and if the B -module $B \otimes_A M$ admits a finite n -presentation, then M admits a finite n -presentation (use exercise 6, b) and exercise 12, p. 169).
- 10) Let E be an A -module. One says that E is pseudo-coherent if every finitely generated submodule of E is of finite presentation; every submodule of a pseudo-coherent module is pseudo-coherent. One says that E is coherent if it is pseudo-coherent and finitely generated (hence of finite presentation).
 - a) Let $0 \rightarrow E' \rightarrow E \rightarrow E'' \rightarrow 0$ an exact sequence of A -modules. Show that, if E is pseudo-coherent (resp. coherent) and E' finitely generated, E'' is pseudo-coherent (resp. coherent). Show that, if E' and E'' are pseudo-coherent (resp. coherent), then so is E . Show that, if E and E'' are coherent, so is E' (use X, p. 11, prop. 6).
 - b) Let E be a coherent A -module, E' a pseudo-coherent (resp. coherent) A -module. Show that, for every homomorphism $u : E \rightarrow E'$, $\text{Im}(u)$ and $\text{Ker}(u)$ are coherent and $\text{Coker}(u)$ pseudo-coherent (resp. coherent) (use a)).
 - c) Show that every direct sum (resp. every finite direct sum) of pseudo-coherent (resp. coherent) modules is a pseudo-coherent (resp. coherent) module.

d) If E is a pseudo-coherent module and if M, N are coherent submodules of E , show that $M + N$ and $M \cap N$ are coherent (use a) and c)).

e) Suppose A is commutative. Show that, if E is a coherent A -module and F a coherent (resp. pseudo-coherent) A -module, $\text{Hom}_A(E, F)$ is a coherent (resp. pseudo-coherent) A -module. (Reduce to the case where F is coherent; consider a finite presentation of E , then use b).)

¶11) a) Let A be a ring. Show that the following properties are equivalent:

α) The A -module A_s is coherent (exercise 10).

β) Every A -module of finite presentation is coherent.

γ) For every finitely presented right A -module M , the dual M^* is a finitely generated A -module.

δ) For every set I , the right A -module A_d^I is flat.

ε) Any product of flat right A -modules is flat.

(To prove that α) implies β), use Exercise 10, b). To see that δ) implies γ), use X, p. 14, th. 1; to show that α) implies ε), use X, p. 170, exercise 18.)

A ring satisfying these properties is said to be left coherent (or simply coherent), and the notion of a right coherent ring is defined similarly.

b) Show that a left Noetherian ring is coherent. Give an example of a right Artinian ring that is not coherent (cf. VIII, § 1, exercise 4).

c) Show that an absolutely flat ring (X, p. 169, exercise 16) is coherent.

d) Show that, if A is a coherent ring and M is an A -module, we have $\lambda(M) = -1$ or $\lambda(M) = 0$ or $\lambda(M) = +\infty$ (exercise 6). In particular, every finitely presented A -module admits a resolution by finitely generated free modules.

e) Let $(A_\alpha, \varphi_{\beta\alpha})$ an inductive system of rings whose index set is filtered, and let $A = \varinjlim A_\alpha$. We assume that for $\alpha \leq \beta$, A_β is a flat right A_α -module. Show that if the A_α are coherent, so is A (Observe that A is a flat right A_α -module for all α and that for every finitely generated left ideal $a \subset A$ there exists an index α and an ideal a_α of A such that

α is isomorphic to $\mathbf{A} \otimes_{\mathbf{A}} a_{\alpha'}$.)

- f) Deduce from e) that every polynomial ring (for any finite or infinite set of indeterminates) over a commutative noetherian ring is coherent. Deduce from this that a quotient ring of a coherent ring is not necessarily coherent.
- g) For $\mathbf{A}$ to be left coherent, it is necessary and sufficient that the left annihilator of every element of $\mathbf{A}$ be finitely generated, and that the intersection of two finitely generated left ideals be finitely generated (use exercise 6, f)).
- 12) Let c be an infinite cardinal. Suppose that every left ideal of the ring $\mathbf{A}$ admits a generating set of cardinality $\leq c$. Show that every $\mathbf{A}$ -module generated by a set of elements of cardinality $\leq c$ admits a left resolution by free $\mathbf{A}$ -modules of rank $\leq c$ (cf. VIII, § 1, exercise 13).
- 13) Let $\mathbf{M}$ be an $\mathbf{A}$ -module. A minimal injective resolution of $\mathbf{M}$ is an injective resolution $(e, 1)$ of $\mathbf{M}$ such that I^n is an injective envelope of $Z^n(I)$ for all $n \geq 0$.
- a) Every $\mathbf{A}$ -module admits minimal injective resolutions; any two such resolutions are isomorphic.
- b) Let I, I' be two injective resolutions of $\mathbf{M}$, with I minimal; let $f : I \rightarrow I'$ and $g : I' \rightarrow I$ two morphisms of resolutions. Then f is injective, g is surjective, and it is the direct sum of the subcomplexes $\text{Im}(f)$ (isomorphic to I) and $\text{Ker}(g)$ (with zero homology).
- 14) Assume that the ring $\mathbf{A}$ is local Noetherian; denote its maximal ideal by $\mathfrak{m}$. Let $\mathbf{L}$ be a complex of finitely generated $\mathbf{A}$ -modules, zero in degrees < 0 , $\mathbf{M}$ a finitely generated $\mathbf{A}$ -module, and $q : \mathbf{L} \rightarrow \mathbf{M}$ a morphism. We make the following hypotheses:
- (i) The homomorphism $1 \otimes q : (\mathbf{A}/\mathfrak{m}) \otimes_{\mathbf{A}} \mathbf{L}_0 \rightarrow (\mathbf{A}/\mathfrak{m}) \otimes_{\mathbf{A}} \mathbf{M}$ is injective.
- (ii) One has $d_i(\mathbf{L}_i) \subset \mathfrak{m}\mathbf{L}_{i-1}$ for $i \geq 1$, and the map

$$\bar{d}_i : \mathbf{L}_i/\mathfrak{m}\mathbf{L}_i \rightarrow \mathfrak{m}\mathbf{L}_{i-1}/\mathfrak{m}^2\mathbf{L}_{i-1} \quad \text{is injective.}$$

Let $(\mathbf{P}, p)$ be a projective resolution of $\mathbf{M}$, $u : \mathbf{L} \rightarrow \mathbf{P}$ a morphism such that $p \circ u = q$. Show

that u is injective and that $u(L)$ is a direct summand of P as a graded A -module (reduce to the case where the resolution P is minimal, and use VIII, § 8, no. 3, cor. 3).

15) Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal, $k = A/\mathfrak{m}$. Let P be a minimal projective resolution of the A -module k ; for $n \geq 0$, we denote by b_n the rank of the free A -module P_n .

- a) Show that we have $b_0 = 1$ and that $b_1 = \dim_k(\mathfrak{m}/\mathfrak{m}^2)$ is the minimal number of generators of $\mathfrak{m}$.
- b) If A is commutative, we have $b_2 \geq \frac{1}{2}b_1(b_1 - 1)$ (consider the exact sequence $P_2 \rightarrow \mathfrak{m}P_1 \rightarrow \mathfrak{m}^2 \rightarrow 0$). c) Assume the ring A is Artinian (not necessarily commutative). Prove the inequality $b_2 \geq \frac{1}{4}b_1^2$ (let $\mathbf{M}_l$ the submodule of P_2 of elements x such that $d_2(x) \in \mathfrak{m}^l P_1$; show that we have

$$P_2 \subset \mathbf{M}_{l+1}$$

and that there exists an exact sequence $0 \rightarrow \mathbf{M}_l/\mathbf{M}_{l+1} \rightarrow \mathfrak{m}^l P_1/\mathfrak{m}^{l+1} P_1 \rightarrow \mathfrak{m}^{l+1}/\mathfrak{m}^{l+2} \rightarrow 0$. Deduce from this that the polynomial $p(t) = \sum_l (\dim_k(\mathfrak{m}^l/\mathfrak{m}^{l+1}))t^l$ satisfies $(b_2 t^2 - b_1 t + 1)p(t) \geq 0$ for $0 \leq t \leq 1$. Conclude

by examining the cases separately $b_1 = 1, b_1 \geq 2$.

- 16) a) Let $0 \rightarrow \mathbf{M}' \rightarrow \mathbf{M} \rightarrow \mathbf{M}'' \rightarrow 0$ an exact sequence of A -modules, (P', p') (resp. (P'', p'')) a projective resolution of $\mathbf{M}'$ (resp. $\mathbf{M}''$). Show that there exists a projective resolution (P, p) of $\mathbf{M}$ and a commutative diagram:

$$\begin{array}{ccccccc} 0 & \rightarrow & P' & \rightarrow & P & \rightarrow & P'' \rightarrow 0 \\ & & \downarrow p' & & \downarrow p & & \downarrow p'' \\ 0 & \rightarrow & M' & \rightarrow & M & \rightarrow & M'' \rightarrow 0 \end{array}$$

where the horizontal lines are exact.

- b) Let $0 \rightarrow \mathbf{N}' \rightarrow \mathbf{N} \rightarrow \mathbf{N}'' \rightarrow 0$ a second exact sequence of A -modules, (Q, q) (resp. (Q', q') , resp. (Q'', q'')) a projective resolution of $\mathbf{N}$ (resp. $\mathbf{N}'$, resp. $\mathbf{N}''$), so that there exists a

commutative diagram with exact rows:

$$\begin{array}{ccccccc} 0 & \rightarrow & Q' & \rightarrow & Q & \rightarrow & Q'' \rightarrow 0 \\ & & \downarrow q' & & \downarrow q & & \downarrow q'' \\ 0 & \rightarrow & N' & \rightarrow & N & \rightarrow & N'' \rightarrow 0 \end{array}$$

Let, on the other hand:

$$\begin{array}{ccccccc} 0 & \rightarrow & M' & \rightarrow & M & \rightarrow & M'' \rightarrow 0 \\ & & \downarrow u' & & \downarrow u & & \downarrow u'' \\ 0 & \rightarrow & N' & \rightarrow & N & \rightarrow & N'' \rightarrow 0. \end{array}$$

a commutative diagram of exact sequences, and let $\tilde{u}' : \mathbf{P}' \rightarrow \mathbf{Q}'$ and $\tilde{u}'' : \mathbf{P}'' \rightarrow \mathbf{Q}''$ morphisms of complexes such that $u' \circ p' = q' \circ \tilde{u}'$ and $u'' \circ p'' = q'' \circ \tilde{u}'$. Show that there exists a morphism $\tilde{u} : \mathbf{P} \rightarrow \mathbf{Q}$ such that $u \circ p = q \circ \tilde{u}$ and that the diagram:

$$0 \rightarrow \mathbf{P}' \rightarrow \mathbf{P} \rightarrow \mathbf{P}'' \rightarrow 0 \quad (*)$$

$$0 \rightarrow Q' \rightarrow Q \rightarrow Q'' \rightarrow 0$$

is commutative.

- c) Let $\tilde{u}'_1 : \mathbf{P}' \rightarrow \mathbf{Q}'$, $\tilde{u}_1 : \mathbf{P} \rightarrow \mathbf{Q}$ and $\tilde{u}''_1 : \mathbf{P}'' \rightarrow \mathbf{Q}''$ three morphisms of complexes such that

$$u' \circ p' = q' \circ \tilde{u}'_1, u \circ p = q \circ \tilde{u}_1, u'' \circ p'' = q'' \circ \tilde{u}''_1$$

making the diagram (*) commutative; let s' (resp. s'') a homotopy connecting $\tilde{u}'$ to $\tilde{u}'_1$ (resp. $\tilde{u}''$ to $\tilde{u}''_1$). Show that there exists a homotopy s connecting $\tilde{u}$ to $\tilde{u}_1$, such that the diagram:

$$\begin{array}{ccccccc}
0 & \rightarrow & P' & \rightarrow & P & \rightarrow & P'' \rightarrow 0 \\
& & \downarrow s' & & \downarrow s & & \downarrow s'' \\
0 & \rightarrow & Q' & \rightarrow & Q & \rightarrow & Q'' \rightarrow 0
\end{array}$$

is commutative.

- d) State and prove the results corresponding to a), b), c) for injective resolutions. ¶17) In this exercise, if $(\mathbf{K}, d', d'')$ is a bicomplex (X, p. 174, exercise 11), we will denote $K_{p,*}$ the complex $(\bigoplus_n K_{p,n}, d'')$. Every complex will be considered as a bicomplex, zero in degree (p, q) for $q \neq 0$.

Let C be a complex of A -modules. A projective resolution of C is a pair (P, p) , where P is a bicomplex and $p : P \rightarrow C$ a morphism of bicomplexes, such that $P_{p,q} = 0$ for $q < 0$ and such that the complex $P_{p,*}$ (resp. $Z'_{p,*}(\hat{P})$, $B'_{p,*}(\mathbf{P})$, $H'_{p,*}(\mathbf{P})$) define a projective resolution of C_p (resp. $Z_p(C)$, $B_p(C)$, $H_p(C)$) for every $p \in \mathbb{Z}$. For (P, p) be a projective resolution of C , it suffices that the complexes $B'_{p,*}(\mathbf{P})$ and $H'_{p,*}(\mathbf{P})$ define, for every p , projective resolutions of $B_p(C)$ and $H_p(C)$.

- a) Show that every A -complex C admits a projective resolution (choose, for every p , projective resolutions $B_{p,*}$ and $H_{p,*}$ of $B_p(C)$ and $H_p(C)$; using Exercise 16, construct projective resolutions $Z_{p,*}$ of $Z_p(C)$ and $P_{p,*}$ of C_p as well as exact sequences:

$$0 \rightarrow B_{p,*} \rightarrow Z_{p,*} \rightarrow H_{p,*} \rightarrow 0; \quad 0 \rightarrow Z_{p,*} \rightarrow P_{p,*} \rightarrow B_{p+1,*} \rightarrow 0).$$

- b) Let C' another complex, $(\mathbf{P}', p')$ a projective resolution of C' , $u : C \rightarrow C'$ a morphism of complexes. Show that there exists a morphism of bicomplexes $\tilde{u} : \mathbf{P} \rightarrow \mathbf{P}'$ such that $u \circ p = p' \circ \tilde{u}$ (for every $p \in \mathbb{Z}$, choose morphisms $\mathbf{B}'_{p',*}(\mathbf{P}) \rightarrow \mathbf{B}'_{p',*}(\mathbf{P}')$ and $\mathbf{H}'_{p',*}(\mathbf{P}) \rightarrow \mathbf{H}'_{p',*}(\mathbf{P}')$; construct $\tilde{u}$ as before, using Exercise 16, b)).
- c) Let $v : C \rightarrow C'$ a morphism of complexes homotopic to u , and let $\tilde{v} : \mathbf{P} \rightarrow \mathbf{P}'$ be a morphism of bicomplexes such that $v \circ p = p' \circ \tilde{v}$. Show that $\tilde{u}$ and $\tilde{v}$ are homotopic (in the sense of X, p. 174, Exercise 11, c); use Exercise 16, c)).

d) Let E be a subset of $\mathbf{Z}$, and C an A -complex such that $C_p = 0$ for $p \in E$. Show that there exists a projective resolution (P, p) of C such that $P_{p,q} = 0$ for $p \in E$.

e) Suppose there exists an integer n such that every A -module admits a projective resolution of length $\leq n$. Show that every A -complex C admits a projective resolution (P, p) such that $P_{p,q} = 0$ for $q > n$.

f) Let C be an A -complex. An injective resolution of C is a pair (e, I) , where I is a bicomplex and $e : C \rightarrow I$ a morphism of bicomplexes, such that $I^{p,q} = 0$ for $q < 0$ and such that the complex $I^{p,*}$ (resp. $Z^{p,*}(I)$, $B^{p,*}(I)$, $H^{p,*}(I)$) defines an injective resolution of C^p (resp. $Z^p(C)$, $B^p(C)$, $H^p(C)$) for every $p \in \mathbf{Z}$. State and prove the analogues of results a) to e) for injective resolutions.

18) Assume the ring A is commutative. A differential graded A -algebra is an A -algebra S , graded of type $\mathbf{Z}$, such that $S_n = 0$ for $n < 0$, equipped with an antiderivation d of degree (-1) , satisfying $d \circ d = 0$. We shall also denote by S the complex of A -modules (S, d) .

a) Let C be a complex of A -modules, zero in degrees < 0 . Show that there exists on $S = T(C)$ (resp. $S = S(C)$, resp. $S = \Lambda(C)$) a structure of differential graded A -algebra, such that the canonical injection of C into S is a morphism of complexes.

b) Show that the multiplication of S induces a structure of graded A -algebra on $H(S)$.

c) Let S, T be two differential graded A -algebras. Set $D(s \otimes t) = ds \otimes t + (-1)^p s \otimes dt$ for $s \in S_p$, $t \in T$. Show that D defines a structure of differential graded A -algebra on the left tensor product $S \otimes_A T$ (III, p. 49).

19) Assume the ring A is commutative.

a) Let (S, d_S) a differential graded A -algebra (exercise 18), M an A -module, $u : M \rightarrow Z_n(S)$ an A -homomorphism. Show that there exists on $S \otimes_A T(M)$ a structure of differential graded A -algebra such that $d(s \otimes 1) = d_S s \otimes 1$ for $s \in S$ and $d(1 \otimes m) = u(m) \otimes 1$ for $m \in M$.

b) Let B be an A -algebra. Show that there exists a differential graded A -algebra S and a

homomorphism of A -algebras $p : S_0 \rightarrow B$ such that (S, p) be a free resolution of the A -module B . (Construct by induction on n , using a), a differential graded algebra $S(n)$, free in each degree, such that $H_i(S(n)) = 0$ for $0 < i < n$ and $H_0(S(n)) = B$.)

- c) If the algebra B is commutative, show that one can find an alternating A -algebra S (III, p. 53) satisfying the conditions of b). If moreover A is noetherian and if B is a quotient of A by an ideal, show that one can find S such that $S_0 = A$ and that S_n is a finitely generated A -module for every n .

¶ 20) Assume that the ring A is an algebra over a commutative ring k . Let $\eta : k \rightarrow A$ be the homomorphism defined by $\eta(\lambda) = \lambda 1_A$ for $\lambda \in k$, and let $\bar{A}$ its cokernel. If a is an element of A , we denote by $\bar{a}$ its class in $\bar{A}$. We set $N_n(A) = A \otimes_k \bar{A}^{\otimes n} \otimes_k A$ for $n \geq 0$, $N_n(A) = 0$ for $n < 0$, and

$$N(A) = \bigoplus_{n \in \mathbb{Z}} N_n(A).$$

- a) Show that there exists an (A, A) -endomorphism d of $N(A)$, graded of degree (-1) , such that

$$d(a \otimes \bar{a}_1 \otimes \cdots \otimes \bar{a}_n \otimes b) = aa_1 \otimes \bar{a}_2 \otimes \cdots \otimes \bar{a}_n \otimes b + \sum_{i=1}^{n-1} (-1)^i a \otimes \cdots \otimes \overline{a_i a_{i+1}} \otimes \cdots \otimes \bar{a}_n \otimes b \\ + (-1)^n a \otimes \bar{a}_1 \otimes \cdots \otimes \bar{a}_{n-1} \otimes a_n b \quad \text{for } a, a_1, \dots, a_n, b \in A$$

- b) Show that $d \circ d = 0$, so that $N(A)$ is a complex of (A, A) -bimodules. If M is a left A -module, we denote by $N(A, M)$ the A -complex $N(A) \otimes_A M$.

- c) Let $B(A, M)$ be the standard resolution of M (X, p. 58), $p : B(A, M) \rightarrow N(A, M)$ the A -homomorphism defined by $p(a_0 \otimes \cdots \otimes a_n \otimes m) = a_0 \otimes \bar{a}_1 \otimes \cdots \otimes \bar{a}_n \otimes m$ for $a_0, \dots, a_n \in A, m \in M$. Show that p is a homotopy equivalence of A -complexes (construct, by induction, a map inverse up to homotopy).

* 21) Let k be a commutative ring, and let g be a k Lie algebra, U its enveloping algebra (cf. LIE, I, § 2). Assume that g is a k -free module.

a) Show that for every $n \geq 0$ an injective U -homomorphism

$$j_n : U \otimes_k \Lambda^n(\mathfrak{g}) \rightarrow U^{\otimes(n+1)} \quad \text{such that} \quad j_n(u \otimes (x_1 \wedge \cdots \wedge x_n)) = \sum_{\sigma \in \mathfrak{S}_n} \varepsilon(\sigma) u \otimes x_{\sigma(1)} \otimes \cdots \otimes x_{\sigma(n)}$$

for $u \in U, x_1, \dots, x_n \in \mathfrak{g}$

We thus define a graded U -homomorphism of degree 0. $j : U \otimes_k \Lambda(\mathfrak{g}) \rightarrow B(U, k)$, where $B(U, k)$ is the standard resolution of the left U -module k (the structure of U -module of k being deduced from the natural homomorphism $U \rightarrow k$). Show that $U \otimes_k \Lambda(\mathfrak{g})$ is identified by j with a subcomplex of $B(U, k)$, the differential being given by the formula :

$$d(u \otimes (x_1 \wedge \cdots \wedge x_n)) = \sum_{1 \leq i \leq n} (-1)^{i+1} u x_i \otimes (x_1 \wedge \cdots \wedge \hat{x}_i \wedge \cdots \wedge x_n) \\ + \sum_{1 \leq i < j \leq n} (-1)^{i+j} u \otimes ([x_i, x_j] \wedge x_1 \wedge \cdots \wedge \hat{x}_i \wedge \cdots \wedge \hat{x}_j \wedge \cdots \wedge x_n) \quad \text{pour} \quad u \in U, x_1, \dots, x_n \in \mathfrak{g}.$$

(where the sign over a letter means that it is to be omitted). We denote by $V(\mathfrak{g})$ the U -complex thus defined. b) Show that $V(\mathfrak{g})$ defines a free resolution of the U -module k (let $(U_n)_{n \geq 0}$ be the natural filtration of U , $F_r V(\mathfrak{g})$ the sub- k -module of $V(\mathfrak{g})$ generated by the elements $u_p \otimes x_q$ for $u_p \in U_p, x_q \in \Lambda^q(\mathfrak{g}), p + q \leq r$. Show that $F_r V(\mathfrak{g})$ is a subcomplex of $V(\mathfrak{g})$, and that the complex $\bigoplus_{r \geq 0} F_r V(\mathfrak{g}) / F_{r-1} V(\mathfrak{g})$

is isomorphic to the complex $S(\mathfrak{g}) \otimes \Lambda(\mathfrak{g})$ defined in X, p. 151). *

Exercises for Section 4: Torsion Products

- 1) We denote by B the ring $A(\varepsilon)$ (X, p. 27); let C be the complex of (B, B) -bimodules such that $C_n = B$ for all $n \in \mathbb{Z}$, $d(x) = \varepsilon x$ for all $x \in C$. Show that the complex C has zero homology, but that $H_n(C \otimes_B C)$ is isomorphic to A for all $n \in \mathbb{Z}$. Deduce that one cannot remove the hypothesis “ E bounded above” in lemma 1 and prop. 4, pp. 66-67.
- 2) Let $\rho : A \rightarrow B$ be a ring homomorphism, M an A -module. Show that the following properties are equivalent:

α) One has $\text{Tor}_1^A(\rho_*(\mathbf{P}), \mathbf{M}) = 0$ for every right B-module P.

β) One has $\text{Tor}_1^A(\rho_*(\mathbf{P}), \mathbf{M}) = 0$ for every monogenic right B-module P.

γ) The B-module $\mathbf{B} \otimes_A \mathbf{M}$ is flat, and one has $\text{Tor}_1^A(\mathbf{B}, \mathbf{M}) = 0$.

(To see that γ) implies α), consider an exact sequence $0 \rightarrow R \rightarrow L \rightarrow P \rightarrow 0$ of right B-modules, where L is free.)

3) Assume the ring A is commutative. Let I be a finite set and $(\mathbf{C}^{(i)}, d^{(i)})_{i \in I}$ a family of complexes of A-modules.

a) We equip the tensor product $\otimes \mathbf{C}^{(i)}$ with its graduation of type $\mathbf{Z}^I$ and with the A-endomorphisms

$$d_i = \bigotimes_{j \in I} d_i^{(j)} \quad \text{with} \quad d_i^{(j)} = 1_{\mathbf{C}^{(j)}} \quad \text{for} \quad j \neq i \quad \text{and} \quad d_i^{(i)} = d^{(i)}.$$

Show that one thus obtains an I-precomplex (X, exercise 13, p. 174); the associated I-complex is called the tensor product I-complex of the complexes $\mathbf{C}^{(i)}$. The complex associated with this I-complex is called the tensor product complex of the $\mathbf{C}^{(i)}$.

b) Show that the choice of a total order on I allows one to define an isomorphism from the tensor product complex of the $\mathbf{C}^{(i)}$ to the complex defined in X, p. 64.

4) Let $\mathfrak{a}$ be a two-sided ideal of A. Let M be an A-module, such that $\text{Tor}_1^A(A/\mathfrak{a}, \mathbf{M}) = 0$ and that the $(A/\mathfrak{a})$ -module $\mathbf{M}/\mathfrak{a}\mathbf{M}$ is projective.

a) Assume either that M is finitely presented and $\mathfrak{a}$ is contained in the radical of A, or that $\mathfrak{a}$ is nilpotent. Show that M is projective (use VIII, § 8, n° 3, cor. 3).

b) Assume that M is finitely generated, that one has $\bigcap \mathfrak{a}^n = 0$ and that the $(A/\mathfrak{a})$ -module $\mathbf{M}/\mathfrak{a}\mathbf{M}$ is free. Show

that the A-module M is free.

c) Give an example of a local ring A, of maximal ideal $\mathfrak{m}$, and of a finitely generated A-module M which is not flat such that $\text{Tor}_1^A(A/\mathfrak{m}, M) = 0$.

5) Assume there exists a (unitary) homomorphism of k -algebras $\pi : \mathbf{A} \rightarrow k$, which endows k with a structure of $\mathbf{A}$ -module. We consider the standard resolution $\mathbf{B}(\mathbf{A}, k)$ (X, p. 58); for $n \geq 0$, we identify $\mathbf{B}_n(\mathbf{A}, k)$ with $\mathbf{A} \otimes_k \mathbf{A}^{\otimes n}$ and we denote by $a[a_1, \dots, a_n]$ the element $a \otimes a_1 \otimes \dots \otimes a_n$ of $\mathbf{A} \otimes_k \mathbf{A}^{\otimes n}$.

a) Show that the complex $\mathbf{B}(\mathbf{A}, k) \otimes_k \mathbf{B}(\mathbf{A}, k)$ defines a resolution of the $(\mathbf{A} \otimes_k \mathbf{A})$ -module k . Show that the homomorphism $(\mathbf{A} \otimes_k \mathbf{A})$ -linear

$$g : \mathbf{B}(\mathbf{A} \otimes_k \mathbf{A}, k) \rightarrow \mathbf{B}(\mathbf{A}, k) \otimes_k \mathbf{B}(\mathbf{A}, k)$$

such that

$$g([x_1 \otimes y_1, \dots, x_n \otimes y_n]) = \sum_{0 \leq p \leq n} \pi(x_{p+1} \dots x_n) [x_1, \dots, x_p] \otimes y_1 \dots y_p [y_{p+1}, \dots, y_n]$$

for $x_1, \dots, x_n, y_1, \dots, y_n$ into $\mathbf{A}$, is a morphism of resolutions.

b) Assume $\mathbf{A}$ is commutative. Show that there exists on $\mathbf{B}(\mathbf{A}, k)$ a structure of graded $\mathbf{A}$ -algebra such that one has for $x_1, \dots, x_n \in \mathbf{A}$:

$$[x_1, \dots, x_p] [x_{p+1}, \dots, x_n] = \sum_{\sigma \in M_{p,n}} \varepsilon(\sigma) [x_{\sigma(1)}, \dots, x_{\sigma(n)}]$$

where $M_{p,n}$ is the subset of S_n formed by the permutations σ such that $\sigma(1) < \dots < \sigma(p)$ and $\sigma(p+1) < \dots < \sigma(n)$.

Show that $\mathbf{B}(\mathbf{A}, k)$ is then a differential graded $\mathbf{A}$ -algebra (X, p. 183, exercise 18).

6) Let P be a flat complex of right $\mathbf{A}$ -modules. For every $\mathbf{A}$ -module M and every integer i , we set

$$T_i^{\mathbb{P}}(M) = H_i(P \otimes_{\mathbf{A}} M);$$

if $u : M \rightarrow N$ is an $\mathbf{A}$ -homomorphism, we denote by $T_i^{\mathbb{P}}(u)$ the k -homomorphism $H_i(1_{\mathbb{P}} \otimes u)$

a) Let r be an integer. Show that the following conditions are equivalent:

α) For every injection $u : \mathbf{M}' \rightarrow \mathbf{M}$, the map $T_r^{\mathbf{P}}(u)$ is injective.

β) For every surjection $v : \mathbf{M} \rightarrow \mathbf{M}''$, the map $T_{r+1}^{\mathbf{P}}(v)$ is surjective.

γ) The right A -module $\mathbf{P}_r/\mathbf{B}_r(\mathbf{P})$ is flat.

δ) There exists a flat complex $\mathbf{P}'$ of right A -modules, whose differential d_{r+1} is zero, and for every A -module M and every integer i a k -isomorphism $\varphi_i^M : T_i^{\mathbf{P}}(M) \rightarrow T_i^{\mathbf{P}'}(M)$ such that one has

$$T_i^{\mathbf{P}'}(u) \circ \varphi_i^M = \varphi_i^N \circ T_i^{\mathbf{P}}(u)$$

for every A -homomorphism $u : M \rightarrow N$

(To prove that γ) implies δ), define $\mathbf{P}'$ by setting $\mathbf{P}'_{r+1} = \mathbf{Z}_{r+1}(\mathbf{P})$ and $\mathbf{P}'_r = \mathbf{P}_r/\mathbf{B}_r(\mathbf{P})$.)

b) We also assume that A is Noetherian and that the A -modules P_i are finitely generated.

Show that the conditions in a) are equivalent to the following condition:

ε) There exists an A -module Q and, for every A -module M , a k -isomorphism $\psi^M : T_r^{\mathbf{P}}(M) \rightarrow \text{Hom}_A(Q, M)$ such that $\psi^N \circ T_r^{\mathbf{P}}(u) = \text{Hom}(1, u) \circ \psi^M$ for every A -homomorphism $u : M \rightarrow N$.

(To prove that δ) implies ε), reduce to the case where $d_{r+1} = 0$ and take for Q the dual of $\mathbf{Z}_r(\mathbf{P})$.) c) Show that the result of b) remains valid under the hypothesis that A is Noetherian, $H_i(\mathbf{P})$ finitely generated for every i and $H_i(\mathbf{P}) = 0$ for $i < i_0$ (using prop. 10, p. 56).

7) Let C be a complex of right A -modules and M a left A -module. The torsion product of C and M is the graded k -module $\mathcal{T}or^A(C, M) = H(C \otimes_A L(M))$. If $f : C \rightarrow C'$ is a morphism of complexes and $g : M \rightarrow M'$ a homomorphism of A -modules, we write $\mathcal{T}or^A(f, g) = H(f \otimes L(g))$. a) If $f : C \rightarrow C'$ is a homomorphism, $\mathcal{T}or^A(f, 1)$ is an isomorphism; if C is flat and bounded on the right, the graded k -module $\mathcal{T}or^A(C, M)$ is isomorphic to $H(C \otimes_A M)$.

b) If $0 \rightarrow C' \rightarrow C \rightarrow C'' \rightarrow 0$ is an exact sequence of complexes of right A -modules and if

M is an A-module, show that there exists an exact sequence

$$\cdots \rightarrow \mathcal{T}or_n^A(C', M) \rightarrow \mathcal{T}or_n^A(C, M) \rightarrow \mathcal{T}or_n^A(C'', M) \rightarrow \mathcal{T}or_{n-1}^A(C', M) \rightarrow \cdots$$

If $0 \rightarrow M' \xrightarrow{i} M \xrightarrow{p} M'' \rightarrow 0$ is an exact sequence of A-modules, show that one has an exact sequence:

$$\cdots \rightarrow \mathcal{T}or_n^A(C, M') \rightarrow \mathcal{T}or_n^A(C, M) \rightarrow \mathcal{T}or_n^A(C, M'') \rightarrow \mathcal{T}or_{n-1}^A(C, M') \rightarrow \cdots$$

(Note that there exists a homotopism from $L(M'')$ to the cone of $L(i)$).

c) Show that there exists a spectral sequence 1E (X, pp. 175–177, exercises 14–17) converging to $\mathcal{T}or^A(C, M)$, such that $E_{pq}^2 = \text{Tor}_p^A(H_q(C), M)$. If moreover the complex C is bounded on the right, show that there exists a spectral sequence E converging to $\mathcal{T}or^A(C, M)$, such that $E_{pq}^2 = H_p(\text{Tor}_q^A(C, M))$ (consider the double complex $C \otimes_A L(M)$).

8) Let A, B be two k-algebras (associative and unital), M a right A-module, P a left B-module, N an (A, B)-bimodule.

a) Show that there exist two spectral sequences of k-modules 1E and 2E , converging to the same k-graded module, such that

$${}^1E_{pq}^2 = \text{Tor}_p^A(M, \text{Tor}_q^B(N, P)); \quad {}^2E_{pq}^2 = \text{Tor}_p^B(\text{Tor}_q^A(M, N), P).$$

(Consider the double complex $L(M) \otimes_A (N \otimes_B L(P))$.)

b) If N is a flat B-module, show that the spectral sequence 2E converges to $\text{Tor}^A(M, N \otimes_B P)$. If N is a flat A-module, the sequence 1E converges to $\text{Tor}^B(M \otimes_A N, P)$.

c) Let $\rho : \mathbf{A} \rightarrow \mathbf{B}$ a homomorphism of k-algebras. Deduce from b) a spectral sequence E

such that

$$E_{pq}^2 = \text{Tor}_p^B \left(\text{Tor}_q^A(\mathbf{M}, \mathbf{B}), \mathbf{P} \right),$$

converging to $\text{Tor}^A(\mathbf{M}, \mathbf{P})$.

- 9) Let $\mathbf{P}$ be a complex of right A -modules, $\mathbf{Q}$ a complex of left A -modules. Suppose either that $\mathbf{P}$ and $\mathbf{Q}$ are bounded below, or that there exists an integer d such that every left or right A -module admits a projective resolution of length $\leq d$.
- a) Show that there exist two spectral sequences $(X, p, 175, \text{exercise } 14)$ $'E$ and $''E$, converging to the same graded k -module, such that

$$'E_{pq}^2 = H_p(\text{Tor}_q^A(\mathbf{P}, \mathbf{Q})); \quad ''E_{pq}^2 = \bigoplus_{q'+q''=q} \text{Tor}_p^A(H_{q'}(\mathbf{P}), H_{q''}(\mathbf{Q}))$$

(where $\text{Tor}_q^A(\mathbf{P}, \mathbf{Q})$ denotes the complex associated with the bicomplex $\bigoplus_r \text{Tor}_q^A(\mathbf{P}_r, \mathbf{Q}_s)$)

(Consider projective resolutions $(X, p, 183, \text{exercise } 17)$ $\mathcal{P}$ and $\mathcal{Q}$ of $\mathbf{P}$ and $\mathbf{Q}$, and the double complex $\mathbf{C}$ such that $C_{p,q} = \bigoplus (\mathcal{P}_{p'q'} \otimes \mathcal{Q}_{p''q''})$.)

- b) If $\mathbf{P}$ is flat, show that the spectral sequence $''E$ converges to $H(\mathbf{P} \otimes_A \mathbf{Q})$. If $H(\mathbf{P})$ is flat, the sequence $'E$ converges to $H(\mathbf{P}) \otimes_A H(\mathbf{Q})$.

Exercises for Section 5: Extension Modules

- ¶ 1) An A -complex $\mathbf{C}$ is said to be pure if for every right A -module $\mathbf{P}$ one has $H(\mathbf{P} \otimes_A \mathbf{C}) = 0$.
- a) A complex homotopic to zero is pure; a complex of flat modules, with zero homology and bounded on the right, is pure. Give an example of a complex of free modules of rank 1, with zero homology but not pure.
- b) Show that the following conditions are equivalent:
- α) $\mathbf{C}$ is pure.
- β) For every finitely presented A -module $\mathbf{M}$, one has $H(\text{Hom}_{gr-A}(\mathbf{M}, \mathbf{C})) = 0$.

γ) There exists a filtered inductive system $(C^{(\alpha)})_{\alpha \in J}$ of complexes homotopic to zero such that $C = \varinjlim C^{(\alpha)}$.

(To see that β) implies γ), reduce to the case where $C_i = 0$ for $i < i_0$ and write C_{i_0} as an inductive limit of finitely presented modules.)

2) Let M be an A -module. We say that a submodule M' of M is pure if the complex defined by the exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M/M' \rightarrow 0$ is pure (Exercise 1).

a) Show that when A is a principal ideal ring, the notion of pure submodule coincides with that of VII, § 2, Exercise 7.

b) Suppose that the A -module M is flat. Show that the following conditions are equivalent:
 α the submodule M' is pure;

β) the quotient M/M' is flat;

γ) for every ideal $\mathfrak{a}$ of A , we have $M' \cap \mathfrak{a}M = \mathfrak{a}M'$.

3) Let M be an A -module, n an integer ≥ 1 . Show that the following conditions are equivalent:

α) M admits a finite n -presentation (X, p. 180, Exercise 6).

β) For every filtered inductive system $(N_\alpha)_{\alpha \in I}$, the canonical homomorphism

$$\varinjlim \text{Ext}_A^i(M, N_\alpha) \rightarrow \text{Ext}_A^i(M, \varinjlim N_\alpha)$$

is bijective for $i < n$.

γ) For every family $(P_\beta)_{\beta \in J}$ of right A -modules, the canonical homomorphism

$$\text{Tor}_i^A \left(\prod_{\beta \in J} P_\beta, M \right) \rightarrow \prod_{\beta \in J} \text{Tor}_i^A(P_\beta, M) \quad \text{is bijective for } i < n.$$

δ) For every set I , the canonical homomorphism $A_d^I \otimes_A M \rightarrow M^I$ is bijective, and we have

$$\text{Tor}_i^A(A_d^I, M) = 0 \quad \text{for } 0 < i < n.$$

(Argue by induction on n , using Exercise 18, p. 170.)

- 4) Let P be a complex of projective A -modules, zero on the right, and n an integer such that $H_i(P) = 0$ for $i < n$. Let M be a left A -module, N a right A -module. Show that we have $H^i(\text{Homgr}_A(P, M)) = 0$ and $H_i(N \otimes_A P) = 0$ for $i < n$, and that the canonical homomorphisms

$$\lambda^n : H^n(\text{Homgr}_A(P, M)) \rightarrow \text{Hom}_A(H_n(P), M), \quad \gamma^n : N \otimes_A H_n(P) \rightarrow H_n(N \otimes_A P)$$

are bijective.

¶ 5) Let P be a projective A -complex, zero on the right, n an integer ≥ 0 . Show that the following conditions are equivalent:

α) There exists a complex P' of finitely generated projective A -modules, such that $P'_i = 0$ for $i > n$ or $i < 0$, and a morphism $f : P' \rightarrow P$ such that $H_i(f)$ is bijective for $i < n$ and surjective for $i = n$.

β) For every filtered inductive system of A -modules $(M_\alpha)_{\alpha \in I}$, the canonical homomorphism

$$\varinjlim H^i(\text{Homgr}_A(P, M_\alpha)) \rightarrow H^i(\text{Homgr}_A(P, \varinjlim M_\alpha))$$

is bijective for $i < n$ and injective for $i = n$.

γ) For any family of right A -modules $(N_\alpha)_{\alpha \in I}$, the canonical homomorphism

$$H_i((\prod_\alpha N_\alpha) \otimes_A P) \rightarrow \prod_\alpha H_i(N_\alpha \otimes P)$$

is bijective for $i < n$ and surjective for $i = n$.

(To prove that α) implies β) and γ), apply Exercise 4 to the cone of f . To prove that β) or γ) implies α), reason by induction on n , applying Exercise 4, Exercise 18, p. 170 and Exercise 10, p. 174.)

- 6) Let A, B be two k -algebras (associative and unital), M a left A -module, P a left B -module, N a (B, A) -bimodule.

a) Show that there exist two spectral sequences $(X, \text{pp. 175-177, exercises 14 to 17})$ of k

-modules 1E and 2E , converging to the same k -graded module, such that

$${}^1E_2^{pq} = \text{Ext}_A^p(\mathbf{M}, \text{Ext}_B^q(\mathbf{N}, \mathbf{P})); \quad {}^2E_2^{pq} = \text{Ext}_B^p(\text{Tor}_q^A(\mathbf{N}, \mathbf{M}), \mathbf{P}).$$

- b) By composing the homomorphisms γ and δ of Exercise 15, p. 176, define a k -homomorphism

$$\mu : \text{Ext}_A(\mathbf{M}, \text{Hom}_B(\mathbf{N}, \mathbf{P})) \rightarrow \text{Hom}_B(\text{Tor}^A(\mathbf{N}, \mathbf{M}), \mathbf{P}).$$

If $\mathbf{P}$ is injective, prove that μ is an isomorphism.

- c) If $\mathbf{N}$ is a flat A -module, show that the sequence 1E converges to $\text{Ext}_B(N \otimes_A M, P)$; if $\mathbf{N}$ is a projective B -module, the sequence 2E converges to $\text{Ext}_A(M, \text{Hom}_B(N, P))$.
- d) Let $\rho : A \rightarrow B$ a homomorphism of k -algebras. Show that there exists a spectral sequence E , such that $E_2^{pq} = \text{Ext}_B^p(\text{Tor}_q^A(B, M), P)$, converging to $\text{Ext}_A(M, P)$ and a spectral sequence $'E$, such that

$${}^{'E}_2^{pq} = \text{Ext}_B^p(\mathbf{P}, \text{Ext}_A^q(\mathbf{B}, \mathbf{M})),$$

converging to $\text{Ext}_A(\mathbf{P}, \mathbf{M})$.

- 7) Let C be an A -complex and M an A -module. One calls the module of extensions of M by C (resp. of C by M) the graded k -module

$$\mathcal{E}xt_A(C, M) = H(\text{Homgr}_A(C, I(M))) \quad (\text{resp. } \mathcal{E}xt_A(M, C) = H(\text{Homgr}_A(L(M), C))).$$

If $f : C \rightarrow C'$ is a morphism of complexes and $g : M \rightarrow M'$ a homomorphism of A -modules, we set $\mathcal{E}xt_A(f, g) = H(\text{Homgr}(f, l(g)))$, $\mathcal{E}xt_A(g, f) = H(\text{Homgr}(L(g), f))$.

- a) If $f : C \rightarrow C'$ is a homomorphism, the k -homomorphisms $\mathcal{E}xt_A(f, 1)$ and $\mathcal{E}xt_A(1, f)$ are bijective. If C is projective and bounded on the right (resp. injective and bounded

on the left), the k -graded module $\mathcal{E}xt_A(C, M)$ (resp. $\mathcal{E}xt_A(M, C)$) is isomorphic to $H(\text{Homgr}_A(C, M))$ (resp. $H(\text{Homgr}_A(M, C))$).

- b) Let $0 \rightarrow C' \rightarrow C \rightarrow C'' \rightarrow 0$ be an exact sequence of A -complexes and $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ an exact sequence of A -modules. Show that there exist exact sequences

$$\begin{aligned} \dots &\rightarrow \mathcal{E}xt_A^n(C'', M) \rightarrow \mathcal{E}xt_A^n(C, M) \rightarrow \mathcal{E}xt_A^n(C', M) \rightarrow \mathcal{E}xt_A^{n+1}(C'', M) \rightarrow \dots \\ \dots &\rightarrow \mathcal{E}xt_A^n(C, M') \rightarrow \mathcal{E}xt_A^n(C, M) \rightarrow \mathcal{E}xt_A^n(C, M'') \rightarrow \mathcal{E}xt_A^{n+1}(C, M') \rightarrow \dots \\ \dots &\rightarrow \mathcal{E}xt_A^n(M, C') \rightarrow \mathcal{E}xt_A^n(M, C) \rightarrow \mathcal{E}xt_A^n(M, C'') \rightarrow \mathcal{E}xt_A^{n+1}(M, C') \rightarrow \dots \\ \dots &\rightarrow \mathcal{E}xt_A^n(M'', C) \rightarrow \mathcal{E}xt_A^n(M, C) \rightarrow \mathcal{E}xt_A^n(M', C) \rightarrow \mathcal{E}xt_A^{n+1}(M'', C) \rightarrow \dots \end{aligned}$$

- c) Show that there exists a spectral sequence “E (X, pp. 175-177, exercises 14 to 17) converging to $\mathcal{E}xt_A(C, M)$ such that $E_2^{pq} = H^q(\text{Ext}_A^p(C, M))$. If C is bounded above, show that there exists a spectral sequence 'E, converging to $\mathcal{E}xt_A(C, M)$, such that $E_2^{pq} = \text{Ext}_A^p(H_q(C), M)$ (consider the bicomplex $\text{Homgr}_A(C, I(M))$).
- 8) Let P, Q be two A -complexes. Suppose either that P is bounded below and Q bounded above, or that there exists an integer d such that every A -module admits a projective resolution of length $\leq d$.

- a) Show that there exist two spectral sequences 'E and “E, converging to the same graded k -module, such that

$${}^{\prime}E_2^{pq} = H^p(\text{Ext}_A^q(P, Q)) \quad {}^{\prime\prime}E_2^{pq} = \bigoplus_{q'+q''=q} \text{Ext}_A^p(H_{q'}(P), H^{q''}(Q))$$

(ou $\text{Ext}_A^q(P, Q)$ denotes the complex associated with the bicomplex $\oplus \text{Ext}_A^q(P_r, Q^s)$).

- b) If P is projective or Q is injective, the spectral sequence “E converges to $H(\text{Homgr}_A(P, Q))$. If $H(P)$ is projective or $H(Q)$ is injective, the spectral sequence 'E converges to $\text{Homgr}_A(H(P), H(Q))$.

Exercises for Section 6: Use of Non-Canonical Resolutions

- 1) Let $\rho : A \rightarrow B$ be a ring homomorphism, M, N two left B -modules, P a right B -module.
- a) Let L_B (resp. L_A) be a projective resolution of the B -module (resp. of the A -module) M . Show that there exists an A -linear morphism of resolutions $L_A \rightarrow L_B$, unique up to homotopy; deduce graded k -homomorphisms of degree zero

$$\alpha : \text{Tor}^A(P, M) \rightarrow \text{Tor}^B(P, M) \quad \beta : \text{Ext}_B(M, N) \rightarrow \text{Ext}_A(M, N).$$

- b) Consider the canonical homomorphisms (X, p. 109)

$$\begin{aligned} \lambda : \text{Tor}^A(P, M) &\rightarrow \text{Tor}^B(P \otimes_A B, M) & \mu : \text{Tor}^A(P, M) &\rightarrow \text{Tor}^B(P, B \otimes_A M) \\ \nu : \text{Ext}_B(B \otimes_A M, N) &\rightarrow \text{Ext}_A(M, N) & \rho : \text{Ext}_B(M, \text{Hom}_A(B, N)) &\rightarrow \text{Ext}_A(M, N). \end{aligned}$$

Let $m : B \otimes_A M \rightarrow M$, $p : P \otimes_A B \rightarrow B$ and $n : N \rightarrow \text{Hom}_A(B, N)$ the B -homomorphisms defined by $m(b \otimes x) = bx$, $p(y \otimes b) = yb$ and $n(z)(b) = bz$ for $b \in B$, $x \in M$, $y \in P$, $z \in N$. Prove the equalities

$$\alpha = \text{Tor}(1_P, m) \circ \mu = \text{Tor}(p, 1_M) \circ \lambda$$

$$\beta = \nu \circ \text{Ext}(m, 1_N) = \rho \circ \text{Ext}(1_M, n).$$

- 2) Let A, B be two k -algebras (associative and unital); let M be a left A -module, P a right B -module, N an (A, B) -bimodule. Consider the homomorphism

$$\sigma : \text{Hom}_B(N, P) \otimes_A M \rightarrow \text{Hom}_B(\text{Hom}_A(M, N), P)$$

such that $\sigma(u \otimes m)(v) = u \circ v(m)$ for $u \in \text{Hom}_B(N, P)$, $v \in \text{Hom}_A(M, N)$, $m \in M$ (II, p. 190, exercise 6). a) By replacing M by a projective resolution, define using σ a k -homomorphism

$$\tau : \text{Tor}^A(\text{Hom}_B(N, P), M) \rightarrow \text{Homgr}_B(\text{Ext}_A(M, N), P).$$

b) Assume henceforth that A is left noetherian and that the A -module M is finitely generated. Show that if the B -module P is injective, τ is an isomorphism.

c) Show that there exist two spectral sequences E and $'E$ (X, pp. 175-177, exercises 14 to 17), converging to the same graded k -module, such that

$$E_2^{pq} = \text{Tor}_{-p}^A(\text{Ext}_B^q(N, P), M) \quad 'E_2^{pq} = \text{Ext}_B^p(\text{Ext}_A^{-q}(M, N), P).$$

3) Let I be an injective k -module; for every A -module M , denote by $D(M)$ the right A -module $\text{Hom}_k(M, I)$. Let M and N be two left A -modules, P a right A -module. Prove that for every $p \geq 0$, the k -modules $\text{Ext}_A^p(P, D(M))$ and $D(\text{Tor}_p^A(P, M))$ are isomorphic; if A is left noetherian and if M is a finitely generated A -module, the k -modules $\text{Tor}_p^A(D(N), M)$ and $D(\text{Ext}_A^p(M, N))$ are isomorphic (use exercise 2, b) and exercise 6, b), p. 188).

4) For every group G , denote by $\varepsilon_G : \mathbf{Z}^{(G)} \rightarrow \mathbf{Z}$ the ring homomorphism such that $\varepsilon_G(e_g) = 1$ for all $g \in G$; set $I_G = \text{Ker}(\varepsilon_G)$.

a) Show that the group $\mathbf{H}_1(G, \mathbf{Z})$ is isomorphic to I_G/I_G^2 .

b) Let $(g_i)_{i \in I}$ be a generating family of G ; show that the elements $(e_{g_i} - 1)$, for $i \in I$, generate the ideal I_G .

c) Suppose that the family $(g_i)_{i \in I}$ is basic (I, p. 86). Show that $(e_{g_i} - 1)_{i \in I}$ is a basis of the left $\mathbf{Z}^{(G)}$ -module I_G .

5) Let X be a set, $G = F(X)$ the free group constructed on X , M a $\mathbf{Z}^{(G)}$ -module. Show that the groups $H^q(G, M)$ and $H_q(G, M)$ are zero for $q \geq 2$ (use exercise 4).

6) Let G be a finite group of order n , M a $\mathbf{Z}^{(G)}$ -module. Show that for every $q \geq 1$, the $\mathbf{Z}$ -modules $H^q(G, M)$ and $H_q(G, M)$ are annihilated by n (consider the homotopy h of $C(G, M)$ into itself defined by

$$hf(g_1, \dots, g_q) = (-1)^q \sum_{g \in G} f(g_1, \dots, g_q, g)$$

for $f \in C^{q+1}(G, M)$ and $g, g_1, \dots, g_q \in G$. If M is a flat Z finitely generated module, the groups $H^q(G, M)$ and $H_q(G, M)$ are finite for $q \geq 1$.

- 7) Let G be a cyclic group of order n , M a $Z^{(G)}$ -module, σ a generator of G . Denote by N the multiplication in M by the element $\sum_{g \in G} e_g$ of $Z^{(G)}$. Show that the Z -module $H^0(G, M)$ (resp. $H_0(G, M)$) is iso-

morphic to $\text{Ker}(1 - \sigma_M)$ (resp. $\text{Coker}(1 - \sigma_M)$); for q odd, the groups $H^q(G, M)$ and $H_{q+1}(G, M)$ are isomorphic to $\text{Ker } N / \text{Im}(1 - \sigma_M)$; for q even ≥ 2 , $H^q(G, M)$ and $H_{q-1}(G, M)$ are isomorphic to $\text{Ker}(1 - \sigma_M) / \text{Im } N$. (Use exercise 1, p. 178.)

- 8) Let F be a free group, R a normal subgroup of F , and $G = F/R$. Show that the group $H_2(G, Z)$ is isomorphic to $((F, F) \cap R) / (F, R)$ (use Exercise 3, p. 179).

- 9) a) Let G be a group, M a G -module, and g an element of the center of G such that the endomorphism $m \mapsto gm - m$ of M is bijective. Show that the groups $H^p(G, M)$ and $H_p(G, M)$ are zero for $p \geq 1$. b) Let V be a vector space over a field k , with $\text{Card}(k) \geq 3$. Prove that the groups $H^p(\text{GL}(V), V)$ and $H_p(\text{GL}(V), V)$ are zero for $p \geq 1$.

- 10) Let G and H be two groups, $u : H \rightarrow G$ a homomorphism, M a $Z^{(G)}$ -module. For every $q \geq 0$, we deduce from the homomorphisms α and β of Exercise 1 homomorphisms

$$u_q^M : H_q(H, M) \rightarrow H_q(G, M) \quad u_M^q : H^q(G, M) \rightarrow H^q(H, M)$$

which we simply denote by u_q and u^q if there is no ambiguity about M .

- a) For $f \in C^q(G, M)$, we denote by f_u the element of $C^q(H, M)$ such that $f_u(h_1, \dots, h_q) = f(u(h_1), \dots, u(h_q))$ for $h_1, \dots, h_q \in H$. Show that the map $f \mapsto f_u$ is a morphism of complexes from $C'(G, M)$ to $C'(H, M)$, which induces on homology the homomorphism $\bigoplus_q u^q$. Define similarly a morphism from $C'(H, M)$

into $C'(G, M)$ which induces $(\bigoplus_q u_q)$.

b) Assume $H = G$ and take for u an inner automorphism of G . Show that the homomorphisms u_q and u^q are equal to the identity (argue by induction on q).

11) Let G be a group, H a distinguished subgroup of G , M a G -module. By Exercise 10, the group G acts by conjugation on the $\mathbb{Z}$ -modules $H^q(H, M)$; since the action of the subgroup H is trivial, these are therefore endowed with a structure of $\mathbb{Z}^{(G/H)}$ -modules.

a) Show that there exists a spectral sequence E (X, p. 175, Exercise 14), converging to $\bigoplus_n H^n(G, M)$,

such that $E_2^{pq} = H^p(G/H, H^q(H, M))$ (consider the spectral sequence of X, p. 188, Exercise 6, d), taking into account the canonical isomorphism (12), p. 109).

b) Show that there exists an exact sequence

$$0 \rightarrow H^1(G/H, H^0(H, M)) \rightarrow H^1(G, M) \rightarrow H^0(G/H, H^1(H, M)) \rightarrow H^2(G/H, H^0(H, M)) \rightarrow H^2(G, M)$$

Describe the homomorphisms of this sequence using the description of $H^i(G, M)$ in terms of $C^i(G, M)$. c) Define a spectral sequence analogous to that of a) for homology.

12) Let G be a group. One says that a $\mathbb{Z}^{(G)}$ -module M is co-induced (resp. induced) if there exists a $\mathbb{Z}$ -module A such that M is isomorphic to $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}^{(G)}, A)$ (resp. $\mathbb{Z}^{(G)} \otimes_{\mathbb{Z}} A$).

a) Every $\mathbb{Z}^{(G)}$ -module is isomorphic to a submodule of a co-induced module and to a quotient of an induced module.

b) Let M be a $\mathbb{Z}^{(G)}$ -module. Show that the following conditions are equivalent:

α) M is a direct summand of a co-induced module.

β) M is injective relative to $\mathbb{Z}$ (exercise 19, p. 170).

γ) There exists a G -average on M (I, p. 136, exercise 8), that is, a $\mathbb{Z}^{(G)}$ -homomorphism

$m : C^1(G, M) \rightarrow M$ (the G -module structure on $C^1(G, M)$ being defined by $(gf)(x) = gf(g^{-1}x)$ for $g \in G, x \in M$) such that if f_x is the constant function with value $x \in M$, one has $m(f_x) = x$.

c) Show that the following conditions are equivalent:

α) M is a direct summand of an induced module.

β) M is projective relative to Z (exercise 19, p. 170).

γ) There exists a Z -endomorphism ρ of M such that for every $x \in M$, the family $(\rho(g^{-1}x))_{g \in G}$ has finite support and one has $x = \sum g\rho(g^{-1}x)$.

d) Let M be an injective (resp. projective) $Z^{(G)}$ -module relative to Z . Show that $H^q(G, M) = 0$ (resp. $H_q(G, M) = 0$) for every $q \geq 1$.

e) If G is finite, show that the notions of induced and co-induced module (resp. projective and injective relative to Z) coincide.

13) Let G be a group, H a subgroup of G , M a left $Z^{(H)}$ -module, and N a right $Z^{(H)}$ -module. We endow the group $\hat{M} = \text{Hom}_{Z^{(H)}}(Z^{(G)}, M)$ (resp. $\check{N} = N \otimes_{Z^{(H)}} Z^{(G)}$) with the structure of $Z^{(G)}$ -module on the left (resp. on the right) deduced from the structure of $Z^{(G)}$ -module on the right of $Z^{(H)}$.

Let $m : \hat{M} \rightarrow M$ and $n : N \rightarrow \check{N}$ be the $Z^{(H)}$ -homomorphisms such that $m(u) = u(1)$ for $u \in \hat{M}$ and

$$n(x) = x \otimes 1 \quad \text{for } x \in N,$$

and let $i : H \rightarrow G$ be the canonical injection. Prove that the composed homomorphisms

$$\hat{i}^q : H^q(G, \hat{M}) \xrightarrow{i^q} H^q(H, \hat{M}) \xrightarrow{H^q(H, m)} H^q(H, M)$$

and

$$\check{i}_q : H_q(H, N) \xrightarrow{H_q(H, n)} H_q(H, \check{N}) \xrightarrow{i_q} H_q(G, \check{N})$$

are isomorphisms for all $q \geq 0$.

14) Let G be a group, and let H be a subgroup of G of finite index. We set $n = (G : H)$.

a) Let Q, M be two $Z^{(G)}$ -modules. Show that there exists a Z -homomorphism

$$t : \text{Hom}_{\mathbf{Z}^{(H)}}(Q, M) \rightarrow \text{Hom}_{\mathbf{Z}^{(G)}}(Q, M)$$

such that, for every system of representatives, $g_1, \dots, g_n$ of the classes of G modulo H ,

$$(tu)(q) = \sum_{i=1}^n g_i u(g_i^{-1} q) \quad \text{for } q \in Q, \quad u \in \text{Hom}_{\mathbf{Z}^{(H)}}(Q, M).$$

If $j : \text{Hom}_{\mathbf{Z}^{(G)}}(Q, M) \rightarrow \text{Hom}_{\mathbf{Z}^{(H)}}(Q, M)$ is the canonical injection, we have $t \circ j = n \cdot \text{Id}$.

- b) Let N be a right $\mathbf{Z}^{(G)}$ -module. With the notation of a), show that there exists a $\mathbf{Z}$ -homomorphism $\theta : N \otimes_{\mathbf{Z}^{(G)}} Q \rightarrow N \otimes_{\mathbf{Z}^{(H)}} Q$ such that

$$\theta(n \otimes q) = \sum_{i=1}^n n g_i^{-1} \otimes g_i q \quad \text{for } n \in N, \quad q \in Q.$$

If $p : N \otimes_{\mathbf{Z}^{(H)}} Q \rightarrow N \otimes_{\mathbf{Z}^{(G)}} Q$ is the canonical map, we have $p \circ \theta = n \cdot \text{Id}$.

- c) Let P be a projective resolution of the $\mathbf{Z}^{(G)}$ -module $\mathbf{Z}$, q an integer. By applying a and b with $Q = P_q$, define homomorphisms $t^q : H^q(H, M) \rightarrow H^q(G, M)$ and $\theta_q : H_q(G, N) \rightarrow H_q(H, N)$. If i denotes the canonical injection of H into G , we have $t^q \circ i^q = n \cdot \text{Id}$ and $i_q \circ \theta_q = n \cdot \text{Id}$.
- d) We consider the $\mathbf{Z}^{(G)}$ -modules $\hat{M} = \text{Hom}_{\mathbf{Z}^{(H)}}(\mathbf{Z}^{(G)}, M)$ and $\check{N} = N \otimes_{\mathbf{Z}^{(H)}} \mathbf{Z}^{(G)}$ (exercise 13). Taking $Q = \mathbf{Z}^{(G)}$, we obtain from the homomorphisms t and θ of a) and b) homomorphisms $t_M : \hat{M} \rightarrow M$ and $\theta_N : N \rightarrow \check{N}$. Show that these homomorphisms are $\mathbf{Z}^{(G)}$ -linear. Show that the composite homomorphism

$$\begin{aligned} H^q(H, M) &\xrightarrow{(i_q)^{-1}} H^q(G, \hat{M}) \xrightarrow{H^q(G, t_M)} H^q(G, M) \\ (\text{resp. } H_q(G, N) &\xrightarrow{H_q(G, \theta_N)} H_q(G, \check{N}) \xrightarrow{(i_q)^{-1}} H_q(H, N)) \end{aligned}$$

is equal to t^q (resp. θ_q).

- e) Let $(g_1, \dots, g_n)$ be a system of representatives of the classes of G modulo H , and let $x \in H^0(H, M)$. Show that we have $t^0(x) = \sum_{i=1}^n g_i x$. Describe t^1

- f) If one identifies the groups $H_1(G, \mathbf{Z})$ and $H_1(H, \mathbf{Z})$ with $G/(G, G)$ and $H/(H, H)$ respectively, the homomorphism t_1 is identified with a homomorphism (“transfer”) $V : G/(G, G) \rightarrow H/(H, H)$. If $s : H \setminus G \rightarrow G$ is a section of the canonical map, show that V is obtained by passage to the quotient from the map $V_s : G \rightarrow H$ such that

$$V_s(g) = \prod_{x \in H \setminus G} s(x)g(s(xg))^{-1} \quad \text{for } g \in G.$$

- 15) Let G be a group and k a commutative field, which we endow with the structures of $\mathbf{Z}^{(G)}$ -module and of $k^{(G)}$ -module deduced from the trivial action of G .
- a) Show that the group $H^q(G, k)$ is isomorphic to $\text{Ext}_{k(G)}^q(k, k)$ for all $q \geq 0$, therefore admits a structure of k -vector space; we denote by $h^q(G, k)$ its dimension over k .
- b) Suppose that the group G admits a presentation (1, p. 86) defined by d generators and r relations. Prove the inequalities

$$d \geq h^1(G, k) \quad \text{and} \quad r \geq h^2(G, k)$$

(use exercise 3, d), p. 179).

¶ 16) Let G be a p -group (1, p. 72, def. 9). We set $d(G) = \dim_{\mathbf{F}_p} H^1(G, \mathbf{F}_p)$ and $r(G) = \dim_{\mathbf{F}_p} H^2(G, \mathbf{F}_p)$ (cf. exercise 15).

- a) Show that $d(G)$ is the minimal number of generators of G (prove that $H^1(G, \mathbf{F}_p)$ is isomorphic to the dual of $G/G^p \mathbf{D}(G)$, and use I, p. 140, exercise 32).
- b) If $\text{Card}(G) = p^n$, show that one has $d(G) \leq n$ and $r(G) \leq \frac{1}{2}n(n+1)$ (prove by induction on n that there exists a presentation of G with n generators and $\frac{1}{2}n(n+1)$ relations). If $G = (\mathbf{Z}/p\mathbf{Z})^n$, show that one has $d(G) = n$ and $r(G) = \frac{1}{2}n(n+1)$.
- c) If G is nontrivial, prove the inequality $r(G) > \frac{1}{4}(d(G))^2$ (“Golod-Shafarevich theorem”: use exercise 15, p. 182, and exercise 22 of VIII, § 8).
- d) Prove that one has $r(G) - d(G) = \dim_{\mathbf{F}_p} (H^3(G, \mathbf{Z}) \otimes_{\mathbf{Z}} \mathbf{F}_p)$.

17) Let G be a finite group. We denote by N the element $\sum_{g \in G} e_g$ of $\mathbf{Z}^{(G)}$. For every $\mathbf{Z}^{(G)}$ -module M , the multiplication

by N defines by passage to the quotient a homomorphism $N_0 : H_0(G, M) \rightarrow H^0(G, M)$. We set $\hat{H}^0(G, M) = \text{Coker}(N_0)$, $\hat{H}^{-1}(G, M) = \text{Ker}(N_0)$, $\hat{H}^q(G, M) = H^q(G, M)$ for $q \geq 1$, and

$$\hat{H}^{-q}(G, M) = H_{q-1}(G, M) \quad \text{for } q \geq 2.$$

a) Let $\mathcal{E} : 0 \rightarrow \mathbf{M}' \rightarrow \mathbf{M} \rightarrow \mathbf{M}'' \rightarrow 0$ an exact sequence of $\mathbf{Z}^{(G)}$ -modules; define an exact sequence

$$\dots \rightarrow \hat{H}^{q-1}(G, \mathbf{M}'') \xrightarrow{\partial^{q-1}(G, \mathcal{E})} \hat{H}^q(G, \mathbf{M}') \rightarrow \hat{H}^q(G, \mathbf{M}) \rightarrow \hat{H}^q(G, \mathbf{M}'') \xrightarrow{\partial^q(G, \mathcal{E})} \hat{H}^{q+1}(G, \mathbf{M}') \rightarrow \dots$$

b) If M is projective relative to $\mathbf{Z}$, show that one has $\hat{H}^q(G, M) = 0$ for all $q \in \mathbf{Z}$ (cf. exercise 12). c) Let H be a subgroup of G , $i : H \rightarrow G$ the canonical injection. For every $\mathbf{Z}^{(G)}$ -module M and every rational integer q one defines homomorphisms $\hat{i}_M^q : \hat{H}^q(G, M) \rightarrow \hat{H}^q(H, M)$ and $\hat{t}_M^q : \hat{H}^q(H, M) \rightarrow \hat{H}^q(G, M)$ (denoted simply $\hat{i}^q$ and $\hat{t}^q$ if there is no ambiguity about M) in the following way: for $q \geq 1$, we set $\hat{i}^q = i^q$ and $\hat{t}^q = t^q$ (exercises 10 and 14); for $q \leq -2$, we set $\hat{i}^q = \theta_{-q-1}$ and $\hat{t}^q = i_{-q-1}$; finally, one defines $\hat{i}^0$ and $\hat{t}^0$ (resp. $\hat{i}^{-1}$ and $\hat{t}^{-1}$) by passage to quotients (resp. to subsets) from i^0 and t^0 (resp. t_0 and i_0). Prove that one has $\hat{i}^q \circ \hat{t}^q = n \cdot \text{Id.}$ for all $q \in \mathbf{Z}$, where n is the index of H in G . Deduce that the groups $\hat{H}^q(G, M)$ are annihilated by the order of G (cf. exercise 6, p. 189).

d) With the notations of b), show that one has $\hat{t}_{M'}^{q+1} \circ \partial^q(G, \mathcal{E}) = \partial^q(H, \mathcal{E}) \circ \hat{i}_{M''}^q$ and

$$\hat{t}_{M'}^{q+1} \circ \partial^q(H, \mathcal{E}) = \partial^q(G, \mathcal{E}) \circ \hat{i}_{M''}^q$$

for all $q \in \mathbf{Z}$.

e) Let p be a prime number and H a p -Sylow subgroup (I, p. 74) of G . Show that the homomorphism $\hat{i}^q : \hat{H}^q(G, M) \rightarrow \hat{H}^q(H, M)$ has as kernel the sum of the l -primary

components of the $\mathbf{Z}$ -module $\hat{H}^q(G, M)$ for $l \neq p$. In particular, if for every prime number p , one chooses a p -Sylow subgroup H_p of G , the map $\hat{H}^q(G, M) \rightarrow \prod \hat{H}^q(H_p, M)$ deduced from the homomorphisms $\hat{i}^q$ is injective.

- 18) Let G be a finite group and M a $\mathbf{Z}^{(G)}$ -module. One says that M is cohomologically trivial if for every subgroup H of G and for every $q \in \mathbf{Z}$, the group $\hat{H}^q(H, M)$ is zero.
- a) Show that a $\mathbf{Z}^{(G)}$ -module projective relative to $\mathbf{Z}$ (in particular an induced module, cf. exercise 12) is cohomologically trivial.
- b) Let n, r, k be three integers such that r divides $(k^n - 1)$; let G be a cyclic group of order n , σ a generator of G . One endows the $\mathbf{Z}$ -module $M = \mathbf{Z}/r\mathbf{Z}$ with the structure of $\mathbf{Z}^{(G)}$ -module defined by $\sigma m = km$ for all $m \in M$. Prove that M is projective relative to $\mathbf{Z}$ if and only if n and r are coprime.
- c) With the notations of b), show that in order for $\hat{H}^q(G, M) = 0$ for all $q \in \mathbf{Z}$ it is necessary and sufficient that one has $(r, k - 1)(r, 1 + k + \dots + k^{n-1}) = r$. In particular, if $r = k^n - 1$, show that M is cohomologically trivial.
- d) From b) and c), deduce an example of a $\mathbf{Z}^{(G)}$ -module cohomologically trivial which is not projective relative to $\mathbf{Z}$, and of a $\mathbf{Z}^{(G)}$ -module M such that $\hat{H}^q(G, M) = 0$ for all $q \in \mathbf{Z}$ which is not cohomologically trivial.

¶ 19) Let p be a prime number, G a p -group, M a $\mathbf{Z}^{(G)}$ -module. We set $A = F_p^{(G)}$.

- a) We assume that $pM = 0$. Show that the following conditions are equivalent:
- α) There exists a rational integer q such that $\hat{H}^q(G, M) = 0$.
- β) M is cohomologically trivial (exercise 18).
- γ) M is an induced $\mathbf{Z}^{(G)}$ -module.
- δ) The A -module M is free.

(To prove that α) implies δ), reduce to the case where $q = -2$; show that one then has

$$\mathrm{Tor}_1^A(\mathbf{F}_p, \mathbf{M}) = 0,$$

and conclude with exercise 4, p. 185 and exercise 22 of VIII, § 8).

b) We assume that multiplication by p is injective in M . Show that the following conditions are equivalent:

α) There exists an integer q such that the groups $\hat{H}^q(G, M)$ and $\hat{H}^{q+1}(G, M)$ are zero.

β) M is cohomologically trivial.

γ) The A -module M/pM is free.

Let G be a finite group, M a $\mathbf{Z}^{(G)}$ -module.

a) We assume that the $\mathbf{Z}$ -module M is free. Show that the $\mathbf{Z}^{(G)}$ -module M is projective if and only if for every Sylow subgroup $H \subset G$, the $\mathbf{Z}^{(H)}$ -module M is cohomologically trivial.

(To prove that the condition is sufficient, consider a resolution $0 \rightarrow R \rightarrow L \rightarrow M \rightarrow 0$ of the $\mathbf{Z}^{(G)}$ -module M , where L is free; show that the $\mathbf{Z}^{(H)}$ -module $Q = \text{Hom}_{\mathbf{Z}}(M, N)$ is cohomologically trivial by observing that Q/pQ is cohomologically trivial and using Exercise 19. Deduce from Exercise 17, e) that we have $H^1(G, Q) = 0$ and conclude from this that M is projective.)

b) Show that the following conditions are equivalent:

α) The $\mathbf{Z}^{(G)}$ -module M is cohomologically trivial.

β) The $\mathbf{Z}^{(G)}$ -module M admits a finite projective resolution.

γ) The $\mathbf{Z}^{(G)}$ -module M admits a projective resolution of length one.

(To show that α) implies γ), consider an exact sequence $0 \rightarrow R \rightarrow L \rightarrow M \rightarrow 0$, where L is a free $\mathbf{Z}^{(G)}$ -module, and apply α) to R .)

c) We assume that the $\mathbf{Z}$ -module M is divisible. Show that the $\mathbf{Z}^{(G)}$ -module M is cohomologically trivial if and only if it is injective (consider an exact sequence $0 \rightarrow M \rightarrow I \rightarrow Q \rightarrow 0$, where I is an injective $\mathbf{Z}^{(G)}$ -module, and show using b) that the $\mathbf{Z}^{(G)}$ -module $\text{Hom}_{\mathbf{Z}}(Q, M)$ is cohomologically trivial).

d) For a $\mathbf{Z}^{(G)}$ -module to be cohomologically trivial, it is necessary and sufficient that it admit an injective resolution of length 1.

¶ 21) Let G be a cyclic group of finite order.

- a) If M is a $\mathbf{Z}^{(G)}$ -module and q an even (resp. odd) rational integer, the group $\hat{H}^q(G, M)$ is isomorphic to $\hat{H}^0(G, M)$ (resp. $\hat{H}^1(G, M)$) (cf. exercise 7).
- b) We denote by $\mathcal{C}$ the set of classes of $\mathbf{Z}^{(G)}$ -modules of finite type T such that the groups $\hat{H}^q(G, T)$ are finite for every $q \in \mathbf{Z}$; if the $\mathbf{Z}^{(G)}$ -module M is of type $\mathcal{C}$, we set

$$h(M) = \text{Card} (H^0(G, M)) / \text{Card} (H^1(G, M)) .$$

Let $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ be an exact sequence of $\mathbf{Z}^{(G)}$ -modules. Show that if two of the modules M' , M, M'' are of type $\mathcal{C}$, the same is true of the third, and that one then has $h(M) = h(M')h(M'')$. In particular, the function h defines a homomorphism from the Grothendieck group $\mathbf{K}(\mathcal{C})$ to the multiplicative group $\mathbf{Q}^*$. c) Let M be a finite $\mathbf{Z}^{(G)}$ -module. Show that M is of type $\mathcal{C}$ and that one has $h(M) = 1$.

- d) Suppose that the order of G is a prime number p . We denote by $\mathcal{C}'$ the set of classes of $\mathbf{Z}^{(G)}$ -modules finitely generated in which multiplication by p has finite kernel and cokernel; if M is of type $\mathcal{C}'$, we set $\varphi(M) = \text{Card} (\text{Coker } p_M) / \text{Card} (\text{Ker } p_M)$. Prove as in b) that φ defines a homomorphism from $\mathbf{K}(\mathcal{C}')$ to $\mathbf{Q}^*$.
- e) Show that $\mathbf{K}(\mathcal{C}')$ is generated by the classes of $\mathbf{Z}^{(G)}$ -modules T possessing one of the following properties:
 - (i) T is finite.
 - (ii) Multiplication by p in T is bijective.
 - (iii) $T = \mathbf{Z}$ endowed with the trivial action of G .
 - (iv) $T = \mathbf{Z}^{(G)}$.
 - (v) $T = \mathbf{Q}_p / \mathbf{Z}_p$ (cf. TG, III, p. 84, exercise 23), endowed with the trivial action of G .
 - (vi) $T = \mathbf{Z}^{(G)} \otimes_{\mathbf{Z}} (\mathbf{Q}_p / \mathbf{Z}_p)$.

(First show that every $\mathbf{Z}^{(G)}$ -module of type $\mathcal{C}'$ can be written in $\mathbf{K}(\mathcal{C}')$ as a sum of a $\mathbf{Z}^{(G)}$ -module finitely generated over $\mathbf{Z}$ and a $\mathbf{Z}^{(G)}$ -module in which multiplication by p is surjective. Then determine the structure of simple modules over the rings $\mathbf{Q}^{(G)}$ and $\mathbf{Q}_p^{(G)}$.)

- g) Deduce from the preceding that one has $\mathcal{C}' \subset \mathcal{C}$ and that if $\mathbf{M}$ is a $\mathbf{Z}^{(G)}$ -module of type $\mathcal{C}'$, we have

$$h(\mathbf{M})^{p-1} = (\varphi(H^0(\mathbf{G}, \mathbf{M})))^p / \varphi(\mathbf{M}).$$

(Verify the formula in the six cases considered in f .)

- 22) Let K be a commutative field, L a finite Galois extension of K , G its Galois group. a)

Show that the exact sequence of G -modules

$$0 \rightarrow \mu_n(L) \rightarrow L^* \xrightarrow{x \mapsto x^n} L^{*n} \rightarrow 0$$

makes it possible to define an isomorphism $k_L : (L^n \cap K^*)/K^{*n} \rightarrow H^1(\mathbf{G}, \mu_n(L))$. When $\mu_n(K)$ has n elements and one identifies $H^1(\mathbf{G}, \mu_n(L))$ with $\text{Hom}(\mathbf{G}, \mu_n(K))$, show that k_L coincides with the homomorphism defined in V, § 11, n° 8.

- b) Let p be a prime number; suppose that K has characteristic p . Show that the exact sequence of G -modules (V, § 11, n° 9)

$$0 \rightarrow \mathbf{F}_p \rightarrow L \xrightarrow{\mathfrak{p}} \mathfrak{p}(L) \rightarrow 0$$

makes it possible to define an isomorphism

$$a_L : (\mathfrak{p}(L) \cap K)/\mathfrak{p}(K) \rightarrow H^1(\mathbf{G}, \mathbf{F}_p)$$

which coincides with the homomorphism defined in loc. cit. if one identifies

$$H^1(\mathbf{G}, \mathbf{F}_p)$$

with Hom

$$(\mathbf{G}, \mathbf{F}_p)$$

- 23) Let G be a group, M a group with operators in G (1, p. 29, def. 2). We denote by ${}^g m$ the composite of $g \in G$ and $m \in M$. We say that M is a G -group if one has ${}^{h(g)m} = {}^{hg} m$ for $m \in M, g, h \in G$, in other words if the map φ from G to $\text{Aut}(M)$ given by $\varphi(g), m = {}^g m$ for $g \in G, m \in M$ is a homomorphism.

If M is a G -group, we denote by $\mathbf{H}^0(\mathbf{G}, \mathbf{M})$ the subgroup of M consisting of the elements $m \in M$ such that ${}^g m = m$ for all $g \in G$, and $\mathbf{Z}^1(\mathbf{G}, \mathbf{M})$ the set of maps h from G to M such that

$$h(gg') = h(g){}^g h(g') \quad \text{for } g, g' \in G.$$

- a) We say that two elements h_1, h_2 of $\mathbf{Z}^1(\mathbf{G}, \mathbf{M})$ are cohomologous if there exists an element m of $\mathbf{M}$ such that $h_2(g) = m^{-1}h_1(g){}^g m$ for all $g \in \mathbf{G}$. Show that the relation thus defined is an equivalence relation on $\mathbf{Z}^1(\mathbf{G}, \mathbf{M})$; the quotient set is denoted by $\mathbf{H}^1(\mathbf{G}, \mathbf{M})$. The class in $\mathbf{H}^1(\mathbf{G}, \mathbf{M})$ of the map $i \in \mathbf{Z}^1(\mathbf{G}, \mathbf{M})$ such that $i(g) = 1_{\mathbf{M}}$ for all $g \in \mathbf{G}$ is denoted by $e_{G, \mathbf{M}}$ or simply $e_{\mathbf{M}}$.
- b) Let $f : \mathbf{M} \rightarrow \mathbf{N}$ be a homomorphism of G -groups (I, p. 30, def. 3); show that f induces a group homomorphism $f^0 : \mathbf{H}^0(\mathbf{G}, \mathbf{M}) \rightarrow \mathbf{H}^0(\mathbf{G}, \mathbf{N})$, and a map $f^1 : \mathbf{H}^1(\mathbf{G}, \mathbf{M}) \rightarrow \mathbf{H}^1(\mathbf{G}, \mathbf{N})$ such that $f^1(e_{\mathbf{M}}) = e_{\mathbf{N}}$.
- c) Let $\mathbf{M} \xrightarrow{i} \mathbf{N} \xrightarrow{p} \mathbf{Q}$ be a sequence of G -groups and homomorphisms, with i injective, p surjective and

$$\text{Im}(i) = \text{Ker}(p).$$

Show that the group $\mathbf{N}' = p^{-1}(\mathbf{H}^0(\mathbf{G}, \mathbf{Q}))$ acts on $\mathbf{Z}^1(\mathbf{G}, \mathbf{M})$ so that $(n, h)(g) = n^{-1}h(g){}^g n$ for $n \in \mathbf{N}', h \in \mathbf{Z}^1(\mathbf{G}, \mathbf{M}), g \in \mathbf{G}$; show that this action induces, by passing to the quotient, an action of $\mathbf{H}^0(\mathbf{G}, \mathbf{Q})$ on $\mathbf{H}^1(\mathbf{G}, \mathbf{M})$. The map $i^1 : \mathbf{H}^1(\mathbf{G}, \mathbf{M}) \rightarrow \mathbf{H}^1(\mathbf{G}, \mathbf{N})$ passes to the quotient and defines a bijection from $\mathbf{H}^1(\mathbf{G}, \mathbf{M})/\mathbf{H}^0(\mathbf{G}, \mathbf{Q})$ onto $\text{Im}(i^1)$.

- d) The action of $\mathbf{H}^0(\mathbf{G}, \mathbf{Q})$ over $\mathbf{H}^1(\mathbf{G}, \mathbf{M})$ allows one to define a mapping $\partial : \mathbf{H}^0(\mathbf{G}, \mathbf{Q}) \rightarrow \mathbf{H}^1(\mathbf{G}, \mathbf{M})$ by $\partial(q) = q \cdot e_M$ for $q \in \mathbf{H}^0(\mathbf{G}, \mathbf{Q})$. Show that we have

$$\text{Im}(i^0) = \text{Ker}(p^0), \text{Im}(p^0) = \partial^{-1}(e_M), \quad \text{Im}(\partial) = (i^1)^{-1}(e_N), \quad \text{Im}(i^1) = (p^1)^{-1}(e_Q).$$

- e) We now assume that $i(\mathbf{M})$ is contained in the center of $\mathbf{N}$. Show that ∂ is a group homomorphism; construct an action of the group $\mathbf{H}^1(\mathbf{G}, \mathbf{M})$ on the set $\mathbf{H}^1(\mathbf{G}, \mathbf{N})$ such that the map p^1 passes to the quotient and defines a bijection from $\mathbf{H}^1(\mathbf{G}, \mathbf{N})/\mathbf{H}^1(\mathbf{G}, \mathbf{M})$ over $\text{Im}(p^1)$.
- f) Let E be the set of maps k from G to N such that $p \circ k \in \mathbb{Z}^1(\mathbf{G}, \mathbf{Q})$. For $k \in E$, show that there exists an element f_k of $\mathbb{Z}^2(\mathbf{G}, \mathbf{M})$ such that $i(f_k(x, y)) = k(x)^x k(y) (k(xy))^{-1}$ for $x, y \in G$. Show that the map $k \mapsto f_k$ defines by passing to the quotient a map $\partial^1 : \mathbf{H}^1(\mathbf{G}, \mathbf{Q}) \rightarrow \mathbf{H}^2(\mathbf{G}, \mathbf{M})$. Show that we have $\text{Im}(p^1) = (\partial^1)^{-1}(0)$.

- 24) Let K be a commutative field, L a finite Galois extension of K , with Galois group G .

Let V (resp. V') be a K -vector space, and t (resp. t') a tensor of type (p, q) on V (resp. V') (III, p. 63). A K -isomorphism (or simply isomorphism) of (V, t) onto (V', t') is a K -linear isomorphism $u : V \rightarrow V'$ such that $(T^p(u) \otimes T^q(u))(t) = t'$. We denote by V_L the L -vector space $L \otimes_K V$ and t_L is the tensor of type (p, q) on V_L deduced from t by extension of scalars. We denote by $F_{L/K}(V, t)$ the set of isomorphism classes of pairs (W, τ) such that (W_L, τ_L) is L -isomorphic to (V_L, t_L) .

- a) Let M be the group of L -automorphisms of (V_L, t_L) . For $m \in M$ and $\sigma \in G$, we set

$$\sigma m = \pi_V(\sigma) \circ m \circ \pi_V(\sigma)^{-1},$$

where $\pi_V : G \rightarrow \text{Aut}_K(V_L)$ denotes the action of G on $V_L = L \otimes_K V$ deduced from the action of G on L . Show that one thus defines a structure of G -group on M .

- b) Let (W, τ) be an element of $F_{L/K}(V, t)$ and let u be an L -isomorphism from (V_L, t_L) onto

(W_L, τ_L) . For $\sigma \in G$, we denote by $f_u(\sigma)$ the element $u^{-1} \circ \pi_W(\sigma) \circ u \circ \pi_V(\sigma)^{-1}$ of M . Show that $f_u \in Z^1(G, M)$ (Exercise 23), and that its class in $H^1(G, M)$ is independent of the choice of u ; it is denoted by $\theta(W, \tau)$.

c) Show that the map $\theta : F_{L/K}(V, t) \rightarrow H^1(G, M)$ is bijective, and that one has $\theta(V, t) = e_M$.

Let K be a commutative field, L a finite Galois extension of K , and G its Galois group. We let G act on the group $GL(n, L)$ by setting $\sigma(a_{ij}) = (\sigma(a_{ij}))$ for $\sigma \in G$, $(a_{ij}) \in GL(n, L)$. Every subgroup of $GL(n, L)$ stable under this action is thereby endowed with a G -group structure (exercise 23). a) Show that the sets $H^1(G, GL(n, L))$ and $H^1(G, SL(n, L))$ are reduced to a single element (cf. V, § 10, no. 5, prop. 9).

b) Show that the set $H^1(G, PGL(n, L))$ is identified with the set of isomorphism classes of central simple algebras S such that the algebra $L \otimes_K S$ is isomorphic to $M_n(L)$ (using Exercise 24). By associating to such an algebra its class in the group $Br(K, L)$ (VIII, § 13) and by identifying this latter group with $H^2(G, L^*)$ (loc. cit.), one obtains a map $H^1(G, PGL(n, L)) \rightarrow H^2(G, L^*)$ which is the opposite of the mapping ∂^1 (Exercise 23, f) deduced from the extension $L^* \rightarrow GL(n, L) \rightarrow PGL(n, L)$. Show that the image of ∂^1 is formed of elements of order n in $H^2(G, L^*)$.

c) Show that the set $H^1(G, Sp(2n, L))$ is reduced to a single element (cf. IX, § 5, no. 1, th. 1).

d) Let Q be a quadratic form on the K -vector space V , Q_L the quadratic form deduced from it on $L \otimes_K V$. Show that $H^1(G, O(Q_L))$ is identified with the set of isomorphism classes of quadratic forms Q' on V such that Q'_L is isomorphic to Q_L .

26) We denote by A^e the k -algebra $A \otimes_k A^0$; every (A, A) -bimodule is naturally equipped with a structure of A^e -module on the left and also a structure of A^e -module on the right. Let M be a (A, A) -bimodule, n be an integer ≥ 0 ; we set:

$$H_n(A, M) = \text{Tor}_n^{A^e}(M, A) \quad H^n(A, M) = \text{Ext}_{A^e}^n(A, M).$$

a) The k -module $\mathbf{H}_0(\mathbf{A}, \mathbf{M})$ is isomorphic to the quotient of $\mathbf{M}$ by the sub- k -module generated by the elements $(am - ma)$ for $a \in \mathbf{A}, m \in \mathbf{M}$; the k -module $\mathbf{H}^0(\mathbf{A}, \mathbf{M})$ is isomorphic to the sub- k -module of $\mathbf{M}$ of elements m such that $am = ma$ for all $a \in \mathbf{A}$. The k -module $\mathbf{H}^1(\mathbf{A}, \mathbf{M})$ is identified with the quotient of the k -module of derivations $\mathbf{D}_k(\mathbf{A}, \mathbf{M})$ by the sub- k -module of inner derivations, that is, the derivations d_m , for $m \in \mathbf{M}$, defined by $d_m(a) = am - ma$ for all $a \in \mathbf{A}$ (cf. III, p. 132).

b) Assume that $\mathbf{A}$ is a projective k -module. Show that the standard resolution $\mathbf{B}(\mathbf{A})$ (X, p. 58) defines a projective resolution of the left $\mathbf{A}^e$ -module $\mathbf{A}$. Deduce that the k -modules $\mathbf{H}^n(\mathbf{A}, \mathbf{M})$ are isomorphic to the homology modules of the complex $\mathbf{C}(\mathbf{A}, \mathbf{M})$, where, for $p \geq 0$, $\mathbf{C}^p(\mathbf{A}, \mathbf{M})$ is the k -module of k -multilinear maps from $\mathbf{A}^p$ to $\mathbf{M}$, $\mathbf{C}^p(\mathbf{A}, \mathbf{M}) = 0$ for $p < 0$, the differential being given by the formula :

for $f \in \mathbf{C}^p(\mathbf{A}, \mathbf{M})$, $x_0, \dots, x_p \in \mathbf{A}$.

Give an analogous expression for the k -modules $\mathbf{H}_n(\mathbf{A}, \mathbf{M})$.

c) Under the hypothesis of b), let $\rho : k \rightarrow k'$ be a homomorphism of commutative rings, $\mathbf{A}'$ the k' -algebra $k' \otimes_k \mathbf{A}$, $\mathbf{M}'$ a $(\mathbf{A}', \mathbf{A}')$ -bimodule. Show that the k -modules $\mathbf{H}_n(\mathbf{A}, \mathbf{M}')$ and $\mathbf{H}_n(\mathbf{A}', \mathbf{M}')$ (resp. $\mathbf{H}^n(\mathbf{A}, \mathbf{M}')$ and $\mathbf{H}^n(\mathbf{A}', \mathbf{M}')$) are isomorphic.

d) Assume henceforth that k is a field. Let $\mathbf{B}$ be a second k -algebra (associative and unital), $\mathbf{M}$ an $(\mathbf{A}, \mathbf{A})$ -bimodule, $\mathbf{N}$ a $(\mathbf{B}, \mathbf{B})$ -bimodule. For every $n \geq 0$, define an isomorphism of k -modules

$$\mathbf{H}_n(\mathbf{A} \otimes_k \mathbf{B}, \mathbf{M} \otimes_k \mathbf{N}) \rightarrow \bigoplus_{p+q=n} (\mathbf{H}_p(\mathbf{A}, \mathbf{M}) \otimes_k \mathbf{H}_q(\mathbf{B}, \mathbf{N})).$$

If one assumes moreover that $\mathbf{A}$ and $\mathbf{B}$ are finite-dimensional over k , show that there likewise exists a k -isomorphism

$$\mathbf{H}^n(\mathbf{A} \otimes_k \mathbf{B}, \mathbf{M} \otimes_k \mathbf{N}) \rightarrow \bigoplus_{p+q=n} (\mathbf{H}^p(\mathbf{A}, \mathbf{M}) \otimes_k \mathbf{H}^q(\mathbf{B}, \mathbf{N})) \quad \text{for every } n \geq 0$$

(use X, p. 76, th. 3).

- e) Let M, N be two left A -modules, P a right A -module, so that the k -modules $\text{Hom}_k(M, N)$ and $M \otimes_k P$ have natural structures of (A, A) -bimodules. Define for every $n \geq 0$ isomorphisms of k -modules:

$$H^n(A, \text{Hom}_k(M, N)) \rightarrow \text{Ext}_A^n(M, N) \quad \text{and} \quad H_n(A, M \otimes_k P) \rightarrow \text{Tor}_n^A(P, M).$$

¶ 27) One calls an augmentation of the k -algebra A a unital homomorphism of k -algebras $\pi : A \rightarrow k$; one says that the pair (A, π) is an augmented k -algebra. The augmentation endows k with a structure of left A -module. Let M be a left A -module, N a right A -module; one sets

$$H_n(A, \pi; N) = \text{Tor}_n^A(N, k) \quad \text{and} \quad H^n(A, \pi; M) = \text{Ext}_A^n(k, M) \quad \text{for every } n \geq 0.$$

- a) We denote by $\mathbf{M}_{(\pi)}$ (resp. $\mathbf{N}_{(\pi)}$) the group $\mathbf{M}$ (resp. $\mathbf{N}$) endowed with the structure of (A, A) -bimodule defined by:

$$ama' = \pi(a')am \quad (\text{resp. } ana' = n\pi(a)a') \quad \text{for } a, a' \in A, m \in \mathbf{M}, n \in \mathbf{N}.$$

Show that the k -module $\mathbf{H}_0(A, \pi; \mathbf{N})$ (resp. $\mathbf{H}^0(A, \pi; \mathbf{M})$, resp. $\mathbf{H}^1(A, \pi; \mathbf{M})$) is isomorphic to the k -module $\mathbf{H}_0(A, \mathbf{N}_{(\pi)})$ (resp. $\mathbf{H}^0(A, \mathbf{M}_{(\pi)})$, resp. $\mathbf{H}^1(A, \mathbf{M}_{(\pi)})$) defined in exercise 26. b) We now assume that A is a projective k -module. Define using the standard resolution (X, p. 58) the k -isomorphisms $\varphi_n : \mathbf{H}_n(A, \mathbf{N}_{(\pi)}) \rightarrow \mathbf{H}_n(A, \pi; \mathbf{N})$ and $\varphi^n : \mathbf{H}^n(A, \mathbf{M}_{(\pi)}) \rightarrow \mathbf{H}^n(A, \pi; \mathbf{M})$ for all $n \geq 0$.

- c) We assume that there exists a homomorphism of k -algebras $\rho : A \rightarrow A^e$ such that A^e is a projective right A -module and such that the left ideal of A^e generated by $\rho(\text{Ker } \pi)$ be the kernel of the homomorphism $A \otimes_k A^* \rightarrow A$ defined by multiplication in A . Show

that for every $(\mathbf{A}, \mathbf{A})$ -bimodule $\mathbf{Q}$ and every integer $n \geq 0$ there are k -isomorphisms $H_n(\mathbf{A}, \pi; \mathbf{Q}) \rightarrow H_n(\mathbf{A}, \mathbf{Q})$ (resp. $H^n(\mathbf{A}, \pi; \mathbf{Q}) \rightarrow H^n(\mathbf{A}, \mathbf{Q})$), $\mathbf{Q}$ being considered as $\mathbf{A}$ -module on the right (resp. on the left) via the homomorphism ρ . (Use the canonical homomorphisms of $\mathbf{X}$, p. 109.)

d) Show that the hypotheses of c) are satisfied in the following cases:

α) $\mathbf{A}$ is the algebra (over k) of a group $\mathbf{G}$, ρ is defined by $\rho(e_g) = e_g \otimes e_{g^{-1}}$.

β) $\mathbf{A}$ is the tensor (resp. symmetric) algebra of a projective k -module $\mathbf{V}$; ρ is defined by

$$\rho(v) = v \otimes 1 - 1 \otimes v.$$

* γ) $\mathbf{A}$ is the enveloping algebra of a Lie algebra $\mathfrak{g}$ over k , free as a k -module; we have

$$\rho(x) = x \otimes 1 - 1 \otimes x \quad \text{for } x \in \mathfrak{g}$$

(using cor. 5 of th. 1 of LIE I, § 2, no. 7).

δ) $\mathbf{A}$ is the ring of functions of a smooth algebraic group G over a field k ; we take

$$\rho(f) = \sum_{i=1}^k f'_i \otimes f''_i, \quad \text{where } f(gh^{-1}) = \sum_{i=1}^k f'_i(g)f''_i(h) \quad \text{for } g, h \in G.*$$

Let $\mathfrak{g}$ be a Lie algebra over the commutative ring k , and U its enveloping algebra. We endow k with the structure of a left U -module corresponding to the trivial representation of $\mathfrak{g}$. Let $\mathbf{N}$ be a right $\mathfrak{g}$ -module, $\mathbf{M}$ a left $\mathfrak{g}$ -module, and n an integer ≥ 0 ; we set

$$H_n(\mathfrak{g}, \mathbf{N}) = \text{Tor}_n^U(\mathbf{N}, k) \quad \text{and} \quad H^n(\mathfrak{g}, \mathbf{M}) = \text{Ext}_{\mathbf{U}}^n(k, \mathbf{M}).$$

a) The k -module $\mathbf{H}_0(\mathfrak{g}, \mathbf{N})$ (resp. $\mathbf{H}^0(\mathfrak{g}, \mathbf{M})$) is isomorphic to $\mathbf{N}/\mathbf{N}\mathfrak{g}$ (resp. the submodule of $\mathbf{M}$ consisting of elements annihilated by $\mathfrak{g}$). We denote by $\mathbf{Z}^1(\mathfrak{g}, \mathbf{M})$ (resp. $\mathbf{B}^1(\mathfrak{g}, \mathbf{M})$) the sub- k -module of $\text{Hom}_k(\mathfrak{g}, \mathbf{M})$ of elements f such that $f([x, y]) = xf(y) - yf(x)$ for $x, y \in \mathfrak{g}$ (resp. the elements f_m , for $m \in \mathbf{M}$, such that $f_m(x) = xm$ for $x \in \mathfrak{g}$). Show that $\mathbf{H}^1(\mathfrak{g}, \mathbf{M})$ is identified with the quotient $\mathbf{Z}^1(\mathfrak{g}, \mathbf{M})/\mathbf{B}^1(\mathfrak{g}, \mathbf{M})$.

- b) We now assume that $\mathfrak{g}$ is a free k -module. Show that the k -modules $\mathbf{H}^n(\mathfrak{g}, \mathbf{M})$ are identified with the homology modules of the complex $\mathbf{C}(\mathfrak{g}, \mathbf{M})$, where for $p \geq 0$, $\mathbf{C}^p(\mathfrak{g}, \mathbf{M})$ is the k -module of alternating multilinear maps from $\mathfrak{g}^p$ to $\mathbf{M}$, $\mathbf{C}^p(\mathfrak{g}, \mathbf{M}) = 0$ for $p < 0$, the differential being given by the formula

$$df(x_1, \dots, x_{p+1}) = \sum_{1 \leq i \leq p+1} (-1)^{i+1} x_i f(x_1, \dots, \hat{x}_i, \dots, x_{p+1}) \\ + \sum_{1 \leq i < j \leq p+1} (-1)^{i+j} f([x_i, x_j], x_1, \dots, \hat{x}_i, \dots, \hat{x}_j, \dots, x_{p+1})$$

for $f \in \mathbf{C}^p(\mathfrak{g}, \mathbf{M})$, $x_1, \dots, x_{p+1} \in \mathfrak{g}$ (where the sign over a letter means that it is to be omitted).

Give an analogous expression for the k -modules $\mathbf{H}_p(\mathfrak{g}, \mathbf{M})$. (Use **X**, p. 184, exercise 21.)

- c) We assume that $\dim_k(\mathfrak{g}) = n < \infty$. Show that $\mathbf{H}^p(\mathfrak{g}, \mathbf{M}) = 0$ for all $p \geq n+1$ and for every $\mathfrak{g}$ -module $\mathbf{M}$, and that there exists a $\mathfrak{g}$ -module $\mathbf{N}$ such that $\mathbf{H}^n(\mathfrak{g}, \mathbf{N}) \neq 0$ (take $\mathbf{N} = \Lambda^n \mathfrak{g}$, where $\mathfrak{g}$ acts by the formula

$$x_* (x_1 \wedge \dots \wedge x_n) = \sum_{i=1}^n x_i \wedge \dots \wedge x_{i-1} \wedge [x, x_i] \wedge x_{i+1} \wedge \dots \wedge x_n;$$

show that $df = 0$ for all $f \in \mathbf{C}^{n-1}(\mathfrak{g}, \mathbf{N})$.

- d) The adjoint representation of $\mathfrak{g}$ and the representation of $\mathfrak{g}$ on $\mathbf{M}$ define a representation θ of $\mathfrak{g}$ in $\mathbf{C}(\mathfrak{g}, \mathbf{M})$, such that

$$(\theta(x)f)(x_1, \dots, x_p) = x f(x_1, \dots, x_p) - \sum_{i=1}^p f(x_1, \dots, [x, x_i], \dots, x_p)$$

for $f \in \mathbf{C}^p(\mathfrak{g}, \mathbf{M})$, $x, x_1, \dots, x_p \in \mathfrak{g}$. Show that, for every $x \in \mathfrak{g}$, $\theta(x)$ is an endomorphism homotopic to zero of the complex $\mathbf{C}(\mathfrak{g}, \mathbf{M})$ (define a homotopy $i(x)$ by setting $(i(x)f)(x_1, \dots, x_p) = f(x, x_1, \dots, x_p)$). * 29) We keep the notation of the preceding exercise; we assume moreover that k is a field of characteristic zero, and that the Lie algebra $\mathfrak{g}$ is semisimple (LIE, I, § 6). Let $\mathbf{M}$ be a $\mathfrak{g}$ -module of finite dimension over k .

- a) If $\mathbf{M}$ is simple and nontrivial, then $\mathbf{H}^p(\mathfrak{g}, \mathbf{M}) = 0$ for every $p > 0$ (the bilinear form on $\mathfrak{g}$ associated with $\mathbf{M}$ is nondegenerate (LIE, I, § 6, n° 1, prop. 1); observe that multiplication

by the corresponding Casimir element (LIE, I, § 3, n° 7) is zero in k and bijective in M).

- b) Show that $\mathbf{H}^1(\mathfrak{g}, \mathbf{M}) = 0$ for every $\mathbf{M}$ (use a) and the fact that $\mathfrak{g} = \mathcal{D}\mathfrak{g}$). Conversely, a k -Lie algebra $\mathfrak{h}$ (of finite dimension) such that $\mathbf{H}^1(\mathfrak{h}, \mathbf{M}) = 0$ for all $\mathfrak{h}$ -module $\mathbf{M}$ of finite dimension is semisimple.
- c) We give k the structure of $\mathfrak{g}$ -module corresponding to the trivial representation. Show that, for every $p \geq 0$, the k -vector space $H^p(\mathfrak{g}, k)$ is isomorphic to the subspace of $C^p(\mathfrak{g}, k)$ consisting of alternating p -linear forms f that are invariant, that is, such that

$$\sum_{i=1}^p f(x_1, \dots, [x, x_i], \dots, x_p) = 0 \quad \text{for all } x, x_1, \dots, x_p \in \mathfrak{g}.$$

(Deduce from the semisimplicity of the representations of $\mathfrak{g}$ and from exercise 28, d) that $\mathbf{H}^p(\mathfrak{g}, k)$ is isomorphic to $\mathbf{H}^p(\mathbf{C}_\theta(\mathfrak{g}, k))$, where $\mathbf{C}_\theta(\mathfrak{g}, k)$ is the subcomplex of $\mathbf{C}(\mathfrak{g}, k)$ annihilated by all the endomorphisms $\theta(x)$ for $x \in \mathfrak{g}$; then show that $\mathbf{C}_\theta(\mathfrak{g}, k)$ has zero differential.)

- d) Deduce from c) that $H^2(\mathfrak{g}, M) = 0$ for every $\mathfrak{g}$ -module M of finite dimension over k . (Show that every invariant bilinear form on $\mathfrak{g}$ is symmetric, by reducing to the case where $\mathfrak{g}$ is simple and k is algebraically closed.) *

Exercises for Section 7: Composition Product

- 1) Let k be a commutative ring, A an associative unital k -algebra, M an (A, A) -bimodule. An extension of A by M is the data of an associative unital k -algebra B and of an exact sequence of k -modules

$$(\mathcal{E}) : 0 \rightarrow M \xrightarrow{i} B \xrightarrow{p} A \rightarrow 0,$$

where p is an algebra homomorphism and where

$$i(p(b)m) = bi(m) \quad \text{and} \quad i(mp(b)) = i(m)b \quad \text{for } b \in B, \quad m \in M.$$

Two extensions $0 \rightarrow M \xrightarrow{i} B \xrightarrow{p} A \rightarrow 0$ and $0 \rightarrow M \xrightarrow{i'} B' \xrightarrow{p'} A \rightarrow 0$ are said to be equivalent if there exists an algebra homomorphism $u : B \rightarrow B'$ such that $p' \circ u = p$ and $u \circ i = i'$; the homomorphism u is then bijective.

- a) We assume that the exact sequence of k -modules $0 \rightarrow \mathbf{M} \xrightarrow{i} \mathbf{B} \xrightarrow{p} \mathbf{A} \rightarrow 0$ is split; let $s : \mathbf{A} \rightarrow \mathbf{B}$ be a k -linear section of p . Show that there exists an element $f \in \mathbf{C}^2(\mathbf{A}, \mathbf{M})$ (X, p. 195, exercise 26, b)) such that $s(a)s(a') - s(aa') = i(f(a, a'))$ for $a, a' \in \mathbf{A}$. Show that f is a 2-cocycle and that its class in $\mathbf{H}^2(\mathbf{A}, \mathbf{M})$ is independent of the choice of s ; we denote it $c(\mathcal{E})$.
 - b) Let $\text{Ex}_k(\mathbf{A}, \mathbf{M})$ be the set of equivalence classes of extensions of $\mathbf{A}$ by $\mathbf{M}$ which are trivial as extensions of k -modules. Show that c defines a bijection from $\text{Ex}_k(\mathbf{A}, \mathbf{M})$ onto $\mathbf{H}^2(\mathbf{A}, \mathbf{M})$.
 - c) Let $\mathcal{E} : 0 \rightarrow \mathbf{M} \xrightarrow{i} \mathbf{B} \xrightarrow{p} \mathbf{A} \rightarrow 0$ and $\mathcal{E}' : 0 \rightarrow \mathbf{M} \xrightarrow{i'} \mathbf{B}' \xrightarrow{p'} \mathbf{A} \rightarrow 0$ be two extensions of $\mathbf{A}$ by $\mathbf{M}$. We denote by $\mathbf{C}$ the subalgebra of $\mathbf{B} \times \mathbf{B}'$ consisting of the elements (b, b') such that $p(b) = p'(b')$, $\mathfrak{c}$ the ideal of $\mathbf{C}$ consisting of the elements $(i(m), -i'(m))$ for $m \in \mathbf{M}$, $\mathbf{B}''$ the algebra $\mathbf{C}/\mathfrak{c}$, $\pi : \mathbf{C} \rightarrow \mathbf{B}''$ the quotient homomorphism. Let $i'' : \mathbf{M} \rightarrow \mathbf{B}''$ and $p'' : \mathbf{B}'' \rightarrow \mathbf{A}$ be the homomorphisms such that $i''(m) = \pi((m, 0))$ for $m \in \mathbf{M}$ and $p''(\pi((b, b'))) = p(b)$ for $b \in \mathbf{B}, b' \in \mathbf{B}'$. Show that the sequence $0 \rightarrow \mathbf{M} \rightarrow \mathbf{B}'' \rightarrow \mathbf{A} \rightarrow 0$ is an extension of $\mathbf{A}$ by $\mathbf{M}$, called the sum extension of $(\mathcal{E})$ and $(\mathcal{E}')$, and that this defines a commutative group law on the set of equivalence classes of extensions of $\mathbf{A}$ by $\mathbf{M}$. In particular, the set $\text{Ex}_k(\mathbf{A}, \mathbf{M})$ is thereby endowed with a group structure, and the map c is a group isomorphism.
- 2) Suppose that the k algebra $\mathbf{A}$ admits an augmentation $\pi : \mathbf{A} \rightarrow k$ (X, p. 196, exercise 27) and that
 - 3) Suppose that the k -algebra $\mathbf{A}$ admits an augmentation $\pi : \mathbf{A} \rightarrow k$ (X, p. 196, exercise 27) and that, moreover, $\mathbf{A}$ is a projective k -module. We denote by $\mathbf{I}$ the kernel of π . Let $\mathbf{M}$ be a left $\mathbf{A}$ -module.
 - a) Let $(\mathcal{E}) : 0 \rightarrow M_{(\pi)} \xrightarrow{i} \mathbf{B} \xrightarrow{p} \mathbf{A} \rightarrow 0$ be an extension of $\mathbf{A}$ by the $(\mathbf{A}, \mathbf{A})$ -bimodule

$M_{(\pi)}$ (X, p. 196, exercise 27); we denote by $\bar{\mathbf{B}}$ the ideal of $\mathbf{B}$ of elements $b \in \mathbf{B}$ such that $p(b) \in I$. We have $i(m)b = 0$ for $m \in \mathbf{M}$ and $b \in \bar{\mathbf{B}}$, so that $\bar{\mathbf{B}}$ is equipped with a structure of left $\mathbf{A}$ -module such that $p(\beta)b = \beta b$ for $\beta \in \mathbf{B}$, $b \in \bar{\mathbf{B}}$. Show that there exists an exact sequence of $\mathbf{A}$ -modules $0 \rightarrow \mathbf{M} \rightarrow \bar{\mathbf{B}} \rightarrow I \rightarrow 0$; we denote by $\theta(\mathcal{E})$ the class of this exact sequence in $\text{Ext}_{\mathbf{A}}^1(I, \mathbf{M})$.

- b) Show that θ defines a map from $\text{Ex}_k(\mathbf{A}, \mathbf{M}_{(\pi)})$ to $\text{Ext}_{\mathbf{A}}^1(I, \mathbf{M})$. Prove the commutativity of the diagram:

$$\begin{array}{ccc} \text{Ex}_k(\mathbf{A}, \mathbf{M}_{(\pi)}) & \xrightarrow{\theta} & \text{Ext}_{\mathbf{A}}^1(I, \mathbf{M}) \\ \downarrow c & & \downarrow -\delta \\ \mathbf{H}^2(\mathbf{A}, \mathbf{M}_{(\pi)}) & \xrightarrow{\varphi} & \text{Ext}_{\mathbf{A}}^2(k, \mathbf{M}) \end{array}$$

where δ is the connecting homomorphism deduced from the exact sequence $0 \rightarrow I \rightarrow A \xrightarrow{\pi} k \rightarrow 0$, and where φ is the isomorphism obtained by computing with the standard resolution (X, p. 196, exercise 27, b)). Deduce that θ is an isomorphism of groups.

- 3) Let G be a group, A the algebra $\mathbf{Z}^{(G)}$, $j : G \rightarrow A$ the canonical injection. One endows A with the augmentation $\pi : A \rightarrow \mathbf{Z}$ defined by $\pi(e_g) = 1$ for all $g \in G$. Let M be a left A -module.

- a) Let $0 \rightarrow M_{(\pi)} \xrightarrow{i} B \xrightarrow{p} A \rightarrow 0$ be an extension of A by the (A, A) -bimodule $M_{(\pi)}$ (X, p. 196, exercise 27). We denote by Γ the subset of B consisting of the elements b of B such that $p(b) \in j(G)$. Show that Γ , endowed with the multiplication induced by that of B , is a group, an extension of G by the G -module M .
- b) The preceding construction defines a map from $\text{Ex}_{\mathbf{Z}}(\mathbf{A}, \mathbf{M}_{(\pi)})$ to $\text{Ex}(\mathbf{G}, \mathbf{M})$ (VIII, § 13). Show that the diagram

$$\begin{array}{ccc} \text{Ex}_{\mathbf{Z}}(\mathbf{A}, \mathbf{M}_{(\pi)}) & \rightarrow & \text{Ex}(\mathbf{G}, \mathbf{M}) \\ \downarrow c & & \downarrow \\ \mathbf{H}^2(\mathbf{A}, \mathbf{M}_{(\pi)}) & \xrightarrow{\varphi^2} & \mathbf{H}^2(\mathbf{G}, \mathbf{M}) \end{array}$$

is commutative.

* 4) Let $\mathfrak{g}$ be a Lie k -algebra, M a $\mathfrak{g}$ -module. An extension of $\mathfrak{g}$ by M is an exact sequence of k -modules $0 \rightarrow M \xrightarrow{i} \mathfrak{h} \xrightarrow{p} \mathfrak{g} \rightarrow 0$ where $\mathfrak{h}$ is a Lie k -algebra, p a homomorphism of Lie algebras, and where $i(p(H)m) = [H, i(m)]$ for $H \in \mathfrak{h}, m \in M$. Two extensions

$$0 \rightarrow M \xrightarrow{i} \mathfrak{h} \xrightarrow{p} \mathfrak{g} \rightarrow 0, 0 \rightarrow M \xrightarrow{i'} \mathfrak{h}' \xrightarrow{p'} \mathfrak{g} \rightarrow 0$$

are said to be equivalent if there exists a homomorphism $u : \mathfrak{h} \rightarrow \mathfrak{h}'$ such that $p' \circ u = p$, $u \circ i = i'$. We denote by $\text{Liex}(\mathfrak{g}, M)$ the set of equivalence classes of extensions of $\mathfrak{g}$ by M . We assume that $\mathfrak{g}$ is a free k -module.

a) Let $(\mathcal{E}) : 0 \rightarrow M \xrightarrow{i} \mathfrak{h} \xrightarrow{p} \mathfrak{g} \rightarrow 0$ be an extension of $\mathfrak{g}$ by M ; we choose a k -linear section $s : \mathfrak{g} \rightarrow \mathfrak{h}$ of p . We define an element $f \in C^2(\mathfrak{g}, M)$ (X, p. 196, exercise 28, b)) by setting

$$i(f(x, y)) = [s(x), s(y)] - s([x, y]) \quad \text{for } x, y \in \mathfrak{g}.$$

Show that f is a 2-cocycle and that its class in $H^2(\mathfrak{g}, M)$ does not depend on the choice of s ; we denote it $\theta(\mathcal{E})$. b) Show that θ defines a bijection from $\text{Liex}(\mathfrak{g}, M)$ onto $H^2(\mathfrak{g}, M)$. Describe directly the group structure on $\text{Liex}(\mathfrak{g}, M)$ obtained by transport via γ .

c) Let U be the enveloping algebra of $\mathfrak{g}$, $M_{(\pi)}$ the (U, U) -bimodule associated with M by means of the augmentation $\pi : U \rightarrow k$ (X, p. 196, exercise 27). Let $0 \rightarrow M_{(\pi)} \rightarrow B \xrightarrow{p} U \rightarrow 0$ be an extension of U by $M_{(\pi)}$ (exercise 1); we denote by $\mathfrak{b}$ the set of elements b of B such that $p(b) \in j(\mathfrak{g})$.

Show that $\mathfrak{b}$ is a Lie k -algebra, an extension of $\mathfrak{g}$ by M . This construction defines a map $\lambda : \text{Ex}_k(U, M_{(\pi)}) \rightarrow \text{Liex}(\mathfrak{g}, M)$. Show that the diagram

$$\begin{array}{ccc} \text{Ex}_k(U, M_{(\pi)}) & \xrightarrow{\lambda} & \text{Liex}(\mathfrak{g}, M) \\ \downarrow c & & \downarrow \gamma \\ H^2(U, M_{(\pi)}) & \xrightarrow{\varphi^2} & H^2(\mathfrak{g}, M) \end{array}$$

(where φ^2 is the isomorphism defined in X, p. 196, exercise 27) is commutative. Deduce

that λ is an isomorphism. *

¶ 5) Let G, F be two groups.

a) For every extension $\mathcal{E} : F \rightarrow E \rightarrow G$ of G by F (I, p. 62), the action of E on F by inner automorphisms defines a homomorphism $E \rightarrow \text{Aut}(F)$, whence, by passing to the quotient, a group homomorphism $\theta : G \rightarrow \text{Aut}(F)/\text{Int}(F)$, called the homomorphism associated with the extension $\mathcal{E}$. Two isomorphic extensions of G by F have the same associated homomorphism.

b) Conversely, we consider a group homomorphism $\theta : G \rightarrow \text{Aut}(F)/\text{Int}(F)$. Let

$$p : \text{Aut}(F) \rightarrow \text{Aut}(F)/\text{Int}(F)$$

the quotient map. We choose a map σ from G to $\text{Aut}(F)$ such that $p \circ \sigma = \theta$, and a map f from $G \times G$ to F such that $\sigma(x)\sigma(y)(\sigma(xy))^{-1} = \text{Int}(f(x, y))$ for $x, y \in G$; set $c(x, y, z) = \sigma(x)(f(y, z))f(x, yz)(f(xy, z))^{-1}(f(x, y))^{-1}$. Show that $c(x, y, z)$ belongs to the center Z of F .

c) We consider Z as a G -module by setting $g.z = \sigma(g)(z)$ for $g \in G, z \in Z$; this definition is independent of the choice of σ . The map c defines an element of $C^3(G, Z)$; show that c is a 3-cocycle, and that its class in $H^3(G, Z)$ is independent of the choices of σ and of f . We denote it $\omega(G, F, \theta)$. d) Show that the vanishing of $\omega(G, F, \theta)$ is a necessary and sufficient condition for there to exist an extension of G by F having as associated homomorphism θ . If $\omega(G, F, \theta) = 0$, show that one can choose f so that the cocycle c is zero, then define a group law on $G \times F$.

d) Assume $\omega(G, F, \theta) = 0$. Show that the set of isomorphism classes of extensions of G by F of the associated homomorphism θ is a principal homogeneous set under the group $H^2(G, Z)$.

e) Let H be a group, M a $\mathbf{Z}^{(H)}$ -module, x be an element of $H^3(H, M)$. Show that there exists a group E with center M and a homomorphism $\varphi : H \rightarrow \text{Aut}(E)/\text{Int}(E)$, inducing on M the given H -module structure, such that $\omega(H, E, \varphi) = x$ (take $E = \mathbf{M} \times \mathbf{L}$, where $\mathbf{L}$ is the free group on $H \times H$).

6) Let M be a right A -module and N a left A -module. For $n \geq 0$, we denote by $\mathcal{T}_n(M, N)$ the set of triples (P, p, q) , where P is a complex of finitely generated free A -modules, such that $P_r = 0$ for $r < 0$ and $r > n$, and where $p : P \rightarrow N$ and $q : P^* \rightarrow M(n)$ are morphisms of A -complexes (we denote by P^* the Homgr complex ${}_A(P, A)$). Two elements (P, p, q) and (P', p', q') of $\mathcal{T}_n(M, N)$ are said to be linked if there exists a morphism $u : P \rightarrow P'$ such that $p = p' \circ u$ and $q' = q \circ u$; we denote by $T_n(M, N)$ the quotient of $\mathcal{T}_n(M, N)$ by the finest equivalence relation for which two related elements are equivalent (cf. II, p. 52, exercise 9). a) Show that $T_n(M, N)$ is identified with $\text{Tor}_n^A(M, N)$.

b) Let $(\mathcal{E}) : 0 \rightarrow M' \xrightarrow{i} M \xrightarrow{\pi} M'' \rightarrow 0$ be an exact sequence of right A -modules. Describe the connecting homomorphism $\partial : T_n(M'', N) \rightarrow T_{n-1}(M', N)$ deduced from $\partial(\mathcal{E}, N)$ using the previous identifications.

(If $(P, p, q) \in \mathcal{T}_n(M'', N)$, show that we have $\partial(P, p, q) = (\bar{P}, \bar{p}, \bar{q})$, where $\bar{P}$ is the complex obtained by replacing $P_{n-1}^* \rightarrow M'$ is equal to u^{n-1} , u being a morphism from P^* into the complex $(\text{Con (i)}) (n)$ such that $\pi \circ u^n = q^n$.)

¶ 7) Let A' be an (associative and unital) k -algebra and $B = A \otimes_k A'$.

a) Let M be a left A -module, Q a right A -module, M' a left A' -module, Q' a right A' -module. Let (P, p) (resp. (P', p')), resp. (R, r) be a projective resolution of the A -module M (resp. of the A' -module M' , resp. of the B -module $M \otimes_k M'$). Show that there exists a morphism of B -complexes

$$u : P \otimes_k P' \rightarrow R,$$

unique up to homotopy, such that $r \circ u = p \otimes p'$. Define, using u , a graded k -homomorphism of degree zero

$$\top : \text{Tor}^A(Q, M) \otimes_k \text{Tor}^{A'}(Q', M') \rightarrow \text{Tor}^B(Q \otimes_k Q', M \otimes_k M').$$

Show that $\top$ does not depend on the choice of resolutions P, P', R and of the morphism u . If $a \in \text{Tor}^A(Q, M)$ and $a' \in \text{Tor}^{A'}(Q', M')$, we set $\top(a \otimes a') = a \top a'$.

- b) We assume moreover that the k -modules $\mathbf{A}$ and $\mathbf{A}'$ are projective and that one has $\text{Tor}_n^k(\mathbf{M}, \mathbf{M}') = 0$ for $n > 0$. Show that $\mathbf{P} \otimes_k \mathbf{P}'$ is then a projective resolution of the $\mathbf{B}$ -module $\mathbf{M} \otimes_k \mathbf{M}'$; deduce from this, for every $\mathbf{A}$ -module $\mathbf{N}$ and for every $\mathbf{A}'$ -module $\mathbf{N}'$, a graded k -homomorphism of degree zero

$$\vee : \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}) \otimes_k \text{Ext}_{\mathbf{A}'}(\mathbf{M}', \mathbf{N}') \rightarrow \text{Ext}_{\mathbf{B}}(\mathbf{M} \otimes_k \mathbf{M}', \mathbf{N} \otimes_k \mathbf{N}').$$

Show that $\vee$ is independent of the choice of resolutions $\mathbf{P}$ and $\mathbf{P}'$. If $b \in \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N})$ and $b' \in \text{Ext}_{\mathbf{A}'}(\mathbf{M}', \mathbf{N}')$, we write $\vee(b \otimes b') = b \vee b'$.

- c) Let $\mathbf{A}''$ be a third k -algebra (associative and unital), $\mathbf{M}''$ and $\mathbf{N}''$ left $\mathbf{A}''$ -modules, $\mathbf{Q}''$ a right $\mathbf{A}''$ -module. We identify the k -algebras $\mathbf{A} \otimes_k (\mathbf{A}' \otimes_k \mathbf{A}'')$ and $(\mathbf{A} \otimes_k \mathbf{A}') \otimes_k \mathbf{A}''$, as well as the k -modules $\mathbf{M} \otimes_k (\mathbf{M}' \otimes_k \mathbf{M}'')$ and $(\mathbf{M} \otimes_k \mathbf{M}') \otimes_k \mathbf{M}''$, $\mathbf{N} \otimes_k (\mathbf{N}' \otimes_k \mathbf{N}'')$ and $(\mathbf{N} \otimes_k \mathbf{N}') \otimes_k \mathbf{N}''$, $\mathbf{Q} \otimes_k (\mathbf{Q}' \otimes_k \mathbf{Q}'')$ and $(\mathbf{Q} \otimes_k \mathbf{Q}') \otimes_k \mathbf{Q}''$. Prove the formulas

$$a \top (a' \top a'') = (a \top a') \top a'' \quad \text{for } a \in \text{Tor}^{\mathbf{A}}(\mathbf{Q}, \mathbf{M}), \quad a' \in \text{Tor}^{\mathbf{A}'}(\mathbf{Q}', \mathbf{M}'),$$

$$a'' \in \text{Tor}^{\mathbf{A}''}(\mathbf{Q}'', \mathbf{M}'')$$

$$b \vee (b' \vee b'') = (b \vee b') \vee b'' \quad \text{for } b \in \text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}), \quad b' \in \text{Ext}_{\mathbf{A}'}(\mathbf{M}', \mathbf{N}'), \\ b'' \in \text{Ext}_{\mathbf{A}''}(\mathbf{M}'', \mathbf{N}'').$$

- d) Let $\sigma : \text{Tor}^{\mathbf{A} \otimes_k \mathbf{A}'}(\mathbf{Q} \otimes_k \mathbf{Q}', \mathbf{M} \otimes_k \mathbf{M}') \rightarrow \text{Tor}^{\mathbf{A}' \otimes_k \mathbf{A}}(\mathbf{Q}' \otimes_k \mathbf{Q}, \mathbf{M}' \otimes_k \mathbf{M})$ and

$$\tau : \text{Ext}_{\mathbf{A} \otimes_k \mathbf{A}'}(\mathbf{M} \otimes_k \mathbf{M}', \mathbf{N} \otimes_k \mathbf{N}') \rightarrow \text{Ext}_{\mathbf{A}' \otimes_k \mathbf{A}}(\mathbf{M}' \otimes_k \mathbf{M}, \mathbf{N}' \otimes_k \mathbf{N})$$

the isomorphisms deduced from the commutation isomorphisms

$$A \otimes_k A' \rightarrow A' \otimes_k A, \quad M \otimes_k M' \rightarrow M' \otimes_k M, \quad N \otimes_k N' \rightarrow N' \otimes_k N, \quad Q \otimes_k Q' \rightarrow Q' \otimes_k Q.$$

Prove the formulas.

$$\begin{aligned} \sigma(a \top a') &= (-1)^{pq} a' \top a \quad \text{for } a \in \text{Tor}_p^A(Q, M), \quad a' \in \text{Tor}_q^{A'}(Q', M') \\ \tau(b \vee b') &= (-1)^{pq} b' \vee b \quad \text{for } b \in \text{Ext}_A^p(M, N), \quad b' \in \text{Ext}_{A'}^q(M', N'). \end{aligned}$$

e) Let $(\mathcal{E}) : 0 \rightarrow \mathbf{M}_2 \rightarrow \mathbf{M}_1 \rightarrow \mathbf{M}_0 \rightarrow 0$ be an exact sequence of A -modules, such that the sequence

$$(\mathcal{E} \otimes M') : 0 \rightarrow M_2 \otimes_k M' \rightarrow M_1 \otimes_k M' \rightarrow M_0 \otimes_k M' \rightarrow 0$$

be exact. Let δ (resp. Δ) denote the connecting homomorphism $\delta(Q, \mathcal{E})$ (resp. $\delta(Q \otimes_k Q', \mathcal{E} \otimes M')$). Show that one has $(\delta a) \top a' = \Delta(a \otimes a')$ for $a \in \text{Tor}^A(Q, M_0)$, $a' \in \text{Tor}^{A'}(Q', M')$.

Prove the analogous formulas for the product $\vee$.

f) If k is a field, show that $\top$ is an isomorphism of graded k -modules; the same holds for $\vee$ if, moreover, A and A' are left noetherian and the A -module M and the A' -module M' are finitely generated. g) If one identifies the modules $\text{Tor}_n^A(Q, M)$ with the sets $T_n(Q, M)$ defined in Exercise 6, prove the formula $(P, p, q) \top (P', p', q') = (P \otimes_k P', p \otimes p', q \otimes q')$.

8) We keep the notation of Exercise 7; we assume that the k -modules A and A' are projective.

a) We assume that one has $\text{Tor}_i^k(M, M') = 0$ for $i > 0$. Show that the map $b \mapsto b \vee 1_{M'}$ defines a graded k -homomorphism of degree zero from $\text{Ext}_A(M, N)$ to $\text{Ext}_B(M \otimes_k M', N \otimes_k M')$, which induces in degree zero the canonical homomorphism $b \mapsto b \otimes 1_{M'}$.

b) We assume that the k -module M' is flat. Let

$$0 \rightarrow N \rightarrow R_n \rightarrow \cdots \rightarrow R_1 \rightarrow M \rightarrow 0$$

be an exact sequence of A -modules, of class $b \in \text{Ext}_A^n(M, N)$. Show that the exact sequence of B -modules

$$0 \rightarrow N \otimes_k M' \rightarrow R_n \otimes_k M' \rightarrow \cdots \rightarrow R_1 \otimes_k M' \rightarrow M \otimes_k M' \rightarrow 0$$

has class $b \vee 1_{M'}$ in $\text{Ext}_B^n(M \otimes_k M', N \otimes_k M')$.

c) We assume that the k -modules M, M', N, N' are flat. Show that we have:

$$b \vee b' = (b \vee 1_{N'}) \circ (1_M \vee b') = (-1)^{pq} (1_N \vee b') \circ (b \vee 1_{M'})$$

for $b \in \text{Ext}_A^p(M, N)$, $b' \in \text{Ext}_A^q(M', N')$.

9) We assume the ring A is commutative. Let B, C be two associative and unital A -algebras; we denote by $m_B : B \otimes_A B \rightarrow B$ the multiplication of B , and m_C the multiplication of C .

a) Show that the homomorphism

$$\text{Tor}(m_B, m_C) \circ \tau : \text{Tor}^A(B, C) \otimes_A \text{Tor}^A(B, C) \rightarrow \text{Tor}^A(B, C)$$

(cf. exercise 7) defines on $\text{Tor}^A(B, C)$ a structure of graded A -algebra, associative and unital.

b) Let S be a differential graded A -algebra and $p : S_0 \rightarrow B$ a homomorphism of A -algebras such that (S, p) is a free resolution of B (cf. X, p. 183, exercise 18). Show that the isomorphism

$$\psi(S, C) : \text{Tor}^A(B, C) \rightarrow H(S \otimes_A C)$$

is an isomorphism of A -algebras if one endows $H(S \otimes_A C)$ with the algebra structure deduced from the differential graded algebra structure of $S \otimes_A C$ (X, p. 183, exercise 18, b) and c)).

c) If the A -algebras B and C are commutative, prove that the graded algebra $\text{Tor}^A(B, C)$ is

alternating (cf. X, p. 183, exercise 19).

10) Let B be a k -bialgebra (III, p. 148), which we regard as an augmented k -algebra (X, p. 196, exercise 27) via the counit $\beta : \mathbf{B} \rightarrow k$; we assume that the k -module B is projective.

a) Let M, N be two left B -modules. The homomorphism $\cup$ of Exercise 7 is identified with a graded homomorphism of degree zero

$$H^\bullet(\mathbf{B}, \gamma; \mathbf{M}) \otimes_k H^\bullet(\mathbf{B}, \gamma; \mathbf{N}) \rightarrow H^\bullet(\mathbf{B} \otimes_k \mathbf{B}, \gamma \otimes \gamma; \mathbf{M} \otimes_k \mathbf{N});$$

by composing with the homomorphism induced by the comultiplication, deduce from this a graded homomorphism of degree zero $\cup : H'(B, \gamma; M) \otimes_k H'(B, \gamma; N) \rightarrow H'(B, \gamma; M \otimes_k N)$.

b) Let C be an associative unital k -algebra, which we equip with the B -module structure induced by γ . Show that the homomorphism $\cup$ defines on $H'(B, \gamma; C)$ a structure of associative unital graded k -algebra; if C is commutative and the bialgebra B is cocommutative, the algebra $H'(B, \gamma; C)$ is anticommutative.

c) We take $C = k$. Show that the product $\cup$ coincides with the composition product on $\text{Ext}_{\mathbf{B}}(k, k)$. d) Let G be a group, M and N two $\mathbf{Z}^{(G)}$ -modules. We endow $M \otimes_{\mathbf{Z}} N$ with the structure of $\mathbf{Z}^{(G)}$ -module defined by $g(x \otimes y) = gx \otimes gy$ for $g \in G, x \in M, y \in N$. Show that the map

$$\cup : H^\bullet(G, M) \otimes_{\mathbf{Z}} H^\bullet(G, N) \rightarrow H^\bullet(G, M \otimes_{\mathbf{Z}} N)$$

is the homomorphism induced on homology by the graded homomorphism of degree zero

$$u : C^*(G, M) \otimes_{\mathbf{Z}} C^*(G, N) \rightarrow C^*(G, M \otimes_{\mathbf{Z}} N)$$

defined by $u(f \otimes h)(g_1, \dots, g_{p+q}) = f(g_1, \dots, g_p) \otimes g_1 \dots g_p h(g_{p+1}, \dots, g_{p+q})$ for

$$f \in C^p(G, M), \quad h \in C^q(G, N), \quad g_1 \dots g_{p+q} \in G.$$

(Use Exercise 5, p. 185.)

¶ 11) Suppose that the bigebra B admits an inversion i (cf. III, p. 198, exercise 4). Let M and N be B -modules. The k -module $M \otimes_k N$ (resp. $\text{Hom}_k(M, N)$) has a natural structure of $B \otimes_k B$ -module (resp. $B \otimes_k B^\circ$ -module); one endows it with the B -module structure deduced from the homomorphism c (resp. $(1_B \otimes i) \circ c$).

- a) Show that the homomorphism u (Exercise 10) defines a graded homomorphism of degree zero

$$\cup : H^*(\mathbf{B}, \gamma; \text{Hom}_k(\mathbf{M}, \mathbf{N})) \otimes_k H^*(\mathbf{B}, \gamma; \mathbf{M}) \rightarrow H^*(\mathbf{B}, \gamma; \mathbf{N}).$$

- b) Let $0 \rightarrow M \rightarrow R_n \rightarrow \cdots \rightarrow R_1 \rightarrow k \rightarrow 0$ be an exact sequence of B -modules, with R_i projective ($1 \leq i \leq n$); let θ be its class in $\text{Ext}_B^n(k, M) = H^n(B, \gamma; M)$. Show that for $p \geq 1$, the map $x \mapsto x \cup \theta$ is a k -isomorphism of $H^p(B, \gamma; \text{Hom}_k(M, N))$ onto $H^{p+n}(B, \gamma; N)$.

(Show that this map factors through

$$H^p(B, \gamma; \text{Hom}_k(M, N)) \rightarrow \text{Ext}_B^p(M, N) \xrightarrow{\theta} \text{Ext}_B^{p+n}(k, N).$$

- c) Let G be a group, N a $\mathbf{Z}^{(G)}$ -module. Let F be a free group, $p : F \rightarrow G$ a surjective homomorphism, $R = \text{Ker}(p)$, $\bar{R} = R/(R, R)$, $\bar{F} = F/(R, R)$. The extension $\bar{R} \rightarrow \bar{F} \rightarrow G$ defines a class $\varepsilon \in H^2(G, \bar{R})$; prove that the map $x \mapsto x \cup \varepsilon$ from $H^p(G, \text{Hom}_{\mathbf{Z}}(\bar{R}, N))$ to $H^{p+2}(G, N)$ is an isomorphism for $p \geq 1$ (use exercise 3, d), p. 179).

Exercises for Section 8: Homological Dimension

- 1) Let $u : \mathbf{A} \rightarrow \mathbf{B}$ be a ring homomorphism, M a B -module. Prove the inequality

$$\text{dp}_{\mathbf{A}}(\mathbf{M}) \leq \text{dp}_{\mathbf{B}}(\mathbf{M}) + \text{dp}_{\mathbf{A}}(\mathbf{B}_s).$$

(Reason by induction on $\text{dp}_{\mathbf{B}}(\mathbf{M})$ or use the spectral sequence of Exercise 6, p. 188.)

- 2) Assume the ring A is noetherian; let M be a finitely generated A -module of projective

dimension n . Show that the k -module $\text{Ext}_A^n(M, A)$ is nonzero.

3) Let n be an integer ≥ 1 . Show that $\text{dh}(\mathbf{M}_n(\mathbf{A})) = \text{dh}(\mathbf{A})$ (cf. VIII, § 4).

4) Let M be an A -module. One calls the injective dimension of M , and denotes it by $\text{di}_A(\mathbf{M})$, the infimum in $\bar{\mathbb{Z}}$ of the lengths of injective resolutions of M .

If n is an integer ≥ 0 , show that the following conditions are equivalent:

α) $\text{di}_A(\mathbf{M}) \leq n$;

β) $\text{Ext}_A^r(\mathbf{N}, \mathbf{M}) = 0$ for every A -module N and every integer $r > n$;

γ) $\text{Ext}_A^{n+1}(\mathbf{N}, \mathbf{M}) = 0$ for every A -module N ;

δ) $\text{Ext}_A^{n+1}(\mathbf{N}, \mathbf{M}) = 0$ for every cyclic A -module N ;

ϵ) for every exact sequence $0 \rightarrow M \rightarrow I^0 \rightarrow \dots \rightarrow I^{n-1} \rightarrow N \rightarrow 0$, where the I^k are injective, N is injective. 5) Let $(M_i)_{i \in F}$ be a family of A -modules.

a) Show that $\text{di}_A(\prod \mathbf{M}_i) = \sup \text{di}_A(\mathbf{M}_i)$.

b) If A is noetherian, show that $\text{di}_A(\bigoplus \mathbf{M}_i) = \sup \text{di}_A(\mathbf{M}_i)$.

c) If $\text{di}_A(\bigoplus_{i \in E} \mathbf{M}_i) = \sup_{i \in E} \text{di}_A(\mathbf{M}_i)$ for every family $(\mathbf{M}_i)_{i \in E}$ of A -modules, A is noetherian (use Exercise 21, p. 170).

6) Let I be the set of left ideals of A . Show that $\text{dh}(A) = \sup \text{dp}_A(A/a)$.

7) Let M be an A -module. We call the flat dimension of M , and denote it by $\text{dpl}_A(M)$, the infimum in $\bar{\mathbb{Z}}$ of the lengths of the flat resolutions of M .

a) If n is an integer ≥ 0 show that the following conditions are equivalent:

α) $\text{dpl}_A(\mathbf{M}) \leq n$.

β) $\text{Tor}_r^A(\mathbf{N}, \mathbf{M}) = 0$ for every right A -module N and every integer $r > n$.

γ) $\text{Tor}_{n+1}^A(\mathbf{N}, \mathbf{M}) = 0$ for every right A -module N .

δ) $\text{Tor}_{n+1}^A(\mathbf{N}, \mathbf{M}) = 0$ for every monogenic right A -module N .

ϵ) For every exact sequence $0 \rightarrow K \rightarrow P_{n-1} \rightarrow \dots \rightarrow P_0 \rightarrow M \rightarrow 0$ where the P_k are flat, K is flat.

- b) Show that one has $\text{dpl}_A(\mathbf{M}) \leq \text{dp}_A(\mathbf{M})$; there is equality if A is noetherian and M is finitely generated.
- c) Let $(N_\alpha)_{\alpha \in I}$ be an inductive system of A -modules and N its inductive limit. Prove the inequality

$$\text{dpl}_A(N) \leq \sup_{\alpha \in I} \text{dpl}_A(N_\alpha).$$

- 8) We call the tor-dimension of A and denote by $\text{td}(A)$ the supremum in $\overline{\mathbb{Z}}$ of the set of integers n for which there exist a left A -module N and a right A -module M such that $\text{Tor}_n^A(\mathbf{M}, \mathbf{N}) \neq 0$.

- a) Let n be an integer ≥ 0 . Show that the following conditions are equivalent:

$\alpha)$ $\text{td}(A) \leq n$.

$\beta)$ For every A -module M , we have $\text{dpl}_A(\mathbf{M}) \leq n$ (Exercise 7).

$\gamma)$ For every monogenic A -module M , we have $\text{dpl}_A(\mathbf{M}) \leq n$.

- b) Show that we have $\text{td}(A) = \text{td}(A^\circ)$ and $\text{td}(A) \leq \text{dh}(A)$; equality holds if A is left noetherian.

¶ 9) a) Show that the following conditions are equivalent:

$\alpha)$ Every finitely generated (left) ideal of A is a projective A -module.

$\beta)$ Every finitely generated submodule of a projective A -module is projective.

$\gamma)$ We have $\text{td}(A) \leq 1$ (Exercise 8), and the ring A is coherent (X, p. 181, Exercise 11).

$\delta)$ For every set I , every submodule of A^I is flat.

- b) Suppose that the ring A satisfies the equivalent conditions of a). Prove that every projective A -module is isomorphic to a direct sum of finitely generated ideals of A . (First treat the case of a finitely generated module; then treat by induction the case of a module admitting a countable family of generators, observing that every element of a projective A -module P is contained in a finitely generated direct summand of P . Deduce the general case from Kaplansky's theorem (II, p. 183, Exercise 2).)

- 10) Assume A is an integral domain; let K be its field of fractions. We say that an ideal $\mathfrak{a}$ of

$\mathbf{A}$ is *invertible* if there exist elements $x_1, \dots, x_n$ of $\mathbf{K}$ such that $x_i \mathfrak{a} \subset \mathbf{A}$ for $1 \leq i \leq n$ and $\sum x_i \mathfrak{a} = \mathbf{A}$.

- a) Show that every invertible ideal is finitely generated.
 - b) Show that a nonzero ideal of $\mathbf{A}$ is a projective $\mathbf{A}$ -module if and only if it is invertible.
 - c) Let $\mathfrak{a}$ be an invertible ideal of $\mathbf{A}$, D a divisible $\mathbf{A}$ -module. Prove that $\text{Ext}_A^i(\mathbf{A}/\mathfrak{a}, D) = 0$ for $i \geq 1$.
- 11) a) Suppose the ring $\mathbf{A}$ is an integral domain. Show that the conditions α to δ of Exercise 9 are still equivalent to the following conditions:

- ϵ) Every finitely generated torsion-free $\mathbf{A}$ -module is projective.
 - ζ) Every torsion-free $\mathbf{A}$ -module is flat.
 - η) Every submodule of a flat $\mathbf{A}$ -module is flat.
- (Use Exercise 13, p. 169.)

An integral domain satisfying these conditions is said to be Prüferian.

- b) Let B be an integral domain, the union of an increasing filtered family of Prüferian subrings. Show that B is Prüferian.
- * c) Show that the ring of algebraic integers is Prüferian. *

- 12) Suppose the ring $\mathbf{A}$ is an integral domain. Show that the following conditions are equivalent:

- α) $\text{dh}(\mathbf{A}) \leq 1$.
 - β) Every divisible $\mathbf{A}$ -module is injective.
 - γ) The ring $\mathbf{A}$ is Prüferian (Exercise 11) and noetherian.
- (To show that α) implies β), use Exercise 10.)

We then say that $\mathbf{A}$ is a Dedekind ring.

- 13) Assume $\mathbf{A}$ is left and right noetherian. Show that the following conditions are equivalent:

- α) $\text{dh}(\mathbf{A}) \leq 2$.

β) For all integers p, q and every A -linear map $u : A^p \rightarrow A^q$, the kernel of u is a projective A -module.

γ) The dual of every finitely generated A -module is projective.

14) Let x be an element of A that is left-cancellable.

a) Show that the ideal xA is two-sided if and only if there exists an endomorphism σ of the ring A such that $ax = x\sigma(a)$ for all $a \in A$.

b) We now assume that the conditions of a) are satisfied. Let M be an A -module such that $xM = 0$. Prove the equality $\text{dp}_A(M) = \text{dp}_{A/xA}(M) + 1$ (one may use the spectral sequence of Exercise 6, d), p. 188).

c) We assume that x belongs to the radical of A , and that A is noetherian. Let N be a finitely generated A -module, such that the homothety of ratio x is injective in N . Show that we have $\text{dp}_A(N) = \text{dp}_{A/Ax}(N/xN)$. d) Under the conditions of c), show that one has $\text{dh}(A) = \text{dh}(A/Ax) + 1$.

15) Let σ be an automorphism of the ring A .

a) We denote by $A_\sigma[X]$ the ring defined in VIII, § 1, no. 3. Show that

$$\text{dh}(A_\sigma[X]) = \text{dh}(A) + 1.$$

(The inequality $\text{dh}(A_\sigma[X]) \geq \text{dh}(A) + 1$ follows from Exercise 14, b); for the opposite inequality, be inspired by the proof of X, p. 142, lemma 3, a.)

b) We denote by $A_\sigma[[X]]$ the ring defined in IV, § 4, Exercise 8. Show that if A is noetherian, then we have $\text{dh}(A_\sigma[[X]]) = \text{dh}(A) + 1$ (use Exercise 14).

16) Let A be a local ring that is noetherian on the right and on the left, and let m be its maximal ideal; we denote by k_s (resp. k_d) the left (resp. right) A -module A/m . Let M be a Tor_{n+1}^A finitely generated left module and let n be an integer. a) Show that one has $\text{dp}_A(M) \leq n$ if and only if $\text{Tor}_{n+1}^A(k_d, M) = 0$ (use Exercise 4, p. 185).

- b) Show that one has $\text{dh}(A) = \text{dp}_A(k_s)$; for $\text{dh}(A) \leq n$ it is necessary and sufficient that one has

$$\text{Tor}_{n+1}^A(k_d, k_s) = 0.$$

- c) Let x be an element of the center of A belonging to m , such that the homothety with ratio x in M is injective. Show that we have $\text{dp}_A(M/xM) = \text{dp}_A(M) + 1$ (cf. also Exercise 14).
- 17) Let M be an A -module, I a well-ordered set, $(\mathbf{M}_\alpha)_{\alpha \in I}$ an increasing family of submodules of M , such that $M = \bigcup M_\alpha$. We set $M'_\alpha = \bigcup M_\beta$ for $\alpha \in I$.
- a) Prove the inequality $\text{dp}_A(M) \leq \sup \text{dp}_A(M_\alpha/M'_\alpha)$.

- b) Suppose there exists an integer n such that $\text{dp}_A(\mathbf{M}'_\alpha) \leq n$ for all $\alpha \in I$. Show that we have

$$\text{dp}_A(\mathbf{M}) \leq n + 1.$$

- 18) a) Let $(\mathbf{M}_\alpha)_{\alpha \in I}$ be an inductive system of A -modules relative to a countable set I , $\mathbf{M}$ its limit, n be an integer ≥ 0 . Show that if $\text{dp}_A(\mathbf{M}_\alpha) \leq n$ for all $\alpha \in I$, we have $\text{dp}_A(\mathbf{M}) \leq n + 1$ (reduce to the case $n = 0$ and $I = \mathbb{N}$, then write an exact sequence $0 \rightarrow \bigoplus_n \mathbf{M}_n \rightarrow \bigoplus_n \mathbf{M}_n \rightarrow \mathbf{M} \rightarrow 0$).

- b) Let P be a flat A -module admitting a presentation $A^{(\mathbb{S})} \rightarrow A^{(\mathbb{T})} \rightarrow P \rightarrow 0$, where the set S is countable. Deduce from a) that one has $\text{dp}_A(P) \leq 1$.

- c) Assume that every left ideal of the ring A is generated by a countable set of elements. Prove that $\text{dp}_A(M) \leq \text{dpl}_A(M) + 1$ for every A -module M generated by a countable family of elements. Deduce the inequalities $\text{dh}(A) \leq \text{td}(A) + 1$ and $|\text{dh}(A) - \text{dh}(A^\circ)| \leq 1$. (Use Exercise 12, p. 181, Exercise 8, p. 202, as well as b).)

- ¶ 19) a) Let n be an integer ≥ 0 , let $(M_\alpha)_{\alpha \in I}$ be an inductive system of A -modules, and let M be its inductive limit. Assume that $\text{Card}(I) \leq N^n$ (E, III, p. 87, exercise 10) and that there

exists an integer r such that $\text{dp}_A(M\alpha) \leq r$ for every $\alpha \in I$. Show that $\text{dp}_A(M) \leq r + n + 1$ (argue by induction on n , using exercise 18, a) and exercise 17).

- b) We assume that every left ideal of A is generated by a family of elements of cardinal $\leq \aleph_n$. Prove that one has $\text{dp}_A(M) \leq \text{dpl}_A(M) + n + 1$ for every A -module M generated by a family of elements of cardinality $\leq \aleph_n$. Deduce from this the inequalities

$$\text{dh}(A) \leq \text{td}(A) + n + 1 \quad \text{and} \quad |\text{dh}(A) - \text{dh}(A^\circ)| \leq n + 1.$$

¶ 20) Let R be a principal ideal ring, K its field of fractions. We denote by B the subring of $M_2(K)$ consisting of the matrices $\begin{pmatrix} a & 0 \\ x & y \end{pmatrix}$ with $a \in R, x, y \in K$.

- a) Show that B is left noetherian but not right noetherian.
b) Show that $\text{dh}(B) = 1$ and $\text{dh}(B^\circ) = 2$.

21) Let r be the radical of A . Show that the following conditions are equivalent:

- α) A/r is semisimple, and for every sequence $(a_n)_{n \geq 1}$ of elements of r , there exists an integer n such that $a_1 \dots a_n = 0$.
 β) Every A -module admits a projective cover.
 γ) For every A -module M , we have $\text{dpl}_A(M) = \text{dp}_A(M)$ (X, p.202, exercise 7).
 δ) For every inductive system $(M_\alpha, f_{\beta\alpha})_{\alpha \in I}$ of A -modules, we have:

$$\text{dp}_A(\varinjlim M_\alpha) \leq \sup_{\alpha \in I} \text{dp}_A(M_\alpha).$$

ε) Every flat A -module is projective.

φ) Every descending sequence of monogenic right ideals of A is stationary.

(For the equivalence of α) and β), cf. VIII, § 8, exercise 26. To show that β) implies γ), note that M admits a minimal projective resolution P and that P_n/rP_n is isomorphic to $\text{Tor}_n^A(A/r, M)$. To prove $\varepsilon) \Rightarrow \varphi$), let $(I_n)_{n \geq 1}$ be a decreasing sequence of monogenic right ideals, such that

$$I_n = a_1 \dots a_n A, \quad \text{with } a_i \in A.$$

In the free module $A^{(N)}$, consider the submodules N_p generated by $e_1 - a_1 e_2, \dots, e_p - a_p e_{p+1}$. Deduce from ε) that $N = \bigcup N_p$ is a direct summand in $A^{(N)}$; conclude from this that $l_{n+1} = l_n$ for n large enough.

For the implication $\varphi) \Rightarrow \alpha)$, use VIII, § 9, exercise 12.)

22) Show that the following conditions are equivalent:

- (α) Every finitely presented A -module of finite projective dimension is projective.
- (β) Every injective A -homomorphism $A^p \rightarrow A^q$ admits a retraction.
- (γ) For every finitely generated right ideal a distinct from A , there exists a nonzero element x of A such that $xa = 0$.
- (δ) Every simple right A -module of finite presentation is isomorphic to a (minimal) right ideal of A .

(Observe that condition γ) is equivalent to condition β in the case $p = 1$.)

¶ 23) Show that the following conditions are equivalent:

- (α) Every A -module of finite projective dimension is projective.
- (β) The ring A satisfies the equivalent conditions of Exercise 21, and for every finitely generated right ideal a distinct from A , there exists an element $x \neq 0$ of A such that $xa = 0$.
- (γ) The ring A satisfies the equivalent conditions of Exercise 21, and every simple A -module is a quotient of an injective A -module.

(Show that α) implies that every descending sequence of monogenic right ideals of A is stationary, by adapting the proof of Exercise 21. Then deduce from Exercise 22 the equivalence of α) and β). To see that α) implies γ), consider a projective cover P of the simple module M , an injective module I containing P , and a projective cover Q of I ; define an A -homomorphism from Q to P , and deduce from it a surjection from I to M . To show that γ) implies α), show that an A -module M such that $\text{dp}_A(M) \leq 1$ is projective by considering a projective cover $p : P \rightarrow M$ and a surjection from $\text{Ker } p$ onto a simple module.)

24) We denote by $\text{dhf}(A)$ (resp. $\text{dif}(A)$, resp. $\text{tdf}(A)$) the supremum of the projective (resp.

injective, resp. flat) dimensions of A -modules for which this dimension is finite. If A has finite homological dimension, then $\text{dhf}(A) = \text{dif}(A) = \text{dh}(A)$ and $\text{tdf}(A) = \text{td}(A)$. If $\text{td}(A) < \infty$, we have $\text{tdf}(A) = \text{td}(A)$. a) Show that $\text{tdf}(A) \leq \text{dif}(A^\circ)$; equality holds if A is left noetherian (use X, p. 189, exercise 3).

b) Assume A is left noetherian. Show that we have $\text{dhf}(A) \leq \text{di}_A(A_s)$. If $\text{di}_A(A_s)$ and $\text{dif}(A)$ are finite, show that they are equal (use Exercise 2, p. 202).

c) Let G be a finite group not reduced to the identity element, and let A be the ring $\mathbf{Z}^{(G)}$. Show that we have

$$\text{dh}(A) = +\infty, \text{dhf}(A) = \text{dif}(A) = \text{tdf}(A) = 1, \text{di}_A(A_s) = +\infty$$

(cf. exercise 20, p. 193).

d) Assume the ring A is self-injective (cf. X, p. 171, exercise 26), not semisimple. Show that every non-projective A -module has infinite projective dimension and infinite injective dimension. Deduce that one has $\text{dh}(A) = +\infty, \text{dhf}(A) = \text{dif}(A) = \text{tdf}(A) = \text{di}_A(A_s) = 0$.

25) We denote by A^e the k -algebra $A \otimes_k A^*$ and we consider A as a left A^e -module. We set

$$\text{da}_k(A) = \text{dp}_{A^e}(A).$$

If n is an integer ≥ 0 , the inequality $\text{da}_k(A) \leq n$ is therefore equivalent to the vanishing of $H^{n+1}(A, M)$ for every (A, A) -bimodule M (X, p. 195, exercise 26). One has $\text{da}_k(A) = \text{da}_k(A^\circ)$.

a) Show that one has $\text{da}_k(\mathbf{M}_n(k)) = 0$ and $\text{da}_k(k[\mathbf{X}_1, \dots, \mathbf{X}_n]) = n$ for all $n \geq 0$.

b) If k is a field, one has $\text{da}_k(A) = 0$ if and only if the k -algebra A is absolutely semisimple (cf. VIII, § 11, n° 3, th. 2).

c) Let M be a free k -module. Prove that one has $\text{da}_k(\mathbf{T}_k(M)) = 1$ (If $(e_i)_{i \in I}$ is a basis of M , prove that the elements $(e_i \otimes 1 - 1 \otimes e_i)$ form a basis of the left ideal of A^e kernel of

multiplication $\mathbf{A} \otimes_k \mathbf{A}^- \rightarrow \mathbf{A}$, with $\mathbf{A} = \mathbf{T}_k(\mathbf{M})$).

- d) We now assume that the k -module A is projective. Let $\rho : k \rightarrow k'$ be a homomorphism of commutative rings; show that one has $\text{da}_k(\mathbf{A} \otimes_k k') \leq \text{da}_k(\mathbf{A})$. If ρ is injective and if $\rho(k)$ is a direct summand of the k -module k' , one has equality (use X, p. 195, exercise 26, c)).

- e) Let B be a second k -algebra (associative and unital), projective over k . Show that

$$\text{da}_k(\mathbf{A} \otimes_k \mathbf{B}) \leq \text{da}_k(\mathbf{A}) + \text{da}_k(\mathbf{B}).$$

If moreover k is a field and A and B are finite-dimensional over k , then equality holds (use X, p. 195, exercise 26, d)).

- f) Assume that k is a field. Prove the inequalities $\text{dh}(\mathbf{A}) \leq \text{da}_k(\mathbf{A})$ and $\text{dh}(\mathbf{A}^\circ) \leq \text{da}_k(\mathbf{A})$ (cf. X, p. 196, exercise 26, e)).

- 26) Let A be an augmented k -algebra (X, p. 196, exercise 27); one assumes that there exists a homomorphism $\rho : A \rightarrow A^e$ which satisfies the conditions of X, p. 196, exercise 27, c). Show that $\text{da}_k(\mathbf{A}) = \text{dp}_A(k)$; if moreover k is a field, one has $\text{da}_k(\mathbf{A}) = \text{dp}_A(k) = \text{dh}(\mathbf{A}) = \text{dh}(\mathbf{A}^\circ)$. (Use Exercise 27, b) and c), p. 196, and Exercise 26, e), p. 196.) In particular, if V is a nonzero k -vector space, we have $\text{dh } \mathbf{T}_k(V) = 1$.

* 27) We assume that k is a field. Let g be a Lie algebra over k of dimension n , U its enveloping algebra. Show that $\text{dh } U = n$. (Use Exercise 26, Exercise 27, d), p. 196, and Exercise 28, c), p. 197.) *

¶ 28) Let G be a group. One calls the cohomological dimension of G , denoted $\text{dc}(G)$, the projective dimension of the $\mathbf{Z}^{(G)}$ -module $\mathbf{Z}$. If n is an integer, we thus have $\text{dc}(G) \leq n$ if and only if $H^q(G, M) = 0$ for every $\mathbf{Z}^{(G)}$ -module M and every $q > n$.

- a) Prove that one has $\text{da}_{\mathbf{Z}}(\mathbf{Z}^{(G)}) = \text{dc}(G)$ (use Exercise 26 and Exercise 27, d), p. 196).

- b) If G is a free group not reduced to the identity element, then $\text{dc}(G) = 1$.

- c) Let H be a subgroup of G ; prove the inequality $\text{dc}(H) \leq \text{dc}(G)$. If $\text{dc}(G) < \infty$ and if H has finite index in G , show that $\text{dc}(H) = \text{dc}(G)$ (show that for $q = \text{dc}(G)$, the homomorphism

$$i^q : H^q(H, M) \rightarrow H^q(G, M)$$

of Exercise 14, p. 191, is surjective).

- d) Show that the cohomological dimension of a finite group not reduced to the identity element is infinite. Deduce that if $\text{dc}(G) < \infty$, every element of G other than the identity element has infinite order (one then says that G is torsion-free).
- e) If G is torsion-free and if H is a subgroup of finite index in G , show that $\text{dc}(H) = \text{dc}(G)$. (If (P, p) is a projective resolution of $\mathbf{Z}^{(H)}$ -module Z and if $(G : H) = n$, define on the graded $\mathbf{Z}^{(H)}$ -module $\mathbf{T}_Z^n(P)$ a structure of $\mathbf{Z}^{(G)}$ -complex so that $(\mathbf{T}_Z^n(P), p^{\otimes n})$ is a projective resolution of the $\mathbf{Z}^{(G)}$ -module Z .)
- f) If H is a normal subgroup of G , show that we have

$$\text{dc}(G) \leq \text{dc}(H) + \text{dc}(G/H).$$

Exercises for Section 9: Koszul Complexes

- 1) Let A be a ring, L an A -module, and K the complex $S(L) \otimes_A \Lambda(L)$ (X, p. 151, example 1).
- a) Show that there exists an A -linear endomorphism s of K , graded of degree zero, such that

$$ds(x) + sd(x) = (p + q)x$$

for all integers $p, q \geq 0$ and every $x \in S^p L \otimes_A \Lambda^q L$.

- b) Assume that A is a $\mathbb{Q}$ -algebra. Show that K defines a resolution of the A -module A .
- 2) Let A be a ring, $u : L \rightarrow M$ a homomorphism of A -modules.

- a) We denote by ε the commutation factor on $\mathbf{Z}^2$ defined by $\varepsilon(\alpha, \beta) = \alpha_2 \beta_1$ for $\alpha, \beta \in \mathbf{Z}^2$, $\alpha = (\alpha_1, \alpha_2)$, $\beta = (\beta_1, \beta_2)$. Show that there exists a unique (A, ε) -derivation (III, p. 118) d of the bigraded algebra $\mathbf{S}(M) \otimes_A \Lambda(L)$ of degree $(1, -1)$, such that one has $d(1 \otimes x) = u(x) \otimes 1$ for $x \in L$. Show that $d \circ d = 0$, so that $\mathbf{S}(M) \otimes_A \Lambda(L)$, equipped with the grading induced by that of $\Lambda(L)$ and with the differential d , defines a complex $\mathbf{K}(u)$.
- b) If $M = L$ and $u = \text{id}_L$, show that $\mathbf{K}(u)$ is equal to the complex defined in no. 3, p. 151.
- c) Let $\mathbf{B} = \mathbf{S}_A(\mathbf{M})$, and let $\bar{u} : \mathbf{B} \otimes_A L \rightarrow \mathbf{B}$ be the $\mathbf{B}$ -homomorphism deduced from u . Show that $\mathbf{K}(u)$ is canonically identified with the complex $\mathbf{K}^{\mathbf{B}}(\bar{u})$.
- d) Let $u' : L' \rightarrow M'$ be a second A -homomorphism. Show that the complexes $\mathbf{K}(u \oplus u')$ and $\mathbf{K}(u) \otimes_A \mathbf{K}(u')$ are isomorphic.
- e) Let $f : L \rightarrow L'$ and $g : M \rightarrow M'$ be two A -homomorphisms such that $g \circ u = u' \circ f$. Show that the homomorphism $\mathbf{S}(g) \otimes \Lambda(f)$ is a morphism of complexes from $\mathbf{K}(u)$ to $\mathbf{K}(u')$. In particular, if $v : M \rightarrow P$ is an A -homomorphism such that $v \circ u = 0$, we obtain a morphism of complexes $h : \mathbf{K}(u) \rightarrow \mathbf{S}_A(P)$.
- f) We assume that the sequence $0 \rightarrow L \xrightarrow{u} M \xrightarrow{v} P \rightarrow 0$ is an exact sequence of flat A -modules. Show that h is an isomorphism (reduce to the case where L, M, P are finitely generated free, and use d)). Deduce from this for every $n \geq 1$ an exact sequence

$$0 \rightarrow \Lambda^n L \rightarrow M \otimes_A \Lambda^{n-1} L \rightarrow \dots \rightarrow S^{n-1} M \otimes_A L \rightarrow S^n M \rightarrow S^n P \rightarrow 0.$$

- 3) Let n be an integer, k a ring, $A = k[X_1, \dots, X_n]$. Show that for every A -module M there exists an isomorphism of graded A -modules

$$\delta_M : \text{Tor}^A(k, M) \rightarrow \text{Ext}_A(k, M)(-n)$$

such that for every homomorphism of A -modules $u : M \rightarrow N$, one has $\text{Ext}(1_k, u) \circ \delta_M = \delta_N \circ \text{Tor}(1_k, u)$. 4) Let A be a ring, L an A -module, $u : L \rightarrow A$ a linear form.

- a) Show that the complex $\mathbf{K}(u)$, endowed with the algebra structure of $\Lambda(L)$, is a graded differential A-algebra (X, p. 183, exercise 18).
- b) Let B be a differential graded A-algebra such that $B_0 = A$ and let $f : L \rightarrow B_1$ be an A-homomorphism such that $d_1 \circ f = u$. Show that there exists a unique morphism of graded differential A-algebras $F : \mathbf{K}(u) \rightarrow \mathbf{B}$ such that $F_0 = 1_A$ and $F_1 = f$.
- 5) Let A be a ring, M an A-module, $x = (x_1, \dots, x_n)$ a sequence of elements of A, $k = (k_1, \dots, k_n) \in \mathbb{N}^n$, $x^k = (x_1^{k_1}, \dots, x_n^{k_n})$. Prove that the sequence x^k is M-regular if and only if the sequence x is; in that case the A-module $M/(x^k)$ admits a composition series whose quotients are isomorphic to $M/(x)$. (Argue by induction on n .)
- 6) Let A be a ring, $x = (x_1, \dots, x_n)$ a sequence of elements of A, $\mathfrak{x}$ the ideal $\sum_i x_i A$. Let

$$0 \rightarrow \mathbf{M}' \xrightarrow{i} \mathbf{M} \rightarrow \mathbf{M}'' \rightarrow 0$$

an exact sequence of A-modules. Suppose that the sequence x is completely secant for M and that the A-modules $\mathbf{M}''/(x_1 \mathbf{M}'' + \dots + x_{i-1} \mathbf{M}'')$ are separated for the x -adic topology ($1 \leq i \leq n$). Show that the following conditions are equivalent:

- α) The sequence x is completely secant for $\mathbf{M}''$.
- β) The homomorphism $\bigoplus_r (x^r \mathbf{M}' / \mathfrak{x}^{r+1} \mathbf{M}') \rightarrow \bigoplus_r (x^r \mathbf{M} / \mathfrak{x}^{r+1} \mathbf{M})$ induced by i is injective.
- γ) The homomorphism $\mathbf{M}' / \mathfrak{x} \mathbf{M}' \rightarrow \mathbf{M} / \mathfrak{x} \mathbf{M}$ induced by i is injective.

- 7) Let A be a ring, M an A-module, $x = (x_1, \dots, x_n)$ a sequence of elements of A.

$$(X, pp. 175 - 177,$$

$$E_{pq}^2 = \text{Tor}_p^A(H_q(x, A))$$

- b) Let p be an integer $\leq n$; set $x' = (x_1, \dots, x_p)$ and $x'' = (x_{p+1}, \dots, x_n)$. Show that there exists a spectral sequence E such that $E_{pa}^2 = H_p(x', H_a(x'', M))$, converging to $H_a(x, M)$.

- 8) Let A be a ring, M an A -module, $x' = (x_1, \dots, x_p)$ and $x'' = (x_{p+1}, \dots, x_n)$ two sequences of elements of A ; we denote by x the sequence $(x_1, \dots, x_n)$, and by M' the A -module $M/(x')M$.
- a) If x' is completely secant for M and x'' completely secant for M' , show that x is completely secant for M .
- b) If the sequences x and x' are completely secant for M , show that the sequence x'' is completely secant for M' .
- c) Give an example of an A -regular sequence (x, y) contained in the radical of A , such that x is completely secant for A/yA , but y is not completely secant for A .
- 9) Let A be a ring, M an A -module, and k an integer ≥ 1 . One says that a family x of elements of A is k -secant for M if one has $H_i(x, M) = 0$ for $1 \leq i \leq k$. A family x is therefore completely secant if and only if it is k -secant for every $k \geq 1$.
- a) Let $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ be an exact sequence of A -modules. Show that if the family x is k -secant for M' and M'' , it is so for M .
- b) If x is k -secant for M and if N is a flat A -module, then x is k -secant for $M \otimes_A N$.
- c) Let $a_1, \dots, a_n$ be integers ≥ 1 . Show that a sequence $(x_1, \dots, x_n)$ is k -secant for M if and only if the same holds for the sequence $(x_1^{a_1}, \dots, x_n^{a_n})$.
- d) For every pair of integers k, n with $1 \leq k < n$, give an example of a ring A , an A -module M , and a sequence $(x_1, \dots, x_n)$ of elements of A that is k -secant but not $(k+1)$ -secant for M (use X, p. 159, remark 4).
- 10) Let A be a ring, M an A -module, $x = (x_1, \dots, x_n)$ a sequence of elements of A , p an integer $\leq n$. We write $x' = (x_1, \dots, x_p)$, $x'' = (x_{p+1}, \dots, x_n)$ and $M' = M/(x')M$.
- a) Let k, k' be two integers such that $k \leq k'$. Suppose that the sequence x' is k' -secant for M (exercise 9). Show that the sequence x is k -secant for M if and only if the sequence x'' is k -secant for M' .

b) Show that if the sequence x is 1-secant for M , then the sequence x'' is 1-secant for M' .

¶ 11) Let A be a noetherian local ring, $\mathfrak{m}$ its maximal ideal, $k = A/\mathfrak{m}$. Let M be a nonzero finitely generated A -module. The depth of M , denoted $\text{prof}(M)$, is defined as the supremum of the integers n for which there exists a sequence $(x_1, \dots, x_n)$ of elements of $\mathfrak{m}$ that is completely secant for M .

- a) Show that $\text{prof}(M) = 0$ if and only if $\mathfrak{m}$ is associated to M (X, p. 172, exercise 28), in other words, if M contains a submodule isomorphic to k (observe that an ideal contained in a union of prime ideals is necessarily contained in one of them).
- b) Show that $\text{prof}(M)$ is the infimum of the integers n for which $\text{Ext}_A^n(k, M)$ is nonzero (argue by induction on n). If $\text{prof}(M) = n$, then every sequence $(x_1, \dots, x_p)$ of elements of $\mathfrak{m}$ that is completely secant for M ($p \leq n$) can be extended to a sequence $(x_1, \dots, x_n)$ in $\mathfrak{m}$ completely secant for M . c) If $(x_1, \dots, x_k)$ is a sequence of elements of $\mathfrak{m}$ completely secant for M , show that

$$\text{prof}(M/(x_1 M + \dots + x_k M)) = \text{prof}(M) - k.$$

- d) Assume that M has finite projective dimension. Prove the equality $\text{dp}_A(M) + \text{prof}(M) = \text{prof}(A)$ (argue by induction on $\text{dp}_A(M)$).
- e) Show that $\text{prof}(A)$ is the supremum of the projective dimensions of A -modules that are finitely generated and of finite projective dimension.

¶ 12) Let A be a noetherian local ring, $\mathfrak{m}$ its maximal ideal.

- a) Show that if $0 < \text{dh}(A) < \infty$ there exists a non-zero-divisor element in $\mathfrak{m} - \mathfrak{m}^2$ (show as in Exercise 11, a) that $\mathfrak{m}$ would otherwise be associated with A whence an exact sequence

$$0 \rightarrow k \rightarrow A \rightarrow N \rightarrow 0;$$

lead to a contradiction by using Exercise 16, p. 203).

- b) For any integer n , show that $\text{dh}(A) = n$ if and only if there exists a sequence $(x_1, \dots, x_n)$ completely secant for A , generating $\mathfrak{m}$ (argue by induction on n , using a) and exercise 14, p. 203).

A local ring is said to be regular if it is noetherian and has finite homological dimension.

- 13) Let A be a regular local ring (Exercise 14) of homological dimension n ; denote its maximal ideal by $\mathfrak{m}$ and let $k = A/\mathfrak{m}$.

- a) Show that one has $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = n$.
- b) Show that the k -algebra $\bigoplus_{r \geq 0} (\mathfrak{m}^r/\mathfrak{m}^{r+1})$ is isomorphic to a polynomial algebra in n indeterminates.
- c) If $n = 0$, A is a field. If $n = 1$, show that A is a principal ideal ring * and therefore a discrete valuation ring * (show that A is an integral domain by observing that $\mathfrak{m} \cap \mathfrak{m}^n = \bigcap \mathfrak{m}^n$, whence $\bigcap \mathfrak{m}^n = 0$).
- d) Show that the ring $A[[T]]$ is regular.

- 14) Let A be a ring, and let $\mathfrak{a}$ be an ideal of A . Consider the graded alternating $(A/\mathfrak{a})$ -algebra $\text{Tor}^A(A/\mathfrak{a}, A/\mathfrak{a})$ (cf. X, p. 201, exercise 9) and the homomorphism of graded $(A/\mathfrak{a})$ -algebras $\theta : \Lambda_{A/\mathfrak{a}}(\mathfrak{a}/\mathfrak{a}^2) \rightarrow \text{Tor}^A(A/\mathfrak{a}, A/\mathfrak{a})$ deduced from the isomorphism $\mathfrak{a}/\mathfrak{a}^2 \rightarrow \text{Tor}_1^A(A/\mathfrak{a}, A/\mathfrak{a})$ (X, p. 73). Show that if the ideal $\mathfrak{a}$ is generated by a completely secant sequence in A , θ is an isomorphism.

- 15) Let A be a noetherian ring, and let $\mathfrak{a}$ be an ideal contained in the radical of A . Suppose that the $(A/\mathfrak{a})$ -module $\mathfrak{a}/\mathfrak{a}^2$ is free and that the homomorphism $\theta_2 : \Lambda_{A/\mathfrak{a}}^2(\mathfrak{a}/\mathfrak{a}^2) \rightarrow \text{Tor}_2^A(A/\mathfrak{a}, A/\mathfrak{a})$ (exercise 14) is surjective. Show that $\mathfrak{a}$ is generated by a completely secant sequence in A . (If x is a minimal family of generators of $\mathfrak{a}$, show with the aid of exercise 7 (or directly) that $\text{Coker}(\theta_2)$ is isomorphic to

$$H_1(x, A) \otimes_A A/\mathfrak{a}.$$

- 16) Let A be a local noetherian ring, $\mathfrak{m}$ its maximal ideal, $k = A/\mathfrak{m}$. Show that the homomorphism of graded k -algebras $\theta : \Lambda_k(\mathfrak{m}/\mathfrak{m}^2) \rightarrow \text{Tor}^A(k, k)$ is injective: if θ_2 is bijective, the ring A is regular (exercise 12).

(If x is a minimal system of generators of $\mathfrak{m}$, show that the complex $K(x, A)$ satisfies the conditions of exercise 14, p. 182; use exercises 4 and 15.)

¶ 17) Let A be a ring such that the A -module $A_{\mathfrak{s}}$ has finite injective dimension (X, p. 202, exercise 4; a regular local ring (exercise 12) satisfies this property).

- a) Let x be a non-zero-divisor and non-invertible element of A . Show that one has

$$\text{di}_{A/Ax}(A/Ax) = \text{di}_A(A) - 1.$$

(Let $0 \rightarrow A \rightarrow I^0 \rightarrow I^1 \rightarrow \dots$ be a minimal injective resolution (X, p. 182, exercise 13) of A ; show that the complex $\text{Hom}_a(A/Ax, I^1) \rightarrow \text{Hom}_a(A/Ax, I^2) \rightarrow \dots$ defines a minimal injective resolution of the (A/Ax) -module A/Ax .)

- b) We henceforth suppose that the ring A is local noetherian. Prove the equality $\text{di}(A) = \text{prof}(A)$ (exercise 11; reason by induction on $\text{di}(A)$).
- c) Let $(x_1, \dots, x_n)$ be a completely secant sequence for A , contained in the maximal ideal of A , with $n = \text{di}(A)$. Show that the ring A is self-injective (X, p. 171, exercise 26).

- 18) Let A be a noetherian ring, C a stable set of classes of A -modules (X, p. 40), $x = (x_1, \dots, x_n)$

a sequence of elements of A , x the ideal generated by x . We denote by $\mathcal{C}_x$ the set of elements of $\mathcal{C}$ which are annihilated by x . Let M be a finitely generated A -module.

- a) Prove that for every $i \geq 0$, there exists an element $\mathbf{M}_i$ of $\mathcal{C}_x$ such that one has

$$\sum_{k \geq 0} (-1)^k [H_{i+k}(x, M)] = [\mathbf{M}_i]$$

in $\mathbf{K}(\mathcal{C}_x)$ (reason by induction on n , using prop. 4, p. 157; in the case $n = 1$, consider the successive powers of the endomorphism $(x_1)_{\mathbb{M}}$).

- b) We assume that $\mathbf{A}/x$ has finite length. Show that there exist positive integers m_i ($i \geq 0$) such that one has $\text{length } H_i(x, \mathbf{M}) = m_i + m_{i+1}$.

Index of Notations

$(C, d) : \text{p.23.}$

$C_n, C^n : \text{p.24.}$

$\mathbf{Z}(C, d), \mathbf{Z}_n(C, d), \mathbf{B}(C, d), \mathbf{B}_n(C, d), \mathbf{H}(C),$

$\mathbf{H}_n(C, d), \mathbf{H}^n(C, d) : \text{p.25.}$

$\mathbf{H}(u), \mathbf{Z}(u), \mathbf{B}(u) : \text{p.26.}$

$\mathbf{C}(p) : \text{p.26.}$

$\mathbf{A}(\varepsilon) : \text{p.27.}$

$\partial_{u,v}, \partial : \text{p.29.}$

$\text{Con}(u), \text{Cyl}(u) : \text{p.36.}$

$\mathbf{K}(\mathcal{C}), [\mathbf{M}]_{\mathcal{C}}, [\mathbf{M}] : \text{ p. 40.}$

$\chi_{\varphi}(\mathbf{M}), \chi(\mathbf{M}) : \text{ p. 41.}$

$\Omega_{A/k}^p : \text{ p. 43.}$

$\nabla : \text{ p.45.}$

$\mathbf{L}(\mathbf{M}), \mathbf{L}(f) : \text{ p.50.}$

$\mathbf{I}(\mathbf{M}), \mathbf{I}(f) : \text{ p.52.}$

$\mathbf{B}(\mathbf{A}, \mathbf{M}) : \text{ p.57.}$

$\mathbf{C} \otimes_{\mathbf{A}} \mathbf{C}' : \text{ p. 61.}$

$\gamma(\mathbf{C}, \mathbf{C}') : \mathbf{H}(\mathbf{C}) \otimes_{\mathbf{A}} \mathbf{H}(\mathbf{C}') \rightarrow \mathbf{H}(\mathbf{C} \otimes_{\mathbf{A}} \mathbf{C}') :$
 p.62.

$\sigma(\mathbf{C}, \mathbf{C}') : \text{ p. 62.}$

$\text{Tor}_n^{\mathbf{A}}(\mathbf{M}, \mathbf{N}) : \text{ p. 67.}$

$\text{Tor}^{\mathbf{A}}(f, g) : \text{ p. 68.}$

$$\psi_{\mathbf{M}}(\mathbf{N}), \overline{\psi}_{\mathbf{N}}(\mathbf{M}) : \text{p. 69.}$$

$$\sigma_{\mathbf{M},\mathbf{N}} : \text{p.71.}$$

$$\text{Homgr}_{\mathbf{A}}(\mathbf{C}, \mathbf{C}') : \text{p. 81.}$$

$$\lambda(\mathbf{C}, \mathbf{C}') : \text{p.82.}$$

$$\text{Ext}_{\mathbf{A}}(\mathbf{M}, \mathbf{N}), \text{Ext}_{\mathbf{A}}^n(\mathbf{M}, \mathbf{N}) : \text{p.86.}$$

$$\varphi_{\mathbf{M}}(\mathbf{N}), \overline{\varphi}_{\mathbf{N}}(\mathbf{M}) : \text{p.88.}$$

$$\delta(\mathbf{M}, \mathcal{E}) : \text{p.90.}$$

$$\delta(\mathcal{F}, \mathbf{N}) : \text{p.92.}$$

$$\psi(\mathbf{S}, \mathbf{R}), \varphi(\mathbf{R}, \mathbf{E}) : \text{p.100.}$$

$$\psi'(\mathbf{S}_1, \mathbf{R}_1), \varphi'(\mathbf{R}_1, \mathbf{E}_1) : \text{p.102.}$$

$$\mathbf{H}^n(\mathbf{G}, \mathbf{M}), \mathbf{H}_n(\mathbf{G}, \mathbf{M}) : \text{p.111.}$$

$$a_{\mathbf{M},\mathbf{N}} : \text{p.113.}$$

$c_{\mathbf{M},\mathbf{N},\mathbf{P}}, u \circ v$: p.114.

$\delta^m(\mathbf{P}, \mathcal{S}), \delta^m(\mathcal{S}, \mathbf{P}), \partial^m(\mathbf{P}, \mathcal{S}), \partial^m(\mathcal{S}_1, \mathbf{M})$

127, 131, 132.

$c_{\mathbf{P},\mathbf{Q};\mathbf{M}}$: p.128.

$c_{\mathbf{P};\mathbf{M},\mathbf{N}}$: p.129.

$\text{dp}_{\mathbf{A}}(\mathbf{M})$: p.134.

$\text{dh}(\mathbf{A})$: p.138.

$\mathbf{K}(u), \mathbf{K}.(u, \mathbf{C}), \mathbf{K}'(u, \mathbf{C})$: p. 147.

$b \vee b', a \top a'$: p.200, exerc. 7.

$\text{di}_{\mathbf{A}}(\mathbf{M})$: p.202, exerc. 4.

$\text{dpl}_{\mathbf{A}}(\mathbf{M})$: p. 202, exerc. 7.

$\text{td}(\mathbf{A})$: p.202, exerc.8.

$\mathrm{da}_k(A) : \text{p.205, exerc.25.}$

$\mathrm{dc}(\mathbf{G}) : \text{p.206, exerc.28.}$

Terminological Index

Absolutely flat (ring —): p. 169, exerc. 16. Acyclic in degree n (complex —): p. 26. x -adic (topology —): p. 159. Augmented algebra: p. 196, exerc. 27. Differential graded algebra: p. 183, exerc. 18. Ascending (degree —): p. 23. Associated (class —) to an exact sequence: p. 117. Augmentation of an algebra: p. 196, exerc. 27. Self-injective (ring): p. 171, exerc. 26. Bicomplex: p. 174, exerc. 11. Bounded above, below: p. 24. Canonical (injective resolution —): p. 52. Canonical (free resolution —): p. 50. Euler-Poincaré characteristic: p. 40. Cellular (complexes —): p. 32. Golod-Shafarevich (theorem of —): p. 192, exerc. 16. Five (lemma of the —): p. 7. Universal coefficients (formula of —): p. 78, 99. Cogenerator (module —): p. 18. Coherent on the left, on the right (ring —): p. 181, exerc. 11. Coherent, pseudo-coherent (module —): p. 181, exerc. 10. Cohomology of groups: p. 111. Cohomological (dimension —): p. 206, exerc. 28. Cohomologous: p. 194, exerc. 23. Co-induced (module —): p. 190, exerc. 12. Commutation (isomorphism of —): p. 71. Completely secant (family —): p. 157. Complexes: p. 23. Koszul complexes: p. 147, 154. Composition (product of —): p. 113. Cone of a morphism of complexes: p. 36. Conical (simplicial scheme —): p. 177, exerc. 19. Connection: p. 45. Convergent (spectral sequence —): p. 175, exerc. 14. Coregular (sequence —): p. 159. Cylinder of a morphism of complexes: p. 36. Dedekind (rings of —): p. 141, 203, exerc. 12. De Rham (complex of —): p. 43. Descending (degree —): p. 23. Commutative diagrams: p. 1. Differential: p. 24. Differential (algebra —): p. 183, exerc. 18.

Exterior differential: p. 44.

Cohomological dimension: p. 206, exerc. 28.

Homological dimension: p. 138.

Injective dimension: p. 202, exerc. 4.

Flat dimension: p. 202, exerc. 7.

Projective dimension: p. 134.

Divisible (module —): p. 17.

Dualizing (module —): p. 172, exerc. 30.

Injective envelope: p. 19.

Euler-Poincaré (characteristic of —): p. 40.

Extensions (modules of —): p. 81, 86.

Extension of an algebra: p. 197, exerc. 1.

Extension of a Lie algebra: p. 198, exerc. 4.

Faithfully flat (module —): p. 169, exerc. 9.

Differential forms: p. 44.

Goldie (theorem of —): p. 171, exerc. 25.

Golod-Chafarevitch (theorem of —): p. 192, exerc. 16.

Hilbert (syzygy theorem of —): p. 146.

Homology (classes of —, module of —) : p. 25.

Homology (exact sequence of —): p. 30.

Group homology: p. 111.

Homological (dimension —): p. 138.

Homologism: p. 26.

Homologous (cycles —): p. 25.

Homomorphisms (complexes of —) : p. 81.

Homotopic to zero (complex —): p. 34.

Homotopies (morphisms —): p. 32.

Homotopic (simplicially —): p. 177, exerc. 19.

Homotopy: p. 32.

Homotopism: p. 33.

Induced (module —): p. 190, exerc. 12.

Injective (modules —): p. 15.

Injective (dimension —): p. 202, exerc. 4.

Iterates (connecting homomorphisms —): pp. 127, 131, 132.
 Koszul (complex of —): pp. 147, 154.
 Künneth (formula of —): p. 79.
 Connecting (homomorphism of —): p. 29.
 Connecting (homomorphism of —) of extension modules: pp. 90, 92.
 Connecting (homomorphism of —) of torsion products: p. 72.
 Length of a complex: p. 24.
 Length (of a presentation): p. 180, exerc. 6.
 Minimal (injective resolution —): p. 182, exerc. 13.
 Minimal (projective resolution —): p. 54.
 Morphism of complexes: p. 25.
 Zero on the right, zero on the left: p. 24.

Flat (absolutely —): p. 169, exerc. 16.
 Flat (faithfully —): p. 169, exerc. 9.
 Flat (dimension —): p. 202, exerc. 7.
 Flat (modules —): p. 8.
 Poincaré (characteristic of Euler- —): p. 40.
 Poincaré polynomial: p. 41.
 Precomplex: p. 174, exerc. 13.
 Presentation, finite presentation: p. 10.
 Presentation of length n : p. 180, exerc. 6.
 Composition product: p. 113,
 Torsion product: pp. 61, 67.
 Projective (dimension —): p. 134.
 Prüferian (ring —): p. 203, exerc. 11.
 Pseudo-coherent (module —): p. 181, exerc. 10.
 Regular (local ring —): p. 208, exerc. 12.
 Regular (sequence —): p. 158.

Regular (spectral sequence —): p. 175, exerc. 14.
 Relatively projective, injective: p. 170, exerc. 19.
 Resolutions: p. 48.
 Canonical injective resolution: p. 52.
 Canonical free resolution: p. 50.
 Graded resolution: p. 56.
 Minimal projective resolution: p. 54.
 Projective, injective resolution of a complex: p. 183, exerc. 17.
 Simplicial scheme: p. 177, exerc. 19.
 Split (completely —): p. 35.
 Splitting: p. 35.
 Secant (completely —): p. 157.
 Snake (— diagram): p. 3.
 Simplex: p. 178, exerc. 19.
 Simplicial (— scheme): p. 177, exerc. 19.
 Simplicially homotopic: p. 177, exerc. 19.
 Singular (— homology): p. 31.
 Spectral (— sequence): p. 175, exerc. 14.
 Standard (— resolution): p. 57.
 Stationary (— spectral sequence): p. 175, exerc. 14.
 Exact sequence of homology: p. 30.
 Spectral sequence: p. 175, exerc. 14.
 Syzygies (theorem of —): p. 146.
 Tensor (— product) of two complexes: p. 61.
 Torsion (— product): p. 61, 67.
 Tor-dimension: p. 202, exerc. 8.
 Topology x -adic: p. 159.
 $(p$ -th) Translate: p. 27.
 Transfer: p. 191, exerc. 14.

Trivial (cohomologically —): p. 192, exerc. 18.

Universal (— coefficients): p. 78, 99.